

An entanglement perspective on generalized symmetries, dualities and anomalies

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Summary

Samenvatting

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Prologue

The quantum many-body problem stands as one of the central problems of physics of the last century. It pertains the description of the collective behaviour of a macroscopic number of quantum constituents. Despite the relative simplicity of the one-particle Schrödinger equation, the exponential growth of the many-body Hilbert space poses a fundamental challenge to analytical and computational approaches, rendering exact solutions essentially intractable for all but the smallest system sizes.

The significance of the many-body problem can hardly be overstated as it lies at the heart of quantum field theory, condensed matter physics, quantum chemistry and quantum computation. Any progress on this front therefore carries implications across a wide spectrum of fields.

A crucial observation however is that the relevant states in the many-body Hilbert space occupy only a tiny manifold, so that a physically motivated approach to the many-body problem should aim to efficiently parametrise this tiny low-complexity manifold and develop methods to project and diagonalise the Hamiltonian within it. This philosophy exactly underlies the tensor network approach, which forms the central technical toolbox of this thesis.

1.1 The exponential wall

The hallmark of a quantum many-body system is the *tensor product structure* of its Hilbert space. By this it is meant that the full many-body Hilbert space $\mathcal{H}$ can be written as a tensor product of constituent spaces hosting localised *quantum modes* or *particles*:

$$\mathcal{H} = \bigotimes_{i=1}^N \mathcal{H}_i. \quad (1.1)$$

Throughout, we will assume the local Hilbert space dimension $\dim \mathcal{H}_i$ to be finite, which is naturally true in many cases of interest for example when $\mathcal{H}_i$ is the Hilbert space of a qubit $\mathbb{C}^2$. Even when $\mathcal{H}_i$ is inherently infinite-dimensional such as in the case of bosonic modes, a truncation of the local basis is often permitted for many physically relevant systems.

Provided an (orthonormal) basis for the local modes, $\mathcal{H}_i := \text{Span}_{\mathbb{C}}\{|i\rangle_i \mid i = 1, 2, \dots, d_i\}$, let us consider the (orthonormal) tensor product basis for $\mathcal{H}$ denoted by

$$|i_1, i_2, \dots, i_N\rangle := |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle. \quad (1.2)$$

In this basis, a many-body wave function $|\Psi\rangle$ can be expressed as

$$|\Psi\rangle = \sum_{\{i_j\}} \Psi_{i_1, i_2, \dots, i_N} |i_1, i_2, \dots, i_N\rangle, \quad (1.3)$$

with amplitudes $\Psi_{i_1, i_2, \dots, i_N} := \langle i_1, i_2, \dots, i_N | \Psi \rangle$. Notably, a generic wave function is specified by a number of complex coefficients $\Psi_{i_1, i_2, \dots, i_N}$ that scales *exponentially* in the number of constituent particles. This dramatic scaling is reminiscent of the exponential growth of the number of classical bit strings with the total number of bits. However, contrary to the classical case, most many-body states cannot be created by any reasonable physical mechanism due to the *superposition principle* of quantum mechanics, which leads to the existence of highly *entangled* many-body states. Consequently, the exact treatment of even a moderate amount of qubits very quickly becomes computationally intractable.

A precise mathematical characterisation of entanglement is as follows. Consider a bipartition of the Hilbert space in a subsystem A and its complement $\bar{A}$, i.e. $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$, and write $d_A = \dim(\mathcal{H}_A)$, $d_{\bar{A}} = \dim(\mathcal{H}_{\bar{A}})$. By the *singular value decomposition* (SVD), any state in $\mathcal{H}$ can be written in the form

$$|\Psi\rangle = \sum_{k=1}^r s_k |\psi_k^A\rangle \otimes |\psi_k^{\bar{A}}\rangle, \quad (1.4)$$

where $|\psi_k^A\rangle$ and $|\psi_k^{\bar{A}}\rangle$, $r \leq \min(d_A, d_{\bar{A}})$, are orthonormal bases for $\mathcal{H}_A$ and $\mathcal{H}_{\bar{A}}$ respectively. The non-negative numbers s_k for $k = 1, 2, \dots, r$ are called *singular values* or *Schmidt numbers*. The state $|\Psi\rangle$ is said to be *entangled* among the parties A and $\bar{A}$ when the Schmidt rank r is strictly larger than 1, and to be *maximally entangled* when all singular values are equal (and non-zero). The state $|\Psi\rangle$ is thus *unentangled* if and only if it is a (tensor) product state $|\Psi\rangle = |\psi^A\rangle \otimes |\psi^{\bar{A}}\rangle$.

Information about the entanglement can also be compressed into one number, the *entanglement entropy* S_A , which is defined as the *Von Neumann entropy* of the reduced density matrix ρ_A (or equivalently $\rho_{\bar{A}}$), and which can simply be expressed as:

$$\rho_A = \text{Tr}_{\bar{A}}(|\Psi\rangle\langle\Psi|) = \sum_k s_k^2 |\psi_k^A\rangle\langle\psi_k^A|, \quad (1.5)$$

such that:

$$S_A = -\text{Tr}(\rho_A \log(\rho_A)) = -\sum_k s_k^2 \log(s_k^2). \quad (1.6)$$

Two of the central results in entanglement theory state that any pair of bipartite states which have the same amount of entanglement, as measured by the entanglement entropy, can be converted into each other by means of only local operations on either subsystem, supplemented with classical communication; moreover any (partially) entangled bipartite state can be created from a maximally entangled resource state such as a spin singlet [1].

More refined information about the entanglement structure of the state is encoded into the *entanglement spectrum* of the state, defined as the eigenvalues $\{s_k^2\}_k$ of the reduced density matrix ρ_A . For many cases of interest — notably the low-energy states of *local gapped* Hamiltonians — the rapid decay of the entanglement spectrum is precisely what permits an efficient *tensor network* parametrisation of the state $|\Psi\rangle$. As they form the technical bedrock of this thesis, chapter 2 is devoted to introduce and discuss them.

From the expression eq. (1.5) we can also conclude that ρ_A only describes a pure state, i.e. $\text{rank}(\rho_A) = 1$, when $|\Psi\rangle$ is a product state; or, in other words, that the subsystem A behaves as a classical mixture if the state $|\Psi\rangle$ is entangled. This neatly illustrates Schrödinger’s observation that [...] *the best possible knowledge of a whole does not necessarily include the best possible knowledge of all its part* [...] [2].

Having defined the notion of entanglement, we can make precise the above claim that most states in the quantum many-body Hilbert space are physically inaccessible. Focussing on the Hilbert space of N qubits, let us consider to this end the following question put forward in ref. [3]: what portion of the many-body Hilbert space can be prepared and explored by means of physical mechanism, specifically by time-evolution under a local time-dependent Hamiltonian for a time that scales at most polynomially in the system size? Following ref. [3], any such Hamiltonian is approximated to arbitrary precision by a polynomial-depth quantum circuit consisting of few-qubit gates drawn from a universal gate set, and ε -balls around the outputs

of these circuits are considered. In this context, an ε -ball around a certain state $|\Psi_0\rangle$ is defined as the collection of all states $|\Psi\rangle$ such that

$$|\langle\Psi|\Psi_0\rangle| \geq 1 - \varepsilon. \quad (1.7)$$

A straightforward counting argument demonstrates that such balls occupy a fraction of the many-body Hilbert space that scales as $\mathcal{O}(\varepsilon^{2^N})$, doubly exponential in ε !

From these considerations we must — perhaps quiet shockingly — conclude that the vast majority of states in the Hilbert space should be deemed unphysical in any operationally meaningful sense. Indeed, there is an overwhelming time barrier to prepare a generic state consisting of any more than a handful of constituent particles.

Any physically motivated approach to the many-body problem should therefore consist of identifying the relevant low-complexity *manifold of states* and developing the techniques needed to deal with them. Perhaps the most well-known such manifold is that of *Slater determinants* of fermions. This parametrisation and its associated self-consistent Hartree-Fock method has been extremely successful for almost-free fermion systems and has elucidated the structure of the periodic table of elements. However, when correlations are strong, perturbative approaches such as Hartree-Fock fail to capture the relevant physics. It turns out that for systems satisfying an *area law* of entanglement entropy, the natural low-complexity manifold is precisely that of tensor network states. In particular, such an area law is known to hold for *local gapped* lattice models.

1.2 Quantum lattice models

Quantum lattice models appear ubiquitously as an effective, compressed description of the low-energy physics of a condensed matter system or as spatial discretisation of a quantum field theory. In the context of this thesis we will roughly think of a *lattice* Λ as a discretisation of the spatial manifold on which the model is defined. It consists of a collection of vertices, edges, plaquettes, ... that we will denote by $V(\Lambda)$, $E(\Lambda)$, $P(\Lambda)$, ...

Provided a lattice, the many-body Hilbert space eq. (1.1) is constructed by assigning a local Hilbert space to the vertices, edges, plaquettes or combination thereof. The model itself is then defined through a microscopic Hamiltonian H consisting of *geometrically local* terms so that it can be written as $H = \sum_i h_{i,K}$ where $h_{i,K}$ denotes a term acting non-trivially on at most K sites in the neighbourhood of site i . Moreover, such Hamiltonians typically depend on a

small number of tunable parameters that are meant to mimic among other things external magnetic fields and chemical potentials.

Throughout this thesis we are primarily interested in the ground state and low-lying excited states of local lattice models in one and two spatial dimensions. Our focus on the low-energy spectrum is physically motivated by the observation that the Fermi temperature of valence electrons in materials of interest are of the order $10^4 K$, well above the temperature scale of experiments. But also in the perturbative computation of cross sections in quantum field theory, the ground state is usually the reference state on top of which excitations are created.

Finding the (approximate) ground state of a generic quantum lattice model is notoriously difficult due to the impossibility to minimise the energy of all Hamiltonian terms simultaneously [4]. This phenomenon is referred to as *quantum frustration* and is intimately related to the concept of *monogamy of entanglement*. As a paradigmatic example, consider the antiferromagnetic spin 1/2 Heisenberg chain:

$$H = \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \quad (1.8)$$

with $\mathbf{S} = (S^x, S^y, S^z)$ the $\mathfrak{su}(2)$ generators in the spin 1/2 representation. On two sites, the unique ground state is the (maximally entangled) spin singlet $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. One might therefore hope that the ground state on any chain length is one for which all reduced density matrices $\rho_{i,i+1}$ are equal to a spin singlet. Such a state would indeed clearly minimise all exchange terms $\mathbf{S}_i \cdot \mathbf{S}_{i+1}$. However, such a state does not exist in virtue of monogamy of entanglement which stipulates that a maximally entangled pair of particles cannot be entangled with a third one [5, 6]. Instead, each spin must optimally distribute its entanglement among its neighbours, and it is precisely this frustration that gives rise to a strongly correlated ground state characterised by intricate entanglement patterns.

The nature of these entanglement structures depends strongly on whether the model is gapped. A model is said to possess an (*energy*) *gap* if the excitation energy of the lowest-lying excited state(s) is non-vanishing in the thermodynamic limit $N \rightarrow \infty$. Such a system is characterised by exponentially decaying *connected correlation functions* to which a finite *correlation length* ξ can be associated, viz.:

$$\langle O(\mathbf{r})O(\mathbf{0}) \rangle_{\text{conn.}} \sim \exp(-|\mathbf{r}|/\xi), \quad (1.9)$$

for a local operator O . The entanglement structure of gapped systems differs vastly from that of arbitrary states in the many-body Hilbert space. In virtue of the *locality* of physical interactions, gapped ground states obey an *area law* of their entanglement entropy. In generic spatial dimensions, the area law of entanglement entropy stipulates that the von Neumann entropy between any region A and its complement does not scale as the volume of its interior — as could be expected for generic quantum states — but rather by the surface area of its boundary ∂A . Specialising to the case of one-dimensional chains, the area of a contiguous region is simply a constant and for this case the area law was formally proven by Hastings by a clever application of Lieb-Robinson bounds [7, 8].

The low-lying spectrum on top of gapped ground states likewise exhibit a characteristic structure. Elementary excitations are typically organised in a few well-defined isolated branches of *quasiparticle* excitations with well-defined energy and momentum [9–13]. At higher energies, quasiparticle combine into *scattering states* whose energy is bounded from below by the two-particle excitation energy.

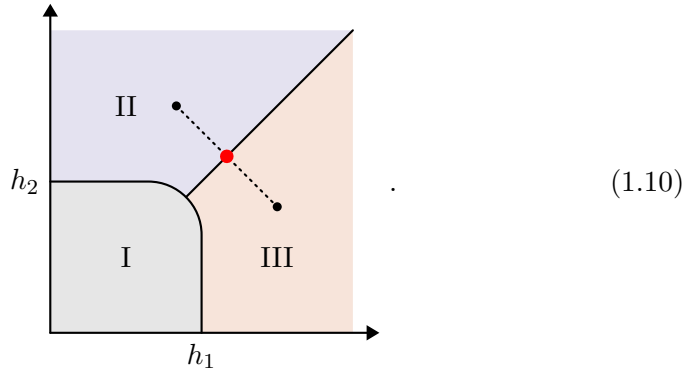
1.3 Symmetries and quantum phases of matter

The central challenge in the many-body problem is to determine and characterise the *phase diagram* of a given microscopic model, provided a *symmetry* of the model.

A symmetry is a collection of transformations that leave invariant the dynamics of the model. A standard distinction is made between *internal* and *spacetime* symmetries. Spacetime symmetries act on the lattice itself — for instance via spatial translations or reflections — or as time-translation or time-reversal. The one-dimensional spacetime symmetries are the topic of ???. In what follows, we restrict attention to internal symmetries, which act directly on the quantum degrees of freedom rather than on the underlying geometry. Concretely, we here assume the internal symmetries to be implemented *globally* via a unitary *on-site* (tensor product) representation of a group G . Crucially, the ground state projector commutes with the symmetry. The ground state space however need not be trivial: even though the ground state space may be one-dimensional, in which case it is *symmetric*, it could also transform in a higher-dimensional representation of G , which signals *spontaneous symmetry breaking* of G . In that case, a reference ground state is stabilised by a maximal subgroup $H \subseteq G$, and one speaks of the *symmetry-breaking pattern* $G \mapsto H$.

A gapped *quantum phase of matter* can roughly be understood as an equivalence class of gapped Hamiltonians whose properties vary smoothly with

the Hamiltonian parameters. More precisely, two models are said to belong to the same phase when they can be interpolated by a smooth path of gapped, bounded-strength, local Hamiltonians that commute with the symmetry group G . From the point of view of quantum information theory, this is equivalent to the existence of a symmetric quantum circuit of sub-linear depth that maps their ground states into each other [14, 15]. Distinct gapped phases are separated by *continuous phase transitions* characterised by *algebraic* decay of correlation functions whose universal scaling behaviour is captured by a *conformal field theory* [16]. For a Hamiltonian with two controllable parameters, a fictitious phase diagram with three gapped phases is sketched below. The dotted line represents a Hamiltonian path interpolating between phases II and III. Along this path the gap closes at the phase boundary.



Built upon intuition about classical many-body systems, *Landau's paradigm* has since long provided the unifying framework of phases and phase transition. This framework states that phases of matter are in one-to-one correspondence with symmetry-breaking patterns of the symmetry. Moreover, the paradigm posits that phase transitions can be diagnosed by local *order parameters* which are charged under the symmetry group that acquire a non-vanishing vacuum expectation value in the symmetry-broken phase. Since Kadanoff and Ceva [17] it has also been understood that *symmetric* phases are analogously recognised via *disorder parameters*.

The canonical example is the one-dimensional transverse field *Ising model*, governed by the Hamiltonian [18]

$$H = -J \sum_i Z_i Z_{i+1} - \lambda \sum_i X_i, \quad (1.11)$$

which possesses a global $\mathbb{Z}_2$ spin-flip symmetry generated by $\prod_i X_i$. In the ordered phase ($J > \lambda$), the spontaneous breaking of this symmetry is

characterised by a non-vanishing expectation value of the local magnetisation $\langle Z_i \rangle \neq 0$, which serves as the local order parameter¹. In the disordered phase ($\lambda > J$), the symmetry is restored and $\langle Z_i \rangle = 0$, but now the disorder parameter $\langle \prod_{i=k}^l X_i \rangle$ attains a non-zero vacuum expectation value.

Since its conception, quantum phases of matter that defy Landau's paradigm have emerged. Perhaps unsurprisingly, these *new* phases are characterised by the distinct ways in which the symmetry can manifest itself in the entanglement patterns of the ground state [19, 20]. Given the prominent role of some of these phases throughout this thesis, let us take some time here to discuss them and some of their properties.

In the presence of an on-site group symmetry, gapped *symmetry-protected topological* or *trivial* (SPT) phases are characterised by the uniqueness of the ground state on periodic boundary conditions and are adiabatically connected to product states, when the symmetry is explicitly broken [21, 22]. On open boundary conditions they exhibit gapless modes that are exponentially localised near the boundaries, and transform *projectively* under the symmetry group. In one dimension, they can be completely described within the framework of one-dimensional tensor network states or *matrix product states*, and we will revisit their classification and physical properties within this framework in section 2.4

Unlike SPT phases, (*intrinsic*) *topological phases* exist only in two or more spatial dimensions, and do not require the presence of a (physical) symmetry. They are characterised by a ground state degeneracy that depends solely on the topology of the spatial manifold and their gapped excitations exhibit *anyonic* braiding statistics. A particular class of topological phases, which admit a microscopic lattice realisation in terms of *string-net models*, form the topic of chapter 3, and are also naturally described in terms of tensor networks. These tensor networks host freely deformable symmetries that act only on the *entanglement degrees of freedom*. Such symmetries furnish representations of *fusion categories*, thereby extending familiar notions from finite group and representation theory to algebraic structures whose multiplication laws do not generally admit inverses.

1.4 Outline of this thesis

¹One typically defines this order parameter by introducing a small longitudinal field $h \sum_i Z_i$ into the Hamiltonian, whose role is to select one of the two symmetry-broken ground states depending on the sign of h , which is then taken to zero, $h \mapsto 0$, after going to the thermodynamic limit $N \mapsto +\infty$.

I

Foundations

2

Tensor networks

Having established the importance of the area law for gapped quantum many-body systems, we now introduce *tensor networks* as the main tool used throughout this thesis. In brief, tensor networks are a class of many-body trial wave functions that due to their structure inherently satisfy an area law for the entanglement entropy with respect to any bipartition. Parametrised with only a polynomial number of variational parameters, they also provide an exponentially more compressed description of the many-body wave functions of interest.

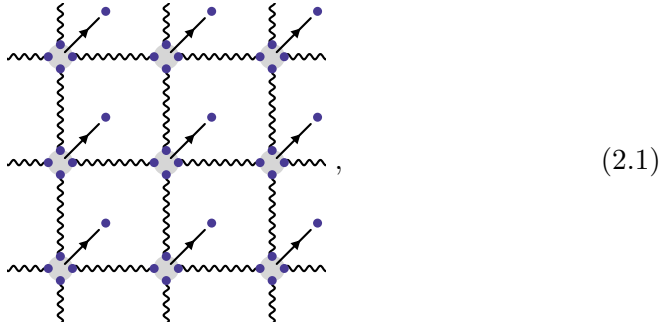
At their core, tensor networks bring a holographic approach to the many-body problem in which the information of the wave function is encapsulated in local constituent tensors encoding both the *physical* and *entanglement* degrees of freedom. All the information — local quantities such as the energy density, as well as non-local ones such as the correlation functions — is stored locally into these tensors. Moreover, the parametrisation of states and operators in terms of tensor networks also allows one to formulate them directly in the thermodynamic limit.

Crucial for our purposes, all symmetries of a given model — both lattice and internal — can typically be easily incorporated in tensor network ansätze. Enforcing symmetry significantly improves both the stability and implementation of numerical algorithms. Moreover, it facilitates the classification of gapped phases of matter. Indeed, in the language of tensor networks, phase classification often effectively boils down to classifying the inequivalent symmetry actions on the constituent tensors. But there is more: in recent years it has been understood that in the absence of any physical symmetry, *intrinsic topological order* can be characterised by the symmetry structure of the entanglement patterns of their ground states. In this way, tensor networks

reveal the *hidden* topological symmetries that characterise the long-range entanglement present in these gapped phases. The algebraic structure of these virtual symmetries is naturally described by a (higher) fusion category, and is inherited by their one-lower-dimensional edge theories. The description of the virtual MPO symmetries of two-dimensional topological orders and their edge Hamiltonians is the content of chapter 3 and chapter 4 respectively.

2.1 Tensor network states

The main lesson of the area law is that distribution of entanglement happens via the boundary of spatial regions. Trial wave functions whose entanglement entropy exhibit such a pattern, and at the same time respect all lattice symmetries can be constructed via a *projected entangled-pair construction*, as seemingly first advocated in ref. [23]. In this framework, maximally entangled pairs of *virtual* spins are first distributed between all pairs of neighbouring spins according to the connectivity of the lattice. Subsequently, one ‘*projects*’¹ onto the physical degrees of freedom of the model by means of a linear *tensor map* on each site. These tensors contain a number of degrees of freedom which play the role of *variational parameters*. Schematically, this construction can be visualised on the two-dimensional square lattice as:



where the nodes connected by wiggly lines represent the virtual pairs, and the arrows denote the local linear projection maps. The virtual degrees of freedom are associated to a Hilbert space $\mathbb{C}^\chi$, where χ is called the *bond dimension*. Though illustrated here for the two-dimensional square lattice, this construction generalises straightforwardly to any dimension, lattice geometry, choice of boundary conditions, or assignment of different tensors on each site. By choosing the tensor identical on every site, this construction provides an explicit way to construct generic quantum states with area law entanglement that enjoy all lattice symmetries.

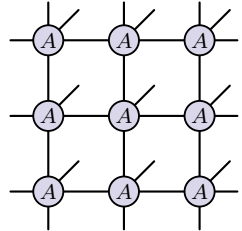
¹The local tensors are usually not idempotent linear maps.

The area law is manifest in this construction, since the entanglement entropy across the boundary of any contiguous region of spins is upper bounded by $\log \chi$ times the number of virtual pairs that cross the boundary, or thus the surface area of the boundary of that region. This will be made more precise in what follows for the one-dimensional case.

Throughout this thesis, we will use a graphical calculus in which tensors are represented by nodes with a number of legs representing their indices. Concretely, a rank 5 tensor A can be expressed as:

$$\begin{aligned}
 \text{Diagram of } A \text{ with 5 legs} &= \sum_{\substack{i,j,k \\ l,m}} j \text{ --- } A \text{ --- } l \begin{matrix} m \\ i \end{matrix} |i, j, k\rangle \langle l, m| \\
 &= \sum_{\substack{i,j,k \\ l,m}} A_{jklm}^i |i, j, k\rangle \langle l, m|
 \end{aligned} \tag{2.2}$$

We adopt the graphical convention that connected legs between tensors are summed over or *contracted*. Using this calculus, the state schematically represented in eq. (2.1) can be written more explicitly as



$$\tag{2.3}$$

In this figure, and similar figures below, it is understood that the tensor networks extend to infinity in all lattice directions.

2.2 Matrix product states

In one dimension, tensor network states boil down to the class of *matrix product states* (MPS) [24]. In the uniform case, they can be parametrised directly in the thermodynamic limit by a single rank 3 tensor $A \in \mathbb{C}^{d \times \chi^2}$ as:

$$|\Psi[A]\rangle = \text{Diagram of } A \text{ with 3 legs} \text{ --- } \text{Diagram of } A \text{ with 3 legs} \text{ --- } \dots \text{ --- } \text{Diagram of } A \text{ with 3 legs} \text{ --- } \text{Diagram of } A \text{ with 3 legs} \tag{2.4}$$

A matrix product state on a finite chain with boundary conditions specified by vectors $|v_L\rangle, |v_R\rangle$ is then defined as

$$|\Psi[A], v_L, v_R\rangle = \langle v_L | \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \dots \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} | v_R \rangle . \quad (2.5)$$

As discussed below, when the MPS describes the ground space of a *symmetry-protected topological phase*, the virtual spaces near the boundaries provide a natural basis for the symmetry-protected *edge modes*.

Continuing along these lines, we can define MPSs with larger unit cells or with distinct tensors $A[i]$ on every site. Notably, every state in the many-body Hilbert state of a finite chain can be represented by an MPS with a bond dimension that scales exponentially in the system size. To every matrix product state one can also associate a *gapped* and *frustration-free parent Hamiltonian* for which the MPS is a ground state. When the MPS has a certain symmetry G , the parent Hamiltonian can be symmetrised so as to commute with G as well. We refer to this as the *symmetric parent Hamiltonian*. Moreover, for a generic MPS it is the unique ground state of its parent Hamiltonian, as detailed below.

We henceforth focus on the case of uniform MPS in the thermodynamic limit [25]. In that case, the number of variational parameters is equal to $d\chi^2$ though not all of these are independent. Indeed, the dimension of the manifold of uniform matrix product states is strictly smaller, since for any invertible matrix X , transformations of the kind

$$A^i \mapsto X A^i X^{-1}, \quad \forall i = 1, 2, \dots, d, \quad (2.6)$$

leave the state $|\Psi[A]\rangle$ unchanged as all occurrences of X and X^{-1} cancel on the bonds. As they describe a redundancy of the MPS description, they are called *gauge transformations*. Taking these into account, a precise counting of the number of independent parameters reveals the dimension of the MPS manifold to be $\chi^2(d-1)$.

Using appropriate gauge transformations matrix product states can be brought in left (right) *canonical form* in which the MPS tensors are left (right) isometric:

$$\sum_i A^{i\dagger} A^i = \mathbb{1}_\chi, \quad \sum_i A^i A^{i\dagger} = \mathbb{1}_\chi. \quad (2.7)$$

Note that even after imposing the left or right canonical form, an MPS still enjoys residual gauge freedom as the MPS tensors can still be conjugated with *unitary* gauge transformations that preserve the canonical form. Writing A_L (A_R) for the MPS tensor A in left (right) canonical form, we can introduce

the matrix C that intertwines them according to $A_L^i C = C A_R^i$, $\forall i$. An MPS that can be written as:

$$|\Psi[A]\rangle = \begin{array}{c} | \\ | \\ | \\ | \end{array} \begin{array}{c} \text{---} A_L \text{---} A_L \text{---} C \text{---} A_R \text{---} A_R \text{---} \end{array} \begin{array}{c} \\ \\ \\ \end{array}, \quad (2.8)$$

is then said to be in the *mixed canonical form*. The upshot of this gauge choice is that the entanglement spectrum of the state (across any bipartition) coincides with the singular values of C , as can be verified by comparing eq. (2.8) with the Schmidt decomposition eq. (1.4).

One of the key properties of MPSs is that correlation functions and expectation values of local operators and the MPS norm can be efficiently computed. Efficiently here means with a cost that scales linear in the distance between operator insertions and polynomially in the bond dimensions, as argued below. An object which plays a pivotal role in these computations is the *transfer matrix* defined as

$$\mathbb{T}_A := \sum_i A^i \otimes \bar{A}^i = \begin{array}{c} \text{---} \bar{A} \text{---} \\ | \\ \text{---} A \text{---} \end{array}, \quad (2.9)$$

and typically interpreted as matrix from the right to the left (virtual) indices. A generic MPS has the property that $\mathbb{T}_A$ has a unique dominant eigenvalue λ_1 in virtue of the *quantum Perron-Frobenius theorem*. By rescaling $A^i \mapsto A^i/\sqrt{\lambda_1}$, this eigenvalue can be made 1. Its corresponding left (right) eigenvectors are denoted by Λ (ρ), both are full rank and can be made positive. Furthermore, they can be normalised to $\text{Tr}(\Lambda\rho) = 1$:

$$\left. \begin{array}{c} \text{---} \\ \text{---} \end{array} \right\} \Lambda \rho = 1, \quad (2.10)$$

which ensures the normalisation of the state in the thermodynamic limit, i.e. $\langle \Psi[A] | \Psi[A] \rangle = 1$.

Similarly, we can express the two-point correlation function of local operators O_1 and O_2 separated by a distance L as:

$$\langle O_1[1]O_2[L+1] \rangle_{|\Psi[A]\rangle} = \text{Diagram 1} \dots \text{Diagram 2} \quad (2.11)$$

By exploiting the tensor network structure of the transfer matrix, this correlation function can be computed with a computational cost that scales as

$\mathcal{O}(L\chi^3)$. Subtracting from eq. (2.11) the disconnected part $\langle O_1 \rangle_{\Psi[A]} \langle O_2 \rangle_{\Psi[A]}$ and invoking the eigendecomposition of $\mathbb{T}_A$ reveals that the dominant decay is exponential, i.e. $\sim \exp(-L/\xi)$, where the *correlation length* ξ is related to the second largest eigenvalue λ_2 of $\mathbb{T}_A$ via $\xi = -1/\log(|\lambda_2|)$.

The uniqueness of the dominant eigenvalue of the transfer matrix is a property called *injectivity*. As mentioned above this is a generic property and a useful equivalent definition is the following. An MPS is injective if there exists a finite *injectivity length* L such that the collection of matrices $\{A^{i_1} A^{i_2} \dots A^{i_L}\}$ constitute a basis for the $\chi \times \chi$ matrix algebra. Injective MPSs have the properties that the implicit choice of boundary conditions in eq. (2.4) in the thermodynamic limit is immaterial and that they are the *unique* ground state of their corresponding parent Hamiltonian. Below, we will interpret non-injective matrix product states as exhibiting spontaneous symmetry-breaking. As such they naturally show up in the classification of gapped phases of matter.

As argued above, MPS satisfy an area law for the entanglement entropy across any bipartition. To make this precise in the case of a uniform MPS, notice that across any entanglement cut, the reduced density matrix with respect to the right-half chain can be depicted as:

$$\rho_R = \begin{array}{c} \text{---} \bigcirc \text{---} \bigcirc \text{---} \\ | \quad | \\ \bigcirc \quad \bigcirc \\ | \quad | \\ \text{---} \bigcirc \text{---} \bigcirc \text{---} \end{array} . \quad (2.12)$$

This demonstrates that the reduced density matrix ρ_R can be interpreted as the product of a $d^L \times \chi$ -dimensional matrix with a $\chi \times d^L$ -dimensional one, where L is formally the number of spins to the right of the cut. Hence, $\text{rank}(\rho_R) \leq \chi$. This means that the entanglement entropy is upper bounded independent of the size of the right subsystem by $S_L \leq \log \chi$, which is the maximal amount of entanglement entropy for a rank- χ state.

Central to the classification of symmetric gapped phases is the following observation. Two MPSs $|\Psi[A]\rangle$ and $|\Psi[B]\rangle$ whose MPS matrices are related via a gauge transformation eq. (2.6) obviously generate the same state. In virtue of the *fundamental theorem of MPS* the converse also holds generically. Two uniform *injective* matrix product states with periodic boundary conditions represent the same state independent of the system size, if and only if the MPS tensors are related by a non-singular gauge transformation on the

virtual level:

$$|\Psi[A]\rangle \sim |\Psi[B]\rangle \iff A^i = e^{i\chi} X B^i X^{-1}, \quad \forall i. \quad (2.13)$$

Despite its relevance, the proof of the fundamental theorem is surprisingly simple and can be formulated as a straightforward application of the Cauchy-Schwarz inequality for which we refer to e.g. ref. [26].

Importantly, two non-injective periodic MPSs can generate the same state while not simply being related by a gauge transformation eq. (2.6). Consider an MPS tensor which in a certain basis reads

$$A^i = \begin{pmatrix} A_1^i & A_o^i \\ 0 & A_2^i \end{pmatrix}, \quad \forall i. \quad (2.14)$$

It follows directly from the block structure of this tensor that any choice of the off-diagonal block A_o^i generates the same state as the trace of the matrices A^i is independent of A_o^i . The upper triangular block structure of the tensor signals the presence of an invariant subspace W_1 , i.e. $A^i W_1 \subseteq W_1, \forall i$. Writing $\mathcal{P}_{W_1}$ ($\mathcal{P}_{W_1^\perp}$) for the projector on W_1 (and its orthogonal complement $W_1^\perp$), we can thus replace

$$A^i \mapsto A_1^i \oplus A_2^i = \mathcal{P}_{W_1} A^i \mathcal{P}_{W_1} + \mathcal{P}_{W_1^\perp} A^i \mathcal{P}_{W_1^\perp} \quad (2.15)$$

so as to bring A_i in block diagonal form. By repeating this procedure a finite number of times we can thus bring any MPS tensor in the form

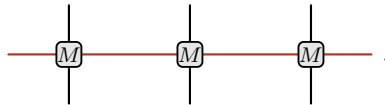
$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i, \quad (2.16)$$

where all $\{A_k^i\}_k$ are injective, and are referred to as the *injective blocks*. Such an MPS is said to be in *canonical form* [24].

Linear operators in turn can naturally be expressed in terms of *matrix product operators* (MPOs) [27, 28]. In full, these are written as

$$\sum_{\{i_k, j_k\}} (\dots M^{i_k, j_k} M^{i_{k+1}, j_{k+1}} \dots) |\dots, i_k, i_{k+1}, \dots \rangle \langle \dots, j_k, j_{k+1}, \dots|, \quad (2.17)$$

and can graphically be depicted as



$$. \quad (2.18)$$

A simple example is the pure density matrix $\rho = |\Psi[A]\rangle \langle \Psi[A]|$ with MPO tensor $M^{ij} = A^i \otimes \bar{A}^j$.

Similar to the case of MPS, an MPO is called *injective* when there exists a finite length for which the MPO matrices span the full matrix algebra.

2.3 Projected-entangled pair states

The natural generalisation of matrix product states to two dimensions is in terms of *projected entangled-pair states* (PEPSs) [29]. In the uniform case, we can parametrise a PEPS on a square lattice in terms of a single rank-5 tensor as follows:

$$|\Psi[A]\rangle = \text{Diagram of a 3x3 grid of tensors } A \text{ with external legs} . \quad (2.19)$$

Unlike the case of MPS, the contraction of a generic PEPS is computationally significantly more costly [30]. To tackle this issue, approximate contraction schemes have been and are being developed and used with success in variational ground state optimisation [31–33].

A further distinction from the one-dimensional case is the absence of a completely general fundamental theorem of PEPS that provides necessary and sufficient conditions for two PEPS tensors to generate the same wave function. For many systems of interest — in particular the *string-net* ground states that are the topic of chapter 3 — PEPS tensors generating the same state $|\Psi[A]\rangle \sim |\Psi[B]\rangle$ are intertwined via an MPO living purely on the virtual level obeying the *pulling-through* equation:

$$\text{Diagram of tensor A with MPO legs} = \text{Diagram of tensor B with MPO legs} , \quad (2.20)$$

together with the analogous relations in all other orientations.

Finally, analogous to matrix product operators, *projected entangled-pair operators* are tensor network representations of linear operators in two dimensions:

$$\text{Diagram of a 3x3 grid of tensors } M \text{ with external legs} . \quad (2.21)$$

2.4 Classification of bosonic one-dimensional phases

Recall that in the classical Landau paradigm phases of matter are classified by their symmetry-breaking pattern $G \rightarrow H \subseteq G$. Additional phases arise in the quantum mechanical setting since the entanglement degrees of freedom can transform in discrete distinct ways under the residual symmetry group H . Concretely, for one-dimensional bosonic systems, these classes are organised in the *second cohomology group* of H over $U(1)$, denoted by $H^2(H, U(1))$ and defined below. Notably, for finite H the second cohomology group $H^2(H, U(1))$ is itself finite², so that the number of gapped phases with residual H symmetry is also finite. One of the great achievements of tensor networks and in particular MPS, is that this one-dimensional classification can be completely carried out and understood in that framework, also in the fermionic case [21, 34, 35].

This classification rests on two pillars. The first is the fact that MPS provide a faithful representation of gapped ground state spaces [4]; the second is the (symmetric) parent Hamiltonian construction of MPS [36]. Indeed, as argued above, two Hamiltonians H_1, H_2 with the same symmetry belong to the same phase if and only if there exists a smooth interpolating Hamiltonian path starting in H_1 and ending in H_2 . Assuming an exact MPS representation of the ground space of the both Hamiltonians, one can restate this as the search for an interpolating path of the *symmetric parent* Hamiltonians. Equivalently, one can rephrase this problem as the search for a smooth interpolating MPS path.

Let us for a moment assume that the on-site symmetry G is the full symmetry group of the model — we return on the case where G is a subsymmetry below. In that case, the ground state corresponds to a non-injective MPS, $A^i = \bigoplus_{k=1}^r A_k^i$, with each of the blocks generating an injective MPS. We denote by k_0 an arbitrary but fixed *reference* block. In a certain basis, the symmetry acts on the virtual level of the MPS via the *induced* representation

$$P_g \bigoplus_{k=1}^r (U(h)^k \otimes \bar{U}(h)^k), \quad \forall g \in G, \quad (2.22)$$

where $h \equiv h_{g,r_\alpha}$ depends on g and k as specified below and the $\{U(h)^k\}_h$ are unitary matrices on the block k . The representation P_g permutes the

²Even though the second cohomology group is also finite for some Lie groups such as $H^2(SO(3), U(1)) \simeq \mathbb{Z}_2$, this is not generically true, not even for compact groups.

injective blocks, and thereby specifies the *stabiliser* subgroup $H \subseteq G$ via

$$H := \{g \in G \mid \pi_g(k_0) = k_0\}, \quad (2.23)$$

where π_g stands for the action of P_g on the labels k . Choosing a different reference block k'_0 would result in a stabiliser group conjugate to H . Let r_α , with $\alpha = 1, 2, \dots, |G|/|H|$, denote a set of representatives of the cosets G/H , with $r_{k_0} = \mathbb{1}$. The group element $h_{g,r_\alpha} \in H$ is then defined as:

$$gr_\alpha = r_\beta h_{g,r_\alpha}. \quad (2.24)$$

In fact, it can be shown that the $U(h)^k$'s furnish a *projective* representation of H , and the corresponding *topological index* is precisely what encodes the SPT phase of the surviving symmetry. We relegate the discussion of this to the next paragraph.

As anticipated above, G could be a subsymmetry of the full symmetry of the model. In that case, the MPS could exhibit an additional block structure, viz. $A^i = \bigoplus_K \bigoplus_k A^i_k$, signalling the breaking of the larger symmetry. This degeneracy is however not protected by the symmetry G . Altogether, this makes precise the above statement that non-injectivity of an MPS is the tensor-network manifestation of symmetry-breaking, and clarifies the role of the subgroup $H \subseteq G$.

We are left to explain the distinct ways an injective MPS can transform under a symmetry group, viz.:

$$U(g)^{\otimes N} |\Psi[A]\rangle \simeq |\Psi[A]\rangle, \quad \forall g \in G. \quad (2.25)$$

We use here G to denote the symmetry group of the MPS $|\Psi[A]\rangle$, but this could be the residual symmetry in the ground state subspace. A direct application of the fundamental theorem eq. (2.13) reveals that the MPS tensor itself has to transform trivially under the symmetry group, namely, for all $g \in G$ we have that:

$$\sum_j U(g)_{ij} A^j = e^{i\chi_g} X_g A^i X_g^\dagger \quad (2.26)$$

for unitary gauge matrices $\{X_g \mid g \in G\}$ and complex phases $\{e^{i\chi_g} \mid g \in G\}$ that may depend on the group element g . We can now act with two distinct group elements g, h on the MPS in two different ways: by acting with $U(gh)$ directly and invoking the fundamental theorem once as in eq. (2.26) or by

sequentially acting with g and h and making use of eq. (2.26) twice. Equating both yields:

$$e^{i\chi_{gh}} X_{gh} A^i X_{gh}^\dagger = e^{i(\chi_g + \chi_h)} X_g X_h A^i X_h^\dagger X_g^\dagger, \quad \forall i, g, h. \quad (2.27)$$

Injectivity of the MPS subsequently implies that $\{e^{i\chi_g} | g \in G\}$ constitutes a one-dimensional representation of G , whereas the X_g 's form a *projective* representation of G , i.e. they multiply according to $X_g X_h = \psi(g, h) X_{gh}$ for $U(1)$ phases $\psi(g, h)$. Imposing associativity of the matrix multiplication constrains these phases and shows that they adhere to a consistency condition known as the *2-cocycle condition*:

$$\psi(g, h) \psi(gh, k) = \psi(g, hk) \psi(h, k), \quad \forall g, h, k \in G. \quad (2.28)$$

Solutions of eq. (2.28) form an abelian group $Z^2(G, U(1))$ under pointwise multiplication. Among these, the trivial solutions, or *coboundaries*, take the form $\psi(g, h) = \frac{\phi_{gh}}{\phi_g \phi_h}$, for any function $\phi : G \rightarrow U(1)$. In the jargon of group cohomology such a trivial solution is referred to as a *coboundary* and they constitute a normal subgroup $B^2(G, U(1)) \trianglelefteq Z^2(G, U(1))$. Notably, rescaling of the gauge matrices with arbitrary phases $X_g \mapsto \phi_g X_g$ maps the corresponding 2-cocycle to one differing by the coboundary ϕ_g , leading to a physical redundancy. Therefore, one *mods out* such solutions in solving eq. (2.28), leading to equivalence classes of 2-cocycles. Two 2-cocycles $\psi \sim \psi'$ are thus considered cohomologically equivalent if they are related via a coboundary and these equivalence classes form the quotient group $H^2(G, U(1)) := Z^2(G, U(1)) / B^2(G, U(1))$ which is the anticipated *second cohomology group* of G over $U(1)$.

An MPS that transforms projectively on the virtual level as per eq. (2.26) is characterised by *symmetry-protected degeneracies* in its entanglement spectrum, as well as the existence of localised *edge modes* on open chains. The existence of both follows from the fact that, contrary to linear representations, there exist no non-trivial one-dimensional projective representations. As an illustrative example, consider the $SO(3)$ -symmetric ground state of the antiferromagnetic spin 1 Heisenberg model. Since the non-trivial representations of $SO(3)$ can be identified with the half-integer representations of its universal cover, $SU(2)$, the entanglement spectrum of the ground state is organised in multiplets of even dimension. This is visualised in fig. 2.1 where the first 48 Schmidt values of the ground state are plotted.

As anticipated above, a natural basis for the edge modes is given by the virtual degrees of freedom of the MPS ground state near the boundaries.

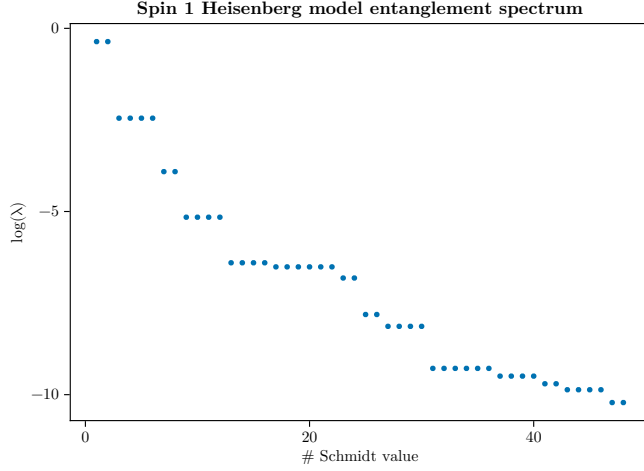


Figure 2.1: The entanglement spectrum of the spin 1 Heisenberg model ground state in the thermodynamic limit with bond dimension 48. Created with MPSKit.jl [37].

Indeed, a uniform MPS on open boundary conditions,

$$|\Psi[A]\rangle = \sum_{\{i_j\}} (A^{i_1} A^{i_2} \dots A^{i_N})_{\alpha\beta} |i_1, i_2, \dots, i_N\rangle, \quad (2.29)$$

corresponds to χ^2 states $|\Psi[A], \alpha, \beta\rangle$ that have the same energy for the parent Hamiltonian. Naively, we would thus count χ^2 ground states on the open chain, but some of these modes can be gapped out by turning on additional terms in the Hamiltonian. However, as long as these additional terms respect the symmetry, a certain amount of degeneracy is persistent; these edge modes transform under the symmetry action according to

$$U(g)^{\otimes N} |\Psi[A], \alpha, \beta\rangle = \sum_{\alpha', \beta'} (X_g)_{\alpha\alpha'} (\bar{X}_g)_{\beta\beta'} |\Psi[A], \alpha', \beta'\rangle, \quad (2.30)$$

so that at least a two-fold degeneracy persists.

This concludes phase classification in the one-dimensional setting where the symmetry acts via a unitary (tensor product) group representation. To summarise, phases are classified by pairs $(H, [\psi])$ where H is a subgroup $H \subseteq G$ up to conjugacy and $[\psi]$ is a class taking values in $H^2(H, \text{U}(1))$ [21, 34].

Remarkably, these data also classify (indecomposable) *module categories* $\mathcal{R}(H, \psi)$ over the *fusion category* Vec_G . Simple objects of the corresponding module category are labelled by cosets $\{r_1 H, r_2 H, \dots\}$ and $\mathcal{R}(H, \psi)$ carries a

natural Vec_G action induced by $g \triangleright rH := (gr)H$. This observation suggests the physical interpretation where the gapped ground states are identified with the simple objects, and the action $\triangleright$ with the action of the on-site G symmetry on the ground state subspace.

This observation is more than a formal analogy and is the first indication of a correspondence between module categories and gapped phases of matter. The *generalised Landau paradigm*, the subject of section 4.3, stipulates that this correspondence persists to one-dimensional (lattice) models with a symmetry described by a generic fusion category $\mathcal{C}$. More precisely, it states that the different gapped phases are given by the different module categories $\mathcal{P}$ over $\mathcal{C}$, with the simple objects labelling the ground states and the *module action* describing the symmetry action on them.

String-net models

In the previous chapter we outlined the classification of one-dimensional bosonic gapped phases of matter, which required the presence of a symmetry to protect a non-trivial phase. This radically changes in one dimension higher since even in the absence of any symmetries, gapped *intrinsic topological phases* arise. These phases are characterised by a robust ground state degeneracy that depends only on the topology of the underlying spatial manifold, *anyonic* quasiparticle excitations and in some cases edge modes.

In this chapter we focus on *non-chiral* topological of phases that admit gapped boundary conditions. It is believed that these phases are fully characterised by their anyon content, corresponding fusion rules and braiding statistics. This brings us in the realm of *category theory* since these data are naturally encapsulated in a *modular fusion category*, specifically the *Drinfeld* or *monoidal centre* $\mathcal{Z}(\mathcal{D})$ of a generically non-braided *input* fusion category $\mathcal{D}$.

Importantly, every such topological order admits a lattice realisation directly in terms of this input $\mathcal{D}$. This is precisely the *Levin-Wen string-net condensation model* that provides a frustration-free Hamiltonian whose torus ground state sectors and excitation spectrum exactly reproduce $\mathcal{Z}(\mathcal{D})$. From the point-of-view of tensor networks, their ground states admit particularly elegant PEPS representations characterised by the existence of MPO symmetries that live purely on the *virtual* level, form a representation of a *Morita equivalent* fusion category $\mathcal{C}$ and that can be continuously deformed through the lattice. As such, these MPO symmetries survive small physical perturbations and characterise the long-range entanglement structure.

In this chapter, we review the string-net models and their PEPS representations. Significant attention is devoted to the classification and construction

of their gapped boundaries. Beyond being interesting in their own right, the data characterising the gapped boundaries also classify *gapped phases* of the string-net boundary Hamiltonians that inherit the (non-invertible) symmetries of the bulk topological order. This is the content of the *generalised Landau paradigm* and is treated in the next chapter.

3

3.1 Topological order & category theory

The description of non-chiral topological orders and their realisation in terms of string-net models requires the language of *fusion categories* and *modular fusion categories*. Whereas the former is used to define the microscopic Hilbert space and Hamiltonian of the string-net models, the description of the anyonic excitations and torus ground state subspace requires the latter.

In what follows we will provide the necessary aspects of (braided) fusion categories, and thereby introduce their graphical calculus. We invite the reader to consult the literature for a more precise and more complete technical treatment of these topics [38–41].

It should be kept in mind that even though fusion categories are strictly richer structures than finite groups and their representation theory, they may in fact be regarded as a natural generalisation of both. Indeed, much of the structure of a fusion category can be understood as the categorical counterpart of familiar concepts from group and representation theory, a perspective that we shall illustrate in detail below. As a result, fusion categories provide a unified framework in which conventional symmetries and topological, generalised symmetries can be treated on equal footing.

3.1.1 Algebraic structure of fusion categories

Simple objects and fusion

Succinctly, a *unitary fusion category* (UFC) $\mathcal{C}$ consists of a collection of objects $\text{ob}(\mathcal{C})$ to which we will refer as *topological defects*, *topological lines*, *sectors* or *charges*, depending on the context. Every $X \in \text{ob}(\mathcal{C})$ admits a decomposition as a direct sum of finitely-many *simple objects* $\mathcal{I}_{\mathcal{C}} = \{X_i\}_i$, i.e.:

$$X = \bigoplus_{X' \in \mathcal{I}_{\mathcal{C}}} N_X^{X'} X', \quad (3.1)$$

for non-negative integers $N_X^{X'}$ counting the multiplicity of each direct summand. The decomposition is unique for every object X , up to a permutation of the factors. The set $\text{ob}(\mathcal{C})$ is endowed with a *monoidal* or *tensor product*

$\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ encoding the *fusion* of defects. We will express this product in terms of an integer rank-3 tensor N via:

$$X_1 \otimes X_2 = \bigoplus_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} X_3, \quad (3.2)$$

for simple objects X_1, X_2 . These fusion rules are associative so that for any four simple objects we have that $\sum_X N_{X_1 X_2}^X N_{X X_3}^{X_4} = \sum_X N_{X_1 X}^{X_4} N_{X_2 X_3}^X$. Generically $N_{X_1 X_2}^{X_3} \neq N_{X_2 X_1}^{X_3}$. Moreover, since the tensor product $\otimes$ distributes over the direct sum $\oplus$, the product of all objects is defined fully by eq. (3.2). Among the simple objects, there exists a unique *monoidal unit* notated as $\mathbb{1}$ that satisfies $\mathbb{1} \otimes X = X \otimes \mathbb{1} = X$, $\forall X \in \text{ob}(\mathcal{C})$. To every object X there corresponds a unique *dual* X^* characterised by $N_{X X^*}^{\mathbb{1}} = 1$.

Below, we also make use of the *quantum dimensions* d_X of the simple objects which are completely specified by

$$d_X > 0, \quad d_{X_1} d_{X_2} = \sum_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} d_{X_3}. \quad (3.3)$$

In particular we have $d_{\mathbb{1}} = 1$ and $d_X = d_{X^*}$, for all $X \in \text{ob}(\mathcal{C})$. Also, the quantum dimensions are additive over the direct sum, so that $d_X = \sum_{X' \in \mathcal{I}_{\mathcal{D}}} N_X^{X'} d_{X'}$ for any $X \in \text{ob}(\mathcal{C})$.

Graphical calculus

The different ways in which simple objects X_1 and X_2 can fuse to X_3 is captured by the *fusion space* $\mathcal{C}(X_1 \otimes X_2, X_3) \simeq \mathbb{C}^{N_{X_1 X_2}^{X_3}}$ for which we will use the shorthand $\mathcal{C}_{X_3}^{X_1 X_2}$. In category theoretic parlance, this Hilbert space for every triple of objects is referred to as *Hom*, *fusion* or *junction space*. Notably, for simple objects X_1, X_2 we have $\mathcal{C}(X_1, X_2) = \delta_{X_1, X_2} \mathbb{C}$. Basis states will be notated as $|X_1 X_2 X_3; i\rangle \in \mathcal{C}_{X_3}^{X_1 X_2}$ with $i = 1, 2, \dots, N_{X_1 X_2}^{X_3}$. These states and their duals can be depicted in terms of strings diagrams as follows:

$$|X_1 X_2 X_3; i\rangle \equiv \begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ \textcircled{i} \\ \diagup \\ X_3 \end{array}, \quad \langle X_1 X_2 X_3; i| \equiv \begin{array}{c} X_3 \\ \diagdown \\ \textcircled{i} \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \end{array}. \quad (3.4)$$

Here and below, all string diagrams are implicitly oriented downwards. We adapt a normalisation of these basis vectors such that the following graphical

identity holds:

$$\left| \begin{array}{c} X_3 \\ \text{---} i \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X'_3 \\ \text{---} j \end{array} \right| = \delta_{X_3, X'_3} \delta_{i,j} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} \left| \begin{array}{c} X_3 \end{array} \right|. \quad (3.5)$$

3

Note that the basis vectors are not normalised to one. Indeed the inner product of basis vectors can be depicted as:

$$\langle X_1 X_2 X_3; i | X_1 X_2 X_3; j \rangle = \left| \begin{array}{c} X_3 \\ \text{---} i \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X_3 \\ \text{---} j \end{array} \right| = \delta_{i,j} \sqrt{d_{X_1} d_{X_2} d_{X_3}}, \quad (3.6)$$

which for the particular case of $X_1 = \mathbb{1}$, $X_2 = X_3 \equiv X$ specialises to

$$\left| \begin{array}{c} X \\ \text{---} \end{array} \right| = d_X. \quad (3.7)$$

This normalisation is referred to as the *isotopy invariant* normalisation convention. This allows us to essentially forget about the orientation of internal lines in the graphical calculus, but other normalisation conventions are used throughout the literature [39]. In this normalisation, the completeness of the basis then boils down to:

$$\left| \begin{array}{c} X_1 \end{array} \right| \left| \begin{array}{c} X_2 \end{array} \right| = \sum_{X_3, i} \sqrt{\frac{d_{X_3}}{d_{X_1} d_{X_2}}} \left| \begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X_3 \quad i \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X_3 \quad i \end{array} \right|. \quad (3.8)$$

Consider now the fusion of three simple objects. In virtue of the associativity of the fusion rules, the associated Hom space capturing the fusion of these four objects decomposes in two distinct but isomorphic ways as follows:

$$\mathcal{C}_{X_4}^{X_1 X_2 X_3} \simeq \bigoplus_{X_5} \mathcal{C}_{X_5}^{X_1 X_2} \otimes \mathcal{C}_{X_4}^{X_5 X_3} \simeq \bigoplus_{X_6} \mathcal{C}_{X_4}^{X_1 X_6} \otimes \mathcal{C}_{X_6}^{X_2 X_3}, \quad (3.9)$$

for all simples X_4 appearing in the decomposition of $X_1 \otimes X_2 \otimes X_3$. The unitary isomorphism between both decompositions is given by the *F-symbol*

or *associator*¹ $F^{X_1 X_2 X_3}$ and its matrix components are conveniently depicted in terms of the *F-move*

$$\begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagup \\ \textcircled{i} \\ \diagup \quad \diagdown \quad \diagup \\ X_5 \quad \textcircled{j} \\ | \\ X_4 \end{array} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, ij}^{X_6, kl} \begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagup \\ \textcircled{k} \\ \diagup \quad \diagdown \quad \diagup \\ \textcircled{l} \quad X_6 \\ | \\ X_4 \end{array} . \quad (3.10)$$

We adapt a convention in which F -symbols are set to zero when not all fusion rules are satisfied. They also satisfy a consistency equation colloquially known as the *pentagon equation* which in full reads:

$$\begin{aligned} & \sum_o \left(F_{X_5}^{X_6 X_3 X_4} \right)_{X_7, lm}^{X_8, no} \left(F_{X_5}^{X_1 X_2 X_8} \right)_{X_6, ko}^{X_9, pq} \\ &= \sum_{X_{10}, rst} \left(F_{X_7}^{X_1 X_2 X_3} \right)_{X_6, kl}^{j, rs} \left(F_{X_5}^{X_1 X_{10} X_4} \right)_{X_7, sm}^{X_9, tq} \left(F_{X_9}^{X_2 X_3 X_4} \right)_{X_{10}, rt}^{X_8, np} . \end{aligned} \quad (3.11)$$

In virtue of *Ocneanu rigidity*, the number of solutions to the pentagon equation, up to basis transformations of the Hom spaces, is finite for any valid set of fusion rules.

Examples

Before proceeding, let us provide a few paradigmatic examples of these rather abstract definitions.

Finite groups: Vec_G^ω Let G be a finite group. The fusion category Vec_G is the category of (complex finite-dimensional) G -graded vector spaces. Its fusion rules are completely defined by

$$(V \otimes W)_g = \bigoplus_{hk=g} V_h \otimes W_k \quad (3.12)$$

$\forall g, h, k \in G$, where V_g stands for the g -graded component of V . Representative simple objects of Vec_G are the one-dimensional vector spaces $\{\mathbb{C}_g\}_{g \in G}$ and

¹The F -symbol has an interpretation as the complex number that $\mathcal{C}$ assigns to the planar projection of a tetrahedron whose edges are decorated by objects in $\text{ob}(\mathcal{C})$ and vertices decorated by junction spaces. In what follows we will assume the F -symbol to be independent of the chosen projection. This property is known as *tetrahedral symmetry*, and not every UFC satisfies this. We refer to ref. [42] for the generalisation of the string-net construction to fusion categories that do not enjoy this symmetry.

are thus in bijection with the group elements of G . Abusing notation, they are sometimes simply denoted by $g \equiv \mathbb{C}_g$. The non-vanishing F -symbols are all equal to one when all fusion rules are fulfilled. Whenever G possesses a non-trivial third cohomology group $H^3(G, \mathbb{U}(1))$, any 3-cocycle ω can be used to construct Vec_G^ω . This fusion category is constructed analogous to Vec_G where now the non-zero components of the F -symbols evaluate to ω . In that case, the pentagon equation is equivalent to the 3-cocycle condition satisfied by ω .

Finite group representations: $\text{Rep}(G)$ Intimately connected to Vec_G is the category $\text{Rep}(G)$, again for a finite group G .² This category encodes as objects the (unitary finite-dimensional) representations of G . Representative *irreducible* representations then constitute the simple objects and their fusion is given by the usual tensor product of G -representations. It can then be deduced that the 6-j symbols are in this case the matrix coefficients of the monoidal associator.

The Fibonacci fusion category: Fib The *Fibonacci* UFC Fib consists of two objects labelled by $\{\mathbb{1}, \tau\}$ whose only non-trivial fusion rule reads $\tau \otimes \tau = \mathbb{1} \oplus \tau$. The quantum dimension of τ is easily found to be the golden ratio $d_\tau = \varphi = \frac{1+\sqrt{5}}{2} = 1.618\dots$, indicative of the fact that Fib is not realised as the representation category of any finite group as this would require d_τ to be a non-negative integer. The non-trivial F -symbols read:

$$F_\tau^{\tau\tau\tau} = \frac{1}{\varphi} \begin{pmatrix} 1 & \sqrt{\varphi} \\ \sqrt{\varphi} & -1 \end{pmatrix}. \quad (3.13)$$

As anticipated above, we come to the intermediary conclusion that fusion categories provide a framework that not only unifies finite group and representation theory — as exemplified by the fusion categories Vec_G and $\text{Rep}(G)$ — but also extends it — e.g. in the form of the Fibonacci category. As a guide to the reader, we compactly summarise this in table 3.1:

²For all G Vec_G and $\text{Rep}(G)$ are *Morita equivalent* as explained below.

Vec_G	$\text{Rep}(G)$	Categorical notion
G -graded vector spaces $V, W, U, \dots$	G -representations (V_i, ρ_i)	Objects $\text{ob}(\mathcal{C})$
Grading-preserving linear maps $\phi : V \rightarrow W$	Intertwiners $f \circ \rho_1 = \rho_2 \circ f$	Junctions $\mathcal{C}(X, X')$
$\text{Vec}_G(\mathbb{C}_{g_1}, \mathbb{C}_{g_2}) =$ $\delta_{g_1, g_2} \mathbb{C}$	$\text{Rep}(G)(\alpha_i, \alpha_j) = \delta_{i,j} \mathbb{C},$ w/ α_i, α_j irreps (Schur)	Simple objects $\mathcal{C}(X_1, X_2) = \delta_{X_1, X_2} \mathbb{C}$
$(V \otimes W)_g =$ $\bigoplus_{hk=g} V_h \otimes W_k$	$\alpha_i \otimes \alpha_j \simeq \bigoplus_k N_{ij}^k \alpha_k$	Monoidal product & fusion rules
$\mathbb{C}_1 \otimes V \simeq V \simeq V \otimes \mathbb{C}_1$	$\mathbb{1} \otimes \rho \simeq \rho \simeq \rho \otimes \mathbb{1},$ w/ $\mathbb{1}$ the trivial irrep	Monoidal unit $\mathbb{1}$
$\mathbb{C}_g \otimes \mathbb{C}_{g^{-1}} \simeq \mathbb{C}_1$	$\alpha \otimes \alpha^* \ni \mathbb{1}$	Dual object
$F : (V \otimes W) \otimes U \xrightarrow{\sim}$ $V \otimes (W \otimes U),$ w/ F trivial	$F : (\alpha_1 \otimes \alpha_2) \otimes \alpha_3 \xrightarrow{\sim}$ $\alpha_1 \otimes (\alpha_2 \otimes \alpha_3),$ w/ F the 6-j symbols	F -symbol (Monoidal associator)

Table 3.1: Comparison of Vec_G , $\text{Rep}(G)$, and their categorical generalisations.

3.1.2 Braiding

Finally, we introduce the *braiding* of two defects diagrammatically as:

$$R^{X_1 X_2} := \begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ X_2 \quad X_1 \end{array}, \quad (R^{X_1 X_2})^{-1} := \begin{array}{c} X_2 \quad X_1 \\ \diagup \quad \diagdown \\ X_1 \quad X_2 \end{array}. \quad (3.14)$$

3

Typically, $R^{X_1 X_2} \circ R^{X_2 X_1} \neq \text{id}_{X_1 \otimes X_2}$ so that in the graphical calculus one should keep track of the distinction between *over* and *underbraiding* of the strands. Categories for which $R^{X_1 X_2} \circ R^{X_2 X_1}$ is the identity map on $X_1 \otimes X_2$, are called *symmetric*.

The components of the braiding eq. (3.14) in our choice of basis then gives rise to the unitary *R-matrix* defined via:

$$\begin{array}{c} X_1 \quad X_2 \\ \diagdown \quad \diagup \\ \text{---} i \text{---} \\ | \\ X_3 \end{array} = \sum_j (R_{X_3}^{X_1 X_2})_{ij} \begin{array}{c} X_1 \quad X_2 \\ \diagup \quad \diagdown \\ j \text{---} \\ | \\ X_3 \end{array}. \quad (3.15)$$

The braiding is required to satisfy a consistency equation called the *hexagon equation* that also involves the *F-symbols*. A fusion category is called *braided* or *pre-modular* when it is endowed with a braiding that satisfies the hexagon axiom. For any braided fusion category we then define the *topological S-matrix* S_{X_1, X_2} via the amplitudes

$$\begin{aligned} S_{X_1, X_2} &:= \frac{1}{\text{FPdim } \mathcal{C}} \begin{array}{c} \bigcirc_{X_1} \quad \bigcirc_{X_2} \end{array} \\ &= \frac{1}{\text{FPdim } \mathcal{C}} \sum_{X_3} N_{X_1 X_2}^{X_3} d_{X_3} \text{Tr}(R_{X_3}^{X_1 X_2} R_{X_3}^{X_2 X_1}), \end{aligned} \quad (3.16)$$

where we have defined the *total quantum dimension* or *Frobenius-Perron dimension* of $\mathcal{C}$ as $\text{FPdim } \mathcal{C} = \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2$. Physically, the amplitudes eq. (3.16) correspond to an interferometry process in which two defects X_1 and X_2 are created from the vacuum, exchanged and annihilated again. Notably, if for every $X_1 \neq \mathbb{1}$ there exists at least one $X_2 \neq \mathbb{1}$ such that their double-braiding $R^{X_1 X_2} \circ R^{X_2 X_1}$ is non-trivial, the S-matrix is non-degenerate [43, Cor. 2.16]. Physically, this means that every non-trivial defect can be distinguished from

the vacuum by double-braiding it with a suitable probe defect. This is called the *principle of remote detectability* [44, 45]. This motivates the definition of a *modular* fusion category, which is a pre-modular fusion category with non-degenerate S -matrix.

Examples

Finite groups: Vec_G Evidently, Vec_G only admits a braiding if G is abelian. In that case the braiding is provided by a choice of *bilinear form* b , meaning a map $b : G \times G \rightarrow \text{U}(1)$ satisfying

$$\begin{aligned} b(g_1, g_3) \cdot b(g_2, g_3) &= b(g_1 g_2, g_3), \\ b(g_1, g_2) \cdot b(g_1, g_3) &= b(g_1, g_2 g_3), \end{aligned} \quad (3.17)$$

for all $g_1, g_2, g_3 \in G$. When the associator belongs to a non-trivial class $[\omega] \in H^3(G, \text{U}(1))$, admissible associator-braiding pairs are in one-to-one correspondence with the elements of the *abelian* cohomology group $H_{\text{ab}}^3(G, \text{U}(1))$. We refer to the relevant literature for a precise definition and construction of the braiding [40].

Finite group representations: $\text{Rep}(G)$ The category $\text{Rep}(G)$ admits a symmetric braiding that amounts to the transposition of representation spaces, i.e. $R : V_1 \otimes V_2 \rightarrow V_2 \otimes V_1$ for $(V_1, \rho_1), (V_2, \rho_2)$ in $\text{ob}(\text{Rep}(G))$.

The Fibonacci fusion category: Fib The Fibonacci category admits a *modular* braiding. Indeed, its S -matrix

$$S = \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}, \quad (3.18)$$

is non-degenerate [46].

3.2 String-net models

3.2.1 Hilbert space and Hamiltonian

The microscopic string-net Hilbert space is constructed from an *input* UFC $\mathcal{D}$ — whose objects we denote by $\{Y\}$ — and an oriented *trivalent lattice* Λ without boundary as follows [47, 48]. A basis state corresponds to an assignment — or *$\mathcal{D}$ -colouring* [48] — of simple objects $Y \in \mathcal{I}_{\mathcal{D}}$ to all the edges $E(\Lambda)$, with the requirement that the fusion rules are satisfied on all vertices $V(\Lambda)$. Conventionally, reversing the orientation of an edge $e \rightarrow -e$ amount

to taking the dual object on that edge. Then, to every vertex with adjacent edges labelled Y_1, Y_2, Y_3 the junction space $\mathcal{C}(Y_1 \otimes Y_2, Y_3)$ is attached. In what follows we specialise to the *hexagonal lattice*. On it, the Hilbert space $\mathcal{H}_{\mathcal{D}}(\Lambda)$ thus is of the form:

$$\mathcal{H}_{\mathcal{D}}(\Lambda) = \text{Span}_{\mathbb{C}} \left\{ \left| \{Y\}, \{i\} \right\} \right\}, \quad (3.19)$$

where it is understood that the picture extends over the entire lattice, and periodic boundary conditions in both spatial directions are assumed. In correspondence with our graphical calculus, we also assume all edges to be oriented downwards.

On every plaquette $p \in P(\Lambda)$ and for every $Y \in \mathcal{I}_{\mathcal{D}}$ we subsequently define a local operator B_p^Y . This operator admits a particularly intuitive definition as inserting a *non-contractible* counter-clockwise loop labelled by the simple object Y in the plaquette and fusing it into the physical lattice with local applications of the resolution of the identity and F -moves eqs. (3.8) and (3.10). In order for this to be consistent with our graphical calculus, and in particular the quantum dimension relation eq. (3.7), we should therefore think of the underlying manifold as being *punctured* in the interior of every plaquette in $P(\Lambda)$. Schematically, the action can therefore be depicted as:

$$B_p^Y \cdot |\Psi\rangle = \xrightarrow{(3.8)} \sum \{d\} \quad (3.20)$$

$$\begin{array}{c}
 \begin{array}{c}
 \text{(3.10)} \\
 \text{(3.5)} \\
 \mapsto
 \end{array}
 \sum \{d\} \{F\}
 \end{array}
 \begin{array}{c}
 \text{Diagram: A hexagonal plaquette with a shaded circle in the center. Six lines extend from the vertices of the hexagon, each ending in a small grey dot. The central circle is filled with diagonal hatching.}
 \end{array}
 \quad . \quad (3.21)$$

3

Herein, we have for simplicity omitted all edge and vertex labels; $\{d\}$ and $\{F\}$ are placeholders for the quantum dimensions and F -symbols obtained via repeated use of eqs. (3.5), (3.8) and (3.10). We refer to [47] for an explicit computation of these matrix elements. We can also infer from application of our graphical calculus that these plaquette operators form a representation of the fusion rules, viz.:

$$B_{\mathbf{p}}^{Y_1} \cdot B_{\mathbf{p}}^{Y_2} = \sum_{Y_3} N_{Y_1 Y_2}^{Y_3} B_{\mathbf{p}}^{Y_3}. \quad (3.22)$$

With these plaquette operators at our disposal, the local Hermitian Hamiltonian terms are constructed as:

$$B_{\mathbf{p}}^{\mathcal{D}} = \sum_{Y \in \mathcal{I}_{\mathcal{D}}} \frac{d_Y}{\text{FPdim} \mathcal{D}} B_{\mathbf{p}}^Y =: \text{Diagram: A dashed circle.} \quad (3.23)$$

From combining eq. (3.22) with the identity

$$d_Y \text{FPdim} \mathcal{D} = \sum_{Y_1, Y_2} N_{Y_1 Y_2}^Y d_{Y_1} d_{Y_2}, \quad (3.24)$$

it directly follows that $B_{\mathbf{p}}^{\mathcal{D}}$ is an idempotent for every $\mathbf{p} \in \mathbf{P}(\Lambda)$, i.e. $B_{\mathbf{p}}^{\mathcal{D}} \cdot B_{\mathbf{p}}^{\mathcal{D}} = B_{\mathbf{p}}^{\mathcal{D}}$, $\forall \mathbf{p}$. Commutation of plaquette operators $B_{\mathbf{p}}^{\mathcal{D}}$ and $B_{\mathbf{p}' \neq \mathbf{p}}^{\mathcal{D}}$ is obvious when $\mathbf{p}$ and $\mathbf{p}'$ are far separated. When $\mathbf{p}$ and $\mathbf{p}'$ share an edge this follows from comparing the matrix elements of both application orders and invoking the pentagon equation eq. (3.11) on their matrix elements. The string-net Hamiltonian is then finally given by

$$H_{\mathcal{D}}(\Lambda) = - \sum_{\mathbf{p} \in \mathbf{P}(\Lambda)} B_{\mathbf{p}}^{\mathcal{D}}. \quad (3.25)$$

Remark that a ground state of $H_{\mathcal{D}}(\Lambda)$ can be constructed by acting with the ground state projector $\prod_{\mathbf{p}} B_{\mathbf{p}}^{\mathcal{D}}$ — which is well-defined since all $B_{\mathbf{p}}^{\mathcal{D}}$'s

commute — on a generic string-net state in $\mathcal{H}_{\mathcal{D}}(\Lambda)$. This will exactly be the approach taken in section 3.4 to write down the ground state PEPS representations.

Importantly, the ground state subspace is one-dimensional when the underlying lattice has the manifold of a two-sphere [48]. However, on the two-torus the ground space is degenerate, and the degeneracy does not depend on the system size. Due to the absence of any global symmetry, this degeneracy cannot be traced back to ordinary symmetry breaking as discussed in section 2.4. Rather, the ground state degeneracy can be understood as the breaking of a more general type of symmetry whose operators are supported on oriented *freely deformable* loops ℓ on the lattice. Such operators constitute an example of a *global 1-form* symmetry [19, 49, 50]. These loop operators are encoded in the *monoidal* or *Drinfeld centre* $\mathcal{Z}(\mathcal{D})$ of $\mathcal{D}$.

3.2.2 Loop operators and ground states

Formally, the centre $\mathcal{Z}(\mathcal{D})$ is a unitary fusion category whose simple objects are pairs $(Z, \gamma_{Z, \bullet})$ where Z is a (generically non-simple) object in $\mathcal{D}$ and $\gamma_{Z, \bullet}$ is a natural isomorphism, called the *half-braiding*. Its matrix elements can be expressed graphically as

$$\begin{array}{c} Y_1 \quad Z \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Z \quad Y_1 \end{array} = \sum_{Y_2} \sum_{i,j} (\Gamma_{Y_2}^{Y_1 Z})_{ij} \begin{array}{c} Y_1 \quad Z \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Z \quad Y_1 \end{array} \quad , \quad (3.26)$$

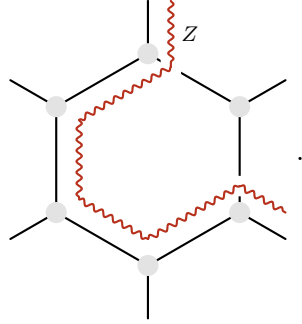
$\forall Y_1 \in \mathcal{I}_{\mathcal{D}}$. The half-braiding satisfies a hexagon consistency axiom involving the associator of $\mathcal{D}$. Consequently, $\mathcal{Z}(\mathcal{D})$ can be endowed with a *modular* braiding, defined directly in terms of $\gamma_{Z, \bullet}$ [51]. Naturality of $\gamma_{Z, \bullet}$ implies the following graphical equality:

$$\begin{array}{c} Y_1 \quad Y_2 \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Y_3 \end{array} = \begin{array}{c} Y_1 \quad Y_2 \\ \diagdown \quad \diagup \\ \text{---} \quad \text{---} \\ \diagup \quad \diagdown \\ Y_3 \end{array} \quad , \quad (3.27)$$

for all admissible junctions $|Y_1 Y_2 Y_3, i\rangle$.

In order to define the action of a loop labelled by $(Z, \gamma_{Z, \bullet})$ on a state, one proceeds in a similar fashion as in the definition of the plaquette operators. One inserts the object Z along the path ℓ , and locally fuses it into the physical

lattice, thereby making use of eq. (3.26) at every crossing with the physical lattice and the usual local recoupling identities eqs. (3.5), (3.8) and (3.10). An illustration of such a loop is:



(3.28)

3

Although drawn as a line, it should be remarked that these operators act (diagonally) on edges adjacent to their path and are therefore more precisely interpreted as *ribbon operators* with a finite width. These coincide with those introduced by Kitaev in ref. [52] and Levin and Wen in ref. [47], see also refs. [53, 54].

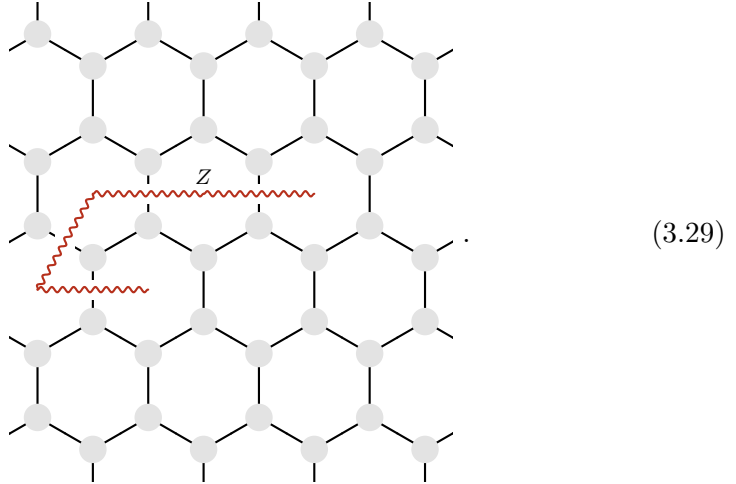
The topological nature of the loop operator follows directly from the naturality condition eq. (3.27), which ensures that Z can slide freely across any junction of the physical lattice without altering the resulting operator. Note that this topological invariance also accounts for the commutativity with the Hamiltonian. When a loop and a plaquette operator have disjoint support, they commute trivially. When they have overlapping support, one can deform the loop away from the plaquette before acting with the latter, and subsequently restore the path of the loop, so that the combined action remains unchanged.

Provided a reference string-net ground state one can construct a complete basis of torus ground states by acting with all simple loop operators $(Z, \gamma_{Z, \bullet}) \in \mathcal{I}_{\mathcal{Z}(\mathcal{D})}$ along a non-contractible cycle of the torus. We therefore conclude that the torus ground state degeneracy is given by $|\mathcal{I}_{\mathcal{Z}(\mathcal{D})}|$. This also implies that orthogonal ground states are locally indistinguishable, as the precise location of the Z loop can be continuously deformed and is thus invisible for any local operator. Acting on the other hand with loop operators along the perpendicular non-contractible cycle provides a different set of ground states that are related to the former via the unitary basis transformation given by the S -matrix of $\mathcal{Z}(\mathcal{D})$ defined in eq. (3.16) [55].

3.2.3 String operators and excited states

Similar to the construction of ground states via loop operators, excited states are constructed by acting on the ground states with *open* string operators, also labelled by objects $(Z, \gamma_Z, \bullet) \in \mathcal{I}_{\mathcal{Z}(\mathcal{D})}$. Such a state corresponds to an anyon–anti-anyon pair, and schematically can be depicted as

3



In this sense, the different ground states constructed in section 3.2.2 can be understood as the result of creating a (Z, Z^*) -pair from a reference ground state, dragging one of the anyons around a non-contractible cycle before annihilating the pair.

Note that in fusing the open string in eq. (3.29) to the physical lattice, one is left with dangling vertical edges on the plaquettes on which the string starts and terminates. Therefore, the obtained state does not belong to the string-net Hilbert space, if Z decomposes as a non-trivial object of $\mathcal{D}$, in which case it is said to possess a non-trivial *charge*. Accommodating this type of excitations naturally leads us to consider the *extended* string-net models, first introduced in ref. [56]. In this framework, the string-net Hilbert space is extended with extra open strands or *tails*, one for every vertex of the original lattice, so that all anyonic excitations, also those with a non-trivial charge, can be considered without the need to leave the Hilbert space or violate the $\mathcal{D}$ fusion rules. Concretely, on the hexagonal lattice, the extended states can

be depicted as

$$(3.30)$$

where we have omitted the junction spaces to avoid cluttering the diagram. In this depiction, an identification between every tail and the closest vertex of the hexagonal lattice is assumed. The extended Hamiltonian consists of the plaquette terms which generalise those introduced in eq. (3.23), and are defined in detail in ref. [56], while additional *vertex terms* $Q_v^{\mathcal{D}}$ for every $v \in V$ are considered. These act by projecting the tail corresponding to v to the identity object. Concretely, locally:

$$Q_v^{\mathcal{D}} \left| \begin{array}{c} Y_1 \quad Y_2 \\ \quad i \\ Y_3 \quad j \text{---} Y \\ Y_4 \quad k \text{---} Y' \\ \quad Y_6 \end{array} \right\rangle = \delta_{Y,1} \left| \begin{array}{c} Y_1 \quad Y_2 \\ \quad i \\ Y_3 \quad 1 \text{---} 1 \\ Y_4 \quad k \text{---} Y' \\ \quad Y_6 \end{array} \right\rangle. \quad (3.31)$$

Collecting everything, the extended string-net Hamiltonian is of the form

$$H_{\mathcal{D}}^{\text{Ext.}}(\Lambda) = - \sum_{p \in P(\Lambda)} B_p^{\mathcal{D}} - \sum_{v \in V(\Lambda)} Q_v^{\mathcal{D}}. \quad (3.32)$$

Notably, in the subspace defined by $Q_v^{\mathcal{D}} = 1$, $\forall v \in V$, the extended string-net model coincides with the original Levin-Wen string-net.

At this point we will not attempt to describe in further detail the action of the loop operators, the explicit ground states or excited states. Rather, we demonstrate in section 3.4 the construction of distinct PEPS representations of

string-net ground states. The upshot of the tensor network approach is that a complete basis of torus ground states can be constructed by inserting a network of matrix product operators living purely on the virtual level that correspond to *central idempotents* of the *tube algebra*. In fact, the representation theory of the tube algebra is often used to construct an explicit realisation of the Drinfeld centre. This description of the torus ground states is significantly simpler than working with the ribbon operators. Also, crucially, since the MPOs live on the virtual level, they survive physical perturbations and hence furnish a representation of the symmetries of the boundary Hamiltonians discussed in section 4.2. In a similar spirit, by introducing a new type of tensors whose support is specified by a central idempotent of the tube algebra, also excited states admit a PEPS representation.

3.3 Gapped boundaries and interfaces

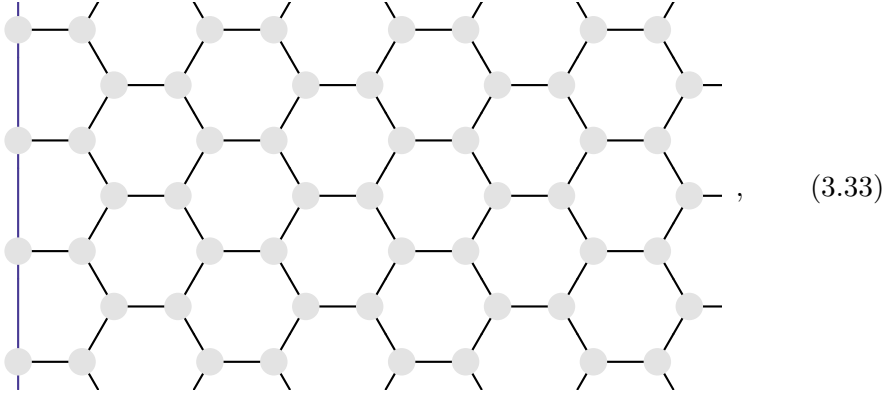
3.3.1 Gapped boundaries

Up until now we have restricted to the case of closed boundary conditions. As anticipated in the introduction, we can define string-net models with different types of boundary conditions while preserving the gap. One choice of boundary is obtained by simply terminating the lattice, in which case the edge degrees of freedom are labelled by objects and junctions in $\mathcal{D}$. This demonstrates that every string-net model has at least one gapped boundary, to which we will refer as the *regular* or *smooth boundary condition*, but depending on the input $\mathcal{D}$, other choices might be possible [57, 58].

The boundary conditions that we will study momentarily are constructed in such a way that the bulk plaquette operators can be extended to the boundary degrees of freedom while retaining their mutual commutativity. Commutation of the plaquette operators follows, similar as in the bulk, from local relations involving the F -symbols of $\mathcal{D}$.

To this end, let us introduce a new set of labels $\mathcal{I}_{\mathcal{R}} = \{R_1, R_2, \dots\}$ for these boundary degrees of freedom, and an action of $\mathcal{D}$ on those labels implemented via $R_1 \triangleleft Y = \sum_{R_2 \in \mathcal{I}_{\mathcal{R}}} N_{R_1 Y}^{R_2} R_2$ so that vertices on the boundary host a new type of vector space with basis vectors $|R_1 Y R_2, i\rangle$, $i = 1, 2, \dots, N_{R_1 Y}^{R_2}$.

Graphically, string-net states in the presence of such a boundary look like:



where we draw edges that carry labels in $\mathcal{I}_{\mathcal{R}}$ in blue. The type of Hom space can be inferred from the colour of the adjacent edges. Similar as in the bulk we define a unitary isomorphism ${}^{\triangleleft}F$, generically distinct from the earlier defined F -symbol, whose matrix entries are defined graphically in terms of an ${}^{\triangleleft}F$ -move:

$$\begin{array}{c}
 \begin{array}{c}
 R_1 \quad Y_1 \quad Y_2 \\
 \diagup \quad \diagdown \\
 \text{---} i \text{---} \\
 \text{---} j \text{---} \\
 \diagdown \quad \diagup \\
 R_3 \quad R_2
 \end{array}
 = \sum_{Y_3, ij} \left({}^{\triangleleft}F_{R_2}^{R_1 Y_1 Y_2} \right)_{R_3, ij}^{Y_3, kl}
 \begin{array}{c}
 \begin{array}{c}
 R_1 \quad Y_1 \quad Y_2 \\
 \diagup \quad \diagdown \\
 \text{---} k \text{---} \\
 \text{---} l \text{---} \\
 \diagdown \quad \diagup \\
 Y_3 \quad R_2
 \end{array}
 \end{array}
 . \quad (3.34)
 \end{array}$$

We will refer to ${}^{\triangleleft}F$ as the (*right*) *module associator*, and explain this terminology below. We require the ${}^{\triangleleft}F$ -symbols to adhere to a consistency axiom involving the F -symbols of $\mathcal{D}$ that is of the form

$${}^{\triangleleft}F {}^{\triangleleft}F = \sum F {}^{\triangleleft}F {}^{\triangleleft}F. \quad (3.35)$$

Its explicit form is straightforward to derive and spelled out in many places, e.g. ref. [59]. For later reference, let us also introduce quantum dimensions of objects in $\mathcal{I}_{\mathcal{R}}$ in the sense of refs. [40, 60] via:

$$\begin{aligned}
 d_R &> 0 \\
 d_R d_Y &= \sum_{R' \in \mathcal{I}_R} N_{RY}^{R'} d_{R'}, \quad (3.36)
 \end{aligned}$$

for all $Y \in \mathcal{I}_{\mathcal{D}}$. The quantum dimensions d_R are defined up to a global rescaling $d_R \mapsto \lambda d_R$, $\forall R \in \mathcal{I}_R$ for any R -independent positive real scalar λ .

To fix it, we will adopt the canonical normalisation

$$\text{FPdim}\mathcal{R} := \sum_{R \in \mathcal{I}_{\mathcal{R}}} d_R^2 = \sum_{Y \in \mathcal{I}_{\mathcal{D}}} d_Y^2 =: \text{FPdim}\mathcal{D}. \quad (3.37)$$

In this normalisation we have the resolution of the identity

$$\left| \begin{array}{c} R \\ \hline Y \end{array} \right| = \sum_{R', i} \sqrt{\frac{d_{R'}}{d_R d_Y}} \left| \begin{array}{c} R \\ \text{---} i \text{---} Y \\ R' \\ \text{---} i \text{---} Y \\ R \end{array} \right|, \quad (3.38)$$

and the inner product

$$\left| \begin{array}{c} R_2 \\ \text{---} i \text{---} Y \\ R_1 \\ \text{---} j \text{---} Y \\ R_2' \end{array} \right| = \delta_{R_2, R_2'} \delta_{i, j} \sqrt{\frac{d_{R_1} d_Y}{d_{R_2}}} \left| \begin{array}{c} R_2 \end{array} \right|. \quad (3.39)$$

Succinctly, provided a unitary fusion category $\mathcal{D}$, the set of boundary labels $\mathcal{I}_{\mathcal{R}}$, together with the action $\triangleleft$ and a solution to the pentagon equation eq. (3.35) form the defining data of a (unitary) *right module category* $\mathcal{R}$ over $\mathcal{D}$ [40, 61]. Similarly, we can define a *left* module category to construct a gapped right boundary condition.

For a generic choice of module category $\mathcal{R}$, the bulk plaquette operators are extended to plaquettes adjacent to the boundary by again inserting a punctured counter-clockwise loop $\bigoplus_{X \in \mathcal{I}_{\mathcal{D}}} \frac{d_X}{\text{FPdim}\mathcal{D}} X$ in the boundary plaquettes and absorbing them into the physical lattice by making use of the different resolutions of the identity, F -moves and normalisations eqs. (3.8), (3.38), (3.10), (3.34), (3.5) and (3.39). It can be verified that the boundary plaquette operators commute with all the other Hamiltonian terms by virtue of eq. (3.35), so that the Hamiltonian remains gapped near the boundary.

3.3.2 Examples

The regular module category Every UFC $\mathcal{D}$ trivially defines a unitary indecomposable module category over itself. We will refer to this as the *regular* module category, and corresponds to the regular boundary condition in the string-net model based on $\mathcal{D}$ defined above.

Vec $_G$ -module categories As anticipated in section 2.4, indecomposable module categories over Vec_G are – up to conjugacy – specified by pairs $(H \subseteq G, \psi)$ where $\psi \in Z^2(H, \text{U}(1))$. In what follows we write the corresponding module category as $\mathcal{R}(H, \psi)$. From this data, the module category is constructed explicitly as follows.

The object set $\mathcal{I}_{\mathcal{R}}$ is a transitive left G -set, hence it can be identified with the cosets G/H endowed with the canonical left G -action. Let $s : G/H \rightarrow G$ be a section so that for any coset $R \in G/H$, $s(R)H = R$. We also assume that the representative of H is $\mathbb{1}$. Notice that in general $s(gR) \neq gs(R)$ but differs by a group element in H that we will write as $h_{g,R}$, i.e. $gs(R) = s(gR)h_{g,R}$. In virtue of the associativity of the group multiplication it holds that $h_{g_1, g_2 R} h_{g_2, R} = h_{g_1 g_2, R}$, $\forall g_1, g_2 \in G, R \in G/H$. The components $\triangleright F$ of the module associator $\triangleright F$ are scalars and adhere to the consistency condition

$$\triangleright F(g_2, g_3)(M) \triangleright F(g_1, g_2 g_3)(M) = \triangleright F(g_1, g_2)(g_3 M) \triangleright F(g_1 g_2, g_3)(M). \quad (3.40)$$

Considering $\text{Fun}(G/H, \text{U}(1))$ as the G -module of functions $G/H \rightarrow \text{U}(1)$, eq. (3.40) exactly expresses that $\triangleright F$ is a representative of a class in $H^2(H, \text{Fun}(G/H, \text{U}(1)))$. We then define $\triangleright F(g, h)(R)$ as the image of ψ under Shapiro's isomorphism

$$H^2(H, \text{U}(1)) \simeq H^2(H, \text{Fun}(G/H, \text{U}(1))) \quad (3.41)$$

induced by

$$\triangleright F_{\psi}(g_1, g_2)(R) = \psi(h_{g_1, g_2 R}, h_{g_2, R}). \quad (3.42)$$

Indeed, it is easy to verify that eq. (3.40) is solved in virtue of the 2-cocycle equation satisfied by ψ .

Notably, for the choice $H = G$, the corresponding module category $\mathcal{R}(G, 1)$ contains only one simple object, so that $\mathcal{R}(G, 1)$ is equivalent to the category Vec of finite-dimensional complex vector spaces.

Vec $_G^{\omega}$ -module categories Analogous to the untwisted case, module categories over Vec_G^{ω} are in one-to-one correspondence with pairs (H, ψ) where now ψ trivialises the 3-cocycle ω on the subgroup H . This imposes constraints on the allowed subgroups $H \subseteq G$.

The object set $\mathcal{I}_{\mathcal{R}}$ can again be identified with the left cosets G/H , and in this case the module associator can be expressed in terms of both ω and ψ , see e.g. ref. [62] for an explicit formula.

Rep(G)-module categories In virtue of the *Morita equivalence* between $\text{Rep}(G)$ and Vec_G — a notion introduced momentarily — pairs (H, ψ) also

Fib The fusion category **Fib** can be explicitly verified to have no non-trivial module categories besides the regular module category.

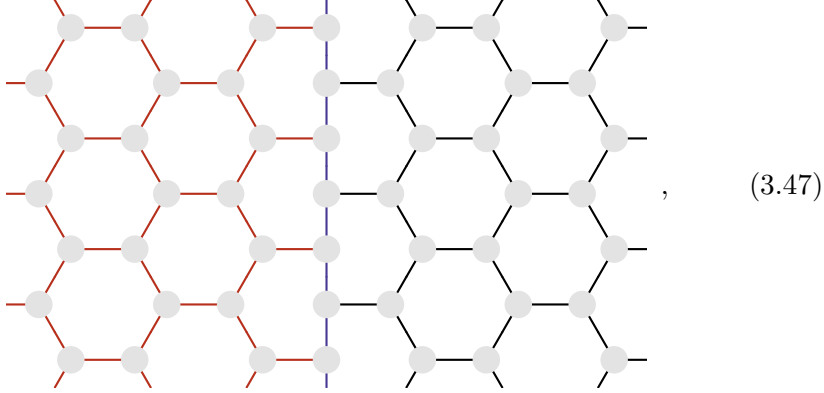
Finally, we would like to similarly define gapped interfaces or *domain walls* between string-net models based on different fusion categories $\mathcal{C}$ and $\mathcal{D}$. From the above considerations it is easy to convince oneself that such an interface $\mathcal{R}$ should in particular be simultaneously a left $\mathcal{C}$ - and right $\mathcal{D}$ -module category. Compatibility of both models in turn implies an additional collection of unitary isomorphisms notated as $\bowtie F$ that allows one to ‘commute’ $\mathcal{C}$ and $\mathcal{D}$ strands in the sense of

where $X \in \mathcal{I}_{\mathcal{C}}$ and $Y \in \mathcal{I}_{\mathcal{D}}$. Together with the left and right module associators these ${}^{\bowtie}F$ -symbols adhere to an additional collection of pentagon equations that are of the form

$$\triangleleft F \bowtie F = \sum \bowtie F \bowtie F \triangleleft F. \quad (3.46)$$

Similar as to the string-net Hilbert space in the presence of an $\mathcal{R}$ gapped boundary, the Hilbert space in the presence of an $\mathcal{R}$ domain wall takes the

form:



where the edges in **red** are valued in $\text{ob}(\mathcal{C})$.

Note that up to this point we have assumed $\mathcal{C}$ and $\mathcal{D}$ to be completely general. In particular we have not assumed the topological orders $\mathcal{Z}(\mathcal{C})$ and $\mathcal{Z}(\mathcal{D})$ to be related in any way. Categories $\mathcal{C}$ and $\mathcal{D}$ for which $\mathcal{Z}(\mathcal{C})$ and $\mathcal{Z}(\mathcal{D})$ are equivalent as braided fusion categories are said to be *Morita equivalent*. Importantly, Morita equivalence admits an equivalent characterisation as the existence of an *invertible* bimodule category $\mathcal{R}$ between $\mathcal{C}$ - and $\mathcal{D}$. Physically, such an invertible bimodule category describes a completely transparent gapped interface between the $\mathcal{C}$ and $\mathcal{D}$ -string-nets for which all anyons in either model can tunnel through the domain wall without *condensing* on it. From the mathematical point of view, invertibility of a bimodule category can be deduced from the explicit form of the bimodule associators ${}^{\mathfrak{M}}F$ as proven in ref. [63]. Interestingly, bimodule invertibility can be thought of as a generalisation of Schur orthogonality to the categorical setting.

Turning things around, for any fusion category $\mathcal{C}$ and module category $\mathcal{R}$, there exists a *unique* fusion category $\mathcal{D}$ for which $\mathcal{R}$ can be endowed with the structure of an invertible $(\mathcal{C}, \mathcal{D})$ -bimodule category. In category theoretic jargon, this fusion category $\mathcal{D}$ is called the *Morita dual* category of $\mathcal{C}$ with respect to $\mathcal{R}$ and is often notated as $\mathcal{D} = \mathcal{C}_{\mathcal{R}}^*$. Vice versa also $\mathcal{D}_{\mathcal{R}}^* = \mathcal{C}$. By definition, the Morita dual $\mathcal{C}_{\mathcal{R}}^*$ is the category of $\mathcal{C}$ -module endofunctors of the module category $\mathcal{R}$, denoted $\text{Func}(\mathcal{R}, \mathcal{R})$.

Example Consider the left Vec_G -module category Vec . The Morita dual $(\text{Vec}_G)_{\text{Vec}}^*$ can be computed to be $\text{Rep}(G)$. Given a representation $(V, \rho) \in \text{ob}(\text{Rep}(G))$, and $g \in \mathbb{C}_g \in \mathcal{I}_{\text{Vec}_G}$, the ${}^{\mathfrak{M}}F$ -symbol evaluates to

$$({}^{\mathfrak{M}}F_{\mathbb{C}}^{g\mathbb{C}\rho})_{\mathbb{C}, 1j}^{\mathbb{C}, i1} = \rho(g)_{ij}, \quad (3.48)$$

for basis vectors i, j in V .

3.4 Tensor network representations

3.4.1 Ground state representations

As anticipated in section 3.2, PEPS representations of string-net ground states are particularly straightforward to construct since the ground state projector can be expressed as a product of local commuting projectors, i.e. $\prod_p B_p^{\mathcal{D}}$. To this end, one can start by considering an admissible reference state in the string-net Hilbert space and acting on it with the ground state projector. This procedure works for generic reference states, but typically the state is chosen as the one with the identity object $1_{\mathcal{D}}$ on every edge $e \in E(\Lambda)$. By carefully working out the action of the ground state projector on the state, one finds that the obtained state is a superposition of all possible string configurations whose wave function amplitudes are a product of F -symbols, one for every vertex $v \in V(\Lambda)$. The local PEPS tensor then simply evaluates to an F -symbol, weighted with the appropriate quantum dimensions. Tensors of this kind are typically referred to as *triple line tensors* due to their sparse internal structure, as will be detailed below. These PEPS ground states are characterised by the existence of corresponding MPOs that live purely on the virtual level and can be freely moved through the lattice. Locally this topological invariance is characterised by a *pulling-through condition* satisfied by the PEPS and MPO tensors. Since pulling these MPOs through the lattice does not implement a physical action, they survive perturbations on the physical level as anticipated above. This construction of string-net ground states is covered in refs. [64–66], but turns out to not be the most general construction.

Indeed, as was shown first in ref. [59], other PEPS representations might exist. String-net ground states are defined by a number of local relations, and equivalently in the tensor network framework by locally deformable MPO symmetries. Therefore, one can find a distinct PEPS representation for any solution of a number of local relations that turn out to be in one-to-one correspondence with right module categories $\mathcal{R}$ over $\mathcal{D}$. For every such choice of $\mathcal{R}$ the corresponding PEPS tensors evaluate to the components of the $\mathcal{R}$ -module associator $\triangleleft F$, while the MPO symmetries now form a representation of the Morita dual category $\mathcal{D}_{\mathcal{R}}^{\star}$, that we will denote for simplicity by $\mathcal{C}$. While the *physical* degrees of freedom are still specified by $\mathcal{D}$, the choice of $\mathcal{R}$ together with $\mathcal{D}$ defines the *virtual* level of the PEPS. Morita equivalence between $\mathcal{D}$ and $\mathcal{C}$ turns out to be crucial since this is equivalent to the notion of MPO injectivity, which is a technical condition that guarantees the system size-independent ground state degeneracy [59, 63, 67, 68].

Let us unpack this construction in more detail. First, we consider for any (right) $\mathcal{D}$ -module category $\mathcal{R}$ following *module* plaquette operators:

$$B_{\mathfrak{p}}^{\mathcal{R}} = \sum_{R \in \mathcal{I}_{\mathcal{R}}} \frac{d_R}{\text{FPdim} \mathcal{R}} B_{\mathfrak{p}}^R =: \text{[Diagram: A dashed blue circle with a central shaded disk]} . \quad (3.49)$$

The action of $B_{\mathfrak{p}}^{\mathcal{R}}$ is now defined via the insertion of a loop labelled by $R \in \mathcal{I}_{\mathcal{R}}$ in the punctured plaquette $\mathfrak{p}$ and fusing it into the physical lattice, thereby making use of eqs. (3.34), (3.38) and (3.39). From these local rules it also follows readily that these module plaquette operators together with the $B_{\mathfrak{p}}^{\mathcal{D}}$'s and $B_{\mathfrak{p}}^{\mathcal{C}}$'s as defined in eq. (3.23) obey

$$B_{\mathfrak{p}}^{\mathcal{R}} \cdot B_{\mathfrak{p}}^{\mathcal{C}} = B_{\mathfrak{p}}^{\mathcal{R}}, \quad B_{\mathfrak{p}}^{\mathcal{D}} \cdot B_{\mathfrak{p}}^{\mathcal{R}} = B_{\mathfrak{p}}^{\mathcal{R}}, \quad (3.50)$$

for every $\mathfrak{p}$. Hence, we can conclude that by acting with the projector $\prod_{\mathfrak{p}} B_{\mathfrak{p}}^{\mathcal{R}}$ on any reference state, this will be a ground state of the $\mathcal{D}$ -string-net model.³ Note however, that the reference state is now necessarily one in $\mathcal{H}_{\mathcal{C}}(\Lambda)$, for example one that corresponds to assigning $\mathbb{1}_{\mathcal{C}}$ to all edges $\mathfrak{e} \in \mathbf{E}(\Lambda)$.

Putting everything together, we can thus summarise the derivation of the ground state PEPS representation of the $\mathcal{D}$ -string-net with a generic module category as:

$$\text{[Diagram: A hexagonal plaquette with a central shaded disk, surrounded by dashed blue and red lines]} \xrightarrow{(3.8)} \sum \{d\} \text{[Diagram: A hexagonal plaquette with a central shaded disk, surrounded by solid blue lines with dots]} \quad (3.51)$$

³By reversing the orientation of the loops in eq. (3.49) — or, equivalently, considering the module plaquette operators $B^{\mathcal{R}^{\text{op}}}$ for the *opposite module category* $\mathcal{R}^{\text{op}}$ — one obtains a ground state projector for the $\mathcal{C}$ -string-net model.

$$\begin{array}{c}
 (3.34) \\
 (3.39) \mapsto \sum \{d\} \{F\}
 \end{array}
 \begin{array}{c}
 \text{Diagram: A hexagon with a shaded circle in the center. Six lines extend from the vertices of the hexagon, forming a larger hexagonal shape. The central circle is shaded with diagonal lines.}
 \end{array}
 \quad (3.52)$$

Notably, all the wave function amplitudes derived via eq. (3.52) are written in terms of $|\mathbf{V}(\Lambda)| \triangleleft F$ -symbols. Their specific form invites us to consider tensors whose non-vanishing components are depicted by:

$$\begin{array}{c}
 \text{Diagram: A tensor with three legs labeled } R_1, R_2, R_3. \text{ The } R_1 \text{ leg has index } i, \text{ the } R_2 \text{ leg has index } j, \text{ and the } R_3 \text{ leg has index } k. \text{ The tensor is represented by a circle with three legs.}
 \end{array}
 \equiv
 \begin{array}{c}
 \text{Diagram: A tensor with three legs labeled } (R_1 Y_1 R_3, i), (R_3 Y_2 R_2, j), \text{ and } (Y_1 Y_2 Y_3, k). \text{ The tensor is represented by a circle with three legs.}
 \end{array}
 \quad (3.53)$$

$$= \frac{1}{\sqrt{d_{R_3}}} \left(\frac{d_{Y_1} d_{Y_2}}{d_{Y_3}} \right)^{1/4} (\triangleleft F_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl},$$

as well as a mirrored version evaluating to the complex conjugate F -symbol. Note that every tensor index corresponds to a triple of objects and a multiplicity index. As per our convention that F -symbols for which not all fusion rules are fulfilled are zero, some of the objects need to be shared among different tensor legs for the component to be non-vanishing. Due to their internal structure, we thus refer to these kind of tensors as *triple line tensors*. Contraction of these tensors happens via identification of objects on legs of concatenated tensors, identification of basis vectors and summing over them. We also adapt the convention that every closed loop in such a contraction is weighted with an additional factor of a quantum dimension.

It follows from the module pentagon equation eq. (3.35) that two distinct ways to contract these tensors are related via an F -symbol as follows:

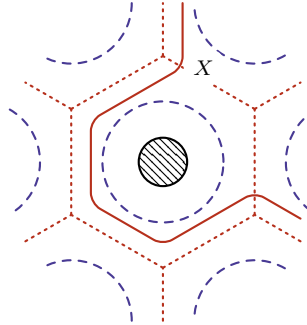
$$\begin{array}{c}
 \text{Diagram: A tensor with three legs labeled } Y_1, Y_2, Y_3. \text{ The } Y_1 \text{ leg has index } i, \text{ the } Y_2 \text{ leg has index } j, \text{ and the } Y_3 \text{ leg has index } k. \text{ The tensor is represented by a circle with three legs.}
 \end{array}
 = \sum_{Y_6, k, l} (F_{Y_4}^{Y_1 Y_2 Y_3})_{Y_5, ij}^{Y_6, kl}
 \begin{array}{c}
 \text{Diagram: A tensor with three legs labeled } Y_1, Y_2, Y_3. \text{ The } Y_1 \text{ leg has index } i, \text{ the } Y_2 \text{ leg has index } j, \text{ and the } Y_3 \text{ leg has index } k. \text{ The tensor is represented by a circle with three legs.}
 \end{array}
 \quad (3.54)$$

The tensors eq. (3.53) can thus be viewed as an explicit representation of the graphical calculus of the fusion category $\mathcal{D}$.

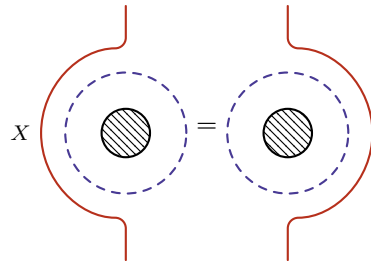
Finally, it can be easily checked that contraction of tensors eq. (3.53) according to the connectivity of the underlying lattice Λ and weighted with the appropriate factors of quantum dimensions exactly reproduces the ground state of eq. (3.52).

3.4.2 MPO symmetries

The PEPS ground state representation enjoys topological MPO symmetries acting only on the virtual level as foreshadowed above. Provided $\mathcal{D}$ and $\mathcal{R}$, these topological symmetries are organised in the Morita dual category $\mathcal{C}$. In order to derive the MPO representation of these topological lines, we consider similarly as in eq. (3.52) the ground state projector acting on an empty string configuration with a loop labelled by an object $X \in \text{ob}(\mathcal{C})$ inserted, for instance:


(3.55)

Note that the intersections of the X -labelled loop with the physical lattice are trivial. Topological invariance of these loops is guaranteed by the following *sliding property* of the ground state projectors:


(3.56)

for every $X \in \text{ob}(\mathcal{C})$. This sliding property is immediately derived from the by now familiar graphical calculus. In a way the ground state projector can thus be interpreted as a defect that hides the punctures from the topological lines and is therefore sometimes referred to as a *cloaking* defect or boundary condition [69, 70], or *Kirby colour* [71].

By carefully working out the recoupling of configurations with strings, one finds that in the presence of a string the resulting PEPS ground state should be modified by inserting triple line tensors of the form

$$\begin{aligned}
 \begin{array}{c} \text{Diagram: A central square node with four legs. Top-left leg (blue) labeled } R_3 \text{ with index } i. \text{ Top-right leg (blue) labeled } R_2 \text{ with index } j. \text{ Bottom-left leg (blue) labeled } R_1 \text{ with index } k. \text{ Bottom-right leg (red) labeled } R_4 \text{ with index } l. \text{ A red line } X \text{ passes through the center from } i \text{ to } l. \text{ A blue line } Y \text{ passes through the center from } j \text{ to } k. \end{array} \equiv (R_1 X R_3, i) \begin{array}{c} \text{Diagram: A square node with four legs. Top leg (blue) labeled } (R_3 Y R_2, j). \text{ Bottom leg (blue) labeled } (R_1 Y R_4, k). \text{ Left leg (red) labeled } (R_1 X R_3, i). \text{ Right leg (red) labeled } (R_4 X R_2, l). \end{array} \\
 &= \frac{1}{\sqrt{d_{R_1} d_{R_2}}} \left(\triangleright F_{R_2}^{X R_1 Y} \right)_{R_3, ij}^{R_4, kl}, \tag{3.57}
 \end{aligned}$$

along the path of the loop at every intersection with the physical lattice. The full MPO is then obtained by contracting these tensor along the ‘horizontal’ direction. In fact, the simple objects $X \in \mathcal{I}_C$ and $Y \in \mathcal{I}_D$ label the *injective blocks* of these topological MPOs in either direction in the sense of chapter 2. It follows from the consistency equation eq. (3.46) that these tensors together with the ground state PEPS tensors satisfy the *pulling-through* or *zipper equation*:

$$\begin{array}{c} \text{Diagram: A horizontal red line } X \text{ with two square nodes. The left node has a blue line } Y_1 \text{ entering from the top and a blue line } Y_3 \text{ exiting to the bottom. The right node has a blue line } Y_2 \text{ entering from the top and a blue line } Y_3 \text{ exiting to the bottom.} \end{array} = \begin{array}{c} \text{Diagram: A horizontal red line } X \text{ with one square node. The node has a blue line } Y_1 \text{ entering from the top and a blue line } Y_3 \text{ exiting to the bottom.} \end{array}, \tag{3.58}$$

without modifying the index i , so that these MPOs can move freely on the virtual level of the PEPS. In fact, by turning things around, one can interpret the PEPS tensors as *intertwiners* for the MPO tensors in the ‘vertical’ direction.

Continuing along these lines, it is easy to convince oneself that two MPOs labelled by X_1 and X_2 supported on homologous loops $\ell_1 \sim \ell_2$ can be fused and decomposed according to $X_1 \otimes X_2 = \sum_{X_3 \in \mathcal{I}_C} N_{X_1 X_2}^{X_3} X_3$. This fusion product can be implemented *locally* in terms of *fusion* tensors defined in terms of the left module associator $\triangleright F$:

$$\begin{aligned}
 \begin{array}{c} \text{Diagram: A Y-junction of three red lines } X_1, X_2, X_3 \text{ meeting at a central node. } X_1 \text{ (left) has index } i, X_2 \text{ (right) has index } k, X_3 \text{ (bottom) has index } j. \text{ Blue lines } R_1, R_2, R_3 \text{ are attached to the legs.} \end{array} \equiv \begin{array}{c} \text{Diagram: A circular node with three red lines entering from the top-left, top-right, and bottom. The lines are labeled } (R_2 X_1 R_3, l), (R_3 X_2 R_1, k), \text{ and } (X_1 X_2 X_3, i). \end{array} \\
 &= \frac{1}{\sqrt{d_{R_3}}} \left(\frac{d_{X_1} d_{X_2}}{d_{X_3}} \right)^{1/4} \left(\triangleright F_{R_2}^{X_1 X_2 R_1} \right)_{X_3, ij}^{R_3, kl}, \tag{3.59}
 \end{aligned}$$

that together with the MPO tensors eq. (3.57) satisfy a pulling-through equation akin to a rotated version of eq. (3.58).

In summary, a $\mathcal{D}$ -string-net PEPS representation fully exploits the structure provided by an invertible $(\mathcal{C}, \mathcal{D})$ -bimodule category $\mathcal{R}$. The bimodule category encodes a collection of five unitary F -symbols in terms of which the PEPS tensors, MPO tensors and their fusion tensors are defined. These F -symbols simultaneously satisfy six coupled pentagon equations. Two of these pentagon equations involve only the associators of $\mathcal{D}$ and $\mathcal{C}$ respectively. Two others — eq. (3.35) and its left counterpart — involve also the left and right module associators and are interpreted as recoupling relations for the PEPS and fusion tensors akin to eq. (3.54). The remaining two involve the bimodule associator ${}^{\mathcal{R}}F$ and are in turn interpreted as intertwining relations of the MPO tensors in either direction, as in eq. (3.58).

3.4.3 Anyonic excitations and the tube algebra

We now proceed to discuss the construction of excited states and complete bases of torus ground states. As mentioned above, both excited states and distinct ground states can be obtained by studying the action of loop and string operators labelled by elements of $\mathcal{Z}(\mathcal{D})$ on a reference ground state. From the perspective of tensor networks it is however more natural to describe these states in terms of the *tube algebra*. In fact, the tube algebra provides a natural framework for computing the Drinfeld centre of a fusion category and the associated half-braidings that appear in the definition of string operators. Formally, this follows from a braided equivalence between the category of representations of the tube algebra and the Drinfeld centre [51, 72–75].

Up to this point, we have been somewhat vague about the precise nature of the anyonic excitations appearing in the string-net models. Physically, anyons can be viewed as quasiparticle excitations whose local energy density is slightly higher than that of the ground state. They are always created in particle-antiparticle pairs on top of a reference torus ground state, a feature that also becomes manifest in the tensor network anyon ansatz discussed below. Since the (extended) string-net Hamiltonian is a local commuting-projector Hamiltonian, anyonic excitations correspond to violations of a small number of plaquette terms, vertex terms, or a mixture of both.

Morally, the idea underlying the tube algebra is to regard anyon excitations as different ground state sectors of the theory defined on a punctured surface, where the punctures coincide with the locations of the anyons, rather than viewing them as defects above a ground state on the unpunctured lattice.

The ground state space on such a punctured surface decomposes into sectors characterised by fixed boundary conditions around the punctures. These sectors form indecomposable *modules* under the action of the tube algebra, and the irreducible representations of the tube algebra precisely label the different anyon species. Since the tube algebra can be endowed with the structure of a (finite-dimensional) C^* algebra, the existence of these irreducible modules is guaranteed by the *Artin-Wedderburn theorem*.

We now proceed with a formal definition of the tube algebra. As an algebra, it is spanned by basis elements that may be represented diagrammatically as cylinders decorated by $\mathcal{C}$ -coloured graphs and containing a number of marked points on each boundary. Different choices for the number of marked boundary points give rise to different tube algebras. Although these algebras are not isomorphic, they are all *Morita equivalent*. Recall that Morita equivalence is a strictly weaker notion of equivalence than algebra isomorphism but guarantees that the corresponding categories of representations are equivalent, implying that all such tube algebras describe the same anyon theory. The multiplication of tubes is defined diagrammatically by stacking two decorated cylinders on top of one another and subsequently applying the graphical calculus to rewrite the resulting diagram as a linear combination of basis tubes. In this way, the structure factors are determined entirely by the associator of the underlying fusion category.

Without loss of generality, we restrict ourselves to the tube algebra with a single boundary label on each side of the cylinder. Within the tensor network formalism, the basis elements of the tube algebra can be represented as matrix product operators on periodic boundary conditions of the following form [66–68]:

$$\mathfrak{T}^{A,A',X,X',i,j} := \text{---} \square \text{---} \square \text{---} \overset{A'}{\underset{i}{\circ}} \text{---} \overset{X'}{\underset{j}{\circ}} \text{---} \square \text{---}, \quad (3.60)$$

for all $A, A', X, X' \in \mathcal{I}_{\mathcal{C}}$ such that $X' \in X \otimes A$, $X' \in A' \otimes X$ and where $i \in \mathcal{C}(X', A' \otimes X)$ and $j \in \mathcal{C}(X \otimes A, X')$. Crucially, the multiplication rules of the tube algebra are independent of the number of MPO tensors appearing in eq. (3.60). This follows from the zipper equation eq. (3.58) that they satisfy. We then compute the product $\mathfrak{T}^{A',A'',X_1,X'_1,i,j} \cdot \mathfrak{T}^{A,\tilde{A},X_2,X'_2,k,l}$, which is

non-vanishing only when $A' = \tilde{A}$, by recoupling the tensor network diagram

which yields:

$$\begin{aligned}
 & \mathfrak{T}^{A', A'', X_1, X_1', i, j} \cdot \mathfrak{T}^{A, \tilde{A}, X_2, X_2', k, l} \\
 &= \sum_{\substack{X_3, X_3' \\ m, n, \tilde{o} \\ p, q}} \sqrt{\frac{d_{X_3'} d_{A'} d_{X_1} d_{X_2}}{d_{X_3} d_{X_1'} d_{X_2'}}} (F_{X_3'}^{X_1 A' X_2})_{X_1', jn}^{X_2', lo} (\bar{F}_{X_3'}^{X_1 X_2 A})_{X_3, mq}^{X_2', lo} \\
 & \quad \times (\bar{F}_{X_3'}^{A'' X_1 X_2})_{X_1', in}^{X_2, mp} \cdot \mathfrak{T}^{A, A'', X_3, X_3', p, q}.
 \end{aligned} \tag{3.62}$$

Invoking the Artin-Wedderburn theorem, we are guaranteed the existence of an isomorphism between the tube algebra and a direct sum of simple matrix algebras, the number of which is equal to the simple objects in $\mathcal{Z}(\mathcal{C})$. Concretely, we can construct linear combinations of tubes that behave as *matrix units* within each block under the multiplication eq. (3.62) [68, 73, 76, 77], viz.:

$$e^{Z, A_i, B_j} \cdot e^{Z', C_k, D_l} = \delta_{Z, Z'} \delta_{B, C} \delta_{j, k} e^{Z, A_i, D_l}, \tag{3.63}$$

where Z, Z' label simple objects of $\mathcal{Z}(\mathcal{C})$, and A, B (C, D) denote the simple objects appearing in the decomposition of Z (Z') as object of $\mathcal{C}$, with i, j (k, l) denoting their multiplicity within this decomposition. Under taking the star, they are transposed according to

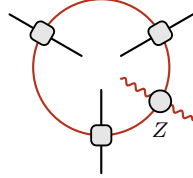
$$(e^{Z, A_i, B_j})^\dagger = e^{Z, B_j, A_i}. \tag{3.64}$$

Notably, matrix units of the form e^{Z, A_i, A_i} are (*simple*) *idempotents*, while off-diagonal elements e^{Z, A_i, B_j} , $A_i \neq B_j$ are *nilpotents*. Summing over all simple idempotents within the same block provides the *minimal central idempotents*

$$\mathcal{P}^Z := \sum_{A_i} e^{Z, A_i, A_i}. \tag{3.65}$$

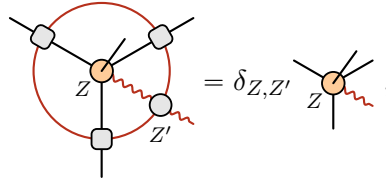
Connecting back to the string operators, it can be shown that the coefficients of the central idempotents when expressed in the basis eq. (3.60) exactly coincide with the coefficients of the half-braiding isomorphisms eq. (3.26) [73, 77, 78].

Let us denote the minimal central idempotents now as follows



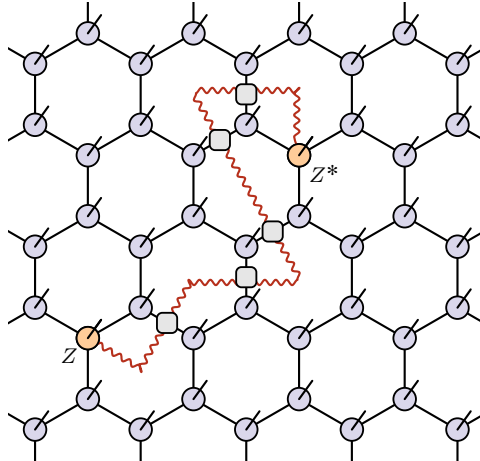
$$(3.66)$$

wherein compared to eq. (3.60) we have explicitly depicted the periodic boundary condition, and conflated the fusion tensors into one tensors labelled by Z . The anyon ansatz of ref. [68] now stipulates that in order to define a (Z, Z^*) anyon–anti-anyon pair, one replaces one of the PEPS ground state tensors by a new type of tensor contained in the image of the corresponding idempotent, i.e.:



$$(3.67)$$

and analogously for Z^* , with the additional requirement that the pair (Z, Z^*) is in the vacuum sector, as spelled out in detail in ref. [68]. Taking everything together, we could depict such a state as:



$$(3.68)$$

Let us stress once again that the location of the string connecting the anyons is not detectable as it can be freely moved through the lattice in virtue of the

pulling-through condition eq. (3.58). With this we conclude our discussion of the anyon ansatz, but let us remark that from it, the anyonic braiding statistics and the fusion rules of the anyons, as encoded in $\mathcal{Z}(\mathcal{C})$, can be computed directly, as demonstrated in ref. [68].

Given these tubes, a complete basis for the $\mathcal{D}$ -string-net torus ground space is also straightforward to construct. For each anyon $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{C})}$ one constructs a ground state with an anyon flux Z threading the ‘vertical’ cycle of the torus by inserting the corresponding tube and closing it. Schematically:

$$\sum_{X \in \mathcal{I}_{\mathcal{C}}} \quad (3.69)$$

In a similar way, one constructs a set of basis states by inserting the tubes in the orthogonal direction. The basis states obtained as such are related to those of eq. (3.69) via the S -matrix.

Note also that we could have used a simple idempotent to construct a ground state. As argued in ref. [68], all simple idempotents contained in the same central idempotent yield the same ground state.

Generalised symmetries and dualities

4.1 MPO representation theory

Reversing the logic of the previous chapter, we are now in a position to construct distinct injective MPO representations of a unitary fusion category $\mathcal{C}$ via a choice of (left) $\mathcal{C}$ -module category $\mathcal{R}$. Indeed, for any module category $\mathcal{R}$ over $\mathcal{C}$, there exists a unique Morita dual $\mathcal{D}$ such that $\mathcal{R}$ is endowed with the structure of an invertible $(\mathcal{C}, \mathcal{D})$ -bimodule category, as discussed in the previous chapter. From the bimodule associators ${}^{\bowtie}F$ we can construct four-valent MPO tensors such that on periodic boundary conditions and independent of system size we have the graphical identities

$$\begin{array}{c} \text{Diagram 1: Two horizontal red lines with three vertical black lines crossing them. The top line has a square box at each crossing, labeled X_1 above the second crossing. The bottom line has a square box at each crossing, labeled X_2 above the second crossing. } \\ \\ \text{Diagram 2: A single horizontal red line with three vertical black lines crossing it. The line has a square box at each crossing, labeled X_3 above the second crossing. } \end{array} = \sum_{X_3} N_{X_1 X_2}^{X_3} \text{Diagram 2}, \quad (4.1)$$

for structure constants N coinciding with the fusion rules of $\mathcal{C}$. In virtue of the fundamental theorem eq. (2.13), there exists $N_{X_1 X_2}^{X_3}$ linearly independent *fusion tensors* that locally implement these fusion rules via

$$\begin{array}{c} \text{Diagram 1: A vertical black line with two horizontal red lines crossing it. The top line has a square box at the crossing, labeled X_1 above. The bottom line has a square box at the crossing, labeled X_2 above. A red line with a circle containing i connects the two boxes. } \\ \\ \text{Diagram 2: A vertical black line with two horizontal red lines crossing it. The top line has a square box at the crossing, labeled X_1 above. The bottom line has a square box at the crossing, labeled X_2 above. A red line with a circle containing i connects the two boxes. } \end{array} = \begin{array}{c} \text{Diagram 3: A vertical black line with two horizontal red lines crossing it. The top line has a square box at the crossing, labeled X_1 above. The bottom line has a square box at the crossing, labeled X_2 above. A red line with a circle containing i connects the two boxes. } \end{array} \quad (4.2)$$

These fusion tensors are constructed from the $\triangleright F$ -symbols, and are themselves recoupled via the F -symbol of $\mathcal{C}$.

Note that in sharp contrast to the group case, generic categorical MPO symmetries are defined on *constrained* Hilbert spaces. These constraints are implemented via the module action $\triangleleft : \mathcal{R} \times \mathcal{D} \rightarrow \mathcal{R}$, as detailed in chapter 8. In what follows we therefore sometimes refer to $\mathcal{R}$ as describing the *kinematical degrees of freedom* of the model. In fact, the $\mathcal{C}$ -symmetric models that we consider below can be thought of as a tensor network representation and a generalisation of the *anyonic spin chain construction* first considered in ref. [79].

4

4.2 Symmetric boundary Hamiltonians and dualities

Having these MPO representations of categorical symmetries at our disposal, we now proceed to constructing local $(1+1)d$ Hamiltonians that commute with said symmetries. Heuristically, one can interpret these Hamiltonians as the boundary Hamiltonians for the string-net models of the previous chapter.

Let us note at this point that also *classical* edge theories of string-net models can be constructed via the *strange correlator* construction. This construction was introduced first as a diagnostic of SPT phases in ref. [80], and later extended to topological string-net wave functions in ref. [81]. In the latter, a classical two-dimensional partition function is constructed as the overlap of a PEPS string-net ground state with a non-topological PEPS, often chosen as a product state. These classical models manifestly inherit all topological symmetries of the string-net model. In the case of critical models, these symmetries are lattice remnants of the topological symmetry lines of the conformal field theory describing their scaling limit [81–87]. In essence, this construction is a lattice incarnation of the celebrated TFT/CFT correspondence due to Fröhlich, Fuchs, Runkel and Schweigert [88–92].

Before proceeding to the construction of Hamiltonians with a genuine categorical symmetry $\mathcal{C}$, let us first specialise to the group case to attain some intuition and put things in a perhaps more familiar context. Considering for a moment an *irreducible* unitary representation (V, α) of a group G , and a rank four tensor — representing for example a local nearest-neighbour Hamiltonian term — whose domain and codomain are the tensor product

space $V \otimes V$. Invariance under the action of G translates graphically to

$$\begin{array}{c} \alpha(g) \quad \alpha(g) \\ | \quad | \\ \text{---} h \text{---} \\ | \quad | \\ \alpha(g)^\dagger \quad \alpha(g)^\dagger \end{array} = \begin{array}{c} | \quad | \\ \text{---} h \text{---} \\ | \quad | \end{array}, \quad \forall g \in G. \quad (4.3)$$

The *Wigner-Eckart* theorem then stipulates that the tensor h can be written as the contraction of two Clebsch-Gordan coefficients implementing the unitary basis transformation

$$\begin{bmatrix} \alpha & \alpha \\ - & - \end{bmatrix}^{\beta, k} : V \otimes V \rightarrow W, \quad k = 1, 2, \dots, N_{\alpha\alpha}^\beta, \quad (4.4)$$

for all irreps (W, β) appearing in the decomposition $\alpha \otimes \alpha \simeq \bigoplus_\beta N_{\alpha\alpha}^\beta \beta$, and *reduced matrix elements* depending only on the irrep labels [93, 94]. Concretely putting things together, we find that h can be expressed in its components as:

$$\begin{array}{c} i \quad i' \\ | \quad | \\ \text{---} h \text{---} \\ | \quad | \\ j \quad j' \end{array} = \sum_{\beta, k} h(\beta, k) \left(\sum_l \begin{bmatrix} \alpha & \alpha \\ i & i' \end{bmatrix}^{\beta, k} \begin{bmatrix} \alpha & \alpha \\ j & j' \end{bmatrix}^{\beta, k} \right)^* |i, i' \rangle \langle j, j'|, \quad (4.5)$$

for complex coefficients $h(\beta, k)$ and $i, i', j, j' = 1, 2, \dots, \dim_{\mathbb{C}} \alpha$. The explicit form of the tensor h in terms of Clebsch-Gordan coefficients invites us to equivalently express it in terms of the fusion tensors eq. (3.53) specialised to the case $\mathcal{C} = \text{Vec}_G$, $\mathcal{R} = \text{Vec}$, $\mathcal{D} = (\text{Vec}_G)_{\text{Vec}}^* = \text{Rep}(G)$:

$$\begin{array}{c} | \quad | \\ \text{---} h \text{---} \\ | \quad | \end{array} = \sum_{\beta, k} h(\beta, k) \begin{array}{c} \alpha \quad \alpha \\ | \quad | \\ \text{---} k \text{---} \\ | \quad | \\ \beta \\ | \quad | \\ \text{---} k \text{---} \\ | \quad | \\ \alpha \quad \alpha \end{array}, \quad (4.6)$$

and where we have left the module labels implicit.¹ Notably, in this more abstract formulation the origin of the invariance of h under the G -action can be traced back to the pulling-through property eq. (3.58). Indeed, remember from eq. (3.48) that the $\approx F$ -symbols for the case at hand precisely evaluate to

¹Note that compared to eq. (3.53) we also dropped factors of quantum dimension for ease of notation.

Continuing along these lines, we ought to consider the triple line tensors eq. (3.53) evaluating to ${}^{\triangleleft}F$ as the elementary $\mathcal{C}$ -symmetric tensors, or *generalised* Clebsch-Gordan coefficients [63]. For these, the symmetry is implemented via the pulling-through equation eq. (3.58) and using them local symmetric Hamiltonians can be constructed. Viewed from this perspective, the topological invariance of the MPO symmetries guaranteed by the pulling-through equation can be seen as the *defining* property of a *generalised, categorical symmetry*. This construction of symmetric Hamiltonians in terms of tensor networks thereby provides an explicit one-dimensional lattice realisation of the viewpoint advocated first in ref. [50] to identify the existence of a collection of topological operators as the definition of a *global* symmetry.

$$h_{i,i+1}^{\mathcal{R}} = \sum_{\{Y\}} \sum_{i,i'} h(\{Y\}, i, i') \cdot \text{diagram} \quad (4.7)$$

Loosely speaking, the local symmetric operators eq. (4.7) constitute the (von Neumann) *bond algebra* of symmetric operators [95, 97–101]. By this it is meant that all local symmetric operators — collectively denoted by $\mathcal{A}$ — can be expressed as a superposition of basis elements $\mathcal{B} = \{\mathbb{1}, h_{i,i+1}^{\mathcal{R}}, h_{i,i+1}^{\mathcal{R}} \tilde{h}_{i+1,i+2}^{\mathcal{R}}, \dots\}$, i.e. $\mathcal{A} = \text{Span}_{\mathbb{C}}\{\mathcal{B}\}$ as a vector space. Being an algebra, $\mathcal{A}$ is moreover endowed with a product that linearly extends the product on the basis vectors. An explicit choice of such a basis is not needed for our discussion, but crucially this product can be fully expressed in terms of structure constants depending

only on the $\mathcal{D}$ F -symbol, viz.:

$$b_A \cdot b_B = \sum_l f_{AB}^C(\{F\}) b_C, \quad \forall b_{A,B,C} \in \mathcal{B}. \quad (4.8)$$

In other words, $\mathcal{D}$ provides a complete description of the algebra of local symmetric operators.

Let us now once more shift our perspective and take the viewpoint where the choice of bond algebra $\mathcal{D}$ and a choice of $\mathcal{R}$ are provided, rather than the symmetry $\mathcal{C}$. From eq. (4.8) we can deduce that the spectrum of a generic Hamiltonian $H^{\mathcal{R}}$ is independent of the choice of module category $\mathcal{R}$. This suggests that upon changing the module category $\mathcal{R} \mapsto \mathcal{R}'$, we perform a *duality* transformation that maps the initial model with the symmetry $\mathcal{D}_{\mathcal{R}}^*$ to one with a *dual* symmetry $\mathcal{D}_{\mathcal{R}'}^*$. On the level of the local symmetric operators eq. (4.7) this boils down to changing the module associators to which the symmetric tensors evaluate, while preserving the reduced matrix elements h . Crucially, the dual symmetry can be a genuine categorical symmetry even when the initial symmetry structure is a group, as exemplified by $(\text{Vec}_G)^*_{\text{Vec}} = \text{Rep}(G)$ for any non-abelian group G . Notwithstanding the abstract nature of the formalism, most well-known duality transformations in the condensed matter and high-energy physics literature — in particular Kramers–Wannier and Kennedy–Tasaki duality — can be rephrased in this language given an appropriate choice of symmetry the model [77, 95, 102–106].

Example: Kramers–Wannier duality — Part I Let us consider the transverse field Ising model encountered above in eq. (1.11). As already mentioned, the model enjoys a $\mathbb{Z}_2$ spin flip symmetry acting on-site and generated by $\prod_i X_i$. Abstractly, the symmetry is encoded in the fusion category $\mathcal{C} = \text{Vec}_{\mathbb{Z}_2}$. It is then straightforward to verify that the Hamiltonian terms can be represented on the Hilbert space $\mathbb{C}[\mathbb{Z}_2]^{\otimes L}$, $\mathbb{C}[\mathbb{Z}_2] = \{|0\rangle, |1\rangle\}$, naturally associated to the choice $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$, and explicitly read:

$$Z_i Z_{i+1} = \sum_{g_i, g_{i+1}} (-1)^{g_{i+1}} \begin{array}{c} g_i \quad g_{i+1} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ g_i \quad g_{i+1} \end{array}, \quad X_i = \sum_{g_i, g_{i+1}} \begin{array}{c} g_{i+1} \quad g_{i+1}+1 \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ g_i \quad g_{i+1} \end{array}. \quad (4.9)$$

From the classification of Vec_G -module categories reviewed in section 3.3.2, it follows that the only non-trivial duality transformation amounts to changing

$\text{Vec}_{\mathbb{Z}_2} \mapsto \text{Vec}$. It is straightforward to derive that upon this change, the Hamiltonian terms transmute according to the prescription

$$Z_i Z_{i+1} \mapsto Z_{i+\frac{1}{2}}, \quad X_i \mapsto X_{i-\frac{1}{2}} X_{i+\frac{1}{2}}. \quad (4.10)$$

Notably, the dual symmetry is generated by $\prod_i Z_{i+1/2}$, described by the symmetry category $\text{Rep}(\mathbb{Z}_2) = (\text{Vec}_{\mathbb{Z}_2})_{\text{Vec}}^*$, in accordance with the prediction of the framework. Also, the order and disorder operators are mapped into each other

$$Z_i Z_j \mapsto \prod_{k=i}^{j-1} Z_{k+\frac{1}{2}}, \quad \prod_{k=i}^j X_k \mapsto X_{i-\frac{1}{2}} X_{j+\frac{1}{2}}, \quad (4.11)$$

as prescribed by the Landau paradigm.

Until now we have not been very explicit about the exact meaning of the map $\mathcal{R} \mapsto \mathcal{R}'$ or expressions like eq. (4.10). Let us note first that eq. (4.10) cannot be implemented by a unitary transformation. Rather, associated to the Kramers-Wannier duality, and more generally for every duality transformation, there is an explicit non-invertible *intertwiner* D that implements the duality via

$$h_{i,i+1}^{\mathcal{R}'} \circ D = D \circ h_{i,i+1}^{\mathcal{R}}, \quad (4.12)$$

for all local symmetric terms. Notably, the Kramers-Wannier duality intertwiner has a non-trivial kernel consisting of all states with a non-trivial charge under the $\mathbb{Z}_2$ spin flip symmetry. Within our framework, such an intertwiner admits an explicit MPO representation. Concretely, these MPO intertwiners are constructed from the data of a $\mathcal{D}$ -module functor between the module categories $\mathcal{R}$ and $\mathcal{R}'$. A $\mathcal{D}$ -module functor between (right) $\mathcal{D}$ -module categories $\mathcal{R}$ and $\mathcal{R}'$ is a functor $\mathfrak{F} : \mathcal{R} \rightarrow \mathcal{R}'$ compatible with the respective module structures, meaning that it is equipped with a natural isomorphism ω whose components take the form $\omega^{RY} : \mathfrak{F}(R \triangleleft Y) \xrightarrow{\sim} \mathfrak{F}(R) \triangleleft' Y$ for objects $R \in \mathcal{I}_{\mathcal{R}}$ and $Y \in \mathcal{I}_{\mathcal{D}}$, with $\triangleleft$ and $\triangleleft'$ the action functors of $\mathcal{R}$ and $\mathcal{R}'$ respectively. This isomorphism adheres to a pentagon axiom involving the module associators of both $\mathcal{R}$ and $\mathcal{R}'$. For each simple module functor $X \equiv (\mathfrak{F}, \omega)$, we write ${}^X \omega$ for its associated isomorphism, whose components are captured diagrammatically by

$$\begin{array}{c}
 X \quad R_1 \quad Y \\
 \diagdown \quad | \quad \diagup \\
 \quad i \\
 R'_1 \quad | \\
 \quad j \\
 \quad | \\
 R'_2
 \end{array}
 = \sum_{R_2, k, l} \left({}^X \omega_{R'_2}^{R_1 Y} \right)_{R'_1, ij}^{R_2, kl}
 \begin{array}{c}
 X \quad R_1 \quad Y \\
 \diagdown \quad | \quad \diagup \\
 \quad k \\
 R_2 \quad | \\
 \quad l \\
 \quad | \\
 R'_2
 \end{array} . \quad (4.13)$$

One can then define the notion of *module natural transformations*, and these form the morphisms in the (semi-simple) category $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ of $\mathcal{D}$ -module functors between the $\mathcal{D}$ -module categories $\mathcal{R}$ and $\mathcal{R}'$ [40]. As anticipated above in section 3.3.3, the Morita dual fusion category $\mathcal{D}_{\mathcal{R}}^*$ corresponds to the category of $\mathcal{D}$ -module *endofunctors* $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R})$ whose fusion structure is given by the composition of module functors. Similarly, the category $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ is naturally endowed with the structure of an invertible $(\mathcal{D}_{\mathcal{R}}^*, \mathcal{D}_{\mathcal{R}'}^*)$ -bimodule category.

Provided a (simple) object X in $\mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')}$, we construct the MPO tensor of the duality intertwiner as

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & j & & \\
 & R'_1 & \uparrow & R'_2 & \\
 i & \leftarrow & X & \rightarrow & l \\
 & R_1 & \downarrow & R_2 & \\
 & & Y & & \\
 & & k & &
 \end{array}
 \end{array} = (X \omega_{R'_2}^{R_1 Y})_{R'_1, ij}^{R_2, kl}. \quad (4.14)$$

The duality transformation eq. (4.12) then follows from the intertwining property

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & R'_5 & & \\
 & & \uparrow & & \\
 & & X & & \\
 & & \downarrow & & \\
 k & \leftarrow & Y_1 & \rightarrow & l \\
 & R'_3 & & R'_4 & \\
 i & \leftarrow & X & \rightarrow & j \\
 & R_1 & & R_2 & \\
 & & Y_2 & & \\
 & & m & & \\
 & & \downarrow & & \\
 & & Y_3 & & \\
 & & n & &
 \end{array}
 \end{array} = \begin{array}{c}
 \begin{array}{ccccc}
 & & R'_5 & & \\
 & & \uparrow & & \\
 & & X & & \\
 & & \downarrow & & \\
 k & \leftarrow & Y_1 & \rightarrow & l \\
 & R'_3 & & R'_4 & \\
 i & \leftarrow & X & \rightarrow & j \\
 & R_1 & & R_2 & \\
 & & Y_2 & & \\
 & & m & & \\
 & & \downarrow & & \\
 & & Y_3 & & \\
 & & n & &
 \end{array}
 \end{array}, \quad (4.15)$$

which is a graphical representation of the aforementioned consistency equation satisfied by ω .

Collecting all these structures in a table:

Physical notion	Categorical notion	Notation
Algebra local symmetric operators	Fusion category	$\mathcal{D}$
Kinematical degrees of freedom	$\mathcal{D}$ -module category	$\mathcal{R}$
Global symmetry original model	Dual fusion category	$\mathcal{C} = \mathcal{D}_{\mathcal{R}}^{\star}$
Duality operators	$(\mathcal{C}, \mathcal{C}')$ -bimodule category	$\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$
Dual kinematical degrees of freedom	$\mathcal{D}$ -module category	$\mathcal{R}'$
Dual global symmetry	Dual fusion category	$\mathcal{C}' = \mathcal{D}_{\mathcal{R}'}^{\star}$

Table 4.1: Dictionary of the key physical concepts, corresponding category theoretical structures, and their notation used throughout the thesis.

4

Example: Kramers-Wannier duality — Part II This leaves us to explain how the Kramers-Wannier duality eq. (4.10) fits into this framework. It follows from $\text{Fun}_{\text{Vec}_{\mathbb{Z}_2}}(\text{Vec}_{\mathbb{Z}_2}, \text{Vec}) \simeq \text{Vec}$ that there exist a unique injective MPO intertwiner implementing the duality $\text{Vec}_{\mathbb{Z}_2} \rightarrow \text{Vec}$.² Removing the trivial lines and Hom spaces from the module structure eq. (4.14) for the case at hand and rewriting the MPO in more conventional tensor network notation, we obtain,

$$D_{\text{KW}} = \text{---} \square \text{---} \square \text{---} \square \text{---} \square \text{---} \square \text{---} \square \text{---} , \quad (4.16)$$

where the constituent tensors are GHZ tensors in the pauli Z and X basis respectively:

$$\text{---} \square \text{---} = |000\rangle + |111\rangle, \quad \text{---} \oplus \text{---} = |+++\rangle + |--\rangle. \quad (4.17)$$

Note in particular that the physical legs on either side of the intertwiner are shifted with half a lattice site with respect to each other, in accordance with the expected mapping of operators eq. (4.10). One subsequently finds that the mapping of symmetric operators eq. (4.10) follows directly from the local symmetry properties

$$\begin{array}{c} \text{---} \square \text{---} \\ | \\ \text{---} \square \text{---} \end{array} \stackrel{Z}{=} \begin{array}{c} \text{---} \square \text{---} \\ | \\ \text{---} \square \text{---} \end{array} \stackrel{Z}{=} \begin{array}{c} \text{---} \square \text{---} \\ | \\ \text{---} \square \text{---} \end{array} \stackrel{X}{=} \begin{array}{c} \text{---} \square \text{---} \\ | \\ \text{---} \square \text{---} \end{array} , \quad (4.18)$$

²It holds more generally that $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$. Concretely, the simple module functors in $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$ are in one-to-one correspondence with simple objects R in $\mathcal{I}_{\mathcal{R}}$ such that $R_{\omega} = {}^{\omega}F^{R--}$.

as well as analogous symmetry properties for the second tensor in eq. (4.17) with $X \leftrightarrow Z$ interchanged.

4.3 The Landau paradigm for categorical symmetries

We are thus finally led to the problem of classifying gapped phases of matter protected by a categorical symmetry in one-dimensional lattice models. Returning to the example of the Ising model of the previous section, one observes that under Kramers–Wannier duality, the symmetric phase ($J < \lambda$) is mapped to a symmetry-broken phase ($J > \lambda$) of the dual Ising model, and conversely. More generally, examples such as the Kennedy–Tasaki duality discussed at length in chapter 10 illustrate that for any gapped phase there exists a *unique* duality transformation that can map it to a dual model whose *dual symmetry* is *spontaneously broken*. This suggests a one-to-one correspondence between module categories and gapped phases of matter.

This observation is formalised by the *generalised Landau paradigm* for one-dimensional gapped phases protected by a symmetry encoded in a UFC $\mathcal{C}$ [20, 62, 96, 107]. The paradigm asserts that gapped phases are in one-to-one correspondence with module categories $\mathcal{P}$ over the symmetry category $\mathcal{C}$. The simple objects of the module category $\mathcal{P}$ label the orthogonal ground states, whilst the module action encodes the action of the symmetry on these ground states. Crucially, for genuinely non-invertible symmetries, the action of a symmetry operator on a given ground state does not in general produce another unique ground state. Instead, it yields a superposition of distinct orthogonal ground states, in sharp contrast with the ordinary invertible case.

The generalised Landau paradigm has been established rigorously in the context of matrix product states in ref. [62], using tensor network techniques closely related to those employed in section 2.4 for the classification of gapped phases protected by a group symmetry. Notably, in the case of the symmetry category $\mathcal{C} = \mathbf{Vec}_G$ this exactly reproduces the conventional Landau classification in terms of subgroups $H \subseteq G$ and 2-cocycles $[\psi] \in H^2(G, \mathbf{U}(1))$ as already remarked and anticipated in section 2.4.

Two of the most important applications of the generalised Landau paradigm are as follows. First, it implies that there always exists a dual description of a given phase in which the entanglement is minimised. Indeed, in the spontaneously broken dual model, degeneracies in the Schmidt spectrum are lifted, thereby maximally reducing the redundancy in the matrix product state description of both the ground states and the low-lying excitations. This has been demonstrated numerically in ref. [96].

Second, in the case of a continuous $SU(2)$ symmetry, where the Mermin–Wagner theorem forbids spontaneous symmetry breaking, a duality transformation can map the system to a dual model with a $\text{Rep}(SU(2))$ symmetry. Since this dual symmetry is discrete rather than continuous, the assumptions underlying the Mermin–Wagner theorem no longer apply, allowing for symmetry-broken phases in the dual description. In fact, this exactly underlies standard numerical tensor network methods with continuous symmetries [108, 109].

4.4 Symmetry-twisted boundary conditions

4

Up until this point, we have considered only periodic boundary conditions. Notably, duality MPOs typically have a non-trivial kernel when defined on periodic boundary conditions and are therefore non-invertible and hence non-unitary. This was already observed in the example of the Kramers–Wannier duality of the Ising model. In that case we noticed that the Kramers–Wannier duality projects out states with non-trivial charge under the spin-flip symmetry.

To access and relate the full spectrum of dual theories, it is necessary to consider *symmetry-twisted boundary conditions* [77, 110–113]. A symmetry twist can heuristically be understood as a spatially localised defect that extends along the time direction. Within our framework, symmetry twists are incorporated as additional degrees of freedom of the Hilbert space. They are conventionally inserted between sites L and $L + 1 \sim 1$, and are labelled by objects in the symmetry fusion category $\mathcal{C}$. In their presence, the symmetric Hamiltonian term across the defect is modified by inserting a rotated symmetry MPO tensor:

$$h_{L,1}^{\mathcal{R},X} = \sum_{\{Y\}} \sum_{i,i'} h(\{Y\}, i, i') \begin{array}{c} Y'_1 \\ | \\ \textcircled{i} \text{---} \square \text{---} \textcircled{i'} \\ | \quad \quad | \\ Y_1 \quad X \quad Y_2 \end{array} \quad (4.19)$$

with $X \in \mathcal{I}_{\mathcal{D}_{\mathcal{R}}^*}$, so that the full Hamiltonian can be written as $H^{\mathcal{R},X} = \sum_{i \neq L} h_{i,i+1}^{\mathcal{R}} + h_{L,1}^{\mathcal{R},X}$. Notably, these symmetry twists are *topological* in the sense that their position can be shifted by one lattice site via a local unitary transformation. A duality still corresponds to a change of module category $\mathcal{R} \mapsto \mathcal{R}'$, and the corresponding intertwining MPOs are now promoted to *bimodule tubes* [76, 114]. Focusing for concreteness on the case where one side corresponds to the regular module category, and using the equivalence

$\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{R}) \simeq \mathcal{R}$, these tubes reduce to

$$\mathfrak{T}_{\mathcal{D}|\mathcal{R}}^{Y,X,R,R',i,j} := \text{---} \square \text{---} \square \text{---} \overset{X}{\underset{i}{\circ}} \text{---} \underset{j}{\circ} \text{---} \square \text{---} \begin{matrix} \boxed{\mathcal{R}} \\ \boxed{\mathcal{D}} \end{matrix}, \quad (4.20)$$

with $R, R' \in \mathcal{I}_{\mathcal{R}}$, $Y \in \mathcal{I}_{\mathcal{D}}$ and $X \in \mathcal{I}_{\mathcal{D}^{\star}}$. All MPO tensors and the rightmost trivalent tensor evaluate to the module associator ${}^{\triangleleft}F$ of $\mathcal{R}$ as right $\mathcal{D}$ -module category, and the left trivalent tensor to the $\mathcal{R} \bowtie F$ -symbol. Eq. (4.20) thus generalises the tubes introduced in eq. (3.60), and we similarly define tubes of the form $\mathfrak{T}_{\mathcal{R}|\mathcal{D}}$. When the duality is between generic $\mathcal{D}$ -module categories $\mathcal{R}$ and $\mathcal{R}'$, the relevant trivalent tensors encode the associativity of the *composition of module functors*. This case is discussed in detail in ref. [77] and reviewed in chapter 8.

The collection of all tubes $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}|\mathcal{R}'}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}\}$ constitutes an enlarged tube algebra which can be shown to also form a finite C^* algebra. Consequently, it admits an Artin-Wedderburn decomposition. In particular, the $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}\}$ and $\{\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}\}$ tubes furnish subalgebras that encode the symmetry of either model in the presence of twists. Their corresponding matrix units coincide with those discussed in section 3.4.3. The associated central idempotents are said to label the *topological sectors* of either model, and in virtue of the Morita equivalence between $\mathcal{D}_{\mathcal{R}}^*$ and $\mathcal{D}_{\mathcal{R}'}^*$, characterised by $\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*) \simeq \mathcal{Z}(\mathcal{D}_{\mathcal{R}'}^*)$, these sectors are in bijection with one another. Physically, these topological sectors label a non-simple boundary condition as well as a charge of the sub-symmetry that preserves that boundary condition. For any Z in $\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*)$ that contains a given $X \in \mathcal{I}_{\mathcal{D}_{\mathcal{R}}^*}$ in its decomposition as object of $\mathcal{D}_{\mathcal{R}}^*$, the Hamiltonian in the topological sector Z with multiplicity i is defined via the simple idempotents as

$$H_{Z,i}^{\mathcal{R},X} := (e_{\mathcal{R}|\mathcal{R}}^{Z,X_i,X_i})^\dagger H^{\mathcal{R},X} e_{\mathcal{R}|\mathcal{R}}^{Z,X_i,X_i}. \quad (4.21)$$

All other twisted Hamiltonians in the same sector, i.e. $H_{Z,j}^{\mathcal{R},X'}$, have the same spectrum as $H_{Z,i}^{\mathcal{R},X}$, since they are related by the isometric nilpotent $e_{\mathcal{R}|\mathcal{R}}^{Z,X',X_i}$ in the same block Z .

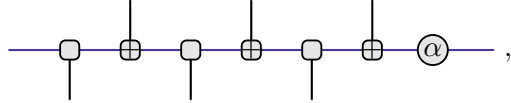
Additionally, the enlarged tube algebra admits matrix units of the type $\mathfrak{T}_{\mathcal{R}|\mathcal{R}'}^{Z, X_i, Z', X'_j}$ and $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}}^{Z', X'_i, Z, X_j}$, where Z' denotes the image of $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{D}_{\mathcal{R}}^*)}$ under the duality transformation. Using them, isometries between the dual models

are constructed via

$$H_{Z',j}^{\mathcal{R}',X'} = (e_{\mathcal{R}|\mathcal{R}'}^{Z,X_i,Z',X'_j})^\dagger H_{Z,i}^{\mathcal{R},X} e_{\mathcal{R}|\mathcal{R}'}^{Z,X_i,Z',X'_j}. \quad (4.22)$$

By identifying the anyonic sectors of the string-net models with the topological sectors of their edge Hamiltonians, we have completed the holographic dictionary relating the bulk and edge. We conclude by exemplifying this correspondence in the case of the Kramers–Wannier duality.

Example: Kramers–Wannier duality — Part III The Ising model admits periodic or antiperiodic boundary conditions, and for each boundary condition decomposes into even and odd charge sectors, resulting in a total of four topological sectors. The intertwining tubes eq. (4.20) can be written as



$$, \quad (4.23)$$

where the boundary matrix α in each sector is given by one of the Pauli matrices, as per the dictionary in table 4.2. Notably, the $(P, +1)$ and $(AP, -1)$ sectors are mapped identically under the Kramers–Wannier duality, while the other two sectors are transposed.

Boundary condition	Charge sector	Dual boundary condition	Dual charge sector	Boundary matrix
P	+1	P	+1	$\mathbb{1}$
P	-1	AP	+1	Z
AP	+1	P	-1	X
AP	-1	AP	-1	Y

Table 4.2: Mapping of topological sectors under the Kramers–Wannier duality. Here, $(A)P$ stands for *(anti)periodic*.

II

Applications

4

Les Houches Lecture Notes on Tensor Networks

5.1 Synopsis

These lecture notes are based on a series of five lectures by Frank Verstraete at the 2025 Les Houches school on *Exact Solvability and Quantum Information*. They cover a variety of topics in the field of tensor networks, with an emphasis on the concept of the MPS manifold, MPO symmetries, integrability and dualities.

5.2 Reference

Les Houches Lecture Notes on Tensor Networks,
B. Vancraeynest-De Cuiper, W. Wiesiolek, F. Verstraete
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5.3 Contributions of the author

I wrote the majority of the manuscript, whose content closely follows that of the lectures on which it is based. Apart from some aspects of the exposition, the content is due to the lecturer.

Les Houches Lecture Notes on Tensor Networks

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Abstract

Tensor networks provide a powerful new framework for classifying and simulating correlated and topological phases of quantum matter. Their central premise is that strongly correlated matter can best be understood by studying the underlying entanglement structure and its associated (generalised) symmetries. In essence, tensor networks provide a compressed, holographic description of the complicated vacuum fluctuations in strongly correlated systems, and as such they break down the infamous many-body exponential wall. These lecture notes provide a concise overview of the most important conceptual, computational and mathematical aspects of this theory.

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Introductory remarks

We are currently witnessing a paradigm shift in the way we understand and do quantum many-body physics. The *quantum many-body problem* has been the central problem in theoretical physics and quantum chemistry for the best of the last century, and its associated exponential wall has precluded progress in describing strongly interacting phases of matter. During the last twenty years however, there has been remarkable progress in simulating and classifying those strongly correlated systems using the theory of *entanglement*, as epitomised by the concept of *tensor networks*.

In essence, tensor networks circumvent the exponential quantum many-body wall by providing a *holographic* description of the many-body wave functions. Instead of describing the wave function in an exponentially large Hilbert space, tensor networks model how entanglement is routed between different parts of the system. Underlying the success of tensor networks lies the fact that all the information needed to calculate any observable of a low-energy wave function can be compressed into a local tensor with so-called *entanglement degrees of freedom* making up a *virtual* Hilbert space. The reason that this can be done is related to the existence of *area laws* for the entanglement entropy: tensor networks are concise and precise representations of the manifold of all states satisfying such an area law. By projecting the

Hamiltonian defined in the exponentially large physical Hilbert space on this low-dimensional manifold, it has become possible to construct numerical algorithms to simulate large-scale quantum many-body systems. DMRG (*Density Matrix Renormalisation Group*), MPS (*Matrix Product State*) and PEPS (*Projected Entangled-Pair State*) algorithms are revolutionising the way we simulate state-of-the-art experiments in atomic physics, current-day quantum computers and the low-energy physics of quantum many-body systems.

But there is more: since the pioneering insights of Landau and Anderson, it is known that many-body physics is all about *symmetries* and *symmetry breaking*. It turns out that the global, local and topological symmetries of the full system are reflected in the symmetries of the local tensors of the tensor network, which provide an extremely effective compression of the full many-body wave function. As a consequence, the classification of gapped *phases of matter*, and in particular *topological phases*, is reduced to the problem of classifying the irreducible ways in which tensors transform under a set of relevant symmetries. As those symmetries act on the virtual (entanglement) degrees of freedom, there is no reason why those symmetries should form a group, and more general associative structures emerge that can be described by *fusion categories*. Indeed, tensor networks provide an explicit way of representing those *generalised symmetries* (and the dualities between them) in terms of MPO (*Matrix Product Operator*) algebras and their higher-dimensional analogues, PEPOs (*projected entangled-pair operators*). Several of the most active research areas in mathematical, condensed matter, statistical and high-energy physics are precisely dealing with the consequences of this insight: a holographic description of strongly correlated and topological phases of matter is the only one which reveals its inner structure.

These lecture notes are based on five lectures given by Frank Verstraete at the Les Houches Summer School of Physics in August 2025 on *Exact Solvability and Quantum Information*, covering a selection of elementary computational and theoretical results of tensor networks. They do not aim to provide a comprehensive review of these topics. Rather, they are meant as an invitation for further study and we can refer to a vast literature of existing reviews and lecture notes: [1–9].

1 Lecture I: Motivation & the manifold of MPS

1.1 The double-exponential wall & the illusion of Hilbert space

Before we dive into these lectures, it is important to recognise the true challenge posed by the quantum many-body problem. As one learns in every course on quantum mechanics, the dimension of the Hilbert scales exponentially in the number of constituent particles. This is in fact not so different from the classical case where the number of possible strings of classical bits also scales exponentially in the number of bits. However, since quantum mechanics allows for *superpositions* of states and consequently *entanglement*, the natural question to ask is not by how many basis states the Hilbert space is spanned, but rather how many of the states in the Hilbert space can actually be explored by some physical mechanism originating from local (few-body) interactions, and in particular with a quantum computer.

To be more precise and concrete, let us consider the following question raised in ref. [10]. Within the Hilbert space of N qubits, which portion of the exponentially large many-body Hilbert space can be reached by running a quantum computer starting from a reference state for a time scaling polynomially in system size? To estimate this, we consider so-called ε -balls of states around points $|\psi\rangle$ in the Hilbert space, i.e. we consider states $|\varphi\rangle$ for which

$$|\langle\psi|\varphi\rangle| \gtrsim 1 - \varepsilon. \quad (1)$$

A simple geometric argument shows that such ε -balls occupy a fraction of the total volume of the Hilbert space which scales according to $\mathcal{O}(\varepsilon^{2^N})$, i.e. a *double-exponentially* (!) small fraction of the total Hilbert space! Due to this scaling, it would take any physical system (including a quantum computer) a time that is at least exponential in the system size to have overlap with a specific ε -ball. We conclude that the vast majority of states is unphysical and is completely irrelevant for the description of physical systems: Hilbert space is a (convenient) illusion, as states with many particles that take exponentially long to create cannot be created by any physically reasonable process.

This dramatic result points to a paradigm shift in our understanding of quantum many-body systems, namely that physical wave functions are those which have low complexity. More specifically, we expect ground states of quantum many-body systems to live on a physically relevant low-complexity *manifold* of states, which for all practical purposes we will want to parametrise with a *small* number of parameters, meaning scaling polynomially in the number of particles. Roughly speaking, a manifold is a space which locally resembles flat Euclidean space. This means that to every point of the manifold one can associate a linear *tangent space* whose dimension is the dimension of the manifold. One of the (many) great insights of Dirac was that one can now actually do physics on such a manifold without ever leaving it by interpreting the manifold as a variational set of states. Let us denote a general variational manifold of states as

$$\mathcal{M} = \{ |\Psi(\mathbf{z})\rangle \mid \mathbf{z} \in \mathbb{C}^n \}. \quad (2)$$

We will write $T_{\mathbf{z}}\mathcal{M}$ for the linear tangent space at any point $\mathbf{z}$. A general state in this tangent space is then of the form

$$\sum_i x^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle, \quad (3)$$

for complex coefficients x^i . Suppose now that we want to compute the time evolution of a certain state $|\Psi(\mathbf{z})\rangle$ subject to some many-body Hamiltonian H :

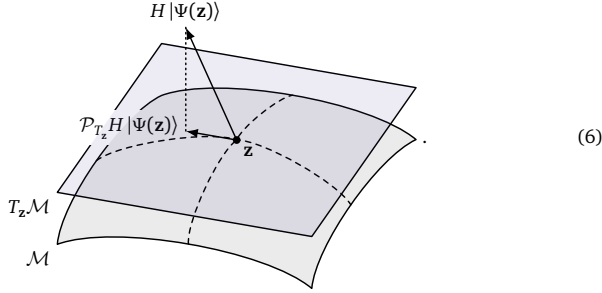
$$i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = H |\Psi(\mathbf{z})\rangle. \quad (4)$$

Then $i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = i \sum_i \dot{z}^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle$ can be thought of as a tangent vector in $T_{\mathbf{z}}\mathcal{M}$. However, for a generic Hamiltonian, $H |\Psi(\mathbf{z})\rangle$ is a vector in the ambient Hilbert space in which $\mathcal{M}$ is embedded that points *out of* the tangent plane, and thus doesn't belong to the variational class anymore. Dirac's proposal was to define an action on this manifold, but for our purposes it is convenient to think about this as the orthogonal projection of $H |\Psi(\mathbf{z})\rangle$ on the tangent space of the manifold $T_{\mathbf{z}}\mathcal{M}$. Geometrically, one can think of this projection as minimising the distance

$$\min_{\{\dot{z}^i\}_i} \left\| i \sum_i \dot{z}^i \frac{\partial}{\partial z^i} |\Psi(\mathbf{z})\rangle - H |\Psi(\mathbf{z})\rangle \right\|. \quad (5)$$

Denoting the orthogonal projector at a point $\mathbf{z}$ by $\mathcal{P}_{T_{\mathbf{z}}}$, we can depict its action on $H |\Psi(\mathbf{z})\rangle$

thus as:



The resulting *time-dependent variational principle* (TDVP) equations

$$i \frac{\partial}{\partial t} |\Psi(\mathbf{z})\rangle = \mathcal{P}_{T_z} H |\Psi(\mathbf{z})\rangle, \quad (7)$$

then describe a collection of highly non-linear (due to the appearance of the tangent space projector) differential equations of the generic many-body state on the manifold $\mathcal{M}$. Importantly, it can be shown that the manifold $\mathcal{M}$ is *symplectic*. This means in particular that the expectation value of the energy, generators of symmetries and other constants of motion are exactly conserved under evolution by the TDVP equations.

This manifold approach to the many-body problem has been applied in many fields with exceptional success. In fact, it is merely a small exaggeration to state that all progress in tackling the quantum many-body problem since the discovery of the Schrödinger equation has been in finding such good manifolds of states in which the relevant physics takes place. *Matrix product states* (MPS) and its higher dimensional generalisations such as two-dimensional *projected entangled-pair states* (PEPS) form such classes, and we will introduce them shortly. But before doing so, let us take some time to discuss some other, perhaps to some readers more familiar, variational classes of states from this manifold point of view.

Free fermions & Hartree-Fock Probably the most well-known variational manifold of many fermions is that of *Slater determinants* [11]. A Slater determinant parametrises an N -fermion wave function in terms of N single-particle orbitals $\psi_i, i = 1, \dots, N$ as a determinant of the form:

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \begin{vmatrix} \psi_1(\mathbf{x}_1) & \dots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \dots & \psi_N(\mathbf{x}_2) \\ \vdots & \ddots & \vdots \\ \psi_1(\mathbf{x}_N) & \dots & \psi_N(\mathbf{x}_N) \end{vmatrix}. \quad (8)$$

By construction, Slater determinants are manifestly anti-symmetric under the exchange of any two fermions, in correspondence with the Pauli exclusion principle. Moreover, note that the number of parameters needed to represent Slater determinants scales *linearly* in the number of electrons and Slater determinants therefore only span an exponentially small submanifold in the Hilbert space. The *Hartree-Fock* method for approximately solving the Schrödinger equation of N fermions is then a self-consistent variational method over this class of states: approximations of both the time-dependent and time-independent Schrödinger equations are obtained by applying the TDVP principle to this manifold. The success of this method can hardly be overstated. It has allowed us to simulate molecules and materials with great accuracy

which would otherwise be completely intractable with exact methods. In fact, Hartree-Fock allowed us to understand the structure of the periodic table of elements and of band structures from first principles [12, 13].

Field coherent states of interacting bosons The bosonic counterpart of Slater determinants also exists in the form of the manifold of *field coherent states*. For the bosonic creation operators¹ $\hat{\psi}^\dagger(\mathbf{x})$, one has for any function $\varphi(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{C}$ the coherent state

$$|\Psi[\varphi]\rangle = \exp\left(\int d^3\mathbf{x} \varphi(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x})\right) |\Omega\rangle, \quad (9)$$

where $|\Omega\rangle$ is the bosonic vacuum characterised by $\hat{\psi}(\mathbf{x})|\Omega\rangle = 0$, $\forall \mathbf{x}$. Recall that coherent states are exact eigenstates of the annihilation operators, i.e. $\hat{\psi}(\mathbf{x})|\Psi[\varphi]\rangle = \varphi(\mathbf{x})|\Psi[\varphi]\rangle$. Let us now apply the TDVP to the Hamiltonian

$$H = \int d^3\mathbf{x} \hat{\psi}^\dagger(\mathbf{x}) \left(-\frac{1}{2m} \nabla^2 + V(\mathbf{x}) \right) \hat{\psi}(\mathbf{x}) + \frac{U}{2} \hat{\psi}^\dagger(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x}) \hat{\psi}(\mathbf{x}) \hat{\psi}(\mathbf{x}), \quad (10)$$

describing a dilute boson gas prone to an external potential V and a contact interaction. One then finds that the TDVP equations give rise to the (time-dependent) *Gross-Pitaevskii* equation:

$$i \frac{\partial}{\partial t} \varphi(\mathbf{x}) = -\frac{\nabla^2}{2m} \varphi(\mathbf{x}) + V(\mathbf{x}) \varphi(\mathbf{x}) + U |\varphi(\mathbf{x})|^2 \varphi(\mathbf{x}). \quad (11)$$

Despite the simplicity of this derivation, the Gross-Pitaevskii has been employed with great success to describe the behaviour of Bose-Einstein condensates [14].

We highlighted here only a few of the most common and successful methods to define and parametrise variational manifolds in the exponentially large Hilbert space. Besides these we have omitted the plethora of post-Hartree-Fock methods such as *coupled cluster methods* [15].

1.2 Quantum spin systems: monogamy & frustration

A very useful and insightful idealisation of a generic quantum many-body system or of a quantum field theory is provided by quantum lattice systems. In this framework, the lattice is a discretisation of space that hosts quantum degrees of freedom on its vertices, and in some cases on the edges. From the point of view of quantum field theory, this is the best controlled way of circumventing UV-divergences [16, 17]. From the point of view of condensed matter and atomic physics, lattice systems provide effective (compressed) descriptions of the relevant degrees of freedom.

In these lecture notes, we will be concerned with the low-energy physics of quantum many-body lattice models in one or two spatial dimensions. Restricting to the low-energy sector is often permitted due the observation that for most realistic materials the Fermi temperature is of the order $10^4 K$, much larger than the typical temperature scale of an experiment (even room temperature is very cold if you consider systems of electrons). But also in quantum field theory the ground state and its low lying excitations are typically the states of interest when computing for example cross sections of scattering experiments. Typically, the Hilbert spaces under consideration are a tensor product of qudits which we will denote by $\mathcal{H} = (\mathbb{C}^d)^N \equiv \bigotimes_{i=1}^N \mathbb{C}^d$, and unless stated otherwise we will consider the thermodynamic limit $N \rightarrow \infty$, while neglecting subtleties regarding order of limits. Note however that the dimension of the corresponding

¹In this paragraph we will exceptionally denote operators with hats to avoid confusion with scalar functions.

Hilbert space is still exponential as a function of the system size, so lattice systems also suffer from the same exponential wall.

Obtaining a good approximation to the ground state of such a model is generically very non-trivial due to *frustration*. In this context, the term frustration refers to the inaptitude of the ground state to simultaneously minimise the energy of all Hamiltonian terms. As an example, consider for instance the paradigmatic spin 1/2 antiferromagnetic Heisenberg chain in $(1+1)d$:

$$H = \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (12)$$

If we would consider only two spins, the ground state would be the singlet state with density matrix $\rho = |\phi\rangle\langle\phi|$ where $|\phi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$. In the case of spin chains with N spins, the total energy would clearly be minimised if all reduced density matrices $\rho_{i,i+1}$ of the ground state are equal to this singlet – but this is clearly not possible: the *monogamy of entanglement* dictates that such a state cannot exist [18, 19]. Indeed, there exists no three-partite state $\rho_{ABC} \geq 0$, such that $\text{Tr}_C(\rho_{ABC}) = \text{Tr}_B(\rho_{ABC}) = |\phi\rangle\langle\phi|$. As such, monogamy implies frustration.

Monogamy of entanglement can also be used to explain why mean field theory becomes increasingly exact when the number of spatial dimensions increases. Consider instead of the one-dimensional chain a lattice with a large coordination number and again a Hamiltonian which is the sum of antiferromagnetic Heisenberg interactions between all neighbouring spins. Intuitively, in that case, for any given spin its entanglement with other spins then needs to be distributed over so many neighbours that the ground state necessarily becomes very close to a separable state $\otimes_i |\psi_i\rangle$ and mean field theory becomes exact. Though presented in a hand waving fashion here, this argument can be made precise by invoking the *quantum de Finetti theorem* [20]. This theorem characterises multi-partite density matrices that are invariant under permutations of the different parties, and plays an important role in proving the security of quantum cryptographic protocols.

From this discussion we have deduced the fact that the ground state cannot be such that all its $\rho_{i,i+1}$ are spin singlets. Now we could rather look for a state whose marginals are, in a sense, *as close as possible* to the spin singlet. To this end, one could for example try the *resonating valence bond state* as an ansatz to the ground state:

$$|\Psi\rangle = |\Psi_+\rangle + |\Psi_-\rangle, \quad \text{where} \quad |\Psi_{\pm}\rangle = \prod_i (|\uparrow\downarrow\rangle_{2i,2i\pm 1} - |\downarrow\uparrow\rangle_{2i,2i\pm 1}). \quad (13)$$

Graphically one could depict this state as

$$|\Psi\rangle = \begin{array}{c} \begin{array}{cccc} 2i-1 & 2i & 2i+1 & 2i+2 \\ \circ & \text{---} & \circ & \text{---} & \circ & \text{---} & \circ \end{array} \\ + \begin{array}{c} \text{---} \circ \quad \circ \text{---} \quad \circ \text{---} \end{array} \end{array}, \quad (14)$$

where the wiggly lines stand for spin singlets between neighbouring sites. In fact, $|\Psi_{\pm}\rangle$ span the two-dimensional ground state subspace of the model introduced by Majumdar and Ghosh in ref. [21]. Note that this state has all the symmetries we could expect: it is $\text{SU}(2)$ symmetric and has translational, time-reversal and reflection invariance. This is important as ground states (or more precisely, ground state subspaces) should inherit all the symmetries of the Hamiltonian. More importantly, it has some non-vanishing entanglement, and we would therefore expect the energy of this state to be much closer to the actual ground state energy than the mean field approximation. Nevertheless, the energy of this state turns out to be still a fairly bad approximation of the true ground state energy of the Heisenberg model as it overestimates the energy quite a lot.

At this point it should be clear that writing down entangled states with the correct symmetries that can serve as ansatz for the ground state of our favourite many-body quantum

Hamiltonian, is very hard. However, the valence bond state suggest one particular way of doing this. Inspired by this Majumdar-Ghosh model, Affleck, Kennedy, Tasaki and Lieb (AKLT) proposed in 1987 a state constructed of pairs of entangled spin $1/2$ particles which are projected onto a spin 1 particle on each lattice site [22]. Their construction goes as follows. One starts from a chain of spin $1/2$ particles where every other pair of neighbouring particles is in a spin singlet. On every site i the rightmost particle is then taken together with the leftmost particle on the next site $i + 1$ to form a *physical* site. The two *virtual* spin $1/2$ particles that make up the physical site are then projected onto their spin 1 subspace. A graphical depiction of this state is more enlightening:

$$|\Psi_{\text{AKLT}}\rangle = \sim (\text{diagram}) \sim, \quad (15)$$

where

$$\text{Oval} = |+\rangle\langle\uparrow\uparrow| + |0\rangle\langle\frac{\langle\uparrow\downarrow| + \langle\downarrow\uparrow|}{\sqrt{2}}| + |+\rangle\langle\downarrow\downarrow|. \quad (16)$$

As shown by the aforementioned authors, this state is the ground state of a Hamiltonian which is very closely related to the spin 1 Heisenberg model, and is believed to belong to the same (symmetry-protected) phase (discussed further in lecture section 3). Since two neighbouring sites exactly involve one spin singlet, the 4 constituent spin 1/2 particles which make up these physical degrees of freedom can not be in a coupled spin 2 state. As such, this state is annihilated by the projector on the spin 2 subspace of two spin 1's. Therefore, this state is the ground state of the AKLT Hamiltonian

$$H = \sum_i \mathbf{s}_i \cdot \mathbf{s}_{i+1} + \frac{1}{3} (\mathbf{s}_i \cdot \mathbf{s}_{i+1})^2, \quad (17)$$

where the local two-body term can be recognised as the spin 2 projector, up to a shift and a prefactor.

1.3 Tensor networks & area laws

A few years after AKLT, in 1992, Fannes, Nachtergaele and Werner [23] introduced the class of one-dimensional *finitely correlated states* which generalises the construction of the AKLT state using ideas of quantum Markov chains. The name *finitely correlated states* stems from the fact that these states probably have exponentially decaying connected correlation functions, as will be demonstrated below in section 2.2. They also proved that all those states are ground states of gapped frustration-free Hamiltonians. The modern point of view, as first advocated in ref. [24], is to construct those states by starting from a chain of maximally entangled *virtual* qubits between neighbouring sites that are projected onto the *physical* spin on each site by a linear map. This linear map is often referred to as *projector* (even though it doesn't have to be an idempotent linear map), which is the origin of the name *projected entangled-pair states*, although this name is now typically used for the higher dimensional generalisations. These projectors contain parameters which play the role of the *variational* degrees of freedom of that state. This class of states coincides with the class of (infinite and translational invariant) uniform *matrix product states* (MPS), and this is the name that is most often used to characterise them. Let us make this more explicit. The construction of MPS thus starts from considering maximally entangled states

$$|I\rangle_i = \sum_{\alpha=1}^{\chi} |\alpha\alpha\rangle_{i,i+1} \quad (18)$$

between each pair of neighbouring sites $i, i+1$, $i \in \mathbb{Z}$. The integer χ is called the *bond dimension* and directly controls the number of variational parameters as we will see. We then consider

on every physical site of the chain following linear map

$$A = \sum_{i,\alpha,\beta} A_{\alpha\beta}^i |i\rangle \otimes |\alpha\rangle \otimes |\beta\rangle \in \mathbb{C}^{d \times d \times d}, \quad (19)$$

on which we act on the entangled pairs, so that the state corresponding to the tensor A is given by:

$$|\Psi[A]\rangle = \left(\bigotimes_{i \in \mathbb{Z}} A \right) \cdot \left(\bigotimes_{i \in \mathbb{Z}} |I\rangle_i \right). \quad (20)$$

We will say that A generates the MPS $|\Psi[A]\rangle$. Expanding now the product in this expression yields the more familiar expression of *uniform* matrix product states, directly in the thermodynamic limit:

$$|\Psi[A]\rangle = \sum_{\{i_j\}} \left(\cdots A^{i_j} A^{i_{j+1}} \cdots \right) | \dots, i_j, i_{j+1}, \dots \rangle. \quad (21)$$

This expression for the wave function explains the name *matrix product* states, as the wave function amplitudes are computed as the products of the matrices $\{A^i\}_i$. In the uniform case the number of variational parameters is $d \chi^2$. Notice however that the dimension of the manifold of states spanned by uniform MPSs is strictly smaller than $d \chi^2$ since for any non-singular matrix X a transformation of the kind

$$A^i \mapsto X A^i X^{-1}, \quad \forall i = 1, 2, \dots, d, \quad (22)$$

called a *gauge transformation*, leaves the state $|\Psi[A]\rangle$ invariant. A more precise characterisation of this manifold is relegated to the next section.

Note that we could also have chosen our tensors (projectors) different at every site, leading to a generic non-uniform MPS with bond dimensions that can vary over the different edges of the lattice. This way we can easily deal with finite spin chains with open, periodic or twisted boundary conditions. Remark that, for the uniform case we did not specify the boundary conditions at infinity. For generic matrix product states, which are *injective* (defined below), the choice of these boundary conditions is irrelevant. Throughout these lecture notes we will focus almost exclusively on matrix product states in the thermodynamic limit for which this injectivity property holds. For completeness let us display here generic MPSs on finite spin chains. A generic MPS on an open chain can be written as:

$$|\Psi[A]\rangle = \sum_{\{i_j\}} \mathbf{v}_L^\dagger A^{i_1} [1] A^{i_2} [2] \cdots A^{i_N} [N] \mathbf{v}_R |i_1, i_2, \dots, i_N\rangle. \quad (23)$$

where the vectors $\mathbf{v}_{L/R}$ encode the boundary conditions, and the integers between square brackets index the physical sites. A translation invariant state on a periodic chain with N sites on the other hand takes the form

$$|\Psi[A]\rangle = \sum_{\{i_j\}} \text{Tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, i_2, \dots, i_N\rangle. \quad (24)$$

As is customary when dealing with tensor networks, we will use a graphical language in which a single MPS tensor is depicted as:

$$\begin{array}{c} i \\ | \\ \alpha - \textcircled{A} - \beta \end{array} = A_{\alpha\beta}^i, \quad \begin{array}{l} i=1,2,\dots,d \\ \alpha,\beta=1,2,\dots,\chi \end{array} \quad (25)$$

and where a leg connecting two tensors indicates a contraction over the corresponding indices. With these conventions in mind, we can depict the MPS (21) as following diagram:

$$|\Psi[A]\rangle = \begin{array}{c} | \\ \text{---} \text{A} \text{---} \end{array} \begin{array}{c} | \\ \text{---} \text{A} \text{---} \end{array} \dots \begin{array}{c} | \\ \text{---} \text{A} \text{---} \end{array} \begin{array}{c} | \\ \text{---} \text{A} \text{---} \end{array} . \quad (26)$$

Often, we will not explicitly indicate the MPS tensor in the figures when they can be deduced from the context.

Now that we have introduced the variational class of matrix product states, let us argue why these states approximate well the ground states of local gapped Hamiltonians [25]. Physically, this follows from the *area law* of entanglement entropy satisfied by gapped ground states. In generic dimensions, an area law for the entanglement entropy means that the bipartite entanglement entropy between a region $\mathcal{A}$ and its complement $\bar{\mathcal{A}}$, defined as the Von Neumann entropy of the reduced matrix $\rho_{\mathcal{A}} = \text{Tr}_{\bar{\mathcal{A}}}(\rho)$, i.e. $S = -\text{Tr}(\rho_{\mathcal{A}} \log(\rho_{\mathcal{A}}))$, does not scale with the size, or *volume*, of the region $\mathcal{A}$ but rather by the size of its boundary $\partial\mathcal{A}$, or thus its *area*. In the one-dimensional setting the boundary of a connected subregion of the chain is just a constant, and in this setting the area law was proven in 2007 by Hastings [26]. As we will now demonstrate MPS inherently satisfy an area law.

To this end, we make use of the graphical calculus of MPS. Herein, let us calculate the reduced density matrix of a uniform MPS when tracing out the left half of the system. This can be depicted as

$$\rho_R = \text{Tr}_L(|\Psi[A]\rangle\langle\Psi[A]|) = \dots \quad \begin{array}{c} \boxed{L} \quad \boxed{R} \\ \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \\ \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \text{A} \text{---} \end{array} \dots, \quad (27)$$

$\mathbb{T}_A$

where the dotted vertical line represents the entanglement cut between the left (L) and right (R) subsystem. An object which plays a pivotal role in the computation of the reduced density matrix, but also appears in e.g. the computation of correlation functions (see section 2.2), is the *transfer matrix*

$$\mathbb{T}_A = \sum_i A^i \otimes \bar{A}^i, \quad (28)$$

which we have indicated as well in the above figure by the dotted box. Let us write its spectral decomposition² as

$$\mathbb{T}_A = \sum_{i=1}^{\chi^2} \lambda_i |\rho_i\rangle\langle\lambda_i|, \quad (29)$$

where the eigenvalues are ordered according to decreasing absolute value. The eigenvectors are clearly invariant subspaces of the completely positive linear maps $\$_L(X) = \sum_i A^i X A^i$ and $\$_R(X) = \sum_i A^i X A^{i\dagger}$. If those maps are ergodic, the quantum Perron-Frobenius theorem dictates that the largest eigenvalue in magnitude will be unique, real and positive. Furthermore, the corresponding left and right eigenvectors $|\rho_1\rangle \equiv |\rho\rangle$ and $|\lambda_1\rangle \equiv |\lambda\rangle$, when reshaped as matrices (they have 2 indices), are positive definite. This property of having a unique dominant eigenvector is equivalent to the notion of injectivity mentioned above. Another equivalent characterisation of injectivity is as follows: an MPS is injective if and only if there exists a finite L , called the *injectivity length*, such that the collection of matrices $\{A^i A^{i_2} \dots A^{i_L}\}$ is a basis

²More generally, Jordan blocks could be needed, but this is not relevant for the following discussion.

To ensure that the state is properly normalised to one we will always normalise the MPS tensor A such that the corresponding eigenvalue λ_1 is equal to one (see also section 2.2). From these considerations, it follows that we can express the reduced density matrix as follows:

As ρ_R is the product of a $d^L \times \chi$ and a $\chi \times d^L$ matrix with L the total number of sites on the right hand side, we conclude that the rank of the reduced density matrix ρ_R is upper bounded by χ . The entanglement entropy is hence bounded by $S \leq \log \chi$, as this is the maximal entropy for a state of rank χ . The theorems in refs. [25, 26] ensure that the converse is also true: whenever a state satisfies an area law for the entanglement entropy, there exists an MPS that will be a faithful approximation of that state. In the next lecture we will study the entanglement properties of general matrix product states in more detail.

Throughout these lecture notes we will also be making use of tensor network representations of linear operators. Uniform *matrix product operators* [27, 28] are written as

and graphically depicted by

In analogy with MPS, the generalisation to open boundary conditions, and the non-uniform case is straightforward.

MPOs have since long been used extensively in the field of statistical physics, where MPOs of a particular type are known as transfer matrices (see [29] for a unifying discussion). Similarly to the case of MPS, we call an MPO injective when the matrices $\{M^{ij}\}_{i,j}$ have no common invariant subspaces and span the entire matrix algebra.

Non-injective MPOs are however also used extensively, for example to encode Hamiltonians of quantum spin chains. Let us e.g. consider the Hamiltonian for the transverse field Ising model with open boundary conditions:

This Hamiltonian can be represented as an MPO with site-independent tensors³ M of bond dimension 3:

³In this notation, the element $M_{\alpha\beta}$ is an operator acting on the physical Hilbert space.

and boundary vectors

$$\mathbf{v}_L = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_R = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (35)$$

Note that since we are dealing with a non-injective MPO here (there are invariant subspaces), the boundary vectors must be defined. This MPO is one of the generators of the Onsager algebra. Its exponential, $\exp(i\lambda H)$, cannot be written anymore as an MPO with a bond dimension independent of the system size – unless $\lambda = 0$, then (exercise) it can be written as an injective MPO with $\chi = 2$.

1.5 The manifold of MPS

Now that we have introduced matrix product states and operators through the lens of the entangled-pair construction, let us investigate in more detail the manifold of uniform matrix product states [4, 30, 31]. To this end we consider all uniform MPSs with a fixed bond dimension χ , $|\Psi[A]\rangle$. As a first observation, note that these states don't span a linear subspace of the full Hilbert space since the sum of two MPSs is typically not an MPS of the same bond dimension. Instead, they live on a surface whose local coordinates are given by the variational parameters of the state, i.e. $A_{\alpha\beta}^i$. Recall from above that to every point of the manifold we can associate a linear tangent space. For the case at hand, the tangent vectors are of the form

$$|\Psi[B, A]\rangle = \sum_I B^I \frac{\partial}{\partial A^I} |\Psi[A]\rangle, \quad (36)$$

where here and below I collectively denotes the physical and virtual indices of the MPS tensors. Graphically, such a tangent vector boils down to

$$|\Psi[B, A]\rangle = \sum_n \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---} \textcircled{B} \text{---} \textcircled{A} \text{---} \textcircled{A} \text{---}, \quad (37)$$

where the sum over n is a sum over all the positions of the B tensor in the infinite chain.

At any given point $|\Psi[A]\rangle$ of this manifold the overlap of two tangent vectors $|\Psi[B, A]\rangle$ and $|\Psi[B', A]\rangle$ provides a metric or *Gram matrix* G via

$$\langle \Psi[\tilde{B}', \tilde{A}] | \Psi[B, A] \rangle = \sum_{I, J} \tilde{B}'^I \underbrace{\left\langle \frac{\partial}{\partial \tilde{A}^I} \Psi[\tilde{A}] \left| \frac{\partial}{\partial A^J} \Psi[A] \right. \right\rangle}_{G(A, \tilde{A})_{I, J}} B^J. \quad (38)$$

Naively one would now assume that the dimension of the manifold is equal to the number of parameters in the tensor B , viz. $d\chi^2$. But note that gauge transformations of the form $B^i \rightarrow B^i + X A^i - A^i X$ leave the state $|\Psi[B, A]\rangle$ invariant so that $d\chi^2$ overcounts the dimension of the MPS manifold. A more careful argument which is made precise in ref. [4] demonstrates that only $(d-1)\chi^2$ of those parameters can be chosen independently. Provided that A_L^i is in the so called *left canonical gauge* (see section 2.2), a faithful parametrisation of B is then $B^i = V^i X$, where the V^i spans the null space of $(A_L)_{\alpha\beta}^i$ (interpreted as matrix from β to α , i) and X encodes the independent degrees of freedom of B . With this parametrisation of B , the metric is locally Euclidean, i.e. the Gram matrix is equal to the identity. From the computational point of view this parametrisation is useful as it vastly simplifies the implementation of algorithms and improves their stability.

One such algorithm is the TDVP applied to the class of MPS. Recall from section 1.1 that TDVP aims to approximately solve the time-dependent Schrödinger equation

$$i \frac{\partial}{\partial t} |\Psi[A]\rangle = H |\Psi[A]\rangle \quad (39)$$

by projecting the right-hand side of this equation onto the tangent plane:

$$i \frac{\partial}{\partial t} |\Psi[A(t)]\rangle = \mathcal{P}_{A(t)} H |\Psi[A(t)]\rangle, \quad (40)$$

where $\mathcal{P}_{A(t)}$ is the projector on the tangent plane. A further manipulation of the TDVP equation eq. (40) reveals that the TDVP is equivalent to a (semi-)classical set of equations of motions governed by a *Poisson bracket* on the MPS manifold. The famous Density Matrix Renormalisation Group (DMRG) algorithm [32] – discussed below in section 2.4.1 – and all its variants can be understood as specific splitting methods for solving this differential equation [33]. Recall that a Poisson bracket is a bilinear, antisymmetric function on phase space which satisfies the Jacobi identity. If we define the Poisson bracket for arbitrary functions f and g that depend on A and $\bar{A}$ as follows:

$$\{f, g\} = -\frac{\partial f}{\partial A_I} (G^{-1})^{IJ} \frac{\partial g}{\partial \bar{A}_J} + \frac{\partial g}{\partial A_I} (G^{-1})^{IJ} \frac{\partial f}{\partial \bar{A}_J}, \quad (41)$$

and write $f(A, \bar{A}) = \langle \Psi[A] | F | \Psi[A] \rangle$, it can be shown that the time evolution of the expectation value f is governed by

$$\partial_t f = i \{f, h\}. \quad (42)$$

This has two significant consequences. For one, it follows from $\{h, h\} = 0$ that the energy expectation value is exactly preserved under time evolution. This was anticipated above and implies that the manifold is symplectic. On the other hand, for any operator K which commutes with the Hamiltonian, and thus constitutes a symmetry generator, as well as with the tangent space projector $\mathcal{P}_A$, it can be shown that $\{K, h\} = 0$ and hence $\partial_t K = 0$. In particular, this is the case for symmetry generators which are tensor products of local on-site unitaries.

Besides TDVP it should be mentioned that the manifold picture has also provided the necessary framework for ground state optimisation directly in the thermodynamic limit, the so-called *variational uniform MPS* (VUMPS) algorithm, as well as the construction of *quasiparticle excitations*. These topics are covered in section 2.4 and section 2.5 respectively, but first we now delve deeper into the calculus of MPS.

1.6 Synopsis of lecture I

The main message of lecture I is the fact that the full many-body Hilbert space is an illusion: to describe relevant many-body states such as ground states of quantum spin systems, it is sufficient to consider a low-dimensional manifold of states which captures all relevant physics. For the particular case of gapped quantum spin chains, the monogamy of quantum entanglement implies the existence of an area law for the entanglement entropy. This implies that the manifold of MPS is in one-to-one correspondence with all states satisfying an area law. Instead of focusing on the full Hilbert space, we can hence project the Hamiltonian on this MPS manifold using the prescription of the time-dependent variational principle; all tensor network algorithms can be understood within this framework.

2 Lecture II: The calculus of MPS & algorithms

In this section we will introduce the calculus of matrix product states. In particular this constitutes the computation of observables, which is simplified greatly by introducing appropriate *normal forms* for the MPS tensors. With these tools at hand we will focus on the renowned *density matrix renormalization group* (DMRG) algorithm for ground state optimisation on finite chains, as well as the *variational uniform matrix product state* (VUMPS) algorithm, which works directly in the thermodynamic limit. We then study quasiparticle excitations on top of the ground state, which are parametrised as the *quasiparticle ansatz*. Let us however kick off with some more examples of matrix product states.

2.1 Examples, injectivity & the parent Hamiltonian

Product states Any product state $|\Psi\rangle = |\psi\rangle^{\otimes N}$, $|\psi\rangle = \sum_i \psi_i |i\rangle$ can be written as a bond dimension 1 MPS of the form

$$A_{11}^i = \psi_i. \quad (43)$$

AKLT The paradigmatic example of a non-trivial matrix product state encountered above in section 1.3 is the Affleck-Kennedy-Lieb-Tasaki (AKLT) state. This state is generated by the MPS matrices:

$$A^i = \sqrt{\frac{1}{3}} \sigma^i, \quad i = X, Y, Z, \quad (44)$$

where the σ^i are the Pauli matrices, and the prefactor ensures that the state is normalised in the thermodynamic limit.

On periodic boundary conditions, this state is the unique gapped ground state of the spin 1 Hamiltonian (17). As will be explained below in section 3.3, this Hamiltonian has a four-fold ground state degeneracy on open chains, characterised by gapless edge modes. This is a hallmark of its *symmetry-protected topological* (SPT) order.

GHZ Another example of a state which can be expressed as a bond dimension 2 MPS is the so called Greenberger–Horne–Zeilinger (GHZ) state [34]:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}). \quad (45)$$

Its MPS representation reads:

$$A^i = \begin{cases} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & i = 0 \\ \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, & i = 1. \end{cases} \quad (46)$$

Note that the GHZ state may be viewed as exhibiting $\mathbb{Z}_2$ symmetry breaking. This is reflected in the fact that the MPS matrices are block diagonal, and the blocks are swapped under the action of $\prod_i \sigma_i^X$. Clearly, this MPS representation of the GHZ-state is non-injective.

This has to be contrasted with the AKLT state, whose MPS matrices span the entire $\mathbb{C}^{2 \times 2}$ matrix algebra, therefore cannot be brought in block diagonal form, and thus define an injective MPS. Note that the injectivity length of the AKLT state is 2, which can be checked by an immediate computation.

It holds generically that every uniform MPS can be brought⁴ in a *block injective* or a *standard* form whereby all MPS matrices are of the form [35, 36]

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i, \quad (47)$$

and where all $\{A_k^i\}_k$, $k = 1, \dots, r$ are injective, and sometimes called *injective blocks*. It should be noted that this notion of injectivity should be revisited in the case of *fermionic* MPS, which is the topic of section 3.5.

Interestingly, every MPS can be regarded as a ground state of a corresponding *parent Hamiltonian*, thereby placing the study of MPS and their associated parent Hamiltonians on the same footing. This parent Hamiltonian is gapped, local and frustration-free, meaning that the energy of each term is minimised by the MPS. Its local terms are the projectors on the null space of the local reduced density matrix on the corresponding contiguous region of spins [5, 35]. For the GHZ state for example, the local parent Hamiltonian term is given by $\frac{1}{2}(\mathbb{1} - \sigma_i^Z \sigma_{i+1}^Z)$. When an MPS is injective, it can be shown that it is the unique ground state of its parent Hamiltonian. More interestingly, when the parent Hamiltonian commutes with an on-site representation of a symmetry group G , as in the case of the GHZ state, the action of G permutes (a minimal irreducible subset of) the different injective blocks. In that case, one finds a maximal subgroup $H \subseteq G$ whose action leaves a particular reference block invariant [37]. The blocks are then labelled by *cosets* in G/H .⁵

Below, in lecture 3 and lecture 5, we will re-encounter the parent Hamiltonian as a tool to classify *quantum phases of matter*, which – roughly speaking – are equivalence classes of Hamiltonians which can be adiabatically transmuted into each other.

2.2 Normal forms, entanglement & observables

As mentioned in section 1.3, the set of matrices $\{A^i\}$ is not a unique representation of a given MPS. This is because matrix product states are invariant under reparametrisations of the form

$$A^i \mapsto X A^i X^{-1}, \quad \forall i = 1, 2, \dots, d, \quad (48)$$

for any non-singular matrix X , since the matrices X and X^{-1} cancel each other on the bonds connecting neighbouring MPS matrices. This redundancy is often called *gauge freedom* of the MPS description. It often turns out that this gauge freedom can be heavily exploited to simplify certain computations and improve the numerical stability of MPS algorithms.

By appropriate gauge transformations, the MPS matrices can be brought in the so-called *left or right canonical forms*, i.e. respectively defined via the conditions:

$$\sum_i A^{i\dagger} A^i = \mathbb{1}_\chi, \quad (49)$$

$$\sum_i A^i A^{i\dagger} = \mathbb{1}_\chi. \quad (50)$$

⁴After getting rid of off-diagonal blocks, as reviewed in e.g. ref. [5].

⁵Recall that a (left) coset gH is the set $gH = \{gh|h \in H\}$. The set of all cosets is denoted G/H and partitions the group G .

Diagrammatically, these conditions can thus be written as follows:

$$\begin{array}{c} \textcircled{A} \\ | \\ \textcircled{A} \end{array} = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right], \quad (51)$$

$$\begin{array}{c} \text{---} \circ \bar{A} \\ | \\ \text{---} \circ A \end{array} \Bigg] = \Bigg] . \quad (52)$$

Note that the conditions eqs. (49) and (50) don't fix the MPS matrices completely as we can still conjugate the matrices with *unitary* gauge matrices. In what follows we will write A_L and A_R for the left and right canonical forms of the MPS tensor A . Let us stress again that A_L and A_R all generate the same state. The gauge transformation or *pivot matrix* that transmutes the left canonical form into the right canonical form is often denoted by C and is thus characterised by $A_L^i C = C A_R^i$, $\forall i$. For future reference let us also introduce the notation $A_L^i C = A_L^i C = C A_R^i$. An MPS is said to be in *mixed canonical form* if it can be written as:

$$|\Psi[A]\rangle = \text{---} \overset{\text{---}}{\underset{\text{---}}{\circ A_L}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\circ A_L}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\circ C}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\circ A_R}} \text{---} \overset{\text{---}}{\underset{\text{---}}{\circ A_R}} \text{---}. \quad (53)$$

As this depiction suggests, the matrix C contains information about the entanglement between the subsystems to the left and right of C . Let us make this more precise.

Consider a Hilbert space that admits a tensor product decomposition $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, with $d_A = \dim(\mathcal{H}_A)$, $d_B = \dim(\mathcal{H}_B)$. In virtue of the *singular value decomposition* (SVD), any state in $\mathcal{H}$ can be written in the form

$$|\Psi\rangle = \sum_k s_k |\psi_k^A\rangle \otimes |\psi_k^B\rangle, \quad (54)$$

where $|\psi_k^A\rangle$ and $|\psi_k^B\rangle$ are orthonormal basis vectors for $\mathcal{H}_A$ and $\mathcal{H}_B$ respectively, and the real numbers s_k for $k = 1, 2, \dots, \min(d_A, d_B)$ are called the *singular values* or *Schmidt numbers* and constitute the *entanglement spectrum* of the state. By tracing out either subsystem we can write the reduced density matrices as

$$\rho_A = \text{Tr}_B(|\Psi\rangle\langle\Psi|) = \sum_{k=1}^{d_A} s_k^2 |\psi_k^A\rangle\langle\psi_k^A|, \quad (55)$$

$$\rho_B = \text{Tr}_A(|\Psi\rangle\langle\Psi|) = \sum_{k=1}^{d_B} s_k^2 |\psi_k^B\rangle\langle\psi_k^B|. \quad (56)$$

A direct comparison with the MPS (53) reveals that the Schmidt spectrum of the state $|\Psi[A]\rangle$ coincides with the singular values of C . The MPS $|\Psi[A]\rangle$ can eventually be brought in the form (54) via the residual unitary gauge freedom $A_L^i \mapsto UA_L^iU^\dagger, A_R^i \mapsto V^\dagger A_R^iV$, with U and V obtained from the SVD $C = USV^\dagger$.

One of the key advantages of MPS is the efficient computation of expectation values of local observables and their correlation functions. *Efficient* means here with a computational cost that scales linear in the distance between the local operators. As an illustration, let us focus on the two-point correlation function of local observables O_1 and O_2 at sites i and j . Due to translation invariance this expectation value depends only on the distance $L = |j - i|$. Defining expectation values of observables as usual:

$$\langle O_1[i]O_2[j] \rangle_{\Psi[A]} := \langle \Psi[A] | O_1[i]O_2[j] | \Psi[A] \rangle, \quad (57)$$

this boils down to the contraction

$$\langle O_1[i] O_2[j] \rangle_{\Psi[A]} = \begin{array}{c} \text{---} \bar{A} \text{---} \bar{A} \text{---} \bar{A} \text{---} \dots \text{---} \bar{A} \text{---} \bar{A} \text{---} \bar{A} \text{---} \\ | \quad | \quad | \quad \quad \quad | \quad | \quad | \\ O_1 \quad \quad \quad O_2 \\ | \quad | \quad | \quad \quad \quad | \quad | \quad | \\ \text{---} A \text{---} A \text{---} A \text{---} \dots \text{---} A \text{---} A \text{---} A \text{---} \end{array} \quad (58)$$

By assuming injectivity and thus again that the transfer matrix has a unique largest eigenvalue 1 with corresponding left and right eigenvectors Λ and ρ , we are able to rewrite this contraction as

$$\begin{array}{c} \bar{A} \quad \bar{A} \\ | \quad | \\ A \quad O_1 \quad A \\ | \quad | \\ A \quad A \end{array} \dots \begin{array}{c} \bar{A} \quad \bar{A} \\ | \quad | \\ A \quad O_2 \quad P \\ | \quad | \\ A \quad A \end{array}, \quad (59)$$

which demonstrates that this correlator can be computed with a cost that scales as $\mathcal{O}(L\chi^3)$. Indeed, the contraction of this tensor network proceeds from left to right, and by choosing the contraction order in a judicious way, the main cost is the multiplication of the $\chi \times \chi$ matrices (see ref. [38] for an algorithm that identifies the optimal contraction order of a generic tensor network). As particular cases of this expression, we have the expectation value of a local operator by taking $O_1 \equiv O$ and $O_2 \equiv 1$:

$$\begin{array}{c} \bar{A} \\ | \\ A \quad O \quad P \\ | \\ A \end{array}, \quad (60)$$

as well as the norm squared of the MPS

$$\langle \Psi[A] | \Psi[A] \rangle := \begin{array}{c} \bar{A} \\ | \\ A \quad P \\ | \\ A \end{array}. \quad (61)$$

If we now define the *connected* two-point correlation function as customary:

$$C(O_1[i], O_2[j]) := \langle O_1[i] O_2[j] \rangle_{\Psi[A]} - \langle O_1[i] \rangle_{\Psi[A]} \langle O_2[j] \rangle_{\Psi[A]}, \quad (62)$$

one can show, by once more making use of the spectral decomposition of the transfer matrix eq. (29) (assuming it exists), that the dominant contribution to this correlation function decays *exponentially* in L , more precisely:

$$C(O_1[i], O_2[j]) \sim \lambda_2^{L-1} \sim \exp(-L/\xi), \quad (63)$$

if we define the *correlation length* ξ as $\xi = -1/\log(|\lambda_2|)$. This exponential decay of correlation functions is a witness of the fact that MPS describe well the ground state physics of *gapped* Hamiltonians, and can thus not reproduce the algebraic decay of correlations present in *critical systems*. Nevertheless, by making use of the theory of *entanglement scaling*, the topic of the next section, precise approximations to critical correlation functions can be achieved.

2.3 Scaling theory

Aside from gapped ground state physics, MPS methods are also a powerful tool to compute properties of *critical systems*. Using uniform methods which work directly in the thermodynamic limit, i.e. on the infinite chain (see Sec. 2.4), it is possible to extract critical exponents and the central charge describing the universality class or CFT of a continuous phase transition. There is a large body of existing research about the topic, originally introduced in refs. [39–42]. Here we will follow the approach developed in the papers [43–45].

Close to a second-order critical point, we expect correlation functions to be of the Ornstein–Zernike form, it is a product of a power law contribution multiplied with an exponential decaying contribution. This can be understood from the Källén–Lehmann representation of correlation functions as a linear combination of a continuum of exponentials [43]. To express the continuum directly using MPS, one would need an infinite bond dimension MPS. Using finite bond dimension MPS is equivalent to a discretisation of this continuum spectrum. We can quantify the ‘discreteness’, or degree of approximation, by the gaps in the transfer matrix spectrum. Let λ_i as before be the eigenvalues of the transfer matrix, with $\lambda_1 = 1$, and define $\epsilon_i = -\ln \lambda_i$. Take a linear combination of the top n spectral variables ϵ_i , excluding the first one:

$$\delta := \sum_{i=2}^n c_i \epsilon_i \quad \text{with} \quad \sum_{i=2}^n c_i = 0. \quad (64)$$

In refs. [43, 44], it was demonstrated that the ϵ_i have a meaning of momenta, and that the difference between consecutive values of ϵ_i therefore introduces an inverse length scale δ which goes to zero if $\chi \rightarrow \infty$, both in the gapped and in the gapless case. In analogy to the finite-size scaling hypothesis of Cardy [46], where the length scale is the system size, it is possible to use this inverse length scale δ to formulate a *scaling hypothesis* for any order parameter m of a second-order phase transition as follows:

$$m(t, \delta) = s^{-\beta/\nu} m(s^{1/\nu} t, s\delta). \quad (65)$$

An analogous scaling hypothesis for the correlation length is:

$$\delta \xi(t, \delta) = \xi(\delta^{-1/\nu} t, 1) = \xi(\delta^{-1/\nu} t). \quad (66)$$

From the forms above, one can also deduce the scaling hypothesis for the entanglement entropy. Using the CFT form for the entanglement entropy S [47], we know that $\exp(\frac{6}{c} S)$ scales as length, so the scaling hypothesis for S is:

$$\exp\left(\frac{6}{c} S(t, \delta)\right) = s \exp\left(\frac{6}{c} S(s^{-1/\nu} t, s\delta)\right), \quad (67)$$

where c is the central charge. Performing simulations using uniform MPS methods with various bond dimensions χ then allows for fitting m, ξ, c using standard optimisation methods. The whole point of this is the fact that many different simulations for many different parameters of the Hamiltonian close to the critical point and for different bond dimensions can be related to each other, and that the correct values of the critical point and critical exponents will lead to a data collapse of all the ensuing curves. We refer to ref. [45] for details. Working with a finite bond dimension is like opening a gap in the system, and its effect is indistinguishable to adding a relevant perturbation to a critical theory. This way, the finite entanglement scaling theory yields the state-of-the-art way of simulating critical theories – the fact that MPS are always gapped is not a bug, it is a feature!

2.4 Variational MPS algorithms

2.4.1 Density matrix renormalisation group

The *density matrix renormalisation group* (DMRG) algorithm, originally due to Steven White [32], is a variational method for finding the ground state of *local one-dimensional Hamiltonians*. It has been applied with tremendous success to a plethora of models and is still the current state of the art for ground state optimisation in (1+1)d for strongly-correlated spin chains [2, 8].

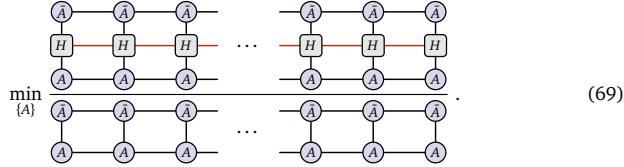
Although DMRG was originally derived and explained in terms of the renormalisation group, its modern interpretation is that of a variational method over the class of matrix product

states, where the optimum is iteratively found by using an alternating least squares method [48]. DMRG minimises the expectation value

$$\min_{\{A\}} \left(\frac{\langle \Psi[A] | H | \Psi[A] \rangle}{\langle \Psi[A] | \Psi[A] \rangle} \right), \quad (68)$$

over all finite MPS of a maximum (bounded) bond dimension $\chi_{\max}$. The larger you choose $\chi_{\max}$, the better the approximation will be. Even with modest computational resources, one can easily achieve bond dimensions $\chi \sim \mathcal{O}(10^3)$ by making use of sparse eigensolvers such as a Lanczos method to solve the effective eigenvalue problem (see below), or even $\mathcal{O}(10^5)$ when symmetries are exploited as well [49–54].

Assuming that the Hamiltonian is written in MPO form (see section 1.4), eq. (68) boils down to:



$$\min_{\{A\}} \left(\frac{\langle \Psi[A] | H | \Psi[A] \rangle}{\langle \Psi[A] | \Psi[A] \rangle} \right). \quad (69)$$

The algorithm is initiated by providing an initial guess for the ground state, typically simply a random MPS, and proceeds with a series of local sequential optimisations of all tensors, sweeping back and forth through the chain, until convergence is reached.

Fixing a site i and bringing all tensors to its left in left canonical form and vice versa for the tensors to its right, the update of the tensor $A[i]$ boils down to replacing it by the eigenvector corresponding to the smallest real eigenvalue of the *effective Hamiltonian*



$$\left[L \right] \left[H \right] \left[R \right]. \quad (70)$$

In this figure, L and R stand for the (site-dependent) *environments* obtained by contracting all MPS and MPO tensors to the left, respectively right, of site i .

As such, this version of the algorithm updates each site individually and is referred to as *one-site DMRG*. The downside of this approach is that there is no convenient way of increasing the bond dimension during the sweeps. This is particularly problematic when symmetries are imposed, since in that case the algorithm can not explore distinct symmetry charges on the virtual level. This shortcoming is remedied by a *two-site* version of the algorithm.

This version of the algorithm proceeds by blocking two sites in every step of every sweep, and subsequently updating the blocked tensor $A'[i, i+1]$ as the minimal eigenvector of the effective two-site Hamiltonian



$$\left[L \right] \left[H \right] \left[A' \right] \left[R \right]. \quad (71)$$

A singular value decomposition is then used to split A' into $A[i], A[i+1]$. It is in this step that the bond dimension can be dynamically controlled by truncating the SVD so as to only keep the largest singular values. The high-level pseudocode of DMRG can be found below.

Algorithm 1: Two-site DMRG

```

1 Initialize the MPS tensors  $A[i]$  (usually to a random state)
2 for  $k = 1$  to num_sweeps do
    // Left-to-right sweep
3   for  $i = 1$  to  $N - 2$  do
4     Optimise and update pair of sites  $i, i + 1$ 
5     Update left environment at site  $i + 1$ 
    // Right-to-left sweep
6   for  $i = N - 1$  to 2 do
7     Optimise and update pair of sites  $i, i + 1$ 
8     Update right environment at site  $i + 1$ 
9 return  $\{A[i]\}$ 

```

2.4.2 Variational uniform matrix product state

The *variational uniform matrix product state* (VUMPS) algorithm is an infinite-chain equivalent of DMRG, and was introduced in ref. [55] (see ref. [4] for a pedagogical review, and [56] for an alternative method). It works directly in the thermodynamic limit, which makes it particularly useful for the simulation of critical systems and extracting their scaling laws. Of course, in the infinite limit the total ground state energy is unbounded, so that VUMPS rather minimises the *energy density*.

Before introducing the algorithm, let us first fix some notation. Recall the central gauge MPS A_C and bond tensor C introduced in section 2.2, alongside the left and right gauges A_L, A_R . We then define the effective single-site Hamiltonian, H_{eff} , as well as the effective environment for the bond tensor C, H_C , as:

$$H_{\text{eff}} := \left[\begin{array}{c} \text{---} L \text{---} \\ | \\ \text{---} H \text{---} \\ | \\ \text{---} R \text{---} \end{array} \right], \quad H_C := \left[\begin{array}{c} \text{---} L \text{---} \\ | \\ \text{---} C \text{---} \\ | \\ \text{---} R \text{---} \end{array} \right]. \quad (72)$$

Here, the left and right environments L, R are the fixed points of the transfer matrices $\mathbb{T}_L^{[H]}, \mathbb{T}_R^{[H]}$ respectively:

$$\mathbb{T}_L^{[H]} := \left[\begin{array}{c} \text{---} A_L \text{---} \\ | \\ \text{---} H \text{---} \\ | \\ \text{---} A_L \text{---} \end{array} \right], \quad \mathbb{T}_R^{[H]} := \left[\begin{array}{c} \text{---} A_R \text{---} \\ | \\ \text{---} H \text{---} \\ | \\ \text{---} A_R \text{---} \end{array} \right]. \quad (73)$$

They can be calculated by solving a system of linear equations.

As derived in ref. [55], the variational minimum of the ground state energy density in the manifold of uniform MPS is characterised by a simultaneous solution to

$$\begin{aligned} H_{\text{eff}}(A_C) &\propto A_C, \\ H_C(C) &\propto C, \\ A_C^i &= A_L^i C = C A_R^i. \end{aligned} \quad (74)$$

The last equation is included so as to avoid having to take $A_L = A_C C^{-1}$, $A_R = C^{-1} A_C$, which is potentially ill-defined if C is close to being singular. Given an initial guess for the ground state, the VUMPS algorithm iteratively approaches this solution until convergence is reached. In each step, the current values of A_C and C are updated by replacing them by the eigenvectors

with minimal real eigenvalue of the current H_{eff} and H_C . Subsequently, approximate $\tilde{A}_L$ and $\tilde{A}_R$ are found from $\tilde{A}_C$ and $\tilde{C}$ by invoking the singular value decomposition for $A_C C$ and $C A_C$:

$$\begin{aligned}\tilde{A}_L &= U_L V_L^\dagger, & U_L \Sigma_L V_L^\dagger &= \tilde{A}_C \tilde{C}, \\ \tilde{A}_R &= U_R V_R^\dagger, & \tilde{C}^\dagger \tilde{A}_C &= U_R \Sigma_R V_R^\dagger.\end{aligned}\quad (75)$$

As such, the 2-norms

$$\begin{aligned}\epsilon_L &= \|\tilde{A}_C - \tilde{A}_L \tilde{C}\|_2, \\ \epsilon_R &= \|\tilde{A}_C - \tilde{C} \tilde{A}_R\|_2,\end{aligned}\quad (76)$$

are minimised.

The VUMPS pseudocode for an MPO Hamiltonian can be found below.

Algorithm 2: VUMPS

```

1 Initialise  $\{A_L, A_R, C\}$  (usually to a random state)
2 Initialise current precision  $\epsilon_{\text{cur}}$ 
3 while  $\epsilon_{\text{cur}} > \epsilon_{\text{tol}}$  do
4   Calculate  $\mathbb{T}_L^{[H]}$  from  $A_L, H$ 
5   Calculate  $\mathbb{T}_R^{[H]}$  from  $A_R, H$ 
6   Calculate the left environment  $L$  from  $\mathbb{T}_R^{[H]}$ 
7   Calculate the right environment  $R$  from  $\mathbb{T}_L^{[H]}$ 
8   Construct the effective infinite Hamiltonian  $H_{\text{eff}}$  from  $L, R, H$ 
9   Construct the effective infinite environment  $H_C$  from  $L, R$ 
10  Find  $A_C$ , the leading eigenvector of  $H_{\text{eff}}$ 
11  Find  $C$ , the leading eigenvector of  $E_{\text{eff}}$ 
12  Calculate  $A_L, A_R$  from  $A_C, C$  as in eq. (75)
13  Set  $\epsilon_{\text{cur}} = \max(\epsilon_L, \epsilon_R)$ 
14 return  $\{A_L, A_R, C\}$ 

```

2.5 Quasiparticle excitations

Up until this point we have mainly focussed on the ground state, and algorithms to variationally approximate it, both in the finite (section 2.4.1) as well as in the infinite (section 2.4.2) setting. A comprehensive understanding of a quantum many-body system should also explain its low-lying excitations. Typically, the spectrum of a gapped quantum many-body Hamiltonian consists of a few isolated branches which have an interpretation in terms of localised *quasiparticle excitations*, as well as a tower of continuum *scattering states* which can be constructed by essentially adding up quasiparticle states. These quasiparticles show up in dynamical correlation functions – which can be probed in e.g. inelastic neutron scattering experiments – but in the case of critical systems also encode universal properties of their effective continuum description⁶.

Matrix product state techniques can be used to capture these quasiparticle excitations and their scattering states by means of the so-called *quasiparticle ansatz*, first introduced in ref. [57] and subsequently generalised in ref. [58] to single-particle excitations in one-dimensional chains and to two-particle states in ref. [59]. Here, we assume a translationally invariant local and gapped Hamiltonian H . For the time being we moreover assume a unique ground state to exclude domain-wall excitations, which will be discussed below. We denote an MPS

⁶Remember that a finite bond dimension gaps out the excitations when approximating critical ground states using MPS.

approximation of the ground state by $|\Psi[A]\rangle$. The ansatz for a quasiparticle with momentum k then takes the form

$$|\Psi[A, B], k\rangle = \sum_n e^{ikn} \text{---} \overset{\text{A}}{\underset{|}{\text{---}}} \text{---} \overset{\text{A}}{\underset{|}{\text{---}}} \underset{\text{B}}{\underset{|}{\text{---}}} \text{---} \overset{\text{A}}{\underset{|}{\text{---}}} \text{---} \overset{\text{A}}{\underset{|}{\text{---}}} \text{---} . \quad (77)$$

That is, the ansatz replaces one of the ground state tensors by a tensor B which contains all variational parameters of the ansatz, and we take a momentum superposition over the position of B in the infinite chain. As such, it can be thought of as a *boosted* version of a tangent vector eq. (37), allowing one to exploit the full arsenal of techniques developed for the tangent space. Also, it is interesting to note that the quasiparticle ansatz can be interpreted as a generalisation of the Feynman-Bijl ansatz, or *single mode approximation* developed to describe collective excitations in liquid Helium [60–62]. Note that due to the finite bond dimension, we should not think about $|\Psi[A, B], k\rangle$ as describing a localised particle on the location of the tensor B , but rather as a lump of increased energy density with respect to the uniform vacuum. Given A , the optimisation of B boils down to a generalised eigenvalue problem for an effective Hamiltonian, similar in spirit to section 2.4.2. A detailed exposition of the algorithm is beyond the scope of these notes, but let us remark that in it, the use of the efficient parametrisation of B , very similar to the one described in section 1.5, is crucial.

It is worth emphasising that the ground state encodes information about the excited states. Indeed, for a generic self-adjoint operator, one of its eigenvectors typically does not reveal anything about the others. However, the locality of the Hamiltonian is the distinguishing feature that ensures that eq. (77) provides a *good* ansatz for quasiparticle excitations. Indeed, in an application of the famous Lieb-Robinson bounds [63–65], it was demonstrated in ref. [66] that every (exact) eigenstate $|\Psi_{k,a}\rangle$ can be approximated arbitrary well via acting with a momentum k superposition of a localised (in space) operator on the ground state. The ansatz eq. (77) should then be recognised as the extreme case where the MPS ground state is acted upon by said operator whose support is truncated to one site.

Interestingly, in the presence of an internal symmetry the excitations are organised in degenerate multiplets transforming in an (irreducible) representation of the symmetry group. This is not compatible with the quasiparticle approach that we presented so far, as in that case the energy window around an excitation energy Δ , $[\Delta - \delta, \Delta + \delta]$, contains more than one excitation for any value of δ . To remedy this, one can modify the ansatz by replacing the operator $\tilde{O}$ by a multiplet of operators $\tilde{O}_m$, which in the MPS picture boils down to specifying the charge of the B tensor [67].

In the case of symmetry breaking and thus a degenerate ground state subspace, the excitations with lowest energy are not of the form eq. (77). Rather, they are *topological domain wall excitations* that form an interface between different ground states. In this context, *topological* refers to the fact that these excitations are created from the ground state by an extensive operator, namely a symmetry operator acting on the half-infinite chain terminated at the location of the domain wall. This can be accounted for by adapting the ansatz so as to encode different ground state tensors on either side of the B tensor. All computational MPS techniques work equally well in this case, but care has to be taken to contract the right environments with each other.

In fig. 1 we depict the low-energy excitation spectrum of the spin 1 antiferromagnetic Heisenberg model. It was obtained by explicitly enforcing the $SU(2)$ symmetry, and we labelled the excitations by their spin. The isolated $S = 1$ branch is clearly visible and reaches its minimum $\Delta \approx 0.41$ at $k = \pi$, famously known as the *Haldane gap* [68, 69]. Astonishingly, this energy gap can be computed using the quasiparticle ansatz with a precision of ca. 10 digits, even with very modest computational resources.

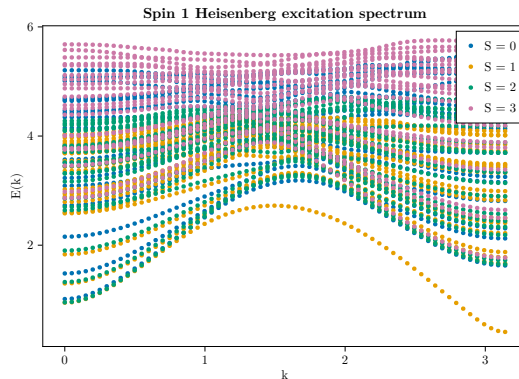


Figure 1: Excitation spectrum of the Heisenberg spin 1 antiferromagnet obtained with the one-site quasiparticle ansatz at bond dimension 48. Excitations are labelled by their $SU(2)$ spin. Note in particular that the *Haldane* gap is situated at momentum π . Created with MPSKit.jl [70].

2.6 Synopsis of lecture II

Lecture II has been dealing with the computational aspects of matrix product states - how to define normal forms, how to minimise the energy on the manifold of MPS using alternating least squares methods (DMRG), how to extend those methods to infinite system sizes (VUMPS), and how to define excitations by projecting the full Hamiltonian on the tangent plane of the manifold of MPS. Furthermore, we touched upon the topic of entanglement scaling - through which critical systems can be studied with unprecedented precision.

The fact that - in all those applications - the computational cost only scales polynomially in both the system size and the bond dimension is a clear witness of the fact that tensor network methods effectively manage to break down the exponential many-body wall - and hence has the potential of resolving central problems in physics and chemistry.

3 Lecture III: Phases of matter

3.1 Gapped phases of matter, entanglement & symmetries

A fundamental challenge in quantum many-body physics is to determine the *phase diagram* of a given model. Two models belong to the same phase if there exists a smooth path of gapped, bounded-strength, local Hamiltonians interpolating between them. From a pragmatic point of view, this is equivalent to the existence of a *sublinear-depth quantum circuit* that maps their ground states into each other [65, 71]. Distinct phases are then separated by phase transitions, which in the case of a second-order phase transition – as encountered above in section 2.3 – is typically described by a conformal field theory [72]. For a model with two parameters $h_{1,2}$ a fictitious phase diagram is sketched below.

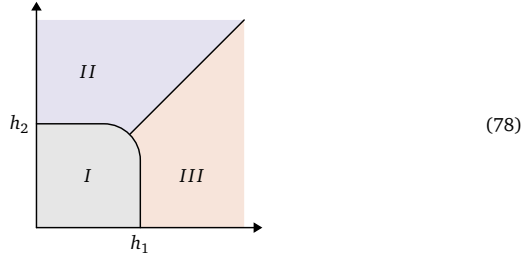
The *Landau paradigm* of phase transitions states that every phase of matter is in one-to-one correspondence with a symmetry breaking pattern [73]. The most famous (classical) example is of course the phase transition of water to ice, in which continuous Euclidean symmetry is broken to one of its 230 subgroups, also known as the *space groups*. Moreover, the paradigm states that this breaking of symmetry can be diagnosed via a local order parameter that is charged under the symmetry group. Vice versa, *symmetric* phases can be identified through *disorder operators*, as argued by Kadanoff and Ceva [74]. The Landau paradigm has since its conception been generalised in a number of ways.

For one, it has since been realised that after breaking its symmetry down to a subgroup, a quantum system can realise its residual symmetry in distinct ways on its entanglement degrees of freedom. These distinguishable entanglement patterns correspond to distinct phases if we require that the adiabatic paths between Hamiltonians commute with the symmetry. Such *symmetry-protected topological* phases are for this reason naturally described and classified in terms of tensor networks [37, 75–77].

The goal of this lecture will be to sketch the classification of one-dimensional quantum phases of matter in the presence of a group symmetry using the framework of MPS. As remarked in section 2.2, symmetry breaking in MPS is manifested by the existence of distinct injective blocks in the ground state MPS. These are labelled by cosets in G/H for a subgroup $H \subseteq G$ and are permuted under the group action. The challenge will thus be to classify the *symmetric phases* realised on each of these injective blocks. To this end, we will first derive the *fundamental theorem* of MPS which states that an MPS is invariant under a symmetry if and only if its local tensors transform trivially under that symmetry. This theorem exemplifies the mantra that tensor networks encode *global* properties of states in the *local* tensors. This will bring us in the realm of *projective representations* whose corresponding cohomology classification provides a *topological index* that classifies distinct symmetric phases in the sense that two injective MPSs belong to the same phase if and only if their topological index is the same. Notably, one-dimensional bosonic systems all belong to the same phase if no symmetry is imposed. For fermions the story is more intricate and non-trivial phases can exist due to the always present fermionic parity symmetry; this will be the topic of section 3.5.

Let us also foreshadow our objectives of subsequent lectures. With the discovery of the fractional quantum Hall effect and other two-dimensional *topological phases of matter*, it also became clear that even when a system does not exhibit any symmetry, it can still belong to a non-trivial quantum phase of matter. These phases of matter can be characterised by a ground state degeneracy which depends on the topology of the underlying space, gapped *anyonic excitations* and in some cases chiral edge modes. They are also not detected by any local order parameter. As we will discuss in the next lecture, these phases owe their stability to the entanglement in these systems being *long range*. In particular, in the case of *non-chiral topological order*, the entanglement patterns are also responsible for the ground state degeneracy, as it

has been appreciated that entanglement degrees of freedom exhibit string-like ‘symmetries’ which are described by a *fusion category*. We will encounter a particularly simple class of topological orders below in section 4.6, and a microscopic construction of the generic non-chiral case in section 5.3. The existence of these *generalised symmetries* on the entanglement degrees of freedom implies in particular that the boundary Hamiltonians of these topological orders, describing the dynamics of their edge modes, inherit also these symmetries. As such, we are inclined to consider a Landau paradigm for these generalised symmetries as well, and this will be one of the topics of the last lecture.



3.2 Fundamental theorem of MPS

As anticipated above, the *fundamental theorem* of MPS [78] is crucial in the classification of one-dimensional symmetric gapped phases of matter and many other applications of MPS. The incarnation of the theorem that we will apply is as follows. Two uniform *injective* matrix product states represent the same state on any system with periodic boundary conditions, independent of the length, if and only if the MPS tensors are related by a non-singular gauge transformation on the virtual level:

$$|\Psi[A]\rangle \sim |\Psi[B]\rangle \iff A^i = e^{i\chi} X B^i X^{-1}, \quad \forall i. \quad (79)$$

Note that we assumed, without loss of generality, that both MPSs have the same bond dimension. When both MPSs are in canonical form, X is unitary.

As an application of some of the key concepts introduced earlier in these lectures, we provide a proof here.

Note first that the $\Leftarrow$ direction is trivial since all appearances of X and X^{-1} cancel on the virtual level. To prove the $\Rightarrow$ direction, let us assume without loss of generality that $|\Psi[A]\rangle$ is in right canonical form, while $|\Psi[B]\rangle$ is in left canonical form. Consider now the following mixed transfer matrix:

Since $|\langle \Psi[A] | \Psi[B] \rangle| = 1$, the leading eigenvalue of the mixed transfer has unit norm [79]. We will denote its corresponding right eigenvector by X , so that $T_{A,B} X = e^{i\phi} X$. Let us now consider the following contraction

First note that on the one hand this expression is equal to $\text{Tr}(X^\dagger X)$ via $\mathbb{T}_{A,B}X = e^{i\phi}X$. On the other hand, interpreting this expression as the inner product of two vectors with respect to the diagonal cut and applying the Cauchy-Schwarz inequality, we obtain that

$$\text{Tr}(X^\dagger X) = e^{-i\phi} \left| \begin{array}{c} \text{Diagram 1: A square with vertices labeled } \bar{X}, \bar{B}, X, A. \text{ A dashed diagonal line connects } \bar{X} \text{ and } X. \end{array} \right| \leq \left| \begin{array}{c} \text{Diagram 2: A square with vertices labeled } X, A, X, A. \end{array} \right|^{1/2} \times \left| \begin{array}{c} \text{Diagram 3: A square with vertices labeled } B, X, B, X. \end{array} \right|^{1/2} = \text{Tr}(X^\dagger X), \quad (82)$$

where we use that A and B are in right and left canonical form. So clearly the equality holds, which implies that $A^\dagger X = e^{i\chi} X B^\dagger$, or since X is full rank thus eq. (79).

3.3 SPT classification of one-dimensional bosonic matter

Let us consider a gapped Hamiltonian H with a unique ground state, and assume it commutes with a symmetry G : $[U(g)^{\otimes L}, H] = 0$, for a representation $\{U(g), g \in G\}$. As H is gapped and has a unique ground state, the ground state must inherit that symmetry. It is guaranteed that this ground state is faithfully represented by an injective MPS. That MPS $|\Psi[A]\rangle$ must therefore transform trivially under G , i.e.:

$$U(g)^{\otimes N} |\Psi[A]\rangle \simeq |\Psi[A]\rangle, \quad \forall g \in G. \quad (83)$$

A direct application of the fundamental theorem eq. (79) reveals that the MPS tensor itself has to transform trivially under the symmetry group, namely:

$$\forall g \in G, \exists X_g, \chi_g : \sum_j U(g)_{ij} A^j = e^{i\chi_g} X_g A^i X_g^\dagger \quad (84)$$

where it should be noted that the gauge matrices and phases themselves depend on the group element g . Let us now consider acting successively with the symmetry operators labelled g and h on the state. We can do this in two different ways: either by acting with $U(gh)$ directly and using the fundamental theorem once or acting sequentially with $U(g)$ and $U(h)$ and applying the fundamental theorem twice. Equating these two results in:

$$e^{i\chi_{gh}} X_{gh} A^i X_{gh}^\dagger = e^{i(\chi_g + \chi_h)} X_g X_h A^i X_h^\dagger X_g^\dagger, \quad \forall i, g, h. \quad (85)$$

As A^i is injective and hence spans the full matrix algebra, this is only possible if both $\chi_{gh} = \chi_g + \chi_h$ and $X_g X_h = \psi(g, h) X_{gh}$ for $U(1)$ phases $\psi(g, h)$. The former means that $\{e^{i\chi_g}, g \in G\}$ constitutes a one-dimensional representation of G . Note that after blocking a finite number of M sites, we obtain an enlarged unit cell for which the phase $e^{iM\chi_g} = 1$ disappears. Therefore we could say that this phase is *unstable against blocking* and we will not take it further into account for phase classification. On the other hand, we have that the set of matrices $\{X_g, g \in G\}$ obeying

$$X_g X_h = \psi(g, h) X_{gh}, \quad \forall g, h \in G, \quad (86)$$

for some phases $\psi(g, h) \in U(1)$ forms a so-called ψ -projective representation of G . Projective representations generalise ordinary linear representations by allowing for the projective phase $\psi(g, h)$ in their multiplication. Due to the associativity of the matrix multiplication the phases $\psi(g, h)$ have to satisfy

$$\psi(g, h)\psi(gh, k) = \psi(g, hk)\psi(h, k), \quad \forall g, h, k \in G, \quad (87)$$

a consistency equation commonly known as the *second cocycle equation*. This condition has trivial solutions of the form $\psi(g, h) = \frac{\phi_{gh}}{\phi_g \phi_h}$ for any function $\phi : G \rightarrow U(1)$. In the context of

group cohomology such a trivial cocycle is called a *coboundary*. In order to classify phases we will want to consider solutions of the cocycle equation *modulo* coboundaries. Indeed, from the fundamental theorem it is clear that we can freely redefine $X_g \mapsto \phi_g X_g$ for any function ϕ . To this end, note that both the cocycles and the coboundaries form abelian groups, which are denoted by $Z^2(G, U(1))$ and $B^2(G, U(1))$ respectively. Since clearly $B^2(G, U(1))$ is a normal subgroup of $Z^2(G, U(1))$, we can consider the quotient $H^2(G, U(1)) := Z^2(G, U(1))/B^2(G, U(1))$, the *second cohomology group* of G over $U(1)$.

Note that the cocycle equation eq. (87) – after taking the logarithm of both sides, and collecting the unknown phases $\psi(g, h)$ in a vector ψ – can be written as a linear system of equations, satisfied modulo 2π :

$$\Omega\psi = 0, \quad \text{mod } 2\pi. \quad (88)$$

Computing the second cohomology group and corresponding representative cocycles can then be done by bringing Ω in *Smith normal form*, a matrix decomposition of matrices with integer coefficients of the form $\Omega = P\Lambda R$, where both P and R are unimodular integer matrices, and Λ is a diagonal matrix. A detailed explanation is relegated to section A. The main observation is that the second cohomology group is always finite and of the form

$$H^2(G, U(1)) \simeq \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_r}, \quad (89)$$

where the d_i are the diagonal entries of Λ . The group structure of $H^2(G, U(1))$ is physically realised by *stacking* two G -symmetric chains and considering the diagonal subgroup of $G \times G$ as the G symmetry of the whole.

One can prove that if two MPSs transform under a G symmetry with projective representations corresponding to distinct cohomology classes in $H^2(G, U(1))$, these MPSs must represent ground states in different phases [37]. In other words, there is no symmetric adiabatic path one can take to transform one into the other without crossing a phase transition. In that case, the distinct cohomology classes are sometimes said to form a *topological obstruction* to do so, while a representative cocycle is sometimes called a *topological index* characterising that phase. If we were to allow paths that break the symmetry, then all MPSs would again belong to the same trivial phase. In particular, the number of symmetric phases is always finite due to eq. (89). The proof is beyond the scope of these lecture notes, but let us remark that the argument relies heavily on the parent Hamiltonian construction of MPS (see section 2.2).

An important property of any projective representation is that its matrix dimension is necessarily strictly bigger than one, when it corresponds to a non-trivial cohomology class. As an example, consider the smallest group with a non-trivial second cohomology group, namely $\mathbb{Z}_2 \times \mathbb{Z}_2$, $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$. If we denote its group elements by tuples (a, b) , $a, b = 0, 1$, a non-trivial projective representation is given by the Pauli matrices as follows: $X_{(a,b)} = (\sigma^x)^a (\sigma^z)^b$, and this representation cannot be further reduced. This fact has an important consequence for the entanglement of an MPS in a non-trivial SPT phase, and also guarantees the existence of gapless symmetry-protected edge modes [41].

Indeed, it turns out that the entanglement spectrum (see section 2.2) is characterised by a degeneracy which is at least the dimension of the smallest irreducible projective representation in that class. This is beautifully illustrated in the spin 1 antiferromagnetic Heisenberg model. In fig. 2 the first 48 Schmidt values are plotted, and the spectrum organises itself visibly in multiplets which are at least twofold degenerate. In this case, the protecting symmetry is $SO(3)$ whose second cohomology group indeed has a non-trivial element, $H^2(SO(3), U(1)) \simeq \mathbb{Z}_2$. All irreps of $SO(3)$ have an odd dimension, so these degeneracies clearly originate from the half-integer irreps of $SU(2)$, which indeed form projective representations of $SO(3)$. In fact, the phase is also protected by a $\mathbb{Z}_2 \times \mathbb{Z}_2$ subgroup of $SO(3)$, so that we can even perturb the

Hamiltonian with terms which break part of the $SO(3)$ symmetry as long as they still commute with $\mathbb{Z}_2 \times \mathbb{Z}_2$, without leaving the SPT phase.

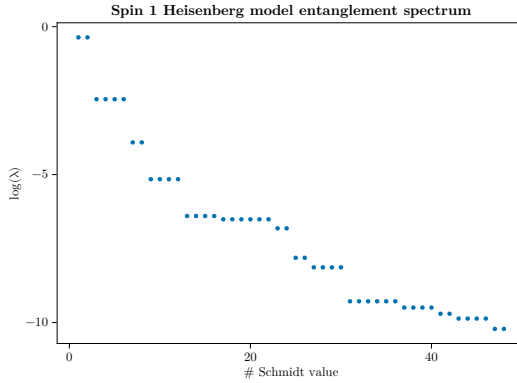


Figure 2: The entanglement spectrum of the spin 1 Heisenberg model ground state in the thermodynamic limit with bond dimension 48. Created with MPSKit.jl [70].

As anticipated in section 2.1, another hallmark of SPT phases is the presence of symmetry-protected gapless *edge modes*, when they are considered on open chains. A feature of matrix product states (and their higher-dimensional generalisations, see section 4.3) is that a tensor product basis for these edge modes is provided by the entanglement degrees of freedom on the boundary. Indeed, suppose we have a uniform MPS $|\Psi[A]\rangle$ of bond dimension χ . We can consider then following χ^2 states on an open chain of N sites:

$$|\Psi[A]\rangle = \sum_{\{i_j\}} (A^{i_1} A^{i_2} \cdots A^{i_N})_{\alpha\beta} |i_1, i_2, \dots, i_N\rangle. \quad (90)$$

All states $|\Psi[A], \alpha, \beta\rangle$, $\forall \alpha, \beta = 1, \dots, \chi$ have the same energy for the parent Hamiltonian, so that seemingly we have χ^2 degenerate edge modes. However, by perturbing the Hamiltonian, some of these edge modes can be gapped out. Nevertheless, as long as we restrict to symmetry-respecting perturbations, the edge modes will transform under the projective representation on the virtual level according to:

$$U(g)^{\otimes N} |\Psi[A], \alpha, \beta\rangle = \sum_{\alpha', \beta'} (X_g)_{\alpha\alpha'} (\bar{X}_g)_{\beta\beta'} |\Psi[A], \alpha', \beta'\rangle, \quad (91)$$

and hence, at least a two-fold degeneracy of the edge modes persists.

This concludes the classification of *bosonic* G -symmetric gapped one-dimensional phases. Provided a symmetry group G , gapped phases are in one-to-one correspondence with a subgroup $H \subseteq G$ which represents the *unbroken* symmetry in the ground state subspace, and a cohomology class $[\psi] \in H^2(H, U(1))$ representing the SPT phase of the surviving H symmetry.

As an aside we note here that such pairs $(H, [\psi])$ also characterise the (indecomposable) *module categories* over Vec_G , the *fusion category* of G -graded vector spaces (see also section B). There is thus a one-to-one correspondence between Vec_G -module categories and gapped G -symmetric phases. Of course this observation is completely superfluous for the simple case

of group symmetries, but in the last lecture we will argue that for generic fusion category symmetries this correspondence continues to hold.

3.4 Spacetime symmetries

The above discussion was focused on internal symmetries of the Hamiltonian, but applies with some slight modifications to spacetime symmetries as well. When dealing with these symmetries, one has to address two issues which are not present in the internal case. First, note that some spacetime symmetries such as translation symmetry and spatial reflection act in a non-local way. On the other hand, *orientation-reversing* symmetries such as time-reversal or reflection symmetry act in an antiunitary way and thus depend on the chosen basis. The latter has strong consequences for the cohomological classification of SPT phases, as we will see below. Nevertheless, MPS provide a suitable setting in which the classification of symmetric phases protected by spatial symmetries can be derived explicitly [41, 80].

Let us illustrate how a similar reasoning as above leads to the classification of phases protected by time-reversal symmetry. In general, time-reversal symmetry acts as UK where U is a unitary operation, $(\sigma^Y)^{\otimes N}$ in the case of spin $1/2$'s as demonstrated by Wigner [81], and K is complex conjugation. We assume here for simplicity that $U = 1$. Acting with complex conjugation on an injective MPS $|\Psi[A]\rangle$ and applying the fundamental theorem results in

$$\bar{A}^i = e^{i\chi} Q A^i Q^{-1}. \quad (92)$$

Applying time-reversal symmetry then a second time and exploiting injectivity of the MPS shows that:

$$Q\bar{Q} = \pm 1. \quad (93)$$

We thus obtain a non-trivial topological index corresponding to the sign, giving rise to a $\mathbb{Z}_2$ classification of time-reversal SPT phases. As an illustration consider the AKLT state eq. (44). It is easily checked that in that case $Q = \sigma^Y$, $Q\bar{Q} = -1$ so that the AKLT state thus belongs to a non-trivial SPT phase w.r.t. time-reversal symmetry.

Note that the Q in eq. (93) cannot be interpreted as a non-trivial projective representation of $\mathbb{Z}_2$ due to the complex conjugation on Q . Consequently, the phase ± 1 in eq. (93) also cannot be interpreted as a non-trivial cocycle for $H^2(\mathbb{Z}_2, \text{U}(1))$, which is in fact trivial. We therefore need a generalised notion of projective representations and cocycles to account for symmetry groups where some group elements act in an antiunitary fashion. It is easy to verify that when a generic symmetry group G has antiunitary elements, the gauge matrices X_g multiply according to

$$X_g \beta_g(X_h) = \psi(g, h) X_{gh}, \quad (94)$$

where $\beta_g(\bullet)$ acts as complex conjugation when g is an antiunitary element and as the identity otherwise. Clearly, this expression is equivalent to eq. (86) when all group elements act unitarily and to eq. (93) when the symmetry group is $\mathbb{Z}_2$ time-reversal. It is straightforward to verify as well that the consistency condition on $\psi(g, h)$ then reads

$$\psi(g, h)\psi(gh, k) = \psi(g, hk)\beta_g(\psi(h, k)), \quad \forall g, h, k \in G. \quad (95)$$

Accordingly, also the notion of coboundaries has to be slightly adapted. Putting things together, this means that in case of antiunitary symmetries, the cohomology classification of SPT phases has to be adapted in a way so as to account for the non-trivial group action $\beta_g(\bullet)$. The corresponding cohomology group is then denoted $H^2(G, \text{U}(1)^\beta)$, and is generically different from $H^2(G, \text{U}(1))$. Importantly, this group can be computed with the same method as outlined above and detailed in section A. In correspondence with the example of the AKLT state, we have indeed that $H^2(\mathbb{Z}_2, \text{U}(1)^\beta) = \mathbb{Z}_2$.

3.5 Fermionic MPS & symmetric phases

Up until now we have focused solely on *bosonic* matrix product states. The goal of this section is to sketch the extension of the MPS framework to *fermionic systems* and highlight some of the technical complications that arise, thereby closely following ref. [82], see also ref. [83, 84].

Crucial is the notion of a *super vector space*, which is a vector space that admits a direct sum decomposition of the form

$$V = V^0 \oplus V^1, \quad (96)$$

where vectors in V^0 , respectively V^1 , are said to possess a *fermion even*, respectively *odd*, parity. Without loss of generality we assume that basis vectors $|i\rangle$ have definite parity, which we will denote by $|i| = 0, 1$. A super vector space is moreover endowed with a *graded tensor product* behaving as follows under a re-arrangement of the tensor product:

$$|i_1\rangle \otimes_{\mathfrak{g}} |i_2\rangle \mapsto (-1)^{|i_1||i_2|} |i_2\rangle \otimes_{\mathfrak{g}} |i_1\rangle. \quad (97)$$

This property is the mathematical manifestation of fermionic particle exchange statistics. The *supertrace* is then defined as

$$C(|i\rangle \otimes_{\mathfrak{g}} |j\rangle) = (-1)^{|i||j|} \delta_{i,j}. \quad (98)$$

A uniform fermionic MPS (fMPS) is then constructed from an *even-parity* tensor

$$A = \sum_{i,\alpha,\beta} A_{\alpha\beta}^i |\alpha\rangle \otimes_{\mathfrak{g}} |i\rangle \otimes_{\mathfrak{g}} |\beta\rangle, \quad (99)$$

i.e. $|i| + |\alpha| + |\beta| = 0 \pmod{2}$ for all indices. Depending on whether the resulting state has even or odd fermionic parity the construction is slightly different.

Even-parity states An even-parity state is constructed from the tensor A as follows

$$|\Psi\rangle_e = C_v(A \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} \cdots \otimes_{\mathfrak{g}} A). \quad (100)$$

In this expression, C_v stands for the supertrace over the virtual indices. We can now rewrite this fMPS in a more explicit form. Note that the entries of a generic even-parity tensor are of the form

$$A^i = \begin{pmatrix} B^i & 0 \\ 0 & C^i \end{pmatrix}, \quad |i| = 0, \quad A^i = \begin{pmatrix} 0 & D^i \\ F^i & 0 \end{pmatrix}, \quad |i| = 1. \quad (101)$$

Defining the *parity matrix*

$$\mathcal{P} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad (102)$$

and carefully working out the expression eq. (100) yields then:

$$|\Psi[A]\rangle_e = \sum_{\{i_j\}} \text{Tr}(\mathcal{P} A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, i_2, \dots, i_N\rangle. \quad (103)$$

Odd-parity states The construction of odd-parity states proceeds analogously to the construction of even states, but requires the insertion of an additional matrix in the supertrace:

$$|\Psi\rangle_o = C_v(Y \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} A \otimes_{\mathfrak{g}} \cdots \otimes_{\mathfrak{g}} A). \quad (104)$$

The requirement to introduce the matrix Y ultimately stems from the fact that accessing odd-parity states necessitates twisted boundary conditions. We will encounter similar considerations in lecture 5.

Similar manipulations as in the previous case allow us to rewrite the odd-parity fMPS as

$$|\Psi[A]\rangle_o = \sum_{\{i_j\}} \text{Tr}(Y A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, i_2, \dots, i_N\rangle. \quad (105)$$

Imposing then translation invariance reveals that Y can be chosen to be of the form

$$Y = \begin{pmatrix} 0 & -\mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad (106)$$

and in this basis the MPS matrices read:

$$A^i = \begin{pmatrix} B^i & 0 \\ 0 & B^i \end{pmatrix}, |i| = 0, \quad A^i = \begin{pmatrix} 0 & B^i \\ -B^i & 0 \end{pmatrix}, |i| = 1. \quad (107)$$

Contrary to the even case, this fMPS is specified by one block B^i .

Note that since the odd fMPS matrices commute with Y , the MPS tensors have a common invariant subspace. Nevertheless, contrary to the bosonic case, the odd-parity states $|\Psi[A]\rangle_o$ define an irreducible matrix product state when the B^i span the full $\mathbb{C}^{N/2 \times N/2}$ matrix algebra. We refer to appendix D of [85] for a detailed technical discussion and proofs on canonical forms of fermionic graded MPS. Physically, the underlying reason can be traced back to the superselection rule for fermionic parity: there exist no superpositions of odd and even-parity states.

A notable example of an odd fMPS state is the ground state of the Kitaev chain of spinless fermions whose Hamiltonian is a simple sum of pairing terms [86]:

$$H = -i \sum_{i=1}^{N-1} \gamma_{2i} \gamma_{2i+1}, \quad (108)$$

for Majorana modes $\gamma_i = \gamma_i^\dagger$, $\{\gamma_i, \gamma_j\} = 2\delta_{ij}$. The translational invariant fMPS ground state is found to have odd parity and its MPS matrices read

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (109)$$

This model is noteworthy for exhibiting Majorana edge modes. Indeed, on open boundary conditions the state takes the form

$$\sum_{\{i_j\}} (A^{i_1} A^{i_2} \dots A^{i_N})_{a\beta} |i_1, i_2, \dots, i_N\rangle. \quad (110)$$

It is then straightforward to check that due to the structure of the tensors, choosing the boundary conditions $\alpha = \beta = 0$ or $\alpha = \beta = 1$ yields one and the same ground state. Similarly, both boundary conditions $\alpha = 0, \beta = 1$ and $\alpha = 1, \beta = 0$ yield a second ground state, resulting eventually in a two-fold ground state degeneracy, which cannot be lifted by adding an extra local interaction to the Hamiltonian.

Similar to the bosonic case, one can extract from fermionic matrix product states certain discrete topological invariants when global internal or spacetime symmetries are imposed. An illustrious example is the case of interacting time-reversal symmetric fermionic SPT phases. In that case there are three distinct topological indices valued in $\{0, 1\}$, one of which is related to whether the fMPS has Majorana edge modes, and two of which are related to the local transformation properties of the fMPS tensors under time-reversal. As carefully demonstrated, in ref. [82] it can be shown that under stacking of the corresponding fMPSs, the $\mathbb{Z}_8$ classification of Fidkowski and Kitaev is recovered [87].

3.6 Synopsis of lecture III

Lecture III explored the manifold of MPS when additional symmetries are imposed on the tensors. The main workhorse of symmetric tensor networks is the *fundamental theorem of MPS*, which states that any *global* symmetry must be reflected in the symmetries of the *local* MPS tensors. Interestingly, those *virtual* symmetries can be represented *projectively*. Any symmetry-preserving adiabatic path between symmetric MPSs which transform according to different projective representations is guaranteed to have a gapless point. As a consequence, all bosonic gapped phases of matter of one-dimensional quantum spin chains with a Hamiltonian that is G -symmetric are in one to one correspondence with all possible subgroups $H \subseteq G$ and its possible projective representations labelled by the second group cohomology $H^2(H, \text{U}(1))$. The latter classify the so-called *SPT phases*. If no symmetry is considered, then all MPSs can adiabatically converted into each other, and therefore the phases are only protected when the symmetry is imposed. We showed that this classification can be generalised to the case of spacetime symmetry groups. Interestingly, when superselection rules are considered such as the parity superselection rule for fermions, genuine topological phases of matter emerge (i.e. a phase that does not need a symmetry to be protected).

4 Lecture IV: MPO algebras: from the algebraic Bethe ansatz to G-injective PEPS

Let us now turn our attention towards integrable models, one of the central topics of this school. It turns out that tensor networks provide a very concise way of rephrasing the *algebraic Bethe ansatz* in terms of matrix product states and operators. In particular, we will see that the *Yang-Baxter equation*, which lies at the heart of its integrability, can be understood as a consequence of the fundamental theorem (section 3.2). Although tensor network techniques are not teaching us anything fundamentally new about the Bethe ansatz, the insights of this lecture will be instrumental in the MPO representation theory of *generalised* or *categorical symmetries* described by *fusion categories*, which is the topic of the final lecture.

After the discussion on the algebraic Bethe ansatz, we will focus in section 4.3 on the two-dimensional generalisation of MPS, namely *projected entangled-pair states*. We will conjecture a generalisation of the fundamental theorem, and discuss the role of MPO symmetries in the tensor network representation of two-dimensional SPT and topological phases, as well as their edge modes.

4.1 Magnon states

Tensor networks can be used as ansätze for analytic solutions of Hamiltonians, as was exemplified for the AKLT model eq. (44). The example of interest in this lecture however, is a famous family of exactly solvable models, called *integrable systems*. An integrable quantum system is roughly speaking a system with an extensive number of conserved charges – local operators commuting with the Hamiltonian – to which they owe their exact solvability. Integrable systems can be solved with the ansatz wave function called the *Bethe ansatz* (see e.g. [88]). There are two formulations of this approach, called the *coordinate Bethe ansatz* and *algebraic Bethe ansatz*, which produce the same set of solutions. Both formulations can be written down using tensor networks.

For the sake of concreteness, we consider the spin 1/2 XXZ Hamiltonian

$$H_{XXZ} = \sum_{i=1}^L \sigma_i^X \sigma_{i+1}^X + \sigma_i^Y \sigma_{i+1}^Y + \Delta \sigma_i^Z \sigma_{i+1}^Z \quad (111)$$

on periodic boundary conditions, i.e. $L + 1 \equiv 1$. The *magnon states* used as a basis when solving the Bethe equations can readily be written down as an MPS. For instance, the *periodic* boundary conditions 1-magnon state can be written down as an *open* boundary conditions MPS (see eq. (23)) with left boundary vector $\mathbf{v}_L^\dagger = \langle 0|$ and right boundary vector $\mathbf{v}_R = |1\rangle$, and MPS tensor:

$$\begin{aligned} A^0(k) &= \begin{pmatrix} e^{ik} & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1(k) = \begin{pmatrix} 0 & e^{ik} \\ 0 & 0 \end{pmatrix} \\ \Rightarrow |\Psi_1[A]\rangle &= \sum_{n=1}^L e^{ikn} |0_0 \dots 0_{n-1} 1_n 0_{n+1} \dots 0_L\rangle \end{aligned} \quad (112)$$

Note that this is precisely how excited states were constructed as boosted tangent vectors on the manifold of MPS in section 2.5. The general n-magnon state can be found in refs. [29, 89, 90], and involves an MPS with a bond dimension that scales exponentially in the number of particles in the system.

4.2 The Yang-Baxter equation as the fundamental theorem of MPS

There is however a completely different way of deriving the exact eigenstates in terms of the algebraic Bethe ansatz (ABA). Before the formulation of both approaches in terms of tensor networks, it was not clear how those two approaches were related to each other; it is now clear that both approaches are related by a non-trivial gauge transformation [91], as has to be because of the fundamental theorem of MPS. In the algebraic Bethe ansatz, all eigenstates of the Hamiltonian can be constructed using the multiplication of MPOs that depend on a continuous *spectral parameter* μ_i :

$$(113)$$

Here, all MPO tensors are of the form

$$A_{\alpha\beta}^{ij}(\mu) = \mu \delta_{\alpha\beta} \delta_{ij} + \delta_{\alpha i} \delta_{\beta j} = \mu \alpha \begin{array}{c} i \\ | \\ \hline | \\ j \end{array} \beta + \alpha \begin{array}{c} i \\ | \\ \hline | \\ j \end{array} \beta. \quad (114)$$

It can easily be checked that every MPO adds one magnon to the system, as the left boundary is $|0\rangle$ and the right boundary $|1\rangle$. The coefficients μ_i are functions of the magnons' momenta, and can be determined by solving the so-called *Bethe equations*. A crucial ingredient of the ABA is the fact that all those MPOs commute with each other.

The underlying logic is as follows. We start from a transfer matrix of a two-dimensional statistical mechanical model that can be written in the form of an injective MPO $T(\mu)$ determined by the same tensor $A(\mu)$ as in equation 114, but this time with periodic boundary conditions. It turns out that this system will be integrable if and only if all those MPOs commute with each other independent of their spectral parameter:

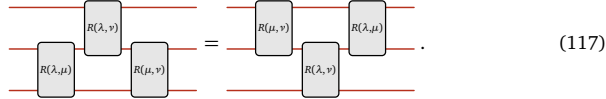
$$\forall \lambda, \mu : [T(\lambda), T(\mu)] = 0. \quad (115)$$

If this is the case, we can invoke the fundamental theorem of MPS eq. (79) to prove the existence of a gauge transform R that transforms those products into each other [29]. Indeed, note that we can merge together two legs of an MPO to form a single leg, so that the fundamental theorem of MPS can be used for rank-4 MPOs. Concretely, there is an intertwiner tensor $R(\lambda, \mu)$ obeying the equation

$$(116)$$

This R -tensor is a local implementation of the global symmetries given by the commuting MPOs.

Now let us take three copies of the R -tensor. Due to the associativity of the MPO algebra, these R -tensors can be fused together in two different ways, giving rise to the following equation:



This is precisely the Yang-Baxter equation – hence the fundamental theorem of MPS reproduces this result in a very concise way. Given any integrable transfer matrix, the corresponding intertwiners have therefore to satisfy the Yang-Baxter equations. It can be checked that the tensor in eq. (114) is precisely one of those solutions. A constructive way of finding integrable models is henceforth to turn the tables around: start with finding a solution of the Yang-Baxter equations (which is typically very hard!), leading to explicit representations of the tensors $R(\lambda, \mu)$. As a next step, solve for the transfer matrix T as a solution of eq. (116). What is surprising is the fact that this equation is equivalent to the one of the Yang-Baxter equation by identifying the MPO tensors to be $A_{\alpha\beta}^{ij}(\mu) := R_{j\beta}^{i\alpha}(\mu, 0)$. This so-called *fundamental representation* of the Yang-Baxter algebra hence allows to construct a one-parameter family of commuting transfer matrices. Note that this is not the only family of MPOs solving eq. (116). One can construct different representations of the same algebra – later on, we will encounter an analogous situation where we will construct different representations of the same symmetric system. As a last step, it is now possible to construct the magnon MPOs of eq. (113) with the same tensors A , but with different boundary conditions.

The key principle and advantage of the tensor network formulation is thus as follows: global conservation laws and commutation relations can be written down as equations involving local degrees of freedom, i.e. MPS and MPO tensors, which are usually easier to solve.

Given the MPO formulation, it is also possible to recover the integrable Heisenberg Hamiltonian. The transfer matrix of the 6-vertex model with all weights equal to each other, the corresponding integrable statistical mechanical model, has exactly the form of eq. (114). By construction, all MPOs $T(\mu)$ with those tensors $A(\mu)$ will commute with each other, so the derivative of this MPO is defined univocally. The Heisenberg Hamiltonian is indeed the logarithmic derivative of this transfer matrix; we leave the proof as a (very interesting) exercise (see [90] for an elaborate discussion):

$$H = T^{-1}(0) \frac{dT(\mu)}{d\mu} \Big|_{\mu=0} = \sum_i \cdots \left| \begin{array}{c} \text{diagram of a vertex with four legs} \end{array} \right| \cdots \quad (118)$$

In conclusion, both Bethe ansatz formulations (coordinate and algebraic) have a natural representation in terms of tensor networks. Tensor networks and Bethe ansatz clearly share the same DNA, but tensor networks turn out to be more generally applicable – they work equally fine for non-integrable systems and higher dimensions.

4.3 Projected entangled-pair states

The natural generalisation of matrix product states to two dimensions are the so-called *projected entangled-pair states* (PEPS) [92]. Similar to the one-dimensional case, a translation invariant PEPS is constructed from a single tensor, which in the case of the square lattice has

one physical and four virtual indices:

$$A = \sum_{i, \alpha, \beta} A_{\alpha\beta\gamma\delta}^i |i\rangle \langle \alpha| \otimes \langle \beta| \otimes \langle \gamma| \otimes \langle \delta| \in \mathbb{C}^{d \times d^4}. \quad (119)$$

Such a tensor is then depicted by

$$\begin{array}{c} \delta \\ | \\ \textcircled{A} \\ | \\ \gamma \end{array} \begin{array}{c} i \\ / \\ \beta \end{array} = A_{\alpha\beta\gamma\delta}^i, \quad \begin{array}{l} i=1,2,\dots,d \\ \alpha,\beta,\gamma,\delta=1,2,\dots,\chi \end{array} \quad (120)$$

so that the full PEPS wave function can be represented by

$$|\Psi[A]\rangle = \begin{array}{c} \begin{array}{ccc} \textcircled{A} & \textcircled{A} & \textcircled{A} \\ | & / & \backslash \\ \textcircled{A} & \textcircled{A} & \textcircled{A} \\ | & / & \backslash \\ \textcircled{A} & \textcircled{A} & \textcircled{A} \end{array} \end{array}. \quad (121)$$

Note that PEPS naturally satisfy an area law since the entanglement entropy of a contiguous region of spins scales linearly with the number of bonds on the boundary of the region, and its perimeter. Therefore PEPS are expected to form a good variational class for low-energy states of local gapped Hamiltonians, which have since long been conjectured to satisfy an area law. Conversely, as for MPS, it can be shown that every PEPS is the exact ground state of a local frustration free gapped parent Hamiltonian. It is also obvious how to generalise this construction to open boundary conditions, arbitrary lattices or to higher dimensions. Similarly, one can generalise MPOs to the two-dimensional setting. Such operators are referred to as *projected entangled-pair operators* (PEPOs).

Considerable effort has been made to generalise DMRG and VUMPS algorithms to the case of PEPS. Although it is in principle straightforward to generalise the manifold picture to the case of PEPS, the actual computational scaling with the bond dimension is much worse than in the 1D case. Nevertheless those PEPS algorithms are providing state of the art variational results for strongly correlated lattice systems in 2D such as for the Hubbard model, and it is a very active research field to come up with better algorithms. We refer to refs. [1, 93–96] for state of the art results. In these lectures, we will focus on the theory of PEPS.

Contrary to the case of matrix product states, there exists no completely general fundamental theorem yet that provides necessary and sufficient conditions for two distinct PEPS tensors A and B to generate the same state for all system sizes. However, it is clear that a sufficient condition for this to hold is the existence of a matrix product operator that *intertwines* the PEPS tensors according to

$$\begin{array}{c} \text{---} \textcircled{A} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \textcircled{B} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \end{array} = \begin{array}{c} \text{---} \textcircled{B} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \textcircled{A} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \end{array}, \quad (122)$$

and all other ways of pulling the MPO through, such as:

$$\begin{array}{c} \text{---} \textcircled{A} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \textcircled{B} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \end{array} = \begin{array}{c} \text{---} \textcircled{B} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \textcircled{A} \text{---} \\ | \quad / \quad \backslash \\ \text{---} \end{array}. \quad (123)$$

In the absence of a proof, we call this the fundamental *conjecture* of PEPS. Assuming periodic boundary conditions, the symmetry action is then implemented by starting from the PEPS $|\Psi[A]\rangle$, nucleating a small bubble of the intertwining MPO, pulling it through the entire lattice and annihilating it with itself so as to end up with the PEPS $|\Psi[B]\rangle$. As we will see in the remainder of these lectures, large classes of PEPS implement their symmetries exactly according to this prescription. In some cases, this MPO is just a tensor product (bond dimension 1), so that the virtual symmetry simply acts as a tensor product on the virtual legs.

In the following sections, we will discuss some particularly interesting PEPS wave functions which represent topological phases of matter. The fact that topological wave functions can easily be constructed using the tensor network formalism is clearly one of the main reasons that they are studied so extensively. Additionally, those tensor network representations identify the long-range entanglement with (MPO) symmetries acting on the virtual degrees of freedom. This is the reason that this fundamental conjecture is so relevant: topological order is characterised by non-trivial MPO symmetries.

4.4 Two-dimensional SPTs: a case study

A paradigmatic example of a PEPS with non-trivial MPO symmetry is the so-called *CZX model* introduced by Chen, Liu and Wen in ref. [76] (see [97–99] for its tensor network representation). Their model is an example of a non-trivial SPT protected by a $\mathbb{Z}_2$ symmetry which acts in an on-site manner $U_g^{\otimes N}$ on the physical level of the PEPS. At the virtual level of the PEPS tensors, this symmetry is implemented via a bond dimension 2 MPO as:

$$(124)$$

for $g \in \{0, 1\}$. The MPOs indexed by g form a non-local representation of $\mathbb{Z}_2$, which graphically can be represented by

$$(125)$$

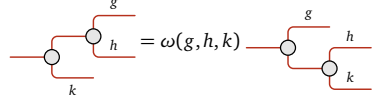
The multiplication eq. (125) can be implemented locally by means of an intertwining tensor satisfying⁷ [76, 98, 100]:

$$(126)$$

Recall that in section 3.3 we argued that in the case of one-dimensional SPT phases, there exists a topological index that takes values in the second cohomology group $H^2(G, \text{U}(1))$. This index is fully captured by the multiplication rules of the virtual matrices which form a projective representation of the symmetry. It is now natural to wonder what the corresponding topological index for two-dimensional SPT phases is and how it is encoded in the MPO algebra. As it turns out, if you consider the multiplication of three MPOs representing the $\mathbb{Z}_2$ symmetry,

⁷The MPO for $g = 0$ is not injective, but can not be decomposed in injective blocks. This can be explained via the underlying algebraic structure, which is not semisimple [100, 101]. As such, the intertwiners cannot be found directly by invoking the fundamental theorem.

then the fusion tensors defined above can be contracted in two different ways. These two ways have to be compatible, so that there must exist a phase ω depending on three group elements satisfying



$$(127)$$

Associativity of the matrix multiplication reveals that ω has to satisfy

$$\omega(g, h, k)\omega(g, hk, l)\omega(h, k, l) = \omega(gh, k, l)\omega(g, h, kl). \quad (128)$$

This condition is known as the *3-cocycle equation* (see also section A). Note that we can freely multiply the fusion tensors with a phase $\varphi(g, h)$ and that under this redefinition the 3-cocycle changes according to the prescription

$$\omega(g, h, k) \mapsto \omega(g, h, k) \frac{\varphi(g, h)\varphi(gh, k)}{\varphi(g, hk)\varphi(h, k)}. \quad (129)$$

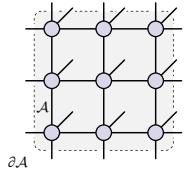
Modding out such redefinitions, we can conclude that the index ω takes values in the third cohomology group $H^3(G, U(1))$. This group can be computed by the methods laid out in section A, and is in particular also finite. For the case of $G = \mathbb{Z}_2$ considered above, we have that $H^3(\mathbb{Z}_2, U(1)) \simeq \mathbb{Z}_2$ so that there exists exactly one non-trivial SPT phase protected by $\mathbb{Z}_2$ symmetry. A representative 3-cocycle for the corresponding non-trivial class reads

$$\omega(g, h, k) = (-1)^{ghk}, \quad (130)$$

which can be immediately deduced from eq. (126) for the fusion tensors of the CZX model. Just as in the case of quantum spin chains, two PEPS wave functions with different virtual symmetries/3-cocycles cannot be converted into each other via symmetry-preserving adiabatic paths without closing the gap.

4.5 Edge Hamiltonians & anomalies

A noteworthy feature of PEPS emerges when they are defined on an open patch $\mathcal{A}$. In that case, the uncontracted bonds define a Hilbert space of dimension $\chi^{|\partial\mathcal{A}|}$ where $|\partial\mathcal{A}|$ is the number of links intersecting the boundary. Graphically:



$$(131)$$

These edge degrees of freedom define a tensor product structure for the edge modes of the corresponding parent Hamiltonian, completely in line with how we identified the SPT edge modes in the one-dimensional case in section 3.3. When the PEPS parent Hamiltonian is subsequently perturbed, this induces an *effective* Hamiltonian on these boundary degrees of freedom, potentially driving them through a phase transition [102]. Interestingly, in the case of PEPS representations of SPTs, these effective boundary Hamiltonians inherit the non-local MPO symmetries of the bulk. We refer to refs. [98, 103] for some examples and their numerical investigation.

Provided an MPO representation of a group raises in turn the question what the phase diagram of a symmetric (edge) Hamiltonian can look like. As we will now demonstrate, the answer to this question depend strongly on whether the 3-cocycle associated to the MPO algebra is trivial or not. As demonstrated in ref. [5, 76] a very elegant and straightforward tensor network argument shows that whenever ω is cohomologically non-trivial, the Hamiltonians that commute with it must either break the symmetry spontaneously or be gapless. The cocycle ω is in that case said to represent the *anomaly* of the symmetry. This observation closely parallels the celebrated Lieb–Schultz–Mattis theorems [63, 104, 105]. The argument goes as follows.

In the argument we assume an *injective* MPO representation of an arbitrary group G . Due to the injectivity we can now invoke the fundamental theorem to proof the existence of fusion tensors implementing the group multiplication locally, as in eq. (126). Assume now that the 3-cocycle defined as per eq. (127) is non-trivial and let us assume the existence of a unique trivially gapped symmetric ground state, represented by an injective MPS. Graphically this boils down to:

$$\text{Diagram (132)} \quad (132)$$

An application of the fundamental theorem of MPS then reveals that there must exist a trivalent *action tensor* (in gray) which satisfies:

$$\text{Diagram (133)} \quad (133)$$

Similar as in the previous section, if we act with two symmetry MPOs on the state, the fusion of the MPO tensors followed by the action on the MPS is up to a phase equal to sequentially acting with the MPOs. Concretely:

$$\text{Diagram (134)} \quad (134)$$

If we now consider the action of three such MPOs on our MPS, and consider the two distinct ways in which we can locally recouple these, we can deduce the following condition

$$\frac{\lambda(g, h)\lambda(gh, k)}{\lambda(h, k)\lambda(g, hk)} = \omega(g, h, k), \quad (135)$$

but the left-hand side can be recognized as a coboundary, which would in turn imply that the 3-cocycle is cohomologically trivial. Since this is in contradiction with the assumption that ω is non-trivial, we must therefore conclude that there is no symmetric injective MPS, and that every MPO symmetric 1-D Hamiltonian must be symmetry broken or gapless. Note that this argument is true independent of whether the Hamiltonian encodes the edge modes of a 2-D SPT phase or of a genuine quantum spin chain.

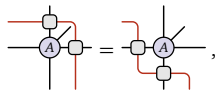
4.6 Topological order: quantum doubles & the tube algebra

Intriguingly, beside SPT phases, there exist in spatial dimensions higher than one robust phases of quantum matter which are not protected by any symmetry. Such phases are said to exhibit *intrinsic topological order* [106–109] which is characterised by a ground state degeneracy that

only depends on the topology of the underlying spatial manifold (sphere, torus,...), gapped excitations with *anyonic* braiding statistics and, depending on its anyon content, a universal negative correction to the area law of entanglement entropy [110–112]. Notably, intimately connected to the absence of symmetry, these phases are not characterised by a local order parameter, and thus fall outside the scope of the traditional Landau paradigm of symmetry breaking [71, 73, 113]. Nevertheless, the fact that there is a ground state degeneracy clearly indicates that some symmetry is broken. It turns out that this symmetry is one that acts purely on the virtual degrees of freedom.

Recall that short-range entangled SPT order gets its stability from the existence of a topological index associated to the transformation properties of the entanglement degrees of freedom under the global symmetry (section 3.3 and 4.4). The robustness of intrinsic topological order on the other hand stems from its long-range entanglement [75]. In a sense, characterising topological orders thus boils down to the classification of equivalence classes of long-range entanglement. Similarly, phase boundaries between these topological phases can be understood from a radical change in the entanglement properties [114, 115]. Having direct access to these entanglement degrees of freedom, as we have in PEPS, suggests that we could further reduce this problem to classifying *local* PEPS tensors that give rise to *global* topological states: the classification of intrinsically topologically ordered systems is equivalent to the construction of MPO-symmetric tensors.

Indeed, in the language of PEPS, the presence of topological order can be traced back to the existence of *virtual* MPO symmetries that commute with the ground state PEPS tensors as:



$$(136)$$

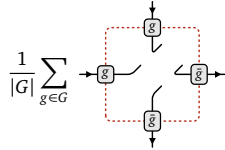
Note that, as opposed to the SPT case, there is no action on the physical degrees of freedom. Consequently, these MPOs are completely deformable – or thus *topological* – on the virtual level. This also implies that the underlying symmetry structure can be much richer than the ones described by groups. As we will see in lecture 5, the symmetry algebra is encoded by a *fusion category*. Since the MPOs are blind to the physical degrees of freedom, such states are resilient to (small) local perturbations on the physical level [115]. However, *anyon condensation* provides a mechanism to drive the system to a trivial phase under sufficiently strong perturbations [116, 117].

It is conjectured that a non-chiral topological order is fully characterised by its anyon content, together with its fusion rules and braiding statistics. Mathematically, this data is encapsulated in a *modular tensor category*, called the *Drinfeld centre* $\mathcal{Z}(\mathcal{D})$ [118–121]. The full extent of this statement, and the microscopic realisations of generic topological orders will be the topic of section 5.3. In this section we will rather focus on the construction of a class of topological orders called (untwisted) *quantum doubles* [110, 122, 123], as a more gentle introduction to PEPS representations of topological order and some of its properties.

The quantum doubles as introduced by Kitaev in ref. [110] are a family of models constructed from a finite group G by assigning $|G|$ -dimensional qudits to the edges of a two-dimensional lattice, subject to a commuting projector Hamiltonian spelled out in the aforementioned reference. They are a lattice Hamiltonian realisation of untwisted Dijkgraaf-Witten topological field theory [122]. For the group $\mathbb{Z}_2$, the model reduces to the *toric code*. The ground states of these quantum double models admit notably simple renormalisation fixed point PEPS representations called *G-injective*, or more precisely *G-isometric*, PEPS. The notion of *G-isometry* essentially captures PEPS tensors that are invertible only on a subspace of the virtual degrees

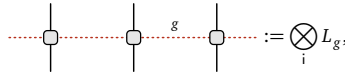
of freedom which is invariant under an action of the group G , and is a particular incarnation of MPO injectivity.

Up to local physical transformations all G -isometric PEPS reduce to a universal form that we will now describe. In this section, we assume a square lattice on a two-dimensional torus. We write L to denote the *left regular representation* of G , acting as $L_g |h\rangle = |gh\rangle$. Define then the projector $1/|G| \sum_g L_g \otimes L_g \otimes L_g^\dagger \otimes L_g^\dagger$, which constitutes the ground state PEPS tensor, graphically:



$$\frac{1}{|G|} \sum_{g \in G} \quad (137)$$

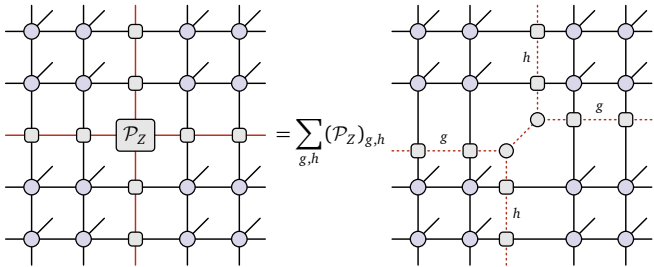
In this figure, the gray boxes represent L_g , and the bar denotes the inverse group element (recall that $L_g^\dagger = L_{\bar{g}}$). Note that the four innermost legs together form the physical level. As the sum in eq. 137 runs over all group elements, this PEPS tensor satisfies the pulling through conditions eq. (136) for the following MPOs labelled by group elements:



$$:= \bigotimes_i L_g, \quad (138)$$

where the dotted line is used to emphasise that the bond dimension is 1, i.e. each MPO acts as a tensor product.

The PEPS tensor eq. (137) gives rise to only one of the ground states of the model. As demonstrated in [124], all the other ones can be obtained unitarily from this one by acting with operators that wrap around either or both of the non-contractible cycles of the torus. The extensive nature of these operators are reflective of the local indistinguishability of the ground states. Equivalently, one can obtain all the ground states by wrapping both cycles of the torus with MPOs on the virtual level and joining them on their intersection by a tensor $\mathcal{P}_Z$ that we need to solve for. In this, we require that the ground states are all translationally invariant and remain normalisable and orthogonal in the thermodynamic limit [125]. Using the fusion tensors for the MPOs eq. (138) we can expand such a state as follows:⁸



$$= \sum_{g,h} (\mathcal{P}_Z)_{g,h} \quad (139)$$

The translation invariance of these states is manifest by the pulling through condition. As demonstrated in refs. [120, 126, 127] a basis of ground states in MPO injective PEPS is in one-to-one correspondence with *minimal central idempotents* or – equivalently – irreducible representations of *Ozawa's tube algebra* [128, 129]. In fact, the Drinfeld centre $\mathcal{Z}(\mathcal{D})$ is in

⁸Note that the fusion tensors in this case simply enforce the group multiplication, i.e. $\sum_{g,h,k} |k\rangle \langle g, h| = \sum_{g,h} |gh\rangle \langle g, h|$.

practice often computed via the tube algebra. For the particular case of quantum doubles, the relevant tube algebra is spanned by closed MPOs of the form:

$$T_x^g := \text{diagram}, \quad (140)$$

thought of as acting from the outside indices to the inside. Their multiplication is defined by stacking them inside each other. Algebraically this boils down to:

$$T_x^g \circ T_y^h = \delta_{x,hyh^{-1}} T_x^{gh}. \quad (141)$$

This multiplication defines an algebra often called the (*untwisted*) *quantum double* $\mathcal{D}(G)$ of the group G [123]. Notably, the tube algebra is endowed with a *star*, which is an antilinear involution that swaps the order of the product. For the case at hand: $(T_x^g)^* := T_{gxg^{-1}}^{g^{-1}}$. As such, the tube algebra is endowed with the structure of a C^* -algebra. Due to a famous result of Artin and Wedderburn, every C^* -algebra can be decomposed in simple matrix algebras. The aforementioned minimal central idempotents then act as projectors on each of these simple blocks.

Let us compute them explicitly. To this end it is useful to focus first, for fixed $x \in G$, on all tubes T_x^g where $g \in Z_x = \{g \in G | gx = xg\} \subseteq G$. These tubes form a representation of the stabiliser group Z_x since

$$T_x^{g_1} \circ T_x^{g_2} = T_x^{g_1 g_2}, \quad \forall g_1, g_2 \in Z_x, \quad (142)$$

as follows directly from eq. (141). Let μ_x denote an irreducible representation of Z_x of dimension d_{μ_x} and with χ^{μ_x} its corresponding character. The operators

$$\mathcal{P}_{(x,\mu)} := \frac{d_{\mu_x}}{|G|} \sum_{k \in Z_x} \chi^{\mu_x}(k) T_x^k, \quad (143)$$

form then hermitian projectors i.e.⁹

$$\mathcal{P}_{(x,\mu_x)} \mathcal{P}_{(y,\nu_y)} = \delta_{x,y} \delta_{\mu_x, \nu_x} \mathcal{P}_{(x,\mu_x)}. \quad (144)$$

These $\mathcal{P}_{(x,\mu)}$ are *simple* idempotents. The central ones are then obtained by additionally summing over all x in a conjugacy class $C_x = \{gxg^{-1} | g \in G\}$:

$$\mathcal{P}_{(C_x, \mu_x)} \equiv \sum_{y \in C_x} \mathcal{P}_{(y, \mu_x)}. \quad (145)$$

This reproduces the results of refs. [111] that torus ground states of the quantum doubles are labelled by a conjugacy class C_x and an irreducible representation of the stabiliser subgroup Z_x of a representative x of C_x .

Importantly, also anyonic excitations are labelled by minimal central idempotents of the tube algebra [130, 131]. At the level of the PEPS, one can create pairs of anyonic excitations by replacing two of the ground state tensors by tensors whose support is contained in the image of the corresponding idempotent labelling the excitation, we refer to [132] for details.

⁹Use Schur orthogonality of the characters $\sum_k \chi^{\mu_x}(k) \chi^{\nu_x}(k^{-1}m) = |Z_x|/d_{\mu} \chi^{\mu}(m)$.

The (untwisted) quantum doubles considered here are generalised in a number of ways. For one, one could consider a *twisted* version of the model where the MPO symmetries form an anomalous representation of G with the anomaly taking values in the third cohomology group $H^3(G, \text{U}(1))$, as described in section 3.3. For this, we refer to refs. [82, 97, 127].

In section 5.3 we will discuss the PEPS representations of so-called string-net models constructed first by Levin and Wen in ref. [133]. These are a class of models that have a spherical fusion category as input, and realise all non-chiral topological orders $\mathcal{Z}(\mathcal{D})$. One way of obtaining the torus ground states and anyonic excitations described by $\mathcal{Z}(\mathcal{D})$ in these models is exactly via the tube algebra associated to $\mathcal{D}$, completely in line with the above construction. For the input $\mathcal{D} = \text{Vec}_G^\omega$ the corresponding Levin-Wen model coincides with the twisted quantum doubles.

4.7 Synopsis of lecture IV

Lecture IV introduced three new topics: symmetries in the form of MPOs which underlie the integrability of special classes of quantum spin chains; the higher dimensional generalisation of MPS to PEPS, and the corresponding manifold which parametrises low-energy states of gapped higher dimensional lattice models; and MPO symmetries that satisfy a so-called *pulling through symmetry* which characterises the different gapped (topological) phases of such systems. Concerning integrability, we demonstrated that the Yang-Baxter equations are a consequence of the fundamental theorem of MPS. It would be wonderful if this point of view leads to more general classes of integrable models, such as the ones satisfying non-trivial categorical symmetries (see lecture V).

We showed that SPT phases in 2D lattice systems are characterised by non-trivial MPO symmetries whose associators encode 3-cocycles, leading to the classification of 2D SPT phases. However, the pulling through symmetries of PEPS open up completely new ways of realising topological phases of matter in 2D systems if no global symmetries are involved. We demonstrated the existence of intrinsic topologically ordered lattice systems in the form of quantum doubles, and showed that those systems exhibit symmetry breaking with respect to the entanglement degrees of freedom. This was the first hint that the Landau paradigm can be saved if you describe many-body phases of matter in terms of the symmetries of the underlying PEPS representation: different phases realise the symmetries of the tensors in a fundamentally different way.

5 Lecture V: Generalised symmetries

5.1 Motivation

In lecture 3 we classified phase diagrams of one-dimensional quantum lattice models with a group symmetry. We also saw that in one-dimensional bosonic systems the presence of a symmetry is *necessary* to have a non-trivial phase diagram. Recall from section 3.5 that the story was more subtle for the case of fermions, which can exhibit intrinsic topological order due to a superselection rule. In recent years, substantial research effort has been made to understand systems with symmetries beyond groups [134–144]. As we foreshadowed in the introduction to lecture 3, generalised symmetries described by fusion categories are encountered as the *virtual symmetries* of non-chiral two-dimensional topological orders and therefore also arise as the symmetries of their boundary Hamiltonians or – equivalently – the symmetries of the effective theories describing their edge modes. Studying how those generalised symmetries influence the phase diagrams of systems has led to a *generalised Landau paradigm*, where different phases of matter were found to be in correspondence with different ways to break a generalised symmetry [9, 53, 135, 139, 144]. With non-group symmetries, one cannot simply say that a symmetry is broken to a subgroup, so we should ask: how can a system (partially) break a generalised symmetry? Is there a map between systems exhibiting different ways of generalised symmetry breaking? These questions will all be expanded upon in this section.

Why are tensor networks a good description for such categorical symmetries? A crucial observation is that to encode symmetries beyond on-site, we need operators that act in a correlated way on neighbouring degrees of freedom. Additionally, we want to be able to define these operators on systems of any size, so we need to encode those correlated symmetries locally. MPOs are exactly the right language to use for this purpose. They are parametrised in terms of a single local tensor, can be defined on any number of sites, and share information between neighbouring physical degrees of freedom through their virtual degrees of freedom. Because of these considerations, we will often use the terms *generalised*, *categorical* and *MPO symmetry* interchangeably.

Indeed, the study of general MPO symmetries revealed that, under mild conditions such as injectivity (discussed further in section 5.2), their underlying structure is equivalent to that of fusion categories. In brief, fusion categories are semisimple associative structures of (simple) objects, very much like a group is an associative multiplication rule of its elements (its simple objects). However, unlike the group case, the multiplication of simple objects of a fusion category may lead to a direct sum of simple objects. A canonical example being the fusion category $\text{Rep}(G)$ whose simple objects are the irreps of a group and whose multiplication rule is dictated by the direct sum decomposition of the tensor product of irreps, e.g. $1/2 \otimes 1/2 = 0 \oplus 1$. Fusion category theory is a very well studied topic in mathematics and mathematical physics, and MPOs turn out to be the building blocks of *representations* of those fusion categories on the lattice, much like matrices represent on-site group symmetries in spin systems.

In this final lecture we will first study the MPO representation theory of categorical symmetries and demonstrate how *module categories* label the distinct representations. We will then provide three concrete applications. To start, we will see how these MPO symmetries arise in the Levin-Wen string-net models, which provide microscopic realisations of all non-chiral two-dimensional topological orders. We then provide two approaches to construct *edge theories* of these string-net models. In section 5.4, we will demonstrate how so-called *strange correlators* of these PEPS representations lead to the construction of two-dimensional *classical* partition functions of critical models [145], generalizing a construction of Aasen, Mong and Fendley [146]. The main merit of this strange correlator construction is that it makes the distinction between *topological* and *geometrical* or *dynamical* features of the critical lattice model

manifest: the (2+0)d partition functions inherit the MPO symmetries from the (2+1)d string-nets, and as such provides the lattice analogue of TFT/CFT correspondence. This holographic approach to the construction of edge theories with categorical symmetries can be interpreted as an explicit lattice realisation of the categorical construction of Kong, Wen and Zheng [147] which provides a concrete foundation for what is now widely referred to as *SymTFT* or *topological holography* [9, 140, 148–150]. Finally, we will provide a systematic way to construct local (1+1)d MPO symmetric Hamiltonians and a generalised duality or gauging procedure for generalised symmetries. We conclude by discussing the anticipated generalised Landau paradigm for categorical symmetries, as well as some of its implications.

5.2 MPO representation theory

In this section we will follow a logic similar to the one in section 4.2. Assuming a collection of injective MPOs representing a fusion algebra, we deduce the consistency equations satisfied by the fusion tensors. We subsequently show how solutions of these equations can be used to construct representations of these MPOs and fusion tensors. It is demonstrated how distinct representations are constructed from the data encoded in *module categories* over the fusion category describing the symmetry.

Concretely, assume a finite collection of injective MPOs labelled by $a, b, c, \dots$ with periodic boundary conditions. We are interested in MPO algebras for which the structure constants N_{ab}^c are independent of the number of sites in the periodic chain:

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ a \quad b \quad c \end{array} = \sum_c N_{ab}^c \begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ c \quad c \quad c \end{array} . \quad (146)$$

Here and in what follows, the sums are over all distinct MPOs appearing in the algebra. Amongst them, there exists one labelled $\mathbb{1}$ for which $N_{a\mathbb{1}}^{a'} = N_{\mathbb{1}a}^{a'} = \delta_{a,a'}$. The non-negative integer tensor N defines a *unital associative fusion* structure, meaning in particular that the structure constants satisfy $\sum_d N_{ab}^d N_{dc}^e = \sum_d N_{bc}^d N_{ad}^e$. Exploiting the fundamental theorem eq. (79), we find that eq. (146) can be implemented locally by means of pulling the following *fusion tensors* through the MPO tensors:

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ c \quad a \quad b \end{array} = \begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ c \quad a \quad b \end{array} , \quad (147)$$

where the multiplicity $i = 1, 2, \dots, N_{ab}^c$ labels the linearly independent fusion tensors. Similarly as in section 4.4, associativity of the MPO multiplication implies the existence of a collection of *unitary* matrices called *F-symbols* or *associators* encoding the basis transformation

$$\begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ d \quad e \quad a \\ \quad \quad \quad b \\ \quad \quad \quad c \end{array} = \sum_{f,k,l} (F_d^{abc})_{e,ij}^{f,kl} \begin{array}{c} \text{---} \square \text{---} \square \text{---} \square \text{---} \\ | \quad | \quad | \\ d \quad a \quad b \\ \quad \quad \quad f \\ \quad \quad \quad c \end{array} . \quad (148)$$

The requirement that the product of four MPOs – which can be recoupled in two distinct ways – is associative, reveals that these *F-symbols* obey a consistency condition colloquially known

as the *pentagon equation*. Schematically, the pentagon equation is a linear multivariate system of equations of the form

$$FF = \sum_i FFF, \quad (149)$$

or, written out in components:

$$\sum_0 (F_e^{cd})_{g,lm}^{h,no} (F_e^{ab})_{f,ko}^{i,pq} = \sum_{i,rs} (F_g^{abc})_{f,kl}^{j,rs} (F_e^{ajd})_{g,sm}^{i,tq} (F_i^{bcd})_{j,rt}^{h,np}. \quad (150)$$

In essence, the fusion rules specified by the N -symbol, together with a unitary solution of the pentagon equation constitute the data encoded in a *unitary fusion category* $\mathcal{C}$.¹⁰ Importantly, for any valid fusion structure, the number of solutions to the pentagon equation is finite, up to basis transformations on the multiplicity indices.

Given a solution to the pentagon equation, how can we find a corresponding MPO representation in the sense of eq. (146)? Completely in line with the logic of section 4.2, we can find a representation by equating the F -symbols to (the components of) the fusion tensors defined in eq. (147) as well as to the MPO tensors. In that case, both eq. (148) as well as eq. (147) are equivalent to the pentagon equation (150) [120, 126, 127, 151]. Concretely, the assignment of non-zero components is as follows¹¹:

$$\begin{array}{c}
 (abe,i) \quad (ecd,j) \\
 \diagdown \quad \diagup \\
 \text{---} \bigcirc \text{---} \\
 \diagup \quad \diagdown \\
 (bcf,k) \quad (afd,l)
 \end{array}
 = (F_d^{abc})_{e,ij}^{f,kl} = (eab,i) \text{---} \begin{array}{c} (ecd,j) \\ \text{---} \square \text{---} \\ (bcf,k) \end{array} \text{---} (daf,l) . \quad (151)$$

Note that every index is labelled by a triple of objects a, b, c and a multiplicity index taking values $1, 2, \dots, N_{ab}^c$. In particular, all triples simultaneously have to satisfy the fusion rules in order for a component to be non-zero. Moreover, the structure of these tensors is such that some of the objects are shared between indices. This internal structure of the tensors invites us to depict these tensors as *triple line tensors*:

[illegible]

Contraction of these tensors takes place via identification of the simple objects on strands of concatenated tensors, pairing indices, and summing over them as detailed in e.g. refs. [132, 152].

Due to the fusion rules, the MPOs act on a Hilbert space which is generically not a tensor product space, but rather a Hilbert space with local *constraints* dictated by the fusion rules. In this sense, the MPOs constructed here and generalised below, form a representation and generalisation of the topological symmetries considered in the context of *anyonic chains* [153–158]. The models considered in the original reference [153] for instance are constructed from the Fibonacci fusion category Fib (see section B).

¹⁰Note the resemblance of the pentagon equation with the 3-cocycle derived in eq. (129) by an analogous reasoning applied to group symmetry. And indeed, a finite group together with a choice of 3-cocycle defines a fusion category called Vec_G^ω , as reviewed in section B, for which the F -symbol coincides with ω .

¹¹We omit a conventional prefactor involving quantum dimensions [132, 152].

$$\triangleright F \triangleright F = \sum F \triangleright F \triangleright F. \quad (153)$$

Example: $\mathcal{C} = \text{Vec}_G$ Formally, a (non-anomalous) finite-group symmetry is described by the fusion category Vec_G of G -graded vector spaces. The simple objects correspond to group elements and the fusion rules are given by the group multiplication, i.e. $N_{g,h}^k = \delta_{gh,k}$. Module categories over Vec_G are in one-to-one correspondence with pairs (H, ψ) where $H \subseteq G$ is a subgroup up to conjugacy and ψ is a $\text{U}(1)$ -valued 2-cocycle of the subgroup H . We refer to section B for details. Let us focus on the case where $H = G$ and ψ is trivial. The corresponding module category, denoted Vec , then results in MPOs

$$\text{Diagram with three vertical lines and a horizontal dashed line} \stackrel{g}{=} \bigotimes_i u_i(g). \quad (154)$$

Note that in the case of on-site group symmetry eq. (154), one can bring all MPO tensors $u_i(\mathbf{g})$ – indexed by the group elements $\mathbf{g} \in G$ – simultaneously in block-diagonal form. The blocks correspond to the irreducible representations contained in the representations u_i . More generally, we could consider the collection of MPO tensors as operator-valued matrices labelled by the blocks $a, b, c, \dots$, and block-decompose the algebra spanned by them in the *vertical* direction. The crucial insight is now that the labels of these injective blocks, and their fusion

rules, i.e.¹²

$$\begin{array}{c} \vdots \\ \text{---} \square \text{---} \\ \alpha \\ \text{---} \square \text{---} \\ \beta \\ \text{---} \square \text{---} \\ \vdots \end{array} = \sum_{\gamma} N_{\alpha\beta}^{\gamma} \begin{array}{c} \vdots \\ \text{---} \square \text{---} \\ \gamma \\ \text{---} \square \text{---} \\ \vdots \end{array}, \quad (155)$$

are governed by another fusion category $\mathcal{D}$, which is generically *distinct* from the symmetry category $\mathcal{C}$. Its simple objects will be labelled by Greek letters $\alpha, \beta, \gamma, \dots$. Given $\mathcal{C}$ and $\mathcal{R}$, $\mathcal{D}$ is fully specified. In other words, fixing the symmetry category and a way to represent it on the lattice via the chosen module category $\mathcal{R}$, $\mathcal{D}$ is determined. Technically, $\mathcal{D}$ corresponds to the *Morita dual* (see also section B) of $\mathcal{C}$ with respect to $\mathcal{R}$, denoted by $\mathcal{D} = \mathcal{C}_{\mathcal{R}}^*$, and equivalently $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^*$. Morita equivalence of $\mathcal{C}$ and $\mathcal{D}$ guarantees in particular that the so-called *Drinfeld centres* of $\mathcal{C}$ and $\mathcal{D}$ are equivalent: $\mathcal{Z}(\mathcal{C}) \simeq \mathcal{Z}(\mathcal{D})$. In section 5.3 we will see that $\mathcal{Z}(\mathcal{D})$ encodes all information about the ground state sectors and anyonic excitations of the string-net with topological virtual symmetries described by $\mathcal{C}$. In section 5.4 and section 5.5, elements of the Drinfeld centre will label the *topological* or *superselection sectors* of the string-net boundary theories.

Being a fusion category, $\mathcal{D}$ comes with its own set of F -symbols that we shall write as ${}^{\circ}F$. These satisfy a pentagon equation analogous to eq. (149). The Morita dual $\mathcal{D}$ is constructed in such a way that $\mathcal{R}$ is also a module category over $\mathcal{D}$, with $\mathcal{D}$ now acting from the *right* on $\mathcal{R}$. As a consequence there are once more corresponding F -symbols, denoted ${}^{\triangleleft}F$, encoding the associativity of the $\mathcal{D}$ action on $\mathcal{R}$. These satisfy the fourth pentagon equation we encounter, now involving those of $\mathcal{D}$ as well:

$${}^{\triangleleft}F {}^{\triangleleft}F = \sum {}^{\circ}F {}^{\triangleleft}F {}^{\triangleleft}F. \quad (156)$$

Beyond being a module category for both $\mathcal{C}$ and $\mathcal{D}$, $\mathcal{R}$ is an (invertible) $(\mathcal{C}, \mathcal{D})$ -bimodule category (see section B). Associated to its bimodule structure is a final collection of F -symbols denoted by ${}^{\bowtie}F$ obeying

$${}^{\bowtie}F {}^{\bowtie}F = \sum {}^{\bowtie}F {}^{\bowtie}F {}^{\bowtie}F, \quad (157)$$

$${}^{\triangleleft}F {}^{\bowtie}F = \sum {}^{\bowtie}F {}^{\bowtie}F {}^{\triangleleft}F. \quad (158)$$

The components of the MPO tensors are equal to the ${}^{\bowtie}F$ -symbols, while the ${}^{\triangleleft}F$ -symbols are exactly what the fusion tensors in the *vertical* direction evaluate to. Gathering everything:

$$\begin{array}{c} \begin{array}{c} B \\ \diagup \quad \diagdown \\ i \quad a \quad b \quad k \\ \diagdown \quad \diagup \\ A \quad C \\ i \end{array} \equiv \begin{array}{c} (AaB, l) \quad (BbC, k) \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ (abc, i) \\ (Ac, j) \end{array} = ({}^{\bowtie}F_A^{abc})_{c, ij}^{B, kl}, \\ \\ \begin{array}{c} C \\ \diagup \quad \diagdown \\ i \quad a \quad \beta \quad j \\ \diagdown \quad \diagup \\ A \quad B \\ i \end{array} \equiv \begin{array}{c} (AaC, i) \quad (C\beta B, j) \\ \diagup \quad \diagdown \\ \bigcirc \\ \diagdown \quad \diagup \\ (a\beta\gamma, k) \\ (A\gamma B, j) \end{array} = ({}^{\triangleleft}F_B^{Aa\beta})_{C, ij}^{\gamma, kl}, \end{array} \quad (159)$$

¹²Note that these MPOs extend vertically – as graphically indicated by the dots – but not horizontally.

$$\begin{array}{c} \text{C} \\ | \\ \text{A} \end{array} \begin{array}{c} \text{D} \\ | \\ \text{B} \end{array} \begin{array}{c} i \\ | \\ a \\ | \\ l \end{array} \equiv \begin{array}{c} (E\alpha D, j) \\ | \\ \text{AaB, } i \\ | \\ \text{D}\alpha\text{B, } l \\ | \\ \text{A}\alpha\text{B, } k \end{array} = (F_D^{aA\alpha})_{C,ij}^{B,kl}.$$

Note that in these triple line tensors, objects in the module category always live on the outer loops. One can then verify that the vertical pulling through condition

$$\begin{array}{c} a \\ | \\ \alpha \end{array} \begin{array}{c} \beta \\ | \\ \gamma \end{array} = \begin{array}{c} \alpha \\ | \\ \gamma \end{array} \begin{array}{c} \beta \\ | \\ a \end{array} \quad (160)$$

is satisfied by virtue of eq. (158), while eq. (147) is equivalent to eq. (157). These fusion tensors in turn are related via a rotated version of the diagram eq. (148) involving these fusion tensors and the ${}^{\circ}F$ -symbol, equivalent to eq. (156).

In the example $\mathcal{C} = \text{Vec}_G$, $\mathcal{R} = \text{Vec}$ considered above, the fusion category $\mathcal{D}$ is $\text{Rep}(G)$, whose simple objects are the irreps of the group G . The corresponding ${}^{\circ}F$ -symbols are equal to the 6j-symbols of G , while the ${}^{\circ}F$ -symbols are equal to the Clebsch-Gordan coefficients. Those Clebsch-Gordan coefficients decompose the MPO tensors $u_i(g)$ in its irreducible blocks, and the data in these blocks is encoded in ${}^{\circ}F$, whose values are given by the irreps of G . In the general (non-group) case, the ${}^{\circ}F$ -symbols will below be interpreted as generalised Clebsch-Gordan coefficients for the MPO symmetries constructed here. At the same time, we will interpret eq. (160) as a generalised Wigner-Eckart theorem that states that MPO symmetric tensors are built from these generalised Clebsch-Gordan coefficients.

In conclusion, the following picture has emerged. Given a symmetry specified by a fusion category $\mathcal{C}$, an MPO representation is constructed from an invertible $(\mathcal{C}, \mathcal{D})$ -bimodule category $\mathcal{R}$. This structure encodes associator data in the form of five F -symbols ($F, {}^{\circ}F, {}^{\circ}F, {}^{\circ}F, {}^{\circ}F$) that solve six pentagon equations. The bimodule associator ${}^{\circ}F$ can be used to construct the injective MPO representation of the fusion category $\mathcal{C}$, whose tensors are locally fused horizontally eq. (147) and vertically eq. (160) via the tensors evaluating to ${}^{\circ}F$ and ${}^{\circ}F$ respectively. Both satisfy a recoupling identity involving F , respectively ${}^{\circ}F$.

5.3 Application 1: string-net ground states & intertwiners

String-net models were first constructed by Levin and Wen in ref. [133]. They constitute a class of RG fixed point lattice models that takes as an input a (spherical) fusion category, denoted by $\mathcal{D}$, and realises the non-chiral topological order given by its Drinfeld centre $\mathcal{Z}(\mathcal{D})$ [131, 159, 160]. By that, it is meant that the ground states and gapped excitations reproduce the topological sectors and anyonic statistics encoded in $\mathcal{Z}(\mathcal{D})$.¹³ The quantum doubles encountered in section 4.6 are reproduced for the choice $\mathcal{D} = \text{Vec}_G$. The name *string-net* stems from the interpretation of the model as describing a condensed phase of coloured strings, with the allowed string-types and their interactions encoded in $\mathcal{D}$. The Hamiltonian is a sum of local commuting projectors on each plaquette, allowing for a straightforward derivation of a PEPS representation of its ground states.

As shown in ref. [152], there exists a ground state PEPS representation for every module category $\mathcal{R}$ over the input $\mathcal{D}$. This observation generalises and unifies earlier references where

¹³Equivalently, they can be thought of as a Hamiltonian realisation of the Turaev–Viro–Barrett–Westbury topological field theory [159, 161, 162].

implicitly the regular module category was assumed [127, 132, 163–165], or in the specific case of the quantum doubles, Vec as module category over Vec_G [124]. From this point of view, a string-net is thus a PEPS with a particular type of *virtual* symmetry. The MPOs which form representations of the underlying Morita dual fusion category $\mathcal{C}$ are its symmetries, and they act purely on the virtual degrees of freedom of the tensor network. In accordance with the generalised Wigner-Eckart theorem discussed above, the PEPS tensors evaluate to ${}^{\mathcal{C}}F$ as in eq. (159), with the physical level taking values in $(\alpha\beta\gamma, k)$.

Akin to section 4.6, the different torus ground state sectors and anyonic excitations, both labelled by elements in $\mathcal{Z}(\mathcal{D})$, can be understood in terms of the *tube algebra*. The tube algebra is constructed from the MPOs in $\mathcal{C}$, and their central idempotents are in one-to-one correspondence with elements in $\mathcal{Z}(\mathcal{C})$ [127, 132]. Recall that $\mathcal{Z}(\mathcal{D})$ is guaranteed to be equivalent to $\mathcal{Z}(\mathcal{C})$ due to the invertibility of $\mathcal{R}$. Indeed, if we consider a string-net on a torus, the different topological sectors can be obtained by winding the MPO symmetries of the previous section around its uncontractable loops. In their intersection, a tensor is placed whose components are determined by the central idempotents. The insertion of a central idempotent associated to the object $Z \in \mathcal{Z}(\mathcal{C})$ in a string-net PEPS can be depicted as:

$$\sum_{\substack{a_\alpha, b_\beta, c_\gamma \\ i, j}} (\mathcal{P}_Z)^{i, j}_{a_\alpha, b_\beta, c_\gamma} \quad \text{(161)}$$

The diagram shows a 2D lattice of purple circular tensors connected by black lines. A central square tensor is highlighted with red lines extending from its four sides. These red lines are labeled a_α (horizontal) and b_β (vertical). A small red circle with the letter 'i' is placed at the intersection of these lines. The entire structure is part of a larger network of similar purple tensors and black lines.

Note that the purple tensors coincide with the fusion tensors of eq. (160), so that the exact location of the central idempotent is undetectable. Here, a_α stands for $a_\alpha \equiv (a, \alpha)$ where a is a simple object in $\mathcal{C}$ and α labels the degeneracy of a in $\mathcal{Z}$.

Given a PEPS representation based on $\mathcal{R}$, one can locally change this to one based on $\mathcal{R}'$ via a new class of MPO *intertwiners* that implement the change of module category. In particular, when $\mathcal{R}' = \mathcal{D}$, these MPO intertwiners are labelled by objects in $\mathcal{R}$ and are constructed from the module associator¹⁴. A change in module category hints at an equivalence between edge Hamiltonians for each of these distinct PEPS representations. Later, in section 5.5, we will see that the edge Hamiltonians corresponding to different representations $\mathcal{R}, \mathcal{R}'$ are related via *duality transformations*, or – equivalently – via gauging (part of) its MPO symmetries. The aforementioned MPO intertwiners play the role of intertwiners between these dual models and can be promoted to isometries, if boundary conditions are adequately accounted for.

Furthermore, the tensor network picture can also be used to explicitly construct *gapped boundaries* of string-net models. This is beyond the scope of these lecture notes and we refer to ref. [152] for details, but let us mention that also in that case module categories show up as labelling different boundary conditions. In conclusion, the complete picture of string-net

¹⁴In the general case, one requires the notion of $(\mathcal{D})$ -module functors between module categories, organised in a category denoted by $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$. Notably, $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ can be endowed with the structure of an invertible $(\mathcal{D}_{\mathcal{R}}, \mathcal{D}_{\mathcal{R}'})$ -bimodule category.

domain walls and boundaries, as developed by Kitaev and Kong in ref. [166], can be recast in tensor network language, making these structures amenable to simulating them on a computer.

5.4 Application 2: classical partition functions as strange correlators

With the tensor network representations of string-net ground states at our disposal, we can now discuss a second application of the general MPO formalism, the so-called *strange correlator* construction. This construction appeared first in ref. [167] as a diagnosis to detect non-trivial SPT phases by considering the matrix element $C(\mathbf{r}, \mathbf{r}') = \langle \Omega | \phi(\mathbf{r}) \phi(\mathbf{r}') | \Psi \rangle$ where $|\Psi\rangle$ is the state under scrutiny, ϕ is a local observable, and $|\Omega\rangle$ denotes a well-chosen short-range entangled trivial state. It was shown for a number of one and two-dimensional states $|\Psi\rangle$ which exhibit SPT order that the quantity $C(\mathbf{r}, \mathbf{r}')$ either decays algebraically in the distance $|\mathbf{r} - \mathbf{r}'|$ or converges to a constant. Hence, this method provides an identification of SPT order in terms of the bulk ground state wave function.

A natural generalisation of this construction amounts to choosing a state $|\Psi\rangle$ which exhibits intrinsic topological order. Focussing on the case of non-chiral topological order in two dimensions, we can thus construct strange correlators from the overlap of string-net ground states of section 5.3 with a suitable state $|\Omega\rangle$. This approach was pioneered in ref. [145]. As we have seen above, the PEPS representations of string-net ground states have exact MPO symmetries acting purely on the virtual level. Consequently, strange correlators constructed from these states inherit these symmetries, regardless of the chosen state $|\Omega\rangle$. However, for many interesting examples, $|\Omega\rangle$ can simply be chosen as a product state. It is then natural to interpret this two-dimensional tensor network as the *partition function* of a *classical statistical model*¹⁵ with topological defect lines/symmetries. Schematically for the case of $|\Omega\rangle$ being a (uniform) product state, the partition function thus takes the form

$$Z(\{\beta_i\}) := \langle \Omega | \{\beta_i\} | \Psi \rangle = \text{[Diagram 1]} = \text{[Diagram 2]}, \quad (162)$$

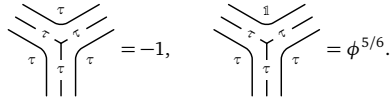
where $\{\beta_i\}_i$ collectively denotes the parameters defining the state $|\Omega\rangle$, and thus indirectly determine the Boltzmann weights of the classical model. For appropriate values of the parameters $\{\beta_i\}_i$, the partition function can in fact describe a critical model. In that case, the MPO symmetries can be interpreted as lattice remnants of the continuum topological defects of the CFT. The strange correlator construction thus provides an explicit lattice representation of the correspondence between conformal field theories and topological field theories as envisioned first by Witten [107], Moore and Seiberg [168] and subsequently developed further by Fröhlich, Fuchs, Runkel and Schweigert in refs. [169–173].

Example: hard hexagon model Perhaps the simplest example is obtained from the string-net model based on the Fibonacci fusion category on the hexagonal lattice. This fusion category contains two simple objects denoted by $\{1, \tau\}$, and the only non-trivial fusion rule reads $\tau \otimes \tau = 1 \oplus \tau$. The strange correlator we will consider is a product state that projects all the physical degrees of freedom of the string-net onto the object τ – or more precisely, in the

¹⁵Note however, that depending on the chosen $|\Omega\rangle$ the classical model may have non-positive Boltzmann weights and requiring these to be positive generically requires fine-tuning the state $|\Omega\rangle$.

notation of eq. (159), to $\phi \cdot (\tau\tau\tau, 1)$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. The degrees of freedom on the loops then take on the role of classical binary variables prone to the constraint that no two neighbouring loops can be in the $\mathbb{1}$ state. It is suggestive to interpret this model as a classical gas of particles on the plaquettes such that a loop labelled by $\mathbb{1}$ signals the presence of a particle on the corresponding plaquette, and τ denotes its absence. This so-called *hard hexagon* model was solved exactly in a tour de force by Baxter in 1980 and demonstrated to exhibit two phases [174]. For a large value of the fugacity the model exhibits order characterised by a full occupancy of either of the three sublattices of the plaquettes of the hexagonal lattice, while for a low fugacity the gas is dilute and homogenous. These phases are separated by a second order phase transition for a critical value z_c of the fugacity, whose scaling limit is described by the $c = 4/5$ minimal Potts model [72].

Focussing again on the strange correlator, the tensor from which the partition function is constructed has the following non-zero components:



$$\begin{array}{c} \tau \\ \tau \end{array} \begin{array}{c} \tau \\ \tau \end{array} = -1, \quad \begin{array}{c} \mathbb{1} \\ \tau \\ \tau \\ \tau \end{array} = \phi^{5/6}. \quad (163)$$

With the above tensor at hand, it is then straightforward to derive that the partition function can be written as

$$Z = \sum_{n=0}^{N/3} z_c^n g(n, N), \quad (164)$$

where N denotes the total number of plaquettes and $g(n, N)$ is a combinatorial factor equal to the number of ways to distribute n particles over N plaquettes subject to the aforementioned constraint [175]. Straightforward algebra reveals that $z_c = \phi^5$, which is exactly the critical fugacity of the hard hexagon model. This simple strange correlator thus immediately yields the critical fugacity, without having to resort to the vastly more advanced techniques employed by Baxter.

From the transfer matrix one can also extract the finite-size CFT spectra. Above, we have highlighted already the significance of the role of the tube algebra to obtain the topological superselection sectors of quantum doubles and string-net models. As detailed in refs. [136, 145, 176] the transfer matrix of the strange correlator can be supplemented with symmetry-twisted boundary conditions labelled by objects of the symmetry fusion category. These boundary conditions are topological in the sense that they preserve lattice translation symmetry up to a unitary transformation. In their presence, the tube algebra can be used to decompose the spectrum of the transfer matrix in topological sectors. For the case of Fibonacci, there are 4 sectors labelled $0, \tau, \bar{\tau}, \tau\bar{\tau}$. Their corresponding spectra are shown in ref. [145], where also the identification with the primaries of the Potts minimal model is explained.

In the previous section we argued that different tensor network representations of string-net ground states correspond to module categories over the input fusion category. Coincidentally, we did not have this freedom for the case study in this section. It turns out that this choice of module category specifies the *modular invariant* of the CFT on the torus as well as its partition function on higher genus surfaces [72, 168, 171]. The so-called *diagonal case* corresponds to choosing the regular module category. Changing this module category is then referred to as *orbifolding* or *gauging*, numerically studied in ref. [177].

There exists a vast literature on tensor network representations of critical partition functions using the strange correlator formalism. We refer to refs. [145, 177–185] for additional case

5.5 Application 3: dualities

For the case of on-site representations of groups, *Clebsch-Gordan coefficients* are the building blocks of symmetric tensors. Indeed, the *Wigner-Eckart theorem* dictates that symmetric tensors factorise as a product of Clebsch-Gordan coefficients and *reduced matrix elements*. The latter notably only depend on the irrep labels, while the coupling of basis states is fully captured by the Clebsch-Gordan coefficients [81, 187]. This result is not only central in the context of perturbation theory and spectroscopy in which they are often introduced, but also forms the cornerstone of symmetric tensor network algorithms [49–54]. Inspired by this perspective, we can deduce from eq. (160) for the more general case of an MPO symmetry $\mathcal{C}$, that generic local operators are constructed from the fusion tensors as follows:

$$h_{i,i+1}^{\mathcal{R}} = h(\alpha, \alpha', \beta, \beta', \gamma, i, i') \quad (165)$$

[illegible]

So far, we have implicitly assumed infinite chains. Completely in line with section 5.4 we can also define these Hamiltonians in the presence of closed periodic and symmetry-twisted boundary conditions [136, 176, 189]. For the latter, an extra degree of freedom is introduced

on the bond between two sites, and the local Hamiltonian term across that bond is modified accordingly, we refer to ref. [189] for details. The importance of including symmetry-twisted boundary conditions will become clear below.

To illustrate the expression eq. (165), consider the spin 1/2 antiferromagnetic Heisenberg model encountered above in eq. (12). By invoking the (traditional) Wigner-Eckart theorem, the local Hamiltonian term $\mathbf{S}_i \cdot \mathbf{S}_{i+1}$ can be constructed from the $SU(2)$ Clebsch-Gordan coefficients. To rephrase this in the current framework, we need to choose $\mathcal{D} = \text{Rep}(SU(2))$, the category of representations of $SU(2)$, with $\mathcal{R} = \text{Vec}$.¹⁶ Spelled out explicitly:

$$\mathbf{S}_i \cdot \mathbf{S}_{i+1} = - \begin{array}{c} \text{2} \quad \text{2} \\ | \quad | \\ \text{0} \\ | \quad | \\ \text{2} \quad \text{2} \end{array} = \sum_{i,j} \sum_{i',j'} \begin{bmatrix} 2 & 2 \\ i' & j' \end{bmatrix} \begin{bmatrix} 2 & 2 \\ i & j \end{bmatrix} |i', j'\rangle \langle i, j|, \quad (167)$$

up to an additive constant. Here, **2** and **0** denote the $SU(2)$ doublet and singlet respectively, and $\begin{bmatrix} 2 & 2 \\ \bullet & \bullet \end{bmatrix}$ stands for the (real) Clebsch-Gordan coefficients associated to

$$\begin{bmatrix} 2 & 2 \\ \bullet & \bullet \end{bmatrix} : 2 \otimes 2 \rightarrow 0 : |i\rangle \otimes |j\rangle = \begin{bmatrix} 2 & 2 \\ i & j \end{bmatrix} |1\rangle, \quad (168)$$

where $i, j, 1$ denote basis vectors of the respective underlying representation spaces. This Hamiltonian term thus essentially projects two neighbouring spin 1/2's on the singlet representation, and the spin 1 representation appearing in the tensor product $\frac{1}{2} \otimes \frac{1}{2}$ is given zero weight by the chosen parameters h .

Let us consider the set of all local symmetric operators (165), as well as their finite products and linear combinations. This constitutes an algebra – sometimes referred to as *bond algebra* [139, 143, 158, 190, 191] – specified by the structure constants

$$h_A^{\mathcal{R}} h_B^{\mathcal{R}} = \sum_C f_{AB}^C({}^{\circ}F) h_C^{\mathcal{R}}, \quad (169)$$

defined for the basis of operators $\{h_A^{\mathcal{R}}\}$. Crucially, the *structure constants* $f_{AB}^C({}^{\circ}F)$ depend on the associators ${}^{\circ}F$ of the fusion category $\mathcal{D}$ only, i.e. they are completely independent of the choice of module category $\mathcal{R}$. In particular, this observation implies that also the spectrum of the Hamiltonian is – up to degeneracies – independent of $\mathcal{R}$. For a fixed $\mathcal{D}$ and choice of coefficients h in eq. (165), changing the module category $\mathcal{R} \rightarrow \mathcal{R}'$ boils down to a *duality transformation*. As such, different module categories over $\mathcal{D}$, or equivalently the symmetry category $\mathcal{C}$, thus organise an entire web of dualities. Although this framework may appear abstract, most dualities of physical interest, including Kramers–Wannier duality, Kennedy–Tasaki duality, and the Jordan–Wigner transformation, naturally fit within it, as will be illustrated further below. Throughout, we will often refer to the (fictional) model of interest as the *original* model, and compare it to its *dual(s)* model(s). The dual model can look radically different from the original one in the following ways.

As mentioned above, the local symmetric operators eq. (165) and their MPO symmetries act on a Hilbert space which is generically constrained if $\mathcal{R}$ has more than one simple object. As a consequence, the Hilbert space of one model is typically completely different from that of any of its dual models and their dimensions don't even have to match. This is closely related to

¹⁶Technically speaking $\text{Rep}(SU(2))$ is a tensor category, rather than a fusion category which is required to have a finite number of simple objects. Readers inclined to more mathematical rigour may wish to apply the formalism to a finite subgroup of $SU(2)$ instead.

the fact that dualities of the kind considered here can be realised by gauging (part of) the original MPO symmetry, so that the constrained Hilbert space arises as an *effective* Hilbert space where (part of) the imposed Gauss constraints are resolved [192–195]. Consequently, also the global symmetries of distinct dual models typically differ as they are encoded in $\mathcal{D}_{\mathcal{R}}^*$ and $\mathcal{D}_{\mathcal{R}'}^*$ respectively. In other words, given the initial symmetry and the duality transformation, the dual symmetry is fully determined. The existence of a duality between models implies also that every observable in one theory is equivalent to a *dual* observable in the dual theory. However, only the *symmetric*, local observables in one theory get mapped to dual local, symmetric operators in the dual theory, whereas charged operators, such as local order parameters, get mapped to non-local *string order parameters*.

One of the upshots of the tensors network approach to dualities is that we can build explicit *intertwiners* between dual models in terms of (non-invertible) matrix product operators [196]. At the level of the Hamiltonian terms they act according to:

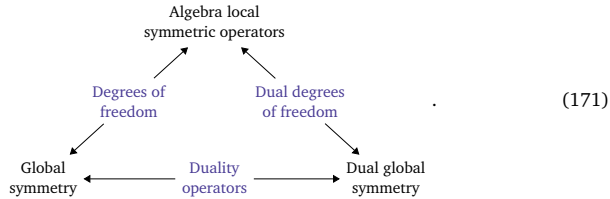
$$h_{i,i+1}^{\mathcal{R}'} \circ D = D \circ h_{i,i+1}^{\mathcal{R}}, \quad (170)$$

from which it follows that acting with D on an eigenstate of $H^{\mathcal{R}}$ results in an eigenstate of $H^{\mathcal{R}'}$ with exactly the same energy.

As demonstrated in ref. [196], such an intertwiner only depends on the categories $\mathcal{D}$, $\mathcal{R}$ and $\mathcal{R}'$ and not on the specific model under scrutiny. In fact, these MPO intertwiners are precisely the MPO intertwiners mentioned in section 5.3 that transmute different PEPS representations of a $\mathcal{D}$ string-net model associated to the module categories $\mathcal{R}, \mathcal{R}'$ into each other.

Another implication of eq. (170) is that given an intertwiner D , we can multiply D from the *right* with symmetry operators of the initial symmetry $\mathcal{C} = \mathcal{D}_{\mathcal{R}}^*$ and from the *left* with the dual symmetry operators in $\mathcal{D}_{\mathcal{R}'}^*$. As such, the different intertwiners are themselves organised in an invertible bimodule category – that we will denote by $\mathcal{M}$ – between the original and dual symmetry categories. The construction is consistent since $\mathcal{D}_{\mathcal{R}'}^* = \mathcal{C}_{\mathcal{M}}^*$.¹⁷

For a pair of dual models, we can collect these structures in a diagram as follows:



In this diagram, every vertex stands for a fusion category. The one on top – $\mathcal{D}$ – encodes the bond algebra generated by local symmetric operators, whereas the other two – $\mathcal{D}_{\mathcal{R}}^*$ and $\mathcal{D}_{\mathcal{R}'}^*$ – encode the global symmetries. The connecting edges represent invertible bimodule categories. The two on the diagonals correspond to $\mathcal{R}$ and $\mathcal{R}'$, encoding the degrees of freedom of either model. The one on the bottom – $\mathcal{M}$ – encodes the MPO intertwiners.

Invertibility of $\mathcal{R}$ and $\mathcal{R}'$ guarantees in particular that the Drinfeld centres of the original symmetry $\mathcal{D}_{\mathcal{R}}^*$ and the dual symmetry $\mathcal{D}_{\mathcal{R}'}^*$ are equivalent. In this context, objects in the Drinfeld centre label *topological sectors* of the theory. These sectors correspond to a boundary condition – which breaks part of the symmetry – together with a symmetry charge of the surviving symmetry. By carefully studying the interplay of the duality operators with these sectors, the

¹⁷This follows from the observation in footnote 14 that duality operators are labelled by $\mathcal{D}$ -module functors organised in $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$.

non-invertible duality operators can be lifted to unitaries [189]. Under a duality transformation the topological sectors are generically permuted, as exemplified in the Kramers-Wannier duality.

Example: Kramers-Wannier duality Let us demonstrate how this formalism reproduces the famous Kramers-Wannier (KW) duality of the transverse field Ising model eq. (33) [197]. To this end, consider the Hamiltonian

$$H_{\text{TFIM}}^{\text{Vec}_{\mathbb{Z}_2}} = - \sum_{i=1}^L \sigma_i^Z \sigma_{i+1}^Z - \lambda \sum_{i=1}^L \sigma_i^X, \quad (172)$$

on periodic boundary conditions, i.e. $L + 1 \equiv 1$. The Hamiltonian clearly commutes with a $\mathbb{Z}_2$ symmetry generated by $\prod_i \sigma_i^x$. The bond algebra corresponds to the fusion category $\mathcal{D} = \text{Vec}_{\mathbb{Z}_2}$ and the module category is $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2}$. The Kramers-Wannier duality can be thought of as gauging the $\mathbb{Z}_2$ symmetry, or thus more abstractly, changing the module category $\mathcal{R} = \text{Vec}_{\mathbb{Z}_2} \rightarrow \mathcal{R}' = \text{Vec}$, corresponding to $\mathcal{M} = \text{Vec}$. The (unique) duality MPO can be depicted as:

$$D_{KW} = \begin{array}{c} \text{---} \square \oplus \square \ominus \square \oplus \square \oplus \square \text{---} \end{array}, \quad (173)$$

where we have explicitly indicated the periodic boundary conditions with the dotted lines, and here and below depict duality MPOs in blue. Note that the physical legs on either side are shifted by half a site. We emphasise again that this duality MPO is the same for every $\mathbb{Z}_2$ symmetric model, and as such is not specific to the Ising model.

The MPO tensors are GHZ tensors in the σ^Z and σ^X eigenbasis and explicitly read

$$\begin{array}{c} \text{---} \square \text{---} \\ | \end{array} = |000\rangle + |111\rangle, \quad \begin{array}{c} | \\ \text{---} \oplus \text{---} \end{array} = |+++ \rangle + |--\rangle. \quad (174)$$

All (global) properties of the intertwiner can be deduced from the symmetry properties of these local tensors. In particular, note that the first tensor is invariant under acting on any two legs with a σ^Z , and acting with σ^X on all its legs simultaneously:

$$\begin{array}{c}
 \text{---} \square \text{---} \\
 | \\
 \text{---} \square \text{---}
 \end{array}
 =
 \begin{array}{c}
 \text{---} \square \text{---} \\
 | \\
 \text{---} \square \text{---}
 \end{array}
 =
 \begin{array}{c}
 \text{---} \square \text{---} \\
 | \\
 \text{---} \square \text{---}
 \end{array}
 =
 \begin{array}{c}
 \text{---} \square \text{---} \\
 | \\
 \text{---} \square \text{---}
 \end{array}
 \quad (175)$$

The second tensor has the same symmetries with σ^Z and σ^X interchanged. Hence, it follows that this MPO maps the local Hamiltonian terms according to

$$\sigma_i^Z \sigma_{i+1}^Z \rightarrow \sigma_{i+\frac{1}{2}}^Z, \quad \sigma_i^X \rightarrow \sigma_{i-\frac{1}{2}}^X \sigma_{i+\frac{1}{2}}^X, \quad (176)$$

so that the dual Hamiltonian reads

$$H_{\text{TFIM}}^{\text{Vec}} = - \sum_{i=1}^L \sigma_{i+\frac{1}{2}}^Z - \lambda \sum_{i=1}^L \sigma_{i-\frac{1}{2}}^X \sigma_{i+\frac{1}{2}}^X. \quad (177)$$

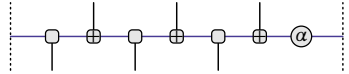
This Hamiltonian has a symmetry generated by $\prod_i \sigma_{i+1/2}^Z$, which corresponds to the symmetry category $\text{Rep}(\mathbb{Z}_2) = (\text{Vec}_{\mathbb{Z}_2})_{\text{Vec}}^*$, in accordance with the prediction of the framework. The

mapping eq. (176) also shows that the order parameter in the original model gets mapped to the disorder parameter in the dual and vice versa:

$$\sigma_i^Z \sigma_j^Z \rightarrow \prod_{k=i}^{j-1} \sigma_{k+\frac{1}{2}}^Z, \quad \prod_{k=i}^j \sigma_k^X \rightarrow \sigma_{i-\frac{1}{2}}^X \sigma_{j+\frac{1}{2}}^X. \quad (178)$$

The MPO eq. (173) has a null space which contains all $\mathbb{Z}_2$ odd states: $\prod_i \sigma_i^X |\Psi\rangle = -|\Psi\rangle$. Similarly, its image consists of only states which are even under the dual symmetry. In order to access the non-trivial charge sectors in either model, we need to adapt the MPO to account for all symmetry-twisted boundary conditions. For the case at hand, this amounts to either periodic or anti-periodic boundary conditions. At the level of the Hamiltonian, changing the boundary condition boils down to flipping the sign of one of the nearest-neighbour interaction terms, conventionally the one between sites L and 1 .

After having specified the boundary condition and the charge sector of the original model, the topological sector of the dual model is specified by the mapping in table 1 [189, 198–200]. Equivalently, after specifying the boundary conditions in both models, the accessible charge sectors are fully determined. Remark in particular the permutation of topological sectors: in essence the boundary condition and charge sector are interchanged under KW duality. For each of these sectors there is a corresponding MPO intertwiner of the form


(179)

where α is a placeholder for one of matrices $\{\mathbb{1}, \sigma^X, \sigma^Y, \sigma^Z\}$ as indicated in table 1 as well.

Boundary condition	Charge sector	Dual boundary condition	Dual charge sector	Boundary matrix
P	+1	P	+1	$\mathbb{1}$
P	-1	AP	+1	σ^Z
AP	+1	P	-1	σ^X
AP	-1	AP	-1	σ^Y

Table 1: Mapping of topological sectors under the Kramers-Wannier duality. Here, (A)P stands for (anti)periodic.

Note that the original model eq. (172) is in the symmetry-broken phase if $0 \leq \lambda < 1$ and in the symmetric phase if $\lambda > 1$; under Kramers-Wannier duality these phases are interchanged in the dual model. Turning the logic around, for any value of the magnetic field strength λ , there is a model, either eq. (172) or eq. (177), that is (fully) symmetry-broken. We could conjecture that this observation holds generally: every phase can be mapped to a completely symmetry-broken phase – with respect to the dual symmetry – by means of a suitable duality mapping. Further evidence for this speculation is provided by the example of the Kennedy-Tasaki duality.

Example: Kennedy-Tasaki duality The *Kennedy-Tasaki* (KT) duality was originally devised to elucidate the ground state physics of the spin 1 antiferromagnetic Heisenberg model, which is now understood to exhibit non-trivial $\mathbb{Z}_2 \times \mathbb{Z}_2$ SPT order. The duality was introduced as a non-unitary mapping on open chains between the Heisenberg model and a model that spontaneously breaks $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry [201, 202]. Recently, it has been reinterpreted in terms of twisted gauging of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry [196, 200, 203–205]. Equivalently, this is described

as the change in module category $\mathcal{R} = \text{Vec} \rightarrow \mathcal{R}' = \text{Vec}^\psi$ over the input $\text{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$, where ψ is a non-trivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$. As such, it is more precise to think about the symmetry as being labelled by characters in $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$, which is equivalent to $\text{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$ as a fusion category. It can be shown that $\mathcal{M} = \text{Rep}^\psi(\mathbb{Z}_2 \times \mathbb{Z}_2)$ so that given a ψ -projective representation π of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the MPO is constructed from the tensor with components:

$$\begin{array}{c} h \\ | \\ a \text{ --- } \boxed{} \text{ --- } \beta \\ | \\ g \end{array} = \delta_{g,h} \pi(g)_{a\beta}. \quad (180)$$

For long it was unclear how to generalise KT from open to closed chains. The MPO framework provides a crisp answer: just take the trace of the MPO to define it on periodic boundary conditions. Also, the original KT transformation on the open chain is obtained by choosing suitable boundary conditions for the MPO eq. (180). Note that the MPO tensors are diagonal in their physical leg, so that KT doesn't change the charge sector, but only permutes the boundary conditions.

Instead of applying it to the Heisenberg model, we focus here on the AKLT state eq. (44), also a non-trivial $\mathbb{Z}_2 \times \mathbb{Z}_2$ SPT. Recall that the AKLT state can be represented as an MPS with matrices $A^i = \sigma^i / \sqrt{3}$. If we take for concreteness the projective representation defined via $\pi((a, b)) = (\sigma^X)^a (\sigma^Z)^b$, we obtain that acting with the duality on the AKLT state gives an MPS with matrices $A^i = \sigma^i \otimes \sigma^i$. This MPS is clearly reducible as all MPS matrices commute. Indeed, by making use the gauge transformation

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}, \quad (181)$$

this MPS can be reduced to a sum of four product states:

$$\begin{aligned} U(\sigma^X \otimes \sigma^X)U^\dagger &= \text{diag}(1, 1, -1, -1), \\ U(\sigma^Y \otimes \sigma^Y)U^\dagger &= \text{diag}(-1, 1, -1, 1), \\ U(\sigma^Z \otimes \sigma^Z)U^\dagger &= \text{diag}(1, -1, -1, 1). \end{aligned} \quad (182)$$

This illustrates that the duality operator is fully responsible for the twofold degenerate entanglement spectrum of the AKLT state. Turning things around, by acting with the duality on the AKLT state – or any other $\mathbb{Z}_2 \times \mathbb{Z}_2$ SPT for that matter – we can get rid of the degeneracies of the entanglement spectra. This observation can be used to our advantage as argued below.

In analogy with the example of KW, one can now define four symmetry-twisted boundary conditions, one for each element of $\mathbb{Z}_2 \times \mathbb{Z}_2$. We relegate the reader to refs. [200, 205] for a detailed discussion. As shown there, the charge sectors are mapped identically as anticipated, but depending on the charge sector the boundary conditions are permuted into each other.

The observation that there seemingly always exists a unique duality transformation that maps a model to a completely symmetry-broken dual model suggests a one-to-one correspondence between module categories and categorically symmetric gapped phases of matter. And indeed, this is exactly the content of the *generalised Landau paradigm*. Provided the symmetry $\mathcal{C}$, the gapped phases of matter are in one-to-one correspondence with $\mathcal{C}$ -module categories $\mathcal{P}$ [53, 77, 135]. This module category $\mathcal{P}$ encodes both the distinct ground states or injective blocks of the

ground state MPS – corresponding to its simple objects – as well as the action of $\mathcal{C}$ on the ground state subspace – via the module action. By making use of the parent Hamiltonian construction, the generalised Landau paradigm was made explicit in the MPS framework in ref. [77]. In this context, the regular module category $\mathcal{P} = \mathcal{C}$ represents the completely symmetry-broken phase where the ground states are in bijection with the symmetry operators. As anticipated already in lecture 3, this generalises the case of a group symmetry where both the module categories and phases of matter are characterised by pairs $(H \subseteq G, \psi)$, $[\psi] \in H^2(H, \text{U}(1))$.

Combining the generalised Landau paradigm and the duality framework has far-reaching implications. For one, note that the symmetry-broken phase is the only gapped phase that is guaranteed to exist.¹⁸ In fact, the existence of the completely symmetric phase is equivalent to the symmetry being anomaly-free and thus fully gaugeable. The corresponding duality operator that interchanges the two then acts by effectively averaging over the symmetry-broken ground states [53, 195]. It is fascinating that, viewed this way, any continuous phase transition can be interpreted as a transition between a fully symmetry-broken phase on one side and a (partially) symmetric phase on the other.

Notably, the symmetry-broken dual model minimises the ground state entanglement entropy. This can be exploited in the DMRG algorithm, and for this reason the symmetry-broken model can be referred to as the *optimal* (dual) model. Indeed, as demonstrated in section 3.3 entanglement spectra can exhibit degeneracies, due to the fact that the entanglement degrees of freedom transform in multiplets of the (projective) virtual symmetry. If this is the case, there is redundancy in the MPS description of the ground state: computational resources are expended to fine-tune some variational parameters to reproduce these spectra. It is therefore beneficial to simulate the optimal dual model that maximally breaks the (dual) symmetry so that no degeneracies remain. In turn, the ground state of the original model is then obtained by acting with the duality operator which reinstates the degeneracies in the entanglement spectrum. A case study was carried out in ref. [53], and the benefit of this approach demonstrated.

Having identified the optimal dual model additionally provides a unifying perspective on the elementary excitations in any gapped phase $\mathcal{P}$. We now reserve the notation $\mathcal{Q}$ to denote the right module category over $\mathcal{D}$ corresponding to the optimal model. As argued in section 2.5, the elementary excitations in this optimal and thus symmetry-broken phase $\mathcal{D}_{\mathcal{Q}}^*$ are topological domain wall excitations. Recall that in the infinite-chain limit such excitations are created by acting on one half of the chain with a symmetry MPO in $\mathcal{D}_{\mathcal{Q}}^*$, while its variational degrees of freedom are encoded in a tensor at its endpoint. The quasiparticle excitations in that case are organised in a fusion category, $\mathcal{E}$, which coincides with the symmetry category, i.e. $\mathcal{E} = \mathcal{D}_{\mathcal{Q}}^*$. Provided this optimal model, its domain wall excitations can in turn be mapped back to the excitations of the original model, which are typically of a completely different nature, such as the magnon excitations in the Heisenberg model (see section 4.1). This situation is depicted as follows:

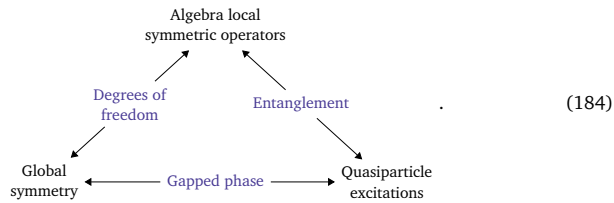
$$(183)$$

Herein, the blue MPO represents a duality operator – coincidentally labelled by an object in $\mathcal{P}$ – between the optimal model corresponding to $\mathcal{Q}$ and the model of interest $\mathcal{R}$ as indicated by the coupons. A domain wall excitation is then depicted on the bottom, and parametrised by the tensor B , similar as in section 2.5. Note that this tensor carries a charge $e \in \mathcal{E}$ that terminates on the duality operator. This picture also reveals that the labels of $\mathcal{Q}$ capture the *entanglement degrees of freedom* of the ground state of interest.

¹⁸For some symmetry categories such as $\mathcal{C} = \text{Fib}$ it is even the only gapped phase.

In this way, the domain wall excitations of the optimal dual model provide an invariant characterisation for excitations of all of its dual models. And the fundamental excitations in the any model can all be parametrised according to eq. (183).

Combined, these structures related to the quasiparticle excitations and the phase of a model are encapsulated in the following diagram, reproduced from ref. [53]:



5.6 Synopsis of lecture V

Lecture V can be seen as a symbiosis of the previous lectures: we encountered the fundamental theorem, MPO algebras, category theory, symmetry-breaking and string-nets models. Viewed from the point of view of tensor networks, topologically ordered systems exhibit symmetry breaking in their entanglement degrees of freedom. Those symmetries are described by MPO algebras which form representations of *fusion categories*.

On the one hand, those symmetries describe the long-range entanglement structure in string-nets, which can themselves readily be expressed in terms of tensor networks. As an application, we introduced the concept of the *strange correlator*. This prescription relates lattice representations of topological field theories to conformal field theories via an explicit holographic construction consisting of an overlap of a string-net with a featureless short-range entangled state. The MPO symmetries of the string-net then become the topological defects in the CFT, thereby providing an explicit lattice realisation of the Fröhlich-Fuchs-Runkel-Schweigert characterisation of topological defects in CFTs.

The same MPO algebras also describe the symmetries of the effective Hamiltonians that govern the physics of the edge modes of those 2+1D systems. In a fascinating twist, the different ways in which those symmetries can be represented – technically described in terms of bimodule categories – lead to a full comprehensive picture for defining dualities between all possible edge Hamiltonians in terms of a set of intertwining MPOs. One of the most fascinating physical consequences of this construction is the fact that every 1D phase of matter is dual to a phase that is completely symmetry broken (with respect to the dual symmetry). Notably, this is irrespective of whether this one-dimensional system is defined on a tensor product structure with on-site symmetries or on a constrained Hilbert space with categorical non-invertible symmetries. Among other things, this means that every possible second order phase transition for such 1D systems is equivalent to one to a completely symmetry broken phase.

In these lectures, we have not touched upon the higher-dimensional versions of MPO symmetries and MPO intertwiners. Interestingly, in higher dimensions, symmetries can act on freely-deformable submanifolds of various dimensionality of the lattice [134, 138, 140, 206–210]. These *higher-form symmetries* are naturally represented in terms of tensor networks as well. In the case of (2+1)D, surface operators are implemented in terms of projected entangled-pair operators (PEPOs), while the 1-form symmetries, living on interfaces between PEPOs, are naturally described by MPOs. As a whole, this symmetry structure and its recoupling theory is encapsulated in a *fusion 2-category* [211, 212]. Similar to the one-dimensional case, distinct ways of representing a fusion 2-categorical symmetry on the lattice are classified by *module 2-categories* [213, 214].

As pioneered in ref. [192], also dualities in 2+1D lattice systems can be implemented on the lattice by PEPOs. Based on this tensor network approach, significant progress in the understanding of higher-categorical symmetries and their dualities has been made [205, 214–219]. Despite these recent advances, plenty of interesting open research directions are still to be explored.

Further research directions include the incorporation of continuous symmetries within this categorical framework. As already touched upon in ref. [53], *Schur-Weyl duality* is completely compatible with the MPO framework for describing dualities. It maps theories with continuous $SU(N)$ symmetries to theories with discrete $\text{Rep}(SU(N))$ symmetries. Since the Mermin-Wagner theorem no longer applies for the latter, the duality map can be used to circumvent it. As such, the Schur-Weyl duality maps interchange the symmetric and symmetry-broken phases, making the DMRG simulation of this dual theory much more efficient.

5

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A Group cohomology

In this section we explain how cohomology groups $H^n(G, U(1)^\beta)$ for any finite group G with a group action β on $U(1)$ can be computed using simple linear algebra methods developed in [80, 220]. As a side product, we will also obtain representative cocycles for every cohomology class.

Let us first introduce our notation and relevant definitions. We denote the set of all functions $G^n \rightarrow U(1)$ by $C^n(G, U(1))$, these functions are called *n-cochains*. The *coboundary maps* $\partial^{n+1} : C^n(G, U(1)) \rightarrow C^{n+1}(G, U(1))$ are then defined via:

$$(\partial^{n+1}\omega)(g_1, g_2, \dots, g_{n+1}) := \beta_{g_1}(\omega(g_2, g_3, \dots, g_{n+1})) \\ \times \prod_{i=1}^n [\omega(g_1, g_2, \dots, g_{i-1}, g_i g_{i+1}, \dots, g_{n+1})]^{(-1)^i} \times [\omega(g_1, g_2, \dots, g_n)]^{(-1)^{n+1}} = 0. \quad (\text{A.1})$$

Here, β denotes the group action of G on (the G -module) $U(1)$. In the case of a symmetry group containing *antiunitary* elements, such as time-reversal and spatial reflection encountered in the main text, $\beta_g(\bullet)$ acts via complex conjugation when g is antiunitary and as the identity otherwise. It is easy to check that these maps are nilpotent, i.e. $\partial^{n+1} \circ \partial^n = 1$. We then define the abelian groups of *n-cocycles* and *n-coboundaries* as respectively:

$$Z^n(G, U(1)^\beta) := \ker(\partial^{n+1}), \quad (\text{A.2})$$

$$B^n(G, U(1)^\beta) := \text{im}(\partial^n). \quad (\text{A.3})$$

From the nilpotency of the coboundary maps, it follows that $B^n(G, U(1)^\beta)$ is a subgroup of $Z^n(G, U(1)^\beta)$, so that we can define the *n-th cohomology group* of G as the quotient

$$H^n(G, U(1)^\beta) = Z^n(G, U(1)^\beta) / B^n(G, U(1)^\beta). \quad (\text{A.4})$$

The key insight in computing $H^n(G, U(1)^\beta)$ and finding representative cocycles is the following. By taking the logarithm of both sides of the *cocycle condition* $\partial^{n+1}\omega = 1$, and collecting the unknown phases $\omega(g_1, \dots, g_n)$ in a vector ω , the condition can be written as a linear system that has to be satisfied modulo 2π :

$$\Omega\omega = 0, \quad \text{mod } 2\pi. \quad (\text{A.5})$$

Here, the coefficient matrix Ω originates from the coboundary map ∂^{n+1} and contains only integers. The dimension of ω is $|G|^n$ and Ω is $|G|^{n+1} \times |G|^n$ -dimensional. Since Ω only contains integers, we can write Ω in its *Smith normal form*:

$$\Omega = P\Lambda R. \quad (\text{A.6})$$

In this decomposition, P and R are respectively $|G|^{n+1} \times |G|^{n+1}$ and $|G|^n \times |G|^n$ matrices that only contain integers and have determinant one (and thus have integer-valued inverses). Λ also only contains integers, is $|G|^{n+1} \times |G|^n$ -dimensional and is of the form

$$\Lambda = \left(\begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0_{r, n-r} \\ \hline 0_{n+1-r, r} & 0_{n+1-r, n-r} \end{array} \right), \quad (\text{A.7})$$

in which the non-zero elements $d_1, \dots, d_r$ along the diagonal, some of which might be one, are in increasing order, $d_1 \leq d_2 \leq \dots$, and every element is a divisor of the next, $d_i | d_{i+1}$. The Smith

normal form Λ is unique. Inserting this decomposition in the system of equations (A.5) gives rise to the solution

$$\omega = 2\pi R^{-1} \Lambda^+ \nu. \quad (\text{A.8})$$

In this equation Λ^+ denotes the (unique) Moore-Penrose pseudoinverse of Λ that satisfies $\Lambda \Lambda^+ \Lambda = \Lambda$ and which is found to be

$$\Lambda^+ = \left(\begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0_{r, n-r} \\ \hline 0_{n+1-r, r} & 0_{n+1-r, n-r} \end{array} \right). \quad (\text{A.9})$$

The vector ν contains arbitrary integers. Writing the solution (A.8) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{\nu_j}{d_j}. \quad (\text{A.10})$$

Because ν can be chosen freely, one can choose subsequently $\nu_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$ to obtain a basis of the solution space that can be written as

$$(\omega_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (\text{A.11})$$

Hence, the n -cocycles ω are found to be the columns of R^{-1} . Since R is full rank, all the solutions ω are linearly independent. In particular, the non-trivial cocycles (below) can not be related by a coboundary, $\omega' = \omega + \Omega' \varphi$, where Ω' denotes the coboundary map of degree $n-1$ and φ is a $|G|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of Λ .

From (A.10) and the fact that R^{-1} contains only integers, it follows that for every diagonal entry $d_i = 1$, a trivial solution $\omega_i = 0 \pmod{2\pi}$ is obtained. The non-trivial solutions are those that correspond to entries $d_i > 1$. From (A.10) and the fact that the solution space is $\mathbb{Z}$ -linear, it follows that the cocycle ω_j corresponding to some d_j generates a cyclic group. Now note that not all elements of ω_j can be divisible by d_j or any of its prime factors as this would be in contradiction with the fact that R^{-1} has determinant one. Hence, the cyclic group generated by ω_j is $\mathbb{Z}_{d_j}$. Finally, the zero entries of Λ can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group G one can write down the matrix representation of the coboundary map Ω that can be brought in Smith normal form $P \Lambda R$. The diagonal entries of Λ , $d_1, d_2, \dots, d_r$, determine the second cohomology group which is then of the form $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$, with the understanding that $\mathbb{Z}_1 = \{e\}$ denotes the trivial group and that all zero diagonal entries can be discarded. The non-trivial n -cocycles in some arbitrary gauge correspond then to the columns of R^{-1} , weighted by the appropriate factor $2\pi/d_j$ as per eq. (A.10).

B Category theory

In this appendix we summarise relevant notions of fusion and (bi)module categories to the extent that they are used in the main text. We refer the reader to the standard references [118, 221, 222] and reviews [111, 121, 223] for a more detailed and technically precise exposition.

B.1 Fusion categories

Succinctly, a (unitary) *fusion category* $\mathcal{D}$ consists of

1. A finite set of *simple objects* (topological defects) $a, b, c, \dots$.
2. Fusion rules $a \otimes b = \bigoplus_c N_{ab}^c c$, together with a unit for the tensor product denoted 1 .
3. Unitary maps called *F-symbols* or *associators* $F : (\bullet \otimes \bullet) \otimes \bullet \xrightarrow{\sim} \bullet \otimes (\bullet \otimes \bullet)$.

The F-symbols adhere to a consistency condition, known as the *pentagon equation*. Spelled out, the pentagon equation boils down to linear multivariate system of equations which schematically are of the form

$$FF = \sum FFF, \quad (\text{B.1})$$

and written out in components read

$$\sum_o (F_e^{fcd})_{g,lm}^{h,no} (F_e^{abh})_{f,ko}^{i,pq} = \sum_{j,rst} (F_g^{abc})_{f,kl}^{j,rs} (F_e^{ajd})_{g,sm}^{i,tq} (F_i^{bcd})_{j,rt}^{h,np}. \quad (\text{B.2})$$

Notably, a fusion category is not necessarily endowed with a *braiding*, i.e. unitary maps $R_{ab} : a \otimes b \xrightarrow{\sim} b \otimes a$. However, for every fusion category one can construct its *Drinfeld centre* $\mathcal{Z}(\mathcal{D})$, which is a *modular fusion category*, meaning in particular that it is endowed with a *non-degenerate braiding*. The centre can be physically realised in the string-net models of section 5.3 as the anyonic excitations of the model. Their exchange statistics are encoded in the braiding of $\mathcal{Z}(\mathcal{D})$.

(Anomalous) Finite group symmetry For any *finite* group G , the fusion category Vec_G is the category of finite-dimensional G -graded vector spaces. In essence, the simple objects of Vec_G correspond to the group elements of G , and the fusion rules are dictated by the group multiplication. The F -symbols are all trivial. Whenever G possesses a non-trivial third cohomology group (see section A), any 3-cocycle ω can be used to construct Vec_G^ω . This fusion category is constructed analogous to Vec_G where now the non-zero components of the F -symbols evaluate to ω . This cocycle encodes the (*t Hooft*) *anomaly*, i.e. an obstruction to gauge to symmetry, or – equivalently – an obstruction for the symmetry to admit a trivial symmetric ground state as encountered in the main text in section 3.3.

In this sense, a fusion category thus encodes a symmetry structure – symmetry operators, their multiplication rules and junctions – as well as their anomaly – encoded in the F-symbol.

Rep(G) symmetry The category $\text{Rep}(G)$, for any finite group G , encodes as objects the representations of G . The irreducible representations then constitute the *simple* objects. The fusion of those objects is then given by the usual tensor product of group representations. It can then be deduced that the F -symbols of $\text{Rep}(G)$ are given by the 6j-symbols.

Fibonacci symmetry The Fibonacci fusion category has two simple objects $\{1, \tau\}$ whose only non-trivial fusion rule reads $\tau \otimes \tau = 1 \oplus \tau$. The F-symbols are spelled out in numerous places such as ref. [223]. The name *Fibonacci* refers to the fact that the multiplicity of τ in subsequent tensor powers of τ generates the Fibonacci series. It can be shown that also the Fibonacci fusion category admits no local symmetric Hamiltonian with trivial symmetric ground state [77].

B.2 Module categories

Let $\mathcal{C}$ be a fusion category. A (unitary left) $\mathcal{C}$ -module category $\mathcal{M}$ consists of

1. A finite set of simple objects $A, B, C, \dots$
2. An action $\triangleright$ of $\mathcal{C}$ on $\mathcal{M}$: $a \triangleright A = \bigoplus_B N_{aA}^B B$, for simple objects B in $\mathcal{M}$.
3. Unitary module ${}^\triangleright F$ -symbols or (left) module associators ${}^\triangleright F : (\bullet \otimes \bullet) \triangleright \bullet \xrightarrow{\sim} \bullet \triangleright (\bullet \triangleright \bullet)$.

These ${}^\triangleright F$ -symbols satisfy in turn a consistency equation involving both the F -symbols of $\mathcal{C}$ and the module ${}^\triangleright F$ -symbols. Schematically:

$${}^\triangleright F {}^\triangleright F = \sum F {}^\triangleright F {}^\triangleright F. \quad (\text{B.3})$$

Module categories of Vec_G Indecomposable module categories of Vec_G are labelled by pairs (H, ψ) , where $H \subseteq G$ is a subgroup (up to conjugacy) and ψ is a 2-cocycle of H . In this case, the simple objects are labelled by cosets $r_1 H, r_2 H, \dots \in G/H$, and the action of Vec_G on them is defined as $g \triangleright rH := (gr)H$. The module associator ${}^\triangleright F$ can be constructed from ψ as detailed in e.g. [224].

Module categories of $\text{Rep}(G)$ Also module categories over $\text{Rep}(G)$ are classified by tuples (H, ψ) due to the Morita equivalence of $\text{Rep}(G)$ and Vec_G . The corresponding module category is $\text{Rep}^\psi(H)$, the category of ψ -projective representations of H . The action of $\text{Rep}(G)$ on $\text{Rep}^\psi(H)$ is given via restriction of G -representations to the subgroup H and subsequently taking the tensor product or (projective) H -representations.

B.3 Bimodule categories

Given a pair of fusion categories $\mathcal{C}, \mathcal{D}$, a $(\mathcal{C}, \mathcal{D})$ -bimodule category $\mathcal{M}$ is a category which is both a left $\mathcal{C}$ - and right $\mathcal{D}$ -module category, together with additional unitary bimodule ${}^\triangleright F$ -symbols or bimodule associators ${}^\triangleright F : (\bullet \triangleright \bullet) \triangleleft \bullet \xrightarrow{\sim} \bullet \triangleright (\bullet \triangleleft \bullet)$.

Vec Vec can be interpreted as a $(\text{Vec}_G, \text{Rep}(G))$ -bimodule category. In that case, the components of the ${}^\triangleright F$ -symbols evaluate to the matrix elements of G -representations.

Throughout this manuscript we exclusively deal with *invertible* $(\mathcal{C}, \mathcal{D})$ -bimodule categories. In particular, this guarantees that $\mathcal{C}$ and $\mathcal{D}$ are *Morita equivalent*, or – equivalently – that their Drinfeld centres $\mathcal{Z}(\mathcal{C})$ and $\mathcal{Z}(\mathcal{D})$ are equivalent modular fusion categories [225]. In ref. [188], invertibility was shown to be equivalent to the notion of MPO injectivity, and a characterisation in terms of the bimodule associators derived.

Given a fusion category $\mathcal{C}$ and $\mathcal{C}$ -module category $\mathcal{M}$, there is a unique fusion category, the *Morita dual* $\mathcal{C}_{\mathcal{M}}^*$, that renders $\mathcal{M}$ an invertible $(\mathcal{C}, \mathcal{C}_{\mathcal{M}}^*)$ -bimodule category. For example: $(\text{Vec}_G)_{\text{Vec}}^* = \text{Rep}(G)$. We refer to the relevant literature for a precise definition of the Morita dual fusion category.

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One-dimensional symmetric phases protected by frieze symmetries

6.1 Synopsis

In section 2.4 we reviewed the classification of one-dimensional SPT phases protected by *internal* symmetries acting in an on-site manner. The present project extends this classification to include quasi-one-dimensional *spatial symmetries*. These are encoded in the so-called frieze groups, of which there are seven. We closely follow an approach analogous to that of section 2.4 and study the action of the symmetry on an injective MPS. We find that these SPT phases are classified by the second cohomology group of the protecting symmetry group over $U(1)$, with the additional feature that *orientation-reversing* group elements — encoding spatial reflection, for instance — act on $U(1)$ via complex conjugation. Additionally, we provide MPS normal forms for each of these phases and explicit renormalisation group fixed-point MPS representations directly in terms of representative cocycles. We also review the SPT classification in the presence of time-reversal symmetry. Finally, we discuss an algorithm that computes cohomology groups of any order and corresponding representative cocycles, including in the presence of non-trivial group action.

6.2 Reference

One-dimensional symmetric phases protected by frieze symmetries

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6.3 Contributions of the author

The project underlying this paper grew out of the Master's thesis of Nicolas Dewolf. Therein the classification of SPT phases protected by frieze symmetries was initiated and the framework established. I completed this classification and their representative cocycles, in full generality, and co-wrote most of the manuscript.

One-dimensional symmetric phases protected by frieze symmetries

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We make a systematic study of symmetry-protected topological gapped phases of quantum spin chains in the presence of the frieze space groups in one dimension using matrix product states. Here, the spatial symmetries of the one-dimensional lattice are considered together with an additional “vertical reflection,” which we take to be an on-site $\mathbb{Z}_2$ symmetry. We identify seventeen distinct non-trivial phases, define canonical forms, and compare the topological indices obtained from the MPS analysis with the group cohomological predictions. We furthermore construct explicit renormalization group fixed-point wave functions for symmetry-protected topological phases with global on-site symmetries, possibly combined with time reversal and parity symmetry. En route, we demonstrate how group cohomology can be computed using the Smith normal form.

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I. INTRODUCTION

Even though there is no intrinsic topological order in gapped one-dimensional quantum spin chains, the phase diagram becomes nontrivial when symmetry constraints are taken into account [1–9]. This gives rise to the well-known paradigm of *symmetry-protected* topological (SPT) order. The lack of topological order in 1D can be understood as follows. Starting from the ground state of a gapped local Hamiltonian, subsequent renormalization group (RG) coarse graining steps do not alter the phase of the system [10]. After sufficiently many steps, the number of which is independent of system size, the state flows towards an RG fixed point exhibiting a valence bond structure [5,9,10]. A tensor product of unitaries on the state then turns this state in a trivial product state, ultimately proving that the state we started from is adiabatically connected to a product state with no topological order. This procedure can be made explicit by writing the state as a matrix product state (MPS) [11,12]. In this formalism, one RG step is equivalent to blocking two sites and acting with an isometry on the blocked site that maximally removes local entanglement inside the block while retaining the entanglement with the rest of the system [10]. When symmetries are taken into account, the RG flow should not break the symmetry. The picture that arises is that the phase diagram, which in the absence of symmetries is simply connected, falls apart in distinct classes that cannot be connected by adiabatic transformations due to topological obstructions.

Chen *et al.* showed that the topological obstructions that prohibit connecting different such SPT phases originate from the fact that physical symmetries can be implemented by *projective* representations of the symmetry group acting on the entanglement degrees of freedom [5–7]. This crucial insight

led Chen *et al.* to a classification of SPT phases in terms of group cohomology [13]. More specifically, the SPT classification corresponds to the second cohomology group $H^2_\beta(\mathbf{G}, \mathbf{U}_1)$, where $\mathbf{G}$ denotes the symmetry group and can contain a global on-site symmetry subgroup, time reversal, parity, or combinations thereof. Here, β denotes the nontrivial action of $\mathbf{G}$ on the $\mathbf{U}_1$ module in case of time reversal or parity symmetry.

A folklore example is one where the global on-site symmetry group is $\mathbb{Z}_2 \times \mathbb{Z}_2$. Since the second cohomology group of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2$ can protect one nontrivial symmetry-protected phase known as the Haldane phase [2,3,14,15]. In translationally invariant systems, the classification is refined to $H^1_\alpha(\mathbf{G}, \mathbf{U}_1) \times H^2_\beta(\mathbf{G}, \mathbf{U}_1)$, where H^1 denotes the first cohomology group [5].

In this paper, we demonstrate that SPT phases can also be protected by quasi-one-dimensional lattice symmetries. The symmetry groups we consider are the seven so-called *frieze groups* [16]. These are defined as being the infinite discrete subgroups of the isometries of a strip, $\text{Isom}([0, 1] \times \mathbb{R})$. Apart from translations, the generators of the frieze groups are reflections in the horizontal or vertical direction, π rotations (equivalent to the composition of a horizontal and vertical reflection) and glide reflections. The seven distinct frieze groups these generators give rise to are denoted by F_0 (only translation), F_V (translation + vertical reflection), F_H (translation + horizontal reflection), F_R (translation + π -rotation), F_G (translation + glide reflection), F_{RG} (translation + π -rotation + glide reflection), and F_{VH} (translation + two reflections). In case of a glide reflection, acting with this glide reflection twice is equivalent to the action of the translation generator.

We derive the SPT classification corresponding to these symmetries by imposing the symmetry on a general injective MPS and identifying topologically distinct ways in which this symmetry can be implemented [17]. Here, any on-site $\mathbb{Z}_2$ symmetry of the system could play the role of the vertical reflection in the frieze groups. However, we preserve the

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geometrical interpretation and represent the vertical reflection as a swap of the two physical degrees of freedom associated with every site (and thus, with every local MPS tensor). The physical models for which this analysis is relevant include ladder systems, systems with two flavors of particles and, notably, the transfer matrix of PEPS. Ultimately, we obtain seventeen nontrivial phases. Furthermore, by imposing the spatial symmetries directly on a generic MPS, the resulting phases are put in one-to-one correspondence with the structure of the local tensors, in particular, their transformation behavior. This structure can in turn be imposed in numerical algorithms. Indeed, by making full use of spatial symmetries and imposing the adequate transformation behavior on the local tensors, for which this paper provides a dictionary, the number of variational degrees of freedom in numerical simulations can be significantly reduced, thus allowing for higher resolution simulations.

A. Outline

In Sec. II, we begin by providing a review of matrix product states, recapitulating the concepts of MPS injectivity, gauge transformations, the transfer matrix and the fundamental theorem of MPS. After reconsidering the implementation of symmetries in MPS and how this leads to the SPT classification of Chen *et al.* in Sec. III, we present in Sec. III A a method to compute group cohomology and explicit cocycles from the Smith normal form of the coboundary map¹ and in Sec. III B construct explicit correlation length zero MPS tensors transforming according to given cohomology classes characterizing an SPT phase. We explicitly construct our ansatz from the nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in Sec. III B 1 and find that it reduces to the fixed point cluster state dressed with a trivial dimer state. In Sec. III C, we generalize our ansatz and argue how an MPS transforming on the physical level in some arbitrary representation of a finite symmetry group can be constructed in such a way that the virtual bond dimension is as small as possible. In Sec. IV, we address the main question of this paper, namely we construct the SPT classification for frieze symmetric MPS and construct canonical forms for most of these phases. Finally, to illustrate the generality of our approach, in Sec. V, we reconsider the problem of imposing time reversal symmetry in MPS and combine time reversal with shifts over one site. Some technical details are relegated to Appendix A, and in Appendix B, we demonstrate our algorithm to compute cocycles of $H_{\rho\rho}^2(\mathbb{Z}_2^P, \mathbf{U}(1))$, the second cohomology of $\mathbb{Z}_2$ with nontrivial group action originating from parity symmetry.

B. Summary of results

In Table I, below we give an overview of the symmetry groups we consider and the SPT classification they give rise to.

Translation symmetry in itself does not give rise to nontrivial SPT phases. It can be shown that every translationally

TABLE I. Overview of the SPT classification of frieze group symmetries as well as of time reversal. F_1^T : independently symmetric under translations over one site and time reversal, F_2^T : translation invariance over two sites and invariance under combined action of translation over one site and time reversal.

Symmetry	SPT classification
F_0	/
F_V	$\mathbb{Z}_2$
F_H (parity)	$\mathbb{Z}_2^{\times 2}$
F_R	$\mathbb{Z}_2^{\times 2}$
F_G	/
F_{RG}	$\mathbb{Z}_2^{\times 2}$
F_{VH}	$\mathbb{Z}_2^{\times 4}$
F_1^T	$\mathbb{Z}_2$
F_2^T	/

invariant MPS admits a uniform representation, as was shown in Ref. [12]. The reflection in the F_V symmetry group can be thought of as an on-site $\mathbb{Z}_2$ symmetry. We rederive that F_V protects one nontrivial SPT phase, and provide an explicit MPS representation in which the local MPS tensors have definite V -parity. The reflection in F_H is equivalent to the parity considered in Ref. [5]. Three nontrivial SPT phases are found, in accordance with [5], and a canonical MPS is constructed in which the tensors have definite H -parity, possibly at the cost of introducing nontrivial *bond tensors* [19]. The same classification is found for F_R symmetry, not previously considered elsewhere. F_G symmetry admits only the trivial phase and we show that every F_G -symmetric MPS can be brought in a manifestly F_G -invariant form. In case of the larger symmetry groups F_{RG} and F_{VH} there are respectively three and seven nontrivial phases. To our knowledge, also the symmetry groups F_G , F_{RG} , and F_{VH} have not previously been considered in the literature.

MPS with independent translation symmetry over one site and time reversal symmetry can protect one nontrivial SPT phase as demonstrated in Ref. [5], if time reversal is implemented linearly. We rederive this result and show that in the nontrivial phase the MPS tensors transform according to a quaternionic representation of $\mathbb{Z}_2^T$. It is argued that injective MPS cannot be invariant under a projective implementation of time reversal symmetry, an MPS interpretation of the Lieb-Schultz-Mattis theorem. Finally, to demonstrate the generality of our approach we also consider MPS which are symmetric under translations over two lattice sites and the combined action of a one-site translation and time reversal, denoted by the symmetry group F_2^T . We find that in this case no nontrivial phases are retained.

II. REVIEW OF MATRIX PRODUCT STATES

In this section, we present a brief review of injective matrix product states. We focus on some key aspects that are used to derive the frieze classification below.

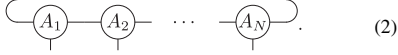
In this paper, we consider bosonic spin systems. The Hilbert space is simply the tensor product of the local d -dimensional Hilbert spaces of the constituent spins, $\mathcal{H} \cong (\mathbb{C}^d)^{\otimes N}$. A matrix product representation of a state in $\mathcal{H}$ with

¹After completion of this manuscript we came to understand that this algorithm appeared previously in Ref. [18].

periodic boundary conditions is of the form

$$|\psi\rangle = \sum_{\{i_k\}} \text{Tr}(A_1^{i_1} A_2^{i_2} \dots A_N^{i_N}) |i_1\rangle |i_2\rangle \dots |i_N\rangle. \quad (1)$$

Such a periodic MPS can be pictorially represented as



The variational degrees of freedom are contained in the local tensors $(A_n^i)_{\alpha\beta} \in \mathbb{C}^D \otimes \mathbb{C}^D \otimes \mathbb{C}^d$, where D is called the *bond dimension*. Every state in $\mathcal{H}$ can be represented with a bond dimension that scales exponentially in the system size, but the power of MPS lies in the fact that ground states of gapped local Hamiltonians can be well approximated by MPS with a bond dimension that scales polynomially in the number of spins [11]. An MPS representation of a state $|\psi\rangle$ is never unique: a *gauge transformation* $A_n^i \mapsto X_n^{-1} A_n^i X_{n+1}$ clearly leaves the state invariant because the gauge tensors X_i cancel on the bonds.

In case of translation invariance, it can be shown that one can always carry out a gauge transformation that brings the translationally invariant MPS in a canonical *uniform* form in which $A_n \equiv A$, $\forall n$ [12]:

$$|\psi(A)\rangle = \dots - \text{A} - \text{A} - \text{A} - \dots \quad (3)$$

Gauge transformations can furthermore be used to bring the MPS parametrization in a left- or right-canonical form, characterized respectively by

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \text{Diagram: A circle with A on top and A-bar on bottom, with two horizontal legs.} = \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right), \quad (4)$$

$$\sum_i A^i (A^i)^\dagger = \mathbb{1}, \quad \text{Diagram: A circle with A on top and A-bar on bottom, with two horizontal legs.} = \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right). \quad (5)$$

We define the *transfer matrix* $\mathbb{E}$ as

$$\mathbb{E} = \sum_i A^i \otimes \bar{A}^i = \text{Diagram: A circle with A on top and A-bar on bottom, with two horizontal legs.} \quad (6)$$

The transfer matrix captures all the relevant information about the entanglement and correlations of the state. Moreover, the transfer matrix determines the MPS uniquely up to a local change in basis. This follows from the observation that the transfer matrix defines a completely positive (CP) map where the local MPS tensors play the role of Kraus operators, combined with the fact that a Kraus decomposition of a CP map is unique up to unitary equivalence [20].

If the matrices $\{A^i | i = 1, \dots, d\}$ generate (via linear combinations and products) the entire $D \times D$ matrix algebra, the MPS is said to be *injective*. In that case, the transfer matrix (interpreted as an $D^2 \times D^2$ matrix from the left pair of indices to the right pair) has a unique eigenvalue of largest magnitude that in an appropriate normalization of the MPS tensors can be

taken to be one. Moreover, when the MPS is in a left or right canonical form, the corresponding eigenvector is $\mathbb{1}$ as follows from (4) and (5).

By far the most important property of injective MPS is that it satisfies the requirements of the *fundamental theorem* of MPS: two injective uniform MPS defined by local tensors A^i and B^i describe the same state $|\psi(A)\rangle \sim |\psi(B)\rangle$ if and only if there is a gauge transformation X and a phase θ that intertwines the two tensors: $A^i = e^{i\theta} X^{-1} B^i X$. If A and B are simultaneously in left (or right) canonical form, then X can be chosen to be unitary. Furthermore, $e^{i\theta}$ is uniquely defined, whereas X is only defined up to an overall scaling. Put differently, for $B^i = A^i$, the relation $A^i = e^{i\theta} X^{-1} A^i X$ implies $\theta = 0 \bmod 2\pi$ and $X = c\mathbb{1}$ for some $c \in \mathbb{C}$, as follows readily from the definition of injectivity.

III. EXPLICIT SYMMETRIC TENSORS

In this section, we review how symmetries are implemented in the tensor network language and discuss how the SPT classification arises from the projective representation of the underlying symmetry group. We present an algorithm to compute group cohomology and explicit cocycles using the Smith normal form and give an MPS ansatz that realizes the SPT phases classified by a given 1- and 2-cocycle. We work out this ansatz for the nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ and demonstrate that this representative MPS is the tensor product of the correlation length zero cluster state and a trivial dimer state. We then discuss how an MPS can be constructed for every possible physical representation of a given finite symmetry group such that the bond dimension is minimal.

If an injective translationally invariant MPS $|\psi(A)\rangle$ is invariant under the action of a unitary on-site symmetry transformation, i.e., $U_g^{\otimes N} |\psi(A)\rangle \sim |\psi(A)\rangle$ for all $g \in \mathbf{G}$, there must exist, for every $g \in \mathbf{G}$, a phase $\varphi(g)$ and a gauge transformation X_g such that

$$\sum_j (U_g)_{ij} A^j = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (7)$$

This equation can only admit solutions if U_g forms a linear unitary representation of the on-site symmetry group $\mathbf{G}$. If the MPS is in either left or right canonical form, the gauge matrices X_g can be chosen to be unitary. Because of the overall scale freedom in how they are determined, it follows that they only need to constitute a projective representation of $\mathbf{G}$, i.e., they form a representation of $\mathbf{G}$ up to phase:

$$X_g X_h = e^{i\omega(g,h)} X_{gh}. \quad (8)$$

Here, the $\omega(g, h)$'s satisfy the well-known 2-cocycle equations

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \omega(h, k) \bmod 2\pi, \quad (9)$$

expressing associativity of the multiplication of the gauge matrices X_g . Solutions to this constraint are called 2-cocycles. The phase $\varphi(g)$, on the other hand, constitutes a one-dimensional linear representation of the symmetry group:

$$\varphi(g) + \varphi(h) = \varphi(gh) \bmod 2\pi. \quad (10)$$

The scale freedom in determining gauge transformations implies that the matrices X_g can be replaced with an equivalent

choice of the form $X_g \mapsto e^{i\gamma(g)} X_g$. Under such a redefinition the cocycles transform according to

$$\omega(g, h) \mapsto \omega(g, h) + \gamma(g) + \gamma(h) - \gamma(gh). \quad (11)$$

Hence, in the classification of projective representations labeled by 2-cocycles, these redefinitions have to be modded out, giving rise to equivalence classes of projective representations. Cocycles ω of the form $\omega(g, h) = \gamma(g) + \gamma(h) - \gamma(gh)$, which are equivalent to the choice $\omega(g, h) = 1$, are called coboundaries. The equivalence classes $[\omega]$, defined by $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \gamma(h) - \gamma(gh)$, are exactly classified by the second cohomology group of $\mathbf{G}$ with respect to $\mathbf{U}_1$, $H^2(\mathbf{G}, \mathbf{U}_1)$. In the context of group cohomology, the one-dimensional linear representation $\varphi(g)$ is referred to as a 1-cocycle and is correspondingly characterized by $H^1(\mathbf{G}, \mathbf{U}_1)$. Chen *et al.* showed that every choice of a 1-cocycle $\varphi \in H^1(\mathbf{G}, \mathbf{U}_1)$ and cohomology class $[\omega] \in H^2(\mathbf{G}, \mathbf{U}_1)$ gives rise to a distinct SPT phase [5].

In case of a finite symmetry group, the second cohomology group is always $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_N}$ for topological indices $d_1, d_2, \dots, d_N$, giving rise to a finite number of SPT phases. Explicit representative 2-cocycles can be obtained by writing the 2-cocycle condition as a linear system modulo 2π and solving it using the Smith normal form, as explained below in Sec. III A.

In case the symmetry group contains time reversal or parity transformations, the above picture has to be modified as follows.

Since the $\mathbb{Z}_2^T$ time reversal is implemented antiunitarily [21], the time reversal operator can be written as a $T = UK$, where K denotes complex conjugation in the basis with respect to which the MPS tensors A^i are defined, and U is a unitary satisfying $U\bar{U} = \pm \mathbb{1}$ depending on whether time reversal is implemented linearly or projectively (Sec. V A). Hence, on the MPS tensors, a symmetry $g \in \mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^T$, $\mathbf{H}$ denoting the global on-site symmetry group, acts according to

$$\sum_j (U_g)_{ij} C_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g. \quad (12)$$

The action of C_g on A^i is taking the complex conjugate only if g contains a time reversal.

From acting with time reversal twice on the MPS tensor, it follows that in this case the matrices X_g form a *generalized* projective representation of the symmetry group, as their multiplication also contains the action C_g :

$$X_g C_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (13)$$

whereas the phases $\exp(i\varphi(g))$ obey

$$e^{i\varphi(g)} C_g(e^{i\varphi(h)}) = e^{i\varphi(gh)}. \quad (14)$$

Hence, $\varphi(g)$ and $\omega(g, h)$ satisfy the 1- and 2-cocycle constraints with a nontrivial group action that takes the conditioned complex conjugation C_g into account. The 1-cocycles constraint reads

$$\varphi(g) + \alpha_g^T(\varphi(h)) = \varphi(gh) \mod 2\pi, \quad (15)$$

whereas 2-cocycles satisfy

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^T(\omega(h, k)) \mod 2\pi, \quad (16)$$

where the group actions α_g^T and β_g^T are multiplication by -1 if g contains time reversal. The SPT classification in this case is given by $H_{\alpha^T}^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^T}^2(\mathbf{G}, \mathbf{U}_1)$. The equivalence classes of $H_{\beta^T}^2$ are given by $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^T(\gamma(h)) - \gamma(gh)$, whereas $H_{\alpha^T}^1$ classifies 1-cocycles up to equivalence of the form $\varphi(g) \mapsto \varphi(g) + \alpha_g^T(c) - c$ for an arbitrary constant $c \in [0, 2\pi)$ as coboundaries of the form $\alpha_g^T(c) - c$ trivially solve the 1-cocycle condition (15).

In case the symmetry group contains global on-site symmetries combined with parity, $\mathbf{G} = \mathbf{H} \rtimes \mathbb{Z}_2^P$, we have that

$$\sum_j (U_g)_{ij} T_g(A^j) = e^{i\varphi(g)} X_g^{-1} A^i X_g, \quad (17)$$

where T_g denotes taking the transpose if g contains the parity transformation. Similarly as in the case of time reversal, the gauge matrices multiply according to a generalized projective representation:

$$X_g P_g(X_h) = e^{i\omega(g, h)} X_{gh}, \quad (18)$$

where P_g is taking the inverse transpose if g contains parity. Note, however, that if g contains the parity transformation, X_g can in general not be chosen unitary. As the conditioned transpose does not affect the phases φ , they form a one-dimensional linear representation. The SPT classification for $\mathbf{H} \rtimes \mathbb{Z}_2^P$ is thus in terms of $H^1(\mathbf{G}, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbf{G}, \mathbf{U}_1)$. The 2-cocycles are given by

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^P(\omega(h, k)) \mod 2\pi, \quad (19)$$

where, similarly as in the case of time reversal, the group action β_g^P is a multiplication by -1 whenever g contains a parity transformation. The equivalence classes in the second cohomology group are then $[\omega] = [\omega'] \iff \omega(g, h) = \omega'(g, h) + \gamma(g) + \beta_g^P(\gamma(h)) - \gamma(gh)$.

A. Computing group cohomology using the Smith normal form

The problem of finding all (generalized) projective representations of a given finite symmetry group $\mathbf{G}$ or, equivalently, computing its second cohomology group $H_{\beta}^2(\mathbf{G}, \mathbf{U}_1)$ can be reduced to a problem in linear algebra that can be solved using the Smith normal form [18,22], as we now demonstrate. Our approach works for both trivial and nontrivial group actions β and the same method can be used to compute other cohomology groups.

First note that the 2-cocycle equation (9) or its generalizations with nontrivial group actions (16) and (19) can be written as the linear system

$$\sum_j \Omega_{ij}^{(2, \beta)} \omega_j = 0, \mod 2\pi, \quad (20)$$

which has to be solved modulo 2π . $\Omega^{(2, \beta)}$ is called the 2-coboundary map, where β again refers to the group action. Every solution $\vec{\omega}$ to this linear system of equations constitutes a valid 2-cocycle.

Since $\Omega^{(2,\beta)}$ only has entries in $\mathbb{Z}$, which forms a principal ideal domain, $\Omega^{(2,\beta)}$ can be written in Smith normal form as follows [23]:

$$P\Lambda R = \Omega^{(2,\beta)}. \quad (21)$$

In this decomposition, P and R are respectively $|\mathbf{G}|^3 \times |\mathbf{G}|^3$ and $|\mathbf{G}|^2 \times |\mathbf{G}|^2$ matrices that only contain integers and have determinant one (and thus have integer-valued inverses). Λ also only contains integers, is $|\mathbf{G}|^3 \times |\mathbf{G}|^2$ -dimensional and is of the form

$$\Lambda = \left(\begin{array}{c|c} \text{diag}(d_1, d_2, \dots, d_r) & 0 \\ \hline 0 & 0 \end{array} \right), \quad (22)$$

in which the nonzero elements $d_1, \dots, d_r$ along the diagonal, some of which might be one, are in increasing order, $d_1 \leq d_2 \leq \dots$, and every element is a divisor of the next, $d_i | d_{i+1}$. The Smith normal form Λ is unique. Inserting this decomposition in the system of equations (20) gives rise to the solution

$$\bar{\omega} = 2\pi R^{-1} \Lambda^+ \bar{v}. \quad (23)$$

Λ^+ denotes the (unique) Moore-Penrose pseudoinverse of Λ that satisfies $\Lambda \Lambda^+ \Lambda = \Lambda$ and which is found to be

$$\Lambda^+ = \left(\begin{array}{c|c} \text{diag}(d_1^{-1}, d_2^{-1}, \dots, d_r^{-1}) & 0 \\ \hline 0 & 0 \end{array} \right). \quad (24)$$

$\bar{v}$ is an arbitrary vector that only contains integers.

Writing the solution (23) in components yields

$$\omega_i = \sum_{j=1}^r 2\pi (R^{-1})_{ij} \frac{v_j}{d_j}. \quad (25)$$

Because $\bar{v}$ can be chosen freely, one can choose subsequently $v_i = \delta_{1,i}, \delta_{2,i}, \dots, \delta_{r,i}$ to obtain a basis of the solution space that can be written as

$$(\bar{\omega}_j)_i = \frac{2\pi}{d_j} (R^{-1})_{ij}, \quad \forall j \in \{1, \dots, r\}. \quad (26)$$

Hence, the 2-cocycles $\bar{\omega}$ are found to be the columns of R^{-1} . Since R is full rank, all the solutions $\bar{\omega}$ are linearly independent. In particular, the nontrivial cocycles (below) can not be related by a coboundary, $\bar{\omega}' = \bar{\omega} + \Omega^{(1,\beta)} \bar{\varphi}$, where $\Omega^{(1,\beta)}$ denotes the 1-coboundary map and $\bar{\varphi}$ is a $|\mathbf{G}|$ -dimensional vector containing arbitrary real numbers. To classify all possible solutions, we now consider the diagonal entries of Λ .

From (26) and the fact that R^{-1} contains only integers, it follows that for every diagonal entry $d_i = 1$, a trivial solution $\omega_i = 0 \pmod{2\pi}$ is obtained. The nontrivial solutions are those that correspond to entries $d_i > 1$. From (26) and the fact that the solution space is $\mathbb{Z}$ -linear, it follows that the cocycle $\bar{\omega}_j$ corresponding to some d_j generates a cyclic group. Now note that not all elements of $\bar{\omega}_j$ can be divisible by d_j (or any of its prime factors) as this would be in contradiction with the fact that R^{-1} has determinant one. Hence, the cyclic group generated by $\bar{\omega}_j$ is $\mathbb{Z}_{d_j}$. Finally, the zero entries of Λ can also be discarded in the cohomology as these correspond to trivial solutions of the cocycle equation that can be multiplied by arbitrary phases and thus correspond to coboundaries.

In conclusion following picture arises. Given some group $\mathbf{G}$ one can write down the coboundary map $\Omega_{ij}^{(2,\beta)}$ that can

be brought in Smith normal form $P\Lambda R$. The diagonal entries of Λ , $d_1, d_2, \dots, d_r$, determine the second cohomology group which is then of the form $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \dots \times \mathbb{Z}_{d_r}$, with the understanding that $\mathbb{Z}_1 = \{e\}$ denotes the trivial group and that all zero diagonal entries can be discarded. The nontrivial 2-cocycles in some arbitrary gauge correspond then to the columns of R^{-1} , weighted by the appropriate factor $2\pi/d_j$.

B. Zero correlation length SPT ansatz

Given some SPT phase characterized by $([\varphi], [\omega]) \in H_\alpha^1(\mathbf{G}, \mathbf{U}_1) \times H_\beta^2(\mathbf{G}, \mathbf{U}_1)$, it is possible to explicitly construct zero-correlation-length MPS tensors that transform according to (7), (12) or (17). Firstly, a projective representation X_g in the class $[\omega]$ and with trivial 1-cocycle can be constructed with virtual dimension $|\mathbf{G}|$, by taking the X_g to form the ω -projective regular representation of $\mathbf{G}$ (27). The dimension of the local physical Hilbert space is then $|\mathbf{G}|^2$. Concretely, this representative state and the regular representation are given by

$$(X_g)_{g_1, g_2} = \delta_{g_1, g g_2} e^{i\omega(g, g_2)} \quad (27)$$

$$(V^{h_1, h_2})_{g_1, g_2} = \frac{1}{|\mathbf{G}|} \delta_{h_1, g_1} \delta_{h_2, g_2} e^{i\omega(h_2, h_1^{-1} h_1)}. \quad (28)$$

Furthermore, the physical group action U_g is given by $L_g \otimes L_g$, with L_g the linear left regular representation. Hence, $(U_g)_{k_1 k_2, h_1 h_2} = \delta_{k_1, g h_1} \delta_{k_2, g h_2}$. Pictorially:

$$\text{Diagram (29): } \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] V \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] L_g \otimes L_g = \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] X_g^\dagger \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] X_g \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] V \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \quad (29)$$

Note that the Kronecker deltas in the definition of the MPS tensor indicate its valence bond structure, which is modified only by a unitary diagonal transformation. Hence, it follows readily that the transfer matrix $\mathbb{E} = \sum_{h_1, h_2} (V^{h_1, h_2})_{g_1, g_2} \otimes (V^{h_1, h_2})_{g'_1, g'_2}$ is idempotent, $\mathbb{E}^2 = \mathbb{E}$, implying that the ansatz (28) has zero correlation length and thus defines an RG fixed point [10].

Similarly, in case of a nontrivial 1-cocycle φ , the ansatz is readily modified to also include this cocycle by adding diagonal matrices to the two physical legs:

$$\text{Diagram (30): } \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] V \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] L_g \otimes L_g \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \sqrt{\varphi} = e^{i\varphi(g)} \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] X_g^\dagger \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] X_g \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] V \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \sqrt{\varphi} \quad (30)$$

where $(\sqrt{\varphi})_{g_1, g_2} = \delta_{g_1, g_2} e^{\frac{i}{2}\varphi(g)}$.

The ansatz can also capture the case of time reversal and parity symmetry. In the first case, we take the physical action to be $L_g \otimes L_g$ combined with the conditioned complex

conjugation:

$$C_g \left[\begin{array}{c} \boxed{V} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ \boxed{L_g \otimes L_g} \end{array} \right] = e^{i\varphi(g)} \left[\begin{array}{c} \boxed{X_g^\dagger} \quad \boxed{X_g} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ \boxed{V} \end{array} \right], \quad (31)$$

where we still have $(X_g)_{g_1, g_2} = \delta_{g_1, g g_2} e^{i\omega(g, g_2)}$.

The ansatz in case of parity symmetry and parity + time reversal requires following conditioned “swap” tensor

$$\text{swap}_g = \begin{cases} \text{X} & , \quad g \text{ contains parity,} \\ \text{I} & , \quad \text{else.} \end{cases} \quad (32)$$

The ansatz in this case then amounts to

$$P_g \left[\begin{array}{c} \boxed{V} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ \boxed{L_g \otimes L_g} \\ \boxed{\text{swap}_g} \end{array} \right] = e^{i\varphi(g)} \left[\begin{array}{c} \boxed{X_g^\dagger} \quad \boxed{X_g} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \\ \boxed{V} \end{array} \right], \quad (33)$$

where P_g acts according to

$$P_g \left[\begin{array}{c} \boxed{V} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \end{array} \right] = C_g \left[\begin{array}{c} \boxed{\text{swap}_g} \\ \boxed{V} \\ \sqrt{\varphi} \quad \sqrt{\varphi} \end{array} \right]. \quad (34)$$

I. Example: $\mathbb{Z}_2 \times \mathbb{Z}_2$

The smallest finite group with a nontrivial 2-cocycle is $\mathbb{Z}_2 \times \mathbb{Z}_2$ as $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{U}_1) = \mathbb{Z}_2$. $\mathbb{Z}_2 \times \mathbb{Z}_2$ is exactly the on-site symmetry group of the AKLT model and the cluster state [10, 24–26]. The latter admits a description as an injective bond dimension two MPS [27]:

$$A^0 = |+\rangle\langle 0| \quad (35)$$

$$A^1 = |-\rangle\langle 1|, \quad (36)$$

where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. This MPS description can be derived from the fact that the cluster state is obtained starting from the product state $|+\rangle^{\otimes N}$, acting on pairs of neighboring spins with controlled Z gates and projecting onto the physical degrees of freedom with the projector $|0\rangle\langle 00| + |1\rangle\langle 11|$. The cluster state is no RG fixed point but the fixed point is obtained

after blocking only two sites [10]. The MPS description of this RG fixed point is then given by

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0| \quad (37)$$

$$A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1| \quad (38)$$

$$A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0| \quad (39)$$

$$A^{11} = \frac{1}{\sqrt{2}}|-\rangle\langle 1|. \quad (40)$$

The normalization of the state is chosen in such a way that the unique nonzero eigenvalue of the transfer matrix is 1. This state is in the nontrivial SPT class with on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry. $\mathbb{Z}_2 \times \mathbb{Z}_2$ acts linearly on the physical level as $S((0, 1)) = \sigma_X \otimes \sigma_X$, $S((1, 0)) = \mathbb{1} \otimes \sigma_X$, and the symmetry is represented projectively on the virtual level by $X((0, 1)) = i\sigma_Y$, $X((1, 0)) = \sigma_X$, $X((1, 1)) = \sigma_Z$.

Since our correlation length zero ansatz (28) that is constructed from a given nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is in the same SPT phase as the cluster state, we can expect that our ansatz is up to a basis transformation and gauge transformations the product of the cluster state fixed point and a trivial state. We now show that our ansatz indeed reduces to the product of the RG fixed point cluster state dressed with a trivial dimer state and demonstrate how to construct this basis transformation and gauge transformation explicitly.

The linear regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ acting on the physical level of our ansatz reads

$$L_{(0,1)} = \mathbb{1} \otimes \sigma_X \quad (41)$$

$$L_{(1,0)} = \sigma_X \otimes \mathbb{1} \quad (42)$$

$$L_{(1,1)} = \sigma_X \otimes \sigma_X. \quad (43)$$

A nontrivial 2-cocycle of $\mathbb{Z}_2 \times \mathbb{Z}_2$ in a particular gauge is given by

$$\omega(g, h) = \begin{matrix} & (0,0) & (1,0) & (0,1) & (1,1) \\ \begin{matrix} (0,0) \\ (1,0) \\ (0,1) \\ (1,1) \end{matrix} & \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \end{matrix}, \quad (44)$$

where g labels the rows. The projective regular representation corresponding to this cocycle (27) reduces to

$$U^\dagger X_{(0,1)} U = \mathbb{1} \otimes i\sigma_Y \quad (45)$$

$$U^\dagger X_{(1,0)} U = \mathbb{1} \otimes -\sigma_X \quad (46)$$

$$U^\dagger X_{(1,1)} U = \mathbb{1} \otimes \sigma_Z, \quad (47)$$

in the basis U given by

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & 0 \end{pmatrix}. \quad (48)$$

We then take this unitary U as a gauge transformation of our representative MPS ansatz. This gauge transformation then intertwines between the projective regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ and the block diagonal projective representation given in (45)–(47). Similarly as in the case of linear representation theory, this illustrates how the projective regular representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ falls apart in projective irreps, given by the Pauli matrices, where each irrep appears with multiplicity equal to its dimension [28].

The physical basis transformation W that brings the representative MPS V in the form of the cluster state fixed point dressed with a trivial dimer state,

(49)

is then immediately found by considering the QR decomposition of our ansatz and the dressed cluster state interpreted as matrix from the physical to virtual level, $QR = V(\bar{U} \otimes U)$, $\bar{Q}DD^\dagger \bar{R} = \mathbb{1} \otimes A \otimes \mathbb{1}$, where D is a diagonal matrix containing only phases that fixes the gauge freedom of the QR decomposition in such a way that $D^\dagger \bar{R} = R$. The basis transformation then reads $W = \bar{Q}DQ^\dagger$. It can then be checked that this W intertwines the physical representation of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry group acting on the dressed cluster state and our ansatz:

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \sigma_X \otimes \mathbb{1})W = L_{(0,1)} \otimes L_{(0,1)} \quad (50)$$

$$W^\dagger(\mathbb{1} \otimes \mathbb{1} \otimes \sigma_X \otimes \mathbb{1})W = L_{(1,0)} \otimes L_{(1,0)} \quad (51)$$

$$W^\dagger(\mathbb{1} \otimes \sigma_X \otimes \mathbb{1} \otimes \mathbb{1})W = L_{(1,1)} \otimes L_{(1,1)}. \quad (52)$$

Notice that in this example we were required to choose the symmetry group of the dimer to be trivial to match the symmetry of our ansatz, even though the dimer has the full U_2 symmetry.

C. A generalized ansatz

As we have demonstrated, given a 1- and 2-cocycle of some finite symmetry group G , one can explicitly construct a bond dimension $|G|$ injective correlation length zero MPS that belongs to the SPT class corresponding to the cohomology classes represented by these cocycles. However, as shown for the explicit example of $\mathbb{Z}_2 \times \mathbb{Z}_2$, our ansatz could be written as a cluster state fixed point and a completely disentangled trivial dimer state that doubles the dimension of the virtual Hilbert space. This redundancy is a consequence of the fact that the physical symmetry action is fixed as being the tensor product of two regular representations of G . Hence, the question rises if an MPS can be constructed for any given representation of the symmetry on the physical level such that the bond dimension is as small as possible. This can in principle be done as follows, possibly at the cost of introducing a correlation length. We restrict to the case of unitary on-site symmetries.

We first note that every projective representation of a finite group G can be lifted to a linear representation of a larger

finite covering group $\tilde{G}$ [29]. This covering group fits in following central exact sequence:

$$1 \rightarrow A \rightarrow \tilde{G} \rightarrow G \rightarrow 1, \quad (53)$$

where $A = H^2(G, U_1) \leq Z(\tilde{G})$ [30]. It should be noted that the covering group is generically not unique. Consider, for example, the case of $\mathbb{Z}_2 \times \mathbb{Z}_2$. Two distinct covering groups of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are the dihedral group $D_4 = \langle r, s | r^4 = s^2 = (rs)^2 = 1 \rangle$ and the quaternion group $Q = \langle a, b | a^4 = 1, a^2 = b^2, b^{-1}ab = a^{-1} \rangle$, both of which are of order 8. Indeed, the two-dimensional projective irrep of $\mathbb{Z}_2 \times \mathbb{Z}_2$ corresponding to the nontrivial class of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U_1) = \mathbb{Z}_2$ given by $\{\sigma_X, i\sigma_Y, \sigma_Z\}$ is lifted to the faithful two-dimensional irrep of D_4 by taking $i\sigma_Y = r$ and $\sigma_Z = s$ as generators, whereas a gauge transformation of this projective representation, $\{\sigma_X, i\sigma_Y, \sigma_Z\} \mapsto \{i\sigma_X, i\sigma_Y, i\sigma_Z\}$, yields an equivalent projective representation which is lifted to the faithful two-dimensional irrep of Q by identifying the generators of the quaternion group as $a = i\sigma_Y, b = i\sigma_Z$.

The classification and construction of the projective irreps of a finite group thus reduces in this way to the linear representation theory of its covering group. To construct the aforementioned MPS that transforms according to a given physical representation Π of the symmetry group G , one chooses the smallest irrep Γ of $\tilde{G}$, which projects down to a projective irrep of G belonging to a certain cohomology class $[\omega]$, such that Π is contained in the tensor product $\bar{\Gamma} \otimes \Gamma$. The explicit MPS tensor is then chosen as the projector of $\bar{\Gamma} \otimes \Gamma$ on the Π -sector, and the virtual symmetry action is Γ .

Consider as example again $\mathbb{Z}_2 \times \mathbb{Z}_2$ and its covering group Q . The irreps of Q are the trivial representation 1 , three nontrivial one-dimensional sign representations, $\Gamma_1, \Gamma_2, \Gamma_3$, and the two-dimensional faithful representation given by the Pauli matrices, Δ . Choosing then a sign representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ as physical symmetry, this representation can be lifted to a one-dimensional sign representation Γ_i of Q . From the fact that $\Delta \otimes \Delta \cong 1 \oplus \Gamma_1 \oplus \Gamma_2 \oplus \Gamma_3$, it follows that the MPS with the smallest bond dimension which transforms according to a sign representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is exactly one that projects $\Delta \otimes \Delta$ onto Γ_i . Since the virtual representation Δ is exactly the projective irrep of $\mathbb{Z}_2 \times \mathbb{Z}_2$, one can immediately conclude that such a state is in the same SPT phase as the cluster state.

This construction also applies to the case of Lie groups. Consider, for example, the symmetry group SO_3 . Choosing the physical symmetry representation to be the 1 of SO_3 , one can choose the virtual representation to be the $\frac{1}{2}$. The projector of $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$ on the 1 subspace then exactly results in the MPS description of the AKLT state which belongs to the nontrivial SPT class of $H^2(SO_3, U_1) = \mathbb{Z}_2$.

Notice that this constructing generically does not give rise to correlation length zero states due to the projection on the correct physical symmetry sector.

IV. FRIEZE SYMMETRIC MPS

In this section, we derive the SPT classification of MPS invariant under frieze symmetries. We do so from starting from a general injective MPS and invoking the symmetry,

which ultimately leads to topological indices that cannot be changed by a symmetry-preserving constant depth quantum circuit.

We introduce following notation:

$$\mathbb{1}_{n,m} := \mathbb{1}_n \oplus -\mathbb{1}_m. \quad (54)$$

This matrix satisfies $\mathbb{1}_{n,m} = \mathbb{1}_{n,m}^\top = \mathbb{1}_{n,m}^{-1}$. Furthermore,

$$\Gamma := \mathbb{1}_n \otimes \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\Gamma^\top = -\Gamma^{-1}. \quad (55)$$

And finally,

$$\sqrt{\Gamma} = \mathbb{1}_n \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad \sqrt{\Gamma}^\top = \sqrt{\Gamma}^{-1}. \quad (56)$$

(1) **F₀**. As mentioned in the introduction, every translationally invariant MPS can be brought in a uniform form by an appropriate gauge transformation [12]. There are no topological obstructions to do so and hence there are no nontrivial SPT phases.

(2) **F_v**. Reflection around the horizontal axis can be thought of as an internal $\mathbb{Z}_2$ transformation of the MPS tensors. In order to impose such a reflection, we consider a uniform MPS with two physical legs that mimic an internal structure of the local degrees of freedom,

$$\cdots - \text{A} - \text{A} - \text{A} - \cdots \quad (57)$$

Under the reflection the tensor A transforms according to

$$A^{ji} = e^{i\theta} X^{-1} A^{ij} X. \quad (58)$$

After applying this symmetry transformation twice we obtain

$$A^{ij} = e^{2i\theta} X^{-2} A^{ij} X^2. \quad (59)$$

From the fundamental theorem it follows that θ is a topological index, $\theta = 0, \pi \pmod{2\pi}$, and $X^2 = e^{i\chi} \mathbb{1}$, however, this phase can be absorbed in X such that X squares to the identity. F_V can thus protect one nontrivial SPT phase characterized by $\theta = \pi$.

If the MPS is in left canonical form, it can be shown that X is unitary, which together with $X^2 = \mathbb{1}$ implies that X can be written as $X = U^\dagger \mathbb{1}_{n,m} U$ for some unitary U and $n + m = D$. With the gauge transformation $A^{ij} \mapsto \tilde{A}^{ij} = U A^{ij} U^\dagger$ the MPS tensors $\tilde{A}$ have definite V -parity:

$$\tilde{A}^{ji} = \pm \mathbb{1}_{n,m} \tilde{A}^{ij} \mathbb{1}_{n,m}, \quad (60)$$

thus tremendously reducing the number of variational degrees of freedom. The signature of $\mathbb{1}_{n,m}$ is irrelevant for the classification of SPT orders. Indeed, notice that the F_V symmetry can be thought of as an on-site $\mathbb{Z}_2$ symmetry acting on the physical level that is implemented by $\sum_{(j_1, j_2)} (U_1)_{(j_1, j_2)}^{(i_1, i_2)} A^{j_1, j_2}$ with $(U_1)_{(j_1, j_2)}^{(i_1, i_2)} = \delta_{j_2}^{i_1} \delta_{j_1}^{i_2}$ and $\mathbb{1}$ denoting the nontrivial element in $\mathbb{Z}_2$. This unitary acting on the physical level thus translates to the $\mathbb{1}_{n,m}$, also a (reducible) representation of $\mathbb{Z}_2$, acting on the virtual level. It was shown in Refs. [5,6] that every such representation of $\mathbb{Z}_2$ at the virtual level gives rise to the same SPT phase because the $\mathbb{1}_{k,l}$ for all k, l belong to the same (trivial) second cohomology class of $H^2(\mathbb{Z}_2, \mathbb{U}_1) = \{e\}$.

In particular, the MPS is in the same phase as the MPS that transforms according to

$$\tilde{A}^{ji} = \pm \tilde{A}^{ij}. \quad (61)$$

This result fits within the well understood SPT classification in terms of group cohomology for the case of an on-site $\mathbb{Z}_2$ symmetry [4–8]. Indeed, we could have implemented the vertical reflection more generally as

$$A^i \mapsto \sum_j R_{ij} A^j, \quad (62)$$

where R_{ij} is a (linear) representation of $\mathbb{Z}_2$, $R^2 = \mathbb{1}$. As mentioned above $H^2(\mathbb{Z}_2, \mathbb{U}_1) = \{e\}$ meaning there are no nontrivial projective representations of $\mathbb{Z}_2$ that can protect the phase; however $H^1(\mathbb{Z}_2, \mathbb{U}_1) = \mathbb{Z}_2$, which is reflected in the fact that $e^{i\theta} = \pm 1$ and as such can be understood as the distinction between V -odd/even tensors $\tilde{A}^{ij}$.

(3) **F_H**. F_H contains apart from a translation generator a reflection in the horizontal direction and as such corresponds to the parity transformation previously considered in Ref. [5]. Without loss of generality we can start from a uniform MPS (3) generated by a tensor A^i . Imposing the $\mathbb{Z}_2$ H -parity symmetry, we can write

$$\sum_j R_{ij} (A^j)^\top = e^{i\theta} X^{-1} A^i X, \quad (63)$$

R being an involutory unitary matrix. Doing the transformation twice results in

$$A^i = e^{2i\theta} X^\top X^{-1} A^i X X^{-\top}, \quad (64)$$

from which it follows that

$$\begin{aligned} \theta &= 0, \pi \pmod{2\pi} \\ X^\top &= e^{i\chi} X. \end{aligned} \quad (65)$$

From the last equation, we conclude that $\chi = 0, \pi \pmod{2\pi}$. Together with the first equation, this leads to a $\mathbb{Z}_2 \times \mathbb{Z}_2$ classification. We now show that every F_H symmetric MPS can be brought in a form in which the local tensors are (skew-)symmetric under H -parity, possibly at the cost of introducing extra bond tensors [19].

First consider the case that X is symmetric. Using the Autonne-Takagi decomposition, we can write $X = C C^\top$ for some complex matrix C . From substitution in (63) it immediately follows that defining $A^i \mapsto \tilde{A}^i = C^{-1} A^i C$ yields a uniform MPS generated by $\tilde{A}^i$ in which the tensors $\tilde{A}^i$ have a definite H -parity given by the topological index θ .

Now assume that X is skew-symmetric, which together with its invertibility requires that $D = 2n$ is even. In that case we similarly can write $X = C \Gamma C^\top$ using the Youla normal form [31]. Again substituting this in (63), we can identify $A^i \mapsto \tilde{A}^i = \sqrt{\Gamma} C^{-1} A^i C \sqrt{\Gamma}$ as the transformation to construct local tensors with a well-defined H -parity. However, this transformation is not an actual gauge transformation in the aforementioned sense since, in order to bring the MPS in the desired form, we should insert $\mathbb{1} = C \sqrt{\Gamma} \Gamma^\top \sqrt{\Gamma} C^{-1}$ between the A -tensors, which after making the transformation $A^i \mapsto \tilde{A}^i$

leaves us with residual bond tensors $\Gamma^\top$:

$$\cdots - \textcircled{\tilde{A}^i} - \textcircled{\Gamma^\top} - \textcircled{\tilde{A}^i} - \textcircled{\Gamma^\top} - \cdots \quad (66)$$

The SPT classification for F_H corresponds to $H^1(\mathbb{Z}_2^P, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2 \times \mathbb{Z}_2$, where the first $\mathbb{Z}_2$ factor corresponds to $\theta = 0, \pi$ and the second $\mathbb{Z}_2$ corresponds to the phase appearing in the generalized projective representation X of the parity symmetry group $\mathbb{Z}_2^P$.

(4) **F_G**. The F_G symmetry group contains a generator of translations T and a glide reflection G which are related through $G^2 = T$. Therefore we will consider an MPS ansatz which is translationally invariant under shifts over two sites. The glide reflection will then be implemented as a shift over one site followed by a reflection around the horizontal axis. Hence, without loss of generality we can take this MPS ansatz to be

$$\cdots - \textcircled{A} - \textcircled{B} - \textcircled{A} - \textcircled{B} - \cdots \quad (67)$$

Invariance under glide reflections relates the tensors A and B up to a gauge transformation which can in general be different on the A - B and B - A bonds:

$$\begin{aligned} A^{ji} &= e^{i\theta_A} X^{-1} B^{ij} Y \\ B^{ji} &= e^{i\theta_B} Y^{-1} A^{ij} X. \end{aligned} \quad (68)$$

Carrying out this transformation twice then results in

$$\begin{aligned} A^{ij} &= e^{i(\theta_A + \theta_B)} X^{-1} Y^{-1} A^{ij} X Y \\ B^{ij} &= e^{i(\theta_A + \theta_B)} Y^{-1} X^{-1} B^{ij} Y X. \end{aligned} \quad (69)$$

Blocking two sites and using the fundamental theorem of MPS yields following conditions on the phases and gauge matrices:

$$\begin{aligned} 2(\theta_A + \theta_B) &= 0 \mod 2\pi \\ X &= e^{i\chi} Y^{-1}. \end{aligned} \quad (70)$$

By absorbing a phase factor $e^{-i\frac{\chi}{2}}$ in both X and Y , X and Y are each other inverses. Substituting $2(\theta_A + \theta_B) = 0 \mod 2\pi$ in (69) shows that only the case $\theta_A + \theta_B = 0 \mod 2\pi$ can survive. We are free to choose, e.g., $\theta_A = 0$. From this, it then follows that $B^{ji} = X A^{ij} X$ such that after redefining $A^{ij} \mapsto \tilde{A}^{ij} = A^{ij} X$, the MPS can be brought in following canonical form:

$$\cdots - \textcircled{\tilde{A}} - \textcircled{\tilde{A}} - \textcircled{\tilde{A}} - \textcircled{\tilde{A}} - \cdots \quad (71)$$

In case of glide reflection symmetry there are thus no nontrivial SPT phases. This is again in line with the cohomological classification. Since the glide reflection should really be thought of as a generalized, dressed translation operator, the group cohomology classifying the SPT classes of glide reflection symmetry is that of the trivial group, which is trivial.²

²There are two distinct consistent ways in which a two-site unit cell structure can be compatible with a $\mathbb{Z}_2$ action. These are classified by $H^1(\mathbb{Z}_2, \mathbb{Z}_2) = \mathbb{Z}_2$, corresponding to a global $\mathbb{Z}_2$ symmetry and to a glide reflection.

(5) **F_R**. Starting from a two-legged uniform MPS ansatz (57), we impose the rotation symmetry as

$$(A^{ji})^\top = e^{i\theta} X^{-1} A^{ij} X. \quad (72)$$

Applying this symmetry twice and using the fundamental theorem, it immediately follows that $\theta = 0, \pi \mod 2\pi$ and $X^\top = e^{i\chi} X$, exactly what was found in case of F_H symmetry. The canonical form of an F_R invariant MPS is thus again one in which the tensors have definite R -parity, again at the cost of introducing extra bond tensors between neighboring sites if X is skew-symmetric.

(6) **F_{RG}**. Different F_{RG} -symmetric SPT phases ought to be classified by $H^1(\mathbb{Z}_2^P, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2 \times \mathbb{Z}_2$, as again the glide reflection should be thought of as a generalized translation. To obtain this classification we first impose glide reflection as in (68) on the ansatz (67), and again we find that under glide reflection the two tensors transform according to

$$\begin{aligned} A^{ji} &= X B^{ij} X \\ B^{ji} &= X^{-1} A^{ij} X^{-1}. \end{aligned} \quad (73)$$

We obtain no topological indices from the glide reflection symmetry alone.

The rotation now acts on the tensors as

$$\begin{aligned} (A^{ji})^\top &= e^{i\psi_A} W^{-1} B^{ij} Y \\ (B^{ji})^\top &= e^{i\psi_B} Y^{-1} A^{ij} W, \end{aligned} \quad (74)$$

where the reflection center lies on an A - B bond.

We then impose $R^2 = 1$ and $RGRG = 1$, and after some lengthy algebra we ultimately find the transformation rules

$$\begin{aligned} (A^{ji})^\top &= e^{-i\sigma} X^\top Y^{-1} X B^{ij} Y \\ (B^{ji})^\top &= e^{i\sigma} Y^{-1} A^{ij} X^{-1} Y X^{-\top} \\ Y^\top &= e^{i\chi} Y, \end{aligned} \quad (75)$$

where $\chi, \sigma = 0, \pi \mod 2\pi$, hence giving rise to the three anticipated nontrivial SPT phases. Here, the phase χ can be identified with the $H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2$, whereas σ corresponds to $H^1(\mathbb{Z}_2^P, \mathbf{U}_1)$. It should be noted that the topological index χ arises purely from the rotational $\mathbb{Z}_2^P$ symmetry and that σ finds its origin in the nontrivial constraint $RGRG = 1$ interlocking the glide reflection and the rotation symmetry.

(7) **F_{VH}**. We consider again the two-legged uniform ansatz (57). First imposing reflection around the horizontal axis yields

$$A^{ji} = e^{i\theta_V} X^{-1} A^{ij} X. \quad (76)$$

Carrying out this symmetry operation twice results in the same conditions on θ_V and X as in the case of F_V : $\theta_V = 0, \pi \mod 2\pi$, $X^2 = \mathbb{1}$. X is again unitary and again we can write $X = U^\dagger \mathbb{1}_{n,m} U$ for some unitary U . As was explained, the signature of $\mathbb{1}_{n,m}$ is irrelevant.

Imposing the second reflection, one obtains

$$(A^{ij})^\top = e^{i\theta_H} Y^{-1} A^{ij} Y, \quad (77)$$

which implies that $\theta_H = 0, \pi \mod 2\pi$, $Y = \pm Y^\top$. Finally we impose that the horizontal and vertical reflection commute on the physical level. Using (76) and (77), we conclude that $Y X^{-\top} = e^{i\psi} X Y$. Using $X^2 = \mathbb{1}$, it follows that $\psi = 0, \pi$. In

this way a $\mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$ classification is obtained, exactly in line with the cohomological classification $H^1(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) \times H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2} \times \mathbb{Z}_2^{\times 2}$. The first factors $\mathbb{Z}_2^{\times 2}$ originating from the first cohomology group correspond to θ_V and θ_H and the last factors $\mathbb{Z}_2^{\times 2}$ correspond to the gauge matrix Y being (anti)symmetric and the generalized (anti)commutation relation among X and Y , $YX^{-T} = \pm XY$. The correspondence between the 2-cocycles of $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$ and the topological indices obtained from the MPS picture can be made a bit more explicit as follows. One can show that there always exists a gauge in which the cocycles of $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$ are of the form

$$\omega(g, h) = \begin{matrix} (0,0) & (1,0) & (0,1) & (1,1) \\ (0,0) & \begin{pmatrix} 1 & 1 & 1 & 1 \\ (1,0) & 1 & 1 & 1 & 1 \\ (0,1) & 1 & \zeta_1 & \zeta_2 & \zeta_1\zeta_2 \\ (1,1) & 1 & \zeta_1 & \zeta_2 & \zeta_1\zeta_2 \end{pmatrix} \end{matrix}, \quad (78)$$

where rows are labeled by g and $\zeta_1, \zeta_2 = \pm 1$. These signs ζ_1, ζ_2 then exactly correspond to the topological indices obtained from the MPS computation above since it follows from (18) that

$$YX^{-T} = e^{i\omega((0,1),(1,0))} YX = \zeta_1 XY \quad (79)$$

$$YY^{-T} = e^{i\omega((0,1),(0,1))} \mathbb{1} = \zeta_2 \mathbb{1}. \quad (80)$$

Note that the results $H_{\beta^P}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2$ and $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbf{U}_1) = \mathbb{Z}_2$ are well known, corresponding to the existence of one nontrivial SPT phase under either parity or under an on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry. The fact that an a single on-site $\mathbb{Z}_2$ symmetry (which in itself does not exhibit nontrivial projective representations) combined with the parity $\mathbb{Z}_2^P$ leads to a richer structure of SPT phases, as expressed by $H_{\beta^P}^2(\mathbb{Z}_2 \rtimes \mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$ is interesting. Using the choice of cocycles in Eq. (78) and the construction from Sec. III B, explicit examples can be constructed for these different phases.

V. TIME REVERSAL AND LATTICE SYMMETRIES

In this section, we study time reversal symmetry and time reversal combined with translations over one site [21]. The most important feature of time reversal is that it is an antiunitary transformation and can hence be written as $T = UK$, where U is a unitary and K denotes complex conjugation in a certain basis. T can be represented linearly or projectively, depending on whether $U\bar{U} = \pm \mathbb{1}$.

We revisit the work by Chen *et al.* on the linear implementation of time reversal in MPS and identify the corresponding SPT classification [5]. We demonstrate that injective MPS

can not be invariant under the projective representation of T , a tensor network manifestation of the Lieb-Schultz-Mattis theorem [32]. Finally we prove that time reversal combined with a shift over one lattice site does not give rise to nontrivial SPT order and construct a canonical form for the trivial phase.

A. F_1^T : time reversal in TI systems

We consider MPS which are independently symmetric under shifts over one lattice site and $\mathbb{Z}_2^T$ implemented through $T = UK$, where U is a unitary satisfying $U\bar{U} = \pm \mathbb{1}$ and K denotes complex conjugation in the basis of the MPS tensors. Together, these symmetries correspond to the group denoted by F_1^T mentioned above in the summary of results.

(1) $\mathbf{T}^2 = \mathbb{1}$ and $\mathbf{U}\bar{\mathbf{U}} = \mathbb{1}$. Consider the uniform ansatz (3). Time-reversal symmetry can then be implemented as

$$\sum_j U_{ij} \bar{A}^j = X^{-1} A^i X. \quad (81)$$

Note that without loss of generality we do not need to consider a phase in this transformation because such a phase can be consistently absorbed in the MPS tensor A^i . For an MPS tensor in left canonical form, X can furthermore be chosen unitary. Doing a second time reversal results in

$$A^i = \bar{X}^{-1} X^{-1} A^i X \bar{X}. \quad (82)$$

By virtue of the fundamental theorem we have that $X\bar{X} = e^{i\chi} \mathbb{1}$, which, combined with unitarity of X , results in $e^{i\chi} = \pm 1$ and thus $X = \pm X^T$.

If X is symmetric, writing $X = VV^T$ (where V is unitary because X is) allows us to bring the MPS in a canonical form by means of the gauge transformation $A^i \mapsto \bar{A}^i = V^\dagger A^i V$, which now transforms according to

$$\sum_j U_{ij} \bar{A}^j = \bar{A}^i. \quad (83)$$

For a skew-symmetric X , we write $X = V\Gamma V^T$, V again being unitary, from which it follows that $\bar{A}^i = V^\dagger A^i V$ transforms according to a quaternionic representation under T , up to multiplication by U_{ij} ³:

$$\sum_j U_{ij} \bar{A}^j = \Gamma^{-1} \bar{A}^i \Gamma. \quad (84)$$

The cohomological classification corresponds to $H_{\alpha^T}^1(\mathbb{Z}_2^T, \mathbf{U}_1) \times H_{\beta^T}^2(\mathbb{Z}_2^T, \mathbf{U}_1) = \{e\} \times \mathbb{Z}_2$, as follows from the Smith normal form (Sec. III A). The $\mathbb{Z}_2$ is understood as X being (skew-)symmetric.

We can now also show that the entanglement spectrum in case of the nontrivial SPT phase for which $X = -X^T$ is at least doubly degenerate [2]. Consider therefore the unique

³Consider A transforming in a quaternionic representation according to $\bar{A} = X^{-1} A X$, X being skew-symmetric and unitary. X can be brought in a skew-symmetric tridiagonal form W by an orthogonal transformation. Unitarity of W implies that X can be written as $X = \bar{Q} \Gamma \bar{Q}^T$ for an orthogonal $\bar{Q}$, showing that $\bar{Q}^T A \bar{Q}$ transforms as $\bar{Q}^T A \bar{Q} \mapsto \Gamma^{-1} \bar{Q}^T A \bar{Q} \Gamma$.

leading right eigenvector ρ of the transfer matrix E . In that case, ρ interpreted as a $D \times D$ matrix is (Hermitian) positive semidefinite by virtue of the quantum Perron-Frobenius theorem [33,34]. Consider some eigenvector $\mathbf{x}$ of ρ with positive eigenvalue λ , then by virtue of $X\rho X^\dagger = \rho$ (proven in Appendix A), $\mathbf{x}^\top X^\dagger$ is a left eigenvector of ρ with the same eigenvalue λ . However, using $X^\top = -X$ it follows that $\text{Tr}(X^\dagger(\mathbf{x} \otimes \mathbf{x})) = 0$, or in other words that $\mathbf{x}^\top X^\dagger$ and $\mathbf{x}$ are orthogonal eigenvectors belonging to the same eigenvalue λ .

(2) $\mathbf{T}^2 = -\mathbb{1}$ and $\mathbf{U}\bar{\mathbf{U}} = -\mathbb{1}$. This case is relevant for, e.g., $U = \sigma_y$, which comes into play in the implementation of time reversal symmetry on spin 1/2 particles [21]. Let us not restrict to this particular example and consider a general unitary U satisfying the aforementioned property $U\bar{U} = -\mathbb{1}$. Starting again from the uniform ansatz (3), considering a projective implementation of time reversal and applying it twice leads to

$$A^i \xrightarrow{T} \sum_j U_{ij} \bar{A}^j \xrightarrow{T} -A^i \stackrel{!}{=} A^i, \quad (85)$$

from which we conclude that translationally invariant injective MPS cannot transform projectively under time reversal symmetry. This can be understood as a tensor network interpretation of the celebrated Lieb-Schultz-Mattis theorem [32] that dictates that the ground state of a system of half-integer spins—in which case time reversal acts projectively—should be either symmetry broken (in contradiction with the assumption that the MPS is symmetric under time reversal) or gapless (in which case the matrix product ansatz does not provide a good description).

B. F_2^T : time reversal combined with a one site shift

We can now break translation invariance over one lattice site to translation invariance over two sites by considering following ansatz:

$$\cdots \text{---} \bigcirc_A \text{---} \bigcirc_B \text{---} \bigcirc_A \text{---} \bigcirc_B \text{---} \cdots \quad (86)$$

The symmetry group we impose, F_2^T , contains translations over two lattice sites, and the combined action of translations over one site and time reversal, implemented via complex conjugation together with a multiplication with a unitary U satisfying $U\overline{U} = \mathbb{1}$. The transformation of the tensors under the latter then reads

$$\begin{aligned}\sum_j U_{ij} \bar{A}^j &= X^{-1} B^t W, \\ \sum_i U_{ij} \bar{B}^j &= W^{-1} A^i X,\end{aligned}\tag{87}$$

where $U\bar{U} = \mathbb{1}$. Note that all phases can be absorbed in the tensors and gauge transformations.

A second transformation results in

$$\begin{aligned} A^i &= \bar{X}^{-1} W^{-1} A^i X \bar{W} \\ B^i &= \bar{W}^{-1} X^{-1} B^i W \bar{X}. \end{aligned} \quad (88)$$

From blocking two tensors, we conclude that $W\overline{X} = e^{ix}\mathbb{1}$. Hence, we can absorb a factor $e^{ix/2}$ in both $\overline{X}$ and W such that they become inverses. In conclusion, there is no nontrivial SPT phase and a canonical form is obtained by writing

$$B^i = \sum_j U_{ij} X \bar{A}^j \bar{X} \quad (89)$$

and defining $C^i = \bar{X}A^i$:

$$\dots - \textcircled{C} - \textcircled{\overline{C}} - \textcircled{C} - \textcircled{\overline{C}} - \dots \quad (90)$$

VI. CONCLUSIONS AND OUTLOOK

In this work, we showed that quasi-one-dimensional spatial symmetries can protect nontrivial SPT phases in quantum spin chains represented by matrix product states, a complete survey of which was lacking in the literature. We identified each of these phases by invoking the symmetries on injective MPS and identifying the topologically distinct ways in which these symmetries can be represented by the local tensors. For most of these phases, we constructed canonical MPS ansätze that are manifestly invariant under the considered symmetries. Finally, we revisited the SPT classification in case of time reversal symmetry and showed that time reversal combined with a translation over one lattice site does not give rise to nontrivial phases.

An important application of the classification is in the simulation of 1 + 1D quantum materials. As we have shown, all frieze SPT phases can be put in one-to-one correspondence with transformation properties of the local MPS tensors, up to the appropriate gauge transformations on the virtual level. This demonstrates that taking into account the spatial symmetries of the spin chains at hand, one can tremendously reduce the number of variational parameters characterizing the state, allowing for simulations with higher bond dimensions.

A natural extension of this work would be to consider the classification of two-dimensional SPT phases protected by space group symmetries. The two-dimensional space groups are known as the wallpaper groups, of which there are seventeen. In this case, the relevant tensor network states are the projected entangled-pair states (PEPS), which form the natural two-dimensional generalization of the MPS considered here. We expect that similarly as in the one-dimensional case, imposing the spatial symmetries directly on the local tensors will also reveal topological obstructions. For each of these phases, canonical ansätze could be constructed that might again prove very useful in numerical simulations of physical systems and materials in which these spatial symmetries are ubiquitous. The framework to investigate these spatial symmetries was laid out in Ref. [35], where it was called the *crystalline equivalence principle*. This principle states that the classification of phases protected by a spatial symmetry group G is the same as that of the SPT phases with

G as global on-site symmetry but acting in a “twisted” way, where orientation-reversing symmetry actions correspond to antiunitary operators and thus to nontrivial group actions. In the tensor network framework, this result was also obtained in Refs. [19,36].

In particular, it would be interesting to investigate whether some of the symmetry transformations in 2D admit an implementation on the virtual level as string-like matrix product operators (MPOs). The physical application of the symmetry is then “gauged away” by pulling these MPOs through the lattice. Similarly, it might be interesting to demonstrate that also time reversal, which, because of the complex conjugation, contains a priori a very nonlocal symmetry, can be implemented using an MPO of finite bond dimension. In this case, we expect that the consistency equation governing such an SPT phase corresponds to the 3-cocycle equation with a nontrivial group action, which can readily be solved numerically using the algorithm proposed in Sec. III A. Time-reversal in itself gives rise to only trivial 3-cocycles, $H_{\beta^r}^3(\mathbb{Z}_2^T, \mathbf{U}_1) = 0$. The smallest group containing time reversal $\mathbb{Z}_2^T$ as a direct summand which gives rise to nontrivial 3-cocycles is $\mathbb{Z}_2 \times \mathbb{Z}_2^T$ since $H_{\beta^r}^3(\mathbb{Z}_2 \times \mathbb{Z}_2^T, \mathbf{U}_1) = \mathbb{Z}_2^{\times 2}$. With the crystalline equivalence principle in mind, another example to consider is $\mathbb{Z}_4 \rtimes \mathbb{Z}_2^T$, $H_{\beta^r}^3(\mathbb{Z}_4 \rtimes \mathbb{Z}_2^T, \mathbb{C}^\times) = \mathbb{Z}_2^{\times 2}$, as the SPT classification of this group is expected to coincide with that of the space group D_4 . We plan to investigate this in future work.

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APPENDIX A: PROOF OF $X\bar{\rho}X^\dagger = \rho$

In this Appendix, we prove the identity $X\bar{\rho}X^\dagger = \rho$ which was used in Sec. V A. Hereto we start from the fact that ρ was defined as the unique right eigenvector of the transfer matrix $\mathbb{E}$ (6) corresponding to the eigenvalue one. Taking the complex conjugate of the eigenvalue equation and exploiting unitarity of U , we can show that $X^\dagger \bar{\rho} X$ is also a right eigenvector with eigenvalue one which because of injectivity and thus nondegeneracy of this eigenvalue has to be equal to ρ : $X\bar{\rho}X^\dagger = \rho$. Pictorially

$$\rho = \text{Tr}(A \bar{A} \rho) \quad (\text{A1})$$

$$\bar{\rho} = \text{Tr}(\bar{A} A \bar{\rho}) \quad (\text{A2})$$

$$\bar{\rho} = \text{Tr}(U \bar{A} U^\dagger \bar{\rho}) \quad (\text{A3})$$

$$\bar{\rho} = \text{Tr}(X A X^\dagger \bar{\rho}) \quad (\text{A4})$$

$$\bar{\rho} = \text{Tr}(\bar{A} X^\dagger \bar{\rho}) \quad (\text{A5})$$

APPENDIX B: EXAMPLE: 2-COCYCLES OF $\mathbb{Z}_2^r$

In this Appendix, we briefly illuminate the algorithm laid out in Sec. III A for computing explicit cocycle representatives. We focus on one of the smallest interesting examples, $H_{\beta^r}^2(\mathbb{Z}_2^P, \mathbf{U}_1)$, which has a nontrivial group action. The cocycle equations to be solved thus read

$$\omega(g, h) + \omega(gh, k) = \omega(g, hk) + \beta_g^P(\omega(h, k)) \mod 2\pi, \quad (\text{B1})$$

where β_g^P is multiplying with -1 when g is the nontrivial element of $\mathbb{Z}_2^P$ and the identity otherwise.

The first step of the algorithm consists of filling up the $\Omega^{(2, \beta^r)}$ -matrix (20), taking into account the nontrivial group action. Denoting the group elements of $\mathbb{Z}_2^P$ by $\{0, 1\}$, we obtain

$$\Omega^{(2, \beta^r)} = \begin{matrix} & \begin{matrix} (0;0) & (0;1) & (1;0) & (1;1) \end{matrix} \\ \begin{matrix} (0;0;0) \\ (0;0;1) \\ (0;1;0) \\ (0;1;1) \\ (1;0;0) \\ (1;0;1) \\ (1;1;0) \\ (1;1;1) \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 1 & -2 \end{pmatrix} \end{matrix} \quad (\text{B2})$$

This matrix is then written in Smith normal form as $P\Omega^{(2, \beta^r)}R = \Lambda$. The basis transformation P contains only integers and can be discarded in solving the cocycle equations as

these are solved modulo 2π . The R matrix reads

$$R = \begin{matrix} & \begin{matrix} (0;0) & (0;1) & (1;0) & (1;1) \end{matrix} \\ \begin{matrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{matrix} & \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}. \quad (\text{B3})$$

From the Λ matrix,

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{B4})$$

the group cohomology is immediately found to be $H_{\text{gr}}^2(\mathbb{Z}_2^P, \mathbf{U}_1) = \mathbb{Z}_2$: the 1 diagonal entries corresponds to trivial cocycles, whereas the entry 2 corresponds to a nontrivial cocycle. We then compute

$$R^{-1} \Lambda^+ \vec{v} = \begin{matrix} & \begin{matrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{matrix} \end{matrix} \begin{pmatrix} v_1 - \frac{v_3}{2} \\ v_2 - \frac{v_3}{2} \\ \frac{v_3}{2} \\ 0 \end{pmatrix}, \quad (\text{B5})$$

where $\vec{v}$ is an arbitrary integer vector. The nontrivial cocycle valued in $\mathbf{U}_1$ that generates the $\mathbb{Z}_2$ cohomology group is then found from (B5) by choosing $v_3 = 1$ and reads

$$\vec{\omega} = \begin{matrix} & \begin{matrix} (0;0) \\ (0;1) \\ (1;0) \\ (1;1) \end{matrix} \end{matrix} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix}. \quad (\text{B6})$$

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Mapping between Morita equivalent string-net states with a constant depth quantum circuit

7.1 Synopsis

As argued in section 1.3, two models belong to the same gapped quantum phase of matter if they are related by an interpolating quantum circuit of sub-linear depth. In this project, we explicitly construct such a constant-depth circuit for string-net models whose input UFCs $\mathcal{D}_1$ and $\mathcal{D}_2$ are *Morita equivalent*. The circuit is built explicitly from the data of an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category, specifically the F -symbols.

Concretely, we first review in detail the framework of *extended string-net models*, mentioned above in section 3.2.3. These extended string-nets provide a framework in which all anyonic excitations encoded in $\mathcal{Z}(\mathcal{D})$, including those violating vertex Hamiltonian terms, can be treated in a unified manner. We then use the aforementioned associator data encoded in an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category to define the quantum circuit and compute its action on both ground and excited states. When restricted to the ground state sector, the circuit reduces precisely to the generalised ground-state projector introduced in section 3.4. Its action on excited states on the other hand requires the introduction of ancillary degrees of freedom that account for mismatches in the dimensions of the central idempotents describing the

anyons in the two theories. In that case, the circuit acts on individual plaquettes via *bimodule tubes*, akin to those introduced in section 4.4.

As an illustrative example, we show that, when applied to the *toric code*, our construction reproduces the familiar *electric–magnetic duality*. More generally, for *quantum double models* based on possibly non-abelian input groups, it recovers a quantum circuit previously introduced in refs. [115, 116].

7.2 Reference

Mapping between Morita equivalent string-net states with a constant depth quantum circuit

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7.3 Contributions of the author

The central idea underlying this project, together with its overall technical framework, was developed by Laurens Lootens in collaboration with Norbert Schuch and Frank Verstraete. My primary contribution was the explicit derivation of the quantum circuit in terms of the bimodule data. In addition, I contributed to the writing of the manuscript, including portions of the review of extended string-net models, the presentation of the quantum circuit, and the examples.

Mapping between Morita-equivalent string-net states with a constant depth quantum circuit

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We construct a constant depth quantum circuit that maps between Morita-equivalent string-net models. Due to its constant depth and unitarity, the circuit cannot alter the topological order, which demonstrates that Morita-equivalent string nets are in the same phase. The circuit is constructed from an invertible bimodule category connecting the two input fusion categories of the relevant string-net models, acting as a generalized Fourier transform for fusion categories. The circuit not only acts on the ground state subspace but also acts unitarily on the full Hilbert space when supplemented with ancillas.

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I. INTRODUCTION

Ever since their conception, the string-net models as originally proposed by Levin and Wen [1] and their subsequent generalizations [2–7] have provided a rich playground for studying microscopic realizations of nonchiral topologically ordered phases of matter in $2 + 1$ dimensions. Taking a unitary fusion category (UFC) $\mathcal{D}$ [8] as input, these exactly solvable models allow for the explicit realization of several key features of topologically ordered systems, such as ground-state degeneracies that depend on the topology of the space and anyonic quasiparticle excitations that satisfy nontrivial braiding statistics. From the category-theoretical side, the different ground states and excitations are described by the monoidal center $Z(\mathcal{D})$ of the input UFC $\mathcal{D}$ [2], which is itself a unitary modular fusion category (UMFC) and describes the topological order. The fact that the UMFC describing the topological order is always the center of some other UFC is the reason why string-net models are only able to describe nonchiral topological order with gappable boundaries; exactly solvable lattice models for chiral topological order have proven more challenging to obtain [9].

An important observation is that the process of going to the center $Z(\mathcal{D})$ of a UFC $\mathcal{D}$ is not an injective operation, or, put differently, that distinct UFCs $\mathcal{D}_1$ and $\mathcal{D}_2$ can have the same center $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$ [2]. This has led to the conjecture that two such string-net models based on different $\mathcal{D}_1$ and $\mathcal{D}_2$ belong to the same topological phase, despite the fact that microscopically they can look very different. UFCs with the same center are said to be Morita equivalent [10], and the collection of UFCs Morita equivalent to $\mathcal{D}$ is called its Morita class. A direct consequence of Morita equivalence is that the topological excitations or anyons of two Morita-equivalent string-net models are in one-to-one correspondence with each other. Kitaev and Kong demonstrated the existence of an invertible domain wall between two Morita-

equivalent string-net models through which anyons can move freely without condensing on the domain wall [2]. Mathematically, the existence of such an invertible domain wall is guaranteed by the fact that, for any two Morita-equivalent UFCs $\mathcal{D}_1$ and $\mathcal{D}_2$, there exists an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$ [8]. The central goal of this paper is to show that this implies the existence of a constant depth quantum circuit which maps the different string-net states onto each other.

From a quantum information point of view, it has been understood that two states are in the same phase if there exists a constant depth quantum circuit that is able to map between the two [11–13]. In this work, we provide such a constant depth circuit for the case of Morita-equivalent string nets $\mathcal{D}_1$ and $\mathcal{D}_2$ that maps the ground states as well as the excitations of one model to those of the other. For the ground state, the quantum circuit can be understood as a generalized version of the Hamiltonian, using the invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$ to intertwine the Hamiltonian of the string-net $\mathcal{D}_1$ to the Hamiltonian of the string-net $\mathcal{D}_2$. Our construction generalizes a previous mapping, which was obtained from Kitaev’s quantum double models [14] for a group G to string-net models $\text{Rep}(G)$ [15,16], to the general case of two Morita-equivalent string nets.

Outline. We begin by providing a brief review of string nets, focusing mainly on a generalization known as the extended string-net models [17]. The reason for this is that these generalizations allow for a more unified treatment of the vertex and plaquette excitations, the two types of excitations in a string-net model. They are understood in the diagrammatic language as idempotents of the tube algebra associated with the string-net UFC $\mathcal{D}$ [3], which is well known to provide a characterization of the center $Z(\mathcal{D})$. Next, we discuss how an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category can be used to provide a map between the tube algebras of the UFCs $\mathcal{D}_1$ and $\mathcal{D}_2$, and thereby show how excitations from one model are mapped to excitations of the other. Taking all this together, we are able to explicitly construct a constant depth unitary quantum circuit which we explain in detail.

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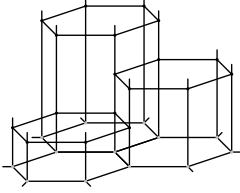


FIG. 1. An illustration of how our circuit acts on the three sublattices of the hexagonal lattice.

For the case of a hexagonal lattice, this circuit is depth three, which follows from the fact that our circuit sequentially acts on three sublattices. This is depicted in Fig. 1 below. The action of the circuit on the three sublattices is worked out in the main text. Each of these local unitaries acts on 18 degrees of freedom associated with a plaquette. We show that the unitary map on the three sublattices is respectively constructed from the different associators of the $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$, which are the solutions to a set of coupled pentagon equations [8, 18]. In this way our construction gives a very concrete quantum information perspective on the categorical notion of Morita equivalence. We also show how the quantum circuit can be generalized to act on the full (extended) string-net Hilbert space and obtain a very similar circuit as in the absence of excitations, the details of which are relegated to the Appendix.

II. STRING NETS

In this section we give a brief review of Levin and Wen's string-net models on the hexagonal lattice. First, we review the ordinary string-net model, herein closely following the original work of Levin and Wen [1], and then we turn to the extended string nets [17]. These can be regarded as a generalization of the ordinary string nets that provide a more natural and convenient setting in which to study and classify excited states.

A. Levin and Wen's string nets

A string-net model is defined by the data of some given input unitary fusion category $\mathcal{D}$. This input data contains a set of simple objects (string types) that we denote $a, b, c, \dots$ with corresponding fusion rules $a \otimes b \simeq \bigoplus_c N_{ab}^c c$, corresponding quantum dimensions $d_a, d_b, d_c, \dots$, and an isomorphism $F : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$. Given some oriented trivalent lattice, such as the hexagonal lattice in this work, the Hilbert space of the string net is defined by configurations of the string types living on the edges of the lattice. Furthermore, for every string type a there exists a corresponding conjugate string type $\bar{a}$ such that $\bar{\bar{a}} \equiv a$ and $\bar{a}$ can be interpreted as a living on the same edge with the opposite orientation. The quantum dimensions obey $d_{\bar{a}} = d_a$. There is a unique self-conjugate vacuum string $\mathbf{1}, \bar{\mathbf{1}} \equiv \mathbf{1}$.

The string-net Hamiltonian consists of two terms:

$$H = - \sum_v A_v - \sum_p B_p. \quad (1)$$

Herein the two sums in the Hamiltonian are over the vertices and plaquettes of the lattice, respectively. The sum over s is a sum over all string types of the input UFC. D is the total quantum dimension defined as $D = (\sum_i d_i^2)^{1/2}$. The vertex operators A_v act on the vertices according to

$$A_v \left| \begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ i \\ \uparrow \\ a \end{array} \right\rangle = \delta_{a,b,\bar{c}} \left| \begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ i \\ \uparrow \\ a \end{array} \right\rangle \quad (2)$$

where $\delta_{a,b,\bar{c}}$ is 1 whenever the fusion of $a \otimes b$ contains $\bar{c}$ and is 0 else. The plaquette operators B_p are defined as

$$B_p = \frac{1}{D} \sum_a d_a B_p^a, \quad (3)$$

where the terms B_p^a acts on a plaquette by inserting a clockwise-oriented string of type a in the plaquette and fusing it to the lattice. This is done by making repeated use of the resolution of the identity for UFCs, namely,

$$\left| \begin{array}{c} a \\ \uparrow \\ a \end{array} \right\rangle \left| \begin{array}{c} b \\ \uparrow \\ b \end{array} \right\rangle = \sum_{c,n} \sqrt{\frac{d_c}{d_a d_b}} \left| \begin{array}{c} a \quad b \\ \swarrow \quad \searrow \\ n \\ \uparrow \\ c \end{array} \right\rangle, \quad (4)$$

the F moves of the given input UFC,

$$\left| \begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ e \quad f \\ \swarrow \quad \searrow \\ a \quad d \end{array} \right\rangle = \sum_{f,m,n} (F_d^{abc})_{e,jk}^{f,mn} \left| \begin{array}{c} b \quad c \\ \swarrow \quad \searrow \\ m \\ \uparrow \\ f \end{array} \right\rangle, \quad (5)$$

and the ‘‘bubble pop’’ identity

$$\left| \begin{array}{c} c' \\ \uparrow \\ m \\ \uparrow \\ n \\ \uparrow \\ c \end{array} \right\rangle \left| \begin{array}{c} c \\ \uparrow \\ c \end{array} \right\rangle = \sqrt{\frac{d_a d_b}{d_c}} \delta_{m,n} \delta_{c,c'} \left| \begin{array}{c} c \\ \uparrow \\ c \end{array} \right\rangle. \quad (6)$$

The matrix elements of this operator are worked out in detail in Refs. [1, 6]. Every operator B_p^a is a 18-body operator in the sense that its action explicitly depends on the six string types living on the edges of the plaquette p , the six multiplicities and the six edges adjacent to the plaquette, but acts diagonally on the latter. It can be shown that both A_v and B_p are Hermitian projection operators, $A_v^2 = A_v$ and $B_p^2 = B_p$, that moreover all commute. Hence, the string-net condensed renormalization

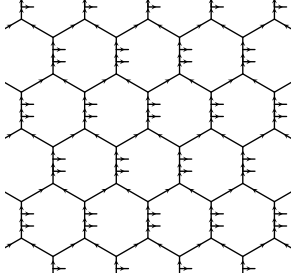


FIG. 2. We adapt the convention that all arrows on the edges point upward and the open legs are attached to the lattice as depicted where every open edge corresponds to the closest vertex of the hexagonal lattice. Multiplicity degrees of freedom are suppressed.

group fixed-point ground states are the simultaneous eigenvectors of all A_v and B_p . Quasiparticle excitations on top of these ground states are obtained by violating at least one of the projector constraints. These excitations are gapped and necessarily come in pairs. Vertices—or electric excitations—are obtained by violating the A_v constraints, whereas plaquettes—or magnetic excitations—violate the B_p constraints. These excitations are treated in a unified way in the language of extended string nets which we now revisit.

B. Extended string nets

The extended string-net models [17] are very similar to the ordinary string-net models except for a few key differences which we discuss in this section.

Given a hexagonal lattice we associate with every vertex an additional open edge that is attached to an edge that is connected to the vertex under consideration. These open edges also carry string degrees of freedom. The choice of which open edge belongs to which vertex is to some extent arbitrary and the open edges can even be moved around the plaquette by a series of F moves but we will adapt the convention denoted in Fig. 2. Furthermore, we demand the fusion rules to be satisfied on every vertex, $\delta_{a,b,c} = 1$. Finally, the Hamiltonian of the extended string net is of the same form as the Hamiltonian of the ordinary string-net model:

$$H = - \sum_v \mathcal{A}_v - \sum_p \mathcal{B}_p, \quad (7)$$

where the $\mathcal{A}_v$ and the $\mathcal{B}_p$ are again mutually commuting Hermitian projectors. $\mathcal{A}_v$ acts on vertices according to

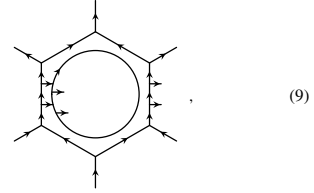
$$\mathcal{A}_v \left| \begin{array}{c} b \\ \nearrow \\ a_2 \\ \downarrow \\ a_1 \end{array} \begin{array}{c} c \\ \searrow \\ d \end{array} \right\rangle = \delta_{d,1} \left| \begin{array}{c} b \\ \nearrow \\ a_2 \\ \downarrow \\ a_1 \end{array} \begin{array}{c} c \\ \searrow \\ d \end{array} \right\rangle. \quad (8)$$

The action of $\mathcal{B}_p$ on the plaquette is explained in detail in Ref. [17] and reduces for $p_1, p_2 = 1$ to the action of the plaquette operator B_p of the ordinary model, where p_1, p_2 denote the open edges of the plaquette p .

The extended string-net model thus deals with vertex excitations in the ordinary string-net model by considering extra degrees of freedom that signal the presence of vertex projector violations.

C. Tube algebra

In the extended string-net models, vertex and plaquette excitations are both characterized by the *tube algebra* [3,17]. The action of a tube $\mathcal{T}_i$ on a plaquette can graphically be depicted as the insertion of the following diagram in a plaquette:



and fusing it to the boundary; here, i runs over all possible configurations that give a nonzero tube. The set of tubes is closed under multiplication,

$$\mathcal{T}_i \mathcal{T}_j = \sum_k f_{ij}^k \mathcal{T}_k, \quad (10)$$

defined by stacking:

$$\begin{array}{c} j \\ \curvearrowright \\ i \end{array} = \sum_k f_{ij}^k \begin{array}{c} k \end{array}, \quad (11)$$

as well as Hermitian conjugation [19,20]:

$$(\mathcal{T}_i)^\dagger = \sum_j u_i^j \mathcal{T}_j. \quad (12)$$

This turns the tube algebra into a finite C^* algebra, for which the Artin-Wedderburn theorem dictates that one can find an isomorphism to a direct sum of simple matrix algebras. Explicitly, this implies the existence of simple idempotents p_{ii}^a and nilpotents $p_{ij}^a, i \neq j$ that satisfy

$$p_{ij}^a = \sum_k x_{ij}^{a,k} \mathcal{T}_k, \quad p_{ij}^a p_{kl}^b = \delta_{ab} \delta_{jk} p_{il}^a, \quad (p_{ij}^a)^\dagger = p_{ji}^a. \quad (13)$$

The elementary excitations of the extended string-net model correspond to the simple idempotents p_{ii}^a of the tube algebra, each of which corresponds to a specific combination of one plaquette excitation and two vertex excitations. These

simple idempotents have to be grouped into central idempotents P_a satisfying

$$P_a = \sum_{i=0}^{n_a-1} P_{ii}^a, \quad [P_a, T_i] = 0, \quad \sum_a P_a = \mathbb{1}, \quad (14)$$

where n_a denotes the dimension of the central idempotent P_a . These central idempotents then correspond to the irreducible representations of the tube algebra and are in one-to-one correspondence with the simple objects of the monoidal center $Z(\mathcal{D})$.

III. BIMODULE CATEGORIES

We consider two string-net models $\mathcal{D}_1$ and $\mathcal{D}_2$ that are supposedly in the same phase; this requires that $\mathcal{D}_1$ and $\mathcal{D}_2$ are Morita equivalent, $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$. From a categorical perspective, this implies the existence of an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$. In this section, we briefly review some properties of a bimodule category, mainly to fix notation,¹ and show how this bimodule category provides a map between the excitations of the two string-net models.

A. Bimodule categories

We start with a spherical UFC $\mathcal{D}_1$, with objects $a, b, c, \dots \in \mathcal{D}_1$ and an associator expressed in a basis of simple objects as

$$\begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ m \\ \swarrow \quad \searrow \\ f \\ \swarrow \quad \searrow \\ a \quad d \end{array} = \sum_{f,mn} (\mathcal{D}_1 F_d^{abc})_{e,jk}^{f,mn} \begin{array}{c} b \quad c \\ \searrow \quad \swarrow \\ j \quad k \\ \swarrow \quad \searrow \\ e \\ \swarrow \quad \searrow \\ a \quad d \end{array}, \quad (15)$$

where in the diagrams we color lines labeled by objects in $\mathcal{D}_1$ in black for the remainder of this work. We also have a second spherical UFC $\mathcal{D}_2$, with objects $\alpha, \beta, \gamma, \dots \in \mathcal{D}_2$ and an associator given by

$$\begin{array}{c} \beta \quad \gamma \\ \searrow \quad \swarrow \\ \mu \\ \swarrow \quad \searrow \\ \alpha \quad \delta \end{array} = \sum_{\nu,mn} (\mathcal{D}_2 F_\delta^{\alpha\beta\gamma})_{\mu,jk}^{\nu,mn} \begin{array}{c} \beta \quad \gamma \\ \searrow \quad \swarrow \\ m \\ \swarrow \quad \searrow \\ \nu \\ \swarrow \quad \searrow \\ \alpha \quad \delta \end{array}, \quad (16)$$

with lines labeled by objects in $\mathcal{D}_2$ drawn in orange. A left $\mathcal{D}_1$ -module category $\mathcal{M}$ is a category $\mathcal{M}$ with objects $A, B, C, \dots \in \mathcal{M}$, a left-action of $\mathcal{D}_1$ on $\mathcal{M}$ given by $a \triangleright A \simeq \bigoplus_B N_{aA}^B B$ and an isomorphism ${}^{\triangleright}F : (a \otimes b) \triangleright A \rightarrow$

$a \triangleright (b \triangleright A)$, which can graphically be expressed as

$$\begin{array}{c} A \\ \uparrow \\ b \quad c \\ \searrow \quad \swarrow \\ j \quad k \\ \swarrow \quad \searrow \\ a \quad B \end{array} = \sum_{C,mn} ({}^{\triangleright}F_B^{abA})_{c,jk}^{C,mn} \begin{array}{c} A \\ \uparrow \\ b \quad m \\ \swarrow \quad \searrow \\ C \\ \swarrow \quad \searrow \\ a \quad n \\ \uparrow \\ B \end{array}, \quad (17)$$

where lines labeled by objects in $\mathcal{M}$ are blue. Similarly, a right $\mathcal{D}_2$ -module category $\mathcal{M}$ has a right-action of $\mathcal{D}_2$ on $\mathcal{M}$ given by $A \triangleleft \alpha \simeq \bigoplus_B N_{A\alpha}^B B$ and an isomorphism ${}^{\triangleleft}F : (A \triangleleft \alpha) \triangleleft \beta \rightarrow A \triangleleft (\alpha \otimes \beta)$, graphically expressed as

$$\begin{array}{c} A \\ \uparrow \\ j \quad k \\ \swarrow \quad \searrow \\ C \\ \swarrow \quad \searrow \\ B \end{array} \begin{array}{c} \alpha \\ \rightarrow \\ m \\ \rightarrow \\ n \end{array} = \sum_{\gamma,mn} ({}^{\triangleleft}F_B^{A\alpha\beta})_{C,jk}^{\gamma,mn} \begin{array}{c} A \\ \uparrow \\ n \quad m \\ \swarrow \quad \searrow \\ B \end{array} \begin{array}{c} \alpha \\ \rightarrow \\ m \\ \rightarrow \\ n \end{array}. \quad (18)$$

A $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$ is then defined as a left- $\mathcal{D}_1$ and right- $\mathcal{D}_2$ module category $\mathcal{M}$ that additionally is equipped with an isomorphism ${}^{\triangleright\triangleleft}F : (a \triangleright A) \triangleleft \alpha \rightarrow a \triangleright (A \triangleleft \alpha)$, graphically depicted as

$$\begin{array}{c} A \\ \uparrow \\ j \quad k \\ \swarrow \quad \searrow \\ C \\ \swarrow \quad \searrow \\ B \end{array} \begin{array}{c} \alpha \\ \rightarrow \\ m \\ \rightarrow \\ n \end{array} = \sum_{D,mn} ({}^{\triangleright\triangleleft}F_B^{aA\alpha})_{C,jk}^{D,mn} \begin{array}{c} A \\ \uparrow \\ m \quad n \\ \swarrow \quad \searrow \\ D \\ \swarrow \quad \searrow \\ B \end{array}. \quad (19)$$

Physically, these bimodule categories arise most evidently in the study of domain walls between different string-net models [2], where the various associativity conditions discussed above amount to the generalizations of the bulk F moves. For a general domain between Morita-equivalent string-net models, the process of pushing excitations through such a domain wall is not reversible, because several excitations can condense on the domain wall. In the special case where excitations can freely move through the domain wall, we are dealing with an *invertible* bimodule category. For every two Morita-equivalent UFCs $\mathcal{D}_1$ and $\mathcal{D}_2$ one can always write down an invertible $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category, and we will use this particular bimodule category in the remainder of this work.

The $(\mathcal{D}_1, \mathcal{D}_2)$ -bimodule category $\mathcal{M}$ does not have an intrinsic duality, but one can define a $(\mathcal{D}_2, \mathcal{D}_1)$ -bimodule category denoted by $\mathcal{M}^{\text{op}}$ or $\overline{\mathcal{M}}$ to contain the dual objects of $\mathcal{M}$. Crucially, this allows us to impose a generalized notion of sphericity [2.21], which we need to perform certain diagrammatic manipulations. Concretely, invertibility of $\mathcal{M}$ can be used to generalize the resolution of the identity (4) of UFCs to a resolution of the identity which allows for the

¹The ${}^{\triangleright}F$, ${}^{\triangleright}F$, ${}^{\triangleright\triangleleft}F$, ${}^{\triangleleft}F$, and ${}^{\triangleright\triangleleft}F$ symbols in this work are denoted as ${}^{\triangleright}F$, ${}^{\triangleright}F$, ${}^{\triangleright}F$, ${}^{\triangleleft}F$, and ${}^{\triangleright}F$ symbols in Ref. [18], respectively.

creation of a $\mathcal{D}_1$ or $\mathcal{D}_2$ line from the fusion of an $\mathcal{M}$ line with an $\mathcal{M}^{\text{op}}$ line:

$$\begin{array}{c} A \\ \downarrow \\ A \end{array} \quad \begin{array}{c} B \\ \downarrow \\ B \end{array} = \sum_{a,n} \sqrt{\frac{d_a}{d_A d_B}} \begin{array}{c} A \quad B \\ \searrow \quad \swarrow \\ n \\ \swarrow \quad \searrow \\ A \quad B \end{array} \quad (20)$$

$$\begin{array}{c} A \\ \downarrow \\ A \end{array} \quad \begin{array}{c} B \\ \uparrow \\ B \end{array} = \sum_{\alpha,n} \sqrt{\frac{d_\alpha}{d_A d_B}} \begin{array}{c} A \quad B \\ \searrow \quad \swarrow \\ n \\ \swarrow \quad \searrow \\ A \quad B \end{array} \quad (21)$$

B. Tube algebra bimodules

Using the bimodule categories $\mathcal{M}$ and $\overline{\mathcal{M}}$ we can write down bimodule tubes for the tube algebras $\mathcal{T}^{\mathcal{D}_1}$ and $\mathcal{T}^{\mathcal{D}_2}$ of $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively. These bimodule tube algebras, denoted by $\mathcal{T}^{\mathcal{M}}$ and $\mathcal{T}^{\overline{\mathcal{M}}}$, are generated by

$$\mathcal{T}_i^{\mathcal{M}} = \begin{array}{c} \text{circle with two arrows entering from the left} \end{array}, \quad \mathcal{T}_i^{\overline{\mathcal{M}}} = \begin{array}{c} \text{circle with two arrows exiting to the right} \end{array} \quad (22)$$

They satisfy

$$\mathcal{T}_i^{\mathcal{M}} \mathcal{T}_j^{\mathcal{D}_1} = \sum_k a_{ij}^k \mathcal{T}_k^{\mathcal{M}}, \quad \mathcal{T}_i^{\mathcal{D}_2} \mathcal{T}_j^{\mathcal{M}} = \sum_k b_{ij}^k \mathcal{T}_k^{\mathcal{M}}, \quad (23)$$

$$\mathcal{T}_i^{\mathcal{D}_1} \mathcal{T}_j^{\overline{\mathcal{M}}} = \sum_k c_{ij}^k \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad \mathcal{T}_i^{\overline{\mathcal{M}}} \mathcal{T}_j^{\mathcal{D}_1} = \sum_k d_{ij}^k \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad (24)$$

$$\mathcal{T}_i^{\overline{\mathcal{M}}} \mathcal{T}_j^{\mathcal{M}} = \sum_k e_{ij}^k \mathcal{T}_k^{\mathcal{D}_1}, \quad \mathcal{T}_i^{\mathcal{M}} \mathcal{T}_j^{\overline{\mathcal{M}}} = \sum_k g_{ij}^k \mathcal{T}_k^{\mathcal{D}_2}, \quad (25)$$

where the last line requires the invertibility of the bimodule category, manifested in the identities (20) and (21). These bimodule tubes are mapped into one another under Hermitian conjugation:

$$\left(\begin{array}{c} A \\ \downarrow \\ A \end{array} \quad \begin{array}{c} A_3 \\ \downarrow \\ A_2 \\ \downarrow \\ A_1 \\ \downarrow \\ A \end{array} \right)^\dagger = \sqrt{\frac{d_{c_1} d_{c_2}}{d_{\gamma_1} d_{\gamma_2}}} \begin{array}{c} A \\ \downarrow \\ A \end{array} \quad \begin{array}{c} A_3 \\ \downarrow \\ A_2 \\ \downarrow \\ A_1 \\ \downarrow \\ A \end{array} \quad (26)$$

This Hermitian conjugation is defined such that the operators obtained from inserting $\mathcal{T}_i^{\mathcal{M}}$ and $(\mathcal{T}_i^{\mathcal{M}})^\dagger$ in a plaquette expressed as matrices acting on the relevant degrees of freedom are related via the usual Hermitian conjugation. Using

these properties, we can define a big C^* algebra generated by $\{\mathcal{T}_i^{\mathcal{D}_1}, \mathcal{T}_j^{\mathcal{M}}, \mathcal{T}_k^{\overline{\mathcal{M}}}, \mathcal{T}_l^{\mathcal{D}_2}\}$, where we take undefined tube multiplications to be zero. This big C^* algebra contains the C^* algebras $\mathcal{T}^{\mathcal{D}_1}$ and $\mathcal{T}^{\mathcal{D}_2}$ as subalgebras. Using the Artin-Wedderburn theorem for this big (finite) C^* algebra, there exists an isomorphism to a direct sum of simple matrix algebras, the number of which equals the number of central idempotents of $\mathcal{T}^{\mathcal{D}_1}$ and $\mathcal{T}^{\mathcal{D}_2}$. In addition to the simple idempotents and nilpotents of $\mathcal{T}^{\mathcal{D}_1}$ and $\mathcal{T}^{\mathcal{D}_2}$, we can define simple bimodules:

$$p_{ij}^{a,\mathcal{M}} = \sum_k \gamma_{ij}^{a,k} \mathcal{T}_k^{\mathcal{M}}, \quad p_{ji}^{a,\overline{\mathcal{M}}} = \sum_k \varepsilon_{ji}^{a,k} \mathcal{T}_k^{\overline{\mathcal{M}}}, \quad (27)$$

$$(p_{ij}^{a,\mathcal{M}})^\dagger = p_{ji}^{a,\overline{\mathcal{M}}}, \quad (28)$$

with $0 \leq i < n_a^{\mathcal{D}_2}$ and $0 \leq j < n_a^{\mathcal{D}_1}$, such that they satisfy

$$\begin{aligned} p_{ij}^{a,\mathcal{D}_1} p_{kl}^{b,\mathcal{D}_1} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_1}, & p_{ij}^{a,\mathcal{D}_2} p_{kl}^{b,\mathcal{D}_2} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_2}, \\ p_{ij}^{a,\mathcal{M}} p_{kl}^{b,\mathcal{D}_1} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{M}}, & p_{ij}^{a,\mathcal{D}_2} p_{kl}^{b,\mathcal{M}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{M}}, \\ p_{ij}^{a,\mathcal{D}_1} p_{kl}^{b,\overline{\mathcal{M}}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\overline{\mathcal{M}}}, & p_{ij}^{a,\overline{\mathcal{M}}} p_{kl}^{b,\mathcal{D}_2} &= \delta_{ab} \delta_{jk} p_{il}^{a,\overline{\mathcal{M}}}, \\ p_{ij}^{a,\overline{\mathcal{M}}} p_{kl}^{b,\mathcal{M}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_1}, & p_{ij}^{a,\mathcal{M}} p_{kl}^{b,\overline{\mathcal{M}}} &= \delta_{ab} \delta_{jk} p_{il}^{a,\mathcal{D}_2}, \end{aligned} \quad (29)$$

where from now on the range of the indices i, j, k, l should be inferred from the context. The simple bimodules will be the essential building blocks of the quantum circuit we aim to construct, providing a map between the ground states and excitations of the string-net model $\mathcal{D}_1$ to the ground states and excitations of the string-net model $\mathcal{D}_2$.

IV. QUANTUM CIRCUIT

Before we construct our quantum circuit, we have to address the point that the total dimension of the Hilbert space need not match between the string-net model $\mathcal{D}_1$ and $\mathcal{D}_2$; the fact that they are Morita equivalent only ensures that number of ground states on some closed manifold is the same. The number of distinct excitations which form a complete basis for the Hilbert space will in general differ between the two models, which is reflected in their tube algebras; since we have $Z(\mathcal{D}_1) \simeq Z(\mathcal{D}_2)$, the number of central idempotents $P_a^{\mathcal{D}_1}$ and $P_a^{\mathcal{D}_2}$ is the same, but their dimensions $n_a^{\mathcal{D}_1}$ and $n_a^{\mathcal{D}_2}$ can differ. This discrepancy in the dimensions of the Hilbert spaces poses an obstacle if we want to construct a unitary quantum circuit that maps the full Hilbert space of one model to the other.

To solve this problem, we will place an ancillary qudit $|j\rangle$ of dimension $n^{\mathcal{D}_2} = \max_a n_a^{\mathcal{D}_2}$ in each plaquette of the string-net model $\mathcal{D}_1$. The original Hilbert space of the string-net model is obtained by fixing these ancillas to $|0\rangle$; we denote states in this Hilbert space as $|\psi^{\mathcal{D}_1}, 0\rangle$. We can then define the following operator:

$$V_{\mathcal{D}_1}^{\mathcal{D}_2} := \left(\sum_{a,i,j} p_{ij}^{a,\mathcal{M}} \otimes |j\rangle\langle i| \right)^{\otimes N}, \quad (30)$$

with N being the number of plaquettes in the system, where $p_{ij}^{a,\mathcal{M}}$ acts on a plaquette as defined before, and $|j\rangle\langle i|$ acts on the corresponding ancilla. The Hermitian conjugate of this

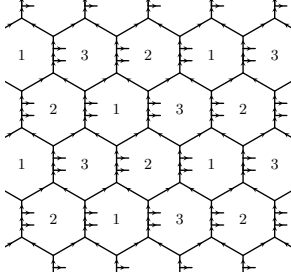


FIG. 3. Division of the hexagonal lattice into three sublattices. We build the quantum circuit by acting on the plaquettes in the order denoted in the figure.

operator is given by

$$(V_{D_1}^{D_2})^\dagger = \left(\sum_{a,i,j} P_{ij}^{a,\overline{\mathcal{M}}} \otimes |j\rangle\langle i| \right)^{\otimes N}, \quad (31)$$

and the product of $V_{D_1}^{D_2}$ with its Hermitian conjugate is

$$\begin{aligned} (V_{D_1}^{D_2})^\dagger V_{D_1}^{D_2} &= \left(\sum_{ab,i',j'} P_{j'i'}^{a,\overline{\mathcal{M}}} P_{ij}^{a,\mathcal{M}} \otimes |i'\rangle\langle j'| |i\rangle\langle i| \right)^{\otimes N} \\ &= \left(\sum_{a,i,j} P_{ij}^{a,D_1} \otimes |i\rangle\langle i| \right)^{\otimes N} \\ &= \left(\sum_a P_a^{D_1} \otimes \mathbb{1}_{n_a} \right)^{\otimes N}. \end{aligned} \quad (32)$$

In general this is a projector implying $V_{D_1}^{D_2}$ is an isometry, but restricting to the original Hilbert space where all ancillas are in the $|0\rangle$ state we get

$$\begin{aligned} (V_{D_1}^{D_2})^\dagger V_{D_1}^{D_2} |\psi^{D_1}, 0\rangle &= \left(\sum_a P_a^{D_1} \otimes \mathbb{1}_{n_a} \right)^{\otimes N} |\psi^{D_1}, 0\rangle \\ &= |\psi^{D_1}, 0\rangle, \end{aligned} \quad (33)$$

meaning that the isometry $V_{D_1}^{D_2}$ is in fact a unitary on the original Hilbert space of the string-net model.

A constant depth quantum circuit is obtained by applying the isometry $V_{D_1}^{D_2}$ in such a way that it is a constant depth operator. For the hexagonal lattice, this can be achieved by dividing the lattice into three sublattices as shown in Fig. 3 and acting sequentially on the first, second, and third sublattice. This implementation of the isometry $V_{D_1}^{D_2}$ turns it into a depth-three quantum circuit. We now discuss this quantum circuit in more detail, first for the ground states, and then for a generic excited state in the original string-net Hilbert space spanned by $|\psi^{D_1}, 0\rangle$.

A. Ground state

The ground states of the string-net model $\mathcal{D}_1$, which we will denote as $|\psi_{\text{GS}}^{D_1}, 0\rangle$, are defined as the eigenvectors of the ground-state projector $P_{\text{GS}}^{D_1}$:

$$\begin{aligned} P_{\text{GS}}^{D_1} |\psi_{\text{GS}}^{D_1}, 0\rangle &= (p_{00}^{0,D_1} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{D_1}, 0\rangle \\ &= |\psi_{\text{GS}}^{D_1}, 0\rangle, \end{aligned} \quad (34)$$

projecting each plaquette onto the simple idempotent p_{00}^{0,D_1} which corresponds to the vacuum; it is equal to the plaquette term of the Hamiltonian,

$$p_{00}^{0,D_1} = \frac{1}{D} \sum_a d_a \mathcal{B}_p^{a,D_1}, \quad (35)$$

which also projects onto the ground state of the vertex term by fixing the extended string-net legs to the trivial string. At first glance, one would expect the simple idempotent p_{00}^{0,D_1} corresponding to the vacuum to be central; the reason this is not the case is that in the extended string-net model, the simple idempotents keep track of one plaquette and two vertex excitations. If the two vertices are excited with a particle-antiparticle pair then together they are in the vacuum, and the corresponding simple idempotent p_{ii}^{0,D_1} , $i \neq 0$ will also be contained in the central idempotent $P_{\mathbb{1}}^{D_1}$ corresponding to the vacuum in $Z(\mathcal{D}_1)$.

Acting with our quantum circuit on the ground state, we obtain

$$V_{D_1}^{D_2} |\psi_{\text{GS}}^{D_1}, 0\rangle = (p_{00}^{0,\mathcal{M}} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{D_1}, 0\rangle. \quad (36)$$

One can readily verify that

$$\begin{aligned} P_{\text{GS}}^{D_2} V_{D_1}^{D_2} |\psi_{\text{GS}}^{D_1}, 0\rangle &= (p_{00}^{0,\mathcal{M}} \otimes |0\rangle\langle 0|)^{\otimes N} |\psi_{\text{GS}}^{D_1}, 0\rangle \\ &= |\psi_{\text{GS}}^{D_2}, 0\rangle, \end{aligned} \quad (37)$$

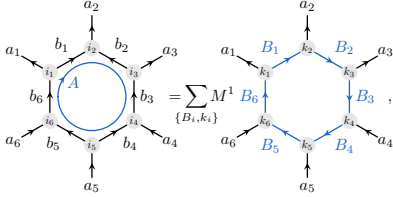
showing that $V_{D_1}^{D_2}$ does indeed provide a unitary map between the ground states of the string-net model $\mathcal{D}_1$ and $\mathcal{D}_2$. The operator $p_{00}^{0,\mathcal{M}}$ can be interpreted as a generalized plaquette term, in the sense that

$$p_{00}^{0,\mathcal{M}} = \frac{1}{D} \sum_A d_A \mathcal{B}_p^{A,\mathcal{M}}, \quad (38)$$

i.e., it is the result of fusing a weighted sum of all loops labeled by $A \in \mathcal{M}$ with the plaquette. When applied to the string-net configuration where all edges are fixed to the trivial label (a product state), $(p_{00}^{0,\mathcal{M}})^{\otimes N}$ projects onto the ground state of the string-net model $\mathcal{D}_2$. This operator can be used to derive a projected entangled pair state (PEPS) representation of the string-net ground state, generalizing the constructions in Refs. [22,23] to the tensor network representations obtained in Ref. [18].

More explicitly, the application of $\mathcal{B}_p^{A,\mathcal{M}}$, $\mathcal{B}_p^{D_1,\mathcal{M}}$, and $\mathcal{B}_p^{G,\mathcal{M}}$ to the three sublattices in the order depicted in Fig. 3 can be worked out by a series of recouplings involving the bimodule

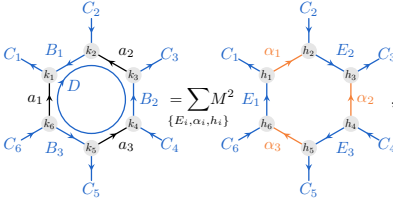
F symbols. For a plaquette of the first kind, we get



where the matrix M^1 depends on $\{a_i, A\}$, which we suppress for convenience, and its components read

$$(M^1)_{[b_i, k_i]}^{[B_i, k_i]} = \sum_{[j_i]} \sqrt{\frac{d_{b_1} d_{b_2} d_{b_3} d_{b_4} d_{b_5} d_{b_6}}{d_{a_1} d_{a_2} d_{a_3} d_{a_4} d_{a_5} d_{a_6}^2}} \left(\triangleright F_{B_6}^{a_1 b_1 A} \right)_{b_6, i_1 j_6}^{B_1, j_1 k_1} \times \left(\triangleright F_{B_1}^{b_1 b_2 B_2} \right)_{a_2, i_2 k_2}^{A, j_1 j_2} \left(\triangleright F_{B_2}^{b_2 a_2 B_3} \right)_{b_3, i_3 j_3}^{B_2, k_2 j_2} \left(\triangleright F_{B_3}^{b_3 a_3 B_4} \right)_{b_4, i_4 j_4}^{B_3, k_3 j_3} \times \left(\triangleright F_{B_4}^{b_4 a_4 B_5} \right)_{a_5, i_5 k_5}^{A, j_4 j_5} \left(\triangleright F_{B_5}^{b_5 a_5 B_6} \right)_{b_6, i_6 j_6}^{B_5, j_5 k_6}. \quad (39)$$

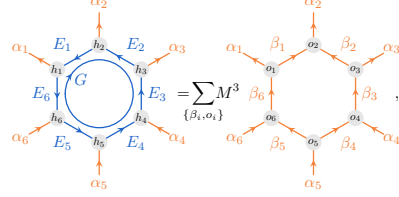
For a plaquette of the second kind, we get



where the matrix M^2 depends on $\{C_i, D\}$ and its components read

$$(M^2)_{[a_i, b_i, k_i]}^{[E_i, \alpha_i, h_i]} = \sum_{[l_i]} \sqrt{\frac{d_{a_1} d_{a_2} d_{a_3} d_{a_4} d_{a_5} d_{a_6}}{d_{C_1} d_{C_2} d_{C_3} d_{C_4} d_{C_5} d_{C_6}^2}} \left(\triangleright \triangleleft F_{E_1}^{a_1 B_1 \alpha_1} \right)_{C_1, k_1 h_1}^{D, j_1 l_1} \times \left(\triangleright \triangleleft F_{D}^{a_2 C_2 \alpha_2} \right)_{B_1, k_2 l_2}^{E_2, h_2 j_2} \left(\triangleright \triangleleft F_{B_2}^{a_3 h_2 \alpha_3} \right)_{D, j_3 l_3}^{C_3, h_3 k_3} \left(\triangleright \triangleleft F_{B_2}^{a_4 E_3 \alpha_4} \right)_{D, j_4 l_4}^{C_4, h_4 k_4} \times \left(\triangleright \triangleleft F_{D}^{a_5 C_5 \alpha_5} \right)_{B_3, k_5 l_5}^{E_3, h_5 j_5} \left(\triangleright \triangleleft F_{E_1}^{a_6 B_6 \alpha_6} \right)_{C_6, k_6 h_6}^{D, l_6 l_1}. \quad (40)$$

Finally, for the third kind of plaquette we get



where the matrix M^3 depends on $\{\alpha_i, G\}$ and its components read

$$(M^3)_{[\beta_i, \alpha_i]}^{[\beta_i, \alpha_i]} = \sum_{[n_i]} \sqrt{\frac{d_{\beta_1} d_{\beta_2} d_{\beta_3} d_{\beta_4} d_{\beta_5} d_{\beta_6}}{d_{\alpha_1} d_{\alpha_2} d_{\alpha_3} d_{\alpha_4} d_{\alpha_5} d_{\alpha_6}^2}} \left(\triangleleft F_{G}^{E_6 \alpha_1 \beta_1} \right)_{E_1, h_1 n_6}^{\beta_6, \alpha_1 n_6} \times \left(\triangleleft F_{E_2}^{E_1 \beta_1 \beta_2} \right)_{G, n_1 n_2}^{\alpha_2, \alpha_2 h_2} \left(\triangleleft F_{E_3}^{G \beta_2 \alpha_3} \right)_{E_2, n_2 n_3}^{\beta_3, \alpha_2 n_3} \left(\triangleleft F_{E_3}^{G \beta_4 \alpha_4} \right)_{E_4, n_4 h_4}^{\beta_5, \alpha_4 n_4} \times \left(\triangleleft F_{E_5}^{E_3 \beta_5 \beta_6} \right)_{G, n_5 n_4}^{\alpha_5, \alpha_5 h_5} \left(\triangleleft F_{G}^{E_6 \alpha_6 \beta_6} \right)_{E_5, h_6 n_5}^{\beta_6, \alpha_6 n_6}. \quad (41)$$

The explicit expression for the operator $(p_{00}^{0, \mathcal{M}})^{\otimes N}$ is then obtained by taking the appropriate linear combinations of the matrices M^i and acting on their respective sublattices in the order described above. While the local Hilbert space of the two string-net models looks different, they have the same number of ground states, and therefore $(p_{00}^{0, \mathcal{M}})^{\otimes N}$ is represented as a square matrix. To show that it is also unitary, we note that

$$(p_{00}^{0, \mathcal{M}})^{\dagger} p_{00}^{0, \mathcal{M}} = p_{00}^{0, \mathcal{D}_1}, \quad (42)$$

implying that the Hermitian conjugate of the matrix representation of $p_{00}^{0, \mathcal{M}}$ should correspond to the matrix representation of $(p_{00}^{0, \mathcal{M}})^{\dagger}$. By virtue of Eq. (26),

$$(p_{00}^{0, \mathcal{M}})^{\dagger} = \frac{1}{D} \sum_A B_A B_p^{A, \overline{\mathcal{M}}}, \quad (43)$$

which implies that the matrix representation of $(p_{00}^{0, \mathcal{M}})^{\dagger}$ is obtained as a linear combination of matrices $\overline{M}$ obtained in similar fashion as the matrices M^i . Using the fact that the different F symbols themselves are unitary matrices since we are dealing with unitary bimodule categories, one can indeed show that $(M^i)^{\dagger} = \overline{M}^i$ through a cumbersome but straightforward computation. Taking everything together, this implies that $(p_{00}^{0, \mathcal{M}})^{\otimes N}$ is represented as a unitary matrix on the ground state.

B. Excited states

When applied to a state with excitations, the mapping provided by our circuit is less obvious since, as stressed before, the dimension of the Hilbert spaces does not match between the string-net model $\mathcal{D}_1$ and $\mathcal{D}_2$. We compensated for this by introducing an ancillary qudit in each plaquette, that we initialize to be in the $|0\rangle$ state. Looking at the action of $V_{\mathcal{D}_1}^{\mathcal{D}_2}$

on the space spanned by $|\psi^{D_1}, 0\rangle$, we find

$$V_{D_1}^{D_2} |\psi^{D_1}, 0\rangle = \left(\sum_{a,j} p_{0j}^{a,\mathcal{M}} \otimes |j\rangle \langle 0| \right)^{\otimes N} |\psi^{D_1}, 0\rangle. \quad (44)$$

Acting on any plaquette of the resulting state with a simple idempotent p_i^{a,D_2} only gives a nonzero result if $i = 0$, meaning that for every central idempotent P_a we only get the excitation corresponding to p_{00}^a . Additionally, this state is no longer part of the original Hilbert space spanned by $|\psi^{D_2}\rangle$, as the ancilla $|j\rangle$ is now entangled with the other degrees of freedom of the string-net model D_2 .

There are a number of ways to disentangle these ancillas, but due to the mismatch in Hilbert space it is impossible to do so in a unique or unitary way. One option is to act with the operator $V_{D_2}^{D_1}$, which gives

$$\begin{aligned} V_{D_2}^{D_1} V_{D_1}^{D_2} |\psi^{D_1}, 0\rangle &= \left(\sum_a \sum_{i=0}^{m_a-1} p_{ii}^{a,\mathcal{M}} \otimes |0\rangle \langle 0| \right)^{\otimes N} |\psi^{D_1}, 0\rangle \\ &= |\psi^{D_2}, 0\rangle, \end{aligned} \quad (45)$$

where $m_a = \min(n_a^{D_1}, n_a^{D_2})$. This state is again part of the original Hilbert space, and the ancilla can be discarded. If $m_a = n_a^{D_1}$, not all excitations of type $a \in Z(D_2)$ in the string-net model D_2 are in the image of this map, while if $m_a = n_a^{D_2}$ some excitations of the string-net model D_1 are projected out, all depending on the choice of enumeration of the simple idempotents. A different approach might be to make sure the energy of the state is conserved when mapping back to the original Hilbert space; we leave the exploration of the different possibilities to future work.

An explicit expression generalizing the matrices M^i to the case of excitations is given in the Appendix. Unitarity of the full circuit can again be shown from unitarity of the F symbols and the fact that the Hermitian conjugate of the bimodule tubes defined in Eq. (26) coincides with the Hermitian conjugate of the matrix representation of the corresponding operator, by definition.

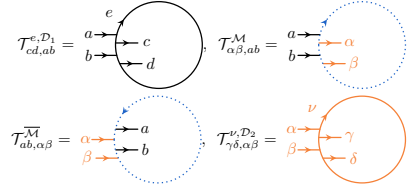
V. EXAMPLES

To illustrate the simple idempotents and simple bimodules that build up our circuit, we write them down for the case of a constant depth circuit that maps the toric code to itself. While this example is almost trivial, it provides enough insight to understand the more general case. We also present the required categorical data for the generic case of a quantum double.

A. Toric code to toric code

The toric code [14] is built from the fusion category $D_1 = \text{Vec}_{\mathbb{Z}_2}$, the category of $\mathbb{Z}_2$ graded vector spaces with two simple objects $\{0, 1\}$ that fuse according to $\mathbb{Z}_2$. It only has one nontrivial module category $\mathcal{M} = \text{Vec}$, the category of finite-dimensional vector spaces with a single simple object $\mathbb{C}$ from which one can construct an invertible bimodule category to $D_2 = \text{Rep}(\mathbb{Z}_2) \simeq \text{Vec}_{\mathbb{Z}_2}$. We can use this bimodule category to construct a constant depth quantum circuit that maps the

toric code to itself, but is not completely trivial. The relevant tubes of this bimodule category can be written as



The central idempotents with their corresponding simple idempotents and nilpotents of the tube algebra of D_1 are labeled by the four excitations in the toric code. These are the vacuum $\mathbb{1}$:

$$\begin{aligned} p_{00}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{00,00}^{0,D_1} + \mathcal{T}_{00,00}^{1,D_1}), & p_{01}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{00,11}^{0,D_1} + \mathcal{T}_{00,11}^{1,D_1}), \\ p_{10}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{11,00}^{0,D_1} + \mathcal{T}_{11,00}^{1,D_1}), & p_{11}^{\mathbb{1},D_1} &= \frac{1}{2}(\mathcal{T}_{11,11}^{0,D_1} + \mathcal{T}_{11,11}^{1,D_1}), \end{aligned}$$

the electric excitation e :

$$\begin{aligned} p_{00}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{01,01}^{0,D_1} + \mathcal{T}_{01,01}^{1,D_1}), & p_{01}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{00,10}^{0,D_1} + \mathcal{T}_{00,10}^{1,D_1}), \\ p_{10}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{10,01}^{0,D_1} + \mathcal{T}_{10,01}^{1,D_1}), & p_{11}^{e,D_1} &= \frac{1}{2}(\mathcal{T}_{10,10}^{0,D_1} + \mathcal{T}_{10,10}^{1,D_1}), \end{aligned}$$

the magnetic excitation m :

$$\begin{aligned} p_{00}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{00,00}^{0,D_1} - \mathcal{T}_{00,00}^{1,D_1}), & p_{01}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{00,11}^{0,D_1} - \mathcal{T}_{00,11}^{1,D_1}), \\ p_{10}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{11,00}^{0,D_1} - \mathcal{T}_{11,00}^{1,D_1}), & p_{11}^{m,D_1} &= \frac{1}{2}(\mathcal{T}_{11,11}^{0,D_1} - \mathcal{T}_{11,11}^{1,D_1}), \end{aligned}$$

and the fermion f :

$$\begin{aligned} p_{00}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{01,01}^{0,D_1} - \mathcal{T}_{01,01}^{1,D_1}), & p_{01}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{01,10}^{0,D_1} - \mathcal{T}_{01,10}^{1,D_1}), \\ p_{10}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{10,01}^{0,D_1} - \mathcal{T}_{10,01}^{1,D_1}), & p_{11}^{f,D_1} &= \frac{1}{2}(\mathcal{T}_{10,10}^{0,D_1} - \mathcal{T}_{10,10}^{1,D_1}). \end{aligned}$$

These central idempotents are all built from two simple idempotents, reflecting the fact that these tube algebras describe the excitations of two vertices and one plaquette. For instance, the simple idempotent $p_{00}^{\mathbb{1},D_1}$ corresponds to the true vacuum, where neither the vertices nor the plaquette is excited. The other simple idempotent $p_{11}^{\mathbb{1},D_1}$ represents the case where both vertices are excited; together, they are in the vacuum, and therefore they also contribute to the vacuum central idempotent. The other central idempotents follow a similar pattern. The simple idempotents and nilpotents of the tube algebra of D_2 are related to those of the tube algebra of D_1 as follows:

$$\begin{aligned} p_{ij}^{\mathbb{1},D_2} &= p_{ij}^{\mathbb{1},D_1}, & p_{ij}^{e,D_2} &= p_{ij}^{m,D_1}, \\ p_{ij}^{m,D_2} &= p_{ij}^{e,D_1}, & p_{ij}^{f,D_2} &= p_{ij}^{f,D_1}. \end{aligned}$$

We note that the electric and magnetic excitations are swapped, which is a manifestation of the e - m duality in the toric code. From these idempotents and nilpotents, one can compute the simple bimodules of $\mathcal{M}$. We find

$$\begin{aligned} p_{00}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,00}^{\mathcal{M}}, & p_{01}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{00,11}^{\mathcal{M}}, \\ p_{10}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{11,00}^{\mathcal{M}}, & p_{11}^{\mathbb{1},\mathcal{M}} &= \frac{1}{\sqrt{2}} \mathcal{T}_{11,11}^{\mathcal{M}}, \end{aligned}$$

for the vacuum,

$$p_{00}^{e,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{00,01}^{\mathcal{M}}, \quad p_{01}^{e,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{00,10}^{\mathcal{M}},$$

$$p_{10}^{e,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{11,01}^{\mathcal{M}}, \quad p_{11}^{e,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{11,10}^{\mathcal{M}},$$

for the electric excitation,

$$p_{00}^{m,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{01,00}^{\mathcal{M}}, \quad p_{01}^{m,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{01,11}^{\mathcal{M}},$$

$$p_{10}^{m,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{10,00}^{\mathcal{M}}, \quad p_{11}^{m,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{10,11}^{\mathcal{M}},$$

for the magnetic excitation and

$$p_{00}^{f,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{01,01}^{\mathcal{M}}, \quad p_{01}^{f,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{01,10}^{\mathcal{M}},$$

$$p_{10}^{f,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{10,01}^{\mathcal{M}}, \quad p_{11}^{f,\mathcal{M}} = \frac{1}{\sqrt{2}} \mathcal{T}_{10,10}^{\mathcal{M}},$$

for the fermion. The simple bimodules of $\overline{\mathcal{M}}$ are obtained by Hermitian conjugation of the simple bimodules of $\mathcal{M}$:

$$p_{ji}^{*,\overline{\mathcal{M}}} = (p_{ij}^{*,\mathcal{M}})^\dagger, \quad p_{ji}^{f,\overline{\mathcal{M}}} = (p_{ij}^{f,\mathcal{M}})^\dagger,$$

$$p_{ji}^{m,\overline{\mathcal{M}}} = (p_{ij}^{m,\mathcal{M}})^\dagger, \quad p_{ji}^{e,\overline{\mathcal{M}}} = (p_{ij}^{e,\mathcal{M}})^\dagger.$$

B. Ground states of Kitaev's quantum doubles to string nets

The previous example of the toric code can now be generalized to a circuit mapping between quantum doubles and string-net ground states. In those examples the group $\mathbb{Z}_2$ is replaced by an arbitrary finite group G . This then amounts to choosing the UFC $\mathcal{D}_1$ to be $\mathcal{D}_1 = \text{Vec}_G$, the category of all finite-dimensional G -graded vector spaces which is the input category of Kitaev's quantum doubles [14]. The simple objects of Vec_G are in one-to-one correspondence with the group elements of G such that the fusion of elements in Vec_G is then equivalent to the group multiplication in G . The ground states of these quantum doubles can then be mapped to string nets with input category $\mathcal{D}_2 = \text{Rep}(G)$, the category of all representations of G in which the simple objects are the irreducible representations. The invertible bimodule category connecting $\mathcal{D}_1$ and $\mathcal{D}_2$ is $\mathcal{M} = \text{Vec}$, as in the case of the toric code. A crucial difference with the previous example is that for a generic group, there is no monoidal equivalence between Vec_G and $\text{Rep}(G)$. The data of this bimodule category is as follows [18]:

(i) $(\mathcal{D}_1 \mathcal{F}_{g_{12}, g_3}^{g_1, g_2, g_3})_{g_{12}, 11} = \omega(g_1, g_2, g_3)$ where ω is a three-cocycle belonging to the trivial class. One can choose a gauge of the $\mathcal{D}_1 \mathcal{F}$ symbol in which this cocycle is identically one, $\omega(g_1, g_2, g_3) \equiv 1, \forall g_1, g_2, g_3 \in G$.

(ii) Writing $(\mathcal{D}_1 \mathcal{F}_{g_{12}, g_3}^{g_1, g_2, g_3})_{g_{12}, 11}^{\mathcal{C}, 11} = \psi(g_1, g_2)$, it can be shown that by virtue of the pentagon equation between $\mathcal{D}_1 \mathcal{F}$ and $\mathcal{D}_2 \mathcal{F}$, ψ is a two-cocycle classified by $H^2(G, U(1))$.

(iii) $(\mathcal{D}_2 \mathcal{F}_{\mathcal{C}, 11}^{g_{\mathcal{C}}, 11})_{\mathcal{C}, 11} = D^a(g)_i^j$ where $D^a(g)$ is the matrix representation of the group element $g \in G$ in the irreducible representation α and where i, j are d_α -dimensional indices in that representation.

(iv) $(\mathcal{C}_{\mathcal{C}, i_1 i_2}^{a_1 a_2 a_3})_{\mathcal{C}, i_1 i_2} = C_{i_1 i_2 i_3, k}^{a_1 a_2 a_3}$ are the Clebsch-Gordan coefficients of the group. In this notation k denotes the possible number of ways in which α_1 and α_2 can fuse to α_3 .

(v) The $\mathcal{D}_2 \mathcal{F}$ -symbols can then be recognized as the Racah W coefficients which are equal to the $6j$ symbols of G . These $6j$ symbols serve as the input of the $\text{Rep}(G)$ string net.

Substituting these $\mathcal{F}$ symbols in the expressions of the quantum circuit then results in a circuit that allows one to map the ground states of all of Kitaev's quantum double models to corresponding string nets. A similar circuit was found in Refs. [15, 16]. In their map the $\text{Rep}(G)$ degrees of freedom where introduced by means of a Fourier transform on G . In our circuit these irreps appear through the $\mathcal{D}_2 \mathcal{F}$ symbol of the bimodel category. We also recover the $6j$ symbols that appeared in their construction in the form of the $\mathcal{D}_2 \mathcal{F}$ symbols. We note that the quantum doubles as defined in Refs. [16, 22] are defined on a triangular lattice and are shown to be equivalent to the string-net model on the hexagonal lattice; our construction does not change the lattice, and a direct comparison is therefore somewhat cumbersome.

VI. CONCLUSION AND OUTLOOK

In this work, we have constructed a constant depth quantum circuit that is able to map between Morita-equivalent ground states. Crucially, the circuit is unitary on the full Hilbert space, meaning that it can be implemented in a truly local way without performing a measurement as would be required to construct the string-net state from a product state. We have shown that the mapping is exact for the ground states, which implies that they are in the same phase as they can be adiabatically connected without closing a gap.

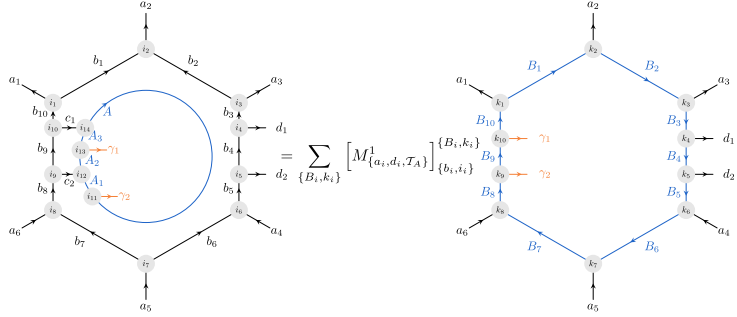
One immediate application of our circuit exploits the recent insight that string-net models can be used to construct partition functions of classical statistical mechanics models [24, 25]. From the tensor network point of view, this is understood as the strange correlator [26], which boils down to taking the overlap of the string-net ground-state PEPS with some unentangled product state to obtain a tensor network representation of a classical partition function. The existence of a quantum circuit between Morita-equivalent string nets allows one to construct different lattice models with the same partition functions. We plan to explore this further in future work.

A natural generalization would be to no longer restrict to closed manifolds but also include the case of boundaries to the vacuum or domain walls between different string-net models. In general this requires the use of a four-object bicategory containing four fusion categories and six invertible bimodule categories [18], the recoupling theory of which has not been written down explicitly in the literature.

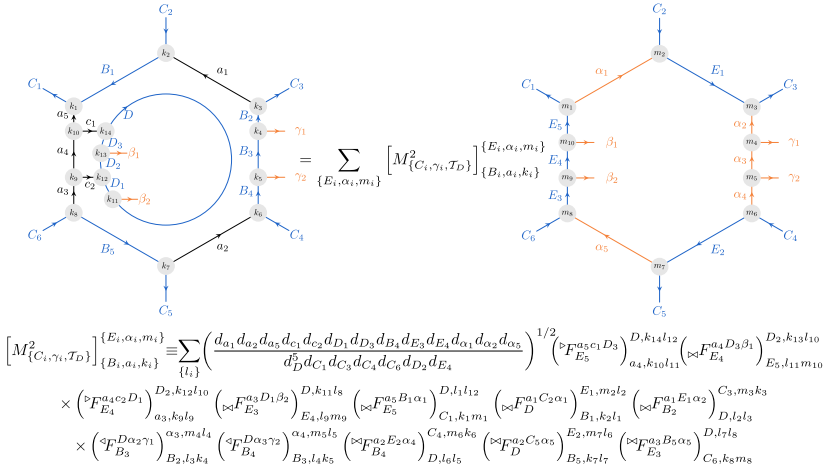
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APPENDIX: THE QUANTUM CIRCUIT FOR THE EXTENDED STRING NET



$$\begin{aligned}
 [M_{\{a_i, d_i, T_A\}}^1]_{\{B_i, k_i\}}^{\{B_i, k_i\}} &\equiv \sum_{\{j_i\}} \left(\frac{d_{b_1} d_{b_2} d_{b_6} d_{b_7} d_{c_1} d_{c_2} d_{A_1} d_{A_3} d_{B_5} d_{B_8}}{d_A^3 d_{a_2} d_{a_5} d_{b_5} d_{b_8} d_{A_2}} \right)^{1/2} \left(\nu F_{B_{10}}^{b_{10} c_1 A_3} \right)_{b_9, i_{10} j_{10}}^{A_1, i_{14} j_{12}} \left(\nu F_{B_9}^{b_9 A_3 \gamma_1} \right)_{B_{10}, j_{11} k_{10}}^{A_2, i_{13} j_{10}} \\
 &\times \left(\nu F_{B_9}^{b_9 c_2 A_1} \right)_{b_8, i_{9} j_9}^{A_2, i_{12} j_{10}} \left(\nu F_{B_8}^{b_8 A_1 \gamma_2} \right)_{B_9, j_9 k_9}^{A_3, i_{11} j_8} \left(\nu F_{B_{10}}^{a_1 b_1 A} \right)_{b_{10}, i_{13} j_{12}}^{B_1, j_1 k_1} \left(\nu F_{B_1}^{b_1 b_2 B_2} \right)_{a_2, i_2 k_2}^{A, j_{12} j_2} \left(\nu F_A^{b_2 a_3 B_3} \right)_{b_3, i_3 j_3}^{B_2, k_3 j_2} \\
 &\times \left(\nu F_A^{b_4 d_2 B_5} \right)_{b_5, i_5 j_5}^{B_4, k_5 j_4} \left(\nu F_A^{b_5 d_1 B_4} \right)_{b_4, i_4 j_4}^{B_5, k_4 j_3} \left(\nu F_A^{b_6 a_4 B_5} \right)_{b_6, i_6 j_6}^{B_6, k_6 j_6} \left(\nu F_{B_7}^{b_7 b_6 B_6} \right)_{a_5, i_7 k_7}^{A, j_6 j_7} \left(\nu F_{B_8}^{a_6 b_7 A} \right)_{b_8, i_8 j_8}^{B_7, j_7 k_8}
 \end{aligned}$$



$$\begin{aligned}
 [M_{\{C_i, \gamma_i, T_D\}}^2]_{\{B_i, a_i, k_i\}}^{\{E_i, \alpha_i, m_i\}} &\equiv \sum_{\{l_i\}} \left(\frac{d_{a_1} d_{a_2} d_{a_5} d_{c_1} d_{c_2} d_{D_1} d_{D_3} d_{B_4} d_{E_3} d_{E_4} d_{\alpha_1} d_{\alpha_2} d_{\alpha_5}}{d_D^5 d_{C_1} d_{C_3} d_{C_4} d_{C_6} d_{D_5} d_{E_4}} \right)^{1/2} \left(\nu F_{E_5}^{a_5 c_1 D_3} \right)_{a_4, k_{10} l_{11}}^{D, k_{14} l_{12}} \left(\nu F_{E_4}^{a_4 D_3 \beta_1} \right)_{E_5, l_{11} m_{10}}^{D_2, k_{13} l_{10}} \\
 &\times \left(\nu F_{E_4}^{a_4 c_2 D_1} \right)_{a_3, k_9 l_9}^{D_2, k_{12} l_{10}} \left(\nu F_{E_3}^{a_3 D_1 \beta_2} \right)_{E_4, l_9 m_9}^{D, k_{11} l_8} \left(\nu F_{E_5}^{a_5 B_1 \alpha_1} \right)_{C_1, k_1 m_1}^{D, l_{11} l_{12}} \left(\nu F_D^{a_1 C_2 \alpha_1} \right)_{B_1, k_2 l_1}^{E_1, m_2 l_2} \left(\nu F_{B_2}^{a_1 E_1 \alpha_2} \right)_{D, l_2 l_3}^{C_3, m_3 k_3} \\
 &\times \left(\nu F_{B_3}^{D \alpha_2 \gamma_1} \right)_{\alpha_3, m_4 l_4}^{a_4, m_5 l_5} \left(\nu F_{B_4}^{D \alpha_3 \gamma_2} \right)_{B_3, l_4 k_5}^{a_4, m_5 l_5} \left(\nu F_{B_4}^{a_5 E_2 \alpha_4} \right)_{D, l_6 l_5}^{C_4, m_6 k_6} \left(\nu F_D^{a_2 C_3 \alpha_5} \right)_{B_5, k_7 l_7}^{E_2, m_7 l_6} \left(\nu F_{E_3}^{a_3 B_5 \alpha_5} \right)_{C_6, k_8 m_8}^{D, l_7 l_8}
 \end{aligned}$$

$$\begin{aligned}
 & \sum_{\{\delta_i, o_i\}} [M^3_{\{\alpha_i, \gamma_i, \tau_G\}}]_{\{E_i, m_i\}} = \sum_{\{\delta_i, o_i\}} [M^3_{\{\alpha_i, \gamma_i, \tau_G\}}]_{\{E_i, m_i\}} \\
 & [M^3_{\{\alpha_i, \gamma_i, \tau_G\}}]_{\{E_i, m_i\}} \equiv \sum_{\{n_i\}} \left(\frac{d_{d_1} d_{d_2} d_{E_1} d_{E_2} d_{E_3} d_{E_5} d_{E_6} d_{E_7} d_{E_8}^2 d_{E_{10}} d_{G_2} d_{\delta_1} d_{\delta_2} d_{\delta_6} d_{\delta_7}}{d_{G_1}^2 d_{G_3} d_{\alpha_2} d_{\alpha_5} d_{\delta_8}} \right)^{1/2} \left(\varphi_{F_G^{d_1} E_9 \delta_{10}} \right)_{G_3, n_9 m_{14}}^{G_3, n_9 m_{14}} \\
 & \quad \times \left(\varphi_{F_{G_2}^{E_9 \delta_{10} \beta_1}} \right)_{G_3, n_9 m_{13}}^{\delta_9, \alpha_{10} n_8} \left(\varphi_{F_{G_2}^{d_2 E_8 \delta_9}} \right)_{E_9, m_9 m_{12}}^{G_1, n_{11} m_{12}} \left(\varphi_{F_G^{E_8 \delta_9 \beta_{12}}} \right)_{G_1, n_{11} m_{11}}^{\delta_8, \alpha_9 n_{10}} \left(\varphi_{F_G^{E_{10} \alpha_1 \delta_1}} \right)_{E_1, m_{11} n_1}^{\delta_{10}, \alpha_1 n_{12}} \left(\varphi_{F_{E_2}^{E_1 \delta_1 \delta_2}} \right)_{G, n_1 n_2}^{\alpha_2, \alpha_2 m_2} \\
 & \quad \times \left(\varphi_{F_{E_3}^{G \delta_2 \alpha_3}} \right)_{E_2, n_2 m_3}^{\delta_3, \alpha_3 n_3} \left(\varphi_{F_{E_4}^{G \delta_3 \gamma_1}} \right)_{E_3, n_3 m_4}^{\delta_4, \alpha_4 n_4} \left(\varphi_{F_{E_5}^{G \delta_4 \gamma_2}} \right)_{E_4, n_4 m_5}^{\delta_5, \alpha_5 n_5} \left(\varphi_{F_{E_5}^{G \delta_6 \alpha_4}} \right)_{E_6, n_6 m_6}^{\delta_5, \alpha_6 n_5} \left(\varphi_{F_{E_6}^{E_7 \delta_7 \delta_6}} \right)_{G, n_7 m_6}^{\alpha_5, \alpha_7 m_7} \left(\varphi_{F_G^{E_8 \alpha_6 \delta_7}} \right)_{E_7, m_8 n_7}^{\delta_8, \alpha_8 n_8}
 \end{aligned}$$

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From gauging to duality in one-dimensional quantum lattice models

8.1 Synopsis

It is well known that gaugings of $\mathcal{C}$ -symmetric two-dimensional quantum field theories are encoded by (Frobenius) *algebra objects* A in $\mathcal{C}$ [88, 117]. These algebra objects specify both a gaugeable collection of topological lines and a generalised notion of *discrete torsion* through their multiplication map. In this project, we extend the gauging map of Haegeman et al. [118] — applicable to non-anomalous group symmetries — to incorporate non-invertible symmetries and show that a choice of algebra object precisely captures the data required for this generalisation. Our construction precisely recovers the results of ref. [118] for the symmetry category Vec_G . We further demonstrate that, up to constant-depth unitary quantum circuits, the resulting gauging maps are equivalent to the duality intertwiners introduced in refs. [77, 95] and discussed in section 4.2. In addition, we consider the inclusion of symmetry-twisted boundary conditions, present several non-trivial examples, and demonstrate the existence of a symmetric, gapped, injective matrix product state when $\mathcal{C}$ describes an anomaly-free symmetry and is thus fully gaugeable.

8.2 Reference

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8.3 Contributions of the author

This project was conceived by José Garre-Rubio and Laurens Lootens. My main contributions consisted of working out the technical details, completing and systematising the derivations, and incorporating the examples. I wrote the majority of the manuscript, with the exception of section 5 which is due to Laurens Lootens.

From gauging to duality in one-dimensional quantum lattice models

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Abstract

Gauging and duality transformations, two of the most useful tools in many-body physics, are shown to be equivalent up to constant depth quantum circuits in the case of one-dimensional quantum lattice models. This is demonstrated by making use of matrix product operators, which provide the lattice representation theory for global (categorical) symmetries as well as a classification of duality transformations. Our construction makes the symmetries of the gauged theory manifest and clarifies how to deal with static background fields when gauging generalised symmetries.



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1 Introduction

Recent years have witnessed an increased interest in generalised notions of finite global symmetry in quantum theories [23, 29]. These generalisations encompass in particular *non-invertible* symmetries in which the algebraic structure governing the symmetry and its 't Hooft anomaly is no longer a group, but instead a (*higher*) *fusion category* [25, 62, 67, 68, 73]. Such symmetries occur ubiquitously in both the continuum [25, 29, 47, 73] as well as in lattice models [3, 35, 50, 56, 58, 69, 70, 74, 78]. The focus of this manuscript is on fusion categorical symmetries in one-dimensional lattice models.

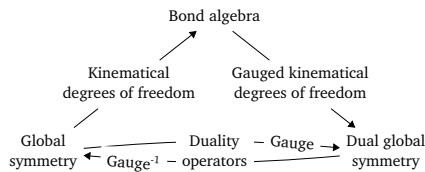
Based on tensor network representations of string-net ground state models [4, 9, 18, 30, 54, 80] – which are a particular case of the Turaev-Viro-Barrett-Westbury state-sum construction with symmetry category $\mathcal{C}$ as input – explicit *matrix product operator* (MPO) representations of fusion categorical symmetries have been classified and realised [9, 12, 69, 70, 74, 82]. Succinctly, for a categorical symmetry $\mathcal{C}$ each choice of *module category* $\mathcal{R}$ over it provides an explicit MPO representation of $\mathcal{C}$ [21, 62]. Subsequently, for the chosen module category, local operators that commute with the categorical symmetry can be built from *generalised Clebsch-Gordan coefficients* [8], thereby providing a systematic machinery to construct local lattice Hamiltonians with a certain categorical symmetry. In particular, the *algebra of local symmetric operators* [15, 42, 44] is encapsulated in the *Morita dual* fusion category $\mathcal{C}_{\mathcal{R}}^*$, and is thus fixed by the choice of symmetry and its representation. Generically these MPOs and local Hamiltonians are represented on Hilbert spaces without local tensor product structure, but which rather obey local constraints [1, 7, 28].

Given such a $\mathcal{C}$ -symmetric model, it was demonstrated in a recent series of papers [50, 51], how different choices of module categories $\mathcal{M}$ over $\mathcal{C}$ lead to distinct *dual* theories. Notable examples of such exact lattice dualities include Kramers-Wannier, Jordan-Wigner and Kennedy-Tasaki duality [45, 48, 49, 61]. Under this operation the symmetry of the dual model is given by the Morita dual of $\mathcal{C}$ with respect to $\mathcal{M}$. Physically, these categorical dualities are characterized by the fact that *local symmetric* operators are mapped to *dual local* operators which commute with the dual symmetry and *charged local* operators are mapped to *string operators* [15, 59]. From the skeletal data of $\mathcal{M}$, MPOs can be constructed which intertwine the dual models.

On the other hand, it is known that gaugings of $\mathcal{C}$ -symmetric models in two spacetime dimensions are given by *special haploid symmetric Frobenius algebra objects* A in $\mathcal{C}$ [25, 62, 66]. Such an algebra object encodes both a collection of symmetry lines that are gauged as well as a choice of *discrete torsion*. On the level of the partition function the gauged model is obtained

from the ungauged partition function by inserting a network of lines labelled by A on a dual triangulation of the underlying manifold [11, 16, 19, 72]. The defining properties of A then guarantee invariance of the gauged partition functions under recouplings of the A defects. In this setting it is understood that the topological defects of the gauged model correspond to *bimodules* over A , encoded in the fusion category $\text{Bimod}_{\mathcal{C}}(A)$ [11, 25]. Notably, the gauging is not determined by A itself, but rather by its *Morita class*. Indeed, given A , the category $\text{Mod}_{\mathcal{C}}(A)$ of A -modules can be constructed, which constitutes a module category over $\mathcal{C}$. As such, different algebra objects can produce the same category of modules, which ultimately dictates the gauging.

In this work, we make this connection between dualities and gauging of categorical symmetries via algebra objects explicit. Given a symmetry $\mathcal{C}$ and MPO representation specified by $\mathcal{R}$, we construct a *gauging map* following the approach taken in refs. [39, 71]. More precisely, from a chosen A we construct a local *Gauss law*, from which in turn a global projector onto the gauge-invariant subspace is made. This point-of-view provides an interpretation of the defects contained in A as being the *gauge degrees of freedom* for the symmetry, in accordance with ref. [24, 71]. We then proceed by showing that the gauging map is equivalent to an MPO duality operator of ref. [50, 51] by means of a constant-depth unitary quantum circuit. This circuit maps the gauging map into an *effective Hilbert space* characterized by local constraints encoded in the module category $\mathcal{R}'$ over $\mathcal{C}_{\mathcal{R}}^*$. As detailed in the main text, $\mathcal{R}'$ is fully specified by the choice of $\mathcal{M}$ and $\mathcal{R}$. As such, the dual global symmetry is manifestly given by $\mathcal{C}_{\mathcal{M}}^*$, which is equivalent to $\text{Bimod}_{\mathcal{C}}(A)$ [11, 53]. The procedure can be summarized in a triangle diagram:



This diagram commutes in the sense that fixing the module category $\mathcal{R}$ representing the global symmetry $\mathcal{C}$ on the lattice – corresponding to the left side of the triangle – and a duality $\mathcal{M}$ – the bottom line – the dual symmetry and its lattice representation is fixed. Also note that the bond algebra of local symmetric operators, encoded in $\mathcal{C}_{\mathcal{R}}^*$, is shared by both the ungauged and the gauged model.

The dualities of ref. [50, 51] are isometries only when the Hilbert space is supplemented with an extra degree of freedom representing the *symmetry-twisted boundary condition*. In the presence of such twists, the Hilbert space decomposes in *topological sectors* which are permuted under the duality mappings [6, 55, 64, 65]. We also provide a version of our gauging map in the presence of such a twist and demonstrate how it maps to a collection of *duality tubes*.

Our results can be extended in a number of ways. For one, the algebra objects considered in this manuscript admit higher-dimensional generalisations. In particular for the case of two spatial dimensions, it is expected that higher-dimensional generalisations of the gauging map [39, 58, 77] can be related to the previously constructed duality maps of ref. [20] via a circuit generalising the one presented here. On the other hand, throughout the manuscript we restrict to internal symmetries. A natural question is how spacetime symmetries such as reflection or time-reversal symmetry fit in the framework. This would require addressing the notion of orientation-reversing domain walls and their Gauss law. We also expect that in this case the tensor network formalism will provide insight, given existing partial progress in this direction [17, 43, 75].

Organisation of the manuscript: We begin sec. 2 by reviewing generic properties of the MPO symmetries considered in this manuscript, before proceeding to the construction of the gauging map from a choice of algebra object. To this end we define a generalised commuting Gauss law from which a global projector onto the gauge-invariant subspace is defined. In sec. 3 we review the construction of duality MPOs and show the unitary equivalence with the gauging map in sec. 4. In section 5 we comment on the case of full gaugeability and how this implies the existence of a short-range entangled symmetric state, and the absence of an anomaly. We then provide a gauging map acting on symmetry-twisted boundary conditions in sec. 6. Algebra objects and their corresponding module categories for the category of representations of the symmetric group S_3 and the Haagerup fusion categories are presented in sec. 7. A review of the relevant category theory used throughout the manuscript is relegated to the appendix. A guide to our notation is provided in the following table.

Table 1: Overview of the key physical concepts, corresponding categorical structures, and their notation used throughout the manuscript.

Physical notion	Categorical notion	Notation
Global symmetry ungauged model	Fusion category	$\mathcal{C}$
Kinematical degrees of freedom	$\mathcal{C}$ -module category	$\mathcal{R}$
Algebra symmetric operators	Morita dual fusion category	$\mathcal{C}_{\mathcal{R}}^*$
Gaugeable subsymmetry	Frobenius algebra in $\mathcal{C}$	A
Duality operators	$\mathcal{C}$ -module category	$\mathcal{M} \simeq \text{Mod}_{\mathcal{C}}(A)$
Gauged kinematical degrees of freedom	$\mathcal{C}_{\mathcal{M}}^*$ -module category	$\mathcal{R}'$
Global symmetry gauged model	Morita dual fusion category	$\mathcal{C}_{\mathcal{M}}^* \simeq \text{Bimod}_{\mathcal{C}}(A)$

2 Gauging procedure for algebra objects

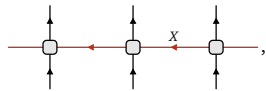
We construct a gauging map for any finite symmetry described by a unitary fusion category $\mathcal{C}$ from a choice of haploid symmetric separable Frobenius algebra object in $\mathcal{C}$. Such an algebra object fully characterises an anomaly-free subset of symmetries in $\mathcal{C}$ together with a choice of generalised discrete torsion. This gauging map directly generalises the one from ref. [39], which was formulated for an anomaly-free invertible symmetry.

In this manuscript we are concerned with generalised finite symmetries described by unitary fusion categories (UFCs). It was shown in ref. [54] that any UFC admits a faithful and internal matrix product operator (MPO) representation of finite bond dimension provided an extra collection of data encoded in a choice of (left) module category over the symmetry category. Since the proposed generalised gauging procedure is oblivious to the concrete lattice realisation of the symmetry, we first review the main properties of the symmetry MPOs before turning to the construction of the gauging map.

Throughout this manuscript we rely heavily on the graphical calculus for tensor networks in which a generic MPO tensor is depicted as

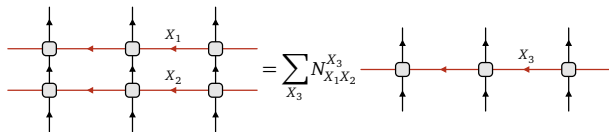
$$\begin{array}{c} \text{---} \circ \text{---} \\ \updownarrow \end{array} \stackrel{X}{=} \sum_{i,j} \sum_{\alpha,\beta} \alpha \begin{array}{c} \text{---} \circ \text{---} \\ \updownarrow \end{array} \stackrel{X}{=} \beta \quad |\alpha\rangle\langle\beta| \otimes |i\rangle\langle j|. \quad (1)$$

We refer to i, j and α, β in the above equation as *physical* and *virtual indices* respectively, borrowing standard tensor network language. All vector spaces we consider are finite-dimensional. In this depiction, and below, the label X is used to both denote the MPO tensor and the virtual space. A uniform and periodic MPO can then be constructed by concatenating individual MPO tensors and contracting basis vectors on the virtual level. In the presence of symmetry twists the symmetry operators have to be modified in a way that is explained below in sec. 6. Graphically, the periodic matrix product operators are



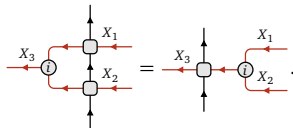
$$(2)$$

with the convention that the left and rightmost virtual indices are identified and summed over so as to implement the periodic boundary condition. Note that the (closed) MPOs are invariant under gauge transformations on the virtual level. From now on we assume without loss of generality that, potentially after carrying out such a gauge transformation, the symmetry MPOs are block-injective, meaning that the MPO tensors are block diagonal and each block generates an injective symmetry MPO [38]. At the level of the virtual spaces, this boils down to a block decomposition of the form $X \simeq \bigoplus_i N_{X_i}^X X_i$, where some of the blocks may appear multiple times given by the integer $N_{X_i}^X$, and the total number of distinct blocks is finite. Furthermore we assume that the MPOs are closed under composition. Concretely, this means that for any triple of blocks X_1, X_2, X_3 there exists an integer denoted by $N_{X_1 X_2}^{X_3}$, referred to as *N-symbol*, such that independent of system size one has:



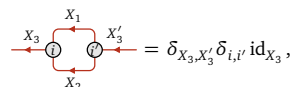
$$(3)$$

Herein, the sum is interpreted so as to run over all injective blocks with the convention that $N_{X_1 X_2}^{X_3} = 0$ whenever X_3 does not appear in the block decomposition of $X_1 \otimes X_2$. Among the injective blocks there is a distinguished one labelled by $\mathbb{1}$ such that $N_{X_1 \mathbb{1}}^{X_2} = N_{\mathbb{1} X_1}^{X_2} = \delta_{X_1, X_2}$ for all X_1, X_2 . As such, N defines a *fusion structure* on the collection of blocks $\{X_i\}_i$. Further requiring that (3) holds for all boundary conditions, the fundamental theorem of MPOs implies there exists an intertwining *fusion tensor* that implements the above decomposition on the level of the individual MPO tensors [9]. Graphically, this intertwining property can be depicted as



$$(4)$$

Herein, $i = 1, \dots, N_{X_1 X_2}^{X_3}$ labels the $N_{X_1 X_2}^{X_3}$ linearly independent fusion tensors and corresponds to the multiplicity of the block X_3 in the decomposition of X_1 and X_2 . We are also guaranteed the existence of a corresponding right inverse, or *splitting tensor*, satisfying the orthogonality condition



$$(5)$$

so that the multiplicity indices i and i' label dual orthogonal bases for the *fusion* and *splitting* space situated on the corresponding tensors, as well as a completeness condition spelled out in [9]. As demonstrated therein, associativity of the multiplication of three injective MPOs requires the existence of a set of invertible matrices, henceforth called *F-symbols*, that encode the two distinct ways in which three injective MPOs can be recoupled. Graphically this statement boils down to the recoupling identity

$$\begin{array}{c} \text{Diagram 1: } X_4 \text{ enters a circle, } X_5 \text{ enters from above, } X_1 \text{ exits to the right, } X_2 \text{ exits from below, } X_3 \text{ exits to the left.} \\ \text{Diagram 2: } X_4 \text{ enters a circle, } X_5 \text{ enters from above, } X_1 \text{ exits to the right, } X_2 \text{ exits from below, } X_3 \text{ exits to the left.} \end{array} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, i j}^{X_6, k l} \begin{array}{c} \text{Diagram 3: } X_4 \text{ enters a circle, } X_5 \text{ enters from above, } X_1 \text{ exits to the right, } X_2 \text{ exits from below, } X_3 \text{ exits to the left.} \end{array}, \quad (6)$$

which has to be satisfied for every choice of the external labels X_1, X_2, X_3 and X_5, i, j . Consistency of the recoupling of four MPOs reveals that the F-symbols need to satisfy a coherence condition known as the *pentagon equation*, which in turn guarantees that every allowed recoupling of fusion tensors is consistent. Crucially, for the symmetries considered in this manuscript, there exists an adequate basis for the junction spaces in eq. (6) in which all F-symbols are unitary as matrices in the sense that

$$\sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, i j}^{X_6, k l} (\overline{F_{X_4}^{X_1 X_2 X_3}})_{X'_5, i' j'}^{X_6, k l} = \delta_{X_5, X'_5} \delta_{i, i'} \delta_{j, j'}, \quad (7)$$

where the overline denotes complex conjugation. Up to basis transformations on the Hom spaces, it turns out that the number of solutions to the pentagon equation is finite for any valid fusion structure as specified by an N-symbol.

The set of labels $\{X_i\}_i$, assuming the existence of a unique trivial label $\mathbb{1}$ and the notion of duals, endowed with a fusion structure and a unitary solution to the pentagon equation exactly constitutes the defining data of a *unitary fusion category* – conventionally denoted by $\mathcal{C}$ – as demonstrated in refs. [9, 54, 82]. Some relevant notions from the theory of fusion categories and the notation we use are summarized in appendix A. In this sense, the MPOs considered in this paper form explicit finite-dimensional lattice representations of the (abstract) graphical calculus of fusion category theory, cfr. eq. (6). In particular this means that we can identify the collection of injective blocks with the set of (representatives of isomorphism classes of) simple objects of $\mathcal{C}$, denoted by $\mathcal{I}_{\mathcal{C}}$, and the F-symbols appearing in the recoupling identity eq. (6) coincide with the matrix elements of the components of the monoidal associator of $\mathcal{C}$, as defined per eq. (A.1). We now proceed to discuss gauging (part of) the $\mathcal{C}$ symmetry.

Following the paradigm of ref. [39], gauging happens by introducing new gauge degrees of freedom between physical sites and enforcing a projector onto the gauge-invariant subspace of this enlarged Hilbert space. This projector is constructed as the product of local commuting projectors that represent a Gauss law. In the invertible symmetry case covered in ref. [39] introducing the gauge degrees of freedom boils down to tensoring the initial – ‘matter’ – Hilbert space with copies of the group algebra $\mathbb{C}[G]$ localized on the links of the lattice. Subsequently, the gauge degrees of freedom are initialized in a configuration which can then be absorbed in the global projection, so as to end up with a gauging map acting on the initial Hilbert space.

As anticipated in the introduction, a choice of *special haploid symmetric Frobenius algebra object* in $\mathcal{C}$ – henceforth often referred to as simply (Frobenius) algebra object – exactly encodes the data needed to generalise the gauging procedure of ref. [39] to fusion-categorical symmetries. Recall that loosely speaking a Frobenius algebra object in $\mathcal{C}$ is a (generically non-simple) object $A \in \bigoplus_{X \in \mathcal{I}_{\mathcal{C}}} N_X^A X$ together with multiplication and comultiplication maps, denoted by $\mu : A \otimes A \rightarrow A$ and $\Delta : A \rightarrow A \otimes A$ respectively, which are (co)associative on the nose and satisfy a number of further criteria spelled out in appendix A. In particular, a finite symmetry $\mathcal{C}$ is called

(fully) gaugeable when the quantum dimensions of all simple objects are integer and the object $\bigoplus_{X \in \mathcal{L}_C} d_X X$ can be endowed with a Frobenius structure. Physically, as an object in $\mathcal{C}$, A exactly encodes the symmetries that are being gauged while the Frobenius structure on A encodes a particular way of gauging them, thereby providing a more general notion of *discrete torsion*. An obstacle arising in the general setting is that the symmetries $\mathcal{C}$ are typically represented on Hilbert spaces that are not naturally equipped with a tensor product structure. Hence, it is not immediately apparent what the appropriate notion of a gauge degree of freedom is in this case. This issue is remedied in the following section where we demonstrate how symmetry twists labelled by A can be inserted in the non-tensor product Hilbert space and how these defects play the role of gauge field in this setting.

In the remainder of this section we will absorb the trivial gauge field directly into the gauging map to obtain the gauging map that acts on the original Hilbert space. Given a Frobenius algebra A in $\mathcal{C}$, we now proceed by proposing a generalised notion of Gauss law [31]. Drawing upon our graphical calculus, we consider the following tensor network operator:

$$\mathcal{P} := \begin{array}{c} A \\ \uparrow \\ \text{---} \bigcirc \text{---} \\ \downarrow \\ A \end{array} \begin{array}{c} A \\ \uparrow \\ \text{---} \bigcirc \text{---} \\ \downarrow \\ A \end{array} \begin{array}{c} A \\ \uparrow \\ \text{---} \bigcirc \text{---} \\ \downarrow \\ A \end{array} . \quad (8)$$

In this we have defined

$$\begin{array}{c} A \\ \uparrow \\ \text{---} \bigcirc \text{---} \\ \downarrow \\ A \end{array} := \sum_{X \in \mathcal{L}_C} N_X^A \begin{array}{c} X \\ \uparrow \\ \text{---} \bigcirc \text{---} \\ \downarrow \\ X \end{array} , \quad (9)$$

and the trivalent tensors are constructed from the fusion and splitting tensors introduced above, with the components of μ and Δ defined per. eq. (A.9) as follows:

$$\begin{array}{c} A \quad A \\ \downarrow \quad \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \mu_{(X_3, i_3), l}^{(X_1, i_1), (X_2, i_2)} \begin{array}{c} X_1 \quad X_2 \\ \downarrow \quad \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ X_3 \end{array} , \quad \begin{array}{c} A \\ \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \quad A \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \Delta_{(X_1, i_1), (X_2, i_2)}^{(X_3, i_3), l} \begin{array}{c} X_3 \\ \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ X_1 \quad X_2 \end{array} . \quad (10)$$

In this definition i_1, i_2, i_3 stand for the multiplicity of X_1, X_2, X_3 in A respectively, and take values in $i_j = 1, 2, \dots, N_{X_j}^A$, $j = 1, 2, 3$. The operator (8) should be interpreted as acting on a physical site and two neighboring gauge fields taking values in the object A . As such, it can be thought of as a localized version of the symmetry MPO (2) restricted to the subset of symmetries encoded in A .

We now show that the $\mathcal{P}$'s are (local) mutually commuting projection operators. Whenever two of these operators overlap in a common gauge degree of freedom, their commutation is graphically depicted as

$$\begin{array}{c} A \quad A \\ \downarrow \quad \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \end{array} \begin{array}{c} A \\ \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \quad A \end{array} \stackrel{(A.8)}{=} \begin{array}{c} A \\ \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \quad A \end{array} \begin{array}{c} A \quad A \\ \downarrow \quad \downarrow \\ \text{---} \bigcirc \text{---} \\ \uparrow \\ A \end{array} . \quad (11)$$

In this figure and those that follow we suppressed A labels when they can be inferred from the context. This commutation property can be directly deduced from the first Frobenius condition (A.8) which is satisfied by the multiplication and comultiplication maps. The fact that $\mathcal{P}$ is a

projector follows in turn from the (co)associativity of μ and Δ together with the intertwining property (4) and the normalization of the Frobenius algebra (A.7):

$$\begin{aligned}
 \mathcal{P}^2 &= \begin{array}{c} \text{Diagram 1: A 2x3 grid of squares. Each square has a horizontal line with a circle (mu) on the left and a triangle (Delta) on the right. Vertical lines have arrows pointing up and down.} \end{array} \stackrel{(A.4)}{=} \begin{array}{c} \text{Diagram 2: A 2x3 grid of squares. The top row has a circle (mu) on the left and a triangle (Delta) on the right. The bottom row has a triangle (Delta) on the left and a circle (mu) on the right.} \end{array} \\
 &\stackrel{(4)}{=} \begin{array}{c} \text{Diagram 3: A 2x3 grid of squares. The top row has a circle (mu) on the left and a triangle (Delta) on the right. The bottom row has a triangle (Delta) on the left and a circle (mu) on the right.} \end{array} \\
 &\stackrel{(A.7)}{=} \begin{array}{c} \text{Diagram 4: A 2x3 grid of squares. Each square has a horizontal line with a circle (mu) on the left and a triangle (Delta) on the right. Vertical lines have arrows pointing up and down.} \end{array} = \mathcal{P}.
 \end{aligned} \tag{12}$$

The fact that these projectors mutually commute allows us to unambiguously define the following (global) projection operator on periodic boundary conditions:

$$\begin{array}{c} \text{Diagram 5: A horizontal chain of three squares. Each square has a horizontal line with a circle (mu) on the left and a triangle (Delta) on the right. Vertical lines have arrows pointing up and down.} \end{array}. \tag{13}$$

In order to define a gauging map acting on the original Hilbert space as anticipated we choose a trivial initial gauge configuration which is acted upon by the gauging map. Such a trivial gauge configuration is provided by the unit structure of the Frobenius algebra. Indeed, the unit provides a vector, denoted by $|\mathbb{1}_A\rangle \in \mathcal{C}(\mathbb{1}, A)$, such that

$$\mathcal{G}_A := \begin{array}{c} \text{Diagram 6: A horizontal chain of three squares. Each square has a horizontal line with a circle (mu) on the left and a triangle (Delta) on the right. Vertical lines have arrows pointing up and down.} \end{array} = \begin{array}{c} \text{Diagram 7: A horizontal chain of three squares. Each square has a horizontal line with a circle (mu) on the left and a triangle (Delta) on the right. Vertical lines have arrows pointing up and down.} \end{array}, \tag{14}$$

is exactly the sought-after gauging map. The action of the gauging map $\mathcal{G}_A$ on a local operator O that commutes with the symmetry MPOs eq. (2) is then given via the prescription

$$O_g \cdot \mathcal{G}_A = \mathcal{G}_A \cdot O, \tag{15}$$

with O_g the local gauged operator.

Despite being explicit, the gauging map (14) suffers from a number of shortcomings. Most notably, it is not obvious from its presentation in eq. (14) that the global symmetries of the gauged model are encoded in the fusion category $\text{Bimod}_{\mathcal{C}}(A)$ of A - A -bimodules in $\mathcal{C}$, as argued in ref. [11]. This is particularly troublesome when trying to consider symmetry-twisted boundary conditions in the gauged model, which is necessary to obtain an invertible version of the gauging map. Moreover, gauging maps constructed from *Morita equivalent* Frobenius algebras should be considered physically equivalent even though their MPO representations (14) are distinct. These weaknesses are remedied by considering a basis in which the above gauging map boils down to the duality matrix product operators constructed in ref. [50, 51] based on the MPO representations of bimodule categories of ref. [54]. In the following section we review the duality matrix product operators pioneered in ref. [50, 51] before demonstrating the unitary equivalence with the gauging maps constructed above.

3 Duality matrix product operators

Given a finite symmetry $\mathcal{C}$ we review the construction of ref. [54] of tensor network representations of the $\mathcal{C}$ symmetry operators, labelled by indecomposable module categories over $\mathcal{C}$, and the local operators invariant under them. We review the category theoretic approach to dualities proposed in ref. [50, 51] in which different lattice realisations of this local operator algebra amount to performing a duality. In preparation for the following section we revisit the matrix product operators intertwining the dual local operator algebras.

From the results of ref. [8] one can infer a generalised Wigner-Eckart theorem which states that local operators that commute with a finite symmetry $\mathcal{C}$ can be decomposed into generalised Clebsch-Gordan coefficients. Distinct collections of such coefficients are in one-to-one correspondence with distinct lattice realisations of the symmetry $\mathcal{C}$ and are classified by (finite indecomposable) (left) module categories $\mathcal{R}$ over $\mathcal{C}$, whose definition is revisited in appendix A.2. More precisely, these coefficients boil down to (the components of) the module associator of $\mathcal{R}$ as a (right) module category over the Morita dual fusion category $\mathcal{C}_{\mathcal{R}}^*$. Recall that $\mathcal{C}_{\mathcal{R}}^*$ is defined as $\text{Func}(\mathcal{R}, \mathcal{C})$, i.e. the category of $\mathcal{C}$ -module endofunctors of $\mathcal{R}$. As such, $\mathcal{R}$ is endowed with the structure of an invertible $(\mathcal{C}, \mathcal{C}_{\mathcal{R}}^*)$ -bimodule category. Below, we denote simple representative objects of $\mathcal{R}$ and $\mathcal{C}_{\mathcal{R}}^*$ by $\{R_m\}_m$ and $\{Y_m\}_m$ respectively. The general framework of tensor network representations of bimodule categories, pioneered in ref. [54], suggests to depict these coefficients in terms of triple line tensors. These tensors generalise those introduced in refs. [4, 30] in the context of PEPS representations of string-net ground states. Concretely, their non-vanishing components evaluate to:

$$\begin{aligned}
 \text{Diagram 1} &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} \text{Diagram 2} |R_1 Y_1 R_3, i\rangle \otimes |R_3 Y_2 R_2, j\rangle \otimes \langle R_1 Y_3 R_2, l| \\
 &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} (F_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl} |R_1 Y_1 R_3, i\rangle \otimes |R_3 Y_2 R_2, j\rangle \otimes \langle R_1 Y_3 R_2, l|, \\
 \text{Diagram 3} &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} \text{Diagram 4} |R_1 Y_3 R_2, l\rangle \otimes \langle R_1 Y_1 R_3, i| \otimes \langle R_3 Y_2 R_2, j| \\
 &\equiv \sum_{\{R_n\}_n} \sum_{i,j,l} (\bar{F}_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl} |R_1 Y_3 R_2, l\rangle \otimes \langle R_1 Y_1 R_3, i| \otimes \langle R_3 Y_2 R_2, j|.
 \end{aligned} \tag{16}$$

Note that every index of these tensors is labelled by a triple of objects (R, Y, R') and a multiplicity label taking values $1, 2, \dots, N_{RY}^{R'}$, and $k = 1, 2, \dots, N_{Y_1 Y_2}^{Y_3}$. More precisely these label basis vectors in the Hom spaces:

$$\begin{aligned}
 |R_1 Y_1 R_3, i\rangle &\in \mathcal{R}(R_1 \triangleleft Y_1, R_3), \\
 |R_3 Y_2 R_2, j\rangle &\in \mathcal{R}(R_3 \triangleleft Y_2, R_2), \\
 |R_1 Y_3 R_2, l\rangle &\in \mathcal{R}(R_1 \triangleleft Y_3, R_2).
 \end{aligned} \tag{17}$$

Throughout, we assume that these bases are so that the F-symbols are unitary when interpreted as linear maps $F_{R_2}^{R_1 Y_1 Y_2} : \bigoplus_{Y_3} \mathcal{C}_{\mathcal{R}}^*(Y_1 \otimes Y_2, Y_3) \otimes \mathcal{R}(R_1 \triangleleft Y_3, R_2) \xrightarrow{\sim} \bigoplus_{R_3} \mathcal{R}(R_1 \triangleleft Y_1, R_3) \otimes \mathcal{R}(R_3 \triangleleft Y_2, R_2)$.

Note that all triples of objects (R, Y, R') as well as (Y_1, Y_2, Y_3) simultaneously have to obey the fusion rules in order for the corresponding tensor component to be non-vanishing. Moreover, the objects $R_{1,2,3}$ are shared between tensor legs as is reflected in the graphical depiction of these tensors.

In order to connect to the previous section, we remark that by combining the triples and corresponding multiplicity index into a single index, one can depict these tensors in a more standard tensor network notation:

$$\begin{array}{c} \text{Diagram 1: A triple line tensor with three horizontal legs. The top leg is labeled } R_3 \text{ and has two vertical strands labeled } i \text{ and } j. \text{ The middle leg is labeled } Y_1 \text{ and } Y_2 \text{ with a central dot. The bottom leg is labeled } R_1 \text{ and } R_2 \text{ with a central dot. A gray patch is at the bottom.} \\ \equiv \\ \text{Diagram 2: A single tensor with a central dot and four legs. Top-left: } (R_1 Y_1 R_3, i), \text{ top-right: } (R_3 Y_2 R_2, j), \text{ bottom-left: } (R_1 Y_3 R_2, l), \text{ bottom-right: } (R_1 Y_3 R_2, l). \end{array} \quad (18)$$

Note that in the right figure the choice of module category is implicit, which would obfuscate the manipulations of the next sections. Therefore we will stick to triple line tensors in what follows.

In eq. (16), and in similar expressions below, we adapt the graphical convention that unlabelled blue strands are summed over, and unlabelled gray patches stand for a sum over basis vectors, i.e.

$$\begin{array}{l} \begin{array}{c} \text{Diagram 1: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \\ \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{I}_{\mathcal{R}}}} \begin{array}{c} \text{Diagram 2: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \\ \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{I}_{\mathcal{R}}}} \sum_i \begin{array}{c} \text{Diagram 3: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \end{array} |R_1 Y R_2, i\rangle, \\ \\ \begin{array}{c} \text{Diagram 4: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \\ \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{I}_{\mathcal{R}}}} \begin{array}{c} \text{Diagram 5: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \\ \equiv \sum_{\substack{R_1, R_2 \\ \in \mathcal{I}_{\mathcal{R}}}} \sum_i \begin{array}{c} \text{Diagram 6: A vertical strand labeled } Y \text{ with a gray patch at the bottom.} \end{array} \langle R_1 Y R_2, i |. \end{array} \end{array} \quad (19)$$

Contraction of these triple line tensors takes place via identification of the simple objects on module strands of concatenated tensors, pairing basis vectors, and summing over them as detailed in ref. [50].

Under the above conventions, a local $\mathcal{C}$ -symmetric two-site operator can be depicted as

$$\tilde{\kappa}_{i,n}^{\mathcal{R}} := \sum_{\{Y_m\}_m} \sum_{i,j} \lambda_{i,n}(\{Y_m\}_m, i, j) \begin{array}{c} \text{Diagram: A tensor with four vertical legs labeled } Y_1, Y_2, Y_3, Y_4 \text{ and a central dot. A gray patch is at the bottom.} \end{array}, \quad (20)$$

where $\{\lambda_{i,n}\}$ is a collection of complex numbers which may depend on all internal labels of the tensor on the right. The underlying Hilbert space can be written as

$$\begin{array}{l} \text{Span}_{\mathbb{C}} \left\{ \begin{array}{c} \text{Diagram: A tensor with four vertical legs labeled } R_{i-1}, R_i, R_{i+1}, R_{i+2} \text{ and a central dot. A gray patch is at the bottom.} \end{array} \left| \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \right\} \\ \equiv \text{Span}_{\mathbb{C}} \left\{ \bigotimes_i |R_i Y_{i+\frac{1}{2}} R_{i+1}, i_{i+\frac{1}{2}}\rangle \left| \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \right\}, \end{array} \quad (21)$$

where periodic boundary conditions are implemented by identifying $R_1 \equiv R_{L+1}$ for some chain length L . Note that this Hilbert space is generically constrained since the Hom space at i depends on those at $i-1$ and $i+1$ via the module objects R_i and R_{i+1} they share. In particular,

the direct sum over $\{R\}$ is absent whenever $\mathcal{R}$ contains a single simple object, and $\mathcal{R}$ is thus equivalent to the category of (finite-dimensional) vector spaces. Hence, in that case, we obtain a tensor product Hilbert space.

A generic symmetric local Hamiltonian acting on this space is then constructed as $\mathbb{H}^{\mathcal{R}} = \sum_i \sum_n \tilde{\kappa}_{i,n}^{\mathcal{R}}$. Note that one can always choose $\mathcal{R} = \mathcal{C}$, in which case the above Clebsch-Gordan coefficients boil down to the F-symbols of $\mathcal{C}$.¹ As such, the lattice models considered in refs. [50, 51] and this manuscript generalise the *anyonic spin chains* first considered in ref. [28].

A *duality* in the sense of ref. [50] then amounts to changing the module category $\mathcal{R}$ while keeping the category $\mathcal{C}_{\mathcal{R}}^*$ and the coefficients $\{\lambda_{i,n}\}$ fixed. Since $\mathcal{C}_{\mathcal{R}}^*$ is kept fixed throughout, we move to a lightweight notation in the expressions below by writing $\mathcal{D}$ for $\mathcal{C}_{\mathcal{R}}^*$, thereby suppressing the dependency of $\mathcal{D}$ on the initial symmetry and choice of module category. Under such a duality transformation the local symmetric operators (20) are mapped to dual local symmetric operators, charged local operators are mapped to disorder operators and the spectrum is preserved up to degeneracies if closed boundary conditions are accounted for as discussed below in sec. 6. The latter can heuristically be understood from the fact that the algebra of local operators, spanned by operators of the form (20), in particular its structure constants, is fully specified by the fixed fusion category $\mathcal{D}$.

Within the tensor network paradigm this change of module category can be implemented explicitly as a matrix product *intertwiner* that transmutes the dual local operators into each other. Indeed, for a generic duality between module categories $\mathcal{R}$ and $\mathcal{R}'$ the intertwiners are, by the result of ref. [50], labelled by *D-module functors* between $\mathcal{R}$ and $\mathcal{R}'$ organized in $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$, which itself is a *right* module category $\mathcal{M}$ over $\mathcal{C}$. To a given simple object $X \in \mathcal{M}$ there corresponds for every $R_1 \in \mathcal{I}_{\mathcal{R}}$, $Y \in \mathcal{I}_{\mathcal{D}}$ and $R'_2 \in \mathcal{I}_{\mathcal{R}'}$ a matrix ${}^X \omega_{R'_2}^{R_1 Y}$ defined in eq. (A.13), from which we construct an MPO tensor whose components are

$$\begin{array}{c} R'_1 \quad R'_2 \\ \downarrow \quad \downarrow \\ i \quad X \quad l \\ \leftarrow \quad \rightarrow \\ R_1 \quad Y \quad R_2 \\ \uparrow \quad \uparrow \\ k \quad \quad \end{array} := ({}^X \omega_{R'_2}^{R_1 Y})_{R'_1, ij}^{kl}. \quad (22)$$

Without loss of generality ${}^X \omega_{R'_2}^{R_1 Y}$ is assumed to be a unitary map ${}^X \omega_{R'_2}^{R_1 Y} : \bigoplus_{R_2} \mathcal{R}(R_1 \triangleleft Y, R_2) \otimes \mathcal{R}'(X \triangleright R_2, R'_2) \xrightarrow{\sim} \bigoplus_{R'_1} \mathcal{R}'(X \triangleright R_1, R'_1) \otimes \mathcal{R}'(R'_1 \triangleleft Y, R'_2)$. Notably, ${}^X \omega$ satisfies a consistency equation involving the F-symbols appearing in eq. (16) for the module categories $\mathcal{R}$ and $\mathcal{R}'$ which can be graphically depicted as the following tensor equality:

$$\begin{array}{c} R'_1 \quad k \quad R'_3 \quad l \quad R'_2 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ i \quad X \quad j \\ \leftarrow \quad \rightarrow \\ R_1 \quad Y_1 \quad Y_2 \quad R_2 \\ \uparrow \quad \uparrow \quad \uparrow \\ m \quad \quad \quad \\ \downarrow \\ Y_3 \quad i \quad n \end{array} = \begin{array}{c} k \quad R'_3 \quad l \\ \downarrow \quad \downarrow \\ Y_1 \quad m \quad Y_2 \\ \downarrow \quad \downarrow \\ R'_1 \quad Y_3 \quad R'_2 \\ \downarrow \quad \downarrow \\ i \quad X \quad j \\ \leftarrow \quad \rightarrow \\ R_1 \quad \quad R_2 \\ \downarrow \\ n \end{array}, \quad (23)$$

¹The term generalised Clebsch-Gordan coefficients stems from the fact that for the case of an invertible symmetry as described by the fusion category $\mathcal{C} = \text{Vec}_G$, the category of G -graded vector spaces, the F-symbols appearing in (16) coincide with the usual Clebsch-Gordan coefficients when the choice $\mathcal{R} = \text{Vec}$ is made. Here, Vec stands for the category of complex finite-dimensional vector spaces. In that case the Morita dual is found to be $(\text{Vec}_G)_{\text{Vec}}^* \simeq \text{Rep}(G)$, the fusion category of complex finite-dimensional representations of G [21, 54].

for all valid labellings of the objects and multiplicity indices. A similar property is satisfied by the second tensor in eq. (16). This equality demonstrates that the MPO constructed from the tensor (22) labelled by the object $X \in \mathcal{M}$ transmutes local operators of one model corresponding to the choice of module category $\mathcal{R}$ to the one of the model based on $\mathcal{R}'$, thereby implementing the duality transformation as foreseen. Importantly, choosing both module categories in eq. (22) equal to $\mathcal{R}$ exactly realises the lattice representation of the symmetry operators encoded in $\mathcal{C}$, here thought of as $\mathcal{C} \equiv \text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}) = \mathcal{D}_{\mathcal{R}}^*$. In that instance it is more precise to refer to (22) as the *bimodule associator* of $\mathcal{R}$ as $(\mathcal{C}, \mathcal{D})$ -bimodule category. Choosing both module categories equal to $\mathcal{R}'$ on the other hand reveals that the global symmetries of the dual model are encoded in the fusion category $\mathcal{D}_{\mathcal{R}'}$, equivalently $\mathcal{C}_{\mathcal{M}}^*$. The global symmetries of the gauged model are thus encoded in the Morita dual of the symmetry category of the ungauged model with respect to the category of duality operators. Note that all duality operators labelled by module functors in $\mathcal{M}$ implement physically the same duality operation. Being itself an invertible $(\mathcal{C}_{\mathcal{M}}^*, \mathcal{C})$ -bimodule category, $\mathcal{M}$ can be acted upon from the *right* by $\mathcal{C}$ and from the *left* by $\mathcal{C}_{\mathcal{M}}^*$. As such, distinct duality operators can be converted into each other by pre- or postcomposing with symmetry operators in $\mathcal{C}$, respectively $\mathcal{C}_{\mathcal{M}}^*$.

Note that in eq. (22) – similar to eq. (18) – we can again conflate the indices so as to obtain a conventional tensor of the form

$$\begin{array}{c} \begin{array}{c} R'_1 \quad j \\ \downarrow \\ i \leftarrow X \rightarrow l \\ \uparrow \\ R_1 \quad k \end{array} \quad \begin{array}{c} R'_2 \\ \downarrow \\ Y \\ \downarrow \\ R_2 \end{array} \end{array} \equiv \begin{array}{c} (R'_1 Y R'_2, j) \\ \downarrow \\ (R_1 X R'_1, i) \rightarrow \boxed{X} \rightarrow (R_2 X R'_2, l) \\ \uparrow \\ (R_1 Y R_2, k) \end{array}, \quad (24)$$

where on the right the choice of module categories is again implicit. Consequently, by specifying to $\mathcal{R} = \mathcal{R}'$, we recover the symmetry MPO tensors of eq. (1).

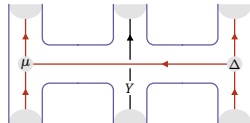
Given such $\mathcal{C}$ symmetry MPOs, what are the fusion and splitting tensors employed in the construction of the gauging map proposed in the previous section? Akin to the definition (16) we can consider tensors evaluating to the components of the *left* module associators of $\mathcal{R}$ over $\mathcal{C}$:

$$\begin{array}{c} \begin{array}{c} l \quad R_2 \quad k \\ \downarrow \quad \downarrow \\ X_1 \leftarrow i \rightarrow X_2 \\ \downarrow \quad \downarrow \\ R_3 \quad X_3 \end{array} \end{array} := (F_{R_3}^{X_1 X_2 R_1})_{X_3, ij}^{R_2, kl} \quad \text{and} \quad \begin{array}{c} \begin{array}{c} R_3 \quad j \quad R_1 \\ \downarrow \quad \downarrow \\ X_1 \leftarrow i \rightarrow X_2 \\ \downarrow \quad \downarrow \\ R_2 \quad X_3 \end{array} \end{array} := (\bar{F}_{R_3}^{X_1 X_2 R_1})_{X_3, ij}^{R_2, kl}. \quad (25)$$

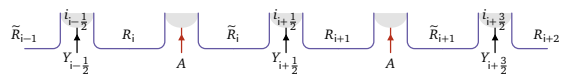
Again by virtue of two mixed pentagon equations these fusion tensors can be shown to satisfy the intertwining property (4) for the symmetry MPO tensors (22) and the recoupling identity (6) involving the $\mathcal{C}$ F-symbol, as showcased in ref. [54]. As such, given an algebra A in $\mathcal{C}$, explicit lattice representations of the multiplication and comultiplication maps read:

$$\begin{array}{c} \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} A \end{array} \end{array} \quad \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} A \end{array} \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \mu_{(X_3, i_3), l}^{(X_1, i_1), (X_2, i_2)} \quad \begin{array}{c} \begin{array}{c} \downarrow \\ X_3 \end{array} \\ \downarrow \\ \begin{array}{c} X_1 \leftarrow i \rightarrow X_2 \end{array} \end{array}, \\ \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} A \end{array} \end{array} \quad \begin{array}{c} \begin{array}{c} \downarrow \\ A \end{array} \\ \downarrow \\ \begin{array}{c} A \end{array} \end{array} := \sum_{\substack{X_1, X_2, X_3 \\ i_1, i_2, i_3, l}} \Delta_{(X_1, i_1), (X_2, i_2)}^{(X_3, i_3), l} \quad \begin{array}{c} \begin{array}{c} \downarrow \\ X_3 \end{array} \\ \downarrow \\ \begin{array}{c} X_1 \leftarrow i \rightarrow X_2 \end{array} \end{array}, \end{array} \quad (26)$$

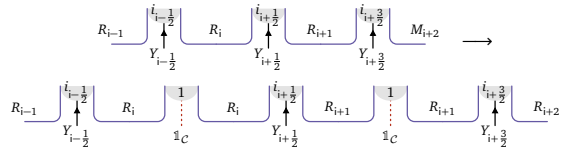
where the sums should be interpreted as explained surrounding eq. (10). Collecting everything, for any choice of module category over the symmetry category $\mathcal{C}$ one finds a corresponding MPO representation of that symmetry so that for any algebra A in $\mathcal{C}$ the generalised Gauss operator is given by:


(27)

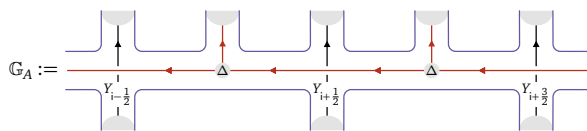
This Gauss operator acts on the Hilbert space (21) supplemented with symmetry twists labelled by the object A . Following ref. [50] states in this enlarged Hilbert space can be depicted as


(28)

where the span is now taken over additional module labels $\tilde{R}_i$. Note that since we have assumed A to be haploid, meaning that the decomposition of A contains a single copy of the monoidal unit $\mathbb{1}_C$, there is a canonical way in which a basis state in the initial Hilbert space (21) can be embedded in this enlarged Hilbert space, namely via


(29)

Equivalently, in line with the previous section, we can incorporate this choice of gauge field directly into the gauging map so as to obtain a gauging map acting on the original Hilbert space (21). The resulting operator can be depicted as:


(30)

As such, this procedure provides an interpretation of the A symmetry twists in the extended Hilbert space (28) as the ‘gauge field’ for the subsymmetry $\mathcal{C}$ which is being gauged, in accordance with ref. [71]. Notably, the insertion of the defects in (28) is independent of the Frobenius structure on A , which only appears in the definition of the Gauss constraint. In that sense, the Frobenius structure can be thought of as generalising the notion of discrete torsion to the categorical setting, as anticipated above.

This can be compared to ref. [39] whose results are obtained as a specific instance of this formalism. Indeed, the results of ref. [39] are recovered by choosing $\mathcal{R} = \text{Vec}$, the category of (complex finite-dimensional) vector spaces, as module category over the fusion category $\mathcal{C} = \text{Vec}_G$ of G -graded vector spaces which describes an invertible G symmetry. In that case the objects $\{Y_i\}$; label representations of G . Gauging the full G symmetry then happens via choosing $A \simeq \mathbb{C}[G]$, here thought of as the regular representation of G . The Hilbert space (28) then effectively boils down to the initial ‘matter’ Hilbert space tensored with copies of the group algebra $\mathbb{C}[G]$ in between the physical sites. Furthermore, one could twist the gauging map by

a choice of 2-cocycle, i.e. $[\psi] \in H^2(G, \text{U}(1))$, by choosing A to be the ψ -twisted group algebra $\mathbb{C}^\psi[G]$ which as a vector space is spanned by states $\{|g\rangle, g \in G\}$ with algebra multiplication given by $|g\rangle \cdot |h\rangle = \psi(g, h)|gh\rangle$ [50, 56, 76].

In preparation for the following section, let us note that the trivalent tensors evaluating to the left module associators (25) are a special instance of a more general kind of tensors which encode the composition of module functors. Indeed, given $\mathcal{M}, \mathcal{N}, \mathcal{O}$, all right module categories over $\mathcal{D}$, the unitary F-symbols introduced in (A.14) and their conjugates can be organised in triple line tensors of the form

$$\begin{array}{c} \text{Diagram 1: A trivalent tensor with three legs labeled } i, j, k \text{ and three regions labeled } O, N, M. \end{array} := (F_O^{X_1 X_2 M})_{X_3, ij}^{N, kl} \quad \text{and} \quad \begin{array}{c} \text{Diagram 2: A trivalent tensor with three legs labeled } i, j, k \text{ and three regions labeled } O, N, M. \end{array} := (\bar{F}_O^{X_1 X_2 M})_{X_3, ij}^{N, kl}, \quad (31)$$

where $M \in \mathcal{I}_{\mathcal{M}}, N \in \mathcal{I}_{\mathcal{N}}, O \in \mathcal{I}_{\mathcal{O}}$ and $X_1 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{N}, \mathcal{O})}, X_2 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{N})}, X_3 \in \mathcal{I}_{\text{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{O})}$. From $\mathcal{D}_{\mathcal{R}}^* \equiv \mathcal{C}$ it follows that for the case where all three module categories are chosen to be equal to $\mathcal{R}$ these tensors indeed reduce to the matrix components of the left module associators of $\mathcal{R}$ over $\mathcal{C}$ as per eq. (25).

4 Equivalence gauging map and duality operator

In this section, we show how the gauging map (30) can be converted to a duality MPO from the previous section by means of a finite sequence of local unitary transformations, i.e. a constant depth quantum circuit.

8

Given a finite symmetry $\mathcal{C}$ represented on the lattice via a choice of module category $\mathcal{R}$, how can the gauging map constructed from a Frobenius algebra object A in $\mathcal{C}$ be related to a duality operator of the kind described in the previous section? By the results of refs. [62, 66] there exists a right $\mathcal{C}$ -module category such that A corresponds to the internal hom of a particular simple module object. More precisely, this module category is exactly equivalent to the category $\mathcal{M} = \text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ of duality operators, for a particular $\mathcal{R}'$. Indeed, $\mathcal{R}'$ is determined by the fact that $\mathcal{M}$ is equivalent to the category $\text{Mod}_{\mathcal{C}}(A)$ of left A -modules [25, 62]. The internal hom reconstruction of A is reviewed in appendix A.3. In what follows X_A denotes the simple object such that $A \equiv \text{Hom}(X_A, X_A)$. The simple module object $X_A \in \mathcal{M}$ is precisely the one labelling the duality MPO to which the gauging map $\mathbb{G}_A$ (30) is unitarily equivalent by means of a unitary constant-depth quantum circuit, as we now demonstrate.

The internal hom reconstruction allows us to ‘inflate’ the comultiplication tensors (26) appearing in the gauging map (30) as follows [26, 66]:

$$\begin{array}{c} \text{Diagram 1: A trivalent tensor with three legs labeled } i_1, i_2, i_3 \text{ and three regions labeled } A, A, A. \end{array} = \sum_{X_1, X_2, X_3} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3}}}{d_{X_A}^2} \begin{array}{c} \text{Diagram 2: A complex diagram involving multiple trivalent tensors and regions labeled } A, X_A, X_1, X_2, X_3. \end{array}, \quad (32)$$

where, as in eq. (25), the sum is over the simple objects appearing in the decomposition of A , and where in virtue of the defining property of the internal hom (A.15),

$N_{X_j}^A = \dim_{\mathcal{C}} \mathcal{C}(X_j, A) = \dim_{\mathcal{C}} \mathcal{M}(X_A \triangleleft X_j, X_A)$ [66]. Also remark that the innermost closed module loop on the right-hand side is valued in $\mathcal{R}'$ over whose simple objects we sum. Furthermore, the normalisation is chosen to match the one in eq. (A.16). In appendix C we demonstrate, based on our graphical calculus for triple line tensors, that the inflated representation of the comultiplication (32) satisfies the defining coalgebra property.

Before applying the aforementioned circuit, we first rewrite the gauging map by making use of eq. (32). We begin by inserting eq. (32) in the gauging map eq. (30), followed by using the orientation-reversing ‘flags’ introduced in appendix B to reverse some of the arrows in eq. (32). We then exploit the unitarity of the trivalent tensors, which in our graphical notation reads

$$\sum_{X,i} \begin{array}{c} \text{Diagram: A trivalent tensor with inputs } k, l \text{ and output } i, \text{ and a loop labeled } X_A - i - \tilde{X}_A. \end{array} = \delta_{R'_1, R'_2} \delta_{k,k'} \delta_{l,l'} \begin{array}{c} \text{Diagram: Two trivalent tensors, one labeled } R'_1 \text{ and one labeled } R'_2. \end{array}, \quad (33)$$

so as to rewrite the gauging map as follows:

$$G_A = \sum_{\{X_i\}, \{j_i\}} d_{X_A}^{-\frac{l}{2}} \begin{array}{c} \text{Diagram: A complex circuit with multiple layers of trivalent tensors and module strands labeled } \mathcal{R}, \mathcal{R}', X_A, \tilde{X}_A, X_i, \tilde{X}_i, Y_i, \text{ etc.} \end{array}, \quad (34)$$

where the letters $\mathcal{R}, \mathcal{R}'$ denote the module categories in which surrounding module strands are valued. Here and below, $\tilde{X}_A \in \mathcal{I}_{\mathcal{M}^{\text{op}}}$, where $\mathcal{M}^{\text{op}} = \text{Fun}_{\mathcal{D}}(\mathcal{R}', \mathcal{R})$, labels the unique ‘opposite’ duality strand specified by the fact that the composition of $\tilde{X}_A$ and X_A exactly yields the algebra object A in $\mathcal{C}$.

Let us stress again that eq. (34) is an exact rewriting of the gauging map. In order to bring it into the form of a duality operator, we now invoke the following unitary gate defined in ref. [53] constructed from the fusion tensors eq. (31):

$$\begin{array}{c} \text{Diagram: A unitary gate with inputs } X_A, \tilde{X}_A \text{ and outputs } X_A, \tilde{X}_A. \end{array} \equiv \sum_X \begin{array}{c} \text{Diagram: A unitary gate with inputs } X_A, \tilde{X}_A \text{ and outputs } X_A, \tilde{X}_A. \end{array} \equiv \sum_{\substack{X, R_1 \\ R_2, R'}} \begin{array}{c} \text{Diagram: A unitary gate with inputs } X_A, \tilde{X}_A \text{ and outputs } X_A, \tilde{X}_A. \end{array}, \quad (35)$$

which should be interpreted as unitary between the Hilbert spaces $\bigoplus_{X, R_1, R_2} \mathcal{R}(X \triangleright R_2, R_1) \otimes \mathcal{C}(X, \tilde{X}_A \otimes X_A) \xrightarrow{\sim} \bigoplus_{R_1, R_2, R'} \mathcal{M}(X_A \triangleright R_1, R') \otimes \mathcal{M}^{\text{op}}(\tilde{X}_A \triangleright R', R_2)$. Graphically, unitarity of this gate can be expressed by a rotated version of eq. (33).

Acting then with a layer of this gate on the strands labelled by objects $\{X_i\}_i$ in eq. (34), eq. (34) can be brought in the following form:

$$G_A \cong d_{X_A}^{-L/2} \cdot \quad (36)$$

At the bottom of eq. (36) we recognize the duality MPO implementing the duality $\mathcal{R} \rightarrow \mathcal{R}'$ labelled by X_A . The 'cups' on top can be removed by means of a local unitary transformation. To this end we also define the following unitary gate whose entries are expressed in terms of a rotated version of the $X_A \omega$ tensors introduced in above eq. (22):

$$\begin{array}{c} \text{Diagram} \end{array} \equiv \sum_Y \begin{array}{c} \text{Diagram} \end{array} \equiv \sum_{\substack{R_1, R_2, \\ R'_1, R'_2}} R_1 \begin{array}{c} \text{Diagram} \end{array} R'_2. \quad (37)$$

This gate can be used to thread the strands labelled by X_A to the left through the physical degrees of freedom $\{Y_{i+1/2}\}_i$. Subsequently, by making use of the conjugate of the gate eq. (35) the $\bar{X}_A$ and X_A strands can be fused together unitarily, in the process picking up an extra overall factor $d_{X_A}^{L/2}$. Due to the presence of the cup in eq. (36) the only allowed fusion outcome is $1_{C_{\mathcal{M}}^*}$ which appears with multiplicity one in $X_A \otimes \bar{X}_A$. As such, one recovers eventually the duality MPO labelled by X_A as anticipated:

$$G_A \cong D_{X_A} := \quad (38)$$

We commented earlier that the global symmetries of the dual model are encoded in the fusion category $C_{\mathcal{M}}^*$. In virtue of the fact that the category $\mathcal{M}$ labelling the duality operators is equivalent to the category $\text{Mod}_{\mathcal{C}}(A)$ of A -modules, $C_{\mathcal{M}}^*$ is indeed equivalent to $\text{Bimod}_{\mathcal{C}}(A)$, in agreement with the results of refs. [11, 25]. This can be verified more explicitly by conjugating the symmetry MPOs labelled by objects in $C_{\mathcal{M}}^*$ with the circuit constructed above. Making use of the inflation trick eq. (32) it is then easy to verify that the resulting operator, labelled by some object in $\mathcal{C}$, indeed is endowed with the structure of an A - A bimodule in $\mathcal{C}$. More precisely, starting from the (non-simple) symmetry operator in the dual model labelled by $X_A \otimes X \otimes \bar{X}_A \in C_{\mathcal{M}}^*$, where $X \in \mathcal{C}$, we can conjugate it with the circuit to obtain the MPO corresponding to the object $A \otimes X \otimes A \in \mathcal{C}$. Interpreting this object as the *induced bimodule* $\text{Ind}_{A|A}(X)$ defined in [27, Def. 5.1], it then follows from [27, Lemma 5.2], that every simple A - A bimodule is obtained in the decomposition of this MPO for a certain X .

We conclude this section with a few additional remarks. First, notice that the depth of the quantum circuit being constant and small is a direct consequence of the fact that the construction of the gauging map involves the introduction of an extensive number of defects labelled

by A . As such it should be compared to the quantum circuit representation of the duality operator showcased in ref. [53], whose depth scales linearly in the system size in the generic case, and is constant in the nilpotent case when supplemented with measurements and feedforward operations.

A further natural question is whether every (simple) duality operator labelled by $X \in \mathcal{I}_{\text{Fun}\mathcal{D}(\mathcal{M}, \mathcal{N})}$ is equivalent to a gauging map for a suitable choice of algebra object. The answer is affirmative and the algebra object is provided by the internal hom $\underline{\text{Hom}}(X, X)$. In that case, the circuit that transmutes the gauging map $\mathbb{G}_{\underline{\text{Hom}}(X, X)}$ to the duality MPO $\mathbb{D}_X$ exactly mimics the one above with $\underline{\text{Hom}}(X, X)$ substituted for A and X for X_A .

As mentioned above, $\text{Fun}\mathcal{D}(\mathcal{M}, \mathcal{N})$ carries the structure of an invertible $(\mathcal{D}_{\mathcal{N}}^*, \mathcal{C})$ -bimodule category. In particular this implies that $\underline{\text{Hom}}(X_A, X_A)$ can be interpreted as an internal hom object in the category $\mathcal{D}_{\mathcal{N}}^*$, namely via $\text{Fun}\mathcal{D}(\mathcal{M}, \mathcal{N})(X \triangleright X_A, X_A) \simeq \mathcal{D}_{\mathcal{N}}^*(X, \underline{\text{Hom}}(X_A, X_A))$. As such, $\underline{\text{Hom}}(X_A, X_A)$ is also endowed with the structure of a Frobenius algebra in $\mathcal{D}_{\mathcal{N}}^*$ which can be expressed in terms of the left $\text{Fun}\mathcal{D}(\mathcal{M}, \mathcal{N})$ module associators, akin to eq. (A.16). This algebra object thus also provides a gauging map which in this case turns out to be equivalent to the duality MPO $\mathbb{D}_{\tilde{X}_A}$ implementing the duality $\mathcal{N} \rightarrow \mathcal{M}$. Note however that $\mathbb{G}_{\tilde{X}_A} \circ \mathbb{G}_{X_A}$ is not equal to the identity operator on the periodic Hilbert space eq. (21). In fact, $\mathbb{G}_{\tilde{X}_A} \circ \mathbb{G}_{X_A}$ yields the symmetry MPO labelled by the algebra object A , which is typically a non-invertible symmetry operator. It turns out that when the original and dual Hilbert spaces are supplemented with symmetry-twists, the duality MPOs can be implemented unitarily. This is the topic of section 6.

5 Anomalies and short-range symmetric states

We comment on the specific case where the symmetry is fully gaugeable and show that in that case we can think of gauging as averaging over a local version of the symmetry. We further argue that full gaugeability implies the existence of a short-range entangled $\mathcal{C}$ -symmetric state, and therefore the absence of an anomaly.

For invertible symmetries described by Vec_G^ω , the 3-cocycle $[\omega] \in H^3(G, U(1))$ poses an obstruction to gauging. Indeed, the possible algebra objects in Vec_G^ω are ψ -twisted group algebras $\mathbb{C}^\psi[K]$, where $\omega|_K$ is trivial, $[\psi] \in H^2(K, U(1))$. Therefore, ω characterizes the anomaly of such a symmetry, and gauging with an algebra object $\mathbb{C}^\psi[K]$ amounts to ψ -twisted gauging of the non-anomalous subgroup K . As argued in [14], the anomaly ω also provides an obstruction to the existence of a short-range entangled state that is symmetric, which is the generally accepted definition for an anomaly in (1+1)d.

For general non-invertible symmetries, the interpretation of the algebra object A as a subset of symmetries that can be gauged is less clear. The reason for this is that in this case, one generally cannot think of the projectors in eq. (8) as an averaging over localised versions of the symmetry operators as in the invertible case. The exception to this is when the symmetry $\mathcal{C}$ is anomaly-free; in this case, it can be gauged completely, even if it is non-invertible. To see this, we rely again on the equivalence between algebra objects in $\mathcal{C}$ and internal Hom objects. Since gapped phases of a $\mathcal{C}$ symmetric theory are classified by module categories $\mathcal{P}$ over $\mathcal{C}$ [32, 73], being anomaly-free implies the existence of Vec as a $\mathcal{C}$ -module category, also referred to as a *fiber functor*. This requires $\mathcal{C}$ to be the representation category of a Hopf algebra $\mathcal{A}$, and we denote $\mathcal{C} = \text{Rep}(\mathcal{A})$. The internal Hom construction provides an algebra object $A \equiv \underline{\text{Hom}}(\mathbb{C}, \mathbb{C})$, with $\mathbb{C}$ the unique simple object in Vec . As an object of $\text{Rep}(\mathcal{A})$, A is the regular representation, $A \simeq \oplus_{X \in \mathcal{I}_{\mathcal{C}}} d_X \cdot X$. It is this choice of algebra object that amounts to the complete gauging of the symmetry $\mathcal{C}$, as we show below.

Using the expression of the local Gauss constraint from section 2, we can write

$$\mathcal{P} = \begin{array}{c} \text{Diagram: A square with four legs labeled } A. \text{ Inside is a circle with a dot.} \end{array} = \sum_{a,i} \begin{array}{c} \text{Diagram: A square with four legs labeled } A. \text{ Inside is a circle with a dot and a label } a. \end{array} =: \sum_a \frac{d_a}{\text{FPdim}\mathcal{M}} \mathcal{P}^a, \quad (39)$$

where the triangular tensor is the projector of A on the block labeled by a and i is the multiplicity of the simple object a in A , cf. eq. (A.9). $\text{FPdim}\mathcal{M} = \text{FPdim}\mathcal{C} = \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2$ denotes the total quantum dimension of $\mathcal{M}$, which in our conventions is equal to that of $\mathcal{C}$. We can show using the graphical calculus and the explicit expressions of the multiplication and comultiplication maps that

$$\mathcal{P}^a \mathcal{P}^b = \sum_c N_{ab}^c \mathcal{P}^c, \quad (40)$$

where the N symbol is that of the symmetry category $\mathcal{C}$. Indeed:

$$\begin{aligned} \mathcal{P}^a \mathcal{P}^b &\stackrel{(A.4)}{=} \frac{(\text{FPdim}\mathcal{M})^2}{d_a d_b} \sum_{i,j} \begin{array}{c} \text{Diagram: Two squares with legs labeled } A. \end{array} \\ &\stackrel{(A.16)}{=} \frac{(\text{FPdim}\mathcal{M})^2}{d_a d_b} \sum_{\substack{c,k, \\ k',l}} \underbrace{\left(\sum_{i,j} \mu_{(c,k),l}^{(a,i),(b,j)} \Delta_{(a,i),(b,j)}^{(c,k'),l} \right)}_{\frac{d_a d_b}{d_c} \frac{1}{\text{FPdim}\mathcal{M}} \delta_{k,k'}} \begin{array}{c} \text{Diagram: A square with legs labeled } A. \end{array} \\ &\stackrel{(7)}{=} \sum_{c,k} N_{ab}^c \frac{\text{FPdim}\mathcal{M}}{d_c} \begin{array}{c} \text{Diagram: A square with legs labeled } A. \end{array} = \sum_c N_{ab}^c \mathcal{P}^c. \end{aligned} \quad (41)$$

This demonstrates that the operators $\mathcal{P}^a$ are local versions of the (non-invertible) symmetry $\mathcal{C}$, and we can interpret the gauging by A as gauging the full symmetry.

The equivalence between these gauging maps and duality operators from the previous section can now also be used to show that if a symmetry can be completely gauged, there must exist a symmetric short-range entangled state. Indeed, it suffices to consider the gauging map $\mathcal{G}_A$, which as we have shown corresponds to the duality operator labelled by X_A up to local unitary transformations. Applied to the above setting, the gauging map for the algebra object A in $\text{Rep}(\mathcal{A})$ is equivalent to the duality operator labelled by $C \in \text{Vec}$. Acting with $\mathcal{G}_A^\dagger$ on a generic product state therefore produces an injective C -symmetric MPS, since $\mathcal{G}_A$ is an injective MPO. This way of producing C -symmetric MPS by ungauging the symmetry $\mathcal{C}$ lies at the heart of symmetric tensor network methods, as shown in [52]. Note that the same procedure can also be applied to the setting where the symmetry category $\mathcal{C}$ contains an anomaly-free subcategory $\mathcal{C}' \subset \mathcal{C}$, despite $\mathcal{C}$ itself being anomalous. In that case one can, by choosing the appropriate algebra object, still construct a short-range entangled $\mathcal{C}'$ -symmetric state. One notable example is the $\text{Vec}_{\mathbb{Z}_2}$ subcategory of Ising [70]; another is the $\text{Vec}_{\mathbb{Z}_3}$ subcategory of the Haagerup category $\mathcal{H}_3$, which is discussed in more detail below.

6 Symmetry-twisted boundary conditions

We propose a generalisation of the gauging map eq. (30) to act on a Hilbert space twisted by a symmetry defect. We demonstrate the equivalence with the symmetry-twisted duality operators introduced in ref. [51].

In the construction of the anyonic spin chains eq. (21), we identified the module labels $R_1 \equiv R_{L+1}$ so as to implement *periodic* boundary conditions. More generally, one can insert a *symmetry defect*, labelled by an object $B \in \mathcal{I}_C$, before closing the chain. This results in a *symmetry-twisted* Hilbert space, given by

$$\text{Span}_{\mathbb{C}} \left\{ \begin{array}{c} R_L \quad \begin{array}{c} \uparrow \\ Y_{L+\frac{1}{2}} \end{array} \quad R_{L+1} \quad \begin{array}{c} \uparrow \\ B \end{array} \quad R_1 \quad \begin{array}{c} \uparrow \\ Y_{\frac{1}{2}} \end{array} \quad R_2 \end{array} \left| \{Y_{i+\frac{1}{2}}\}_i, \{R_i\}_i, \{i_{i+\frac{1}{2}}\}_i \right. \right\}. \quad (42)$$

The symmetry twist is *topological* in the sense that the Hilbert space obtained by moving the twist by one site is related to the original one by a unitary transformation given by the bimodule associator of $\mathcal{R}$. Note that for the choice $B = \mathbb{1}$ the periodic chain eq. (21) is retrieved.

Local operators acting on the twisted Hilbert space away from the twist are still given by ladder diagrams of the form eq. (20), whereas local operators that act across the defect are given by a modified version of eq. (20) in which a rotated version of the tensor eq. (22) with $\mathcal{R} = \mathcal{R}'$ is inserted, as detailed in ref. [51]. As before, a duality amounts to changing the choice of module category $\mathcal{R} \rightarrow \mathcal{R}'$. The duality MPOs defined above in eq. (38) now need to be promoted to *tubes*, given by matrix product operators of the form

$$\mathfrak{T}_{\mathcal{R}|\mathcal{R}'}^{B,B',X,X',k,k'} := \begin{array}{c} \begin{array}{c} \uparrow \\ B' \end{array} \\ \begin{array}{c} \leftarrow X \quad \leftarrow k' \quad \leftarrow X' \quad \leftarrow k \quad \leftarrow X \end{array} \\ \begin{array}{c} \uparrow \\ B \end{array} \end{array}, \quad (43)$$

where $X, X' \in \mathcal{I}_{\mathcal{M}}$, $B' \in \mathcal{I}_{\mathcal{C}'}^{\mathcal{M}}$ and X' is contained in both $B \otimes X$ and $X \otimes B'$. The trivalent tensors used to fuse the twists with the duality strand are those defined in eq. (31) for the appropriate choice of module categories. Similar to the untwisted case, tubes of the type $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$ and $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$ commute with the aforementioned local operators so that they constitute the symmetry operators of the original and the dual model. Together, the collection of tubes $\{\mathfrak{T}_{\mathcal{R}|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}, \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}, \mathfrak{T}_{\mathcal{R}|\mathcal{R}'}\}$ is endowed with a multiplication defined via stacking of tubes as well as a dagger. As such they span a finite C^* -algebra [51, 57, 60]. This algebra admits an Artin-Wedderburn decomposition in terms of simple matrix algebras which are identified with the *topological sectors* of the model. In particular, this algebra consists of two subalgebras spanned by tubes of the type $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$ and $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$ respectively, whose blocks are in one-to-one correspondence with simple objects of the *Drinfeld center* $\mathcal{Z}(\mathcal{C})$ labelling the superselection sectors of the original and dual model. Such a (simple) sector $Z \in \mathcal{I}_{\mathcal{Z}(\mathcal{C})}$ provides a certain (typically non-simple) boundary condition as well as a charge leaving that boundary condition invariant. In virtue of the Morita equivalence between $\mathcal{C}$ and $\mathcal{C}_{\mathcal{M}}^*$, $\mathcal{Z}(\mathcal{C})$ is equivalent to $\mathcal{Z}(\mathcal{C}_{\mathcal{M}}^*)$ as a braided fusion category, and projectors on the topological sectors in both models are given by the minimal central idempotents of the $\mathfrak{T}_{\mathcal{R}|\mathcal{R}}$ and $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}'}$ tube algebras. The duality $\mathcal{R} \rightarrow \mathcal{R}'$ then induces a permutation of the topological sectors, i.e. $\mathcal{Z}(\mathcal{C}) \xrightarrow{\sim} \mathcal{Z}(\mathcal{C}_{\mathcal{M}}^*)$, in accordance with the fact that not all boundary conditions in the dual model are compatible with a certain boundary condition in the original model, as is apparent from eq (43).

Let us now proceed by proposing a gauging map acting on the twisted Hilbert space eq. (42) which is unitarily equivalent to a (linear combination) of intertwining tubes eq. (43) [81]. Requiring that the gauging map is constructed from a product of local commuting projectors acting on an initial trivial gauge field and that it acts diagonally on the boundary condition suggests following definition:

$$\mathbb{G}_A^B := \sqrt{d_A} \text{ (diagrammatic expression) }, \quad (44)$$

where, as above, every red strand is labelled by the algebra object A . The unitary circuit that converts $\mathbb{G}_A^B$ in a linear combination of tubes $\mathfrak{T}_{\mathcal{R}'|\mathcal{R}}$ parallels the one from the previous section. We begin by inflating all occurrences of the multiplication tensor in $\mathbb{G}_A^B$ using eq. (32). Subsequently we apply the gates defined in eq. (35). This results again in a collection of ‘cups’ which in turn can be removed by a depth 2 circuit, as discussed around eq. (37). As such, one ends up with

$$\mathbb{G}_A^B \cong \sqrt{d_A} \text{ (diagrammatic expression) }. \quad (45)$$

Finally, the boundary condition B can be fused to the duality strands X_A and $\bar{X}_A$ by sequentially applying a variation of the gate eq. (35). The following linear combination of tubes is obtained:

$$\mathbb{G}_A^B \cong \sqrt{d_{X_A}} \sum_{\substack{X', B' \\ k, k'}} \text{ (diagrammatic expression) }. \quad (46)$$

The orientation reversing flag can then be absorbed in the fusion tensor to its right, at the cost of introducing extra quantum dimensions as per eq. (B.5). In the end, we obtain that

$$\mathbb{G}_A^B \cong \sum_{\substack{X', B' \\ k, k'}} \sqrt{\frac{d_{X_A} d_{X'}}{d_{B'}}} \times \mathfrak{T}_{\mathcal{R}'|\mathcal{R}}^{B, B', X, X', k, k'}. \quad (47)$$

7 Examples

We illustrate our formalism by means of two examples corresponding to symmetries encoded in the fusion category of representations of S_3 and the Haagerup categories.

7.1 Rep(S_3)

As a first example, consider the fusion category $\text{Rep}(S_3)$ of (finite-dimensional complex) representations of the permutation group S_3 . An example Hamiltonian exhibiting S_3 symmetry is the spin-1/2 XXZ model. This model and all its dualities were considered in ref. [51].

The simple objects of $\text{Rep}(S_3)$ are denoted by $\mathcal{I}_{\text{Rep}(S_3)} = \{\underline{1}, \underline{1}^*, \underline{2}\}$, and correspond to the trivial, sign and 2-dimensional irreducible representation respectively. Its non-trivial fusion rules are given by

$$\begin{aligned}\underline{1}^* \otimes \underline{1}^* &\simeq \underline{1}, \\ \underline{1}^* \otimes \underline{2} &\simeq \underline{2} \oplus \underline{1}^* \simeq \underline{2}, \\ \underline{2} \otimes \underline{2} &\simeq \underline{1} \oplus \underline{1}^* \oplus \underline{2}.\end{aligned}\quad (48)$$

Recall that the inequivalent indecomposable module categories of $\text{Rep}(S_3)$ are in one-to-one correspondence with subgroups of S_3 up to conjugacy, of which there are four: S_3 , $\mathbb{Z}_3$, $\mathbb{Z}_2$ and $\{1\}$. Given such a subgroup $H \subseteq S_3$ the corresponding module category is $\text{Rep}(H)$ which is acted upon by $\text{Rep}(S_3)$ via $\lambda \triangleleft \rho \simeq \lambda \otimes \text{Res}_H^{S_3}(\rho)$, where $\text{Res}_H^{S_3}$ denotes the restriction of S_3 -representations to the subgroup H and $\otimes$ stands for the usual tensor product of H -representations. For each of these module categories their Morita duals are computed to be

$$\begin{aligned}\text{Rep}(S_3)_{\text{Rep}(S_3)}^* &\simeq \text{Rep}(S_3), & \text{Rep}(S_3)_{\text{Rep}(\mathbb{Z}_3)}^* &\simeq \text{Vec}_{S_3}, \\ \text{Rep}(S_3)_{\text{Rep}(\mathbb{Z}_2)}^* &\simeq \text{Rep}(S_3), & \text{Rep}(S_3)_{\text{Vec}}^* &\simeq \text{Vec}_{S_3}.\end{aligned}\quad (49)$$

We now proceed to discuss the algebra objects in $\text{Rep}(S_3)$ to which each of these module categories correspond.

Let us denote the irreps of $\mathbb{Z}_2$ by $\mathcal{I}_{\text{Rep}(\mathbb{Z}_2)} = \{\underline{1}_{\mathbb{Z}_2}, \underline{1}_{\mathbb{Z}_2}^*\}$ where $\underline{1}_{\mathbb{Z}_2}^*$ stands for the sign representation of $\mathbb{Z}_2$. The restriction functor for this subgroup applied to the S_3 -irreps yields $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{1}) \simeq \underline{1}_{\mathbb{Z}_2}$, $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{1}^*) \simeq \underline{1}_{\mathbb{Z}_2}^*$ and $\text{Res}_{\mathbb{Z}_2}^{S_3}(\underline{2}) \simeq \underline{1}_{\mathbb{Z}_2} \oplus \underline{1}_{\mathbb{Z}_2}^*$. Making use of def. (A.15) and the $\text{Rep}(\mathbb{Z}_2)$ fusion rules it is then straightforward to find that at the level of the objects, the internal Homs correspond to:

$$\begin{aligned}\underline{\text{Hom}}(\underline{1}_{\mathbb{Z}_2}, \underline{1}_{\mathbb{Z}_2}) &\simeq \underline{1} \oplus \underline{2}, \\ \underline{\text{Hom}}(\underline{1}_{\mathbb{Z}_2}^*, \underline{1}_{\mathbb{Z}_2}^*) &\simeq \underline{1} \oplus \underline{2}.\end{aligned}\quad (50)$$

All components of their respective multiplication morphisms evaluate to one except those given in terms of the F-symbols

$$\begin{aligned}(F_{\underline{1}_{\mathbb{Z}_2}}^{1_{\mathbb{Z}_2} 22})_{\underline{1}_{\mathbb{Z}_2}, 11}^{1, 11} &= (F_{\underline{1}_{\mathbb{Z}_2}}^{1_{\mathbb{Z}_2} 22})_{\underline{1}_{\mathbb{Z}_2}, 11}^{1, 11} = \frac{1}{\sqrt{2}}, \\ (F_{\underline{1}_{\mathbb{Z}_2}}^{1_{\mathbb{Z}_2} 22})_{\underline{1}_{\mathbb{Z}_2}, 11}^{2, 11} &= \frac{1}{\sqrt{2}}, \quad (F_{\underline{1}_{\mathbb{Z}_2}}^{1_{\mathbb{Z}_2} 22})_{\underline{1}_{\mathbb{Z}_2}, 11}^{2, 11} = -\frac{1}{\sqrt{2}}.\end{aligned}\quad (51)$$

These algebra structures are readily checked to be isomorphic. Equivalently, both inner Homs can be converted into each other by an automorphism of the module category given by the module functor $-\triangleleft \underline{1}^*$.

We write $\mathcal{I}_{\text{Rep}(\mathbb{Z}_3)} = \{\underline{1}_{\mathbb{Z}_3}, \omega_{\mathbb{Z}_3}, \omega_{\mathbb{Z}_3}^2\}$ for the three one-dimensional representations of $\mathbb{Z}_3$. Restrictions of the S_3 -irreps to $\mathbb{Z}_3$ are computed to be $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{1}) \simeq \underline{1}_{\mathbb{Z}_3}$, $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{1}^*) \simeq \underline{1}_{\mathbb{Z}_3}$ and $\text{Res}_{\mathbb{Z}_3}^{S_3}(\underline{2}) \simeq \omega_{\mathbb{Z}_3} \oplus \omega_{\mathbb{Z}_3}^*$. The internal Homs are given by $\underline{\text{Hom}}(\lambda, \lambda) \simeq \underline{1} \oplus \underline{1}^*$ for all $\lambda \in \mathcal{I}_{\text{Rep}(\mathbb{Z}_3)}$. The components of the multiplication map all evaluate to $\mu_{\lambda_1 \otimes \lambda_2}^{\lambda_1, \lambda_2} = \frac{1}{\sqrt{2}}$ for all $\lambda_1, \lambda_2 \in \{\underline{1}, \underline{1}^*\}$.

Finally, considering Vec as module category, one finds as anticipated the regular representation of S_3 as corresponding algebra object:

$$\underline{\text{Hom}}(\mathbb{C}, \mathbb{C}) \simeq \underline{1} \oplus \underline{1}^* \oplus \underline{2} \cdot \underline{2}.\quad (52)$$

Its algebra morphism is then specified by the F-symbol $F_C^{C\lambda_1\lambda_2}$ whose components evaluate to the (unitary) Clebsch-Gordan coefficients according to the prescription:

$$(F_C^{C\lambda_1\lambda_2})_{c,ij}^{\lambda_3,k1} = \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 \\ i & j & k \end{bmatrix},\quad (53)$$

where $\lambda_3 \in \lambda_1 \otimes \lambda_2$.

7.2 Haagerup categories

The Haagerup Morita class consists of three distinct unitary fusion categories denoted by $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_3$. The first two arise from the *Haagerup subfactor* proposed and constructed in refs. [2, 34], while in ref. [33] the existence of $\mathcal{H}_3$ was proven by providing an appropriate algebra object in $\mathcal{H}_1$ whose corresponding category of bimodules was hitherto unknown. The interest in the Haagerup subfactor stems from Jones' conjectured correspondence between subfactors and conformal field theories [41]. Since the conjecture does not provide an explicit construction of the CFT, this has led to a plethora of recent studies in which both one-dimensional quantum models [10, 13, 37, 40, 50] as well as classical models [79] with $\mathcal{H}_3$ symmetry have been constructed and investigated using analytical and numerical methods, so as to shed light on the alleged Haagerup CFT.

The unitary fusion categories $\mathcal{H}_2$ and $\mathcal{H}_3$ are both rank six and have the same simple objects, denoted by $\mathcal{I}_{\mathcal{H}_2} = \mathcal{I}_{\mathcal{H}_3} = \{1, \alpha, \alpha^2, \rho, \alpha\rho, \alpha^2\rho\}$. They satisfy the same fusion rules, the non-trivial ones being given by

$$\begin{aligned} \alpha \otimes \alpha &\simeq \alpha^2, & \alpha^i \otimes \rho &\simeq \alpha^i \rho \simeq \rho \otimes \alpha^i, \\ \alpha^2 \otimes \alpha &\simeq 1, & \rho \otimes \rho &\simeq 1 \oplus \rho \oplus \alpha\rho \oplus \alpha^2\rho. \end{aligned} \quad (54)$$

The two categories differ in their unitary F-symbols, which can not be transmuted into each other via basis transformations on their multiplicity spaces. Both $\mathcal{H}_2$ and $\mathcal{H}_3$ possess a $\text{Vec}_{\mathbb{Z}_3}$ subcategory consisting of the simple objects $\{1, \alpha, \alpha^2\}$.

The fusion category $\mathcal{H}_1$ on the other hand has four simple objects denoted by $\mathcal{I}_{\mathcal{H}_1} = \{1, \mu, \eta, \nu\}$. Its fusion rules are symmetric, and the non-trivial ones read

$$\begin{aligned} \mu \otimes \mu &\simeq 1 \oplus \nu, & \nu \otimes \nu &\simeq 1 \oplus 2\nu \oplus 2\eta \oplus \mu, \\ \mu \otimes \nu &\simeq \mu \oplus \nu \oplus \eta, & \nu \otimes \eta &\simeq 2\nu \oplus \eta \oplus \mu, \\ \mu \otimes \eta &\simeq \nu \oplus \eta, & \eta \otimes \eta &\simeq 1 \oplus \nu \oplus \eta \oplus \mu. \end{aligned} \quad (55)$$

The F-symbols of $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_3$ were first calculated in refs. [5, 36, 63] respectively.

Since these three fusion categories are Morita equivalent, there exist invertible bimodule categories between them [33]. Let us denote these bimodule categories by $\mathcal{H}_{i,j}$, $i, j = 1, 2, 3$, such that $(\mathcal{H}_i)_{\mathcal{H}_{i,j}}^* \simeq \mathcal{H}_j$, with the understanding that $\mathcal{H}_{j,i} \equiv \mathcal{H}_{i,j}^{\text{op}}$ and $\mathcal{H}_{i,i} \equiv \mathcal{H}_i$.

The aforementioned quantum models with $\mathcal{H}_3$ symmetry can be obtained by choosing $\mathcal{M} = \mathcal{D} = \mathcal{H}_3$ as input to the models defined in eq. (20), and appropriate choices for the coefficients $\lambda_{i,n}$. Dual models with symmetries encoded in $\mathcal{H}_1$, $\mathcal{H}_2$ are then obtained by choosing the module categories $\mathcal{H}_{1,3}$, respectively $\mathcal{H}_{2,3}$, over the input category whose module associators are attached to ref. [5]. Coincidentally, in virtue of the fact that $\text{Fun}_{\mathcal{D}}(\mathcal{D}, \mathcal{M}) \simeq \mathcal{M}$, the category of duality operators is in those respective cases also given by $\mathcal{H}_{1,3}, \mathcal{H}_{2,3}$.

Let us first consider $\mathcal{H}_{1,3}$, whose simple objects we write as $\{\Gamma, \Gamma\alpha, \Gamma\alpha^2, \Lambda\}$. It follows from $N_{\Gamma, \alpha\rho}^{\Gamma} = N_{\Gamma, \alpha^2\rho}^{\Gamma} = 1$ that $\text{Hom}(\Gamma, \Gamma) \simeq 1 \oplus \alpha\rho \oplus \alpha^2\rho$, as an object. The subset of F-symbols appearing in the algebra structure on $\text{Hom}(\Gamma, \Gamma)$ are given in ref. [5]. In similar fashion one constructs the algebra objects $\text{Hom}(\Gamma\alpha, \Gamma\alpha)$, $\text{Hom}(\Gamma\alpha^2, \Gamma\alpha^2)$, which are related to $\text{Hom}(\Gamma, \Gamma)$ by conjugation with the invertible objects α^2 and α of $\mathcal{H}_3$ respectively. Finally one constructs also the algebra object $\text{Hom}(\Lambda, \Lambda)$ which turns out to be equivalent to $\text{Hom}(\Lambda, \Lambda) \simeq 1 \oplus \alpha\rho \oplus \alpha^2\rho$.

$\mathcal{H}_{2,3}$ has two simple objects $\mathcal{I}_{\mathcal{H}_{2,3}} \simeq \{\Omega, \Omega\rho\}$. Its fusion rules can be inferred from $\Omega \triangleleft \alpha^i \simeq \Omega$ and $\Omega \triangleleft \rho \simeq \Omega\rho$. From the first set of fusion rules it follows that the duality $\mathcal{H}_{2,3}$ amounts to gauging the $\text{Vec}_{\mathbb{Z}_3}$ symmetry generated by α , as indeed $\text{Hom}(\Omega, \Omega) \simeq 1 \oplus \alpha \oplus \alpha^2$. The other module object gives rise to the Morita equivalent algebra object $\text{Hom}(\Omega\rho, \Omega\rho) \simeq 1 \oplus \alpha \oplus \alpha^2 \oplus 3 \cdot (\rho \oplus \alpha\rho \oplus \alpha^2\rho)$. Likewise, one shows that $\mathcal{H}_3$ is obtained from gauging the $\text{Vec}_{\mathbb{Z}_3}$ subsymmetry inside $\mathcal{H}_2$.

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A Categorical prerequisites

In this section we introduce the necessary category theoretic tools, notations and conventions used throughout the manuscript. We refer to the standard references [21, 25] for a more elaborate exposition.

A.1 Fusion categories and algebra objects

Succinctly, a *unitary fusion category* (UFC) $\mathcal{C}$ consists of a collection of objects labeling topological defects whose set of (representatives of isomorphism classes of) simple objects is denoted $\mathcal{I}_{\mathcal{C}}$. The collection of defects is endowed with a monoidal product $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ encoding their fusion. It is useful to introduce an integer-valued rank 3 tensor N encoding the fusion rules as $X_1 \otimes X_2 \simeq \bigoplus_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} X_3$. Among the simple objects there exists a unique trivial defect $\mathbb{1}$ satisfying $\mathbb{1} \otimes X \simeq X \otimes \mathbb{1} \simeq X$, $\forall X \in \mathcal{I}_{\mathcal{C}}$. The fusion rules are associative up to an isomorphism which satisfies a consistency axiom known as the *pentagon equation*. Given an appropriate choice of basis for the junction spaces $\mathcal{C}(X_1 \otimes X_2, X_3)$, the components of this isomorphism can be expressed as unitary *F-symbols* defined graphically via:

$$\begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagdown \\ \quad \text{I} \quad \quad \quad \text{I} \\ \quad \diagup \quad \diagdown \quad \diagup \\ X_5 \quad \text{J} \\ \quad \quad \quad \diagdown \\ \quad \quad \quad X_4 \end{array} = \sum_{X_6, k, l} (F_{X_4}^{X_1 X_2 X_3})_{X_5, ij}^{X_6, kl} \begin{array}{c} X_1 \quad X_2 \quad X_3 \\ \diagdown \quad \diagup \quad \diagdown \\ \quad \quad \quad \text{I} \\ \quad \quad \quad \diagup \quad \diagdown \\ \quad \quad \quad \text{K} \\ \quad \quad \quad \diagup \quad \diagdown \\ X_4 \quad \quad \quad X_6 \end{array} . \quad (\text{A.1})$$

In the unitary gauge we additionally assume that

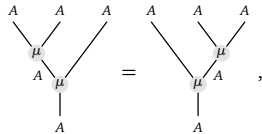
$$(F_X^{X\bar{X}X})_{\mathbb{1}, \mathbb{1}}^{\mathbb{1}, \mathbb{1}} = \frac{\varkappa_X}{d_X}, \quad (\text{A.2})$$

where $\varkappa_X = \pm 1$ denotes the *Frobenius-Schur indicator* of X , which satisfies $\varkappa_{\bar{X}} = \varkappa_X^*$, and the *quantum dimension* d_X of the defect X is the unique solution to

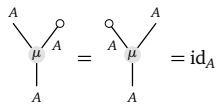
$$d_X > 0, \quad d_{X_1} d_{X_2} = \sum_{X_3 \in \mathcal{I}_{\mathcal{C}}} N_{X_1 X_2}^{X_3} d_{X_3}, \quad (\text{A.3})$$

for all $X, X_1, X_2 \in \mathcal{I}_C$.

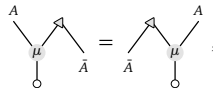
An *algebra object*, henceforth often called simply *algebra*, in a UFC $\mathcal{C}$ is a triple (A, μ, η) consisting of an object $A \simeq \bigoplus_{X \in \mathcal{I}_C} N_X^A X$, with $N_X^A = \dim_{\mathcal{C}} \mathcal{C}(X, A)$, together with a *multiplication morphism* $\mu \in \mathcal{C}(A \otimes A, A)$ that satisfies following associativity condition on the nose


(A.4)

and *unit map* $\eta \in \mathcal{C}(\mathbb{1}, A)$, denoted in our graphical calculus by a white node, satisfying

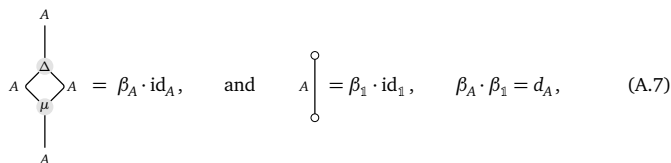

(A.5)

It is a useful fact that $N_X^A \leq [d_X]$ for the algebra objects considered in this manuscript. We also assume that all algebra objects are *haploid*, meaning that A contains the monoidal unit $\mathbb{1}$ of $\mathcal{C}$ with multiplicity 1, or thus $N_{\mathbb{1}}^A = 1$. Indeed, by virtue of [62, Thm. 3.1] every Morita class of (semisimple) indecomposable algebra objects contains at least one haploid representative. An algebra object (A, μ, η) is called *symmetric* if there exists a morphism $\varepsilon \in \mathcal{C}(A, \mathbb{1})$ so that


(A.6)

where the flags depict respectively the right and left coevaluation morphisms of A . It can be shown [25, Cor. 3.10] that Haploid algebras are symmetric for all $\varepsilon \in \mathcal{C}(A, \mathbb{1})$ in virtue of their defining property $\dim_{\mathcal{C}} \mathcal{C}(\mathbb{1}, A) = 1$.

One can then define an analogous *coalgebra object* as a triple (A, Δ, ε) where A is an object in $\mathcal{C}$, $\Delta \in \mathcal{C}(A, A \otimes A)$ is a *comultiplication map* satisfying a coassociativity condition akin to eq. (A.4) and $\varepsilon \in \mathcal{C}(A, \mathbb{1})$ is a *counit map* which together with Δ is required to obey a mirrored version of eq. (A.5). A *special algebra object* is an object that is both an algebra (A, μ, η) and coalgebra (A, Δ, ε) for which


(A.7)

with the quantum dimension of the algebra object A given by $d_A = \sum_{X \in \mathcal{I}_C} N_X^A d_X$. To reduce the occurrence of quantum dimensions in the main text, we adopt without loss of generality the convention that $\beta_A = 1$, $\beta_{\mathbb{1}} = d_A$.

Proceeding as such, a *Frobenius algebra object* is a quintuple $(A, \mu, \eta, \Delta, \varepsilon)$ where (A, μ, η) constitutes an algebra and (A, Δ, ε) a coalgebra, such that the multiplication and comultipli-

cation map jointly obey the *Frobenius relations*

$$\begin{array}{c} A \\ | \\ \Delta \\ / \quad \backslash \\ A \quad A \\ | \quad | \\ \mu \quad \mu \\ | \quad | \\ A \quad A \end{array} = \begin{array}{c} A \quad A \\ | \quad | \\ \mu \quad \mu \\ | \quad | \\ A \quad A \\ | \\ \Delta \\ A \end{array} \quad (A.8)$$

Considering an algebra (A, Δ, μ) in $\mathcal{C}$, the multiplication morphism can be written in components by means of

$$\begin{array}{c} X_1 \quad X_2 \\ | \quad | \\ i \quad j \\ | \quad | \\ A \quad A \\ | \quad | \\ \mu \quad \mu \\ | \quad | \\ A \quad A \\ | \\ X_3 \end{array} = \sum_l \mu_{(X_3,k),l}^{(X_1,i),(X_2,j)} \begin{array}{c} X_1 \quad X_2 \\ | \quad | \\ l \quad l \\ | \quad | \\ A \quad A \\ | \\ X_3 \end{array} \quad (A.9)$$

where i, j, k and l label orthonormal basis vectors in the vector spaces $\mathcal{C}(X_1, A)$, $\mathcal{C}(X_2, A)$, $\mathcal{C}(A, X_3)$ and $\mathcal{C}(X_1 \otimes X_2, X_3)$ respectively. Analogously one defines the tensor $\Delta_{(X_1,i),(X_2,j)}^{(X_3,k),l}$ for the comultiplication map.

Two algebra objects are *isomorphic* when they are related by unitary basis transformations on the multiplicity spaces $\mathcal{C}(X_i, A)$ and $\mathcal{C}(A, X_i)$.

A.2 Module categories and module functors

Given a UFC $\mathcal{D}$, a (unitary finite indecomposable semisimple) *right module category* $\mathcal{R}$ over $\mathcal{D}$ is a semisimple category whose set of (representative objects of isomorphism classes of) simple objects is written as $\mathcal{I}_{\mathcal{R}}$ and which is equipped with a right action bifunctor $\triangleleft: \mathcal{R} \times \mathcal{D} \rightarrow \mathcal{R}$. Up to an isomorphism the action is associative. In an appropriate basis of the junction spaces, this isomorphism can once more be expressed as following unitary F-symbols:

$$\begin{array}{c} R_1 \quad Y_1 \quad Y_2 \\ | \quad | \quad | \\ R_3 \\ | \\ R_2 \end{array} = \sum_{Y_3, ij} (F_{R_2}^{R_1 Y_1 Y_2})_{R_3, ij}^{Y_3, kl} \begin{array}{c} R_1 \quad Y_1 \quad Y_2 \\ | \quad | \quad | \\ R_3 \\ | \\ R_2 \end{array} \quad (A.10)$$

These F-symbols satisfy a mixed pentagon equation which also involves the F-symbols of $\mathcal{D}$. We choose a basis for the Hom spaces such that

$$(F_{R_2}^{R_1 X_2})_{R_1, 1i}^{X_2, 1j} = (F_{R_2}^{R_1 X_1 1})_{R_2, i1}^{X_1, 1j} = \delta_{i,j}. \quad (A.11)$$

Also for the module objects $R \in \mathcal{I}_{\mathcal{R}}$ we define quantum dimensions, which are uniquely specified by [22]:

$$\begin{aligned} d_R &> 0, \\ d_{R_1} d_X &= \sum_{R_2 \in \mathcal{I}_{\mathcal{R}}} N_{R_1 X}^{R_2} d_{R_2}, \\ \sum_{R \in \mathcal{I}_{\mathcal{R}}} d_R^2 &= \sum_{X \in \mathcal{I}_{\mathcal{C}}} d_X^2, \end{aligned} \quad (A.12)$$

$$\begin{array}{c} X \\ \textcolor{red}{\diagdown} \\ \textcolor{blue}{i} \\ \textcolor{blue}{\diagup} \\ R_1 \\ \textcolor{blue}{\vdots} \\ R'_1 \end{array} \begin{array}{c} R_1 \\ \textcolor{blue}{\vdots} \\ \textcolor{blue}{j} \\ \textcolor{blue}{\diagdown} \\ Y \end{array} = \sum_{R_2, k, l} ({}^X \omega_{R'_2}{}^{R_2, kl})_{R'_1, ij} \begin{array}{c} X \\ \textcolor{red}{\diagdown} \\ \textcolor{blue}{k} \\ \textcolor{blue}{\diagup} \\ R_1 \\ \textcolor{blue}{\vdots} \\ R'_1 \end{array} \begin{array}{c} R_1 \\ \textcolor{blue}{\vdots} \\ R_2 \\ \textcolor{blue}{\diagdown} \\ Y \end{array} . \quad (\text{A.13})$$
$$\begin{array}{c}
X_1 \quad X_2 \quad M \\
\diagdown \quad \diagup \quad | \\
i \quad j \\
\diagup \quad \diagdown \quad | \\
X_3 \quad j \\
| \\
Q
\end{array}
= \sum_{N,kl} (F_O^{X_1 X_2 M})_{X_3, ij}^{N, kl}
\begin{array}{c}
X_1 \quad X_2 \quad M \\
\diagdown \quad \diagup \quad | \\
\quad \quad k \\
\diagup \quad \diagdown \quad | \\
\quad \quad l \\
| \\
Q
\end{array}, \quad (\text{A.14})$$

8

Given a right module category $\mathcal{M}$ over $\mathcal{C}$, an *internal Hom object* [62] for any two objects M_1, M_2 is an object $\text{Hom}(M_1, M_2)$ in $\mathcal{C}$ together with a collection of isomorphisms

satisfying a number of naturality conditions specified in ref. [66]. Crucially, for any simple object $M \in \mathcal{I}_M$, $\text{Hom}(M, M)$ is guaranteed to be a special haploid symmetric Frobenius algebra object, as demonstrated in ref. [66]. Let us spell this out explicitly. Notating $A \equiv \text{Hom}(M, M)$, it readily follows from (A.15) that the decomposition of A in simple objects C is given by $A \simeq \bigoplus_{X \in \mathcal{I}_M} N_{M \times X}^M M_{M \times X}$, where $N_{M \times X}^M \equiv \dim_c \mathcal{M}(M \triangleleft X, M)$. The components of the multiplication and comultiplication maps can be related to the right module associator (A.10) via

$$\begin{aligned}\mu_{(X_3,k),l}^{(X_1,i),(X_2,j)} &= \frac{1}{d_M} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_M^{MX_1X_2})_{M,ij}^{X_3,lk}, \\ \Delta_{(X_1,i),(X_2,j)}^{(X_3,k),l} &= \frac{1}{d_M} \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (\bar{F}_M^{MX_1X_2})_{M,ij}^{X_3,lk},\end{aligned}\tag{A.16}$$

in which once more the isomorphisms (A.15) were leveraged. In this normalisation of the multiplication and comultiplication map and the F-symbol eq. (A.11), the unique component of the unit and counit labelled by η_1 and ε_1 respectively is given by

$$\eta_1 = \varepsilon_1 = d_M, \quad (\text{A.17})$$

which is verified to be compatible with the normalization in eq. (A.7).

B Orientation-reversing flags

Let X be a simple object in $\text{Fun}_{\mathcal{D}}(\mathcal{R}, \mathcal{R}')$ in what follows. Orientation-reversing ‘flags’ for X are defined in terms of the fusion (splitting) tensors by terminating them in the vacuum label $\mathbb{1}$:

$$\begin{aligned} \begin{array}{c} R' \\ \leftarrow X \rightarrow \bar{X} \rightarrow \\ R \end{array} &\equiv \sum_{i,j} \begin{array}{c} R' \\ \leftarrow X \rightarrow \bar{X} \rightarrow j \\ R \end{array} |RXR', i\rangle \otimes |R'\bar{X}R, j\rangle \\ &\equiv \sum_{i,j} \sqrt{d_X} \times \begin{array}{c} R' \quad \mathbb{1} \quad R' \\ \downarrow \quad \downarrow \quad \downarrow \\ \leftarrow X \rightarrow \bar{X} \rightarrow \\ R \end{array} |RXR', i\rangle \otimes |R'\bar{X}R, j\rangle, \end{aligned} \quad (\text{B.1})$$

and

$$\begin{aligned} \begin{array}{c} R \\ \leftarrow X \rightarrow \bar{X} \leftarrow \\ R' \end{array} &\equiv \sum_{i,j} \begin{array}{c} R \\ \leftarrow X \rightarrow \bar{X} \leftarrow j \\ R' \end{array} \langle RXR', i| \otimes \langle R'\bar{X}R, j| \\ &\equiv \sum_{i,j} \sqrt{d_X} \times \begin{array}{c} i \quad R \quad j \\ \downarrow \quad \downarrow \quad \downarrow \\ \leftarrow X \rightarrow \bar{X} \leftarrow \\ R' \end{array} \langle RXR', i| \otimes \langle R'\bar{X}R, j|. \end{aligned} \quad (\text{B.2})$$

In these definitions, $\mathbb{1}$ denotes the monoidal unit in the fusion category $\mathcal{C}_{\mathcal{R}'}$, and 1 stands for the sole basis vector in $\mathcal{R}'(\mathbb{1} \triangleright R, R) \simeq \mathbb{C}$.

The Frobenius-Schur indicator, which for module objects is defined in a way analogous to eq. (A.2), can be used to reverse the orientation of the flags, concretely:

$$\begin{aligned} \begin{array}{c} R' \\ \leftarrow X \rightarrow \bar{X} \rightarrow \\ R \end{array} &= \chi_X^* \begin{array}{c} R' \\ \leftarrow X \rightarrow \bar{X} \leftarrow \\ R \end{array}, \\ \begin{array}{c} R \\ \leftarrow X \rightarrow \bar{X} \leftarrow \\ R' \end{array} &= \chi_X \begin{array}{c} R \\ \leftarrow X \rightarrow \bar{X} \rightarrow \\ R' \end{array}. \end{aligned} \quad (\text{B.3})$$

The flags satisfy a number of properties generalising those spelled out in refs. [9, 54]. Most notably, they satisfy a collection of ‘snake axioms’ stating that oppositely paired flags cancel in the sense that eg.

$$\begin{array}{c} \leftarrow X \rightarrow \bar{X} \rightarrow \\ \leftarrow X \rightarrow \bar{X} \leftarrow \end{array} = \begin{array}{c} \leftarrow X \rightarrow \end{array}. \quad (\text{B.4})$$

As such, these flags implement the evaluation and coevaluation maps of X , $\text{ev}_X : \bar{X} \otimes X \rightarrow \mathbb{1}$, respectively $\text{coev}_X : \mathbb{1} \rightarrow X \otimes \bar{X}$, in terms of triple line tensors.

Moreover, these flags can be used to bend lines on the fusion and splitting tensors as follows:

$$\begin{aligned}
 & \text{Diagram 1} = \sum_j \underbrace{\sqrt{d_{X_1}} (\bar{F}_{X_2}^{\bar{X}_1 X_1 X_2})_{\mathbb{1}, 11}^{X_3, ij}}_{\sqrt{\frac{d_{X_3}}{d_{X_2}}} (\bar{A}_{X_3}^{X_1 X_2})_{ij}} \text{Diagram 2}, \\
 & \text{Diagram 3} = \sum_j \underbrace{\sqrt{d_{X_2}} (F_{X_1}^{X_1 X_2 \bar{X}_2})_{X_3, ij}^{\mathbb{1}, 11}}_{\sqrt{\frac{d_{X_3}}{d_{X_1}}} (B_{X_3}^{X_1 X_2})_{ij}} \text{Diagram 4}.
 \end{aligned} \tag{B.5}$$

The unitary matrices A and B are defined as [46]

$$\begin{aligned}
 (A_{X_3}^{X_1 X_2})_{ij} &= \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_{X_2}^{\bar{X}_1 X_1 X_2})_{\mathbb{1}, 11}^{X_3, ij}, \\
 (B_{X_3}^{X_1 X_2})_{ij} &= \sqrt{\frac{d_{X_1} d_{X_2}}{d_{X_3}}} (F_{X_1}^{X_1 X_2 \bar{X}_2})_{X_3, ij}^{\mathbb{1}, 11}.
 \end{aligned} \tag{B.6}$$

C Proof (co)algebra multiplication

Here, we showcase that the inflated comultiplication tensor (32) satisfies the comultiplication axiom by making use of our graphical calculus [26, 66]. Invoking the orientation-reversing flags from appendix B one finds:

$$\sum_{\substack{\{X_j\}_j, \\ \{i_j\}_j}} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4} d_{X_5}}}{d_{X_A}^4} \text{Diagram} \tag{C.1}$$

$$\begin{aligned}
&= \sum_{\substack{\{X_j\}_j, \\ \{i_j\}_j}} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4}}}{d_{X_A}^3} \\
&= \sum_{\substack{\{X_j\}_j, \\ \{i_j\}_j}} \frac{\sqrt{d_{X_1} d_{X_2} d_{X_3} d_{X_4} d_{X_6}}}{d_{X_A}^4}
\end{aligned}$$

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Systematic construction of stabilizer codes via gauging abelian boundary symmetries

9.1 Synopsis

In this project, we generalise a construction of the second author, published in ref. [123], to arbitrary spatial dimensions. Starting from a generic d -dimensional quantum state with an *abelian* symmetry, we iteratively gauge the symmetry and its dual to obtain a state in one dimension higher. The admissible initial symmetries include global 0-form, higher-form, *subsystem* and *fractal symmetry*.

The resulting $(d + 1)$ -dimensional state naturally inherits this original symmetry at its boundary and can be shown to be the ground state of a local commuting projector Hamiltonian. The construction relies on an extension, developed in this work, of Williamson's gauging procedure of *submanifold symmetries* [124]. We formulate the framework in terms of tensor networks, where gauging maps are implemented as tensor network operators akin to those of ref. [118]. The composition of these then provides a tensor network representation of the one-higher-dimensional state.

We illustrate the approach through several examples, including three-dimensional *Clifford-deformed surface codes* and *type-I fracton phases* obtained by gauging linear subsystem and *Sierpinski* fractal symmetries.

9.2 Reference

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9.3 Contributions of the author

The current project is a generalisation of the construction developed in ref. [123] by José Garre-Rubio to arbitrary dimensions. My main contribution was in generalising the gauging procedure of Williamson, which in its original form was formulated for the symmetry group $\mathbb{Z}_2$, to arbitrary abelian groups. This extension constitutes the main technical ingredient of the work. The remaining constructions follow by applying the procedure to the relevant symmetry structures.

I co-wrote the majority of the manuscript and worked out the example of the three-dimensional Clifford-deformed surface codes. The other examples are fully due to José Garre-Rubio.

Systematic construction of stabilizer codes via gauging abelian boundary symmetries

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We propose a systematic framework to construct a $(d+1)$ -dimensional stabilizer model from an initial generic d -dimensional abelian symmetry. Our approach builds upon the *iterative gauging procedure*, developed by one of the authors in [J. Garre-Rubio, *Nature Commun.* **15**, 7986 (2024)], in which an initial symmetric state is repeatedly gauged to obtain an emergent model in one dimension higher that supports the initial symmetry at its boundary. This method not only enables the construction of emergent states and corresponding commuting stabilizer Hamiltonians of which they are ground states, but it also provides a way to construct gapped boundary conditions for these models that amount to spontaneously breaking part of the boundary symmetry.

In a detailed introductory example, we showcase our paradigm by constructing three-dimensional Clifford-deformed surface codes from iteratively gauging a global 0-form symmetry that lives in two dimensions. We then provide a proof of our main result, hereby drawing upon a slight extension of the gauging procedure of Williamson. We additionally provide two more examples in $d=2$ in which different type-I fracton orders emerge from gauging initial linear subsystem and Sierpinski fractal symmetries. En passant, we provide explicit tensor network representations of all of the involved gauging maps and the emergent states.

Introduction

Different incarnations of *bulk-boundary correspondence* are ubiquitous in condensed matter physics. Generically, a bulk-boundary correspondence establishes a relation between the non-trivial physics in the bulk of a certain physical system and the physics constrained to its boundary. One example of such a relation is the conjecture that the bulk of certain *anomaly-free* topological orders is fully determined by

their gapped boundary theory by means of an appropriate notion of the categorical *center* [1–3]. A bulk-boundary correspondence has also proven to be the key for finding cohomological invariants that differentiate certain symmetry-protected topological phases (SPTs) protected by an invertible symmetry, whereby these invariants pertain to the transformation properties of the edge modes under the symmetry group. For the particular case of $(1+1)d$, this approach has led to a complete classification of gapped one-dimensional phases with group-like symmetry [4, 5]. More recently, it has also been argued how the boundaries of particular fracton phases host *emergent* subsystem symmetries whose explicit realization can be traced back to the characteristics of the bulk excitation spectrum [6]. Contrary to the case of intrinsic topological orders, it was demonstrated that the *bulk reconstruction* of the fracton orders under scrutiny depends in a non-trivial way on the relative orientation of the boundary with respect to the bulk. In a certain sense all of the above examples of bulk-boundary correspondence can thus be brought together under the umbrella term of a *holographic principle*.

In another recent series of developments an opposite avenue has been pursued in which a given quantum field theory (QFT) with a (higher) *categorical symmetry* is regarded as an interval compactification of a topological field theory in one dimension higher, coined ‘SymTFT’, with a *topological* gapped boundary condition determining the symmetry of the QFT and a not necessarily gapped *physical* boundary condition that dictates the dynamics of the QFT [7, 8]. In this picture the focus is very much on the understanding of certain features of the symmetric QFT, such as (dis)order parameters, gapped phases and anomalies, from the higher-dimensional SymTFT. Aspects of this paradigm have previously appeared under the name of ‘strange correlator’, which admits an explicit lattice realization in terms of tensor network representations of the higher-dimensional topological order [9–11].

In this manuscript we realize an explicit bulk-boundary correspondence for generic abelian *boundary symmetries* in any dimension by making use of the *iterative gauging procedure* proposed by one of the authors in [12] for *one-dimensional abelian bound-*

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ary symmetries. That work builds on the observation that gauging an *initial* abelian (anomaly-free global 0-form) symmetry gives rise to *dual* on-site global symmetry generators labeled by the one-dimensional representations of the underlying symmetry group. Since this emergent symmetry is on-site, a tensor product of local unitaries, it is free of any 't Hooft anomaly, and so it can be gauged, thereby recovering the initial G symmetry. The crux of the paradigm put forward in [12] is that these gauging procedures can be repeated indefinitely and that in this way a state emerges which naturally lives in one dimension higher than the initial symmetry. Such an *emergent state* was shown in [12] to be stabilized by certain local operators whose origin can be traced back to the Gauss constraints of the involved gauging procedures. In this picture, the initial symmetry which is being gauged can be thought of as a *boundary symmetry* for the emergent state. This idea is made very tangible by making use of the explicit *gauging map* for quantum states proposed in [13] and generalized later in [14]. One of the main advantages of using this approach is that a tensor network description of the gauging maps is readily available, which can in turn be used to write down an explicit tensor network representation of the emergent state, facilitating the analytical study of its properties and its numerical simulation.

In the current manuscript, Ref. [12] is extended to arbitrary dimensions and generic abelian symmetries. Specifically, we propose a slight generalization of the gauging procedure of [14] to arbitrary abelian groups, and demonstrate that the iterative gauging procedure of [12] in *any dimension* gives rise to emergent states with corresponding *local stabilizer* Hamiltonians. The abelian symmetries which are being gauged in the framework of [14] are completely specified by a *complete collection* of ‘checks’ which commute with all symmetry generators. As such, that framework includes the gauging of *fractal* and *subsystem* symmetries as specific cases. In our general framework we are also able to relate inequivalent choices of *gapped boundary stabilizers* to different spontaneous symmetry breaking patterns of the boundary symmetry.

To showcase our framework we provide three detailed worked examples where the initial boundary symmetry is two-dimensional. Gauging a global on-site 0-form symmetry in $(2+1)d$ gives rise to emergent 1-form – or *Wilson loop*– symmetries. In this case the iterative gauging framework gives rise to three-dimensional *Clifford-deformed surface codes* [15], which are up to a local basis change equivalent to the three-dimensional abelian quantum doubles [16]. Two-dimensional linear subsystem symmetries map to dual linear subsystem symmetries on the dual lattice. Applying the framework in this case gives rise to foliated anisotropic type-I fracton orders [17]. Finally, we leverage the full potential of our framework and the gauging procedure of [14] to

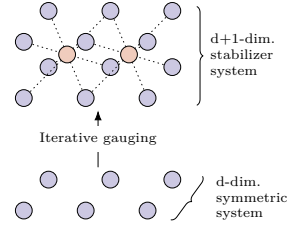


Figure 1: Schematic overview of the iterative gauging procedure. Given a d -dimensional symmetric state, iterative gauging of its symmetry and the emerging dual symmetries give rise to a $(d+1)$ -dimensional stabilizer model with the initial state as boundary.

gauging fractal boundary symmetry, specifically of Sierpinski type, and obtain the fracton model proposed by Caselnovi and Chamon in [18] for the study of quantum glassiness. For each of these examples we also provide a tensor network description of the involved gauging maps, and hence the emergent state, which is based on the tensor network description of the gauging map for global on-site 0-form symmetries of [13].

Our manuscript is organized as follows. In section 1 we provide an introductory example to explicitly demonstrate the iterative gauging procedure applied to an initial two-dimensional global on-site symmetry. All the details of our framework including the study of symmetry-breaking boundary conditions and the tensor network representation of the emergent Clifford-deformed CSS codes are discussed in detail. This section also serves as an introduction to many of the conventions and notations used throughout this manuscript. We then proceed by putting these observations on systematic footing in section 2. First, we slightly generalize the gauging procedure of [14] to generic abelian groups and then prove that the iterative gauging procedure in any dimension gives rise to emergent states with a corresponding commuting stabilizer Hamiltonian of which it is a ground state. We then provide a way to add boundary stabilizers to the emergent Hamiltonian which commute with all terms in the bulk and among each other, thereby preserving the gap. We argue that different gapped boundary stabilizers correspond to certain symmetry breaking patterns of the boundary symmetry. We then conclude with two more examples in section 3 and section 4 that pertain to the derivation of different type-I fracton models from the iterative gauging of linear subsystem boundary symmetry and Sierpinski fractal symmetry respectively.

1 Introductory example: Clifford-deformed surface codes from abelian higher-form symmetries

In this section we review the gauging procedure of abelian on-site global 0-form symmetries proposed in [13] for two-dimensional symmetric states. We argue that the gauged state is in the even sector of a dual global 1-form $\widehat{G}$ symmetry acting non-trivially solely on the introduced gauge degrees of freedom. We use an explicit gauging procedure for this 1-form symmetry, following refs. [19, 14], that yields in turn again the original G symmetry. For both gauging maps we construct a tensor network representation of bond dimension $|G|$. We show that the concatenation of these 2D gauging maps as in [12] produces 3D Clifford-deformed surface codes [15], which are unitarily equivalent to abelian G quantum doubles [16].

Let us consider the two-dimensional torus endowed with a lattice Λ , whose sets of vertices, oriented edges and plaquettes will be denoted by $V(\Lambda)$, $E(\Lambda)$ and $P(\Lambda)$ respectively. The starting point of the gauging procedure laid out in [13] is then a two-dimensional state $|\psi\rangle$ living in a tensor product Hilbert space of the so-called *matter* degrees of freedom supported on the vertices of Λ , $|\psi\rangle \in \bigotimes_{v \in V(\Lambda)} \mathcal{H}^v$. The local Hilbert space $\mathcal{H}^v$ forms a unitary representation of G , which is assumed to be abelian throughout the manuscript, and whose matrix representation is written as $\{u_g | g \in G\}$. We adapt the notation $\bar{g}$ to denote the inverse group element of $g \in G$ and 1 to denote the identity element. The state $|\psi\rangle$ is then required to transform trivially under the tensor product representation $\{\bigotimes_{v \in V(\Lambda)} u_g^v | g \in G\}^1$, i.e.:

$$\left(\bigotimes_{v \in V(\Lambda)} u_g^v \right) |\psi\rangle = |\psi\rangle, \quad \forall g \in G. \quad (1.1)$$

Notice that the symmetry operators considered here act on all matter degrees of freedom $\mathcal{H}^v$, $v \in V(\Lambda)$, simultaneously. Such a global G symmetry is in modern jargon referred to as an *invertible global 0-form* symmetry [20].

Here and below, for $q = 0, 1$, a q -form operator is a *topological* operator that acts non-trivially only on a codimension q submanifold of the spatial manifold, i.e. the two-dimensional torus. The case of the $\bigotimes_v u_g^v$ symmetry operators then naturally corresponds to $q = 0$.

For the sake of being concrete we will henceforth make an explicit choice of local Hilbert space and its rep-

resentation.² To this end we consider the character group $\widehat{G} := \text{Hom}(G, \text{U}(1))$ of G , which is isomorphic to G here since G is abelian. We then choose the local Hilbert space as $\mathcal{H}^v = \mathbb{C}[\widehat{G}]$ where $\mathbb{C}[\widehat{G}] := \text{Span}_{\mathbb{C}}\{|\chi\rangle | \chi \in \widehat{G}\}$ is endowed with the inner product $\langle \chi_1 | \chi_2 \rangle = \delta_{\chi_1, \chi_2}$. This local Hilbert space is then assumed to carry the G -representation $u_g = Z_g$ defined as $Z_g := \sum_{\chi \in \widehat{G}} \chi(g) |\chi\rangle \langle \chi|$, $\forall g \in G$. Let us also introduce the shorthand $\mathbb{U}_g := \bigotimes_{v \in V(\Lambda)} Z_g^v$.

1.1 Review: gauging two-dimensional abelian 0-form symmetries

Gauging a global on-site 0-form G symmetry à la [13]³ begins with supplementing the Hilbert space with gauge degrees of freedom supported on the edges $E(\Lambda)$ and valued in $\mathbb{C}[G] := \text{Span}_{\mathbb{C}}\{|g\rangle | g \in G\}$, $\langle g_1 | g_2 \rangle = \delta_{g_1, g_2}$. The group algebra $\mathbb{C}[G]$ is naturally endowed with the left regular representation given by:

$$X_g := \sum_{h \in G} |gh\rangle \langle h|, \quad \forall g \in G. \quad (1.2)$$

On this enlarged Hilbert space

$$\left(\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\widehat{G}] \right) \otimes \left(\bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \quad (1.3)$$

we then define a *gauge transformation* for every vertex $v \in V(\Lambda)$ and every $g \in G$:

$$\mathcal{G}_g^v := Z_g^v \bigotimes_{e(\rightarrow v)} X_{\bar{g}}^e \bigotimes_{e(\leftarrow v)} X_g^e. \quad (1.4)$$

Here, $e(\rightarrow v)$, respectively $e(\leftarrow v)$, stands for the set of edges of Λ that point into, respectively point away from, the fixed vertex v .

Notice that the operators $\mathcal{G}_g^v$ in eq. (1.4) form a *local* representation of G in the sense that they are supported on a small neighborhood around v and obey $\mathcal{G}_{g_1}^v \cdot \mathcal{G}_{g_2}^v = \mathcal{G}_{g_1 g_2}^v$ for every $v \in V(\Lambda)$ and for every $g_1, g_2 \in G$. Moreover, notice that the global symmetry operators are recovered by taking the product of the local symmetry operators on all vertices, namely

$$\prod_{v \in V(\Lambda)} \mathcal{G}_g^v = \mathbb{U}_g, \quad \forall g \in G. \quad (1.5)$$

²It should be noted that the gauging procedure which follows can be carried out for any on-site realization of the symmetry [13]. However, in the construction of the emergent state below, the choice of the representation of the G symmetry on the boundary does not influence the bulk stabilizers of the emergent state.

³Notice that the framework of [13] applies more generally to non-abelian groups as well. The gauged theory in that case enjoys Wilson loop symmetries labeled by representations of the symmetry group. Wilson loops corresponding to higher-dimensional representations of the group can be realized on the lattice by non-trivial *matrix product operators* [21–23].

¹Indeed, if $|\psi\rangle$ transforms non-trivially under G , the gauging map defined below annihilates $|\psi\rangle$. However, a different gauging map can be defined for each symmetry sector.

Next, a *local* projector is obtained by averaging $\mathcal{G}_g^\nu$ over G :

$$\mathcal{P}^\nu := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^\nu. \quad (1.6)$$

Indeed it is straightforward to verify that $\mathcal{P}^\nu \cdot \mathcal{P}^\nu = \mathcal{P}^\nu$, $\forall \nu \in V(\Lambda)$ by a relabeling of summation variables. The commutation of the local projectors $[\mathcal{P}^\nu, \mathcal{P}^{\nu'}] = 0$ allows us to unambiguously define the *global* projector onto the gauge invariant subspace of the enlarged Hilbert space:

$$\mathcal{P} := \prod_{\nu \in V(\Lambda)} \mathcal{P}^\nu. \quad (1.7)$$

The state $|\psi\rangle$ is then gauged by applying the projector $\mathcal{P}$ to the initial state $|\psi\rangle$ coupled to a trivial flat gauge field $\bigotimes_{e \in E(\Lambda)} |1\rangle^e$. Hence, we define the *gauging map* $\mathcal{G}^0$ for the global 0-form G symmetry as:

$$\begin{aligned} \mathcal{G}^0 &:= \mathcal{P} \left(\bigotimes_{e \in E(\Lambda)} |1\rangle^e \right) : \\ &\left(\bigotimes_{\nu \in V(\Lambda)} \mathbb{C}[\widehat{G}] \right) \rightarrow \left(\bigotimes_{\nu \in V(\Lambda)} \mathbb{C}[\widehat{G}] \right) \otimes \left(\bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right), \end{aligned} \quad (1.8)$$

and the *gauged state* is by definition $\mathcal{G}^0 |\psi\rangle$.

In order to elucidate the above procedure and facilitate the construction of the emergent state below, let us present a tensor network representation of the gauging map $\mathcal{G}^0$. To this end we introduce two types of tensors. For every vertex of the lattice with m incoming edges and n outgoing edges, we introduce the rank- $(m+n)$ *vertex tensor*, or *Z-type tensor*, defined as follows:

$$\begin{aligned} &\text{Diagram: A central blue circle with four legs. Two red legs (incoming) and two blue legs (outgoing).} \\ &:= \sum_{\substack{\chi_1, \chi_2 \\ \{g_k\}_k}} \frac{1}{|G|} g_{m+n} \text{ (red leg)} \otimes |g_{m+1}, \dots, g_{m+n}\rangle \langle g_1, \dots, g_m| \\ &:= \sum_{g \in G} \frac{1}{|G|} Z_g \otimes |g, \dots, g\rangle \langle g, \dots, g|. \end{aligned} \quad (1.9)$$

In this and subsequent figures, the dots are a placeholder for tensor legs which are omitted for clarity.

In order to implement the flatness of the gauge field supported on the edges of the lattice we consider the rank-3 *+type* tensors given by:

$$\begin{aligned} &\text{Diagram: A red circle with three legs. Two red legs (incoming) and one blue leg (outgoing).} \\ &:= \sum_{\substack{g, h, k \\ g \in G}} g \text{ (red leg)} \otimes |g, h\rangle \langle k| \\ &:= \sum_{g, h \in G} |g, h\rangle \langle gh|, \end{aligned} \quad (1.10)$$

and

$$\begin{aligned} &\text{Diagram: A red circle with three legs. One red leg (incoming) and two blue legs (outgoing).} \\ &:= \sum_{\substack{g, h, k \\ g \in G}} k \text{ (red leg)} \otimes |k\rangle \langle g, h| \\ &:= \sum_{g, h \in G} |gh\rangle \langle g, h|. \end{aligned} \quad (1.11)$$

Using these tensors, the local projector $\mathcal{P}^\nu$ introduced in eq. (1.6) admits the simple tensor network representation

$$\mathcal{P}^\nu = \text{Diagram: A blue circle with four legs. Two red legs (incoming) and two blue legs (outgoing).} \quad (1.12)$$

where the orientation of the horizontal red legs is assumed to be in correspondence with the orientation of the edges of the lattice. Note that the commutation of two local projectors $\mathcal{P}^\nu, \mathcal{P}^{\nu'}$ on different vertices $\nu \neq \nu'$ follows immediately from the definition of the $+$ -type tensors. Combining this observation with the fact that projecting one of the incoming legs of the $+$ -type tensor eq. (1.11) onto $|1\rangle$ effectively removes that leg, it follows that the gauging map $\mathcal{G}^0$ admits a representation as projected entangled-pair operator (PEPO) of bond dimension $|G|$, whose unit cell can be depicted as:

$$\text{Diagram: A unit cell of the PEPO, consisting of a blue circle (Z-type tensor) and a red circle (+-type tensor) connected by red legs.} \quad (1.13)$$

In this construction it should be understood that the full PEPO on the torus is obtained by placing copies of this unit cell on all vertices of the lattice and contracting along the virtual red legs according to the connectivity of the lattice.

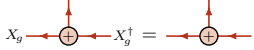
Let us proceed by dissecting the symmetry properties of the gauging map $\mathcal{G}^0$. One of the upshots of the PEPO representation is that these symmetry properties automatically follow from the symmetry properties of the local tensors from which the PEPO is made up. It follows readily from its definition that the vertex tensor satisfies

$$\text{Diagram: A blue circle with four legs.} = x_g^\dagger \text{ (red leg)} \otimes \text{Diagram: A blue circle with four legs.} = x_g \text{ (blue leg)} \otimes \text{Diagram: A blue circle with four legs.} \quad (1.14)$$

whereas the edge tensor obeys the relations

$$\text{Diagram: A red circle with three legs.} = x_g \text{ (red leg)} \otimes \text{Diagram: A red circle with three legs.} = x_g \text{ (blue leg)} \otimes \text{Diagram: A red circle with three legs.} \quad (1.15)$$

Consider first gauge invariance of the gauging map, that is $\mathcal{G}_g^\vee \cdot \mathcal{G}^0 = \mathcal{G}^0$ for all $g \in G$ and $\mathbf{v} \in \mathbf{V}(\square)$. Note that it is exactly in this sense that this gauging procedure *localizes* the global 0-form G symmetry. From the PEPO representation this property can be deduced by combining the second relation in eq. (1.14) with both identities in eq. (1.15) and noticing that all $X_g^{(\dagger)}$'s on the virtual level exactly cancel with each other. Note that algebraically this follows directly from a renaming of summation variable in the definition of $\mathcal{P}^\vee$ eq. (1.6). One also has that $\mathcal{G}^0 \cdot \mathbb{U}_g = \mathcal{G}^0$ for all $g \in G$. Physically this means that the gauging map projects out states which transform non-trivially under the global symmetry G . As such, $\mathcal{G}^0$ has a non-trivial kernel spanned by states in the original matter Hilbert space $\bigotimes_{\mathbf{v} \in \mathbf{V}(\Lambda)} \mathbb{C}[\widehat{G}]$ which are charged under the G symmetry. This can be seen from application of the first identity in eq. (1.14) on every vertex together with the invariance of the edge tensors, namely



$$X_g \leftarrow \bigoplus \leftarrow X_g^\dagger = X_g^\dagger \leftarrow \bigoplus \leftarrow X_g, \quad (1.16)$$

which is deduced from eq. (1.15).

As mentioned in the introduction, the gauged states exhibit at least a global 1-form $\widehat{G}$ symmetry as a consequence of the imposed Gauss laws. This 1-form symmetry is the topic of the next section.

To conclude this paragraph, a comment which is of importance for what follows is in place. As argued in [13] it is possible to disentangle the initial matter degrees of freedom from the newly introduced gauge degrees of freedom by means of a finite depth unitary quantum circuit, rendering the Gauss constraints $\mathcal{G}_g^\vee \stackrel{!}{=} 1$ trivial and effectively freezing the matter degrees of freedom to the $+1$ eigenstate of Z_g . As showcased below, we will construct a three-dimensional state by concatenating gauging operators, wherein the ‘matter’ degrees of freedom in every layer will play the role of physical degrees of freedom of the three-dimensional state. We therefore refrain from applying this unitary circuit.

1.2 The dual 1-form symmetry

Let us consider $\ell \in Z_1(\Lambda, \mathbb{Z})$ to be an oriented closed path supported on the edges of the lattice Λ , and introduce

$$Z_\chi := \sum_{g \in G} \chi(g) |g\rangle\langle g|, \quad \forall \chi \in \widehat{G}, \quad (1.17)$$

which together with X_g obeys the following commutation relation:

$$Z_\chi X_g = \chi(g) X_g Z_\chi. \quad (1.18)$$

We define then following *Wilson loop* operators acting exclusively on the gauge degrees of freedom:

$$\mathbb{U}_\chi[\ell] := \bigotimes_{e \in \ell} Z_{o_\ell^\epsilon(\chi)}. \quad (1.19)$$

Here, $o_\ell^\epsilon(\chi)$ is defined so that $o_\ell^\epsilon(\chi) = \chi$ if the orientation of $\mathbf{e}$ and ℓ coincide and $\bar{\chi}$ otherwise. As we will momentarily show, it holds that

$$\mathbb{U}_\chi[\ell] \cdot \mathcal{G}^0 = \mathcal{G}^0, \quad (1.20)$$

$\forall \chi \in \widehat{G}, \ell \in Z_1(\Lambda, \mathbb{Z})$. This implies that the gauged state $\mathcal{G}^0 |\psi\rangle$ transforms trivially under $\mathbb{U}_\chi[\ell]$ for every valid choice of ℓ and every character $\chi \in \widehat{G}$. Making use of the fact that

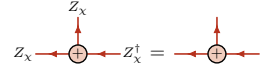
$$\chi_1(g)\chi_2(g) = (\chi_1 \otimes \chi_2)(g), \quad (1.21)$$

$\forall \chi_1, \chi_2 \in \widehat{G}, g \in G$, it follows that the operators $\mathbb{U}_\chi[\ell]$, with same ℓ , multiply according to the product of characters in $\widehat{G}$, i.e.:

$$\mathbb{U}_{\chi_1}[\ell] \cdot \mathbb{U}_{\chi_2}[\ell] = \mathbb{U}_{\chi_1 \otimes \chi_2}[\ell]. \quad (1.22)$$

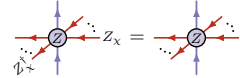
Also note that the operators $\mathbb{U}_\chi[\ell]$ are invariant under continuous deformations of ℓ , rendering them *topological*. These observations demonstrate that after gauging a global 0-form G symmetry, a *global 1-form* $\widehat{G}$ symmetry emerges.⁴ Alternatively, one could think of (1.20) as a *kinematical* constraint satisfied by all allowed states in the gauge invariant Hilbert space.

In order to verify eq. (1.20) we can again resort to the PEPO representation eq. (1.13). The $+$ -type tensor eq. (1.11) satisfies following invariance property



$$Z_\chi \leftarrow \bigoplus \leftarrow Z_\chi^\dagger = Z_\chi^\dagger \leftarrow \bigoplus \leftarrow Z_\chi, \quad (1.23)$$

whereas the Z -type tensor eq. (1.9) obeys



$$Z_\chi \leftarrow \bigoplus \leftarrow Z_\chi^\dagger = Z_\chi^\dagger \leftarrow \bigoplus \leftarrow Z_\chi. \quad (1.24)$$

Both these invariance properties once more follow directly from the definitions of these respective tensors. From them, the claim eq. (1.20) follows readily as again all occurrences of $Z_\chi^{(\dagger)}$ that appear on the virtual level after application of $\mathbb{U}_\chi[\ell]$ cancel.

For the sake of completeness, let us also verify this more explicitly algebraically. To this end, one checks that for every such loop ℓ this line operator commutes with the local gauge transformations:

$$\mathbb{U}_\chi[\ell] \cdot \mathcal{G}_g^\vee = \mathcal{G}_g^\vee \cdot \mathbb{U}_\chi[\ell], \quad \forall g \in G, \mathbf{v} \in \mathbf{V}(\Lambda). \quad (1.25)$$

To prove this, first notice that if $\mathbb{U}_\chi[\ell]$ and $\mathcal{G}_g^\vee$ overlap, they overlap in an even number of edges. Let us

⁴In fact, one should think of the emergent 1-form $\widehat{G}$ symmetry, or more precisely $\text{Rep}(G)$ symmetry, as being the invertible component of the full symmetry structure of the gauge theory. The complete symmetry structure was shown to constitute the 1-form symmetry together with all its 0-form *condensation descendants* and is organized in the *fusion 2-category* $2\text{Rep}(G)$ [24–27, 23, 28].

consider just the case of two overlapping edges that we call e_1, e_2 . Then for the given vertex v , if both e_1, e_2 have the same orientation as ℓ or both have the opposite orientation as ℓ , commuting $\mathbb{U}_\chi[\ell]$ through $\mathcal{G}_g^v$ yields two cancelling phases $\chi(g)$ and $\bar{\chi}(g)$ by virtue of the definition of $\mathcal{G}_g^v$ and eq. (1.18). If on the other hand ℓ has the same orientation as e_1 but has the opposite orientation as e_2 , then we also obtain, due to the relative difference in orientation, phases $\chi(g)$ and $\bar{\chi}(g)$ that cancel. From eq. (1.25) together with the fact that

$$\mathbb{U}_\chi[\ell] \left(\bigotimes_{e \in E(\Lambda)} |1\rangle^e \right) = \bigotimes_{e \in E(\Lambda)} |1\rangle^e \quad (1.26)$$

we can conclude eq. (1.20).

1.3 Gauging the dual 1-form symmetry

Since the emergent 1-form symmetry is realized on-site, it is anomaly-free in the sense of [29] and can hence be gauged. To this end we use a similar procedure as in [13, 19, 14]. After gauging this 1-form symmetry one expects the original 0-form G symmetry to re-emerge [20, 19].

Following [19, 14, 30], gauging of the 1-form symmetry happens via introduction of gauge degrees of freedom on the *vertices* $V(\Lambda)$ of the lattice valued in the group algebra $\mathbb{C}[\hat{G}]$.⁵ A local 1-form gauge transformation for every edge $e \in E(\Lambda)$ is then constructed as follows:

$$\mathcal{G}_\chi^e := X_\chi^{v(\leftarrow e)} \otimes Z_\chi \otimes X_{\bar{\chi}}^{v(\rightarrow e)}, \quad (1.27)$$

where we used an analogue notation to the one in eq. (1.4) denoting by $v(\leftarrow e)$ the vertex which is the target of e and by $v(\rightarrow e)$ the source vertex of the edge e . Naturally, X_χ is defined as

$$X_\chi := \sum_{\eta \in \hat{G}} |\chi\eta\rangle\langle\eta|. \quad (1.28)$$

Akin to the case of the gauging of a global 0-form symmetry, these operators form a *local* representation of $\hat{G}$ supported near e , $\mathcal{G}_{\chi_1}^e \cdot \mathcal{G}_{\chi_2}^e = \mathcal{G}_{\chi_1\chi_2}^e$, $\forall \chi_1, \chi_2 \in \hat{G}$. Notice that the global 1-form symmetries can be recovered by taking the (oriented) product of the 1-form gauge transformations along the loop ℓ , i.e.:

$$\mathbb{U}_\chi[\ell] = \prod_{e \in \ell} \mathcal{G}_{\chi|_\ell}^e, \quad (1.29)$$

$\forall \chi \in \hat{G}$, $\ell \in Z_1(\Lambda, \mathbb{Z})$, analogous to eq. (1.5).

Group-averaging over $\hat{G}$ allows us to construct the local projectors

$$\mathcal{P}^e := \frac{1}{|\hat{G}|} \sum_{\chi \in \hat{G}} \mathcal{G}_\chi^e, \quad (1.30)$$

⁵By invoking a form of Poincaré duality, one can identify the vertices of the primal lattice Λ with the plaquettes of the dual lattice $\Lambda^\vee$, i.e. $V(\Lambda) \equiv P(\Lambda^\vee)$. In that sense the gauge field can indeed be thought of as a 2-form gauge field for the global 1-form symmetry [30].

where we made use of $|G| = |\hat{G}|$. From these we can again unambiguously construct the global projector onto the gauge invariant subspace as:

$$\mathcal{P} := \prod_{e \in E(\Lambda)} \mathcal{P}^e. \quad (1.31)$$

Gauging the 1-form symmetry then ultimately happens in a similar fashion as in the 0-form case. We construct a gauging map by initializing all newly introduced gauge degrees of freedom in the trivial representation $1 \in \hat{G}$ followed by acting with the projector onto the gauge invariant subspace:

$$\mathcal{G}^1 := \mathcal{P} \left(\bigotimes_{v \in V(\Lambda)} |1\rangle^e \right) : \quad (1.32)$$

$$\left(\bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \rightarrow \left(\bigotimes_{e \in E(\Lambda)} \mathbb{C}[G] \right) \otimes \left(\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}] \right).$$

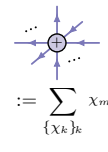
Let us draw attention to the fact that without loss of generality the gauging map $\mathcal{G}^1$ has as its domain only the edge degrees of freedom. This can be traced back to the fact that the emergent 1-form symmetry generators $\mathbb{U}_\chi[\ell]$ only act (non-trivially) on these degrees of freedom.

Let us now proceed by showcasing the tensor network representation of the gauging map $\mathcal{G}^1$, and its symmetry properties. In a similar vein as for the gauging map of the global 0-form G symmetry, we introduce tensors that will be placed on the edges of the lattice that read:



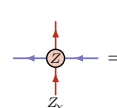
$$\begin{aligned} &:= \sum_{\substack{g_1, g_2 \\ \chi_1, \chi_2}} \frac{1}{|\hat{G}|} \chi_2 \otimes \chi_1 |g_2\rangle\langle g_1| \otimes |\chi_2\rangle\langle\chi_1| \\ &:= \sum_{\chi \in \hat{G}} \frac{1}{|\hat{G}|} Z_\chi \otimes |\chi\rangle\langle\chi|. \end{aligned} \quad (1.33)$$

Tensors incorporating the flatness of the (2-form) gauge field placed on the vertices read then



$$\begin{aligned} &:= \sum_{\{\chi_k\}_k} \chi_{m+1} \chi_{m+2} \dots \chi_{m+n} |\chi_{m+1}, \dots, \chi_{m+n}\rangle \otimes \langle\chi_1, \dots, \chi_m| \\ &:= \sum_{\{\chi_k\}_k} \delta_{\prod_{k=1}^m \chi_k, \prod_{k=m+1}^{m+n} \chi_k} |\chi_{m+1}, \dots, \chi_{m+n}\rangle \otimes \langle\chi_1, \dots, \chi_m|. \end{aligned} \quad (1.34)$$

Their relevant symmetry properties are:



$$= X_\chi^\dagger \otimes X_\chi = \otimes X_\chi, \quad (1.35)$$

and

$$(1.36)$$

Making use of these tensors then results in following PEPO representation of the Gauss law $\mathcal{P}^e$:

$$(1.37)$$

From this, the PEPO representation of the full gauging map $\mathcal{G}^1$ then readily follows. One unit cell is given by

$$(1.38)$$

where, as alluded to above, the blue $+$ -type tensors are placed in alignment with the vertices of the primal lattice.

Gauge invariance which in this case reads $\mathcal{G}_\chi^e \cdot \mathcal{G}^1 = \mathcal{G}^1$ for all $\chi \in \hat{G}$ and $e \in E(\Lambda)$ is a direct consequence of the second equality in eq. (1.35) combined with the identities of the vertex tensor in eq. (1.36). Also the property $\mathcal{G}^1 \cdot \mathcal{U}_\chi[\ell] = \mathcal{G}^1$ holds for all closed paths ℓ and $\chi \in \hat{G}$. In other words, the gauging map $\mathcal{G}^1$ annihilates states which are not in the even sector of the global 1-form $\hat{G}$ symmetry. This follows from the first equation in eq. (1.35) and the invariance properties of the vertex tensor, one of which reads:

$$(1.39)$$

This is not an independent property as it can be derived immediately from eq. (1.36).

Finally and crucially, after gauging the 1-form symmetry the initial global 0-form G symmetry reappears as anticipated. Indeed, it is found that

$$\mathcal{U}_g \cdot \mathcal{G}^1 = \mathcal{G}^1, \quad (1.40)$$

for all $g \in G$.

For future reference we demonstrate this here algebraically. To this end, first note that for two vertices v, v' which share an edge e , $Z_g^v \otimes Z_g^{v'}$ commutes with the gauge transformation $\mathcal{G}_\chi^e$, $\forall g \in G$ by virtue of the identity

$$Z_g X_\chi = \chi(g) X_\chi Z_g. \quad (1.41)$$

Combined with the observation that

$$\mathcal{U}_g \left(\bigotimes_{v \in V(\Lambda)} |1\rangle^v \right) = \bigotimes_{v \in V(\Lambda)} |1\rangle^v, \quad (1.42)$$

we recover indeed eq. (1.40).

In order to verify this using the PEPO representation eq. (1.38) on the other hand, the relevant invariance properties this time read:

$$(1.43)$$

and

$$(1.44)$$

From these properties, eq. (1.40) follows readily since after application of $\mathcal{U}_g$ on the above discussions Λ all occurrences of $Z_g^{(f)}$ again cancel on the virtual bonds.

1.4 3D local symmetries from iterative gauging

Constructing a three-dimensional state from the above gauging maps now happens as follows. We start by choosing a two-dimensional state $|\psi\rangle$ that will serve as the boundary condition of the three-dimensional state and fixing a choice of abelian group G . We assume, as anticipated above, that this state lives in $\bigotimes_{v \in V(\Lambda)} \mathbb{C}[\hat{G}]$ and transforms trivially under the abelian 0-form G symmetry $\mathcal{U}_g$, i.e. $\mathcal{U}_g |\psi\rangle = |\psi\rangle$, $\forall g \in G$. Throughout the above discussions Λ was kept completely general, and as such the construction that follows also can be applied to this most general case.

On $|\psi\rangle$ we can thus first act with the gauging operator $\mathcal{G}^0$ defined explicitly in eq. (1.8), thereby introducing degrees of freedom valued in $\mathbb{C}[G]$ and supported on the edges of Λ . Indeed, recall that $\mathcal{G}^0$ acts according to $\mathcal{G}^0 : \mathbb{C}[\hat{G}]^{\otimes V} \rightarrow \mathbb{C}[\hat{G}]^{\otimes V} \otimes \mathbb{C}[G]^{\otimes E}$, in which $\otimes E$ and $\otimes V$ stand for tensor products over the vertices and edges of the lattice. The gauged state $\mathcal{G}^0 |\psi\rangle$ has then as argued in section 1.2 an emergent dual 1-form $\hat{G}$ symmetry which is completely supported on the edges of Λ and that in turn can be gauged using the procedure outlined in section 1.3. Notice that the gauging map for this 1-form symmetry, $\mathcal{G}^1$ defined in (1.32), acts only on the $\mathbb{C}[G]$ -valued degrees of freedom supported on the edges, leaving the original ‘matter’ degrees of freedom on the vertices $v \in V(\Lambda)$ valued in $\mathbb{C}[\hat{G}]$ untouched. With slight abuse of notation: $\mathcal{G}^1 : \mathbb{C}[G]^{\otimes E} \rightarrow \mathbb{C}[G]^{\otimes E} \otimes \mathbb{C}[\hat{G}]^{\otimes V}$. In turn we can then again gauge the emergent 0-form G symmetry that arises after the application of $\mathcal{G}^1$ using the gauging map $\mathcal{G}^0$. In this step $\mathcal{G}^0$ does not act on the degrees of freedom in $\mathbb{C}[\hat{G}]$ living on $E(\Lambda)$. By repeatedly concatenating the gauging maps $\mathcal{G}^0$ and $\mathcal{G}^1$ we thus in every step are left with degrees of freedom in $\mathbb{C}[\hat{G}]$ and $\mathbb{C}[G]$ which are not acted upon by the next gauging map. These are the degrees of freedom which, following the paradigm of Ref. [12], are

interpreted as *physical* degrees of freedom of a *foliated* three-dimensional state $|\Psi[\psi]\rangle$ whose 2D layers are made up of the ‘matter’ degrees of freedom in every application of the gauging maps. Formally $|\Psi[\psi]\rangle$ reads:

$$|\Psi[\psi]\rangle := \cdots \circ \mathcal{G}_2^1 \circ \mathcal{G}_2^0 \circ \mathcal{G}_1^1 \circ \mathcal{G}_1^0 |\psi\rangle. \quad (1.45)$$

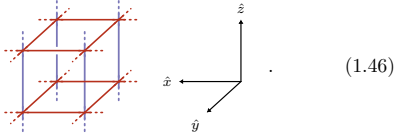
The exact meaning of the subscripts will be explained momentarily.

For concreteness we specialize to the square lattice $\square = \Lambda$ henceforth. It can then be argued that the physical degrees of freedom of $|\Psi[\psi]\rangle$ naturally organize themselves on the edges of a three-dimensional cubic lattice that we will denote by $\mathcal{C}$, and where the initial two-dimensional state serves as a physical boundary of the lattice.

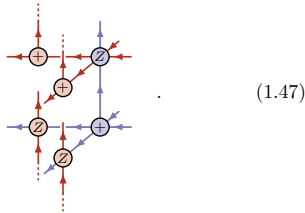
We therefore consider the cubic lattice as being made up of alternating layers of vertical edges followed by a layer of horizontal edges, that we label by $\frac{1}{2}, \frac{3}{2}, \dots$ and $1, 2, \dots$ respectively. The cubic lattice hence has a physical boundary made up of vertical edges.

For every layer of vertical edges we associate these edges with the vertices of a copy of $\square$ and the corresponding $\mathbb{C}[\hat{G}]$ -valued ‘matter’ degrees of freedom that are not acted upon by the gauging maps $\mathcal{G}^1$. Analogously, we associate with every horizontal edge the $\mathbb{C}[G]$ -valued degrees of freedom that are untouched by the $\mathcal{G}^0$ ’s.

Using a blue color for edges that carry a copy of the local Hilbert space $\mathbb{C}[\hat{G}]$ and red for those with a copy of $\mathbb{C}[G]$, one can depict the cubic lattice away from the boundary as follows:



One unit cell of the state $|\Psi[\psi]\rangle$ away from the boundary then can be depicted by combining eq. (1.13) and eq. (1.38) as follows:



In this figure it is understood that all horizontal legs are contracted with neighboring unit cells and the dots represent a contraction with unit cells in the vertical direction. The physical degrees of freedom

of the state then correspond to the uncontracted vertical legs.

By virtue of the local symmetries of all the gauging maps appearing in eq. (1.45), the obtained state $|\Psi[\psi]\rangle$ also inherits those local symmetries in a slightly modified way. More precisely, the emergent state $|\Psi[\psi]\rangle$ is stabilized by plaquette operators for every plaquette of the cubic lattice, coming from the localization of the 1-form symmetry, and vertex operators for every vertex, from gauging the global 0-form symmetry. We denote the plaquette stabilizers which are labeled by $\chi \in \hat{G}$ by $\mathbb{b}^\chi$:

$$\mathbb{b}_\chi^{p_{xz}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.48)$$

$$\mathbb{b}_\chi^{p_{yz}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.49)$$

$$\mathbb{b}_\chi^{p_{xy}} |\Psi[\psi]\rangle = |\Psi[\psi]\rangle, \quad (1.50)$$

and the vertex stabilizers $\mathbb{o}^g$ carry a group label $g \in G$:

$$\mathbb{o}_g^v |\Psi[\psi]\rangle = |\Psi[\psi]\rangle. \quad (1.51)$$

Explicitly, the plaquette and vertex terms are given by

$$\mathbb{b}_\chi^{p_{xz}} := \begin{array}{c} \text{---} Z_x^\dagger \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \\ | \\ \text{---} Z_x \text{---} \end{array}, \quad (1.52)$$

$$\mathbb{b}_\chi^{p_{yz}} := \begin{array}{c} \text{---} Z_x^\dagger \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \\ | \\ \text{---} Z_x \text{---} \end{array}, \quad (1.53)$$

$$\mathbb{b}_\chi^{p_{xy}} := \begin{array}{c} \text{---} Z_x \text{---} \\ | \\ X_x \text{---} \text{---} X_x^\dagger \\ | \\ \text{---} Z_x^\dagger \text{---} \end{array}, \quad (1.54)$$

$$\mathbb{o}_g^v := \begin{array}{c} \text{---} Z_g^\dagger \text{---} \\ | \\ X_g \text{---} \text{---} X_g^\dagger \\ | \\ \text{---} Z_g \text{---} \end{array}. \quad (1.55)$$

Let us first show algebraically that $\mathbb{b}_\chi^{p_{xz}}$ and $\mathbb{b}_\chi^{p_{yz}}$ stabilize $|\Psi[\psi]\rangle$ before residing to our graphical PEPO representation of the state. To this end we single out one of those plaquette term. From the commutation relation eq. (1.41) together with the definition of $\mathcal{G}^0$ in terms of the local gauge transformations (1.4), it follows that

$$\begin{aligned} & (X_\chi^{v(\leftarrow e)} \otimes X_\chi^e \otimes X_\chi^{v(\rightarrow e)}) \cdot \mathcal{G}^0 \\ &= \mathcal{G}^0 \cdot (X_\chi^{v(\leftarrow e)} \otimes X_\chi^{v(\rightarrow e)}). \end{aligned} \quad (1.56)$$

Recognizing then $X_\chi^{v(\leftarrow e)} \otimes Z_\chi \otimes X_\chi^{v(\rightarrow e)}$ as the local 1-form gauge transformation (1.27), which leaves $\mathcal{G}^1$ invariant, together with the fact that $Z_\chi \cdot \mathcal{G}^1 = \mathcal{G}^1 \cdot Z_\chi$ proves that $\mathbb{b}_\chi^{p_{xz}}$ and $\mathbb{b}_\chi^{p_{yz}}$ are stabilizers of the state. Let us now give an interpretation of these stabilizers in terms of our PEPO representation of the state

eq. (1.47). Note first that from the commutation relation eq. (1.41) it follows that

$$\text{Diagram 1} = \text{Diagram 2} = \text{Diagram 3}, \quad (1.57)$$

from which, together with eq. (1.35), one finds that in the xz -plane eq. (1.56) can be depicted as:

$$\text{Diagram 1} = \text{Diagram 2}, \quad (1.58)$$

and similar in the yz -plane. Considering now a slice of the tensor network representation of our state eq. (1.47) along the xz -plane, we compute that:

$$\mathbb{b}_\chi^{\text{P}^{xz}} |\Psi[\psi]\rangle = \text{Diagram 1} = \text{Diagram 2} = |\Psi[\psi]\rangle, \quad (1.59)$$

where in the second equality we made use of the fact that the operator $Z_\chi^\dagger$ on top can be freely *pulled through* the Z -type tensor on which it acts, together with eq. (1.58). In the final equality we then recognized the gauge invariance of $\mathcal{G}^1$. The yz -plaquette term can be derived in similar fashion.

The final xy -plaquette terms can be recognized as the (minimal) global 1-form symmetries of the $\mathcal{G}^0$ gauging map. In a similar vein to eq. (1.56), one can check that

$\mathfrak{o}_g^\vee$ applied to $\mathcal{G}^1$ maps to $\mathcal{G}_g^\vee$ which is exactly the local symmetry of $\mathcal{G}^0$, from which it follows that $\mathfrak{o}_g^\vee$ is also a stabilizer.

These observations allow us to consider the state $|\Psi[\psi]\rangle$ a ground state of the commuting projector bulk Hamiltonian (a stabilizer code):

$$\mathbb{H}^{\text{blk}} := - \sum_{v \in V(\mathcal{C})} \mathbb{P}^v - \sum_{p_{xz} \in P(\mathcal{C})} \mathbb{P}^{p_{xz}} - \sum_{p_{yz} \in P(\mathcal{C})} \mathbb{P}^{p_{yz}} - \sum_{p_{xy} \in P(\mathcal{C})} \mathbb{P}^{p_{xy}}. \quad (1.60)$$

In this expression the sum over $V(\mathcal{C})$ is over all vertices of the cubic lattice, and the sums over $P(\mathcal{C})$ are over plaquettes in the xz , yz and xy planes respectively. The projectors are explicitly given by

$$\mathbb{P}^v := \frac{1}{|G|} \sum_{g \in G} \mathfrak{o}_g^\vee, \quad (1.61)$$

and

$$\mathbb{P}^{p_{xz}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{b}_\chi^{\text{P}^{xz}}, \quad (1.62)$$

$$\mathbb{P}^{p_{yz}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{b}_\chi^{\text{P}^{yz}}, \quad (1.63)$$

$$\mathbb{P}^{p_{xy}} := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathbb{b}_\chi^{\text{P}^{xy}}. \quad (1.64)$$

It turns out that the Hamiltonian in eq. (1.60) is unitarily equivalent to the G abelian quantum double [16]. In its displayed form (1.60) it appeared previously in [15] under the name of *Clifford-deformed surface code*. It can be brought in its more familiar form [16] defined on $\bigotimes_{e \in E(\mathcal{C})} \mathbb{C}[G]$ by applying the generalized Hadamard transformation

$$\mathcal{U} := \sum_{g \in G, \chi \in \widehat{G}} \frac{\chi(g)}{\sqrt{|G|}} |g\rangle\langle\chi| \quad (1.65)$$

to every vertical edge carrying a copy of $\mathbb{C}[\widehat{G}]$, as it can be checked that

$$\mathcal{U} Z_g \mathcal{U}^\dagger = X_g^\dagger, \quad (1.66)$$

$$\mathcal{U} X_\chi \mathcal{U}^\dagger = Z_\chi. \quad (1.67)$$

1.5 Gapped boundary conditions

The local stabilizers analyzed so far act on the bulk of the three-dimensional state $|\Psi[\psi]\rangle$. In this section we study the boundary stabilizers. These naturally are given by a subset of truncated bulk stabilizers commuting with the bulk stabilizers $\mathbb{b}$ and $\mathfrak{o}$ defined above, where the subset depends on the choice of the boundary state $|\psi\rangle$, in particular the way in which $|\psi\rangle$ is invariant under the G 0-form symmetry. Indeed, for

every choice of subgroup $H \leq G$ one can construct explicitly a G -symmetric boundary state $|\psi\rangle_H$ for which $|\Psi[\psi_H]\rangle$ is a ground state of the bulk Hamiltonian $\mathbb{H}^{\text{blk}}$ (1.60) plus additional boundary terms which are projectors and which all commute with the terms in $\mathbb{H}^{\text{blk}}$ (1.60) and among themselves.

Let us proceed by constructing these boundary states and the corresponding boundary Hamiltonian terms explicitly. To this end we first define the following operators near the boundary $\partial\mathfrak{B}$ (effectively truncating $\mathbb{B}$) for any $\chi \in \widehat{G}$:

$$\mathfrak{d}_\chi^{\text{p}_{zz}} := \begin{array}{c} \text{---} Z_\chi^\dagger \text{---} \\ | \\ X_\chi^\dagger \quad X_\chi \\ | \\ \text{---} \end{array}, \quad (1.68)$$

$$\mathfrak{d}_\chi^{\text{p}_{yz}} := \begin{array}{c} \text{---} Z_\chi^\dagger \text{---} \\ | \\ X_\chi \quad X_\chi^\dagger \\ | \\ \text{---} \end{array}. \quad (1.69)$$

These terms commute both with the bulk vertex operators $\mathfrak{e}_g^\vee$ and plaquette operators $\mathbb{L}_\chi^\text{p}$. Naturally, they also commute all among each other. From the explicit form of the gauging maps $\mathcal{G}_\frac{2}{3}^0$ and $\mathcal{G}_1^1$ it follows that given an irrep $\chi \in \widehat{G}$ the operators (1.68-1.69) are stabilizers of $|\Psi[\psi]\rangle$ if and only if $(X_\chi^\vee \otimes X_\chi^\vee)|\psi\rangle = |\psi\rangle$, equivalently

$$\begin{aligned} \mathfrak{d}_\chi |\Psi[\psi]\rangle &= |\Psi[\psi]\rangle \iff \\ \langle \psi | (X_\chi^\vee \otimes X_\chi^\vee) | \psi \rangle &= 1 \end{aligned} \quad (1.70)$$

for neighboring vertices $\mathbf{v}, \mathbf{v}' \in \mathbf{V}(\square)$. Notice that $X_\chi^\vee \otimes X_\chi$ is an order parameter for the (partial) breaking of the global 0-form G symmetry. Its expectation value is a two-point symmetric correlation function. In what follows we show that given a choice of subgroup $H \leq G$, $\mathfrak{d}_\chi$ is a stabilizer of $|\Psi[\psi_H]\rangle$ for an appropriate boundary state $|\psi\rangle_H$ depending on that subgroup, provided that χ is trivial when restricted to H . It is a fact that characters in $\widehat{G}$ that are trivial when restricted to H form a subgroup of $\widehat{G}$ isomorphic to $\widehat{G/H}$. From now on we will abuse notation and denote this subgroup simply by $\widehat{G/H} \leq \widehat{G}$. For the chosen subgroup $H \leq G$ we then propose following boundary state $|\psi\rangle_H$. First, we define the normalized states

$$|+\rangle_H := \frac{1}{\sqrt{|\widehat{G/H}|}} \sum_{[\chi] \in \widehat{G/H}} |\chi\rangle. \quad (1.71)$$

Notice that $|+\rangle_H$ is stabilized by H in the sense that $Z_h |+\rangle_H = |+\rangle_H$ for all $h \in H$. Given $|+\rangle_H$ we define

$$|\psi\rangle_H := \frac{1}{\sqrt{|\widehat{G/H}|}} \sum_{[g] \in \widehat{G/H}} \left(\bigotimes_{\mathbf{v} \in \mathbf{V}(\square)} Z_g^\vee |+\rangle_H^\vee \right). \quad (1.72)$$

By construction $|\psi\rangle_H$ is symmetric under the 0-form G symmetry and, as anticipated, is symmetric under

$\{X_\chi^\vee \otimes X_\chi^\vee, \chi \in \widehat{G/H}\}$ since $X_\chi^\vee$ leaves invariant $|+\rangle_H$ and $X_\chi^\vee \otimes X_\chi^\vee$ commutes with $Z_g^\vee \otimes Z_g^\vee$. Notice one can think of $|\psi\rangle_H$ as an equal weight superposition of the $|\widehat{G/H}|$ ground states of a G symmetric Hamiltonian whose symmetry is spontaneously broken down to H . It is therefore meaningful to refer to these boundaries as *symmetry breaking* boundaries. We also define the following operators acting on the boundary:

$$\mathfrak{e}_g^{e_z} := \begin{array}{c} | \\ Z_g \\ | \end{array}, \quad (1.73)$$

which are symmetries of $|\Psi[\psi_H]\rangle$, and which commute with $\mathfrak{d}_\chi$ only if $g \in H$. Defining then

$$\mathbb{Q}_H^{\text{p}_{zz}} := \frac{1}{|\widehat{G/H}|} \sum_{\chi \in \widehat{G/H}} \mathfrak{d}_\chi^{\text{p}_{zz}}, \quad (1.74)$$

$$\mathbb{Q}_H^{\text{p}_{yz}} := \frac{1}{|\widehat{G/H}|} \sum_{\chi \in \widehat{G/H}} \mathfrak{d}_\chi^{\text{p}_{yz}}, \quad (1.75)$$

$$\mathbb{Q}_H^{e_z} := \frac{1}{|H|} \sum_{h \in H} \mathfrak{e}_h^{e_z}, \quad (1.76)$$

the Hamiltonian in the presence of such boundary terms reads

$$\begin{aligned} \mathbb{H}_H &= \mathbb{H}^{\text{blk}} + \sum_{\mathbf{p}_{zz} \in \mathbf{P}(\partial\mathfrak{B})} \mathbb{Q}_H^{\text{p}_{zz}} \\ &+ \sum_{\mathbf{p}_{yz} \in \mathbf{P}(\partial\mathfrak{B})} \mathbb{Q}_H^{\text{p}_{yz}} + \sum_{\mathbf{e}_z \in \mathbf{E}(\partial\mathfrak{B})} \mathbb{Q}_H^{e_z}, \end{aligned} \quad (1.77)$$

where the dependence on the choice of subgroup $H \leq G$ is explicit. As an example, one can consider the extreme cases of $H = G$ and $H = \{1\}$ that can be identified with *Dirichlet* and *Neumann* boundary conditions respectively. In the context of lattice gauge theory as of here it is also customary to refer to these cases as realizing the *smooth* and *rough* boundary conditions.⁶

As a final comment we consider an analogous construction of a three-dimensional state with a corresponding stabilizer Hamiltonian, which is constructed by starting from a state $|\hat{\psi}\rangle$ invariant under a 1-form $\widehat{G}$ symmetry generated by the Wilson loop operators (1.19):

$$|\hat{\Psi}[\hat{\psi}]\rangle := \cdots \circ \mathcal{G}_2^1 \circ \mathcal{G}_\frac{2}{3}^0 \circ \mathcal{G}_1^1 |\hat{\psi}\rangle. \quad (1.78)$$

⁶Notice that we don't recover *twisted* boundary conditions that amount to pasting an SPT phase on top of the symmetry broken boundary phase [31–35]. It is expected that all such boundary conditions, which are in one-to-one correspondence with pairs $(H \leq G, [\psi] \in H^3(H, \text{U}(1)))$, are recovered by appropriately twisting the boundary terms akin as in the lower-dimensional setting [12, 32]. This is most easily facilitated for by the triangular lattice.

The boundary terms in that case are given by truncating $\mathfrak{e}_g^\vee$ and $\mathfrak{b}_\chi^\vee$ as follows:

$$\mathfrak{e}_g^\vee := \begin{array}{c} \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \end{array} \begin{array}{c} Z_g^\dagger \\ X_g^\dagger \\ X_g^\dagger \end{array} \text{---} , \quad (1.79)$$

$$\mathfrak{f}_\chi^\vee := \text{---} Z_\chi \text{---} , \quad (1.80)$$

$$\mathfrak{f}_\chi^\vee := \text{---} Z_\chi \text{---} . \quad (1.81)$$

Given a $g \in G$, $\mathfrak{e}_g^\vee$ is a symmetry of $|\hat{\Psi}[\hat{\psi}]\rangle$ if and only if

$$\begin{array}{c} \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \\ \text{---} X_g \text{---} \end{array} \begin{array}{c} X_g^\dagger \\ X_g^\dagger \\ X_g^\dagger \end{array} |\hat{\psi}\rangle = |\hat{\psi}\rangle . \quad (1.82)$$

Similarly as in the previous case one could define $|\hat{\psi}\rangle_{\hat{H}}$ hosting a (partial) symmetry breaking phase of the 1-form symmetry with unbroken symmetry group $\hat{H} \leq \hat{G}$. Commuting boundary stabilizers for such a state are given by

$$\mathbb{R}_H^\vee := \frac{1}{|H|} \sum_{h \in H} \mathfrak{e}_h^\vee , \quad (1.83)$$

$$\mathbb{R}_H^\vee := \frac{1}{|G/H|} \sum_{\chi \in G/H} \mathfrak{f}_\chi^\vee . \quad (1.84)$$

Notice thus in particular that $|\hat{\psi}\rangle$ can host non-trivial 2D topological order. It is understood that besides these boundaries and the G symmetry breaking ones considered above, it is also possible to define additional gapped boundaries which amount to (trivially) stacking a 2D topological order or stacking a 2D topological order while condensing composites of boundary anyons of the emergent state and (bulk) anyons from the stacked topological order [34].

2 General framework

What are the main takeaways from this introductory example? From iteratively gauging a global abelian 0-form symmetry and its dual 1-form symmetry we obtain stabilizer codes (local commuting projector Hamiltonians) in one dimension higher. Moreover, the obtained codes can be readily extended to include gapped boundary terms, and we have written down the gauging maps and states explicitly in terms of tensor networks.

The goal of this section is to provide a general framework for iteratively gauging a state $|\psi\rangle$ which is symmetric under a collection of abelian symmetries defined via certain *check operators* which commute with them. Hereby we extend the formalism of [14] to arbitrary abelian groups. We argue that in this framework the concatenated gauging maps generically exhibit *local symmetries* originating from the Gauss constraints

of the symmetries under scrutiny. These local symmetries can in turn be organized in an *emergent* local commuting projector Hamiltonian that naturally lives in one dimension higher than the initial state $|\psi\rangle$ which is being gauged. Towards the end of this section we comment on the inclusion of the boundary terms in the emergent Hamiltonian which can be identified with certain spontaneous symmetry breaking patterns of the boundary symmetry realized by $|\psi\rangle$. Since these gauging maps can be defined in any dimension, we recover the results from [12] as a specific case.

2.1 Gauging abelian symmetries

To set the scene, we assume a lattice Λ in which each vertex $\mathbf{v} \in V(\Lambda)$ supports a copy of $\mathbb{C}[\hat{G}]$. The tensor product Hilbert space $\bigotimes_{\mathbf{v} \in V(\Lambda)} \mathbb{C}[\hat{G}]$ will be written as $\mathcal{H} := \bigotimes_{\mathbf{v} \in V(\Lambda)} \mathbb{C}[\hat{G}]$ henceforth. Notice that we don't make any assumption about the underlying topology or dimension of the spatial manifold on which Λ is defined.

Throughout this section we deal with symmetry operators that are fully specified by a *complete collection of check operators*, labeled $c \in \mathcal{C}$, that commute with all of them. The formal gauging procedure of Williamson [14] then happens via the introduction of a gauge degree of freedom for every such check. Formally, consider for any $g \in G$ the operators

$$\mathbb{U}_g[\Omega] := \bigotimes_{\mathbf{v} \in \Omega} Z_g^\vee , \quad g \in G, \quad (2.1)$$

where Ω is a subset of the vertices, i.e. $\Omega \subseteq V(\Omega)$. Borrowing the notation of [14], we will denote by η a function

$$\begin{aligned} \eta : \mathcal{C} &\rightarrow P(V(\Lambda)) \\ c &\rightarrow \eta(c) \subseteq V(\Lambda) , \end{aligned} \quad (2.2)$$

from $\mathcal{C}$ to the power set $P(V(\Lambda))$. Physically, η maps a given constraint c from the collection $\mathcal{C}$, that will below be identified with the gauge degrees of freedom, to a corresponding set of vertices $\eta(c) \subseteq V(\Lambda)$. Furthermore we introduce the involution $\alpha_c^\vee(\chi)$ which evaluates to χ or $\bar{\chi}$ depending on the *relative orientation* of the constraint c and the vertex $\mathbf{v}$. An operator $\mathbb{U}_g[\Omega]$ is then a *valid symmetry generator* if it commutes with the checks

$$X_\chi := \bigotimes_{\mathbf{v} \in \eta(c)} X_{\alpha_c^\vee(\chi)}^\vee \quad (2.3)$$

for all $c \in \mathcal{C}$ and all $\chi \in \hat{G}$. Below, we will sometimes refer to the symmetry structure generated by these valid symmetry generators as the *initial* symmetry.

To connect with the previous section, let us remark that the global 0-form G symmetry corresponds to the case $\Omega = V(\Lambda)$, and thus $\mathbb{U}_g[V(\Lambda)] \equiv \mathbb{U}_g$. For

these symmetry generators the checks can be chosen so that there is exactly one check per edge of the lattice, i.e. $\mathcal{C} \equiv \mathcal{E}(\Lambda)$. Explicitly, the checks then read $X_\chi^e = X_\chi^{v(\rightarrow e)} \otimes X_\chi^{v(\leftarrow e)}$, where we refer to the previous section for a detailed explanation of the notation. An important observation is that the local constraints can be interpreted as local Hamiltonian terms that commute with the symmetry [14]. As such, they can be used to construct *generalized Ising models* and when perturbed, for example with single-vertex Z_g terms, they host symmetry breaking phases of such symmetries.

We now proceed by gauging the symmetry generated by the operators $\mathbb{U}_g[\Omega]$, essentially mimicking the procedure used in the previous section based on [13, 14] adapted to this more general setting. To this end we first enlarge the ‘matter’ Hilbert space $\mathcal{H}$ with a $\mathbb{C}[G]$ -valued *gauge degree of freedom* for every check in $\mathcal{C}$. We will denote the physical support of the gauge degrees of freedom by $\mathbf{c} \in \mathcal{C}(\Lambda)$ where it has to be understood that there is bijection between $\mathcal{C}(\Lambda)$ and $\mathcal{C}$. Typically, $\mathcal{C}(\Lambda)$ can be identified with a (subset) of k -cells of the lattice Λ rendering the aforementioned bijection canonical.

Having again the example from the previous section in mind, $\mathcal{C}(\Lambda)$ can thus be identified with $\mathcal{E}(\Lambda)$ as anticipated above.

Writing $\tilde{\mathcal{H}} := \bigotimes_{\mathbf{c} \in \mathcal{C}} \mathbb{C}[G]$, the first step of the gauging procedure thus entails enlarging $\mathcal{H}$ according to $\mathcal{H} \rightarrow \mathcal{H} \otimes \tilde{\mathcal{H}}$. Given the map η defined above, we define $\eta^{-1}(\mathbf{v})$ as the preimage of $\mathbf{v}$ under η . In other words, η^{-1} maps the matter degree of freedom localized on $\mathbf{v}$ to its associated gauge degrees of freedom. In light of the above identification of gauge degrees of freedom and constraints, we will also interpret the set of constraints $\eta^{-1}(\mathbf{v})$ as its corresponding subset of $\mathcal{C}(\Lambda)$. For every $g \in G$ we then define the *gauge transformation* $\mathcal{G}_g^\mathbf{v}$ on the enlarged Hilbert space $\mathcal{H} \otimes \tilde{\mathcal{H}}$ as:

$$\mathcal{G}_g^\mathbf{v} := Z_g^\mathbf{v} \bigotimes_{\mathbf{c} \in \eta^{-1}(\mathbf{v})} X_{o_\mathbf{c}^\mathbf{v}(g)}^\mathbf{c}. \quad (2.4)$$

Out of these gauge transformations, a *local projector* is constructed by taking the G -average:

$$\mathcal{P}^\mathbf{v} := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^\mathbf{v}, \quad (2.5)$$

$\forall \mathbf{v} \in \mathcal{V}(\Lambda)$. Indeed remark that from a redefinition of summation variables it follows that $\mathcal{P}^\mathbf{v} \cdot \mathcal{P}^\mathbf{v} = \mathcal{P}^\mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}(\Lambda)$.

The *gauging map* is then finally defined as the operator obtained by acting with the local projectors on the trivial gauge field $\bigotimes_{\mathbf{c} \in \mathcal{C}(\Lambda)} |1\rangle^\mathbf{c}$:

$$\mathcal{G} := \left(\prod_{\mathbf{v} \in \mathcal{V}(\Lambda)} \mathcal{P}^\mathbf{v} \right) \left(\bigotimes_{\mathbf{c} \in \mathcal{C}(\Lambda)} |1\rangle^\mathbf{c} \right) : \quad (2.6)$$

$$\mathcal{H} \rightarrow \mathcal{H} \otimes \tilde{\mathcal{H}}.$$

Let us summarize the most important properties of $\mathcal{G}$ before proceeding. We refer to [14] for a more detailed digression. For a given $\mathbf{v} \in \mathcal{V}(\Lambda)$, the gauge transformations $\mathcal{G}_g^\mathbf{v}$ form by construction a G -representation supported on $\eta^{-1}(\mathbf{v})$. It then follows from the definitions that $\mathcal{G}_g^\mathbf{v} \cdot \mathcal{G} = \mathcal{G}$. By a relabeling of summation variable in the definition of $\mathcal{P}^\mathbf{v}$ it also follows immediately that $\mathcal{G} \cdot \mathbb{U}_g[\Omega] = \mathcal{G}$ for all allowed symmetry operators. As illustrated in section 1 this property implies that only states $|\psi\rangle$ transforming trivially under the symmetry, i.e. $\mathbb{U}_g[\Omega] |\psi\rangle = |\psi\rangle$, $\forall g \in G$ and for all allowed Ω , are *gaugeable*. Given such a state, we coin $\mathcal{G} |\psi\rangle$ the *gauged state*. From $\mathcal{G}_g^\mathbf{v} \cdot \mathcal{G} = \mathcal{G}$ it readily follows that the gauged state is gauge invariant as expected. Even though we only deal with gauging *states* throughout this manuscript, it should be noted that also operators can be gauged using this prescription. In particular, *local* and *symmetric* operators in the initial matter theory are mapped to *dual* local gauge invariant operators under the gauging procedure in such a way that the matrix elements of these symmetric operators w.r.t. a symmetric state are preserved.

As an example, consider once more the case of a 0-form symmetry. Notice that with the choices of relative orientations $o_\mathbf{c}^\mathbf{v}$ implicitly defined above, we recover the Gauss constraints defined above in eq. (1.4), from which eventually the gauging map given in eq. (1.8) of [13] is recovered.

Crucially, the gauge theory obtained via the aforementioned proposal exhibits global *emergent* symmetries that, by construction, are anomaly-free and hence gaugeable [29]. Let us continue by identifying these emergent symmetries. First notice that even though the constraints $X_\chi^\mathbf{c}$ do not commute with the gauging map, the *dressed constraints* $Z_\chi^\mathbf{c} \otimes X_\chi^\mathbf{c}$ do commute with all Gauss operators $\mathcal{G}_g^\mathbf{v}$ and, a fortiori, with the gauging map $\mathcal{G}$. Indeed, this follows from a direct computation. For fixed $\mathbf{v} \in \mathcal{V}(\Lambda)$ and $\mathbf{c} \in \mathcal{C}$ let us assume without loss of generality that $\mathbf{v} \in \eta(\mathbf{c})$, which necessarily implies that $\mathbf{c} \in \eta^{-1}(\mathbf{v})$:

$$\begin{aligned} & \left(Z_\chi^\mathbf{c} \otimes X_\chi^\mathbf{c} \right) \cdot \mathcal{G}_g^\mathbf{v} \\ &= \left(Z_\chi^\mathbf{c} \bigotimes_{\mathbf{w} \in \eta(\mathbf{c})} X_{o_\mathbf{w}^\mathbf{v}(\chi)}^\mathbf{w} \right) \cdot \left(Z_g^\mathbf{v} \bigotimes_{\mathbf{d} \in \eta^{-1}(\mathbf{v})} X_{o_\mathbf{d}^\mathbf{v}(g)}^\mathbf{d} \right) \\ &= o_\mathbf{c}^\mathbf{v}(\bar{\chi})(g) \cdot \chi(o_\mathbf{c}^\mathbf{v}(g)) \cdot \mathcal{G}_g^\mathbf{v} \cdot \left(Z_\chi^\mathbf{c} \otimes X_\chi^\mathbf{c} \right) \\ &= \mathcal{G}_g^\mathbf{v} \cdot \left(Z_\chi^\mathbf{c} \otimes X_\chi^\mathbf{c} \right). \end{aligned} \quad (2.7)$$

In the second line the definition of $X_\chi^\mathbf{c}$ and $\mathcal{G}_g^\mathbf{v}$ was employed. In the third line we used the commutation relations eq. (1.18) and eq. (1.41). Combined with the fact that $Z_\chi^\mathbf{c}$ evaluates to 1 on the trivial gauge field $\bigotimes_{\mathbf{c} \in \mathcal{C}(\Lambda)} |1\rangle^\mathbf{c}$, this implies that the constraint $X_\chi^\mathbf{c}$ for every $\mathbf{c}$ and $\chi \in \hat{G}$ is the preimage of $Z_\chi^\mathbf{c} \otimes X_\chi^\mathbf{c}$ under

the gauging map $\mathcal{G}$. In particular, it follows that for a choice of constraints $\Gamma \subseteq \mathcal{C}$ and choice of involution $\delta^c(\chi)$, $c \in \Gamma$ such that

$$\prod_{c \in \Gamma} X_{\delta^c(\chi)}^c = \text{id}, \quad (2.8)$$

there exists a corresponding operator in the gauge theory

$$\mathbb{U}[\Gamma, \delta^c] = \prod_{c \in \Gamma} Z_{\delta^c(\chi)}^c \quad (2.9)$$

which is an emergent symmetry of $\mathcal{G}$ in the sense that

$$\mathbb{U}[\Gamma, \delta^c] \cdot \mathcal{G} = \mathcal{G}. \quad (2.10)$$

Applying the above to the example of $\mathcal{G}^0$ (1.8) considered in section 1, we note that eq. (2.8) is exactly satisfied for any choice Γ which coincides with a closed loop ℓ supported on the edges $\mathbf{E}(\Lambda)$ of the lattice. With the above choice of δ^c , δ^c exactly takes the relative orientation of ℓ with respect to the orientation of the edges into account.

In order to gauge the emergent symmetry operators we adapt a similar approach as for the initial symmetry (2.1). Notice that the symmetry generators defined in eq. (2.9) commute with the local checks

$$X_g^v := \bigotimes_{c \in \eta^{-1}(v)} X_{\delta^c(g)}^c \quad (2.11)$$

which are naturally localized around the vertices $\mathbf{V}(\Lambda)$. Therefore we will identify the set of these checks with the set of vertices $\mathbf{V}(\Lambda)$. To prove that these checks indeed commute with the emergent symmetry generators, consider some vertex $v \in \eta(\Gamma)$. The identity (2.8) guarantees the existence of constraints $c, d \in \eta^{-1}(v)$ such that $\delta^c(\alpha_v^c(\chi)) = \delta^d(\alpha_v^d(\chi))$. It follows immediately that $Z_{\delta^c(\chi)}^c \otimes Z_{\delta^d(\chi)}^d$ commutes with $X_{\alpha_v^c(g)}^c \otimes X_{\alpha_v^d(g)}^d$, from which the claim follows. In order to define the corresponding dual gauging map $\widehat{\mathcal{G}}$, we once more enlarge the Hilbert space according to $\widehat{\mathcal{H}} \rightarrow \widehat{\mathcal{H}} \otimes \mathcal{H}$, on which we define the Gauss operators

$$\mathcal{G}_\chi^c := Z_\chi^c \bigotimes_{v \in \eta(c)} X_{\delta^c(\chi)}^v. \quad (2.12)$$

Notice the complex conjugate appearing in δ^c . Corresponding local projectors localized around c follow from averaging over $\widehat{G}$:

$$\mathcal{P}^c := \frac{1}{|\widehat{G}|} \sum_{\chi \in \widehat{G}} \mathcal{G}_\chi^c \quad (2.13)$$

Ultimately the dual gauging map reads

$$\widehat{\mathcal{G}} := \left(\prod_{c \in \mathcal{C}(\Lambda)} \mathcal{P}^c \right) \left(\bigotimes_{v \in \mathbf{V}(\Lambda)} |\mathbb{1}\rangle^v \right). \quad (2.14)$$

It follows that after gauging these emergent symmetries using the gauging map $\widehat{\mathcal{G}}$ the original G symmetry is recovered, as expected. This is quickly demonstrated by noticing that all symmetry generators $\mathbb{U}_g[\Omega]$ commute by definition with the checks X_χ^c , and by extension with the Gauss constraints $\mathcal{G}_\chi^c$, so that $\mathbb{U}_g[\Omega] \cdot \widehat{\mathcal{G}} = \widehat{\mathcal{G}}$.

Finalizing the example of the previous section, one verifies immediately that the 1-form Gauss constraints eq. (1.27) coincide precisely with the definition provided in eq. (2.12) applied to the case at hand.

2.2 Concatenation of gauging maps

Let us now consider the composition of the gauging operators defined above. With slight abuse of notation we consider in this section thus the composed operators

$$\widehat{\mathcal{G}} \circ \mathcal{G} : \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \rightarrow \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathcal{C}|} \otimes \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|},$$

and

$$\mathcal{G} \circ \widehat{\mathcal{G}} : \mathbb{C}[G]^{\otimes |\mathcal{C}|} \rightarrow \mathbb{C}[G]^{\otimes |\mathcal{C}|} \otimes \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathcal{C}|}.$$

Herein, $|\mathbf{V}|$ stands for the number of vertices of the lattice Λ and by definition $|\mathcal{C}|$ is equal to the number of constraints defining the initial symmetry (2.1), i.e. $|\mathcal{C}| = |\mathcal{C}|$. As anticipated above, these concatenated gauging maps exhibit local symmetries that can be traced back to the Gauss constraints enforced by $\mathcal{G}$ and $\widehat{\mathcal{G}}$. Crucial in the derivation of the local symmetries are following identities which can be derived in a similar fashion as eq. (2.7):

$$\left(Z_g^v \bigotimes_{c \in \eta^{-1}(v)} X_{\delta^c(g)}^c \right) \cdot \widehat{\mathcal{G}} = \widehat{\mathcal{G}} \cdot \left(\bigotimes_{c \in \eta^{-1}(v)} X_{\alpha_v^c(g)}^c \right), \quad (2.15)$$

$$\left(Z_\chi^c \bigotimes_{v \in \eta(c)} X_{\delta^c(\chi)}^v \right) \cdot \mathcal{G} = \mathcal{G} \cdot \left(\bigotimes_{v \in \eta(c)} X_{\alpha_v^c(\chi)}^v \right). \quad (2.16)$$

Given these identities, it is now straightforward to argue that the mutually commuting operators

$$\alpha_g^v := Z_g^v \left(\bigotimes_{c \in \eta^{-1}(v)} X_{\delta^c(g)}^c \right) \otimes Z_g^v, \quad (2.17)$$

$$\in \text{End}(\mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathcal{C}|} \otimes \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|})$$

and

$$\mathbf{b}_\chi^c := Z_\chi^c \left(\bigotimes_{v \in \eta(c)} X_{\delta^c(\chi)}^v \right) \otimes Z_\chi^c \quad (2.18)$$

$$\in \text{End}(\mathbb{C}[G]^{\otimes |\mathcal{C}|} \otimes \mathbb{C}[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes \mathbb{C}[G]^{\otimes |\mathcal{C}|})$$

are symmetries of the concatenated gauging maps $\widehat{\mathcal{G}} \circ \mathcal{G}$ and $\mathcal{G} \circ \widehat{\mathcal{G}}$ respectively. Indeed:

$$\alpha_g^\vee \cdot (\widehat{\mathcal{G}} \circ \mathcal{G}) = \widehat{\mathcal{G}} \circ \mathcal{G}, \quad (2.19)$$

$$\mathbb{b}_\chi^\vee \cdot (\mathcal{G} \circ \widehat{\mathcal{G}}) = \mathcal{G} \circ \widehat{\mathcal{G}}, \quad (2.20)$$

for all $\mathbf{v} \in V(\Lambda)$, $\mathbf{c} \in C(\Lambda)$, $g \in G$, $\chi \in \widehat{G}$. The iterated gauging procedure put forward in [12] now dictates that given any state $|\psi\rangle \in \mathcal{H}$ stabilized by the initial G symmetry, that is

$$\mathbb{U}_g[\Omega] |\psi\rangle = |\psi\rangle \quad (2.21)$$

for every *valid* symmetry operator $\mathbb{U}_g[\Omega]$, one can define the *emergent* state $|\Psi[\psi]\rangle$ as

$$\begin{aligned} |\Psi[\psi]\rangle &:= \cdots \mathcal{G}_{\frac{1}{2}} \circ \widehat{\mathcal{G}}_1 \circ \mathcal{G}_{\frac{1}{2}} |\psi\rangle \\ &\in \bigotimes_{\frac{1}{2}} C[\widehat{G}]^{\otimes |\mathbf{V}|} \bigotimes_i C[G]^{\otimes |\mathbf{C}|}, \end{aligned} \quad (2.22)$$

where the big tensor products run over all appearances of the gauging maps in the definition.

This state is then by construction stabilized by the operators $\alpha_g^\vee$ and $\mathbb{b}_\chi^\vee$ defined above. Given these stabilizers one can subsequently construct a stabilizer Hamiltonian for which $|\Psi[\psi]\rangle$ is a ground state. Remark that additional stabilizers are given by the emergent symmetries of every gauging map. For the examples that follow we will refrain from including these stabilizers as they are in those cases non-local operators. However, note that this is by no means always the case. Recall that in the example of section 1, the emergent 1-form symmetries around individual plaquettes gave rise to the plaquette terms defined in eq. (1.54). In fact, we rather would like to think of these terms as kinematical constraints on the Hilbert space on which the emergent state is defined and as such we are allowed to include them in the Hamiltonian.

Given the foliated way in which the emergent state (2.22) is constructed it is now a sensible thing to consider this foliation as an additional dimension so that $|\Psi[\psi]\rangle$ is a state that lives in exactly one dimension higher than the initial state $|\psi\rangle$. In this interpretation $|\psi\rangle$ behaves as a boundary condition for the emergent state. The lattice on which $|\Psi[\psi]\rangle$ is supported can then also be thought of as *emerging* from Λ on which $|\psi\rangle$ is defined. More precisely, the lattice is made up of layers which each carry a copy of Λ . In every other layer $C[\widehat{G}]$ -valued degrees of freedom are placed on $V(\Lambda)$ and $C[G]$ -valued degrees of freedom supported on $C(\Lambda)$. This is most easily demonstrated by concrete examples.

2.3 Gapped boundary conditions

In the interpretation of $|\Psi[\psi]\rangle$ as a state living in one dimension higher than $|\psi\rangle$, the stabilizers $\alpha_g^\vee$ and $\mathbb{b}_\chi^\vee$ are interpreted as acting on the *bulk* of the state

$|\Psi[\psi]\rangle$. The question of how to include boundary terms that stabilize $|\Psi[\psi]\rangle$ poses itself. As it turns out, the answer to this question is intimately interwoven with the exact way in which the initial G symmetry is realized on the boundary state $|\psi\rangle$.

Let us begin by considering the simplest local candidate stabilizer. It can be verified readily that for a given $g \in G$ and $\mathbf{v} \in V(\Lambda)$

$$\begin{aligned} &Z_g^\vee \otimes \text{id} \\ &\in \text{End}(C[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes C[G]^{\otimes |\mathbf{C}|}) \end{aligned} \quad (2.23)$$

commutes with the gauging map $\mathcal{G}$ since it obviously commutes with the local gauge transformations $\mathcal{G}_g^\vee$. Hence, it is a stabilizer of the emergent state exactly when it is one of the boundary condition, i.e. $Z_g^\vee |\psi\rangle = |\psi\rangle$. Note that one could think of this operator as a ‘truncated’ vertex term $\alpha_g^\vee$.

Consider likewise the truncated $\mathbb{b}_\chi^\vee$ term

$$\begin{aligned} &\text{id} \otimes Z_\chi^\vee \bigotimes_{\mathbf{v} \in \eta(\mathbf{c})} X_{\alpha_g^\vee(\chi)}^\vee \\ &\in \text{End}(C[\widehat{G}]^{\otimes |\mathbf{V}|} \otimes C[G]^{\otimes |\mathbf{C}|} \otimes C[\widehat{G}]^{\otimes |\mathbf{V}|}) \end{aligned} \quad (2.24)$$

Recall that the preimage of this operator under $\widehat{\mathcal{G}} \circ \mathcal{G}$ was shown to be

$$\bigotimes_{\mathbf{v} \in \eta(\mathbf{c})} X_{\alpha_g^\vee(\chi)}^\vee, \quad (2.25)$$

in eq. (2.16). This observation thus implies that the truncated term (2.24) is a stabilizer of the emergent state, whenever $|\psi\rangle$ is stabilized by the check X_χ^c . Since these local constraints X_χ^c can be understood as local symmetric order parameters for the global G symmetry (2.1), inequivalent boundary conditions realizing different symmetry breaking patterns of this symmetry can be devised. More formally, consider for any subgroup $H \leq G$ the state $|+\rangle_H$ which we recall was defined in section 1.5 as $|+\rangle_H := \frac{1}{\sqrt{|G/H|}} \sum_{[\chi] \in \widehat{G/H}} |\chi\rangle$. Constructing then the tensor product state

$$\bigotimes_{\mathbf{v} \in V(\Lambda)} |+\rangle_H^\vee, \quad (2.26)$$

it is clear that this reference state is stabilized by $Z_h^\vee$ and X_χ^c provided that $h \in H$, and $\chi \in \widehat{G/H}$. Remember indeed from section 1.5 that $\widehat{G/H}$ stands for the subgroup of characters in $\widehat{G}$ which are trivial when restricted to H . From this it also follows that the corresponding boundary stabilizers of the emergent state $|\Psi[\psi_H]\rangle$ commute. More generally, we could equally well consider boundary states of the form

$$\left(\sum_{[g] \in G/H} \mathbb{U}_g[\Omega] \right) \left(\bigotimes_{\mathbf{v} \in V(\Lambda)} |+\rangle_H^\vee \right) \quad (2.27)$$

for any allowed Ω , which enjoy the same stabilizers.

Also remark that, depending on the specifics of the G symmetry generators, even more generic boundary conditions can be defined, for example by also (partially) breaking the translational invariance of the boundary state. An exhaustive study of gapped boundary conditions for the models under scrutiny will be the topic of future work.

3 Foliated type-I fracton order from 2D linear subsystem symmetries

In this section we showcase the gauging of abelian linear subsystem symmetries and their dual emergent symmetries, which are themselves linear subsystem symmetries on the dual lattice, following the prescription of section 2. Concatenation of these gauging maps results in foliated type-I fracton codes, generalizing the anisotropic models studied in [17] defined for cyclic groups to arbitrary finite abelian groups.

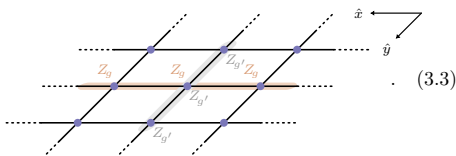
3.1 Gauging 2D abelian linear subsystem symmetries and their emergent symmetries

Throughout this section we consider again the two-dimensional torus, now endowed with a square lattice denoted by $\square$. The initial $\mathbb{C}[\hat{G}]$ -valued ‘matter’ degrees of freedom are located on the vertices $v \in V(\square)$ as customary. It will be useful to express the vertices in terms of their Cartesian coordinates as $v \equiv (i, j) \in \mathbb{Z}_{L_x} \times \mathbb{Z}_{L_y}$. Naturally, L_x and L_y stand for the linear system size in the x and y direction respectively. The linear subsystem symmetry (LSS) generators are then represented on the matter Hilbert space $\mathcal{H} := \bigotimes_{v \in V(\square)} \mathbb{C}[\hat{G}]$ as

$$Z_g^{\hat{x},j} := \bigotimes_{i \in \mathbb{Z}_{L_x}} Z_g^{(i,j)}, \quad (3.1)$$

$$Z_g^{\hat{y},i} := \bigotimes_{j \in \mathbb{Z}_{L_y}} Z_g^{(i,j)}. \quad (3.2)$$

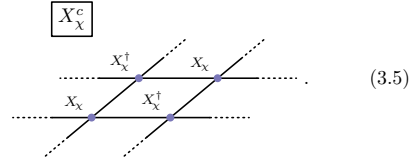
There thus exists an extensive number of symmetry generators, one for every j in the $\hat{x}$ direction of the square lattice, $Z_g^{\hat{x},j}$, and one for every i in the $\hat{y}$ direction, $Z_g^{\hat{y},i}$. Notice however that not all of the symmetries generators are independent since clearly $\prod_{j \in \mathbb{Z}_{L_y}} Z_g^{\hat{x},j} = \prod_{i \in \mathbb{Z}_{L_x}} Z_g^{\hat{y},i}$, $\forall g \in G$. Graphically, an example of symmetry operators $Z_g^{\hat{x},j}$ and $Z_g^{\hat{y},i}$ on a portion of the square lattice looks like:



In order to employ the recipe of the general framework laid out in section 2 we need to consider the checks that fully specify these symmetry generators. An adequate choice is given by

$$X_\chi^c = X_\chi^{(i,i)} \otimes X_\chi^{(i+1,j)} \otimes X_\chi^{(i,j+1)} \otimes X_\chi^{(i+1,j+1)}, \quad (3.4)$$

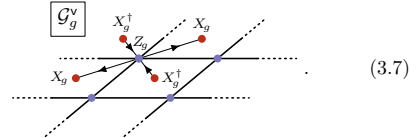
$\forall \chi \in \hat{G}, \forall i, j$. These operators are naturally associated with the plaquettes $P(\square)$, and we will therefore identify the constraints with the plaquettes, i.e. $\mathcal{C} \ni c \equiv p \in P(\square)$. Indeed, graphically, one such operator can be depicted as



Let us continue by gauging this symmetry. To this end we introduce thus $\mathbb{C}[\hat{G}]$ -valued gauge fields on the plaquettes $p \in P(\square)$ and thereby extend the Hilbert space according to $\mathcal{H} \rightarrow \mathcal{H} \otimes \hat{\mathcal{H}}$, where $\hat{\mathcal{H}} := \bigotimes_{p \in P(\square)} \mathbb{C}[\hat{G}]$. The local gauge transformations in the neighborhood of the vertex $v \in V(\square)$ read then

$$\mathcal{G}_g^v := Z_g^v \bigotimes_{p \rightarrow v} X_g^p \bigotimes_{p \leftarrow v} X_g^p, \quad (3.6)$$

for every $g \in G$. The tensor products appearing in this definition are over the sets of plaquettes $p \rightarrow v$, respectively $p \leftarrow v$, which point into, respectively point away from, the vertex v according to the following orientations:



As such, the definition of $\mathcal{G}_g^v$ is in accordance with eq. (2.4). Notice that here the same blue and red color code of the previous sections for depicting degrees of freedom in respectively $\mathbb{C}[\hat{G}]$ and $\mathbb{C}[G]$ is recycled.

As prescribed by our general gauging procedure, we define the following local projector:

$$\mathcal{P}^v := \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g^v, \quad (3.8)$$

for every vertex $v \in V(\Lambda)$, out of which we construct the projector onto the gauge invariant subspace as:

$$\mathcal{P} := \prod_{v \in V(\square)} \mathcal{P}^v. \quad (3.9)$$

Gauging of the LSS then happens by initializing the gauge fields in the trivial tensor product configuration $\bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}}$ and applying the above projector $\mathcal{P}$ so as to obtain

$$\mathcal{G}^{(0,1)} := \mathcal{P} \left(\bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}} \right). \quad (3.10)$$

The notation $\mathcal{G}^{(0,1)}$, following [19], is motivated by the fact that there is a sense in which linear subsystem symmetries can be thought of as ‘(0,1)-form’ symmetries: 0-form symmetries acting on a codimension 1 foliation of the spatial manifold. From the above definition of the gauging map it follows that it satisfies:

$$\mathcal{G}^{(0,1)} \cdot Z_g^{\hat{x},j} = \mathcal{G}^{(0,1)}, \quad (3.11)$$

$$\mathcal{G}^{(0,1)} \cdot Z_g^{\hat{y},i} = \mathcal{G}^{(0,1)}, \quad (3.12)$$

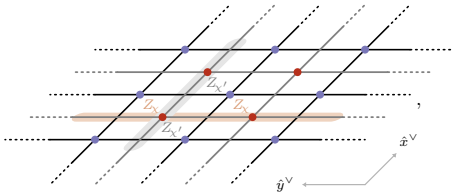
$\forall g \in G, i \in \mathbb{Z}_{L_y}, j \in \mathbb{Z}_{L_x}$. Hence, in line with the expectation from the general framework, states transforming non-trivially under one or more of the global linear subsystem symmetry generators are annihilated by the gauging map $\mathcal{G}^{(0,1)}$.

Let us now present the emergent symmetries. These happen to be themselves linear subsystem symmetries supported on the dual square lattice $\square^\vee$ and are labeled by characters $\chi \in \hat{G}$. Similarly as in the case of the initial LSS it proves useful to define the vertices of the dual square lattice, $\mathbf{v}^\vee \in \mathbf{V}(\square^\vee)$, in terms of their Cartesian coordinates, i.e. $\mathbf{v}^\vee \equiv (i^\vee, j^\vee) \in \mathbb{Z}_{L_y} \times \mathbb{Z}_{L_x}$. Also notice that the dual vertices can be identified under Poincaré duality with the plaquettes of the primal square lattice, formally: $\mathbf{V}(\square^\vee) \ni \mathbf{v}^\vee \equiv \mathbf{p} \in \mathbf{P}(\square)$. We then define for any character $\chi \in \hat{G}$ the *dual* LSS generators:

$$Z_\chi^{\hat{x},j^\vee} := \bigotimes_{i^\vee \in \mathbb{Z}_{L_y}} Z_\chi^{(i^\vee, j^\vee)}, \quad (3.13)$$

$$Z_\chi^{\hat{y},i^\vee} := \bigotimes_{j^\vee \in \mathbb{Z}_{L_x}} Z_\chi^{(i^\vee, j^\vee)}. \quad (3.14)$$

Examples of $Z_\chi^{\hat{x},j^\vee}$ and $Z_\chi^{\hat{y},i^\vee}$ then look like



wherein the dual lattice $\square^\vee$ is drawn in gray on top of the primal lattice $\square$ drawn in black.

The emergent symmetries are apparent from

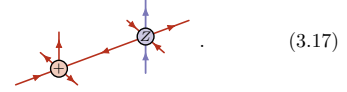
$$Z_\chi^{\hat{x},j^\vee} \cdot \mathcal{G}^{(0,1)} = \mathcal{G}^{(0,1)}, \quad (3.15)$$

$$Z_\chi^{\hat{y},i^\vee} \cdot \mathcal{G}^{(0,1)} = \mathcal{G}^{(0,1)}, \quad (3.16)$$

for all $\chi \in \hat{G}, i^\vee \in \mathbb{Z}_{L_y}, j^\vee \in \mathbb{Z}_{L_x}$.

Despite these identities being a mere application of eq. (2.10) to the case at hand, let us nevertheless derive these emergent symmetries explicitly for completeness. To this end, note that the operators $Z_\chi^{\hat{x},j^\vee}, Z_\chi^{\hat{y},i^\vee}$ commute with the projector in eq. (3.9) and leave invariant the initial trivial gauge field on the plaquettes $\bigotimes_{\mathbf{p} \in \mathbf{P}(\square)} |1\rangle^{\mathbf{p}}$. The former is satisfied because along dual horizontal and vertical directions $\hat{x}^\vee/\hat{y}^\vee$, the operators $Z_\chi^{\hat{x},j^\vee}, Z_\chi^{\hat{y},i^\vee}$ overlap in two faces with the projectors $\mathcal{P}_g^\vee$ appearing in the definition of $\mathcal{G}^{(0,1)}$ that act as $X_g \otimes X_g^\dagger$ on them.

To conclude the construction of the $\mathcal{G}^{(0,1)}$ gauging map, let us provide its PEPO representation. To this end we use tensors which slightly generalize those introduced in section 1. Their symmetry properties are therefore also completely analogous and are omitted here. Again restricting to one unit cell of the PEPO, which in this case is made up of one Z-type tensor on the vertex $\mathbf{v} \in \mathbf{V}(\square)$ of the unit cell and one +-type tensor on an adjacent plaquette $\mathbf{p} \in \mathbf{P}(\square)$, we claim that the PEPO is constructed from:



Indeed, let us verify its relevant properties. In the first place it follows from eq. (1.14) combined with eq. (1.16) that the gauging map absorbs the initial LSSs supported on the vertices of the primal lattice, i.e. (3.11-3.12). Analogously, the gauge invariance of the gauging map, i.e. $\mathcal{G}_g^\vee \cdot \mathcal{G}^{(0,1)}, \forall \mathbf{v} \in \mathbf{V}(\square), g \in G$, is proven by making use of eq. (1.14) and (1.15). Finally, the existence of the emergent symmetries is a direct consequence of the flatness of the gauge field enforced by the +-tensors, combined with the observation that the Z-type tensors are diagonal in its virtual indices.

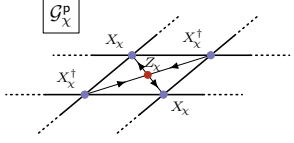
3.2 Iterative gauging of linear subsystem symmetries

Due to the structural similarity between the dual LSS generators (3.13, 3.14) and the initial ones (3.1, 3.2), the gauging procedure of the latter can be readily adopted to gauge the emergent LSS symmetries. To this end, we introduce $\mathbb{C}[\hat{G}]$ -valued gauge fields on the plaquettes $\mathbf{p}^\vee$ of the dual lattice $\square^\vee$, which, as we mentioned, can equivalently be thought of as the vertices $\mathbf{v} \in \mathbf{V}(\square)$. As is custom by now, we initiate these gauge fields as $\bigotimes_{\mathbf{v} \in \mathbf{V}(\square)} |1\rangle^\vee$. On this enlarged Hilbert space we define the local gauge transformation

$$\mathcal{G}_\chi^\mathbf{p} := Z_\chi^\mathbf{p} \bigotimes_{\mathbf{v} \rightarrow \mathbf{p}} X_\chi^\mathbf{v} \bigotimes_{\mathbf{v} \leftarrow \mathbf{p}} X_\chi^\mathbf{v}. \quad (3.18)$$

The orientations determining the sets of ‘incoming’ and ‘outgoing’ plaquettes on the dual square lattice, respectively $\mathbf{v} \rightarrow \mathbf{p}$ and $\mathbf{v} \leftarrow \mathbf{p}$, are dual to those on

the primal lattice as per the convention in (3.7). From the perspective of the primal lattice this gauge transformation can hence be depicted as



$$(3.19)$$

Using the local projectors

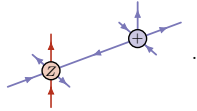
$$\mathcal{P}^p := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathcal{G}_\chi^p, \quad (3.20)$$

the following gauging map is then constructed:

$$\widehat{\mathcal{G}}^{(0,1)} := \left(\prod_{p \in P(\square)} \mathcal{P}^p \right) \left(\bigotimes_{v \in V(\square)} |1\rangle^v \right). \quad (3.21)$$

After the application of $\widehat{\mathcal{G}}^{(0,1)}$ LSSs on the primal lattice represented by (3.1-3.2) are reattained.

The PEPO representation of $\widehat{\mathcal{G}}^{(0,1)}$ in turn is constructed from the unit cell



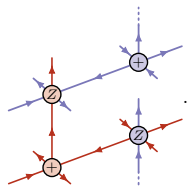
$$(3.22)$$

In this depiction it has to be understood that the dual Z-type tensor is placed on a plaquette $p \in P(\square)$ of the lattice whereas the dual + -type tensor is located on a vertex $v \in V(\square)$. The symmetries of this PEPO are immediately found from dualizing those of the above PEPO defined in eq. (3.17); from them the properties of $\widehat{\mathcal{G}}^{(0,1)}$ are recovered immediately.

An initial state $|\psi\rangle$ that is in the even sector of the LSSs (3.1-3.2) can now be iteratively gauged by using the usual prescription as follows:

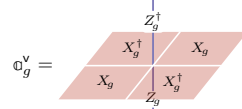
$$|\Psi[\psi]\rangle := \dots \circ \mathcal{G}_{\frac{1}{2}}^{(0,1)} \circ \widehat{\mathcal{G}}_1^{(0,1)} \circ \mathcal{G}_{\frac{1}{2}}^{(0,1)} |\psi\rangle. \quad (3.23)$$

Following our general paradigm we then interpret $|\Psi[\psi]\rangle$ as a three-dimensional state living on a cubic lattice $\mathfrak{C}$ endowed with a copy of $\mathbb{C}[\widehat{G}]$ to every edge in the $\hat{z}$ direction (blue) and a local Hilbert space $\mathbb{C}[G]$ (red) associated to every plaquette in the xy -plane. Putting the PEPO representations eqs. (3.17) and (3.22) together, it then follows that the emergent state away from the boundary can be represented by the PEPO unit cell

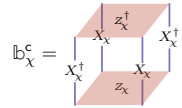


$$(3.24)$$

From the general procedure outlined in section 2, it then follows that the bulk stabilizers of the state $|\Psi[\psi]\rangle$ can be pictorially represented as



$$\mathfrak{o}_g^v = \quad (3.25)$$



$$\mathfrak{b}_\chi^c = \quad (3.26)$$

for every vertex $v \in V(\mathfrak{C})$ and every cube $c \in \mathbb{C}(\mathfrak{C})$ respectively. The state (3.23) is thus a ground state of the commuting projector Hamiltonian

$$\mathbb{H}^{\text{blk}} := - \sum_{v \in V(\mathfrak{C})} \mathbb{P}^v - \sum_{c \in \mathbb{C}(\mathfrak{C})} \mathbb{P}^c, \quad (3.27)$$

in which

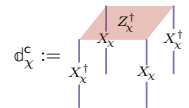
$$\mathbb{P}^v := \frac{1}{|G|} \sum_{g \in G} \mathfrak{o}_g^v, \quad (3.28)$$

$$\mathbb{P}^c := \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \mathfrak{b}_\chi^c. \quad (3.29)$$

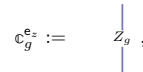
This generalizes the anisotropic model introduced in [17], which exhibits foliated type-I fracton order, to arbitrary finite groups. Our representation can be brought in the form showcased in [17] by applying the local basis transformation defined in eq. (1.65) to the red plaquettes. It is expected that our approach will facilitate the classification and construction of these fracton orders.

3.3 Gapped boundary conditions

Let us now also extend the bulk Hamiltonian to the boundary of the three-dimensional cubic lattice. To this end we consider following operators near the boundary:



$$\mathfrak{d}_\chi^c := \quad (3.30)$$



$$\mathfrak{e}_g^e := \quad (3.31)$$

for all $\chi \in \widehat{G}$ and $g \in G$.

Being truncated versions of the bulk stabilizers $\mathfrak{b}_\chi^c$ and $\mathfrak{o}_g^v$, they naturally commute with all bulk terms appearing in the bulk Hamiltonian $\mathbb{H}^{\text{blk}}$. Note that,

similar as to the boundary terms defined for the introductory example in section 1.5, $\mathfrak{d}_\chi^\pm$ and $\mathfrak{c}_g^\pm$ only commute for particular choices of χ and g such that $\chi(g) = 1$. This statement will be refined below. From the general framework it follows that $\mathfrak{d}_\chi^\pm$ stabilizes the three-dimensional state $|\Psi[\psi]\rangle$ if the boundary condition $|\psi\rangle$ is symmetric under the operator

(3.32)

for the boundary plaquette $\mathbf{p} \in \mathbf{P}(\square)$ corresponding to c . One recognizes these operators as the checks X_X^c defined above. For illustrative purposes, let us construct one particular reference boundary before proceeding to the more general case.

A suitable choice for $|\psi\rangle$ is any of the extensive number of ground states of the two-dimensional *classical G Ising plaquette model*. Indeed, recall that the Hamiltonian of this model is exactly given by $\mathbb{H}^{\text{Ising}} := -\sum_c \sum_{\{x_c\}} X_c^x$. One such ground state is for example constructed by starting from the product state

$$\bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v, \quad (3.33)$$

where the notation introduced in eq. (1.71) was used, and acting on this reference state with all LSS generators according to

$$|\psi\rangle_{\{1\}} := \left(\sum_{j=0}^{L_y-1} \sum_{\substack{\{g_j\}_j \\ \in G}} Z_{g_j}^{\hat{x},j} \right) \cdot \left(\sum_{i=0}^{L_x-1} \sum_{\substack{\{g_i\}_i \\ \in G}} Z_{g_i}^{\hat{y},i} \right) \cdot \left(\bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^v \right). \quad (3.34)$$

We stress that in this definition every summation over group elements is over the entire group G . From the observation that $X_\chi^\pm(\bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^\vee) = \bigotimes_{v \in V(\square)} |+\rangle_{\{1\}}^\vee$ for every $\chi \in \widehat{G}$, together with the fact that the checks commute by definition with all LSS generators, it indeed follows that $|\psi\rangle_{\{1\}}$ is a good boundary condition for the emergent state $|\Psi[\psi_{\{1\}}]\rangle$ for which additional boundary stabilizers are given by

$$Q_{\{1\}}^c := \sum_{\chi \in \widehat{G}} d_{\chi}^c, \quad \forall c \in C(\partial \square). \quad (3.35)$$

Similar as to the introductory example in section 1, and as per the general paradigm in section 2, we can also define more general boundary conditions for different symmetry breaking patterns of the LSS on the boundary. Indeed, given again the state $|+\rangle_H$ defined

in eq. (1.71), it is natural to define

$$|\psi\rangle_H := \left(\sum_{j=0}^{L_y-1} \sum_{\substack{\{[g_j]\}_j \\ \in G/H}} Z_{g_j}^{\hat{x},j} \right) \cdot \left(\sum_{i=0}^{L_x-1} \sum_{\substack{\{[g_i]\}_i \\ \in G/H}} Z_{g_i}^{\hat{y},i} \right) \left(\bigotimes_{v \in V(\square)} |+\rangle_v^H \right). \quad (3.36)$$

Given this choice of boundary condition, appropriate commuting boundary stabilizers are given by:

$$\mathbb{Q}_H^c := \frac{1}{|G/H|} \sum_{\chi \in \widehat{G/H}} \mathbb{d}_\chi^c, \quad (3.37)$$

$$\mathbb{Q}_H^{\mathbf{e}_z} := \frac{1}{|H|} \sum_{h \in H} \mathbb{C}_h^{\mathbf{e}_z}. \quad (3.38)$$

Indeed given that $h \in H$, and thus $\chi(h) = 1$ for any $\chi \in \widehat{G/H}$, the boundary operators ϕ_h^χ and c_h^χ defined in (3.30, 3.31) commute. These boundary terms are then added to the bulk Hamiltonian (3.27) to obtain the Hamiltonian

$$\mathbb{H}_H := \mathbb{H}^{\text{blk}} + \sum_{c \in C(\partial \square)} \mathbb{Q}_H^c + \sum_{e_z \in E(\partial \square)} \mathbb{Q}_H^{e_z}. \quad (3.39)$$

Before proceeding we note that the boundary conditions presented here are by no means exhaustive. In particular it should be pointed out that the boundary states defined in eq. (3.36) are manifestly translationally invariant, which is a condition which can be relaxed [6]. A complete study of all (gapped) boundary conditions for these models will be presented elsewhere.

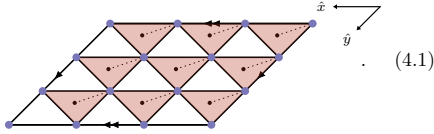
4 Type-I fracton order from 2D Sierpinski fractal symmetry

In this section we consider $\mathbb{Z}_2$ Sierpinski fractal symmetry as initial symmetry to our iterative gauging procedure. It is shown that the original Castelnovo-Chamon fractal model proposed in [18] emerges.

4.1 Gauging Sierpinski fractal symmetry and its emergent dual fractal symmetry

Let us begin by defining a set of generators for the fractal Sierpinski symmetry. Therefore we consider the two-dimensional torus endowed with a triangular lattice that we will denote by Δ . Given a vertex $v \in V(\Delta)$, we will denote its rectangular coordinates in the basis $\{\hat{x}, \hat{y}\}$, indicated in the figure below, by $v \equiv (i, j)$, $i, j \in \mathbb{Z}_L$, L being the *linear system size*. Every vertex v carries a copy of the local Hilbert space

$\mathbb{C}[\mathbb{Z}_2] \simeq \mathbb{C}^2 = \text{Span}_{\mathbb{C}}\{|0\rangle, |1\rangle\}$ on which the Pauli operators X and Z are defined.⁷ We will consider a subset $P(\Delta)^+ \subset P(\Delta)$ of the plaquettes whose member plaquettes are shaded in red in the figure below, foregrounding that we will put gauge degrees of freedom on these plaquettes when gauging the fractal symmetry. Note that the number of such plaquettes equals the number of vertices, i.e. $|P(\Delta)^+| = |V(\Delta)|$.⁸ This allows us to assign to every $\mathbf{p}^+ \in P(\Delta)^+$ a *base point* $\text{bp}(\mathbf{p}^+)$ which we conventionally choose as dictated by the dotted lines in the figure:



As pointed out in [36] the number of symmetry generators depends in a non-trivial way on the (linear) system size L . In this section we will for demonstrative purposes restrict to the case of $L = 2^m - 1$ for an integer m , in which case the number of symmetry generators is exactly equal to $L - 1$. The Sierpinski symmetry generators are then of the form

$$\mathbb{U}[S_q] := \prod_{i,j} (Z^{(i,j)})^{S_q(i,j)}, \quad (4.2)$$

where S_q is a function $S_q : \mathbb{Z}_L \times \mathbb{Z}_L \rightarrow \mathbb{F}_2$ defined such that $\mathbb{U}[S_q]$ commutes with the checks $X^{\mathbf{p}^+}$ which are given by

$$X^{\mathbf{p}^+} := X^{(i,j)} \otimes X^{(i+1,j)} \otimes X^{(i,j+1)}, \quad (i,j) = \text{bp}(\mathbf{p}^+), \quad (4.3)$$

for all $\mathbf{p}^+ \in P(\Delta)^+$. A check thus consists of three Pauli X operators acting on the three vertices surrounding a plaquette $\mathbf{p}^+ \in P(\Delta)^+$.

As follows from a direct computation, imposing commutation of the generators (4.2) with all these checks requires S_q to satisfy the recursive identity

$$S_q(i, j+1) := S_q(i+1, j) + S_q(i, j), \quad (4.4)$$

$\forall i, j \in \mathbb{Z}_L$. A *valid symmetry generator* is hence fully specified by a *valid choice* of ‘initial condition’ at $j = 0$:

$$S_q(i, 0) := q(i) \in \mathbb{F}_2^L. \quad (4.5)$$

The function $S_q(i, j)$, for $j \neq 0$ follows then from applying the recursion relation to this choice of q , and as

⁷In fact, in the remainder of this section we will identify $\mathbb{C}[\mathbb{Z}_2]$ and $\mathbb{C}[\widehat{\mathbb{Z}}_2]$ by making use of the obvious isomorphism $\mathbb{Z}_2 \simeq \widehat{\mathbb{Z}}_2$.

⁸Indeed, this follows from the fact that the Euler characteristic of the torus is zero, which can be expressed as $|V(\Delta)| - |E(\Delta)| + |P(\Delta)| = 0$, combined with the facts that $|P(\Delta)| = 2|P(\Delta)^+|$ and $|E(\Delta)| = 3|V(\Delta)|$.

such the corresponding generator $\mathbb{U}[S_q]$ commutes by construction with all checks. The $L - 1$ independent choices for q which give rise to the $L - 1$ independent Sierpinski symmetry generators that we expect can be chosen as

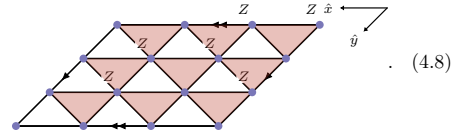
$$q^{(\alpha)}(i) = \delta_{i,\alpha} + \delta_{i,\alpha+1}, \quad \alpha = 1, 2, \dots, L - 1. \quad (4.6)$$

Let us mention as a side remark that it turns out that one could think of S_q as a *valid history* of a *Sierpinski cellular automaton* in the following sense. For fixed j , $S_q(i, j)$ corresponds to a state valued in $\mathbb{F}_2^L$. Given this state, the state at $j + 1$, i.e. $S_q(i, j + 1)$ for fixed $j + 1$, is fully specified by the above recursion relation from $S_q(i, j)$. Therefore j can be thought of as labeling discrete ‘time’ steps, where this notion of time thus flows in the direction of increasing j . In this picture $q(i)$ serves as an initial condition for the automaton, a posteriori justifying our jargon introduced above.

As a concrete example take $L = 3$, and $q^{(1)}(i) = [1 \ 1 \ 0]_i$. The corresponding history S_q is then computed to be

$$S_q(i, j) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{j,i}, \quad (4.7)$$

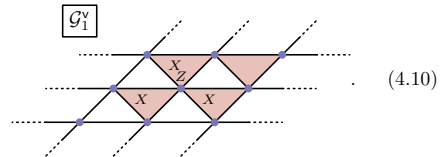
whose matching lattice operator $\mathbb{U}[S_q]$ can be depicted as:



Let us now gauge the Sierpinski symmetry using the general paradigm of section 2. To this end we introduce qubit gauge degrees of freedom on the plaquettes in $P(\Delta)^+$. A local gauge transformation for every $\mathbf{v} \in V(\Delta)$ reads then

$$\mathcal{G}_1^{\mathbf{v}} := Z^{\mathbf{v}} \bigotimes_{\mathbf{p}^+ \in P(\Delta)^+ | \mathbf{v}} X^{\mathbf{p}^+}, \quad (4.9)$$

wherein the tensor product is understood to be over plaquettes $\mathbf{p}^+ \in P(\Delta)^+ | \mathbf{v}$ that contain $\mathbf{v}$. Graphically:



From the local projector

$$\mathcal{P}^{\mathbf{v}} := \frac{1}{2} (\text{id} + \mathcal{G}_1^{\mathbf{v}}), \quad (4.11)$$

one constructs, as is by now routine, the global projector $\mathcal{P} = \prod_{v \in V(\Delta)} \mathcal{P}^v$ from which one obtains the gauging map by trivially initializing the gauge field:

$$\mathcal{G} := \mathcal{P} \left(\bigotimes_{p \in P(\Delta)^+} |0\rangle^p \right). \quad (4.12)$$

Evidently, one has $\mathcal{G}_1^\vee \cdot \mathcal{G} = \mathcal{G}$. Let us also verify explicitly that this gauging map indeed projects out states transforming non-trivially under the Sierpinski symmetry, i.e. $\mathcal{G} \cdot \mathbb{U}[S_q] = \mathcal{G}$, for all *valid* symmetry generators defined in eq. (4.2). To this end, we could expand the local projectors appearing in the definition of the gauging map $\mathcal{G}$:

$$\prod_{v \in V(\Delta)} \prod_{\{g_v\}_v} (\mathcal{G}_1^\vee)^{g_v}. \quad (4.13)$$

From the definition of $\mathcal{G}_1^\vee$ in eq. (4.9) it then follows that three $\mathcal{G}_1^\vee$'s in the expansion (4.13) on vertices $(i+1, j)$, (i, j) and $(i, j+1)$ overlap in exactly one plaquette $p^+ \in P(\Delta)^+$, namely the one such that $(i, j) = \text{bp}(p^+)$. Locally, this means that the property $\mathcal{G} \cdot \mathbb{U}[S_q] = \mathcal{G}$ is equivalent to S_q satisfying (4.4), which it has to in order to define a genuine Sierpinski symmetry generator.

We now turn to the emergent symmetries. As a matter of fact, after gauging we obtain a *dual* Sierpinski fractal symmetry supported on the gauge degrees of freedom. Given a plaquette $p^+ \in P(\Delta)^+$, we define its *rectangular coordinates* as $(i^\vee, j^\vee) := \text{bp}(p^+)$. Mimicking the definition of S_q , let us define $S_q^\vee : \mathbb{Z}_L \times \mathbb{Z}_L \rightarrow \mathbb{F}_2$ by means of the recursive identity

$$S_q^\vee(i^\vee, j^\vee) := S_q^\vee(i^\vee, j^\vee + 1) + S_q^\vee(i^\vee - 1, j^\vee + 1), \quad (4.14)$$

$$S_q^\vee(i^\vee, 0) := q(i^\vee), \quad (4.15)$$

for all $i^\vee, j^\vee \in \mathbb{Z}_L$. By comparison with the definition of S_q in (4.4) we can infer that $S_q^\vee$ defines a Sierpinski fractal symmetry in which the ‘time steps’ labeled by j flow in the opposite direction compared to the one in (4.4) and in which the ‘spatial direction’ i is reflected as well. It can then be shown that $S_q^\vee$ exactly specifies the emergent fractal symmetries after gauging. Namely, defining

$$\mathbb{U}^\vee[S_q^\vee] := \prod_{i^\vee, j^\vee} (Z^{(i^\vee, j^\vee)} S_q^\vee(i^\vee, j^\vee)), \quad (4.16)$$

it follows that

$$\mathbb{U}^\vee[S_q^\vee] \cdot \mathcal{G} = \mathcal{G}. \quad (4.17)$$

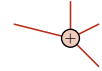
Indeed, this follows from the fact that exactly for this choice of $S_q^\vee$ the operator $\mathbb{U}^\vee[S_q^\vee]$ commutes with every $\mathcal{G}_1^\vee$, $\mathbb{U}^\vee[S_q^\vee] \cdot \mathcal{G}_1^\vee = \mathcal{G}_1^\vee \cdot \mathbb{U}^\vee[S_q^\vee]$, together with the trivial observation that the initial flat gauge configuration $\bigotimes_{p^+ \in P(\Delta)^+} |0\rangle^{p^+}$ is a +1 eigenstate of $\mathbb{U}^\vee[S_q^\vee]$.

Since we are dealing with the symmetry group $\mathbb{Z}_2$ here, we can in fact significantly simplify the tensor network calculus introduced in section 1 by removing the arrows and using a **red** color for all tensor legs since we have identified $\mathbb{C}[\mathbb{Z}_2]$ and $\mathbb{C}[\widehat{\mathbb{Z}_2}]$ above. By virtue of these simplifications, we will also only need two type of tensors for both gauging maps which are given by:



$$, \quad (4.18)$$

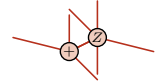
which in the construction of the $\mathcal{G}$ -PEPO will be placed on every vertex $v \in V(\Delta)$ and in the case of the $\widehat{\mathcal{G}}$ -PEPO on every plaquette in $P(\Delta)^+$, and



$$, \quad (4.19)$$

placed on the plaquettes $P(\Delta)^+$ when it appears in the PEPO representation of $\mathcal{G}$ and on the vertices $v \in V(\Delta)$ when it appears in the $\widehat{\mathcal{G}}$ -PEPO.

For the PEPO representations of both gauging maps we make a choice of unit cell such that a plaquette tensor on a given $p^+ \in P(\Delta)^+$ is combined with a vertex tensor placed on the corresponding base point $\text{bp}(p^+)$. For the gauging map $\mathcal{G}$, the PEPO unit cell is then explicitly given by:



$$, \quad (4.20)$$

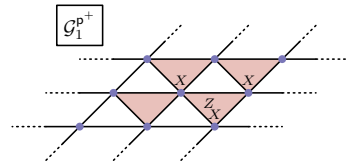
where, as mentioned above, the + -type tensor is placed on a certain plaquette $p^+ \in P(\Delta)^+$ and the Z -type tensor on the corresponding base point $\text{bp}(p^+)$.

4.2 Iterative gauging of Sierpinski fractal symmetry

Gauging of the dual Sierpinski symmetry happens via the introduction of qubits on the vertices $V(\Delta)$. In the neighborhood of a plaquette $p^+ \in P(\Delta)^+$, following gauge transformation is defined:

$$\mathcal{G}_1^{p^+} := Z^{p^+} \bigotimes_{v \in V(\Delta)|p^+} X^v, \quad (4.21)$$

which forms a local $\mathbb{Z}_2$ -representation. In this, $V(\Delta)|p^+$ stands for the set of three vertices surrounding p^+ . Concretely:



$$. \quad (4.22)$$

Based on the local projector, defined as

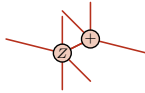
$$\mathcal{P}^{p^+} := \frac{1}{2} \left(\text{id} + \mathcal{G}_1^{p^+} \right), \quad (4.23)$$

the corresponding gauging map reads:

$$\hat{\mathcal{G}} := \mathcal{P} \left(\bigotimes_{v \in V(\Delta)} |0\rangle^v \right), \quad (4.24)$$

in which $\mathcal{P} := \prod_{p^+ \in P(\Delta)^+} \mathcal{P}^{p^+}$. The expected properties $\mathbb{U}[S_q] \cdot \hat{\mathcal{G}} = \hat{\mathcal{G}}$ and $\hat{\mathcal{G}} \cdot \mathbb{U}^V[S_q^V] = \hat{\mathcal{G}}$, for all valid symmetry generators $\mathbb{U}[S_q]$ and $\mathbb{U}^V[S_q^V]$, hold by construction.

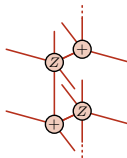
Under the aforementioned conventions, the unit cell of the PEPO representation of $\hat{\mathcal{G}}$ can be depicted as



$$, \quad (4.25)$$

where now the Z-type tensor is placed on a plaquette $p^+ \in P(\Delta)^+$ and the second tensor on the corresponding base point.

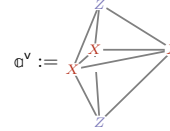
What is the emergent state obtained from these gauging procedures? It turns out that the emergent state can be interpreted as living on the vertices of a *hexagonal close-packed* (HCP) lattice, which consists of layers of triangular lattices whose vertices are slightly displaced with respect to each other [37]. A slightly different realization of this lattice which is closer to our setup is the following. We consider also layers of two-dimensional triangular lattices Δ , but in such a way that the vertices of all layers align along the direction perpendicular to the stack of layers. In every other layer we then place qubits on the vertices of the triangular lattice, i.e. on $V(\Delta)$, and on the plaquettes $P(\Delta)^+$. The PEPO representation of the emergent state for one unit cell is then given by:



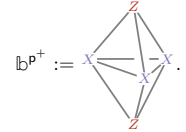
$$, \quad (4.26)$$

Using a blue color for the degrees of freedom on $V(\Delta)$ and a red color for those on the plaquettes $P(\Delta)^+$, the bulk terms of the emergent Hamiltonian are then

the following:



$$, \quad (4.27)$$



$$. \quad (4.28)$$

With slight abuse of notation, v refers to the vertex of the emergent lattice on which the bottom blue qubit appearing in the o operator is supported. On the other hand, p^+ refers to the support of the red qubit on the bottom of the depiction of the b^{p^+} operator. It is clear that these operators all mutually commute. We can construct projectors out of these operators as

$$\mathbb{P}^v := \frac{1}{2} \left(\text{id} + o^v \right), \quad (4.29)$$

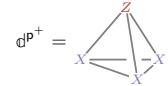
$$\mathbb{P}^{p^+} := \frac{1}{2} \left(\text{id} + b^{p^+} \right). \quad (4.30)$$

The emergent bulk Hamiltonian then reads

$$\mathbb{H}^{\text{blk}} := - \sum_{v \in V} \mathbb{P}^v - \sum_{p^+ \in P^+} \mathbb{P}^{p^+}, \quad (4.31)$$

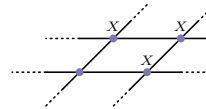
where the sums are over all (bulk) vertices and p^+ -plaquettes of the HCP lattice. This model was proposed by Caselново and Chamon in [18], is covered in the family of models by Yoshida [38] and was recently studied in detail in [37].

If we start the iterative gauging by $\mathcal{G}$, the possible boundary terms correspond to the truncation of (4.27) near the boundary or individual Z terms. Explicitly the truncated (4.27) terms read:



$$. \quad (4.32)$$

which also have been identified and studied in [37]. The term o^{p^+} acts on the boundary state $|\psi\rangle$, which is symmetric under the fractal Sierpinski symmetry, as:



$$, \quad (4.33)$$

which can be recognized as the local checks X^{p^+} and serve as a local order parameter for the initial fractal symmetry. As such, there are two possible boundaries depending on whether the fractal symmetry is spontaneously broken by $|\psi\rangle$ or not. In the first case, the

boundary stabilizers are given by

$$\mathbb{Q}^{p^+} := \frac{1}{2} \left(\text{id} + X^{p^+} \right), \quad (4.34)$$

for all $p^+ \in P(\Delta)^+$ on the boundary Δ , whereas in the symmetry breaking case the stabilizers are given by

$$\mathbb{Q}^{p^+} := \frac{1}{2} \left(\text{id} + Z^v \right). \quad (4.35)$$

5 Conclusion and outlook

In this work we have put the iterative gauging paradigm of [12] for abelian symmetries on a systematic footing and extended it to arbitrary dimensions and generic types of abelian symmetries. We provided explicit examples where the emergent states are three-dimensional. Concretely we have shown the emergence of surface codes, foliated type-I fracton codes and non-foliated type-I fracton codes from the iterative gauging of 0-form, 1-form, linear subsystem and Sierpinski fractal symmetry respectively.

The method proposed in this work resembles the *sandwich* construction of SymTFT. In particular, we constructed states of the following form:

$$\langle \hat{\psi} | \mathcal{G} \circ \hat{\mathcal{G}} \circ \dots \circ \mathcal{G} \circ \hat{\mathcal{G}} \circ \mathcal{G} | \psi \rangle,$$

made up of the gauging operator $\mathcal{G}$ of a G symmetry, its dual $\hat{\mathcal{G}}$ and states $|\psi\rangle, |\hat{\psi}\rangle$ symmetric under the initial abelian symmetry and its dual. For the case where the boundary supports an abelian 1-form symmetry, our method exactly recovers the quantum double in correspondence with the SymTFT proposal. The symmetries of this sandwich state are considered as stabilizer local Hamiltonian terms giving rise to the Hamiltonian:

$$\mathbb{H}^{\text{blk}} + \mathbb{H}^{\text{bdry}},$$

where the bulk terms only depend on the symmetry G and the boundary terms depend on the quantum phase of $|\psi\rangle$.

We predict a number of ways in which this framework can be further refined and propose the following open problems. From the case of the emergent surface codes, it is clear that further refinement is required to accommodate boundary conditions hosting SPT phases on top of the partial symmetry breaking. It is plausible that akin to the lower-dimensional case [12] this requires modification of the boundary stabilizers. A similar question arises regarding the boundary conditions of the emergent anisotropic models studied in section 3. In particular, it would be revealing to study boundaries that (potentially partially) break translation symmetry. It is plausible to expect that by including boundary states which realize different symmetry breaking patterns in distinct regions, we could realize domain walls in the emergent orders such as the electromagnetic domain wall

in the three-dimensional toric code [22]. One could also wonder to what extent our framework is exhaustive in producing classes stabilizer codes.

The main assumption of this work is the symmetry group being abelian, which results in abelian higher-dimensional codes. As such we leave the study of non-abelian, and more general categorical, symmetries in the iterative gauging framework for future work. As a first step, we will generalize the one-dimensional gauging map of [13] to generalized symmetries described by unitary fusion categories in forthcoming work [39]. An obvious family of models not treated in this work are type-II fracton models, which differ from type-I fracton models in the mobility of their excitations. Adapting our approach to accommodate these fracton orders could prove to be a splendid challenge.

We further foresee that the explicit tensor network constructions of the states provided here will enhance their numerical study. Having direct access to the boundary states in this framework could facilitate the study of boundary phase transitions of these emergent models.

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Twisted gauging and topological sectors in (2+1)d abelian lattice gauge theories

10.1 Synopsis

An open problem in the study of three-dimensional quantum field theories is the construction of genuinely *self-dual* models — that is, models that are mapped to themselves under some duality transformation. Necessarily, such a self-duality should preserve the symmetry structure of either model. A central observation of ref. [119] is that a specific class of gauging and twisted gauging transformations appears to be the only mechanism capable of producing such self-dual theories in this dimension.

The duality transformations we study in this project intertwine (2+1)-dimensional *twisted Abelian lattice gauge theories* whose symmetry structure is encoded by the fusion 2-category $2\text{Rep}(G)$, which simultaneously captures Wilson line operators as 1-form symmetry operators, their condensation defects as 0-form symmetry operators, as well as topological interfaces between those membrane operators. The duality transformation amounts to the change in module 2-category $2\text{Vec} \mapsto 2\text{Vec}^\alpha$ over the input 2Vec_G , with $[\alpha] \in H^3(G, \text{U}(1))$ encoding the aforementioned twist or discrete torsion. Physically, this map amounts to the gauging of the 1-form symmetry of the theory, followed by a twisted gauging of the resulting 0-form 2Vec_G symmetry. We provide lattice representations of these symmetry operators and duality intertwiners in the form of topological matrix product operators and projected entangled-pair operators. We characterise the 1-form symmetry-

twisted boundary conditions and charge sectors, and construct the unitary duality transformations that relate the dual theories to one another.

We first illustrate the mechanism in one spatial dimension, where the construction for the case of $G = \mathbb{Z}_2 \times \mathbb{Z}_2$, with the twist encoded in $[\psi] \in H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathrm{U}(1)) \simeq \mathbb{Z}_2$, reduces to the Kennedy-Tasaki duality [103] and which serves as an introduction to the higher-dimensional case. Turning to the setting of two spatial dimensions, we work out an explicit realisation of the duality between the toric code and the double semion model corresponding to the string-net model with $\mathrm{Vec}_{\mathbb{Z}_2}$, respectively $\mathrm{Vec}_{\mathbb{Z}_2}^\alpha$, as input, $[\alpha]$ being the non-trivial class in $H^3(\mathbb{Z}_2, \mathrm{U}(1)) \simeq \mathbb{Z}_2$ [47, 52, 120]. Finally, we show that the sum of these two models admits an interpretation as a *higher gauge theory*, in which the duality transformation is promoted to a genuine internal symmetry.

10.2 Reference

Twisted gauging and topological sectors in (2+1)d abelian lattice gauge theories

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10.3 Contributions of the author

This project was conceived by Clement Delcamp, who developed the overarching conceptual framework based on earlier work [121, 122], carried out the majority of the technical derivations, and wrote the majority of the manuscript. My contributions focused primarily on the derivations related to the one-dimensional models, with additional contributions to parts of the two-dimensional models. I also participated throughout in technical discussions that helped refine and sharpen the overall framework, and co-wrote parts of the paper.

Twisted gauging and topological sectors in (2+1)d Abelian lattice gauge theories

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Abstract

Given a two-dimensional quantum lattice model with an Abelian gauge theory interpretation, we investigate a duality operation that amounts to gauging its invertible 1-form symmetry, followed by gauging the resulting 0-form symmetry in a twisted way via a choice of discrete torsion. Using tensor networks, we introduce explicit lattice realisations of the so-called condensation defects, which are obtained by gauging the 1-form symmetry along submanifolds of spacetime, and employ the same calculus to realise the duality operators. By leveraging these tensor network operators, we compute the non-trivial interplay between symmetry-twisted boundary conditions and charge sectors under the duality operation, enabling us to construct isometries relating the dual Hamiltonians. Whenever a lattice gauge theory is left invariant under the duality operation, we explore the possibility of promoting the self-duality to an internal symmetry. We argue that this results in a symmetry structure that encodes the 2-representations of a 2-group.



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1 Introduction

Given a quantum theory, a modern viewpoint identifies the existence of a collection of *topological defects* as the definition of a global *symmetry* [40]. Aside from producing notions of symmetry that go beyond the ordinary one—and its axiomatisation in terms of abstract *groups*—this viewpoint has led to a calculus of symmetries that takes full advantage of well-established methods from topological quantum field theory [35]. In particular, this calculus greatly facilitates the *gauging* of finite symmetries and the study of the consequences of such an operation, both in the continuum [1, 14, 18, 23, 39, 44] and in the discrete [6, 27, 48, 59].

In this manuscript, we are interested in two-dimensional quantum lattice models that result from gauging an ordinary finite Abelian 0-form symmetry. These are defined on a constrained Hilbert space of states satisfying magnetic *Gauß constraints*. As such, these can be interpreted as Abelian *lattice gauge theories*. It is well established that theories obtained in this way can be coupled to topological lines labelled by the *Pontrjagin dual* of the initial Abelian group, which define a 1-form symmetry. But, it was recently understood that in addition to these topological lines, such theories also host topological surfaces referred to as *condensation defects* [56, 65], which have the peculiarity of being typically non-invertible as linear maps on the constrained Hilbert space [16, 19, 27]. The resulting complete symmetry structure is axiomatised in terms of a higher mathematical structure encoding the so-called *2-representations* of the group [4, 5, 13, 22, 27].

Given a (2+1)d Abelian lattice gauge theory, we consider an operation that amounts to ungauging its original 0-form symmetry, before gauging it back in a twisted way via a choice of *discrete torsion*, which is classified by the third cohomology of the group. The (untwisted) gauging operation alone is intimately related to the celebrated *Kramers–Wannier transformation* [47, 75], which is known to be a *non-invertible* operation on a fixed Hilbert space. Indeed, assuming closed periodic boundary conditions, gauging projects both theories onto the *singlet* sectors of their respective symmetries. Therefore, turning the Kramers–Wannier transformation into a duality requires addressing the non-trivial interplay between closed boundary conditions and charge sectors. An isometry acting on the resulting total Hilbert space that relate the spectra of the dual models can then be found by promoting the choice of boundary condition to a dynamical degree of freedom [49, 57, 59, 70]. Calculating the mapping of topological sectors under the combination of untwisted and twisted gaugings described above is the main objective of this work.

One merit of the dualities investigated in this manuscript is the ability to define them in any spacetime dimension. More precisely, one can apply the same strategy to any quantum theory that results from gauging the finite invertible 0-form symmetry of another theory. For

instance, it was demonstrated in ref. [48] that in $(1+1)d$ the so-called *Kennedy–Tasaki* transformation was precisely of this type [45,63]. Another attractive trait of these dualities is that they preserve the symmetry structure. This is especially relevant in $(2+1)d$ where most gauging procedures are found to modify the symmetry structure. As a matter of fact, it was conjectured in ref. [27] that most non-trivial dualities in $(2+1)d$ sharing this feature are of the type investigated in this manuscript, later corroborated in ref. [28]. Since the symmetry structure is preserved under the duality, one can ask for models that are left invariant under the transformation. Whenever this is the case, one can further ask to promote the resulting self-duality to an internal symmetry, which mathematically amounts to performing a *group extension* of the corresponding symmetry structure [30,33,73]. In $(2+1)d$, we argue that this results into the emergence of lattice *higher gauge theories* [8,9,11,25,26,46,77] whose symmetry operators are organised into higher mathematical structures encoding the 2-representations of 2-groups.

Concretely, given a finite symmetry acting on a one-dimensional quantum lattice model, a framework was developed in ref. [48] in order to systematically gauge any of its subsymmetries, effectively realising the approach of ref. [14,34] in the discrete. This framework relies on a generalisation of the *anyonic chain* construction [10,36,37], which, in addition to clearly distinguishing an abstract symmetry from its explicit realisation in a physical system, makes *tube algebra* techniques [61,62] amenable to the computation of *topological sectors* [2,49,53]. In this formulation, the action of a finite symmetry, which is encoded into a so-called *fusion category* [33], requires in particular a choice of *module category* [31]. Two physical systems that only differ in such a choice of module category are related by some gauging procedure. Then, the type of duality considered in the present manuscript relates two physical systems that differ in a choice of module category that is equivalent to the category of complex vector spaces. A generalisation of this framework to $(2+1)d$ was initiated in ref. [27]. In this higher-dimensional context, the action of a finite symmetry, which is now encoded into a so-called *fusion 2-category* [24,38], requires in particular a choice of *module 2-category* [21,22,28,29]. The type of duality we are interested in then relates two Abelian lattice gauge theories that only differ in a choice of module 2-category that is equivalent to the 2-category of 2-vector spaces. A general strategy to realise on the lattice the various symmetry operators evoked above was presented in ref. [27,42]. Following ref. [49], the same strategy can be employed to compute the lattice operators that transmute the dual Hamiltonians onto one another. Here, we realise these various operators very explicitly in the form of *tensor network operators* using the calculus outlined in ref. [22], partially extending to higher dimensions the results of ref. [3,20,51,66,74]. Using these operators, we are then able to compute the mapping of topological sectors under the dualities. In Table 1, we have summarised the symmetry structures, dualities and permutation of the topological sectors under scrutiny in this manuscript, as a guide to the reader.

Organisation of the manuscript: We begin by reviewing in sec. 2 the one-dimensional scenario. After introducing a tensor network operator implementing the duality that amounts to the gauging of the Abelian invertible symmetry followed by the twisted gauging of the dual symmetry, we derive the interplay between symmetry-twisted boundary conditions and charge sectors. This section is concluded by exploring concrete examples. In sec. 3, we begin by introducing families of two-dimensional quantum lattice models with a lattice gauge theory interpretation. We then unpack the symmetry encoded into the fusion 2-category of 2-representations of the gauge group, realising in particular the condensation defects in the form of tensor networks. We then employ the same calculus to realise the duality operator, which we leverage to compute the permutation of topological sectors under the duality. Finally, we discuss lifting the self-duality to a genuine internal symmetry through the lens of a simple theory built from the so-called toric code and the double semion model.

Table 1: Summary of the main results. We consider a duality operation in (1+1)d and (2+1)d that amounts to gauging a 0-form and 1-form $\text{Rep}(G)$ symmetry, respectively, followed by a twisted gauging of the resulting dual invertible 0-form symmetry. Both in (1+1)d and (2+1)d, the symmetry structures are the same on both sides of the duality.

	Sec. 2 — (1+1)d	Sec. 3 — (2+1)d
Discrete torsion	$[\psi] \in H^2(G, \text{U}(1))$	$[\lambda] \in H^3(G, \text{U}(1))$
Symmetry structure	$\text{Rep}(G) \simeq \text{Mod}(\mathbb{C}[G])$	$2\text{Rep}(G) \simeq \text{Mod}(\text{Vec}_G)$
Duality operators	$\pi \in \text{Rep}^\psi(G)$	$\mathcal{M}(A, \phi) \in 2\text{Rep}^A(G)$
Topological sectors	$(g, \eta) \in G \times \widehat{G}$	$(g_1, g_2, \eta) \in G \times G \times \widehat{G}$
Permutation sectors	$(g, \eta) \mapsto (g, t_g(\psi)\eta)$	$(g_1, g_2, \eta) \mapsto (g_1, g_2, t_{g_1, g_2}^2(\lambda)\eta)$

2 Twisted gauging of (1+1)d spin chains

Given an arbitrary one-dimensional quantum lattice model with an Abelian invertible symmetry, we review the non-trivial interplay between closed symmetry-twisted boundary conditions and charge sectors under dualities that amount to the (untwisted) gauging of the symmetry, followed by the twisted gauging of the dual symmetry. Given pairs of compatible boundary conditions and charge sectors, the unitary operators relating the spectra of dual Hamiltonians are explicitly constructed in the form of tensor network operators.

2.1 Symmetric Hamiltonians and closed symmetry-twisted boundary conditions

Let G be a finite Abelian group G and ψ a normalised representative 2-cocycle of a cohomology class $[\psi]$ in $H^2(G, \text{U}(1))$. We begin by constructing a family of Hamiltonians on a closed one-dimensional lattice that commutes with symmetry operators labelled by characters in the Pontrjagin dual $\widehat{G} = \text{Hom}(G, \text{U}(1))$ of G . Given a closed one-dimensional lattice, we denote the sets of vertices and oriented edges by V and E , respectively. Notice that $|V| = |E|$. Choosing a total ordering of the vertices induces an orientation of the edges. For any edge $e \in E$ we write its source and target vertex as $\partial_- e$ and $\partial_+ e$, respectively, so that we identify $e \equiv (\partial_- e, \partial_+ e)$. Throughout this section, we assume without loss of generality that all the edges share the same orientation. To every edge, we assign a copy of $\mathbb{C}[G] = \text{Span}_{\mathbb{C}}\{|g\rangle | g \in G\}$, with $\langle g_1 | g_2 \rangle = \delta_{g_1, g_2}$, so that the microscopic Hilbert space of the model is given by $\bigotimes_{e \in E} \mathbb{C}[G]$. Given a function $g : E \rightarrow G$, we notate via $|g\rangle = \bigotimes_{e \in E} |g(e)\rangle$ the function g regarded as a basis element of the microscopic Hilbert space. Moreover, for every vertex $v \in V$, we introduce a function $r_v : V \rightarrow G$ such that $r_v(v_1) = \mathbb{1}$, whenever $v_1 \neq v$. The set of such functions is naturally isomorphic to G .¹ Finally, to every $v \in V$, we assign the local operators

$$\mathbb{B}_{v,n}^{\psi} := \sum_{g, r_v} h_n(g, r_v) \psi(g, r_v) |g\rangle \langle g|, \quad (1)$$

wherein n is some label, $h_n(g, r_v)$ are complex coefficients, $g, r_v : E \rightarrow G$ is the function such that

$$(g, r_v)(e) := \begin{cases} r_v(v)g(e), & \text{if } e \equiv (v, v_2), \\ g(e)r_v(v)^{-1}, & \text{if } e \equiv (v_1, v), \\ g(e), & \text{otherwise,} \end{cases} \quad (2)$$

¹Note that by summing over r_v , we are summing over all possible values in G such a function can assign to v . Although this notation may seem excessive at this point, we introduce it here in preparation for the higher-dimensional study.

and $\psi(\mathfrak{g}, \mathfrak{r}_v)$ is the phase factor

$$\psi(\mathfrak{g}, \mathfrak{r}_v) := \frac{\psi(\mathfrak{g}(v_1 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v))}{\psi(\mathfrak{r}_v(v), \mathfrak{g}(v v_2))}, \quad (3)$$

where $v_1 < v < v_2$, so that $(v_1 v)$ and $(v v_2)$ are the edges whose target and source vertices are v , respectively.

For any possible choice of complex coefficients $h_n(\mathfrak{g}, \mathfrak{r}_v)$, for every $\mathfrak{g} : E \rightarrow G$ and $\mathfrak{r}_v : V \rightarrow G$, and cohomology class $[\psi]$, it follows from the defining property of group characters that local operators (1) commute with (0-form) symmetry operators of the form

$$\mathbb{U}^\chi := \sum_{\mathfrak{g}} \left(\prod_{e \in E} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (4)$$

for any $\chi \in \widehat{G}$. From $(\chi_1 \otimes \chi_2)(\mathfrak{g}) = \chi_1(\mathfrak{g})\chi_2(\mathfrak{g})$ we readily infer that the symmetry operators compose according to $\mathbb{U}^{\chi_1} \circ \mathbb{U}^{\chi_2} = \mathbb{U}^{\chi_1 \otimes \chi_2}$. Although it is somewhat superfluous for an Abelian group, we can state that symmetry operators are organised into the fusion (1-)category $\text{Rep}(G)$ of (finite-dimensional complex) representations of G . Finally, we construct arbitrary $\widehat{G}$ -symmetric models as $\mathbb{H}^\psi = \sum_{v \in V} \sum_n \mathbb{H}_{v,n}^\psi$. Notice that we restricted to local operators acting on nearest neighbours for simplicity but nothing prevents us from defining local operators acting on additional sites.

Up until this point we have implicitly assumed periodic boundary conditions. Symmetry-twisted closed boundary conditions are then obtained by inserting symmetry twists, which—in contrast to symmetry operators that extend over the whole space—are localised at one spatial point and extend in the time direction. The resulting symmetry-twisted boundary conditions preserve the translation invariance of the system, in the sense that there is an isomorphism between Hilbert spaces obtained by moving the defect by one site [2, 49, 53, 67]. More specifically, we promote symmetry-twisted boundary conditions to genuine physical degrees of freedom by tensoring the microscopic Hilbert space $\text{Span}_{\mathbb{C}}\{|\mathfrak{g}\rangle \mid \mathfrak{g} : E \rightarrow G\}$ with a copy of $\mathbb{C}[\widehat{G}]$ associated with a vertex conventionally denoted by v_0 . Fixing the boundary condition to be that labelled by $\eta \in \widehat{G}$ then amounts to setting the $\mathbb{C}[\widehat{G}]$ degree of freedom to $|\eta\rangle$, and we denote the resulting kinematical Hilbert space by $\mathcal{H}^\eta$. Local operators across the symmetry defect labelled by $\eta \in \widehat{G}$ are acted upon by the symmetry action as follows:

$$\mathbb{H}_{v_0,n}^{\psi,\eta} := \sum_{\mathfrak{g}, \mathfrak{r}_{v_0}} h_n(\mathfrak{g}, \mathfrak{r}_{v_0}) \psi(\mathfrak{g}, \mathfrak{r}_{v_0}) \eta(\mathfrak{r}_{v_0}) |\mathfrak{g} \mathfrak{d} \mathfrak{r}_{v_0}\rangle \langle \mathfrak{g}|, \quad (5)$$

with $\eta(\mathfrak{r}_{v_0}) \equiv \eta(\mathfrak{r}_{v_0}(v_0))$. The Hamiltonian in the presence of this symmetry twist hence reads

$$\mathbb{H}^{\psi,\eta} := \sum_{\substack{v \in V \\ v \neq v_0}} \sum_n \mathbb{H}_{v,n}^\psi + \sum_n \mathbb{H}_{v_0,n}^{\psi,\eta}. \quad (6)$$

In order to accommodate the presence of a symmetry twist labelled by $\eta \in \widehat{G}$, the symmetry operators (4) need to be modified as follows:

$$\mathbb{U}_\eta^\chi := \sum_{\mathfrak{g}} \left(\prod_{e \in E} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}| \otimes |\eta\rangle \langle \eta|. \quad (7)$$

Again, it readily follows from the defining property of group characters that $[\mathbb{H}_{v_0,n}^{\psi,\eta}, \mathbb{U}_\eta^\chi] = 0$. Given a choice of symmetry-twisted closed boundary condition $\eta \in \widehat{G}$, it follows from the $\widehat{G}$ -symmetry that a Hamiltonian $\mathbb{H}^{\psi,\eta}$ decomposes into charge sectors labelled by group elements

$g \in G$ corresponding to the holonomy of $\mathfrak{g}$ around the spatial manifold. Pairs $(g, \eta) \in G \times \widehat{G}$ label the (simple) *topological sectors* of the theory that decompose the action of the algebra of operators (7) onto the microscopic Hilbert space. The projector onto the topological sector (g, η) is then given by $\sum_{\chi \in \widehat{G}} \chi^\vee(g) \cup_\eta^\chi$, where $\chi^\vee$ denotes the complex conjugate character of χ and where we are using the fact

$$\sum_{\chi \in \widehat{G}} \chi^\vee(g) \prod_{e \in E} \chi(g(e)) = \sum_{\chi \in \widehat{G}} \chi^\vee(g) \chi(\text{hol}(\mathfrak{g})) = \delta_{g, \text{hol}(\mathfrak{g})}, \quad (8)$$

with $\text{hol}(\mathfrak{g}) := \prod_{e \in E} g(e)$.

2.2 Duality operators

Given a finite Abelian group G , we constructed in sec. 2.1 families of one-dimensional quantum lattice models with $\text{Rep}(G)$ symmetry, which are parameterised by normalised representatives ψ of a cohomology class $[\psi]$ in $H^2(G, \text{U}(1))$ as well as sets of complex coefficients $\{h_n\}_n$. We claim that models constructed in this way that only differ in the choice of cohomology class $[\psi]$ are dual to one another. One way to confirm this statement would be to verify that (von Neumann) algebras of local operators generated by the Hamiltonian terms are isomorphic, which follows from the 2-cocycle condition of ψ . Another possibility is to explicitly compute the lattice operators transmuting Hamiltonians associated with distinct choices of $[\psi]$ onto one another. Without loss of generality, let us focus on the relation between Hamiltonian models $\mathbb{H}^\psi$ and $\mathbb{H}$ with and without non-trivial 2-cocycle. We know from the results of ref. [48] that operators transmuting local symmetric operators of two Hamiltonians that only differ in such a choice of 2-cocycle ψ onto one another are labelled by projective representations of G with Schur's multiplier ψ . Formally, this is the statement that these operators are encoded into the category $\text{Rep}^\psi(G)$ of ψ -projective representations of G .

Let us verify this statement explicitly. To this end, let π denote the representative of an isomorphism class of simple objects in $\text{Rep}^\psi(G)$. The operator transmuting the Hamiltonian $\mathbb{H}$ into $\mathbb{H}^\psi$ with periodic boundary conditions explicitly reads

$$\mathbb{D}^\pi := \sum_{\mathfrak{g}} \text{tr} \left(\prod_{e \in E} \pi(g(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (9)$$

where the product over edges $e \in E$ is ordered. The operator $\mathbb{D}^\pi$ acts diagonally in the computational basis. We claim that $\mathbb{D}^\pi$ transmutes $\mathbb{H}$ into $\mathbb{H}^\psi$, i.e., $\mathbb{D}^\pi \circ \mathbb{H} = \mathbb{H}^\psi \circ \mathbb{D}^\pi$. Indeed letting $v \in V$ and $\mathfrak{r}_v : V \rightarrow G$ such that $\mathfrak{r}_v(v_1) = 1$, whenever $v_1 \neq v$, it follows from the definition of a projective representation that

$$\begin{aligned} \pi((g \, d\mathfrak{r}_v)(v_1 v)) &= \pi(g(v_1 v) \mathfrak{r}_v(v)^{-1}) = \frac{1}{\psi(g(v_1 v), \mathfrak{r}_v(v)^{-1})} \pi(g(v_1 v)) \pi(\mathfrak{r}_v(v)^{-1}), \\ \pi((g \, d\mathfrak{r}_v)(v v_2)) &= \pi(\mathfrak{r}_v(v) g(v v_2)) = \frac{1}{\psi(\mathfrak{r}_v(v), g(v v_2))} \pi(\mathfrak{r}_v(v)) \pi(g(v v_2)). \end{aligned} \quad (10)$$

Bringing everything together, one obtains

$$\prod_{(v_1 v_2)} \pi((g \, d\mathfrak{r}_v)(v_1 v_2)) = \psi(g, \mathfrak{r}_v) \cdot \prod_{(v_1 v_2)} \pi(g(v_1 v_2)), \quad (11)$$

where $\psi(g, \mathfrak{r}_v)$ was defined in eq. (3). This guarantees that $\mathbb{D}^\pi \circ \mathbb{H} = \mathbb{H}^\psi \circ \mathbb{D}^\pi$, as desired. Notice how this operation only depends on ψ and not the specific choice of simple object in $\text{Rep}^\psi(G)$.

Before proceeding further, let us introduce a tensor network representation of the duality operator (9). Given the tensor

$$\begin{array}{c} \uparrow \\ \leftarrow \boxed{\pi} \rightarrow \\ \downarrow \end{array} \equiv \sum_{g_1, g_2 \in G} \sum_{i_1, i_2=1}^{\dim_{\mathbb{C}} \pi} i_1 \leftarrow \boxed{\pi} \begin{array}{c} \uparrow g_2 \\ \leftarrow i_2 \\ \downarrow g_1 \end{array} |g_2\rangle\langle g_1| \otimes |i_1\rangle\langle i_2| \equiv \sum_{\substack{g \in G \\ i_1, i_2}} \pi(g)_{i_1 i_2} |g\rangle\langle g| \otimes |i_1\rangle\langle i_2|, \quad (12)$$

one can realise the duality operator $\mathbb{D}^\pi$ with periodic boundary conditions as a tensor network of the form

$$\mathbb{D}^\pi \equiv \begin{array}{c} \uparrow \\ \leftarrow \boxed{\pi} \rightarrow \leftarrow \boxed{\pi} \rightarrow \leftarrow \boxed{\pi} \rightarrow \\ \downarrow \end{array}, \quad (13)$$

where opposite indices in the horizontal direction are contracted so as to implement periodic boundary conditions. Restricting to the charge sector labelled by $g \in G$, it follows from the defining property of the projective representation π that the duality operator explicitly depends on the complex number $\text{tr } \pi(g) = \text{tr } \pi(\text{hol}(g))$. Indeed, repeatedly using the defining property of ψ -projective representation π one finds

$$\mathbb{D}^\pi = \sum_g \psi(g(v_1 v_2), g(v_2 v_3)) \psi(g(v_1 v_2), g(v_2 v_3), g(v_3 v_4)) \cdots \psi(g(v_1 v_2), g(v_2 v_3) \cdots g(v_{|V|-1} v_{|V|}), g(v_{|V|} v_1)) \text{tr } \pi(\text{hol}(g)) |g\rangle\langle g|, \quad (14)$$

where $v_1 < v_2 < \cdots < v_{|V|}$. Graphically, this follows from

$$\begin{array}{c} \uparrow \\ \leftarrow \boxed{\pi} \rightarrow \leftarrow \boxed{\pi} \rightarrow \\ \downarrow \end{array} = \sum_{g_1, g_2 \in G} \begin{array}{c} g_1 \quad g_2 \\ \searrow \quad \swarrow \\ \boxed{\psi} \\ \swarrow \quad \searrow \\ g_1 \quad g_2 \end{array}, \quad (15)$$

where we introduced the tensors

$$\begin{array}{c} \searrow \quad \swarrow \\ \boxed{\psi} \\ \swarrow \quad \searrow \end{array} \equiv \sum_{g_1, g_2, g_3 \in G} \begin{array}{c} g_1 \quad g_2 \\ \searrow \quad \swarrow \\ \boxed{\psi} \\ \swarrow \quad \searrow \\ g_3 \end{array} |g_1, g_2\rangle\langle g_3| \equiv \sum_{g_1, g_2 \in G} \psi(g_1, g_2) |g_1, g_2\rangle\langle g_1 g_2|, \quad (16)$$

$$\begin{array}{c} \uparrow \\ \searrow \quad \swarrow \\ \boxed{\psi} \\ \swarrow \quad \searrow \end{array} \equiv \sum_{g_1, g_2, g_3 \in G} \begin{array}{c} g_3 \\ \swarrow \quad \searrow \\ \boxed{\psi} \\ \swarrow \quad \searrow \\ g_1 \quad g_2 \end{array} |g_3\rangle\langle g_1, g_2| \equiv \sum_{g_1, g_2 \in G} \frac{1}{\psi(g_1, g_2)} |g_1 g_2\rangle\langle g_1, g_2|,$$

for any normalised representative ψ of $[\psi] \in H^2(G, \text{U}(1))$. Besides, we can intuitively infer from this graphical identity that duality operators transmuting Hamiltonians $\mathbb{H}^{\psi'}$ into $\mathbb{H}^\psi$ would be labelled by simple objects in $\text{Rep}^{\psi/\psi'}(G)$.

The duality operator acting diagonally in the computational basis, any charge sector $g \in G$ is preserved as we transmute $\mathbb{H}$ into $\mathbb{H}^\psi$. But, notice that

$$\text{tr } \pi(g) = \text{tr } \pi(g x x^{-1}) = \frac{\psi(g, x)}{\psi(x, g)} \text{tr } \pi(g), \quad (17)$$

for all $x \in G$. This implies that the duality operator forces both $\mathbb{H}$ and $\mathbb{H}^\psi$ onto charge sectors $g \in G$ for which $\frac{\psi(g, -)}{\psi(-, g)} = 1$. This follows from the fact that we are implicitly assuming periodic

boundary conditions for both models. Accessing other charge sectors requires modifying the duality operator, which in turn forces one of the models to have a different symmetry-twisted boundary condition. This is the statement that topological sectors are not mapped identically under the duality. We compute the corresponding permutation of topological sectors in the following.

2.3 Permutation of topological sectors and non-invertibility

In order to lift the duality operator $\mathbb{D}^\pi$ to a unitary transformation, we need to characterise the interplay between symmetry-twisted closed boundary conditions and charge sectors. First, we need to modify the duality operator so as to accommodate the presence of a symmetry twist. Let π be a (unitary) irreducible ψ -projective representation and $\chi \in \widehat{G}$. Naturally, the tensor product $\chi \otimes \pi$ is isomorphic to a ψ -projective representation $\tilde{\pi}$ of the same dimension as π via an intertwining unitary map

$$\begin{bmatrix} \chi & \pi \\ 1 & - \end{bmatrix} \in \text{Hom}_{\text{Rep}^\psi(G)}(\chi \otimes \pi, \tilde{\pi}), \quad (18)$$

satisfying

$$(\chi \otimes \pi)(g)_{i_1 i_2} = \sum_{i_3, i_4} \begin{bmatrix} \chi & \pi \\ 1 & - \end{bmatrix}^*_{i_3 i_4} \tilde{\pi}(g)_{i_3 i_4} \begin{bmatrix} \chi & \pi \\ 1 & - \end{bmatrix}_{i_2 i_4}. \quad (19)$$

A similar definition holds for $\pi \otimes \chi$. From these intertwining maps, we construct the following tensor

$$\begin{aligned} & \begin{array}{c} \eta_2 \\ \updownarrow \\ \text{---} \bigcirc \text{---} \\ \updownarrow \\ \eta_1 \end{array} \equiv \sum_{i_1, i_2} i_1 \begin{array}{c} \eta_2 \\ \updownarrow \\ \text{---} \bigcirc \text{---} \\ \updownarrow \\ \eta_1 \end{array} i_2 |\eta_2\rangle\langle\eta_1| \otimes |i_1\rangle\langle i_2| \\ & \equiv \sum_{i_1, i_2, i_3} \begin{bmatrix} \pi & \eta_2 \\ i_1 & 1 \end{bmatrix} \begin{bmatrix} \tilde{\pi}^* \\ i_3 \end{bmatrix} \begin{bmatrix} \eta_1 & \pi \\ 1 & i_2 \end{bmatrix} \begin{bmatrix} \tilde{\pi} \\ i_3 \end{bmatrix} |\eta_2\rangle\langle\eta_1| \otimes |i_1\rangle\langle i_2|, \end{aligned} \quad (20)$$

for any $\eta_1, \eta_2 \in \widehat{G}$, out of which we construct the duality operator $\mathbb{D}^\pi_{\eta_1 \rightarrow \eta_2} : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$ depicted as

$$\mathbb{D}^\pi_{\eta_1 \rightarrow \eta_2} \equiv \begin{array}{c} \eta_2 \\ \updownarrow \\ \boxed{x} \end{array} \begin{array}{c} \eta_2 \\ \updownarrow \\ \text{---} \bigcirc \text{---} \\ \updownarrow \\ \eta_1 \end{array} \begin{array}{c} \eta_2 \\ \updownarrow \\ \boxed{x} \end{array} \begin{array}{c} \eta_2 \\ \updownarrow \\ \boxed{x} \end{array}. \quad (21)$$

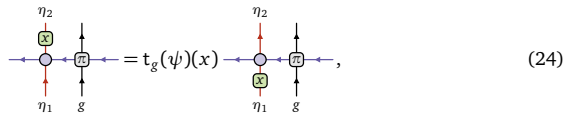
By definition of the map (18) implementing an isomorphism of projective representations, the rank 4 tensor (20) satisfies the following invariance property:

$$\begin{array}{c} \eta_2 \\ \updownarrow \\ \boxed{x} \end{array} \begin{array}{c} \eta_2 \\ \updownarrow \\ \text{---} \bigcirc \text{---} \\ \updownarrow \\ \eta_1 \end{array} = \begin{array}{c} \eta_2 \\ \updownarrow \\ \text{---} \bigcirc \text{---} \\ \updownarrow \\ \eta_1 \end{array} \begin{array}{c} \eta_2 \\ \updownarrow \\ \boxed{x} \end{array}, \quad (22)$$

for all $x \in G$, where we introduced the matrices

$$\begin{array}{c} \eta \\ \updownarrow \\ \boxed{x} \end{array} \equiv \sum_{\eta \in \widehat{G}} \begin{array}{c} \eta \\ \updownarrow \\ \boxed{x} \end{array} |\eta\rangle\langle\eta| \equiv \sum_{\eta \in \widehat{G}} \eta(x) |\eta\rangle\langle\eta|, \quad \begin{array}{c} i_1 \\ \updownarrow \\ \boxed{x} \end{array} \equiv \sum_{i_1, i_2=1}^{\dim_\mathbb{C} \pi} \begin{array}{c} i_1 \\ \updownarrow \\ \boxed{x} \end{array} |i_1\rangle\langle i_2| \equiv \sum_{i_1, i_2=1}^{\dim_\mathbb{C} \pi} \pi(x)_{i_1 i_2} |i_1\rangle\langle i_2|. \quad (23)$$

It follows from eq. (10) in combination with the invariance property (22) that $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi \circ \mathbb{H}^{1, \eta_1} = \mathbb{H}^{\psi, \eta_2} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi$. As suggested above, such a duality operator mapping boundary condition $\eta_1 \in \widehat{G}$ onto $\eta_2 \in \widehat{G}$ puts some constraints on the charge sectors allowed, which must be the same on both side of the duality. Conversely, a choice of boundary condition $\eta_1 \in \widehat{G}$ and charge sector $g \in G$, fully specifies the boundary condition $\eta_2 \in \widehat{G}$. Concretely, invoking eq. (14), one can consider the duality operator effectively acting on a single site whose degree of freedom is labelled by the charge sector g , which combined with the invariance property (22) and $\pi(x)\pi(g) = \frac{\psi(x, g)}{\psi(g, x)}\pi(g)\pi(x)$ results for every $x \in G$ in



$$\text{Diagram} = t_g(\psi)(x), \quad (24)$$

where in the spirit of eq. (15), the diagrams should here be interpreted as tensor networks whose opposite indices in the horizontal direction are contracted so as to impose periodic boundary conditions. Here, we have made use of $t_g : Z^2(G, \text{U}(1)) \rightarrow Z^1(G, \text{U}(1))$ defined according to

$$t_g(\psi)(x) := \frac{\psi(x, g)}{\psi(g, x)}, \quad (25)$$

for all $x, g \in G$ [76]. Since the previous derivation holds for any $x \in G$, we find that the boundary condition η_2 is related to η_1 via

$$\eta_2 \stackrel{!}{=} t_g(\psi) \eta_1. \quad (26)$$

Putting everything together, upon our duality operation, the group of topological sectors undergo the following automorphism in $\text{Aut}(G \times \widehat{G})$ [60]:

$$(g, \eta) \mapsto (g, \eta_g^\psi), \quad \text{with} \quad \eta_g^\psi := t_g(\psi) \eta. \quad (27)$$

In particular, it follows from the normalisation of ψ that in the singlet sector $g = \mathbb{1}$ the boundary condition is always preserved.

Before considering some examples, let us comment on the *non-invertibility* of the duality operations studied in this section. We already noticed that fixing periodic boundary conditions on both sides of the duality, the operator $\mathbb{D}_{1 \rightarrow 1}^\pi$ projects onto certain charge sectors, and as such it is non-invertible. Nevertheless, restricting to such a sector, acting with the corresponding projector on $\mathbb{D}^\pi \equiv \mathbb{D}_{1 \rightarrow 1}^\pi$ produces the unitary map relating the spectra of the dual Hamiltonians. More generally, given compatible sectors as per eq. (27), projecting the duality operator performing the mapping $\eta \mapsto \eta_g^\psi$ onto the charge sector $g \in G$ produces a unitary map. More specifically about the non-invertibility, instead of transmuting $\mathbb{H}$ into $\mathbb{H}^\psi$, we can transmute $\mathbb{H}^\psi$ into $\mathbb{H}$ via a duality operator labelled by a projective representation in $\text{Rep}^{1/\psi}(G)$. Given a projective representation π with Schur's multiplier ψ , its contragredient representation $\pi^\vee$ is a projective representation with Schur's multiplier $1/\psi$. Indeed, $\pi^\vee$ is defined as $\pi^\vee(g) := ((\pi(g)^{-1})^\vee : V^\vee \rightarrow V^\vee)$, where $\pi(g)^\vee$ is the transpose of $\pi(g)$ and $V^\vee := \text{Hom}(V, \mathbb{C})$. It follows from $\pi(g)^{-1} = \frac{1}{\psi(g, g^{-1})}\pi(g^{-1})$ that

$$\pi^\vee(g_1)\pi^\vee(g_2) = \frac{1}{\psi(g_1, g_2)}\pi^\vee(g_1 g_2), \quad (28)$$

for all $g_1, g_2 \in G$, as expected. In particular, we have that $\pi^\vee \otimes \pi$ is a linear representation of G , where $\otimes$ is the usual tensor product, and thus labels a symmetry operator of $\mathbb{H}$.

Defining duality operators $\mathbb{D}_{\eta_2 \rightarrow \eta_1}^{\pi^\vee}$ in the same vein as $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi$, it is clear for instance that $\mathbb{D}_{\eta_2 \rightarrow \eta_1}^{\pi^\vee} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^\pi = \mathbb{D}_{\eta_1 \rightarrow \eta_1}^{\pi^\vee \otimes \pi}$ does not act as the trivial symmetry operator in general. Nevertheless, it is possible to find a combination of duality operators that boils down to the trivial symmetry operator. For every $\eta \in \widehat{G}$ such that $\pi \otimes \eta \cong \pi$, consider for instance duality operators $\mathbb{D}_{1 \rightarrow \eta}^\pi$. But,

$$\dim_{\mathbb{C}} \text{Hom}_{\text{Rep}^\psi(G)}(\pi \otimes \eta, \pi) = \dim_{\mathbb{C}} \text{Hom}_{\text{Rep}(G)}(\eta, \pi^\vee \otimes \pi) \leq 1, \quad (29)$$

for every $\eta \in \widehat{G}$. In particular, characters in $\widehat{G}^\pi := \{\eta \in \widehat{G} \mid \dim_{\mathbb{C}} \text{Hom}_{\text{Rep}(G)}(\eta, \pi^\vee \otimes \pi) = 1\}$ form a subgroup of $\widehat{G}$ such that $|\widehat{G}^\pi| = (\dim_{\mathbb{C}} \pi)^2$. It follows from the unitarity of the intertwinning map

$$\left[\begin{array}{cc|c} \pi^\vee & \pi & - \\ - & - & 1 \end{array} \right] \in \text{Hom}_{\text{Rep}(G)} \left(\pi^\vee \otimes \pi, \bigoplus_{\eta \in \widehat{G}^\pi} \eta \right), \quad (30)$$

that the operator $\sum_{\eta \in \widehat{G}^\pi} \mathbb{D}_{\eta \rightarrow 1}^{\pi^\vee} \circ \mathbb{D}_{1 \rightarrow \eta}^\pi$ does act like the trivial symmetry operator on the microscopic Hilbert space associated with periodic boundary conditions.

2.4 Examples

Let us illustrate our formalism by specialising to the group $G = \mathbb{Z}_p \times \mathbb{Z}_p$, with p prime. We begin by introducing some notations: Throughout this part, we denote group elements of $\mathbb{Z}_p \times \mathbb{Z}_p$ by $g \equiv (g^{(1)}, g^{(2)})$ where $g^{(1)}, g^{(2)} \in \{0, 1, \dots, p-1\}$ and we define the multiplication rule of $\mathbb{Z}_p$ as addition modulo p . Introducing the primitive p^{th} root of unity $\omega^{(p)} := \exp(\frac{2\pi i}{p})$, a normalised representative 2-cocycle ψ generating the second cohomology group $H^2(\mathbb{Z}_p \times \mathbb{Z}_p, \text{U}(1)) \cong \mathbb{Z}_p$ is given by²

$$\psi(g_1, g_2) := (\omega^{(p)})^{g_1^{(2)} g_2^{(1)}}. \quad (31)$$

It turns out that for any cohomology class in $H^2(\mathbb{Z}_p \times \mathbb{Z}_p, \text{U}(1))$, there is exactly one irreducible projective representation, up to equivalence, which is of dimension p [17]. In other words, the category $\text{Rep}^\psi(\mathbb{Z}_p \times \mathbb{Z}_p)$ admits one isomorphism class of simple objects. Given the generalised Pauli matrices

$$X^{(p)} := \sum_{g=0}^{p-1} |g+1\rangle\langle g|, \quad \text{and} \quad Z^{(p)} := \sum_{g=0}^{p-1} (\omega^{(p)})^g |g\rangle\langle g|, \quad (32)$$

which satisfy the commutation relation $Z^{(p)} X^{(p)} = \omega^{(p)} X^{(p)} Z^{(p)}$, the projective representation π_k associated with the cocycle $\psi^k(g_1, g_2)$ with $k = 0, \dots, p-1$ explicitly reads

$$\pi_k(g) := (X^{(p)})^{g^{(1)}} (Z^{(p)})^{k g^{(2)}}, \quad (33)$$

for all $g \equiv (g^{(1)}, g^{(2)}) \in \mathbb{Z}_p \times \mathbb{Z}_p$. Indeed, one can readily check that

$$\pi_k(g_1) \pi_k(g_2) = (\omega^{(p)})^{k g_1^{(2)} g_2^{(1)}} \pi_k(g_1 + g_2).$$

Elements of the character group of $\mathbb{Z}_p \times \mathbb{Z}_p$ are denoted by $\chi_l \equiv \chi_{(l^{(1)}, l^{(2)})}$ with $l^{(1)}, l^{(2)} \in \{0, 1, \dots, p-1\}$ and are defined via

$$\chi_l : g \mapsto (\omega^{(p)})^{g^{(1)} l^{(1)} + g^{(2)} l^{(2)}}. \quad (34)$$

²In defining $\psi(g_1, g_2)$, we are implicitly promoting $\mathbb{Z}_p$ to the corresponding Galois field $\mathbb{F}_p$.

The tensor product of two characters simply reads

$$\chi_{l_1} \otimes \chi_{l_2} \cong \chi_{(l_1^{(1)}+l_2^{(1)}, l_1^{(2)}+l_2^{(2)})}, \quad (35)$$

where the sums are again modulo p . Finally, bases of intertwiners realising the isomorphisms $\chi_l \otimes \pi_k \xrightarrow{\sim} \pi_k$ and $\pi_k \otimes \chi_l \xrightarrow{\sim} \pi_k$ are provided by coefficients

$$\begin{bmatrix} \pi_k & \chi_l & \pi_k \\ i_1 & 1 & i_2 \end{bmatrix} = \begin{bmatrix} \chi_l & \pi_k \\ 1 & i_1 \end{bmatrix} \begin{bmatrix} \pi_k \\ i_2 \end{bmatrix} = \left[(X^{(p)})^m (Z^{(p)})^{-l^{(1)}} \right]_{i_2 i_1}, \quad (36)$$

for all $k, l^{(1)}, l^{(2)} \in \{0, \dots, p-1\}$ and $i_1, i_2 \in \{1, \dots, p\}$, where m is a solution of $mk = l^{(2)} \pmod{p}$, which is guaranteed to exist since p is prime. Let us confirm the mapping (27) of topological sectors using the explicit formula (36) for the intertwiners. For concreteness, let us assume periodic boundary conditions for the Hamiltonian $\mathbb{H}$. Given the charge sector $g \in \mathbb{Z}_p \times \mathbb{Z}_p$, let us compute the dual symmetry-twisted boundary condition for the Hamiltonian $\mathbb{H}^{\psi, k}$. We find that the duality operator explicitly depends on the complex number

$$\text{tr} \left[(Z^{(p)})^{l^{(1)}} (X^{(p)})^{-m} (X^{(p)})^{g^{(1)}} (Z^{(p)})^{kg^{(2)}} \right] = \text{tr} \left[(X^{(p)})^{g^{(1)}-m} (Z^{(p)})^{kg^{(2)}+l^{(1)}} \right], \quad (37)$$

where $mk = l^{(2)} \pmod{p}$. But, since p is prime, this factor is non-vanishing if and only if $m = g^{(1)}$ and $l^{(1)} = -kg^{(2)} \pmod{p}$. This indicates that the boundary condition of the dual theory must be provided by the character

$$g \mapsto \left(\omega^{(p)} \right)^{-g^{(1)}kg^{(2)}+g^{(2)}g^{(1)}k}, \quad (38)$$

which indeed coincides with $\psi^k(-, g)/\psi^k(g, -)$.

It is instructive to further specialise to the case $p = 2$. Let us consider a representative of the gapped phase where the $\widehat{\mathbb{Z}}_2 \times \widehat{\mathbb{Z}}_2$ is spontaneously broken in the ground state subspace for periodic boundary conditions. To make contact with the literature, we think of the edges of the lattice as the sites of a chain of length $|\mathbb{E}|$, which we enumerate $i = 1, \dots, |\mathbb{E}|$. The microscopic Hilbert space is then given by $\bigotimes_{i=1}^{|\mathbb{E}|} \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$ and the Hamiltonian encodes a pure ferromagnet:

$$\mathbb{H} = - \sum_{i=1}^{|\mathbb{E}|} \left[(X\mathbb{I})_i (X\mathbb{I})_{i+1} + (\mathbb{I}X)_i (\mathbb{I}X)_{i+1} \right], \quad (39)$$

where $X \equiv X^{(2)}$ is the usual Pauli matrix. In the notation of eq. (1), this is the Hamiltonian $\mathbb{H} = \sum_v \mathbb{h}_{v,1}$ obtained by choosing non-vanishing coefficients $h_1(g, \mathbf{r}_v) = -1$ for any $g : \mathbb{E} \rightarrow G$ and $\mathbf{r}_v(v) \equiv (1, 0)$ or $(0, 1)$. Replacing the implicit trivial 2-cocycle in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$ by the normalised representative $\psi : (g_1, g_2) \mapsto (-1)^{g_1^{(2)}g_2^{(1)}}$ in the unique non-trivial class yields the dual model

$$\mathbb{H}^\psi = - \sum_{i=1}^{|\mathbb{E}|} \left[(XZ)_i (X\mathbb{I})_{i+1} + (\mathbb{I}X)_i (ZX)_{i+1} \right], \quad (40)$$

which we recognise as the (1+1)d *cluster state* model [12]. This model is the stable renormalisation group fixed point of the non-trivial $\widehat{\mathbb{Z}}_2 \times \widehat{\mathbb{Z}}_2$ symmetry protected topological (SPT) phase. Within this context, the duality operator $\mathbb{D}^\pi$ implements the celebrated *Kennedy–Tasaki* duality transformation, which was first introduced in [45] as a means to elucidate the Haldane phase of the antiferromagnetic spin-1 Heisenberg XXZ model.³ We count four charge sectors

³The fact that the duality as formulated in this manuscript coincides with the Kennedy–Tasaki duality in the context of the spin-1 Heisenberg XXZ model was demonstrated in ref. [48].

and four boundary conditions that amount to imposing periodic/antiperiodic boundary conditions with respect to each copy of $\mathbb{Z}_2$. The mapping of the 16 topological sectors given by eq. 27 coincides with that obtained in ref. [54].⁴

As another application of our formalism, let us determine the fate of the $\widehat{\mathbb{Z}_2} \times \widehat{\mathbb{Z}_2}$ ferromagnetic order parameters under the duality. The symmetric ferromagnetic order parameter acting on the first $\mathbb{Z}_2$ component on sites $i < j$ away from the boundary condition reads $(X\mathbb{I})_i(X\mathbb{I})_j$. The duality operator $\mathbb{D}^\pi$ transmutes this local order parameter into the string order parameter $(XZ)_i(\prod_{k=i+1}^{j-1}(\mathbb{I}Z)_k)(X\mathbb{I})_j$, where the product is over all sites between i and j . One proceeds similarly for the second $\mathbb{Z}_2$ component. Graphically, the mapping simply follows from

$$\begin{array}{c} \uparrow \\ \textcircled{x} \textcircled{g} \\ \downarrow \\ g \end{array} = \frac{\psi(g, x)}{\psi(x, g)} \begin{array}{c} \uparrow \\ \textcircled{g} \textcircled{x} \\ \downarrow \\ g \end{array}, \quad (41)$$

as well as the symmetry

$$\begin{array}{c} \uparrow \\ \textcircled{g} \textcircled{x} \\ \downarrow \\ g \end{array} = \psi(g, x) \begin{array}{c} \uparrow \\ \textcircled{x} \textcircled{g} \\ \downarrow \\ g \end{array} = \psi(x, g) \begin{array}{c} \uparrow \\ \textcircled{g} \textcircled{x} \\ \downarrow \\ g \end{array}, \quad \text{with} \quad \begin{array}{c} \uparrow \\ \textcircled{x} \\ \downarrow \end{array} \equiv \sum_{g \in \mathbb{Z}_2} |g\rangle\langle g+x|, \quad (42)$$

for any $x \in \mathbb{Z}_2 \times \mathbb{Z}_2$.

2.5 Category theoretic underpinnings

In preparation for the following section, let us delve a little bit into the mathematical formalism underlying the results presented above, following the approach of ref. [48–50]. Consider a one-dimensional quantum lattice model with a non-anomalous Abelian symmetry group G . In modern parlance, the model is said to admit topological lines valued in G [40]. The requirement that we should be able to construct junctions (in spacetime) of such topological lines invites us to promote the group G to a higher mathematical structure [14], namely the *fusion category* Vec_G of G -graded vector spaces. In particular, simple objects in Vec_G are one-dimensional vector spaces $\mathbb{C}_g$, for every $g \in G$, such that $(\mathbb{C}_{g_1})_{g_2} = \delta_{g_1, g_2} \mathbb{C}$, $\text{Hom}_{\text{Vec}_G}(\mathbb{C}_{g_1}, \mathbb{C}_{g_2}) \cong \delta_{g_1, g_2} \mathbb{C}$ and $\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2} = \mathbb{C}_{g_1 g_2}$. Moreover, the *monoidal associator* evaluates to the identity so that $1_{\mathbb{C}_{g_1 g_2 g_3}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \otimes \mathbb{C}_{g_3} \xrightarrow{\sim} \mathbb{C}_{g_1} \otimes (\mathbb{C}_{g_2} \otimes \mathbb{C}_{g_3})$. Given a non-anomalous symmetry G —or rather, a symmetry Vec_G —it is always possible to gauge a subsymmetry A , where A is a subgroup of G . There are several ways to perform such a gauging, classified by the second cohomology group $H^2(A, \text{U}(1))$. Whenever the whole symmetry is gauged, the resulting theory possesses a $\text{Rep}(G)$ -symmetry, i.e., a symmetry whose operators are labelled by representations of the group G and compose according to the tensor product of representations. Since the group G is assumed to be Abelian, we further have the equivalence $\text{Rep}(G) \simeq \text{Vec}_{\widehat{G}}$.

Mathematically, a pair (A, ψ) consisting of a subgroup $A \subseteq G$ and a normalised representative ψ of a cohomology $[\psi] \in H^2(A, \text{U}(1))$ specifies a so-called (finite semisimple indecomposable) *module category* over Vec_G [64]. Let $\mathcal{M}(A, \psi)$ be the category whose simple objects are

⁴Note that we can add to the Hamiltonian $\mathbb{H}$ symmetric terms of the form $-\sum_i (Z\mathbb{I})_i$. The resulting model would describe a symmetry-breaking ferromagnetic interaction supported on the second copy of $\mathbb{Z}_2$ and a decoupled critical transverse-field Ising chain supported on the first copy of $\mathbb{Z}_2$. The corresponding dual Hamiltonian would then be given by $\mathbb{H}^\vee - \sum_i (Z\mathbb{I})_i$. An effective model describing the low-energy physics of this Hamiltonian was first examined in ref. [69] and was termed a *gapless SPT phase*. In ref. [55], this and other gapless SPT phases were constructed by making use of the Kennedy–Tasaki duality. It is plausible that our approach would facilitate the construction of other examples of gapless SPT phases as well as the study of their properties.

one-dimensional vector spaces $\mathbb{C}_{rA}$, for every $rA \in G/A$, equipped with the (left) Vec_G -module structure provided by the G -action of left cosets in G/A , i.e. $\mathbb{C}_g \triangleright \mathbb{C}_{rA} := \mathbb{C}_{g \triangleright rA} = \mathbb{C}_{(gr)A}$, and module associator $\alpha_\psi^\triangleright$ specified by isomorphisms

$$\alpha_\psi^\triangleright(g_1, g_2)(rA) \cdot 1_{\mathbb{C}_{(g_1 g_2) \triangleright rA}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \triangleright \mathbb{C}_{rA} \xrightarrow{\sim} \mathbb{C}_{g_1} \triangleright (\mathbb{C}_{g_2} \triangleright \mathbb{C}_{rA}), \quad (43)$$

for every $g_1, g_2 \in G$ and $rA \in G/A$. The 2-cochain $\alpha_\psi^\triangleright \in C^2(G, \text{Fun}(G/A, U(1)))$ is defined in terms of the 2-cocycle ψ via

$$\alpha_\psi^\triangleright(g_1, g_2)(rA) := \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA}), \quad (44)$$

where given $g \in G$ and $rA, sA \in G/A$ such that $g \triangleright rA = sA$, $a_{g, rA}$ is the unique group element in A satisfying $g = s a_{g, rA} r^{-1}$. It follows from the 2-cocycle condition satisfied by ψ and the property $a_{g_1, g_2 \triangleright rA} a_{g_2, rA} = a_{g_1 g_2, rA}$, for every $g_1, g_2 \in G$ and $rA \in G/A$, that

$$\frac{\alpha_\psi^\triangleright(g_2, g_3)(rA) \alpha_\psi^\triangleright(g_1, g_2 g_3)(rA)}{\alpha_\psi^\triangleright(g_1 g_2, g_3)(rA) \alpha_\psi^\triangleright(g_1, g_2)(g_3 \triangleright rA)} = 1, \quad (45)$$

for every $g_1, g_2, g_3 \in G$ and $rA \in G/A$. Identity (45) guarantees the so-called *pentagon axiom* of $\mathcal{M}(A, \psi)$.

Performing the ψ -twisted gauging of the symmetry G amounts to picking the Vec_G -module category $\mathcal{M}(G, \psi) \equiv \text{Vec}^\psi$, which is equivalent to Vec as a category but whose Vec_G -module structure depends on ψ . The symmetry structure of the theory resulting from the ψ -twisted gauging of the subsymmetry A is provided by the fusion category $(\text{Vec}_G)_{\mathcal{M}(A, \psi)}^\vee := \text{Fun}_{\text{Vec}_G}(\mathcal{M}(A, \psi), \mathcal{M}(A, \psi))$ defined as the category of Vec_G -module functors from $\mathcal{M}(A, \psi)$ to itself, with fusion structure provided by the composition of Vec_G -module functors [31]. In particular, we have $(\text{Vec}_G)_{\text{Vec}^\psi}^\vee \simeq \text{Rep}(G)$, for any $[\psi] \in H^2(G, U(1))$, as expected.

We can think of the $\widehat{G}$ -symmetric models defined in sec. 2.1 as resulting from the ψ -twisted gauging of G -symmetric models. Therefore, two models with Hamiltonians $\mathbb{H}$ and $\mathbb{H}^\psi$ —only differing in a choice of 2-cocycle ψ —are associated with the Vec_G -module categories Vec and Vec^ψ , respectively. This can be made very explicit employing the formalism of ref. [48, 49], from which it also follows that a duality operator transmuting $\mathbb{H}$ into $\mathbb{H}^\psi$ is associated with a simple object in $\text{Fun}_{\text{Vec}_G}(\text{Vec}, \text{Vec}^\psi) \simeq \text{Rep}^\psi(G)$, as expected. For future reference, it is useful to observe that the category $\text{Rep}^\psi(G)$ of ψ -projective representations of G is equivalent to the category $\text{Mod}(\mathbb{C}[G]^\psi)$ of modules over the ψ -twisted group algebra $\mathbb{C}[G]^\psi$. Explicitly, $\mathbb{C}[G]^\psi$ is the algebra with underlying vector space $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in G\}$ and multiplication $|g_1\rangle \cdot |g_2\rangle = \psi(g_1, g_2) |g_1 g_2\rangle$, for every $g_1, g_2 \in G$. In this context, topological sectors of a $\widehat{G}$ -symmetric model correspond to simple objects in the so-called *Drinfel'd center* $\mathcal{Z}(\text{Vec}_{\widehat{G}})$ of the symmetry fusion category $\text{Vec}_{\widehat{G}}$ [49, 53], which are indeed labelled by group elements in $G \times \widehat{G}$.

2.6 Self-duality

As announced in the introduction, an appealing feature of the dualities we consider is that they strictly preserve the symmetry structure. Therefore, it is sensible to ask for (1+1)d models that would be left invariant under such duality transformations. Supposing we have such a model, one can further ask to promote the self-duality to a genuine internal symmetry, which amounts to performing a group extension of the symmetry fusion category.

Let us illustrate this procedure for the case $G = \mathbb{Z}_2 \times \mathbb{Z}_2$. The Hamiltonian $\mathbb{H} + \mathbb{H}^\psi$ constructed from the Hamiltonians (39) and (40), respectively, is left invariant under the action

of the duality operator (9). Out of the possible extensions of $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$ by $\mathbb{Z}_2$ computed in ref. [73], promoting the self-duality operator to a symmetry operator results in the fusion category $\text{Rep}(\mathbb{D}_8)$ of representations of the dihedral group $\mathbb{D}_8$ of order eight [72], which, as a category, is equivalent to $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2) \boxplus \text{Rep}^\psi(\mathbb{Z}_2 \times \mathbb{Z}_2)$. One could verify this fact by computing tensors implementing junctions between the symmetry operators in $\text{Rep}(\mathbb{Z}_2 \times \mathbb{Z}_2)$ and the self-duality operator, and show that these correspond to morphisms in $\text{Rep}(\mathbb{D}_8)$. Instead, we will show that the Hamiltonian $\mathbb{H} + \mathbb{H}^\psi$ can be constructed from local operators that are manifestly $\text{Rep}(\mathbb{D}_8)$ symmetric.

We begin by noticing that the dihedral group $\mathbb{D}_8$ with presentation $\langle r, s \mid r^4 = s^2 = (rs)^2 = \mathbb{1} \rangle$ can be realised as the *central extension*

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{D}_8 \cong \mathbb{Z}_2 \times_\phi (\mathbb{Z}_2 \times \mathbb{Z}_2) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow 1, \quad (46)$$

specified by the cohomology class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_2) \cong \mathbb{Z}_2^3$ with normalised representative ϕ such that $\phi(g_1, g_2) = g_1^{(2)} + g_2^{(1)} \in \mathbb{Z}_2$ for every $g_1, g_2 \in \mathbb{Z}_2 \times \mathbb{Z}_2$. In particular, we notice the expected close similarity between the definitions of the 2-cocycles ψ and ϕ . Identifying $\mathbb{1} \equiv (0, (0, 0))$, $s \equiv (0, (0, 1))$ and $r \equiv (0, (1, 1))$, one finds that the set $\mathbb{Z}_2 \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$ equipped with the multiplication rule

$$(n_1, q_1) \cdot (n_2, q_2) := (n_1 + n_2 + \phi(q_1, q_2), q_1 + q_2), \quad (47)$$

for every $q_1, q_2 \in \mathbb{Z}_2 \times \mathbb{Z}_2$ and $n_1, n_2 \in \mathbb{Z}_2$, is indeed isomorphic to $\mathbb{D}_8$. Families of one-dimensional quantum lattice models that possess a $\text{Rep}(\mathbb{D}_8)$ symmetry can be easily constructed mimicking the definitions of sec. 2.1. Consider the microscopic Hilbert space $\bigotimes_{e \in E} \mathbb{C}[\mathbb{D}_8]$ spanned by states $|g\rangle$ where $g: E \rightarrow \mathbb{D}_8$. Defining functions $r_v: V \rightarrow \mathbb{D}_8$ as in sec. 2.1, for every $v \in V$, consider the following local operators acting on the kinematical Hilbert space:

$$\mathbb{h}_{v,n} := \sum_{g, r_v} h_n(g, r_v) |g\rangle \langle g|, \quad (48)$$

wherein n is some label, $h_n(g, r_v)$ are complex coefficients, and $g \, dr_v$ was defined in eq. (2). For any possible choice of complex coefficients $h_n(g, r_v)$, for every $g: E \rightarrow \mathbb{D}_8$ and r_v , it follows from the defining property of group representations that local operators (48) commute with symmetry operators of the form

$$\mathbb{U}^\rho := \sum_g \text{tr} \left(\prod_{e \in E} \rho(g(e)) \right) |g\rangle \langle g|, \quad (49)$$

where the product over edges $e \in E$ is ordered, for any simple object ρ in $\text{Rep}(\mathbb{D}_8)$. By virtue of $(\rho_1 \otimes \rho_2)(g) = \rho_1(g) \otimes \rho_2(g)$, for every $g \in \mathbb{D}_8$, the symmetry operators compose according to $\mathbb{U}^{\rho_1} \circ \mathbb{U}^{\rho_2} = \sum_{\rho_3 \in \rho_1 \otimes \rho_2} \mathbb{U}^{\rho_3}$. In particular, symmetry operators associated with higher-dimensional representations are non-invertible. Finally, we construct arbitrary $\text{Rep}(\mathbb{D}_8)$ -symmetric models as $\sum_{v \in V} \sum_n \mathbb{h}_{v,n}$.

Let us now reveal how the Hamiltonian $\mathbb{H} + \mathbb{H}^\psi$ is effectively constructed from local operators of the form (48). By definition, the Hamiltonian $\mathbb{H} + \mathbb{H}^\psi$, which acts on the kinematical Hilbert space $\bigotimes_i \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$, is made of local operators that act as $-(\mathbb{1}X)_i(\mathbb{1}X)_{i+1} - (\mathbb{1}X)_i(ZX)_{i+1}$ or $-(X\mathbb{1})_i(X\mathbb{1})_{i+1} - (XZ)_i(X\mathbb{1})_{i+1}$. We argue that there is a choice of complex coefficients $h_1(g, r_v)$ such that local operators of the form (48) effectively act in the same way. The crux is to restrict the microscopic Hilbert space to states $|g\rangle \equiv |(n, q)\rangle$, with $q: E \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$ and $n: E \rightarrow \mathbb{Z}_2$, such that n assigns the identity group element to every edge/site, effectively reducing the microscopic Hilbert space to $\bigotimes_i \mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$. This can be enforced via a choice of complex coefficients $h_1(g, r_v) \equiv h_1((n, q), r_v)$. One can further choose these coefficients such that the only

non-vanishing ones are such that $\mathfrak{r}_v(v) = (0, (0, 1)) \equiv s$ or $\mathfrak{r}_v(v) = (0, (1, 0)) \equiv rs$ and evaluate to -2 . Proceeding as such, it follows from $\phi((0, 1), (1, 0)) = 1$ that the local operator acting at site i is a sum of two operators, effectively acting as $-2(\mathbb{I}X)_i(\mathbb{I}X)_{i+1}$ and $-2(X\mathbb{I})_i(X\mathbb{I})_{i+1}$, respectively, while projecting out states $|g\rangle$ such that $g(i+1) \notin \{\mathbb{I}, s\}$ and $g(i) \notin \{\mathbb{I}, rs\}$, respectively. On the effective Hilbert space, this recovers the local operators constitutive of $\mathbb{H} + \mathbb{H}^\psi$. One can now confirm that the two-dimensional representation ρ of $\mathbb{D}_8$, which is such that $\rho(r) = XZ$ and $\rho(s) = Z$ reproduces the action of the self-duality operator.

3 Twisted gauging of (2+1)d lattice gauge theories

Given an arbitrary two-dimensional lattice gauge theory with an Abelian invertible 1-form symmetry, we study the non-trivial interplay between symmetry-twisted closed boundary conditions and charge sectors under the duality that amounts to the (untwisted) gauging of the 1-form symmetry, followed by the twisted gauging of the dual 0-form symmetry. Given pairs of compatible boundary conditions and charge sectors, the unitary operators relating the spectra of dual Hamiltonians are explicitly constructed in the form of tensor network operators.

3.1 Local Hamiltonians

Let us begin by constructing a family of two-dimensional lattice gauge theories on the two-torus $\mathbb{T}^2$. Let $\mathbb{T}_\Delta^2$ be a triangulation of $\mathbb{T}^2$, whose sets of vertices, edges and plaquettes are denoted by $V(\mathbb{T}_\Delta^2)$, $E(\mathbb{T}_\Delta^2)$ and $P(\mathbb{T}_\Delta^2)$, respectively. Given an edge $e \in E(\mathbb{T}_\Delta^2)$, $-e$ denotes the edge with opposite orientation. For simplicity, we shall often assume that $\mathbb{T}_\Delta^2$ is the triangular lattice, but our results typically hold more generally. We equip $\mathbb{T}_\Delta^2$ with a total ordering of its vertices, which induces a relative orientation for its edges and plaquettes. Given an oriented edge $e \in E(\mathbb{T}_\Delta^2)$, $\partial_- e$ and $\partial_+ e$ still denote its source and target vertices, respectively, and we identify $e \equiv (\partial_- e \partial_+ e)$. Given a finite Abelian group G ,⁵ we consider a differential on the spaces of simplicial G -valued forms:

$$C^0(\mathbb{T}_\Delta^2, G) \xrightarrow{d^{(0)}} C^1(\mathbb{T}_\Delta^2, G) \xrightarrow{d^{(1)}} C^2(\mathbb{T}_\Delta^2, G), \quad (50)$$

such that for any $e \in E(\mathbb{T}_\Delta^2)$ and $p \in P(\mathbb{T}_\Delta^2)$

$$(d^{(0)}\mathfrak{r})(e) := \mathfrak{r}(\partial_- e)\mathfrak{r}(\partial_+ e)^{-1}, \quad \text{and} \quad (d^{(1)}\mathfrak{g})(p) := \prod_{e \in \partial p} \mathfrak{g}(e), \quad (51)$$

where the product $\prod_{e \in \partial p}$ is over edges in the boundary of p , whose orientations are assumed to be induced by that of p . The differential is clearly nilpotent, i.e., $d \circ d = 0$, and we define the groups $Z^n(\mathbb{T}_\Delta^2, G)$ of closed G -valued forms and $B^n(\mathbb{T}_\Delta^2, G)$ of exact G -valued forms as $\ker d^{(n)}$ and $\text{im } d^{(n-1)}$, respectively.

To every edge $e \in E(\mathbb{T}_\Delta^2)$, we assign a microscopic degree of freedom valued in $\mathbb{C}[G]$. Given $g \in C^1(\mathbb{T}_\Delta^2, G)$ such that $g(-e) = g(e)^{-1}$, for every $e \in E(\mathbb{T}_\Delta^2)$, we notate via $|g\rangle$, the function g regarded as an element of the tensor product Hilbert space $\bigotimes_e \mathbb{C}[G]$. The kinematical Hilbert space of the two-dimensional lattice gauge theories on the two-torus is chosen to be the subspace of the microscopic Hilbert space $\bigotimes_e \mathbb{C}[G]$ that is spanned by vectors $|g\rangle$,

⁵Although we focus on finite Abelian groups throughout this manuscript, and occasionally use formalism specific to Abelian groups, most of our results readily generalise to the non-Abelian case, sometimes requiring little to no modification in the derivations.

where $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$. In other words, at every plaquette $p \in P(\mathbb{T}_\Delta^2)$, we impose the kinematical constraint that $\prod_{e \in \partial p} \mathfrak{g}(e) = \mathbb{1}$, promoting $\mathfrak{g}$ to a (1-form) flat gauge field. Since the group G is Abelian, we can view these kinematical constraints as Gauß constraints for a ‘magnetic’ gauge field defined on the Poincaré dual lattice, so that the resulting theory does have the interpretation of a lattice gauge theory.

Let λ be a normalised representative 3-cocycle of a cohomology class $[\lambda] \in H^3(G, \mathbb{U}(1))$. Given a vertex $v \in V(\mathbb{T}_\Delta^2)$, we denote by $\mathfrak{r}_v \in C^0(\mathbb{T}_\Delta^2, G)$ a function $V(\mathbb{T}_\Delta^2) \rightarrow G$ such that $\mathfrak{r}_v(v_1) = \mathbb{1}$, whenever $v_1 \neq v$. We then consider the following local operators acting on the kinematical Hilbert space:

$$\mathbb{H}_{v,n}^\lambda := \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \sum_{\mathfrak{r}_v} h_n(\mathfrak{g}, \mathfrak{r}_v) \lambda(\mathfrak{g}, \mathfrak{r}_v) |\mathfrak{g} d^{(0)} \mathfrak{r}_v\rangle \langle \mathfrak{g}|, \quad (52)$$

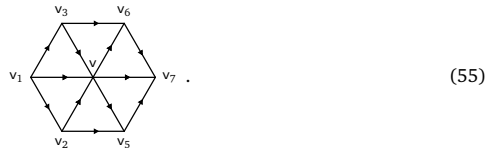
where n is some label, $h_n(\mathfrak{g}, \mathfrak{r}_v)$ are arbitrary complex coefficients, $\mathfrak{g} d^{(0)} \mathfrak{r}_v$ is by definition the function $E[\mathbb{T}_\Delta^2] \rightarrow G$ such that

$$(\mathfrak{g} d^{(0)} \mathfrak{r}_v)(e) = \begin{cases} \mathfrak{r}_v(v) \mathfrak{g}(e), & \text{if } e \equiv (v v_1), \\ \mathfrak{g}(e) \mathfrak{r}_v(v)^{-1}, & \text{if } e \equiv (v_1 v), \\ \mathfrak{g}(e), & \text{otherwise,} \end{cases} \quad (53)$$

and $\lambda(\mathfrak{g}, \mathfrak{r}_v)$ is the phase factor

$$\begin{aligned} \lambda(\mathfrak{g}, \mathfrak{r}_v) &:= \prod_{(v_1 v_2 v)} \lambda(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v))^{e(v_1 v_2 v)} \\ &\cdot \prod_{(v_1 v v_2)} \lambda(\mathfrak{g}(v_1 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v), \mathfrak{g}(v v_2))^{-e(v_1 v v_2)} \\ &\cdot \prod_{(v v_1 v_2)} \lambda(\mathfrak{r}_v(v), \mathfrak{g}(v v_1), \mathfrak{g}(v_1 v_2))^{e(v v_1 v_2)}. \end{aligned} \quad (54)$$

In the definition (54) of the phase factor $\lambda(\mathfrak{g}, \mathfrak{r}_v)$, the products are over plaquettes adjacent to v such that $v_1 < v_2 < v$, $v_1 < v < v_2$ and $v < v_1 < v_2$, respectively, while $e(p) = \pm 1$ depends on the relative orientation of the plaquette p . Notice that $\mathbb{H}_{v,n}^\lambda$ only acts non-trivially within a neighbourhood of the vertex v . Concretely, suppose $\mathbb{T}_\Delta^2$ is the triangular lattice, and choose a total ordering of the vertices such that edges of the hexagonal subcomplex centered around v are oriented as follows:



Orientations above are compatible for instance with a total ordering such that $v_1 < v_2 < v_3 < v < v_5 < v_6 < v_7$. Under these assumptions, the phase factor (54) explicitly reads

$$\lambda(\mathfrak{g}, \mathfrak{r}_v) = \frac{\lambda(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v)) \lambda(\mathfrak{g}(v_2 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v), \mathfrak{g}(v v_5)) \lambda(\mathfrak{r}_v(v), \mathfrak{g}(v v_5), \mathfrak{g}(v_5 v_7))}{\lambda(\mathfrak{g}(v_1 v_3), \mathfrak{g}(v_3 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v)) \lambda(\mathfrak{g}(v_3 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v), \mathfrak{g}(v v_6)) \lambda(\mathfrak{r}_v(v), \mathfrak{g}(v v_6), \mathfrak{g}(v_6 v_7))}. \quad (56)$$

Finally, arbitrary combinations of local operators (52) can be organised into translation invariant local Hamiltonians $\mathbb{H}^\lambda = \sum_{v \in V(\mathbb{T}_\Delta^2)} \mathbb{H}_v^\lambda \equiv \sum_{v \in V(\mathbb{T}_\Delta^2)} \sum_n \mathbb{H}_{v,n}^\lambda$. By construction, the Hamiltonian $\mathbb{H}^\lambda$ assumes periodic boundary conditions.

3.2 Symmetry operators

Given a choice of normalised representative 3-cocycle in a cohomology class $[\lambda] \in H^3(G, \mathbb{U}(1))$, consider a local Hamiltonian $\mathbb{H}^\lambda$ defined in terms of local operators (52). Regardless of the choices of complex coefficients $h_n(\mathfrak{g}, \mathfrak{r}_v)$, for every $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$ and $\mathfrak{r}_v$, and cohomology class $[\lambda]$, Hamiltonians $\mathbb{H}^\lambda$ share the same symmetry structure. First of all, the model with Hamiltonian $\mathbb{H}^\lambda$ possesses a 1-form $\widehat{G}$ symmetry—or rather, $\text{Rep}(G)$ symmetry—generated by the Wilson loop operators

$$\chi(\ell) \equiv \bigotimes_{e \in \ell} \chi(e) := \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \left(\prod_{e \in \ell} \chi(\mathfrak{g}(e)) \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|, \quad (57)$$

where $\chi \in \widehat{G}$, $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$ is any oriented (simplicial) 1-cycle of $\mathbb{T}_\Delta^2$ and the product $\prod_{e \in \ell}$ is over edges along ℓ whose orientations are assumed to be induced by that of ℓ . In particular, whenever the orientation of $e \subset \ell$ does not agree with that of ℓ , it contributes a factor $\chi(\mathfrak{g}(-e)) = \chi(\mathfrak{g}(e)^{-1}) = \chi^\vee(\mathfrak{g}(e))$ to the product. Invoking that $\chi_1(g)\chi_2(g) = (\chi_1 \otimes \chi_2)(g)$, for every $g \in G$ and $\chi_1, \chi_2 \in \widehat{G}$, the composition of operators $\chi_1(\ell) \circ \chi_2(\ell)$ equals $(\chi_1 \otimes \chi_2)(\ell)$. Clearly, it follows from the definition of the kinematical constraints that $\chi(\ell)$ is invariant under continuous deformations of ℓ , making it a topological line operator. In particular, it implies that $\chi(\ell) = 1$, whenever ℓ is a contractible 1-cycle, i.e., ℓ is homologically trivial. Whenever ℓ is a non-contractible 1-cycle, the commutation relation $[\mathbb{H}^\lambda, \chi(\ell)] = 0$ follows from the fact that

$$\prod_{e \in \ell} \chi(\mathfrak{g} d^{(0)} \mathfrak{r}_v)(e) = \prod_{e \in \ell} \chi(\mathfrak{g}(e)), \quad (58)$$

for every vertex $v \in V(\mathbb{T}_\Delta^2)$ and function $\mathfrak{r}_v \in C^0(\mathbb{T}_\Delta^2, G)$ as defined above. This is essentially the ‘gauge invariance’ of Wilson loop operators. It follows from the 1-form $\text{Rep}(G)$ symmetry that the Hamiltonian $\mathbb{H}^\lambda$ decomposes into charge sectors labelled by $(g_1, g_2) \in G \times G$ corresponding to holonomies of the flat gauge field $\mathfrak{g}$ along both non-contractible 1-cycles of the two-torus.

Importantly, we can also construct topological junctions of topological Wilson lines.⁶ Consider any number of oriented Wilson line operators meeting at any given vertex $v \in V(\mathbb{T}_\Delta^2)$. As long as the Wilson lines are chosen so that the tensor product of their corresponding representations, or their duals depending on the orientations, is isomorphic to the trivial representation, then it is a valid symmetry operator. As a matter of fact, since G is Abelian, it simply boils down to a composition of Wilson loop operators. Therefore, any network of topological Wilson lines thus constructed is a valid symmetry operator. More formally, consider the non-degenerate pairing

$$\begin{aligned} \langle -, - \rangle : \widehat{G} \times G &\longrightarrow \mathbb{U}(1), \\ (\chi, g) &\longmapsto \langle \chi, g \rangle := \chi(g). \end{aligned} \quad (59)$$

It enables us to define the following *cup product* [7]:

$$\begin{aligned} \langle -, \smile - \rangle : C^1(\mathbb{T}_\Delta^2, \widehat{G}) \times C^1(\mathbb{T}_\Delta^2, G) &\longrightarrow C^2(\mathbb{T}_\Delta^2, \mathbb{U}(1)), \\ : (\mathfrak{X}, \mathfrak{g}) &\longmapsto ((\mathfrak{X}, \smile \mathfrak{g}) : (v_1 v_2 v_3) \longmapsto \langle \mathfrak{X}(v_1 v_2), \mathfrak{g}(v_2 v_3) \rangle). \end{aligned} \quad (60)$$

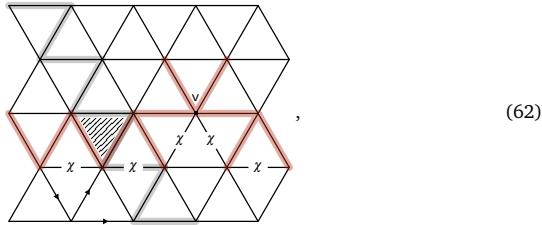
Given the differential on the spaces of simplicial $\widehat{G}$ -valued forms defined as in eq. (50), we can

⁶This ability to construct junctions of topological lines is precisely why it is more accurate to talk about a 1-form $\text{Rep}(G)$ symmetry rather than a 1-form $\widehat{G}$ symmetry, as a higher mathematical structure is required to encompass both lines and their junctions.

consider groups $Z^n(\mathbb{T}_\Delta^2, \widehat{G})$ and $B^n(\mathbb{T}_\Delta^2, \widehat{G})$. Let $\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})$ and consider the operator

$$\mathfrak{X}(\mathbb{T}_\Delta^2) := \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \int_{\mathbb{T}_\Delta^2} \langle \mathfrak{X}, \smile \mathfrak{g} \rangle |\mathfrak{g}| \equiv \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \left(\prod_{p \in P(\mathbb{T}_\Delta^2)} \langle \mathfrak{X}, \smile \mathfrak{g} \rangle(p) \right) |\mathfrak{g}| |\mathfrak{g}|. \quad (61)$$

It follows from the definition of the cup product that $\mathfrak{X}(\mathbb{T}_\Delta^2)$ precisely amounts to inserting a certain combination of Wilson loops as defined in eq. (57). Let us consider a concrete example in the case where $\mathbb{T}_\Delta^2$ is the triangular lattice. Let $\chi(\ell)$ be a Wilson loop operator with $\chi \in \widehat{G}$ and $\ell \in Z_1(\mathbb{T}_\Delta^2, Z)$. One can always associate to it a 1-cocycle $\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})$ such that the action of $\mathfrak{X}(\mathbb{T}_\Delta^2)$ coincides with that of $\chi(\ell)$. We depict such a configuration below:⁷



where the bold red lines represent the only edges to which $\mathfrak{X}$ assigns non-trivial elements in $\widehat{G}$, to wit χ or $\chi^\vee$ depending on the orientations. We also depicted in bold gray lines a choice of gauge field $\mathfrak{g}$. Finally, the hashed plaquette is the only one that contributes non-trivially to $\langle \mathfrak{X}, \smile \mathfrak{g} \rangle$. It follows from the properties of the cup product that $\langle \mathfrak{X}, \smile \mathfrak{g} \rangle$ yields the same result regardless of the choices of representatives in $[\mathfrak{g}]$ and $[\mathfrak{X}]$, respectively. In particular, picking a different representative in $[\mathfrak{X}]$ simply amounts to performing a continuous deformation of ℓ .

It was recently pointed out that the 1-form $\text{Rep}(G)$ symmetry is only a component of the total symmetry structure [4, 13, 22]. In particular, topological surface operators commuting with the Hamiltonian can also be defined. These are often referred to as *condensation operators* in the literature [56, 65]. A lattice implementation of condensation operators was first presented in ref. [27], employing the tensor network calculus developed in ref. [22]. We shall provide here particularly explicit expressions for these operators in the form of tensor networks. First of all, we define the topological surface operator $\mathbb{U}^{(1)}(\mathbb{T}_\Delta^2)$, which acts on the whole kinematical Hilbert space, by summing over all possible insertions of Wilson loops. Another way to describe this operator, is that we insert a network of Wilson line operators associated with the representation $\bigoplus_{\chi \in \widehat{G}} \chi$, in such a way that every junction implements the tensor product of representations—a process that amounts to gauging the $\text{Rep}(G)$ symmetry along the space-time submanifold $\mathbb{T}^2$ [65]. The notation $\mathbb{U}^{(1)}(\mathbb{T}_\Delta^2)$ then stems from the fact that the induced representation $\text{Ind}_{\{1\}}^G(\mathbb{O}_{\{1\}})$ of the trivial representation $\mathbb{O}_{\{1\}}$ of $\{1\} \subset G$ in G is isomorphic to the regular representation $\mathbb{C}^G \cong \bigoplus_{\chi \in \widehat{G}} \chi$. Invoking eq. (61), we can equivalently express this operator as follows:

$$\mathbb{U}^{(1)}(\mathbb{T}_\Delta^2) := \frac{|Z^0(\mathbb{T}_\Delta^2, \widehat{G})|}{|Z^1(\mathbb{T}_\Delta^2, \widehat{G})|} \sum_{\mathfrak{X} \in Z^1(\mathbb{T}_\Delta^2, \widehat{G})} \mathfrak{X}(\mathbb{T}_\Delta^2). \quad (63)$$

⁷Notice how deforming the Wilson loop so as to 'straighten' it amounts to multiplying $\mathfrak{X}$ by a 1-coboundary. Specifically, the coboundary is given by b_v wherein v stands for the vertex highlighted in eq. (62) and which is defined as $b_v(v_i) = 0$ if $v_i \neq v$, 0 being the trivial representation of G , and $b_v(v) = \chi$.

But, whenever $\mathfrak{X}$ is cohomologically trivial, $\mathfrak{X}(\mathbb{T}_\Delta^2) = 1$. Decomposing the sum over $Z^1(\mathbb{T}_\Delta^2, \widehat{G})$ into a sum over $H^1(\mathbb{T}_\Delta^2, \widehat{G}) = Z^1(\mathbb{T}_\Delta^2, \widehat{G})/B^1(\mathbb{T}_\Delta^2, \widehat{G}) \cong \widehat{G} \times \widehat{G}$ and a sum over $B^1(\mathbb{T}_\Delta^2, \widehat{G})$ yields the alternative formula

$$\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2) = \frac{1}{|\widehat{G}|} \sum_{[\mathfrak{X}] \in H^1(\mathbb{T}_\Delta^2, \widehat{G})} \mathfrak{X}(\mathbb{T}_\Delta^2), \quad (64)$$

where we used the fact that $|\widehat{G}| = |G|$, together with $|Z^0(\mathbb{T}_\Delta^2, \widehat{G})| = |\widehat{G}|$. What is the effect of such a condensation operator? It acts as a projector onto the singlet sector of the 1-form $\text{Rep}(G)$ -symmetry. Indeed, denoting by (γ_1, γ_2) a generator of $H_1(\mathbb{T}_\Delta^2, \mathbb{Z}) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, i.e. two inequivalent non-contractible 1-cycles along the one-skeleton of $\mathbb{T}_\Delta^2$, it follows from $\sum_{\chi \in \widehat{G}} \chi(g) = |G| \delta_{g, \mathbb{1}}$, for every $g \in G$, that

$$\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2) = \frac{1}{|G|} \left(\sum_{\chi_1 \in \widehat{G}} \chi_1(\gamma_1) \right) \circ \left(\sum_{\chi_2 \in \widehat{G}} \chi_2(\gamma_2) \right), \quad (65)$$

enforces holonomies $\text{hol}_{\gamma_1}(\mathfrak{g})$ and $\text{hol}_{\gamma_2}(\mathfrak{g})$ of the gauge field $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$ along γ_1 and γ_2 , respectively, to evaluate to the identity element in G . In other words, it projects onto the subspace of the kinematical Hilbert space spanned by vectors $|d^{(0)}\mathfrak{m}\rangle$, where $\mathfrak{m} \in C^0(\mathbb{T}_\Delta^2, G)$. Accordingly, we have $\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2) \circ \chi(\ell) = \chi(\ell) \circ \mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2) = \mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2)$, for every $\chi \in \widehat{G}$ and $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$. Moreover, it is immediate from eq. (65) that acting twice with $\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2)$ amounts to acting with $|G| \cdot \mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2)$.

In a similar vein, for every subgroup $A \subseteq G$, one defines a condensation operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ obtained by introducing a network of Wilson line operators associated with the representation $\text{Ind}_A^G(\mathbb{0}_A) \cong \mathbb{C}^{G/A}$. But, by Frobenius reciprocity, the representations that appear in $\text{Ind}_A^G(\mathbb{0}_A)$ are precisely the representations in G that are trivial when restricted to A , so that the set of such representations is isomorphic to $\widehat{G/A}$.⁸ Bringing everything together, one defines

$$\mathbb{U}^A(\mathbb{T}_\Delta^2) := \frac{1}{|G/A|} \left(\sum_{\chi_1 \in \text{Ind}_A^G(\mathbb{0}_A)} \chi_1(\gamma_1) \right) \circ \left(\sum_{\chi_2 \in \text{Ind}_A^G(\mathbb{0}_A)} \chi_2(\gamma_2) \right). \quad (66)$$

It follows from $\sum_{\chi \in \text{Ind}_A^G(\mathbb{0}_A)} \chi(g)$ being equal to $|G/A|$ whenever $g \in A$, and zero otherwise,⁹ that $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ has the effect of projecting onto charge sectors where holonomies along both γ_1 and γ_2 evaluate to group elements in A . Moreover, we have

$$\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = |G/A| \cdot \mathbb{U}^A(\mathbb{T}_\Delta^2). \quad (67)$$

Naturally, $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ boils down to $\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_\Delta^2)$ when specialising to $A = \{\mathbb{1}\}$. By construction, Wilson lines labelled by representations in $\text{Ind}_A^G(\mathbb{0}_A)$ condense on this condensation operator in the sense that $\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \chi(\ell) = \chi(\ell) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = \mathbb{U}^A(\mathbb{T}_\Delta^2)$, for every $\chi \in \text{Ind}_A^G(\mathbb{0}_A)$ and $\ell \in Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$, whereas Wilson lines generating the $\text{Rep}(A)$ 1-form subsymmetry survive on it. In other words, even while acting with $\mathbb{U}^A(\mathbb{T}_\Delta^2)$, the model still possesses a faithful $\text{Rep}(A)$ 1-form symmetry. Importantly, these are not the only topological line operators that can equip

⁸Since G is Abelian, every subgroup is normal and thus the quotient G/A is always a group. As a matter of fact, since G is finite Abelian, G admits a subgroup that is isomorphic to G/A , for any $A \subseteq G$.

⁹Given any subgroup $R \subseteq G$, there exists $A \subseteq G$ and a group isomorphism $\varphi : G/A \xrightarrow{\sim} R$. Let $g \in G$ such that $g \notin A$; it follows from $G = \bigsqcup_{rA \in G/A} rA$ that $g = ra$ with $a \in A$ and $r \neq \mathbb{1}$. Therefore, $\sum_{\chi \in \text{Ind}_A^G(\mathbb{0}_A)} \chi(g) = \sum_{\chi \in \text{Ind}_A^G(\mathbb{0}_A)} \chi(r) = \sum_{\chi \in \widehat{R}} \chi(\varphi(rA)) = 0$, as desired.

the topological surface operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)$. Indeed, one can also decorate $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ with networks of 't Hooft lines labelled by left cosets in G/A so that the resulting surface operator projects onto charge sectors where holonomies along both γ_1 and γ_2 evaluate to any group element in G . Topological interfaces between condensation operators $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ associated with different choices of $A \subseteq G$ can be constructed in a similar fashion [22, 27].

Before introducing the remaining surface operators, it is very useful to provide a microscopic realisation of $\mathbb{U}^A(\mathbb{T}_\Delta^2)$, for every $A \subseteq G$, in terms of tensor networks, applying the general framework outlined in ref. [22]. First of all, notice that the condensation operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ can be equivalently expressed as follows:

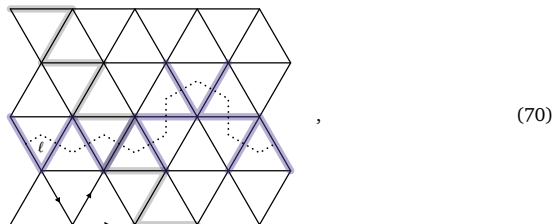
$$\mathbb{U}^A(\mathbb{T}_\Delta^2) = \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left(\prod_{(v_1 v_2)} \delta_{g(v_1 v_2) \triangleright m(v_2), m(v_1)} \right) |g\rangle \langle g|. \quad (68)$$

Indeed, since for every $e \equiv (v_1 v_2) \in E(\mathbb{T}_\Delta^2)$, there is a Kronecker delta imposing $g(v_1 v_2) \triangleright m(v_2) = m(v_1)$, it follows from the comments in sec. 2.5 that there exists a unique group element $a_{g(v_1 v_2), m(v_2)} \in A$ such that $g(v_1 v_2) = \text{rep}(m(v_1)) \cdot a_{g(v_1 v_2), m(v_2)} \cdot \text{rep}(m(v_2))^{-1}$, where $\text{rep} : G/A \rightarrow G$ assigns to every left coset in G/A its representative in G . Moreover, by definition, we have $a_{g_1, g_2 \triangleright rA} \cdot a_{g_2, rA} = a_{g_1 g_2, rA}$, for every $g_1, g_2 \in G$ and $rA \in G/A$. Consequently, defining $a_{g, m} \in C^1(\mathbb{T}_\Delta^2, A)$ via $a_{g, m}(v_1 v_2) := a_{g(v_1 v_2), m(v_2)}$, for every $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$ such that $g(v_1 v_2) \triangleright m(v_2) = m(v_1)$, one finds $a_{g, m}(v_1 v_2) a_{g, m}(v_2 v_3) = a_{g, m}(v_1 v_3)$, for every $(v_1 v_2 v_3) \in P(\mathbb{T}_\Delta^2)$, and thus $a_{g, m} \in Z^1(\mathbb{T}_\Delta^2, A)$. This confirms that $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ acts as a projector onto the subspace spanned by states $|a d^{(0)} \text{rep}(m)\rangle$, where $a \in Z^1(\mathbb{T}_\Delta^2, A)$ and $m \in C^0(\mathbb{T}_\Delta^2, G/A)$. Furthermore, we recover the fact that $\mathbb{U}^A(\mathbb{T}_\Delta^2) \circ \mathbb{U}^A(\mathbb{T}_\Delta^2) = |G/A| \cdot \mathbb{U}^A(\mathbb{T}_\Delta^2)$, as following from the fact that the Cartesian product $G/A \times G/A$ decomposes into a disjoint union of $|G/A|$ copies of G/A under the diagonal action of G .

Invoking (68), one can explain how to construct the surface operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)[f]$ that is decorated by a network f of 't Hooft lines. Let us consider a concrete example. Let ℓ be an oriented 1-cycle along the Poincaré dual $\mathbb{T}_\Delta^{2V}$ of $\mathbb{T}_\Delta^2$. Consider the 't Hooft loop labelled by $fA \in G/A$ with support ℓ . One defines the 1-cocycle $f \in Z^1(\mathbb{T}_\Delta^2, G/A)$ associated with this 't Hooft loop as follows:

$$f(e) = \begin{cases} fA, & \text{if } \iota(e) \subset \ell, \\ f^{-1}A, & \text{if } \iota(-e) \subset \ell, \\ \mathbb{1}, & \text{otherwise,} \end{cases} \quad (69)$$

for every oriented edge $e \in E(\mathbb{T}_\Delta^2)$, where $\iota : E(\mathbb{T}_\Delta^2) \xrightarrow{\sim} E(\mathbb{T}_\Delta^{2V})$. When defining the corresponding surface operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)[f]$, we replace the Kronecker deltas entering the definition (68) by $\delta_{g(v_1 v_2) \triangleright m(v_2), f(v_1 v_2) m(v_1)}$ for every $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$. Supposing that $\ell \equiv \gamma_1$ is one of the generators of $H_1(\mathbb{T}_\Delta^{2V}, \mathbb{Z})$, we depict such a configuration on the triangular lattice below



(70)

where the bold purple lines represent the only edges to which f assigns non-trivial elements in G/A . There is a choice of $m \in C^0(\mathbb{T}_\Delta^2, G/A)$ such that a gauge field satisfying $g(v_1 v_2) \triangleright m(v_2) = f(v_1 v_2) m(v_1)$, at every edge $(v_1 v_2)$, is non-vanishing along the bold gray lines only. It follows from the various definitions that $\text{hol}_{\gamma_2}(g) \in fA$, for any continuous deformation of the 't Hooft loop.

Still specialising to the case where $\mathbb{T}_\Delta^2$ is the triangular lattice, let us invoke eq. (68) to express $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ as a tensor network. One begins by introducing a rank- $(p+q)$ tensor in $\mathbb{C}[G/A]^{\otimes p} \rightarrow \mathbb{C}[G/A]^{\otimes q}$ of the form $\sum_{\{r_i A\}_i} (\prod_{i=1}^{p+q-1} \delta_{r_i A, r_{i+1} A}) \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A|$, graphically depicted as

$$\begin{aligned} \text{Diagram} &\equiv \sum_{\{r_i A\}_{i=1}^{p+q}} \text{Diagram} \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A| \\ &\equiv \sum_{\{r_i A\}_{i=1}^{p+q}} \left(\prod_{i=1}^{p+q-1} \delta_{r_i A, r_{i+1} A} \right) \otimes_{i=p+1}^{p+q} |r_i A\rangle \otimes_{i=1}^p \langle r_i A|, \end{aligned} \quad (71)$$

where the sums are over left cosets in G/A . Henceforth, we refer to such a tensor as a 'Kronecker delta tensor'. We also require a tensor implementing the G -action on G/A :

$$\begin{aligned} \text{Diagram} &\equiv \sum_{\substack{g \in G \\ r_1 A, r_2 A \in G/A}} r_1 A \xrightarrow{g} \text{Diagram} \xrightarrow{g} r_2 A |g\rangle \langle g| \otimes |g\rangle \langle g| \otimes |r_1 A\rangle \langle r_2 A| \\ &\equiv \sum_{\substack{g \in G \\ rA \in G/A}} |g\rangle \langle g| \otimes |g\rangle \langle g| \otimes |g\rangle \langle g| \triangleright rA |rA\rangle, \end{aligned} \quad (72)$$

as well as a tensor implementing the flatness of the gauge field:

$$\begin{aligned} \text{Diagram} &\equiv \sum_{g_1, g_2, g_3 \in G} \text{Diagram} |g_1, g_2\rangle \langle g_3| \equiv \sum_{g_1, g_2 \in G} |g_1, g_2\rangle \langle g_1 g_2|, \\ \text{Diagram} &\equiv \sum_{g_1, g_2, g_3 \in G} \text{Diagram} |g_3\rangle \langle g_1, g_2| \equiv \sum_{g_1, g_2 \in G} |g_1 g_2\rangle \langle g_1, g_2|. \end{aligned} \quad (73)$$

Bringing everything together, we can realise $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ as a tensor network whose unit cell is given by

$$\text{Diagram} \quad (74)$$

Let us dissect this tensor network. First of all, to every vertex of the triangular lattice, one assigns such a unit cell, in such a way that the Kronecker delta tensor coincides with the vertex,

and the purple lines with the edges of the lattice. Comparing with eq. (68), the Kronecker delta tensors perform the summation over $C^0(\mathbb{T}_\Delta^2, G/A)$, tensors (72) impose the constraints encoded into the Kronecker delta tensors, whereas the remaining tensors enforce the flatness of the gauge field. Notice that when realising $\mathbb{U}^{A_\psi}(\mathbb{T}_\Delta^2)$, it follows from $d^{(1)}(d^{(0)}m) = 0$, for every $m \in C^0(\mathbb{T}_\Delta^2, G)$, that the tensors imposing the flatness of the gauge field are superfluous and can thus be omitted.

It turns out that the model with Hamiltonian $\mathbb{H}^\lambda$ hosts as many topological surface operators as there are (indecomposable) module categories over the category Vec_G . We reviewed in sec. 2.5 that indecomposable Vec_G -module categories are labelled by pairs (A, ψ) consisting of a subgroup $A \subseteq G$ and a normalised representative ψ of a cohomology class $[\psi] \in H^2(A, \text{U}(1))$. Whenever the 2-cocycle ψ is trivial, the topological surface operator associated with the Vec_G -module category $\mathcal{M}(A, 1)$ is the condensation operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)$, as defined above [22]. A non-trivial 2-cocycle ψ modifies it in the following way:

$$\mathbb{U}^{A_\psi}(\mathbb{T}_\Delta^2) := \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left(\prod_{(v_1 v_2)} \delta_{g(v_1 v_2) \triangleright m(v_2), m(v_1)} \prod_{(v_1 v_2 v_3)} \psi(a_{g, m}(v_1 v_2), a_{g, m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} \right) |g\rangle \langle g|, \quad (75)$$

where we recognise the 2-cochain $\alpha_\psi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$ defined in eq. (44) the module associator of $\mathcal{M}(A, \psi)$ evaluates to. The commutation relation $[\mathbb{H}^\lambda, \mathbb{U}^{A_\psi}(\mathbb{T}_\Delta^2)] = 0$ follows from the 2-cocycle condition satisfied by ψ . Indeed, letting $v \in V(\mathbb{T}_\Delta^2)$ and $r_v \in C^0(\mathbb{T}_\Delta^2, G)$ such that $r_v(v_1) = 1$, whenever $v_1 \neq v$, the 2-cocycle condition ensures that

$$\prod_{(v_1 v_2 v_3)} \psi(a_{g \, d^{(0)} r_v, r_v \triangleright m}(v_1 v_2), a_{g \, d^{(0)} r_v, r_v \triangleright m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} = \prod_{(v_1 v_2 v_3)} \psi(a_{g, m}(v_1 v_2), a_{g, m}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)}, \quad (76)$$

for every $g \in Z^1(\mathbb{T}_\Delta^2, G)$ and $m \in C^0(\mathbb{T}_\Delta^2, G/A)$ such that $g(v_1 v_2) \triangleright m(v_2) = m(v_1)$ at every edge $(v_1 v_2) \in E(\mathbb{T}_\Delta^2)$, and where $(r_v \triangleright m)(v_1) := r_v(v_1) \triangleright m(v_1)$ for every $v_1 \in V(\mathbb{T}_\Delta^2)$. Introducing the tensors

$$\begin{aligned} \text{Diagram 1} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram 2} |g_1, g_2\rangle \langle g_3| \otimes \langle rA| \\ &\equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA}) |g_1, g_2\rangle \langle g_1 g_2| \otimes \langle rA|, \end{aligned} \quad (77)$$

and

$$\begin{aligned} \text{Diagram 3} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram 4} |g_3\rangle \langle g_1, g_2| \otimes |rA\rangle \\ &\equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \psi(a_{g_1, g_2 \triangleright rA}, a_{g_2, rA})^{-1} |g_1 g_2\rangle \langle g_1, g_2| \otimes |rA\rangle, \end{aligned} \quad (78)$$

we can realise $\mathbb{U}^{A_\psi}(\mathbb{T}_\Delta^2)$ on the triangular lattice as a tensor network whose unit cell is given

by¹⁰

where, in order to obtain the r.h.s., we used the fact that contracting Kronecker delta tensors results in another Kronecker delta tensor. As a special case, the topological surface operator associated with $\mathcal{M}(G, \psi)$ reads

$$\begin{aligned} \mathbb{U}^{G_\psi}(\mathbb{T}_\Delta^2) &= \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \left(\prod_{(v_1, v_2, v_3)} \psi(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v_3))^{\epsilon(v_1 v_2 v_3)} \right) |\mathfrak{g}\rangle \langle \mathfrak{g}| \\ &= \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)} \frac{\psi(\text{hol}_{\gamma_1}(\mathfrak{g}), \text{hol}_{\gamma_2}(\mathfrak{g}))}{\psi(\text{hol}_{\gamma_2}(\mathfrak{g}), \text{hol}_{\gamma_1}(\mathfrak{g}))} |\mathfrak{g}\rangle \langle \mathfrak{g}|, \end{aligned} \quad (79)$$

where we invoked the 2-cocycle condition of ψ to go from the first line to the second line.

We commented earlier that the condensation operator (68) could be obtained by summing over networks of Wilson lines valued in a subgroup of $\widehat{G}$ that is isomorphic to $\widehat{G/A}$, which should be thought as topological lines decorating the identity surface operator $\mathbb{U}^G(\mathbb{T}_\Delta^2)$. Conversely, the same operator $\mathbb{U}^A(\mathbb{T}_\Delta^2)$ can be obtained, starting from $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$, by summing over all possible networks of topological lines labelled by group elements in $A \subseteq G$, in such a way that the group multiplication in A —or rather the multiplication rule of the group algebra $\mathbb{C}[A]$ —is implemented at every junction. In this formulation, the operator $\mathbb{U}^{G_\psi}(\mathbb{T}_\Delta^2)$ is obtained starting from $\mathbb{U}^{\{1\}}(\mathbb{T}_\Delta^2)$ by summing over all possible networks of topological lines in G such that the multiplication rule of the twisted group algebra $\mathbb{C}[G]^\psi$ is implemented at every junction.

We have established that for every Vec_G -module category, one can construct a topological surface operator. Moreover, these topological surfaces can be decorated with networks of topological lines. In particular, the surface operator $\mathbb{U}^{A_\psi}(\mathbb{T}_\Delta^2)$ associated with the Vec_G -module category $\mathcal{M}(A, \psi)$ can be decorated with lines labelled by simple objects in the category $(\text{Vec}_G)_{\mathcal{M}(A, \psi)}^\vee$ defined in sec. 2.5, which happens to be equivalent to $\text{Rep}(A)^{\boxplus G/A}$ since the group G is Abelian, as expected. Bringing everything together, this indicates that the symmetry structure of Hamiltonians $\mathbb{H}^\lambda$ is provided by the fusion 2-category $\text{Mod}(\text{Vec}_G)$ of Vec_G -module categories and Vec_G -module functors. By analogy with $\text{Rep}(G) \simeq \text{Mod}(\mathbb{C}[G])$, it is enlightening to write $2\text{Rep}(G) \equiv \text{Mod}(\text{Vec}_G)$, where Vec_G is thought as a categorification of $\mathbb{C}[G]$ obtained by promoting the ring $\mathbb{C}$ to the fusion (1-)category Vec , and refer to objects in $2\text{Rep}(G)$ as 2-representations of G .

3.3 Symmetry twists

As mentioned earlier, the $2\text{Rep}(G)$ -symmetric Hamiltonians $\mathbb{H}^\lambda = \sum_{\mathbf{v}} \sum_n \mathbb{H}_{\mathbf{v}, n}^\lambda$ defined in terms of local operators (52) assume periodic boundary conditions. Let us now consider symmetry-twisted closed boundary conditions obtained by inserting symmetry twists at various loci

¹⁰Notice that we can unambiguously omit certain arrows for visual convenience.

of the spatial manifold. The same way we distinguish two types of symmetry operators, namely topological surfaces and topological lines, we distinguish two types of symmetry twists. These are localised along specific spatial 1-cycles and 0-cycles, respectively, while extending in the time direction. Henceforth, we refer to the resulting boundary conditions as being 0-form symmetry-twisted or 1-form symmetry-twisted, respectively. When implementing such symmetry-twisted boundary conditions, the kinematical Hilbert space typically requires to be supplemented with appropriate degrees of freedom, but not always. In any case, the resulting symmetry-twisted boundary conditions preserve the translation invariance of the system, in the sense that there is a unitary isomorphism between Hilbert spaces associated with symmetry twists that are related by continuous deformations. Generally, the presence of 0-form and 1-form symmetry-twisted boundary conditions can be witnessed by the corresponding point-like and line-like charges by moving them along 1- and 2-cycles, respectively. Throughout this section, the focus will be on 1-form symmetry-twisted boundary conditions, since the 0-form ones can be deduced from them, the same way the topological surface operators descend from the topological line operators.

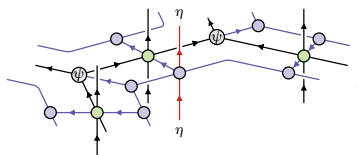
Starting from the kinematical Hilbert space $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in Z^1(\mathbb{T}_{\Delta}^2, G)\}$, consider inserting at a vertex $v_0 \in V(\mathbb{T}_{\Delta}^2)$ the topological line defect labelled by an irreducible representation $\eta : G \rightarrow \text{U}(1)$. Concretely, we perform this operation by considering the tensor product between the kinematical Hilbert space and $\mathbb{C}[\widehat{G}]$ before enforcing the degree of freedom in $\mathbb{C}[\widehat{G}]$ to be $|\eta\rangle$. We denote the resulting kinematical space by $\mathcal{H}^{\eta}$, which is still isomorphic to $\text{Span}_{\mathbb{C}}\{|g\rangle \mid g \in Z^1(\mathbb{T}_{\Delta}^2, G)\}$. Local symmetric operators acting at v_0 are modified in the following way:

$$\mathbb{H}_{v_0, n}^{\lambda, \eta} := \sum_{g \in Z^1(\mathbb{T}_{\Delta}^2, G)} \sum_{\mathbf{r}_{v_0}} h_n(g, \mathbf{r}_{v_0}) \eta(\mathbf{r}_{v_0}) \lambda(g, \mathbf{r}_{v_0}) |g d^{(0)}_{\mathbf{r}_{v_0}}\rangle \langle g|, \quad (81)$$

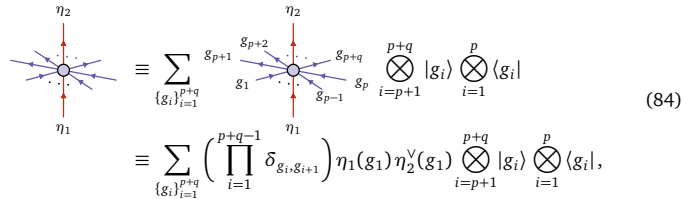
where $\eta(\mathbf{r}_{v_0}) := \eta(\mathbf{r}_{v_0}(v_0))$. All the other local operators are left intact so that the resulting local Hamiltonians read

$$\mathbb{H}^{\lambda, \eta} = \sum_{\substack{v \in V(\mathbb{T}_{\Delta}^2) \\ v \neq v_0}} \sum_n \mathbb{H}_{v, n}^{\lambda} + \sum_n \mathbb{H}_{v_0, n}^{\lambda, \eta}. \quad (82)$$

Since the group G is Abelian, the presence of the symmetry twist does not alter the symmetry structure of the Hamiltonian $\mathbb{H}^{\lambda, \eta}$, in the sense that symmetry operators are still organised in $2\text{Rep}(G)$. Nonetheless, they need to be adapted so as to accommodate the presence of the symmetry twist. This is very simple in the case of topological surface operators $\mathbb{U}_{\eta}^{G_{\psi}}(\mathbb{T}_{\Delta}^2) : \mathcal{H}^{\eta} \rightarrow \mathcal{H}^{\eta}$ as it simply requires taking the tensor product of the vanilla operator (80) with $|\eta\rangle\langle\eta|$. However, specialising to the case where $\mathbb{T}_{\Delta}^2$ is the triangular lattice, the topological surface operator $\mathbb{U}_{\eta}^{\{1\}}(\mathbb{T}_{\Delta}^2) : \mathcal{H}^{\eta} \rightarrow \mathcal{H}^{\eta}$ is obtained by replacing the unit cell of its tensor network parametrisation located at v_0 by the following one:

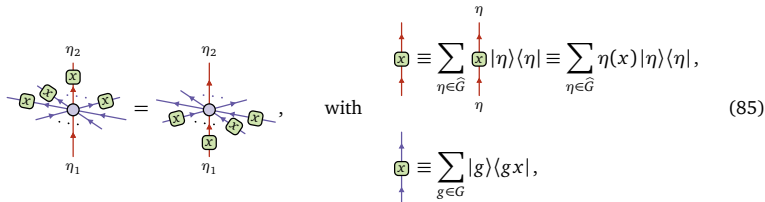

(83)

where we introduced the tensor



$$\begin{aligned} & \equiv \sum_{\{g_i\}_{i=1}^{p+q}} \eta_1(g_1) \eta_2(g_2) \bigotimes_{i=p+1}^{p+q} |g_i\rangle \bigotimes_{i=1}^p \langle g_i| \\ & \equiv \sum_{\{g_i\}_{i=1}^{p+q}} \left(\prod_{i=1}^{p+q-1} \delta_{g_i, g_{i+1}} \right) \eta_1(g_1) \eta_2(g_2) \bigotimes_{i=p+1}^{p+q} |g_i\rangle \bigotimes_{i=1}^p \langle g_i|, \end{aligned} \quad (84)$$

for every $\eta_1, \eta_2 \in \widehat{G}$. More generally, for an arbitrary triangulation of $\mathbb{T}_{\Delta}^2$, it suffices to replace the tensor of the form (71) located at v_0 by the corresponding one of the form (84). Importantly, tensors (84) satisfy the following invariance property:



$$\begin{aligned} & \equiv \sum_{\eta \in \widehat{G}} |\eta\rangle \langle \eta| \equiv \sum_{\eta \in \widehat{G}} \eta(x) |\eta\rangle \langle \eta|, \\ & \equiv \sum_{g \in G} |g\rangle \langle gx|, \end{aligned} \quad (85)$$

for every $x \in G$. Together with (76), this symmetry condition guarantees that the symmetry operator $\mathbb{U}^{(1)}(\mathbb{T}_{\Delta}^2)$ commutes with local operators $\mathbb{H}_{v_0, n}^{\lambda, \eta}$ acting at the locus of the 1-form symmetry twist. The remaining symmetry operators can be treated in a similar fashion but the tensors (84) take a more complicated form. For any choice of 1-form symmetry-twisted boundary condition $\eta \in \widehat{G}$, it follows from the $2\text{Rep}(G)$ symmetry—more specifically, the 1-form $\text{Rep}(G)$ symmetry—that Hamiltonians $\mathbb{H}^{|\lambda, \eta}$ still decompose into charge sectors labelled by $(g_1, g_2) \in G \times G$. Triples $(g_1, g_2, \eta) \in G \times G \times \widehat{G}$ label the *topological sectors* of the theory.

We mentioned above that we could also consider 0-form symmetry-twisted boundary conditions. For instance, consider inserting along a non-contractible cycle $\gamma_1 \in Z_1(\mathbb{T}_{\Delta}^2, \mathbb{Z})$ the topological surface defect labelled by simple 2-representations of the form $\mathcal{M}(G, \psi) \simeq \text{Vec}^{\psi} \in 2\text{Rep}(G)$, whose corresponding symmetry operator $\mathbb{U}^{G, \psi}(\mathbb{T}_{\Delta}^2)$ was defined in eq. (80). Similarly to the 1-form symmetry-twisted boundary conditions, this specific type of topological surface defect results in boundary conditions that do not alter the kinematical Hilbert space of the system. Local symmetric operators acting at vertices $v \subset \gamma_1$ are modified in the following way:

$$\mathbb{H}_{v, n}^{\lambda, \psi} := \sum_{g \in Z^1(\mathbb{T}_{\Delta}^2, G)} \sum_{\mathbf{r}_v} h_n(g, \mathbf{r}_v) \psi(g, \mathbf{r}_v) \lambda(g, \mathbf{r}_v) |g d^{(0)} \mathbf{r}_v\rangle \langle g|, \quad (86)$$

where

$$\psi(g, \mathbf{r}_v) := \prod_{(v_1 v) \subset \gamma_1} \psi(g(v_1 v) \mathbf{r}_v(v)^{-1}, \mathbf{r}_v(v)) \prod_{(v v_2) \subset \gamma_1} \psi(\mathbf{r}_v(v), g(v v_2))^{-1}. \quad (87)$$

All the other local operators are left intact so that the resulting local Hamiltonians read

$$\mathbb{H}^{\lambda, \psi} = \sum_{\substack{v \in V(\mathbb{T}_{\Delta}^2) \\ v \notin \gamma_1}} \sum_n \mathbb{H}_{v, n}^{\lambda} + \sum_{\substack{v \in V(\mathbb{T}_{\Delta}^2) \\ v \subset \gamma_1}} \sum_n \mathbb{H}_{v, n}^{\lambda, \psi}. \quad (88)$$

The remaining symmetry twists can be implemented in a similar fashion.

3.4 Duality operators

Given a finite Abelian group G , we constructed in sec. 3.1 families of two-dimensional quantum lattice models with $2\text{Rep}(G)$ symmetry, which are parametrised by normalised representatives λ of a cohomology class $[\lambda]$ in $H^3(G, \text{U}(1))$ as well as sets of complex coefficients $\{h_n\}_n$. We claim that models constructed in this way that only differ in the choice of cohomology class $[\lambda]$ are dual to one another. For now, let us assume periodic boundary conditions and construct the operator transmuting Hamiltonians $\mathbb{H}$ and $\mathbb{H}^\lambda$ into each other. Postponing more formal justifications to sec. 3.7, let us first motivate the algebraic structure encoding such duality operators by analogy with the one-dimensional case. In (1+1)d, we showed that Hamiltonians of $\text{Rep}(G)$ -symmetric models only differing in a choice of 2-cocycle ψ could be transmuted into each other by any duality operator encoded into the (1-)category $\text{Rep}^\psi(G) \simeq \text{Mod}(\mathbb{C}[G]^\psi)$ of ψ -projective representations of G . Thus we naively expect Hamiltonians of $2\text{Rep}(G)$ -symmetric models only differing in a choice of 3-cocycle λ to be transmutable into each other by any duality operator encoded into the 2-category $2\text{Rep}^\lambda(G)$ of ‘ λ -projective 2-representations of G ’. Consider the fusion category Vec_G of G -graded vector spaces, whose associator isomorphism, we recall, is given by the identity. Now, considering instead the associator

$$\lambda(g_1, g_2, g_3) \cdot 1_{\mathbb{C}_{g_1 g_2 g_3}} : (\mathbb{C}_{g_1} \otimes \mathbb{C}_{g_2}) \otimes \mathbb{C}_{g_3} \xrightarrow{\sim} \mathbb{C}_{g_1} \otimes (\mathbb{C}_{g_2} \otimes \mathbb{C}_{g_3}), \quad (89)$$

the resulting fusion category is denoted by Vec_G^λ . We then define the fusion 2-category $2\text{Rep}^\lambda(G)$ as the 2-category $\text{Mod}(\text{Vec}_G^\lambda)$ of Vec_G^λ -module categories. Indecomposable module categories over Vec_G^λ are defined similarly to module categories over Vec_G with the following difference: Only subgroups A such that $\lambda|_{A \times A \times A}$ is cohomologically trivial are admissible, and the module associator (43) now evaluates to a 2-cochain $\alpha_\phi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$, associated with a choice of 2-cochain $\phi \in C^2(A, \text{U}(1))$ such that $d^{(2)}\phi = \lambda^{-1}|_{A \times A \times A}$,¹¹ that satisfy

$$d^{(2)}\alpha_\phi^\triangleright(g_1, g_2, g_3)(rA) := \frac{\alpha_\phi^\triangleright(g_2, g_3)(rA) \alpha_\phi^\triangleright(g_1, g_2 g_3)(rA)}{\alpha_\phi^\triangleright(g_1 g_2, g_3)(rA) \alpha_\phi^\triangleright(g_1, g_2)(g_3 \triangleright rA)} = \lambda(g_1, g_2, g_3)^{-1}, \quad (90)$$

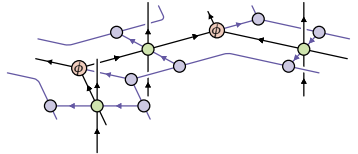
for every $g_1, g_2, g_3 \in G$ and $rA \in G/A$. The condition (90) guarantees the pentagon axiom, which now involves the monoidal associator (89). We denote the resulting Vec_G^λ -module category by $\mathcal{M}(A, \phi)$.

Given the data of a Vec_G^λ -module category $\mathcal{M}(A, \phi)$, consider the tensors

$$\begin{aligned} \text{Diagram 1} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram 2} |g_1, g_2\rangle \langle g_3| \otimes \langle rA| \equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \alpha_\phi^\triangleright(g_1, g_2)(rA) |g_1, g_2\rangle \langle g_1 g_2| \otimes \langle rA|, \\ \text{Diagram 3} &\equiv \sum_{\substack{g_1, g_2, g_3 \in G \\ rA \in G/A}} \text{Diagram 4} |g_3\rangle \langle g_1, g_2| \otimes \langle rA| \equiv \sum_{\substack{g_1, g_2 \in G \\ rA \in G/A}} \alpha_\phi^\triangleright(g_1, g_2)(rA)^{-1} |g_1 g_2\rangle \langle g_1, g_2| \otimes \langle rA|, \end{aligned}$$

¹¹Since we do not require it, we shall omit to write any explicit relation between the 2-cochains $\phi \in C^2(A, \text{U}(1))$ and $\alpha_\phi^\triangleright \in C^2(G, \text{Fun}(G/A, \text{U}(1)))$. Nonetheless, it is important to keep in mind that even if ϕ is trivial, the corresponding 2-cochain $\alpha_\phi^\triangleright$ may not be. For instance, whenever $A = \{1\}$, there is no choice of ϕ , and yet we have $\alpha_1^\triangleright(-, -)(-) = \lambda(-, -, -)$.

and denote by $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ the tensor network operator whose unit cell on the triangular lattice with periodic boundary conditions is given by


(91)

We claim that the operator $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ transmutates $\mathbb{H}$ into $\mathbb{H}^\lambda$, i.e., $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) \circ \mathbb{H} = \mathbb{H}^\lambda \circ \mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$. This computation is best carried out invoking the explicit expression

$$\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) := \sum_{\substack{g \in Z^1(\mathbb{T}_\Delta^2, G) \\ m \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left(\prod_{g(v_1 v_2) \triangleright m(v_2), m(v_1)} \prod_{(v_1 v_2 v_3)} \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v_3))(m(v_3))^{\epsilon(v_1 v_2 v_3)} \right) |g\rangle \langle g|. \quad (92)$$

Indeed, letting $v \in V(\mathbb{T}_\Delta^2)$ and $r_v \in C^0(\mathbb{T}_\Delta^2, G)$ such that $r_v(v_1) = \mathbb{1}$, whenever $v_1 \neq v$, it immediately follows from condition (90) that

$$\begin{aligned} & \alpha_\phi^\triangleright((g d^{(0)} r_v)(v_1 v_2), (g d^{(0)} r_v)(v_2 v)) (r_v \triangleright m)(v) \\ &= \frac{\alpha_\phi^\triangleright(g(v_2 v) r_v(v)^{-1}, r_v(v)) (m(v)) \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v)) (m(v)) \lambda(g(v_1 v_2), g(v_2 v) r_v(v)^{-1}, r_v(v))}{\alpha_\phi^\triangleright(g(v_1 v) r_v(v)^{-1}, r_v(v)) (m(v))}, \\ & \alpha_\phi^\triangleright((g d^{(0)} r_v)(v_1 v), (g d^{(0)} r_v)(v v_2)) (r_v \triangleright m)(v_2) \\ &= \frac{\alpha_\phi^\triangleright(g(v_1 v), g(v v_2)) (m(v_2)) \alpha_\phi^\triangleright(g(v_1 v) r_v(v)^{-1}, r_v(v)) (g(v v_2) \triangleright m(v_2))}{\alpha_\phi^\triangleright(r_v(v), g(v v_2)) (m(v_2)) \lambda(g(v_1 v) r_v(v)^{-1}, r_v(v), g(v v_2))}, \\ & \alpha_\phi^\triangleright((g d^{(0)} r_v)(v v_1), (g d^{(0)} r_v)(v_1 v_2)) (r_v \triangleright m)(v_2) \\ &= \frac{\alpha_\phi^\triangleright(g(v v_1), g(v_1 v_2)) (m(v_2)) \alpha_\phi^\triangleright(r_v(v), g(v v_2)) (m(v_2)) \lambda(r_v(v), g(v v_1), g(v_1 v_2))}{\alpha_\phi^\triangleright(r_v(v), g(v v_1)) (g(v_1 v_2) \triangleright m(v_2))}. \end{aligned} \quad (93)$$

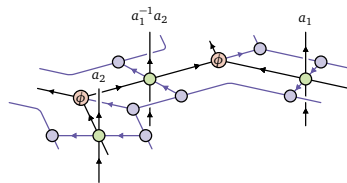
Bringing everything together, one obtains

$$\begin{aligned} & \prod_{(v_1 v_2 v_3)} \alpha_\phi^\triangleright((g d^{(0)} r_v)(v_1 v_2), (g d^{(0)} r_v)(v_2 v_3)) (r_v \triangleright m)(v_3))^{\epsilon(v_1 v_2 v_3)} \\ &= \lambda(g, r_v) \cdot \prod_{(v_1 v_2 v_3)} \alpha_\phi^\triangleright(g(v_1 v_2), g(v_2 v_3)) (m(v_3))^{\epsilon(v_1 v_2 v_3)}, \end{aligned} \quad (94)$$

where $\lambda(g, r_v)$ was defined in eq. (54). This guarantees that $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2) \circ \mathbb{H} = \mathbb{H}^\lambda \circ \mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$, as desired. Notice that the above derivation only relies on (90), which is verified for any Vec_G^λ -module category, and thus does not depend on a specific choice $\mathcal{M}(A, \phi)$. From this case, it would be easy to infer the duality operators transmuting Hamiltonians $\mathbb{H}^\lambda$ and $\mathbb{H}^{\lambda'}$ into each other.

Suppose λ is a non-trivial 3-cocycle. It follows from (90) that it is not possible to choose the whole group G to define a Vec_G^λ -module category, as otherwise it would require ϕ to be a 2-cochain in $C^2(G, \text{U}(1))$ trivialising λ , which would be a contradiction. Therefore, a duality operator $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ is necessarily associated with a Vec_G^λ -module category $\mathcal{M}(A, \phi)$ with A a proper subgroup of G . This implies in particular that any duality operator $\mathbb{D}^{A_\phi}(\mathbb{T}_\Delta^2)$ necessarily

projects the Hamiltonians $\mathbb{H}$ and $\mathbb{H}^\lambda$ onto specific charge sectors. Consider for instance the duality operator $\mathbb{D}^{(\mathbb{1})}(\mathbb{T}_\Delta^2)$ associated with the Vec_G^λ -module category $\mathcal{M}(\{\mathbb{1}\}, 1) \simeq \text{Vec}_G^\lambda$, whose module associator evaluates to λ so that (90) merely follows from $d^{(3)}\lambda = 1$. Similarly to the 0-form symmetry operator $\mathbb{U}^{(\mathbb{1})}(\mathbb{T}_\Delta^2)$, the duality operator $\mathbb{D}^{(\mathbb{1})}(\mathbb{T}_\Delta^2)$ enforces holonomies along both non-contractible cycles to be trivial, and thus acts as a projector onto the singlet sector of the 1-form $\text{Rep}(G)$ symmetry, making this duality operator non-invertible, and a fortiori non-unitary. Nonetheless, upon restricting to the singlet sector, the operator $\mathbb{D}^{(\mathbb{1})}(\mathbb{T}_\Delta^2)$ does become invertible and relates the spectra of the dual Hamiltonians. In order to access additional charge sectors, one option is to decorate the topological surface with topological lines labelled by elements in G , in the same vein as for the symmetry operator $\mathbb{U}^{(\mathbb{1})}(\mathbb{T}_\Delta^2)$ (see sec. 3.5). However, this is insufficient as the duality operator typically imposes additional constraints on the charge sectors that depend on the choice of λ . In the same vein, employing a different duality operator associated with a distinct $\mathcal{M}(A, \phi)$ could a priori permit to access additional charge sectors, as it would not necessarily project onto the singlet sector, but there would still be additional constraints that depend on λ . Indeed, suppose we are restricting to charge sectors $(a_1, a_2) \in A \times A$. It follows from the cocycle condition (90) that the duality explicitly depends on the complex number¹²



$$\text{Diagram} = \sum_{rA \in G/A} \frac{\alpha_\phi^\triangleright(a_1, a_1^{-1}a_2)(rA)}{\alpha_\phi^\triangleright(a_1^{-1}a_2, a_1)(rA)}, \quad (95)$$

where the l.h.s. should be interpreted as a single unit cell of the tensor network whose opposite indices are contracted so to as implement periodic boundary conditions. Consider now shifting the summation variable as follows: $rA \mapsto x \triangleright rA$, for any $x \in G$. But, repeatedly using the cocycle conditions (90) and $d^{(3)}\lambda = 1$, as well as the fact that $a \triangleright rA = rA$ for every $a \in A$ and $rA \in G/A$, one finds that

$$\frac{\alpha_\phi^\triangleright(a_1, a_1^{-1}a_2)(x \triangleright rA)}{\alpha_\phi^\triangleright(a_1^{-1}a_2, a_1)(x \triangleright rA)} = \frac{t_x(\lambda)(a_1, a_1^{-1}a_2)}{t_x(\lambda)(a_1^{-1}a_2, a_1)} \cdot \frac{\alpha_\phi^\triangleright(a_1, a_1^{-1}a_2)(rA)}{\alpha_\phi^\triangleright(a_1^{-1}a_2, a_1)(rA)}, \quad (96)$$

where $t_x : Z^3(G, U(1)) \rightarrow Z^2(G, U(1))$ is defined via [76]

$$t_x(\lambda)(g_1, g_2) := \frac{\lambda(x, g_1, g_2)\lambda(g_1, g_2, x)}{\lambda(g_1, x, g_2)}, \quad (97)$$

for every $g_1, g_2, x \in G$. Additional manipulations reveal that

$$\frac{t_x(\lambda)(a_1, a_1^{-1}a_2)}{t_x(\lambda)(a_1^{-1}a_2, a_1)} = t_{a_1, a_1^{-1}a_2}^2(\lambda)(x) = t_{a_1, a_2}^2(\lambda)(x), \quad (98)$$

where $t_{x,y}^2 : Z^3(G, U(1)) \rightarrow Z^1(G, U(1))$ is defined via

$$t_{x_1, x_2}^2(\lambda)(g) := \frac{t_{x_1}(\lambda)(x_2, g)}{t_{x_1}(\lambda)(g, x_2)}, \quad (99)$$

¹²To establish this fact, we are employing a higher dimensional version of the trick presented in eq. (15).

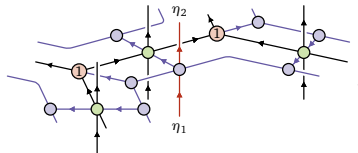
for every $g, x_1, x_2 \in G$. Bringing everything together, we showed that within the charge sector $(a_1, a_2) \in A \times A$, $\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = t_{a_1, a_2}^2(\lambda)(x) \cdot \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2)$, for every $x \in G$. Since it is true for every $x \in G$ and $t_{a_1, a_2}^2(\lambda) \in \widehat{G}$, we further have

$$\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = \left(\frac{1}{|G|} \sum_{x \in G} t_{a_1, a_2}^2(\lambda)(x) \right) \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2) = \begin{cases} \mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2), & \text{if } t_{a_1, a_2}^2(\lambda) = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (100)$$

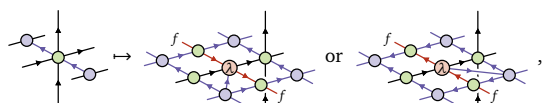
which means that charge sectors $(a_1, a_2) \in A \times A$, for which $t_{a_1, a_2}^2(\lambda)$ is not the trivial representation are projected out. In order to access all the charge sectors—regardless of the choice of duality operator—it is required to allow for the duality operator to modify the closed boundary conditions. We compute in the following the relevant permutation of topological sectors.

3.5 Permutation of topological sectors

As alluded to above, in order to lift a duality operator $\mathbb{D}^{A_\Phi}(\mathbb{T}_\Delta^2)$ to a unitary transformation, it is required to characterise the interplay between boundary conditions and charge sectors. First, we need to modify the duality operators so as to accommodate the presence of symmetry twists, the same way we modified the symmetry operators. Since for every group G and λ , the only duality operator whose existence we can guarantee is $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$, we shall focus on this case throughout the remainder of this section. Suppose a 1-form symmetry twist is inserted at the vertex v_0 . Specialising to the triangular lattice, we define the duality operator $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2) : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$ by replacing the unit cell (91) of its tensor network parametrisation located at v_0 by the following one:


(101)

where we are employing the tensors defined in eq. (84). It is clear from the previous derivations that $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2) \circ \mathbb{H}^{\lambda, \eta_1} = \mathbb{H}^{\lambda, \eta_2} \circ \mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$. Now, recall that the duality operator $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$ projects out states $|g\rangle$ such that $\text{hol}_{\gamma_1}(g)$ and $\text{hol}_{\gamma_2}(g)$ are not the identity element. We can lift this limitation by decorating $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{1\}}(\mathbb{T}_\Delta^2)$ by appropriate topological lines in G , thereby allowing to consider any charge sector $(g_1, g_2) \in G \times G$. More formally, the duality operator $\mathbb{D}^{\{1\}}(\mathbb{T}_\Delta^2)$ being associated with the Vec_G^λ -module category $\mathcal{M}(\{1\}, 1) \simeq \text{Vec}_G^\lambda$, it can be decorated with 't Hooft lines labelled by simple objects in the category $(\text{Vec}_G^\lambda)_{\mathcal{M}(\{1\}, 1)}^\vee \simeq \text{Vec}_G^\lambda$. Therefore, although the 't Hooft lines are effectively labelled by group elements in G , their action is more subtle than in (70) and depends in particular on λ . Correspondingly, it is not enough to implement the group multiplication at junctions in order to ensure topological invariance of network of lines. Since any network of 't Hooft lines boils down to a collection of single 't Hooft loops, by virtue of the group G being Abelian, let us focus for now on a single 't Hooft loop labelled by $f \in G$. Consider an oriented closed loop in $Z_1(\mathbb{T}_\Delta^2, \mathbb{Z})$. For every edge in this closed loop, we modify the corresponding local patch of the tensor network as follows


(102)

depending on the orientation of the edge relative to that of the loop, where we are using tensors adapted from eq. (72) in an obvious way and

$$\begin{aligned} \text{Diagram 1} &\equiv \sum_{f_1, f_2, g_1, g_2, x \in G} f_1 \text{Diagram 2} f_2 |f_2\rangle\langle f_1| \otimes |g_2\rangle\langle g_1| \otimes |x\rangle\langle x| \equiv \sum_{f, g, x \in G} \lambda(g, x, f) |f\rangle\langle f| \otimes |g\rangle\langle g| \otimes |x\rangle\langle x|, \\ \text{Diagram 3} &\equiv \sum_{f_1, f_2, g_1, g_2, x \in G} f_1 \text{Diagram 4} f_2 |f_1\rangle\langle f_2| \otimes |g_2\rangle\langle g_1| \otimes |x\rangle\langle x| \equiv \sum_{f, g, x \in G} \lambda(g, x, f)^{-1} |f\rangle\langle f| \otimes |g\rangle\langle g| \otimes |x\rangle\langle x|. \end{aligned}$$

One can check that the cocycle condition $d^{(3)}\lambda = 1$ ensures that the resulting duality operator is left invariant under continuous deformations of the closed loop. Although the definition of these operators may seem somewhat ad hoc, it is actually a reinterpretation of another tensor network calculus that is closely related to that of fusion 2-categories [22]. For instance, given the data of a Vec_G^λ -module endofunctor of $\mathcal{M}(A, \phi)$, there is a systematic way of constructing a tensor network operator, which boils down to that defined in eq. (102) whenever $\mathcal{M}(A, \phi) \simeq \text{Vec}_G^\lambda$. In order to construct the operator $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^1(\mathbb{T}_\Delta^2)[g_1, g_2]$ projecting onto a charge sector labelled by $[f] \in H^1(G, \mathbb{T}_\Delta^2)$ with holonomies $\text{hol}_{\gamma_1}(f) = g_1$ and $\text{hol}_{\gamma_2}(f) = g_2$, it suffices to insert two loop operators as defined above along both non-contractible cycle of the torus. However, as mentioned above, the junction of these two loops operators requires some care so as to be topologically invariant. Immediately specialising to our configuration of interest, one finds that the duality operator $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{\{\frac{1}{2}\}}(\mathbb{T}_\Delta^2)[g_1, g_2] : \mathcal{H}^{\eta_1} \rightarrow \mathcal{H}^{\eta_2}$ explicitly depends on the complex number

$$\begin{aligned} &\text{Diagram (103)} \\ &= \sum_{x \in G} \frac{\lambda(g_1, g_1^{-1}g_2, g_1x) \lambda(g_2, x, g_1) \lambda(xg_1, g_2g_1^{-1}, g_1)}{\lambda(g_1^{-1}g_2, g_1, x) \lambda(g_1, x, g_2) \lambda(x, g_1, g_1^{-1}g_2)} \eta_1^\vee(g_1x) \eta_2(g_1x) \\ &= \sum_{x \in G} t_{g_1, g_2}^2(\lambda)(x) \eta_1(g_1x) \eta_2^\vee(g_1x), \end{aligned} \tag{103}$$

where, as in eq. (95), the l.h.s. is interpreted as a single unit cell of the tensor network whose opposite indices are contracted so to as implement periodic boundary conditions. In addition to the unit cell (101) and tensors of the form (102), one introduced the following tensors that implement the fusion of topological lines in Vec_G^λ :

$$\text{Diagram 5} \equiv \sum_{f_1, f_2, f_3, x \in G} f_1 \text{Diagram 6} f_2 f_3 |f_1, f_2\rangle\langle f_3| \otimes |x\rangle\langle x| \equiv \sum_{f_1, f_2, x \in G} \lambda(x, f_2, f_1) |f_1, f_2\rangle\langle f_1f_2| \otimes |x\rangle\langle x|,$$

$$\text{Diagram} \equiv \sum_{\substack{f_1, f_2 \\ f_3, x \in G}} \text{Diagram} |f_3\rangle \langle f_1, f_2| \otimes |x\rangle \equiv \sum_{f_1, f_2, x \in G} \lambda(x, f_2, f_1)^{-1} |f_1 f_2\rangle \langle f_1, f_2| \otimes |x\rangle.$$

It simply follows from the orthogonality of representations in G that, in order for the operator $\mathbb{D}_{\eta_1 \rightarrow \eta_2}^{(\mathbb{1})}(\mathbb{T}_{\Delta}^2)[g_1, g_2]$ not to vanish, we must have $\eta_2(-) = \mathbf{t}_{g_1, g_2}^2(\lambda)(-) \eta_1(-)$. Bringing everything together, one obtains the following mapping of topological sectors:

$$(g_1, g_2, \eta) \mapsto (g_1, g_2, \eta_{g_1, g_2}^{\lambda}) \quad \text{with} \quad \eta_{g_1, g_2}^{\lambda}(-) := \mathbf{t}_{g_1, g_2}^2(\lambda)(-) \eta(-). \quad (104)$$

In particular, it follows from the normalisation of λ that the topological sectors of the form $(\eta, \mathbb{1}, \mathbb{1})$ are mapped to themselves under $\mathbb{D}^{A_{\Phi}}(\mathbb{T}_{\Delta}^2)$ for any $\eta \in \widehat{G}$. Finally, promoting the 1-form symmetry twist to a dynamical degree of freedom, the invertible duality operators between compatible topological sectors constructed above organise into a unitary operator onto the resulting total Hilbert space.

3.6 Examples

Let illustrate our formalism with a couple of examples. Let $G = \mathbb{Z}_2$. As in sec. 2.4, we write the multiplication rule of $\mathbb{Z}_2 = \{0, 1\}$ as addition modulo 2. A normalised representative of the cohomology class $[\lambda]$ generating $H^3(\mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$ is provided by $\lambda(g_1, g_2, g_3) = (-1)^{g_1 g_2 g_3}$, for every $g_1, g_2, g_3 \in \mathbb{Z}_2$. Given a two-dimensional quantum lattice model with a 0-form $\mathbb{Z}_2$ symmetry, we have two gauging procedures at our disposal, namely the untwisted one and the one twisted by the representative λ defined above. The two Hamiltonians resulting from these two gauging procedures possess a $2\text{Rep}(\mathbb{Z}_2)$ symmetry. Moreover, since they only differ in the choice of 3-cocycle λ , the corresponding duality operator is encoded into the unique simple object in the 2-category $2\text{Rep}^{\lambda}(\mathbb{Z}_2)$, which is $\text{Vec}_{\mathbb{Z}_2}^{\lambda}$ as a module category over itself. Let us consider the special Hamiltonian whose local operators (52) are obtained by choosing the non-vanishing complex coefficients $h(\mathfrak{g}, \mathfrak{r}_v)$ to be equal to -1 , for every $\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ and $\mathfrak{r}_v \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ such that $\mathfrak{r}_v(v) = 1$, i.e.,

$$\mathbb{H} = - \sum_v \prod_{e \supset v} X_e. \quad (105)$$

Recall that the kinematical Hilbert space is spanned by states $|\mathfrak{g}\rangle$, where $\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$. Making the kinematical constraint $\ker d^{(1)} \mathfrak{g} = 0$ dynamical would result in the celebrated (2+1)d toric code model [43]. Let us briefly comment on the $2\text{Rep}(\mathbb{Z}_2)$ symmetry structure. By definition, there are two topological surface operators, namely the identity operator and the condensation operator¹³

$$\mathbb{U}^{\{\mathbb{1}\}}(\mathbb{T}_{\Delta}^2) = \sum_{\substack{\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ \mathfrak{m} \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{\mathfrak{g}, d^{(0)} \mathfrak{m}} |\mathfrak{g}\rangle \langle \mathfrak{g}| = \frac{1}{2^{P(\mathbb{T}_{\Delta}^2)}} \left(\bigotimes_v (|+\rangle) \right) \prod_e (\mathbb{1} + Z_{\partial_e} Z_e Z_{\partial_e}) \left(\bigotimes_v (|+\rangle) \right), \quad (106)$$

where $|+\rangle := \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Decorating the trivial surface operator $\mathbb{U}^{\mathbb{Z}_2}(\mathbb{T}_{\Delta}^2)$ with the network of Wilson lines encoded into the 1-cocycle $\mathfrak{X} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ results into the surface operator¹⁴

$$\mathbb{U}^{\mathbb{Z}_2}(\mathbb{T}_{\Delta}^2)[\mathfrak{X}] \equiv \mathfrak{X}(\mathbb{T}_{\Delta}^2) = \sum_{\mathfrak{g} \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)} \exp \left(i\pi \int_{\mathbb{T}_{\Delta}^2} \mathfrak{X} \smile \mathfrak{g} \right) |\mathfrak{g}\rangle \langle \mathfrak{g}|. \quad (107)$$

¹³Recall that $P(\mathbb{T}_{\Delta}^2) = E(\mathbb{T}_{\Delta}^2) - V(\mathbb{T}_{\Delta}^2)$ since the Euler characteristic of $\mathbb{T}^2$ equals 0.

¹⁴Here, we are exploiting the isomorphism $\mathbb{Z}_2 \cong \widehat{\mathbb{Z}_2}$ in order to simplify the definition of the cup product (60).

Decorating $\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ with the network of 't Hooft lines encoded into the 1-cocycle $f \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ results into the surface operator

$$\mathbb{U}^{\{1\}}(\mathbb{T}_{\Delta}^2)[f] = \sum_{\substack{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{g, d^{(0)}m} |g\rangle \langle g| = \frac{1}{2^{P(\mathbb{T}_{\Delta}^2)}} \left(\bigotimes_v (+) \right) \prod_e \left(\mathbb{I} + (-1)^{f(e)} Z_{\partial_e} Z_e Z_{\partial_e} \right) \left(\bigotimes_v (+) \right).$$

Replacing in the definition of the local operators the trivial 3-cocycle by λ , while keeping everything else the same, yields the Hamiltonian $\mathbb{H}^{\lambda}$. Note that $\mathbb{H}^{\lambda}$ does not admit a particularly compact expression. However, specialising to the scenario where $\mathbb{T}_{\Delta}^2$ is the triangular lattice, one can perform the unitary transformation [68]

$$U := \sum_g \int_{\mathbb{T}_{\Delta}^2} i^g |g\rangle \langle g|, \quad (108)$$

so as to obtain

$$U \mathbb{H}^{\lambda} U^{\dagger} = \sum_v \prod_{e \supset v} X_e \prod_{\substack{(v_1 v_2) \\ (v_1 v_2 v_3) \\ (v_1 v_2 v_3)}} S_{(v_1 v_2)}, \quad (109)$$

where $S = |0\rangle \langle 0| + i|1\rangle \langle 1|$. At this point, making the kinematical constraint $\ker d^{(1)}g = 0$ dynamical would result in the Hamiltonian of the so-called *double semion* model [52, 58], which is the Hamiltonian realisation of the Turaev–Viro–Barrett–Westbury state-sum invariant with input fusion category $\text{Vec}_{\mathbb{Z}_2}^{\lambda}$ [15, 71]. Assuming periodic boundary conditions for both $\mathbb{H}$ and $\mathbb{H}^{\lambda}$, the duality operator $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ transmuting the Hamiltonians $\mathbb{H}$ and $\mathbb{H}^{\lambda}$ into each other takes the following form:

$$\begin{aligned} \mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2) &= \sum_{\substack{g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2) \\ m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)}} \delta_{g, d^{(0)}m} \left(\prod_{(v_1 v_2 v_3)} \lambda(g(v_1 v_2), g(v_2 v_3), m(v_3))^{\epsilon(v_1 v_2 v_3)} \right) |g\rangle \langle g| \\ &= \sum_{m \in C^0(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)} \left(\prod_{(v_1 v_2 v_3)} (-1)^{m(v_1)m(v_3)(m(v_2)+1)} \right) |d^{(0)}m\rangle \langle d^{(0)}m|. \end{aligned} \quad (110)$$

One can now verify that $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2) \circ \mathbb{H} = \mathbb{H}^{\lambda} \circ \mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$. What about the mapping of topological sectors? Since $\lambda(g_1, g_2, g_3) = \lambda(g_2, g_3, g_1) = \lambda(g_3, g_1, g_2)$, for every $g_1, g_2, g_3 \in \mathbb{Z}_2$, it turns out that $t_{g_1, g_2}^2(x) = 1$, for every $x \in \mathbb{Z}_2$. Therefore, any 1-form symmetry twisted boundary condition is exceptionally preserved under the duality transmuting $\mathbb{H}$ into $\mathbb{H}^{\lambda}$. Naturally, $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ is not a unitary operator on the total kinematical Hilbert space, since it projects both $\mathbb{H}$ and $\mathbb{H}^{\lambda}$ onto the singlet sector of the 1-form $\text{Rep}(\mathbb{Z}_2)$ -symmetry. In order to obtain the unitary operator relating the full spectra of $\mathbb{H}$ and $\mathbb{H}^{\lambda}$, it is required to decorate $\mathbb{D}^{\{1\}}(\mathbb{T}_{\Delta}^2)$ with networks of 't Hooft lines, following the procedure described in sec. 3.5.

Since the previous example is not accompanied with a non-trivial permutation of topological sectors, let us briefly consider a slightly more complicated example. Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \ni (g^{(1)}, g^{(2)}, g^{(3)}) \equiv g$. There is a normalised representative of a cohomology class in $H^3(\mathbb{Z}_2^3, \text{U}(1))$ defined by

$$\lambda(g_1, g_2, g_3) = \exp(i\pi g_1^{(1)} g_2^{(2)} g_3^{(3)}), \quad (111)$$

for every $g_1, g_2, g_3 \in \mathbb{Z}_2^3$. It follows from the general derivation that upon restricting to the charge sector $(g_1, g_2) \in \mathbb{Z}_2^3 \times \mathbb{Z}_2^3$, the duality operator $\mathbb{D}_{1 \rightarrow \eta}^{\{1\}}(\mathbb{T}_\Delta^2)[g_1, g_2]$ would be non-vanishing if and only if $\eta(-) = \mathbb{t}_{g_1, g_2}^2(-)$ with

$$\mathbb{t}_{g_1, g_2}^2(x) = \exp \left(i\pi \begin{vmatrix} g_1^{(1)} & g_2^{(1)} & x^{(1)} \\ g_1^{(2)} & g_2^{(2)} & x^{(2)} \\ g_1^{(3)} & g_2^{(3)} & x^{(3)} \end{vmatrix} \right), \quad (112)$$

for every $x \in \mathbb{Z}_2^3$. Therefore, for a generic charge sector, the 1-form symmetry twisted boundary condition of the Hamiltonian $\mathbb{H}^\lambda$ would need to be non-trivial.

3.7 Higher category theoretic underpinnings

Transposing sec. 2.5, let us sketch the mathematical formalism underlying the results presented above, following the approach of ref. [27]. Consider a two-dimensional quantum lattice model with a non-anomalous Abelian symmetry group G , which admits topological surfaces valued in G . The requirement that we should be able to construct topological interfaces between surfaces and topological junctions of interfaces invites us to rather consider a higher category theoretic structure, namely the *fusion 2-category* 2Vec_G of G -graded 2-vector spaces, where a 2-vector space is defined to be a finite semisimple $\mathbb{C}$ -linear (1-)category. In particular, simple objects in 2Vec_G are 2-vector spaces Vec_g , for every $g \in G$, such that $(\text{Vec}_{g_1})_{g_2} = \delta_{g_1, g_2} \text{Vec}$ with hom-categories $\text{Hom}_{2\text{Vec}_G}(\text{Vec}_{g_1}, \text{Vec}_{g_2}) \cong \delta_{g_1, g_2} \text{Vec}$, while the monoidal structure reads $\text{Vec}_{g_1} \otimes \text{Vec}_{g_2} = \text{Vec}_{g_1 g_2}$. Moreover, the monoidal associator evaluates to the identity 1-morphism, and it satisfies the pentagon axiom up to an invertible modification, known as the *monoidal pentagonator*, which evaluates to the identity 2-morphism. Given a non-anomalous symmetry G in (2+1)d—or rather, a symmetry 2Vec_G —it is always possible to gauge a sub-symmetry A , where A is a subgroup of G . As in (1+1)d, we identify several ways to perform such a gauging, which are now labelled by cohomology classes in $H^3(A, \text{U}(1))$. Whenever the whole symmetry is gauged, the resulting theory possesses a $2\text{Rep}(G)$ -symmetry, i.e., a symmetry structure encompassing both 0-form symmetry operators labelled by 2-representations of the group G , which compose according to the fusion of Vec_G -module categories [41], as well as 1-form symmetry operators labelled by Vec_G -module functors. Crucially, even though the group is Abelian, 2Vec_G and $2\text{Rep}(G)$ are clearly not equivalent, even as 2-categories.

Mathematically, a pair (A, λ) consisting of a subgroup $A \subseteq G$ and a normalised representative λ of a cohomology class $[\lambda] \in H^3(A, \text{U}(1))$ specifies a so-called *module 2-category* over 2Vec_G [21, 22]. Let $\mathcal{M}(A, \lambda)$ be the (finite semisimple) 2-category whose simple objects are 2-vector spaces Vec_{rA} , for every $rA \in G/A$. It is equipped with the (left) 2Vec_G -module structure provided by the G -action of left cosets in G/A , i.e. $\text{Vec}_g \triangleright \text{Vec}_{rA} := \text{Vec}_{g \triangleright rA} \simeq \text{Vec}_{(gA)rA}$, such that the module associator evaluates to the identity 1-morphisms, and it satisfies the pentagon axiom up to an invertible modification $\pi_\lambda^\triangleright$ referred as the *module pentagonator*. Components of the module pentagonator $\pi_\lambda^\triangleright$ evaluate to the 3-cochain $\pi_\lambda^\triangleright \in C^3(G, \text{Fun}(G/A, \text{U}(1)))$ defined in terms of the 3-cocycle λ via

$$\pi_\lambda^\triangleright(g_1, g_2, g_3)(rA) := \lambda(a_{g_1, g_2, g_3 \triangleright rA}, a_{g_2, g_3 \triangleright rA}, a_{g_3, rA}), \quad (113)$$

for every $g_1, g_2, g_3 \in G$ and $rA \in G/A$. It follows from the 3-cocycle condition satisfied by λ that

$$\frac{\pi_\lambda^\triangleright(g_2, g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2, g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2, g_3)(g_4 \triangleright rA)}{\pi_\lambda^\triangleright(g_1, g_2, g_3, g_4)(rA) \pi_\lambda^\triangleright(g_1, g_2, g_3, g_4)(rA)} = 1, \quad (114)$$

for every $g_1, g_2, g_3, g_4 \in G$ and $rA \in G/A$. Identity (114) guarantees the so-called *associahedron axiom* of $\mathcal{M}(A, \lambda)$ [21, 22].

Performing the λ -twisted gauging of a symmetry 2Vec_G amounts to picking the 2Vec_G -module 2-category $\mathcal{M}(G, \lambda) \equiv 2\text{Vec}^\lambda$, which is equivalent to 2Vec as a 2-category but whose 2Vec_G -module structure depends on λ . The same data encodes a choice of fiber 2-functor $2\text{Vec}_G \rightarrow 2\text{Vec}$. The symmetry structure of the theory resulting from the λ -twisted gauging of the subsymmetry A is provided by the fusion 2-category $(2\text{Vec}_G)_{\mathcal{M}(A, \lambda)}^\vee := 2\text{Fun}_{2\text{Vec}_G}(\mathcal{M}(A, \lambda), \mathcal{M}(A, \lambda))$ defined as the category of 2Vec_G -module 2-functors from $\mathcal{M}(A, \lambda)$ to itself, with fusion structure provided by the composition of 2Vec_G -module 2-functors [4, 5, 22, 27]. In particular, we have $(2\text{Vec}_G)_{2\text{Vec}^\lambda}^\vee \simeq 2\text{Rep}(G)$, for any $[\lambda] \in H^3(G, \text{U}(1))$, as expected.

We can think of the $2\text{Rep}(G)$ -symmetric models defined in sec. 3.1 as resulting from the λ -twisted gauging of G -symmetric models. Therefore, two models with Hamiltonians $\mathbb{H}$ and $\mathbb{H}^\lambda$ —only differing in a choice of 3-cocycle λ —are associated with the 2Vec_G -module categories 2Vec and 2Vec^λ , respectively. This can be made very explicit employing the formalism of ref. [27], from which it also follows that a duality operator transmuting $\mathbb{H}$ into $\mathbb{H}^\lambda$ corresponds to a choice of simple object in $2\text{Fun}_{2\text{Vec}_G}(2\text{Vec}, 2\text{Vec}^\lambda)$, which can be shown to be equivalent to $2\text{Rep}^\lambda(G)$, as expected.

3.8 Self-duality and lattice higher gauge theories

As in (1+1)d, a particularly attractive feature of the dualities considered in this section is that they preserve the symmetry structure, namely $2\text{Rep}(G)$ for some finite Abelian group G . As a matter of fact, it seems that all non-trivial dualities of bosonic two-dimensional quantum lattice models preserving symmetry structures are of this flavour, i.e., dualities between models only differing in a choice of fiber 2-functor over some input fusion 2-category [28]. This property suggests that it may be possible to define a model that is self-dual with respect to such a duality. Given such a self-dual model, one could then consider promoting the self-duality to a genuine internal symmetry, which mathematically amounts to considering an extension of the symmetry fusion 2-category.

Let us illustrate this principle with the first duality studied in sec. 3.6. There, we constructed two manifestly $2\text{Rep}(\mathbb{Z}_2)$ -symmetric Hamiltonians $\mathbb{H}$ and $\mathbb{H}^\lambda$, which are associated with fiber 2-functors $2\text{Vec}_{\mathbb{Z}_2} \rightarrow 2\text{Vec}$ and $2\text{Vec}_{\mathbb{Z}_2} \rightarrow 2\text{Vec}^\lambda$, respectively. Clearly, the Hamiltonian $\mathbb{H} + \mathbb{H}^\lambda$ is left invariant under the action of the duality operator (110), making $\mathbb{H} + \mathbb{H}^\lambda$ self-dual. We claim that promoting this self-duality to a genuine internal symmetry results in the fusion 2-category $2\text{Rep}(\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]})}$ of 2-representations of a 2-group $\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]}}$ with homotopy group $\mathbb{Z}_2$ both in degree one and two. Generally, a 2-group is defined as a monoidal category such that every object admits a weak inverse and every morphism is invertible. In particular, given two finite Abelian groups G and H , and a normalised representative 3-cocycle μ of a cohomology class in $H^3(G, H)$, the 2-group $G^{[0] \bullet_\mu H^{[1]}}$ is the monoidal category such that the objects form the group G , automorphisms of the monoidal unit forms the group H , while the monoidal associator is provided by μ .¹⁵ The fusion 2-category $2\text{Rep}(G^{[0] \bullet_\mu H^{[1]})}$ is then defined as the fusion 2-category of pseudofunctors from the so-called delooping of the 2-group to 2Vec .¹⁶ The underlying 2-category was studied in detail in ref. [32]:

$$2\text{Rep}(G^{[0] \bullet_\mu H^{[1]})} \simeq \bigsqcup_{\chi \in \hat{H}} 2\text{Rep}^{\chi \circ \mu}(G). \quad (115)$$

We are now ready to show that the self-dual Hamiltonian $\mathbb{H} + \mathbb{H}^\lambda$ has a $2\text{Rep}(\mathbb{Z}_2^{[0] \bullet_\mu \mathbb{Z}_2^{[1]})}$ symmetry structure, where $\mu(g_1, g_2, g_3) = 1 \in \mathbb{Z}_2$ if $g_1 = g_2 = g_3 = 1$, and 0 otherwise.

¹⁵More generally, the group H can be endowed with the structure of a G -module, but we choose the module structure to be trivial.

¹⁶Similarly, the fusion 2-category of 2-representations of a finite group G can be defined as the fusion 2-category of pseudofunctors from the delooping of the group to 2Vec .

Our strategy is to demonstrate that $\mathbb{H} + \mathbb{H}^\lambda$ has a lattice *higher gauge theory* interpretation [9, 9, 11, 25, 26, 46, 77] and results from gauging a two-dimensional quantum lattice model with a $2\text{Vec}_{\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]}}$ symmetry [27].

We begin by constructing a family of two-dimensional quantum lattice models with a lattice higher gauge theory interpretation that possess a $2\text{Rep}(G^{[0]} \bullet_\mu H^{[1]})$ symmetry, before specialising to $2\text{Rep}(\mathbb{Z}_2^{[0]} \bullet_\mu \mathbb{Z}_2^{[1]})$. The kinematical Hilbert space is spanned by vectors $|\mathfrak{g}, \mathfrak{h}\rangle$, where $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$ and $\mathfrak{h} \in C^2(\mathbb{T}_\Delta^2, H)$. Let μ be a normalised representative 3-cocycle of a cohomology class $[\mu] \in H^3(G, H)$. Given a vertex $v \in V(\mathbb{T}_\Delta^2)$, we denote by $\mathfrak{r}_v \in C^0(\mathbb{T}_\Delta^2, G)$ a function $V(\mathbb{T}_\Delta^2) \rightarrow G$ such that $\mathfrak{r}_v(v_1) = \mathbb{1}$ whenever $v_1 \neq v$. Moreover, we denote by $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, H)$ a function $E(\mathbb{T}_\Delta^2) \rightarrow H$ such that $\mathfrak{z}_v(v_1 v_2) = \mathbb{1}$ whenever $v \notin \partial(v_1 v_2)$. We consider the following local operators acting on the kinematical Hilbert space:

$$\mathbb{h}_{v,n} := \sum_{\substack{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G) \\ \mathfrak{h} \in C^2(\mathbb{T}_\Delta^2, H)}} \sum_{\mathfrak{r}_v, \mathfrak{z}_v} h_n(\mathfrak{g}, \mathfrak{h}, \mathfrak{r}_v, \mathfrak{z}_v) |\mathfrak{g} d^{(0)} \mathfrak{r}_v, \mathfrak{h} d^{(1)} \mathfrak{z}_v \mu(\mathfrak{g}, \mathfrak{r}_v)^{-1}\rangle \langle \mathfrak{g}, \mathfrak{h}|, \quad (116)$$

where n is some label, $h_n(\mathfrak{g}, \mathfrak{h}, \mathfrak{r}_v, \mathfrak{z}_v)$ are arbitrary complex coefficients, $\mathfrak{h} d^{(1)} \mathfrak{z}_v \mu(\mathfrak{g}, \mathfrak{r}_v)^{-1}$ is by definition the function $P[\mathbb{T}_\Delta^2] \rightarrow G$ such that

$$(\mathfrak{h} d^{(1)} \mathfrak{z}_v \mu(\mathfrak{g}, \mathfrak{r}_v)^{-1})(p) = \begin{cases} \mathfrak{h}(p) \mathfrak{z}_v(v_2 v) \mathfrak{z}_v(v_1 v)^{-1} \mu(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v))^{-1}, & \text{if } p \equiv (v_1 v_2 v), \\ \mathfrak{h}(p) \mathfrak{z}_v(v v_2) \mathfrak{z}_v(v_1 v) \mu(\mathfrak{g}(v_1 v) \mathfrak{r}_v(v)^{-1}, \mathfrak{r}_v(v), \mathfrak{g}(v v_2)), & \text{if } p \equiv (v_1 v v_2), \\ \mathfrak{h}(p) \mathfrak{z}_v(v v_1) \mathfrak{z}_v(v v_2)^{-1} \mu(\mathfrak{r}_v(v), \mathfrak{g}(v v_1), \mathfrak{g}(v_1 v_2))^{-1}, & \text{if } p \equiv (v v_1 v_2), \\ \mathfrak{h}(p), & \text{otherwise,} \end{cases} \quad (117)$$

where we defined the 2-cochain $\mu(\mathfrak{g}, \mathfrak{r}_v)$ analogously to the phase factor (54).

We claim that any Hamiltonian built from local operators (116) possesses a $2\text{Rep}(G^{[0]} \bullet_\mu H^{[1]})$ symmetry. Let us focus on the 0-form symmetry operators. It follows from eq. (115) that simple objects $2\text{Rep}(G^{[0]} \bullet_\mu H^{[1]})$ are labelled by pairs $(\chi \in \widehat{H}, \mathcal{M}(A, \phi) \in 2\text{Rep}^{\chi \circ \mu}(G))$. In particular, whenever χ is the trivial representation of H , the topological surface operator is labelled by a 2-representation of G , acts as in eq. (75), and commutes with local operators $\mathbb{h}_{v,n}$ for the same reasons as before. More generally, let us denote by $\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2)$ the topological surface operator associated with the pair $(\chi, \mathcal{M}(A, \phi))$:

$$\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2) := \sum_{\substack{\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G) \\ \mathfrak{h} \in C^2(\mathbb{T}_\Delta^2, H) \\ \mathfrak{m} \in C^0(\mathbb{T}_\Delta^2, G/A)}} \left(\prod_{(v_1 v_2)} \delta_{\mathfrak{g}(v_1 v_2) \triangleright \mathfrak{m}(v_2), \mathfrak{m}(v_1)} \right. \\ \left. \cdot \prod_{(v_1 v_2 v_3)} [\chi(\mathfrak{h}) \alpha_\phi^\triangleright(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v_3))(\mathfrak{m}(v_3))]^{\epsilon(v_1 v_2 v_3)} \right) |\mathfrak{g}, \mathfrak{h}\rangle \langle \mathfrak{g}, \mathfrak{h}|. \quad (118)$$

Letting $\mathfrak{r}_v \in C^0(\mathbb{T}_\Delta^2, G)$ and $\mathfrak{z}_v \in C^1(\mathbb{T}_\Delta^2, H)$ defined as above, it follows from eq. (94) as well as $\prod_{(v_1 v_2 v_3)} \chi(d^{(1)} \mathfrak{z}_v) = 1$ that

$$\prod_{(v_1 v_2 v_3)} [\chi(\mathfrak{h} d^{(1)} \mathfrak{z}_v \mu(\mathfrak{g}, \mathfrak{r}_v)^{-1}) \alpha_\phi^\triangleright((\mathfrak{g} d^{(0)} \mathfrak{r}_v)(v_1 v_2), (\mathfrak{g} d^{(0)} \mathfrak{r}_v)(v_2 v_3))(\mathfrak{r}_v \triangleright \mathfrak{m})(v_3))]^{\epsilon(v_1 v_2 v_3)} \\ = \prod_{(v_1 v_2 v_3)} [\chi(\mathfrak{h}) \alpha_\phi^\triangleright(\mathfrak{g}(v_1 v_2), \mathfrak{g}(v_2 v_3))(\mathfrak{m}(v_3))]^{\epsilon(v_1 v_2 v_3)}, \quad (119)$$

for every $\mathfrak{g} \in Z^1(\mathbb{T}_\Delta^2, G)$ and $\mathfrak{h} \in C^2(\mathbb{T}_\Delta^2, H)$, thereby confirming that $\mathbb{U}^{A, \phi, \chi}(\mathbb{T}_\Delta^2)$ commutes with local operators $\mathbb{h}_{v,n}$.

Let us now specialise to the 2-group $\mathbb{Z}_2^{[0]} \bullet_{\mu} \mathbb{Z}_2^{[1]}$. By definition, the Hamiltonian $\mathbb{H} + \mathbb{H}^{\lambda}$ constructed from eq. (105) and eq. (109) is made of local operators that acts as $-\prod_{e \supset v} X_e$, for every $v \in V(\mathbb{T}_{\Delta}^2)$, while projecting out states $|g\rangle$ satisfying $\lambda(g, r_v) = -1$. We argue that there is a choice of complex coefficients $h_1(g, h, r_v, \mathfrak{z}_v)$ such that local operators of the form (116) effectively act in the same way. Similarly to the example of sec. 2.6, the crux is to restrict the kinematical Hilbert space to states $|g, h\rangle$, for which the 2-cochain $h \in C^2(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$ assigns the identity group element to every plaquette. This can be enforced via a choice of coefficients $h_1(g, h, r_v, \mathfrak{z}_v)$. One can further choose these coefficients such that the only non-vanishing ones are equal to -1 , for every $g \in Z^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$, $\mathfrak{z}_v \in C^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$, and $r_v(v) = 1$. Proceeding as such, $(h d^{(1)}_{\mathfrak{z}_v} \mu(g, r_v)^{-1})(p) = 0$ enforces that $(d^{(1)}_{\mathfrak{z}_v} \mu(g, r_v)^{-1})(p) = 0$, for every $p \in P(\mathbb{T}_{\Delta}^2)$. For any $\mathfrak{z}_v \in C^1(\mathbb{T}_{\Delta}^2, \mathbb{Z}_2)$, the previous condition enforces that $\mu(g, r_v)(p)$ is distinct from the identity element in $\mathbb{Z}_2$ for an even number of plaquettes, precisely constraining g to be such that $\lambda(g, r_v) = 1$, as desired. Bringing everything together, this confirms that promoting the self-duality of $\mathbb{H} + \mathbb{H}^{\lambda}$ to an internal symmetry does result in a $2\text{Rep}(\mathbb{Z}_2^{[0]} \bullet_{\mu} \mathbb{Z}_2^{[1]})$ symmetry.

Naturally, one can construct much more general lattice higher gauge theories with a $2\text{Rep}(\mathbb{Z}_2^{[0]} \bullet_{\mu} \mathbb{Z}_2^{[1]})$ symmetry than the one discussed above. Given such a theory, one can then ask for a duality relation of the same type as those considered so far, i.e. combining twisted and untwisted gauging operations. This is possible since the fusion 2-category $2\text{Vec}_{\mathbb{Z}_2^{[0]} \bullet_{\mu} \mathbb{Z}_2^{[1]}}$ admits two inequivalent fiber 2-functors, which differ in a choice of 2-group 3-cocycle. Indeed, representatives of cohomology classes in $H^3(\mathbb{Z}_2^{[0]} \bullet_{\mu} \mathbb{Z}_2^{[1]}, U(1))$ are labelled by pairs (ϵ, γ) consisting of a 3-cochain $\gamma \in C^3(\mathbb{Z}_2, U(1))$ and a homomorphism $\epsilon : \mathbb{Z}_2 \rightarrow \widehat{\mathbb{Z}_2}$ such that $d\gamma = \langle \epsilon, \smile \mu \rangle$ [46]. The unique non-trivial solution is provided by choosing ϵ to be the non-trivial character of $\mathbb{Z}_2$ and $\gamma(g_1, g_2, g_3) = i^{g_1 g_2 g_3}$. Given a model that would happen to be self-dual under the resulting duality, lifting the self-duality to an internal symmetry would result in one of the Tambara–Yamagami fusion 2-categories considered in ref. [28].

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Conclusions and outlook

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